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Essays in the Political Economy of Growth, Distribution, and Crisis in the Center and Periphery

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Abstract

This dissertation investigates capitalist reproduction when capital accumulation, productive capacity, and macroeconomic stability cannot be assumed to move proportionally or independently of distribution, institutions, and external constraints. Productive capacity is not directly observed, yet its identification determines the denominator of capacity utilization. Three essays connect this measurement problem to the structural conditions of capacity formation and to the historical operation of monetary and external constraints in the periphery.

The first essay critically replicates an accounting-based capacity measure for the US corporate sector over 1947–2011. The full-sample bivariate output–capital system does not sustain the required long-run properties. A relation survives when the rate of exploitation and historical breaks enter the system, with retained transformation elasticities below unity. This result challenges the treatment of productive capacity as an autonomous technical benchmark.

The second essay develops the principal structural mechanism connecting accumulation to capacity through distribution, choice of technique, internally divided firms, and capital composition. Long-run estimates for the United States over 1931–2024 and Chile over 1943–2010 indicate non-unitary transformation. In Chile, the estimated wage-share elasticity of the capital composition gap falls from approximately 3.29 to 0.14 when the external constraint binds. The comparison is analytically asymmetric: import dependence and foreign-exchange availability alter the feasible transformation process in the periphery.

The third essay examines the crisis of Chile’s Unidad Popular through dependency theory, historical reconstruction, an open-economy central-bank balance-sheet framework, and monthly dynamic evidence. It distinguishes global monetary change, commodity-price movements, Chile-specific financial pressure, domestic conflict, and central-bank agency. In the adverse solvency-growth state, reverse monetary accommodation is more persistent and cumulatively larger than forward transmission, which remains significant. These findings are consistent with endogenous accommodation under external stress. They establish directional precedence and conditional responses, not experimental counterfactual causal identification.

Together, the essays connect capacity measurement to the relations governing its formation and show why peripheral reproduction also requires attention to international money and institutional choice. Their different designs do not statistically validate one another or establish a universal crisis law. They provide complementary explanations of how accumulation and macroeconomic reproduction depend on historically specific distributive, productive, and international relations.

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Introduction

The Problem of Productive Capacity and Capitalist Reproduction

Actual output is observed; productive capacity is not directly observed. Capacity utilization, $\mu \equiv Y/Y^p$, relates realized output Y to a productive ceiling Y^p whose identification requires an account of what the accumulated capital stock can produce. Surveys, engineering benchmarks, and statistical estimates supply different answers to that question. Treating their denominator as a neutral technical datum leaves unexplained the economic and institutional relations through which investment becomes usable capacity.

This dissertation investigates capitalist reproduction when accumulation, productive capacity, and macroeconomic stability do not move together automatically. The three essays move from the identification of capacity, through the structural determinants of its formation, to the historical operation of external and monetary constraints during Chile's Unidad Popular. They find, respectively, that an autonomous output-capital benchmark does not survive the full-sample system tests, that the transformation of accumulation into capacity is non-unitary and distributionally conditioned, and that monetary transmission in the Chilean case depends on external solvency conditions. Their common concern is how production can be reproduced under distributive conflict and institutional and international constraints; each addresses a different part of that problem.

The starting point is the distinction between capital accumulation and capacity formation. If productive capacity is assumed to expand proportionally with capital, the long-run capacity-to-capital ratio is fixed before it is measured. Chapters 1 and 2 instead investigate the transformation elasticity, $\theta \equiv \partial \ln Y^p / \partial \ln K$, where K denotes the capital stock. The benchmark $\theta = 1$ expresses proportional expansion. An elasticity below unity means that, within the specified relation, capacity grows less than proportionally with capital. The essays call this tendency structural overaccumulation. That definition does not by itself establish a complete theory of crisis.

Opening the transformation relation brings the balanced- and unbalanced-growth problem into the measurement of utilization. Decentralized investment decisions need not generate proportional reproduction, while the choice of technique depends on distribution and on how firms organize production. Machinery, structures, and intellectual property also enter the capital stock in different proportions. These are distinct dimensions: functional income distribution concerns wages and profits; capital composition concerns the assets accumulated; labor-process conflict concerns the organization of control and cooperation inside production. Productive capacity is formed through these relations. Utilization measures actual output relative to the resulting capacity.

For an import-dependent economy, the transformation also requires access to foreign exchange and external finance. A firm can retain an incentive to mechanize while facing limits on its ability to import machinery. During a peripheral crisis, however, the problem extends beyond long-run capacity formation to the reproduction of payments, enterprise liquidity, and international settlement. These questions require a historical account of institutions and agency. They provide the connection to Chapter 3 without making its monetary analysis another estimate of θ .

Fields of Inquiry and Research Questions

The dissertation enters four connected fields. The first is *capacity utilization, accumulation, and heterodox macroeconomics*: Post-Keynesian and Kaleckian debates over normal and realized utilization, Sraffian approaches to distribution and normal capital productivity, and classical-Marxian accounts of accumulation and profitability. As the first two essays discuss, a disagreement over the adjustment of utilization also raises a prior question about the denominator being adjusted (Kurz, 1986; Shaikh, 2016). The dissertation brings that identification problem into the debate over balanced and unbalanced growth.

The second field is *technical change, choice of technique, and the political economy of the firm*. Chapter 2 connects decentralized accumulation to the internally divided organization that converts investment into operating capacity, drawing on the frameworks of Okishio (1961, 2022) and Vidal (2019, 2022). Mechanization responds to distributive pressure, but its capacity consequences depend on capital composition and the organization of production. This field supplies the structural mechanism left open by the first essay's measurement results.

The third is *structuralist and dependency political economy*. Import dependence, uneven development, and the balance-of-payments constraint enter the explanation of why the conditions of accumulation differ between center and periphery. Chapter 2 asks how foreign-exchange availability conditions the choice of technique; Chapter 3 investigates the broader relations of financial subordination and international settlement. A balance-of-payments constraint limits import financing, whereas financial subordination describes an asymmetric international relation. The latter helps explain the conditions under which the former operates, but the two are not interchangeable.

The fourth is *historical and international monetary political economy*, centered on the Unidad Popular. Chapter 3 compares competing interpretations of the Chilean crisis and develops dependency theory as a research programme (Kvangraven, 2021; Lakatos, 1978). World money, international currency hierarchy, central-bank balance sheets, and endogenous monetary accommodation enter an inquiry that also requires historical contingency and institutional agency. History of economic thought is a transversal methodological dimension: Chapter 1 examines the premises embedded in capacity measurement, while Chapter 3 asks how competing research programmes select and interpret the historical evidence.

Three questions organize this inquiry:

1. **Measurement and identification.** How can productive capacity be identified empirically when it is unobserved and the long-run output–capital relation is historically and distributionally conditioned?
2. **Accumulation and structural mechanism.** What determines how accumulated capital becomes productive capacity, and how do distribution, choice of technique, capital composition, and external constraints condition that transformation?
3. **Peripheral crisis and historical political economy.** How do external finance, foreign-exchange solvency, distributive conflict, international monetary conditions, and institutional agency shape macroeconomic transmission during a peripheral structural transformation such as the Unidad Popular?

The United States and Chile belong together through an analytically asymmetric comparison. The US evidence provides a setting for examining capacity measurement and long-run relations among accumulation, distribution, technique, and capital composition within a center economy. Chile introduces conditions that cannot simply be inferred from that setting: dependence on im-

ported capital goods, limits on foreign-exchange availability, financial subordination, and peripheral monetary institutions. Center-periphery therefore changes the analytical problem rather than merely supplying a second country observation. The comparison does not establish a universal two-country law. In particular, Chapter 2 uses different specifications for the two economies, so its comparative findings concern mechanisms and the direction of relationships rather than directly interchangeable coefficient estimates.

From Measurement to Mechanism to Historical Crisis

Chapter 1 asks whether the long-run output-capital relation can sustain an autonomous empirical benchmark for productive capacity. Its critical replication identifies the distributional and historical conditioning of the relation that survives system testing. This raises the question subsequently examined in Chapter 2: how does distribution enter the conversion of accumulated capital into capacity? The second essay develops the dissertation's principal structural account and extends it to the external constraint facing peripheral accumulation.

Chapter 3 examines the historically specific operation of external and monetary constraints during the Unidad Popular. The question now concerns crisis transmission and institutional response, including how international money and the central bank's balance sheet condition domestic reproduction. The chapter inherits the broader problem of external constraint from Chapter 2, but it introduces a different object of explanation and a different evidentiary design. Its historical argument and dynamic estimates do not statistically validate the long-run transformation model.

Contributions of the Dissertation

Capacity Measurement and Heterodox Macroeconomics

The dissertation treats the productive-capacity denominator as an object requiring identification. Within Chapter 1's full-sample US replication, output and capital alone do not sustain the required long-run system properties. A relation survives when the rate of exploitation and historical breaks enter the system. The contribution is to challenge the interpretation of the output-capital coefficient as an insulated technical constant. It also qualifies the use of a fixed denominator in debates about normal utilization: the conditions of capacity formation must be examined before deviations from capacity can be interpreted.

Accumulation, Technical Change, and Capacity Formation

Chapter 2 turns the non-unitary output-capital result into a structural question and an estimable relation. Its account links distribution and cost-minimizing technique to decentralized investment and conflict within the firm, then incorporates heterogeneous capital composition. Accumulation can therefore expand without generating a proportional increase in productive capacity. This contribution connects the political economy of production to aggregate capacity measurement. The estimates characterize the resulting macroeconomic relation; they do not separately identify every workplace mechanism invoked by the theory.

Center–Periphery Political Economy

The Chilean extension shows why the external constraint changes the feasible transformation process when it binds. An incentive to substitute machinery for labor is insufficient if the means of acquiring imported equipment are restricted. In the estimated constrained state, the mechanization response is sharply attenuated. Foreign exchange thus enters the choice-of-technique mechanism itself, rather than serving only as another regression control. Read alongside Chapter 3, this result connects the material conditions of investment to the monetary and financial relations through which peripheral reproduction is sustained or disrupted.

Historical Political Economy, Dependency, and International Money

Chapter 3 reopens the Unidad Popular debate by linking a comparison of research programmes to historical reconstruction and an open-economy central-bank balance-sheet framework. It formalizes peripheral solvency in terms of foreign reserve assets relative to domestic monetary liabilities, while distinguishing international structural constraints from institutional choices. Its dynamic evidence is consistent with endogenous monetary accommodation under adverse external conditions and challenges accounts that treat money creation as an autonomous explanation of Chilean inflation ([Dornbusch and Edwards, 1990](#); [Edwards, 2024](#)). The contribution is an empirically grounded alternative interpretation whose causal reach remains bounded by the historical and time-series evidence.

Data, Measurement, and Empirical Strategy

These intellectual contributions rest on reconstructed historical capital stocks, archival monthly monetary and external series, staged critical replication, long-run cointegrating estimation, and threshold estimation of state-dependent relations. The data and methods make different aspects of reproduction observable and contestable. Their value lies in the questions they permit the essays to answer, rather than in a common estimator applied across all three.

Research Design, Evidence, and Levels of Analysis

Chapter 1 uses critical replication and system testing to answer an identification question. A single-equation fit can recover a plausible capacity path without establishing that the underlying variables sustain a stable long-run relation. The chapter therefore examines whether the benchmark survives changes in specification and joint-system tests. Chapter 2 combines theoretical construction with long-run comparative estimation to investigate the mechanism connecting accumulation to capacity. Historical shifts and, in Chile, an estimated external-constraint threshold enter that design.

Chapter 3 combines historical reconstruction, archival evidence, directional-precedence testing, and non-linear dynamics to investigate historically specific crisis transmission. Its monthly estimates use a broader historical sample to interpret the Unidad Popular episode; they are not estimates based only on the government’s three years. They identify systematic dynamic patterns and state-dependent conditional responses. Unobserved common shocks and structural breaks remain limitations, and the analysis does not provide experimental micro-level counterfactual causal identification.

Methodological plurality does not imply that evidence in one essay statistically validates the model or findings of another. Nor do the different estimated utilization paths or transformation

elasticities constitute a single harmonized series. The evidentiary division of labor supports a common argument about reproduction while preserving the identification conditions and limits of each essay.

The Three Essays

Essay 1: Measurement and Identification

Critical Replication of Shaikh's Capacity Utilization Measure asks whether an accounting-based output–capital relation can supply a capacity benchmark independent of distribution. It enters the heterodox measurement debate through a critical replication of [Shaikh \(2016\)](#) for the US corporate sector, 1947–2011. The design reconstructs the baseline, opens the single-equation specification space, and tests the relation at system level.

The full-sample bivariate system does not sustain cointegration and dynamic stability. The retained trivariate specifications include the rate of exploitation, defined as the ratio of profit share to wage share, together with historical controls; their transformation elasticity is below unity. The finding makes distributional conditioning part of the identification problem. It does not establish that this measured distributive ratio directly captures all labor-process conflict. The aggregate, single-sector framework and time-invariant parameterization limit the interpretation, and the mechanism of capacity formation remains the question for the next essay.

Essay 2: Structural Mechanism and Comparative Extension

The Transformation of Accumulation into Productive Capacities: Capacity Utilization in the Center and Periphery asks what determines the conversion of accumulation into capacity. It connects classical choice of technique to the political economy of the firm and then to the external constraint. The theoretical analysis and long-run estimation cover the United States over 1931–2024 and Chile over 1943–2010.

For the United States, the estimated elasticity remains below unity and varies with effective wage pressure and capital composition. Intellectual property enters as a component of capital, not as a functional income share. For Chile, the wage-share elasticity of the capital composition gap falls from approximately 3.29 in the unconstrained state to 0.14 in the constrained state. This is evidence of a sharply attenuated mechanization response, not an assertion that all investment ceases. The chapter supplies the principal structural mechanism of the dissertation while explicitly stopping short of a complete crisis theory, a fully specified profitability theory, or an exhaustive dependency framework.

Essay 3: Historical-Institutional Crisis Analysis

Re-visiting the Political Economy of the Rise and Fall of the Unidad Popular: Towards a Global Political Economy Approach asks how external finance, distributive conflict, international money, and central-bank solvency interacted during Chile's attempted socialist transformation. It places competing interpretations of the episode in dialogue with dependency theory, historical reconstruction, annual evidence, and monthly monetary and external data covering 1960–1980.

The historical account distinguishes five connected processes: the breakdown of Bretton Woods and global monetary instability; movements in world commodity prices and import costs; Chile-

specific bilateral and commercial financial pressure; domestic distributive and political conflict; and the institutional response of the Central Bank of Chile (BCCCh). Global monetary change altered the environment without mechanically causing the Chilean crisis. Targeted financial pressure narrowed the available policy space. Within that space, the BCCCh chose to accommodate enterprise liquidity needs rather than impose immediate deflationary adjustment. Constraint did not eliminate agency.

The threshold estimates distinguish monetary transmission across solvency-growth states. In the adverse state, the reported response of base-money growth to inflation is more persistent and cumulatively larger than the inflation response to base-money growth. Forward monetary transmission remains significant, so accommodation does not mean that money has no effect on prices. These conditional responses support an interpretation of endogenous accommodation under external stress; they do not isolate an experimental counterfactual or establish an inevitable policy response. The essay's contribution is to explain why the historical crisis requires an account of international settlement and institutional choice beyond the long-run capacity relation.

Structure of the Dissertation

The three essays follow the progression from measurement and identification, to structural mechanism and comparative extension, to historical-institutional crisis analysis. Each retains its own bibliography and appendices so that its argument and evidence can be evaluated on their own terms. A consolidated conclusion then draws together their implications and limits: capacity measurement depends on the relations governing capacity formation, while the reproduction of a peripheral economy also depends on external and monetary conditions that require their own historical explanation.

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Essay 1

Critical Replication of Shaikh's Capacity Utilization Measure

Abstract

Identifying a productive capacity ceiling requires estimating an unobserved benchmark, a task that risks conflating technical engineering limits with the institutional realities of capitalist accumulation. This chapter critically replicates Anwar Shaikh's accounting-based capacity measure for the post-war US corporate sector (1947–2011), reinterpreting the long-run output–capital coefficient as a capacity transformation elasticity (θ) under conditions of unbalanced growth. Testing this relation across a combinatorial grid of 500 single-equation specifications and a system-level Vector Error Correction Model (VECM), we demonstrate that the standalone bivariate output-capital relation fails to cointegrate. System-level stability emerges exclusively when the rate of exploitation (e_t) enters the cointegrating vector, yielding an estimated transformation elasticity ($\hat{\theta} < 1$) indicative of structural overaccumulation. This evidence suggests that the productive ceiling is not a neutral engineering constraint. Rather, it is a contingent outcome of decentralized accumulation, structurally conditioned by the distributive conflict and the labor process within and beyond the workplace, which physically dictate the conversion of accumulated capital into active productive capacity.

Keywords: Capacity Utilization; Critical Replication; Shaikh; Transformation Elasticity; Cointegration; ARDL; VECM; Distribution.

JEL Codes: B51; C32; E01; E22; E32; P16.

1.1 Introduction

The unobserved productive ceiling is not a neutral engineering constraint; it is the central macroeconomic variable linking capital accumulation, profitability, and class struggle. While actual output is driven by effective demand, productive capacity reflects the institutional limits of exploitation and the physical intensity of the labor process. [Shaikh \(2016\)](#) offers a portable, accounting-based solution to measure this ceiling via the long-run relation between aggregate output and capital stock. However, treating the estimated output–capital coefficient as a stable, physical law of production risks masking the historical and distributive conflicts that govern the workplace. This chapter critically replicates and extends Shaikh’s strategy to determine whether this empirical coefficient represents a stable technical relation or a historically contingent reflection of class struggle.

Distinct from the baseline approach of [Shaikh \(2016\)](#), this chapter reframes the aggregate output–capital coefficient as a capacity transformation elasticity (θ). We define the latent capacity ceiling in log-linear form as:

$$\ln Y_t^p = \theta \ln K_t + \ln R_0^n, \quad (1.1)$$

where Y_t^p is potential capacity, K_t is the capital stock, and R_0^n is a scale constant. Under the standard balanced-growth assumption of post-Keynesian models, this elasticity is constrained to unity ($\theta = 1$), meaning capacity expands in lockstep with capital. If $\theta \neq 1$, however, capacity formation decouples from accumulation. An elasticity below unity ($\theta < 1$) characterizes a structural overaccumulation regime where capital growth outpaces capacity expansion, while $\theta > 1$ leads to chronic excess capacity. By analyzing θ directly, we move beyond Shaikh’s focus on short-run residual fluctuations to investigate the long-run structural stability of the output–capital relation.

Evaluating this relationship reveals that Shaikh’s baseline results are approximately recoverable, yet the underlying elasticity is highly sensitive to model selection. In replicating the single-equation ARDL model of [Shaikh \(2016\)](#) using US corporate sector data (1947–2011), we come close to but do not exactly reproduce the original estimates due to undocumented lag structures and price deflator choices. Interrogating this fragility across a combinatoric specification space of 500 ARDL models reveals "disciplined non-uniqueness." Rather than converging to a single technical parameter, the estimated elasticity ranges from 0.65 to 0.95. This parameter volatility demonstrates that the recovered capacity path is a statistical artifact of researcher-chosen lag lengths and historical dummy variables, rather than a unique physical law.

Shifting the analysis to a joint system framework reveals that the bivariate output–capital relationship is spurious. In a bivariate Vector Error Correction Model (VECM), the output–capital relation fails cointegration and dynamic stability tests across all lag profiles. Cointegration and system-level stability emerge only when the logged rate of exploitation ($\ln e_t = \ln \pi_t - \ln \omega_t$) enters the state vector in a trivariate VECM. This cointegrating survival provides the central empirical finding of the chapter: following the classical choice-of-technique framework ([Kurz, 1986](#)), changes in functional distribution alter cost-minimizing techniques and normal capital productivity. When renormalized, the cointegrating vector reveals a reserve-army feedback elasticity ($\rho = -0.050$), showing that capacity utilization and distributive conflict adjust jointly within a single long-run attractor.

These findings define the historical limits of the accounting-based capacity measurement of [Shaikh \(2016\)](#). While the output–capital ratio remains a useful anchor to construct capacity utilization, the resulting ceiling cannot be treated as an insulated engineering benchmark. Cointegration survives only when the model is conditioned by the rate of exploitation and stabilized by historical

dummy controls that mark structural breaks in post-war US accumulation. Ultimately, the long-run output–capital relation cannot be estimated in isolation from the institutional conflicts of the capitalist labor process, which physically govern how capital is converted into potential output.

To establish this argument, the chapter unfolds along a narrative that historicizes the measurement problem before executing the econometric tests. We first reconstruct the historical evolution of capacity metrics in Section 2, tracing how survey indicators and output-gap routines shifted from tracking industrial output to enforcing price-level and wage discipline. In Section 3, we develop the conceptual Leontief framework, showing how omitting distribution introduces an identity trap that biases the capital coefficient, and detailing how Basu (2022)’s viability condition governs endogenous technical change. Section 4 presents the staged empirical stress tests—reconstructing the baseline, mapping the ARDL specification space, and testing system-level VECM cointegration. Section 5 concludes by detailing the theoretical implications of structural overaccumulation ($\hat{\theta} < 1.0$) for profit rate crisis theory, evaluating the interpretation of these distributive coefficients, and outlining the Chapter 2 point of departure for Global South economies.

1.2 A Historical Trace of Capacity Utilization Measurement

2.1 This section traces the capacity utilization index not as a neutral technical indicator, but as a historical product of macroeconomic statecraft and contested political economy. The metric emerged to make idle capital legible for Keynesian demand management, transformed into a sentinel for inflation control during the crisis of Fordism, and now structures neoliberal policy through technocratic output-gap benchmarks. Throughout these historical phases, the unobserved productive ceiling has remained conceptually underidentified. We trace this trajectory through four steps: the bureaucratic standardization of survey measures, the transition to output-gap governance, the theoretical debates within the post-Keynesian tradition between Sraffian and Kaleckian economists over long-run utilization, and Shaikh’s accounting-based framework.

1.2.1 From Survey Measures to an Inflation Sentinel: Capacity Utilization during the Fordist Era

2.2 Before capacity utilization could serve as a policy target, early researchers had to resolve a fundamental contradiction in its definition. The mass idleness of the Great Depression forced the state to quantify unused resources, exposing a tension between engineering and economic definitions of productive capacity (Burns, 1954; Klein and David, 1958). Engineering capacity represents the absolute physical limit of machine operation. Economic capacity, by contrast, defines the profit-maximizing level of output given prevailing wages, input costs, machine speeds, and other factors of technological development. From its inception, measuring capacity was not a neutral descriptive exercise, but an act of statecraft designed to identify the normal conditions of operation for capitalist firms, shaping economic understanding amidst the rise of American Keynesianism.

2.3 During the post-war expansion, Alvin Hansen’s secular stagnation thesis warned that chronic underutilization of capital was an inherent drag on long-run growth, forcing the state to confront the productive ceiling directly (Barber, 1987; Hansen, 1936, 1941). As John Hicks and Paul Samuelson codified these concerns into the IS-LM synthesis, demand management became the core paradigm of macroeconomic policy (Hicks, 1937; Samuelson, 1947, 1948). To stabilize this post-war compromise,

the state apparatus required quantitative indicators to detect productive bottlenecks and effective-demand shortfalls. Capacity utilization emerged as the primary metric to render capital idleness legible, transforming abstract stagnation concerns into concrete policy triggers.

2.4 The McGraw-Hill plant-and-equipment surveys standardized this legibility by asking managers to report their operating rates relative to preferred capacity benchmarks. Emerging from the private sector, these surveys served as a central information infrastructure for corporate planning and monopoly capital business strategy, facilitating coordination and forecasting before public initiatives took root (Baran and Sweezy, 1988; Munroe, 2007). The surveys folded heterogeneous, firm-level engineering realities into a uniform statistical reporting stream (Butler, 1958; Phillips, 1963). This private institutionalization depoliticized the capacity concept, replacing structural debates over the efficiency of capitalist production with a standardized language of business forecasting. When the Federal Reserve Board subsequently integrated these survey responses into its official Industrial Production index, it institutionalized these subjective manager estimates as the default benchmark for macroeconomic policy (Board of Governors of the Federal Reserve System, 2017). Alternative, objective physical measures—such as the workweek of capital—never secured comparable bureaucratic adoption (Foss, 1963; Taubman and Gottschalk, 1971).

2.5 The stagflation crisis of the 1970s dismantled the post-war demand-stabilization framework. Faced with falling profit rates and rising inflation, policymakers repurposed the capacity index: it was no longer an indicator of effective-demand shortfalls to be corrected, but a sentinel for inflation control (Shaikh et al., 1999). High capacity utilization was redefined as a warning of upcoming price pressures, prompting monetary tightening to cool the labor market. Academic and policy debates increasingly evaluated capacity measures by their ability to predict inflation, subordinating employment targets to price-level stability (Klein et al., 1973). In these debates, Alan Greenspan argued that capacity measures must reflect systemic shortages rather than administrative averages, asserting that "if the numbers do not indicate that the economy is currently at capacity, then I suggest that the numbers are wrong, and that the correct concept of capacity must reflect the situation" (Klein et al., 1973, p. 758). This shift did not resolve the unobserved nature of the productive ceiling, but it established the technocratic foundation for modern output-gap governance.

1.2.2 From Capacity Utilization to Output-Gap Governance

2.6 During the late twentieth century, the rise of global trade and financialization reduced the policy relevance of national manufacturing surveys. Central banks and international financial institutions demanded portable, rule-based indicators to monitor economic activity across national borders (Jessop, 2002, 2003; Konings, 2007). This demand institutionalized the "output gap"—the difference between actual output and potential output. This transition replaced discretionary demand management with technocratic, rule-based governance models, such as inflation targeting, Taylor policy functions, and central bank independence (Johnson, 2024; Taylor, 1993). Under the New Consensus Macroeconomics, potential output was redefined as the non-inflationary limit of economic activity, serving as an anchor to enforce wage discipline through NAIRU estimates (Saad-Filho, 2018). Yet this policy turn did not resolve the measurement problem: potential output remained a latent variable whose shape depended entirely on modeling assumptions. Like other technocratic indicators that mask political-economic conflicts behind fragile statistical choices

(Ash et al., 2017; Herndon et al., 2014), potential output is treated by policymakers as a neutral, engineering-like benchmark that depoliticizes underlying distributional conflicts.

2.7 Potential output, however, is unobservable. Hence, its estimation is contingent on specific paradigms. For example, mainstream economics typically relies on statistical filters that smooth observed output trends (Baxter and King, 1999; Beveridge and Nelson, 1981; Hamilton, 2018; Hodrick and Prescott, 1997), aggregate production functions (Alichi, 2015; Congressional Budget Office, 2004), or structural vector autoregressions (Blanchard and Quah, 1989). These aggregate methodologies abstract from institutional reality, treating capacity formation as a friction-free engineering exercise. In reality, the conversion of accumulation demand into productive capacity is institutionally and distributionally mediated, shaped by conflict at the workplace and competitive pressures in the market. The identification of potential output depends on the intensity of labor and the organization of the workplace, where class struggle over work schedules, machine speed, and supervisor discipline mediates how intensively machines run. In the market, capital strikes and shifts in the reserve army of labor regulate whether this potential capacity is realized. When low unemployment strengthens labor's bargaining power, profit squeeze dynamics prompt capital to idle capacity rather than expand production.

2.8 The 2008 Global Financial Crisis exposed the structural limits of this technocratic framework. Post-crisis revisions routinely flipped the estimated sign of output gaps, and downward-revised potential-output estimates were used to justify fiscal austerity and compress public investment in the European periphery (Heimberger, 2020; Heimberger and Kapeller, 2017). While supranational research divisions, such as those of the IMF, documented severe hysteresis and revision volatility (Cerra et al., 2023; Cerra and Saxena, 2017), policy surveillance accelerated the use of high-frequency nowcasting and forward-guidance models to preserve the output-gap apparatus as adequate for real-time policymaking (Aastveit and Trovik, 2014; Foster, 2024). This revision-prone, closure-dependent policy framework remains entrenched, providing the institutional backdrop against which heterodox debates over capacity utilization must be understood.

1.2.3 Capacity-Utilization Debates in Political Economy

2.9 Within the debates in political economy, the "capacity utilization controversy" defines clear dividing lines between contested approaches under a central inquiry: is the long-run normal rate of utilization endogenous to demand, or is it anchored in cost structures and classical competition? The Neo-Kaleckian tradition treats the normal rate as endogenous, arguing that firms adjust their long-run operating rates in response to demand shocks, micro-level shift choices, or historical hysteresis (Lavoie, 1995; Nikiforos, 2013; Setterfield and David-Avritzer, 2020). In contrast, Sraffian-Keynesian economists argue that cost minimization and classical competition anchor the normal rate of utilization in the long run, with capital investment adjusting capacity to meet changes in effective demand (Ciccone, 1986; Kurz, 1986).

2.10 Both heterodox positions face conceptual and empirical challenges when identifying the productive ceiling. In the Sraffian supermultiplier framework, convergence to a normal rate requires a component of aggregate demand to remain permanently autonomous, a condition that is difficult to reconcile with the debt dynamics of stock-flow consistent models (Nikiforos, 2018). On the other hand, the Neo-Kaleckian claim of endogenous normal utilization depends on capacity utilization

exhibiting non-stationarity (or path dependency) in the long run. This empirical premise has triggered a revival of the "capacity utilization controversy" in political economy. Critics like [Gahn and González \(2020\)](#) comment on Nikiforos's data, and [Gahn and González \(2022\)](#) present cross-country evidence, arguing that capacity utilization is stationary and mean-reverting, suggesting that actual utilization indeed converges to a structurally determined normal rate. However, these empirical tests remain highly contested, with their results depending entirely on the choice of data and unit-root testing specifications ([Nikiforos, 2020](#)).

2.11 By focusing primarily on manufacturing, both the FRB and AWW series fail to capture the structural shifts of modern, service-dominated economies ([Haluska, 2023](#)). Identifying the long-run dynamics of capacity formation therefore requires an alternative empirical approach that bypasses these survey-based and smoothing constraints entirely.

1.2.4 Shaikh's Structural Identification of Capacity Utilization

2.12 Rather than relying on survey measures or statistical filters that assume a symmetric business cycle and enforce mean reversion around a smooth potential trend, [Shaikh \(2016\)](#) shifts the identification of the capacity ceiling to classical profitability accounting. This methodology addresses a core conceptual limit of conventional filters: by smoothing out fluctuations, standard statistical routines treat expansions and contractions as mirror images, thereby obscuring the fundamental asymmetry of the business cycle—where slow, grinding booms are interrupted by rapid, violent contractions. This critique of symmetric filters aligns with Latin American neo-structuralist accounts by [Ffrench-Davis \(2010, 2012\)](#), who argue that deep contractions induce a persistent "recessive gap"—a demand-driven deficit that permanently scars productive capacity through hysteresis rather than mean-reverting. Both perspectives reject smoothing filters, showing that capacity utilization measurement must capture these asymmetrical paths. Shaikh's framework anchors capacity in the dynamics of capital accumulation and profitability. Using a Generalized Perpetual Inventory Method (GPIM) to construct the capital stock, he derives the potential-output path from the accounting identities linking actual profit rates to the maximum potential rate. Under this approach, capacity utilization is identified as the residual deviation of actual output from this long-run, profit-rate-determined capacity ceiling, preserving the historical and asymmetrical traces of accumulation and scrapping.

2.13 This accounting-based approach is methodologically significant because of its reproducibility and portability. By relying strictly on national accounts, alongside a theoretically grounded distinction of net capital stocks (capturing valuation dynamics) versus gross capital stocks (reflecting productive assets in operation), it provides a transparent and replicable framework for independent researchers to analyze accumulation dynamics without relying on country-specific survey designs or arbitrary statistical filters. However, treating capacity utilization as a deterministic residual assumes that output and capital share a stable, long-run cointegrating vector. If this underlying econometric relation is misspecified or unstable, the resulting capacity index will be highly sensitive to minor specification choices. This chapter critically replicates Shaikh's cointegration approach to systematically evaluate its stability. In doing so, it serves as a necessary stepping stone: by mapping the empirical boundaries and sensitivities of Shaikh's baseline model, we lay the groundwork to overcome its limitations in subsequent chapters.

1.3 Conceptual Framework

3.1 The conceptual framework of capacity utilization rests on a fundamental accounting challenge: while aggregate output is directly observed, the productive ceiling remains a latent, unobservable variable. This measurement problem is deeply entangled with the critique of aggregate production functions. Rather than trying to identify physical laws of production or marginal productivities through neoclassical functions, we follow Shaikh’s methodology which estimates capacity utilization from the long-run relation between aggregate output and capital stock. Because this estimated path serves as the denominator for the capacity utilization index, the properties of the resulting series depend entirely on the stability of the output–capital coefficient driving the long-run relation between them. To analyze this capacity ceiling, we ground this relationship in an aggregate Leontief production structure with fixed labor-capital proportions, which is the standard toolkit of the political economy of growth and distribution modeling. This framework treats capacity utilization not as a constant, but as a variable output-capital ratio that deviates from its normal ceiling.¹

3.2 To map this conceptual framework, this section proceeds as follows. First, we analyze the capital-to-capacity relationship within an aggregate Leontief production structure. Second, we evaluate the algebraic identity trap highlighted by the accounting identity critique (Shaikh, 1974; ?), demonstrating the misspecification problem of omitting distribution when estimating value-based macroeconomic relations. Third, we relax the balanced-growth assumption to define accumulation regimes and analyze capital accumulation instability. Finally, we analyze the trend-stabilized closure, recalling Basu’s viability criteria and the double-misspecification problem in econometrics specification to show how exogenous trends mask the endogeneity of technical change and distribution.

1.3.1 Production Function and Productive Capacities

The empirical validity of a capacity-utilization series depends strictly on the stability of the parameter that generates the fitted capacity path. While Shaikh (2016) avoids direct physical proxies, his empirical output-capital relation can be grounded in a standard classical Leontief production function:

$$Y_t = \min\{A_t L_t, \mu_t R_t^n K_t\}, \quad (1.2)$$

3.3 where Y_t is aggregate output, L_t is employment, K_t is the capital stock, A_t is labor productivity, R_t^n is the unobserved normal capacity-capital ratio (which is Shaikh’s original symbol for capital productivity at normal capacity utilization, Y_t^P/K_t), and μ_t is capacity utilization. In a capital-constrained regime where labor is not binding due to the reserve army of labor, this production function simplifies to $Y_t = \mu_t R_t^n K_t$. Taking natural logarithms yields:

$$\ln Y_t = \ln K_t + \ln R_t^n + \ln \mu_t. \quad (1.3)$$

This aggregate Leontief specification hides a concrete workplace-level transmission mechanism. At the shop-floor level, the conversion of the capital stock K_t into potential capacity ($Y_t^P = R_t^n K_t$)

¹Although we treat the capacity transformation elasticity, θ , as a technological parameter in this chapter, subsequent chapters will endogenize θ to show how distributive struggles and changes in the wage share shape the long-run capacity ceiling.

is not a passive technical constant. Rather, it is determined by the speed of machinery, the organization of shifts, and the physical intensity of labor. This conversion rate, captured in the normal capacity-capital ratio R_t^n , changes over time as capitalists introduce labor-saving technical modifications to increase labor discipline. Once potential capacity is established, the production of actual output Y_t is mediated by capacity utilization μ_t , which reflects the pace of market demand. Class struggle over the length and intensity of the workday directly moderates these conversions, determining both the maximum potential output capitalists can extract from a given capital stock and the actual rate of utilization they can sustain.

3.4 Isolating the long-run capacity path requires modeling the unobserved normal capacity-capital ratio ($\ln R_t^n$). [Shaikh \(2016\)](#) proposes a linear function of a deterministic time trend and the observed capital stock:

$$\ln R_t^n = bt + c \ln K_t. \quad (1.4)$$

3.5 Substituting this definition into the Leontief-derived log relation yields the estimable single-equation level relation:

$$y_t = a + bt + d k_t + \varepsilon_t, \quad d \equiv 1 + c, \quad (1.5)$$

3.6 In this regression, y_t and k_t represent the natural logarithms of unscaled output and capital, while d represents the long-run output-capital elasticity, which this chapter reinterprets as the unbalanced growth transformation elasticity θ . The constant a absorbs initial conditions and fixed measurement discrepancies in the levels of output and capital. The linear trend component (bt) is conventionally interpreted as a measure of autonomous technical change. Whereas Shaikh focuses on the error-correction term, interpreting it as the capacity utilization rate, this chapter focuses instead on the underlying long-run behavior driven by the relation between capital accumulation and productive capacity creation ($g_{Y^p} = \theta g_K$). Consequently, the deterministic trend included in a cointegrating level relation might obscure an omitted variable bias imposed by this theoretical misspecification.

3.7 The fitted path of productive capacity follows directly from the recovered long-run components:

$$\hat{y}_t^p = \hat{a} + \hat{b}t + \hat{d} k_t. \quad (1.6)$$

3.8 Capacity utilization is then constructed as the deviation of realized output from this long-run capacity path:

$$\ln \hat{\mu}_t = y_t - \hat{y}_t^p, \quad \hat{\mu}_t = \frac{Y_t}{\hat{Y}_t^p}. \quad (1.7)$$

3.9 This estimation sequence establishes a direct functional dependency, where the constructed utilization series inherits its level, timing, and variance from the estimated parameter $\hat{d}$. Consequently, different single-equation specifications yield divergent $\hat{d}$ estimates, producing contradictory trajectories for both capacity and utilization. Because the model forces a dynamic accumulation process into a rigid accounting identity, the empirical sensitivity of this parameter raises an immediate identification problem. Structural measurement therefore rests on evaluating whether this long-run multiplier captures a genuine behavioral law of production, or whether it operates merely as an algebraic placeholder that absorbs omitted historical variation.

1.3.2 Laws of Algebra and Laws of Production

3.10 The estimated long-run coefficient d operates as the transformation elasticity ($\theta \equiv \partial \ln Y^p / \partial \ln K$) anchoring the profit-rate decomposition of Shaikh (2016). Because the normal rate of profit relies directly on the constructed capacity–capital ratio, the structural integrity of the entire empirical operationalization of his theoretical contributions rests on this specific parameter. While the aggregate accounting identity deterministically organizes these profit magnitudes, it leaves the underlying behavioral mechanism translating capital accumulation into productive capacity undefined. Following the analytical standard of Shaikh (1974), an accounting identity that leaves its core behavioral mechanism undefined mechanically mimics a structural production law. As previously stated, this entails a misspecification risk: the inclusion of a trend variable might partially absorb historical and institutional variation overlooked in Shaikh’s formulation.

3.11 Evaluating Shaikh’s capacity regression reveals that it is vulnerable to the very critique he leveled against Solow’s aggregate production function. In his seminal critique, Shaikh (1974) showed that the high fit of aggregate production functions is a mathematical byproduct of the underlying value-added accounting identity ($p_t Y_t \equiv W_t + P_t = w_t L_t + r_t K_t$) under a constant wage share, rather than proof of physical substitution laws. As elaborated by ?, aggregate regressions using constant-price monetary data do not identify physical production laws, but instead approximate this value identity. Applying this critique to Shaikh’s own capacity regression reveals a similar limitation: since the output–capital ratio is linked by identity to the rate of profit (r_t) and the profit share (π_t) via $Y_t/K_t = r_t/\pi_t$, taking logarithms yields $y_t - k_t = \ln r_t - \ln \pi_t$. When we estimate the level relation over a period of shifting profit shares and class struggle, the deterministic trend bt does not necessarily identify an autonomous technological change component, but instead reflects distributive variation. The long-run coefficient d must adjust to satisfy the accounting identity, mimicking a stable physical relation while masking changes in the rate of exploitation.

3.12 To ground this value identity trap in post-war US history using the original data tables by Shaikh (2016), we observe that while real gross value added grew at an average annual rate of 3.3% and real gross capital stock grew at 4.4%, the output-to-capital ratio exhibited long-run stability but fluctuated significantly around a historical average of 0.53. More importantly, this variation is closely tied to shifts in the rate of exploitation ($e_t = \text{NOScorp}/\text{ECcorp}$), which acts as the empirical proxy for the rate of surplus value. During the Post-War Golden Age (1947–1973), the rate of exploitation was high and stable, averaging 0.319 (corresponding to a net profit share of 21.78%). In the subsequent stagflation transition (1974–1982), a structural profit squeeze caused the exploitation rate to collapse to a historical low of 0.256 (profit share of 18.54%). Finally, during the Neoliberal Era (1983–2011), the containment of real wage growth restored the exploitation rate to an average of 0.286 (profit share of 20.13%). Because the single-equation ARDL model excludes these distributional shifts, the estimated long-run technical coefficient is contaminated by this omitted historical variation, appearing stable only because it mathematically averages across these distinct regimes of class struggle. This confirms that the output–capital relationship is not a static engineering baseline, but is dynamically conditioned by the rate of exploitation.

3.13 The capital stock series K_t , reconstructed via Shaikh’s Generalized Perpetual Inventory Method (GPIM) to correct for infinite service-life biases, converges to a 99.6% average ratio with the BEA current-cost measure (1947–2005); the remaining measurement discrepancy is therefore largely

transitory and stationary. This discrepancy is not mere statistical noise; it reflects the material imprint of workplace dynamics—deferred maintenance, extended machine running times, and labor speed-ups—that allow physical capital to outlast its book-value depreciation schedule. Because this error is stationary, standard cointegration theory dictates it does not induce systematic asymptotic bias in the estimated cointegrating vector, though any finite-sample distortion cannot be signed *a priori*. We interpret our empirical estimates of θ conservatively, recognizing that the true structural conversion of capital into active productive capacity may be either somewhat more or less responsive than the baseline data suggests.

1.3.3 Balanced vs. Unbalanced Growth: Accumulation Regimes and Capital Accumulation Instability

3.14 Marx’s schemas of reproduction provide the analytical benchmark for proportional macroeconomic growth (Okishio, 2022). Smooth capitalist reproduction requires strict material proportions: commodities must sell, surplus value must be realized, and the material composition of output must match the input requirements of continuous capital accumulation (Basu, 2022). Because real capitalist economies routinely violate these proportions, smooth reproduction remains an analytical abstraction rather than a historical description. To isolate the supply-side relation between capital and capacity, the long-run classical closure ($\mu = 1$) holds demand instability constant (Foley, 1985). Alternatively, canonical post-Keynesian models enforce a balanced-growth baseline by assuming a unitary transformation elasticity ($\theta = 1$) in the long run. Combined with demand fluctuations, this unitary assumption generates the characteristic properties of the Harrodian knife-edge.

3.15 Introducing an unbalanced growth closure ($\theta \neq 1$) achieves long-run identification by treating proportional accumulation as suggested by Okishio (2022), as a restrictive special case of equilibrium accumulation trajectory. Relaxing the unitary assumption shifts the analytical focus away from the classical closure ($\mu = 1$). Under a non-unitary parameter, systematic instability is an inherent feature of capacity formation, distinct from temporary deviations driven by demand. The transformation elasticity (θ) governs the long-run development of productive capacity independently of short-run demand stabilization. In this setup, θ acts as a structural parameter, where the interest of this critical replication lies. Its endogenization to distributive conflict is beyond the scope of this chapter, but will be tackled in the next chapter of this dissertation.

3.16 Defining the transformation elasticity as $\theta \equiv \partial \ln Y^p / \partial \ln K$, the analytical properties of the accumulation system depend on whether this parameter deviates from unity. Table 1.1 defines three macroeconomic regimes bounded by these conditions.

Table 1.1. Accumulation Regimes and the Productive-Capacity Relation

Regime	Parameter Condition	Long-Run Relation ($\hat{Y}^p, \hat{K}$)	System Tendency
Balanced Growth	$\theta = 1.0$	Capital and capacity expand proportionally	Knife-edge baseline
Overaccumulation	$\theta < 1.0$	Capital expands faster than productive capacity	Accumulation decelerates toward stagnation
Excess Capacity	$\theta > 1.0$	Productive capacity expands faster than capital	Accumulation accelerates without supply-side limits

Note: This table defines the theoretical regimes governing the system.

3.17 This regime classification uses a single-sector macroeconomic model. While this simplifies the analysis of stagnation tendencies driven by overaccumulation ($\theta < 1$), it abstracts from inter-sectoral flows. In Marx’s reproduction schemas, disproportionality involves imbalances between Department I (means of production) and Department II (means of consumption). By setting aside inter-sectoral composition effects, our single-sector setup defines disproportionality as the divergence between capital accumulation and productive capacity creation when the transformation elasticity is non-unitary ($\theta \neq 1$). This provides an analytical baseline for our empirical replication while acknowledging that sectoral shifts are not modeled here.

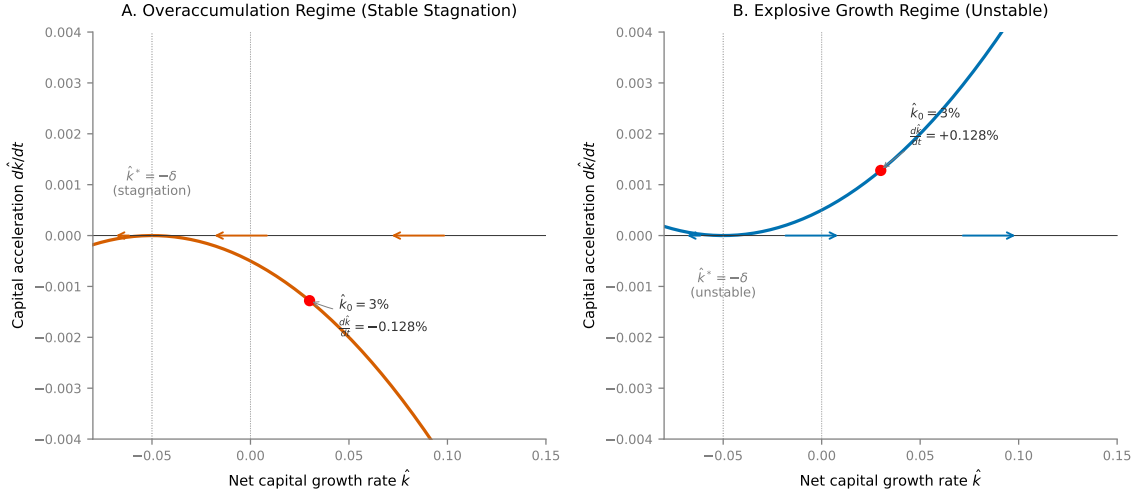
3.18 These regimes describe the dynamic tendencies of the accumulation process. When capital and capacity grow at different rates without autonomous technical change, the path of the net capital growth rate ($\hat{k} \equiv \dot{K}/K - \delta$) is described by an ordinary differential equation where the sign of $d\hat{k}/dt$ depends on $(\theta - 1)$ (the mathematical derivation and stability proofs are provided in Appendix 1.A). Net accumulation decelerates under $\theta < 1$, accelerates under $\theta > 1$, and remains stationary under $\theta = 1$. This instability arises from the accounting relation between capital and capacity when $\theta \neq 1$, rather than from Harrodian demand fluctuations.

3.19 The term $d\hat{k}/dt$ is a capital growth acceleration term, representing the rate of change of the net capital accumulation rate over time. Because the net accumulation rate cannot accelerate or decelerate indefinitely, institutional responses keep it within bounds over long periods. When accumulation accelerates ($\theta > 1$), rising wages, higher interest rates, and debt limits dampen further expansion. When accumulation decelerates ($\theta < 1$), the resulting stagnation triggers bankruptcies, capital devaluation, and wage reductions that eventually stabilize profitability, keeping the accumulation rate from falling without limit.

3.20 To illustrate this dynamic, consider a stylized example where $\theta = 0.8$. Suppose the gross capital stock expands by 10% ($g_K = 0.10$). With $\theta = 0.8$, the maximum productive capacity expands by only 8% ($g_{Y^p} = 0.08$). Two factors explain why capitalists continue investing without immediate market corrections. First, productive capacity is a latent variable; managers adjust to short-run demand shifts through shift work, machine speedups, and deferred maintenance, which obscures the underlying decline in capital productivity. Second, competition compels firms to invest: an individual enterprise that curtails capital spending to protect capital productivity loses market share and scale economies to competitors. Firms therefore continue investing even as

industry-wide profit margins decline. Assuming a baseline net accumulation rate of $\hat{k}_0 = 3\%$ (0.03) and depreciation of $\delta = 5\%$ (0.05), this capacity mismatch ($\theta - 1 = -0.2$) reduces normal capital productivity ($\hat{R}^n = -1.6\%$), causing net accumulation to decelerate at $d\hat{k}/dt = -0.128\%$ per year toward stagnation. Conversely, if the transformation elasticity is $\theta = 1.2$, capacity expands at 9.6%, causing normal capital productivity to rise ($\hat{R}^n = +1.6\%$) and the net accumulation rate to accelerate at $d\hat{k}/dt = +0.128\%$ per year, driving the system into an explosive trajectory (as detailed in Appendix 1.A). Figure 1.1 illustrates these phase dynamics.

Figure 1.1. Capital accumulation dynamics under unbalanced growth



Note: Panel A plots the overaccumulation regime ($\theta = 0.8 < 1.0$) where capital acceleration is negative ($d\hat{k}/dt < 0$ for $\hat{k} > -\delta$), pushing the net accumulation rate toward the stable stagnation equilibrium at $\hat{k}^* = -\delta$. Panel B plots the explosive growth regime ($\theta = 1.2 > 1.0$) where capital acceleration is positive ($d\hat{k}/dt > 0$ for $\hat{k} > -\delta$), causing the accumulation rate to diverge. Both panels show the initial condition $\hat{k}_0 = 3\%$ where $d\hat{k}/dt = \mp 0.128\%$, consistent with the numerical example in §1.3.3.

3.21 While effective demand and distributive conflict are crucial to the broader macroeconomic picture, we abstract from them in this initial formulation to isolate the supply-side mechanics of accumulation. Institutional and macroeconomic forces certainly influence the speed and direction of capital accumulation, but they do not generate the underlying dynamic itself. The differential equations in Appendix 1.A demonstrate how the system behaves under different accumulation regimes. However, the empirical single-equation model we will estimate restricts the transformation elasticity (θ) to a constant value over time. The next step is to test whether this assumption of a time-invariant θ actually holds in the historical data, thereby revealing the limits of this fixed-parameter approach.

1.3.4 Trend Stabilization, Viability Condition, and the Double-Misspecification Bias

3.22 We have previously argued that, econometrically, the inclusion of a time trend might absorb variation arising from distributive conflict and distinct historical configurations. Hence, we analyze the inclusion of a time trend under an unbalanced growth closure ($g_{Y^p} = \theta g_K$) and its consequences for dynamic stability. As established in the phase dynamics, a system experiencing overaccumulation

($\theta < 1$) forces the net accumulation rate to decay to zero. Using the same accounting identities that drive dynamic stability, we can show that introducing a deterministic level trend (b) under an unbalanced growth closure ($g_{Y^p} = \theta g_K + b$) stabilizes the system.

3.23 Substituting this extended closure into the acceleration identity transforms the system into a quadratic ordinary differential equation:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta) [(\theta - 1)\hat{k} + b]. \quad (1.8)$$

3.24 This specification shifts the equilibrium structure of the accumulation process. The stagnation limit ($\hat{k}^* = 0$) is replaced by a new interior root:

$$\hat{k}^* = \frac{b}{1 - \theta}. \quad (1.9)$$

3.25 Under overaccumulation ($\theta < 1$) with positive autonomous technical progress ($b > 0$), this interior equilibrium is locally stable. The depreciation rate (δ) anchors the negative root, while the ratio $b/(1 - \theta)$ scales autonomous progress against the structural imbalance, preventing stagnation. Conversely, in the excess capacity regime ($\theta > 1$), the deterministic trend b fails to provide a stabilizing anchor. Because $(1 - \theta)$ is negative, the interior root $\hat{k}^* = b/(1 - \theta)$ becomes negative, meaning no economically meaningful positive steady-state accumulation rate exists. Instead of stabilizing the system, the trend b merely compounds the dynamic divergence, exacerbating the explosive trajectory where productive capacity continuously outpaces capital accumulation and reinforcing the chronic excess capacity that ultimately requires exogenous autonomous demand to resolve.

3.26 The analysis of technical change by [Basu \(2022\)](#) demonstrates why treating the trend-stabilized steady state as a smooth, exogenously determined path is fundamentally problematic. Basu establishes that under capitalism, technical change is not a neutral, exogenous drift but is governed by the strict filter of *capitalist viability*: producers only adopt new techniques if they reduce costs at prevailing prices and real wages. Crucially, the trajectory and specific type of technical change—whether it is capital-using, labor-saving (CU-LS), or capital-saving, labor-saving (CS-LS)—are not predetermined. Instead, they are endogenously shaped by the balance of class struggle over the real wage relative to productivity growth. Therefore, the deterministic trend coefficient (b) in the capacity-capital ratio does not represent autonomous technological progress. Rather, it is the econometric residue of the feedback loop between distributional conflict and the choice of technique. In other words, imposing a time trend masks how class struggle actively and endogenously reshapes the ceiling of productive capacities.

3.27 The statistical validity of the capacity utilization series relies on establishing a stable cointegrating relation between output, capital, and the deterministic trend. Omitting distributive conflict while including the exogenous trend introduces a double misspecification. As [Basu \(2020\)](#) demonstrates, when an econometric model excludes a relevant variable—in this case, distributive conflict and its influence on the choice of technique ([Kurz, 1986](#))—while including an irrelevant proxy like a deterministic time trend, the estimator for the capital coefficient suffers from compound

bias. Because the trend mathematically absorbs the unobserved shifts in distribution, this indirect bias channel compromises the recovered elasticity parameter θ , making it a passive statistical fit rather than a stable production relation.

3.28 Estimating this relation over Shaikh’s sample imposes rigid temporal constraints. This restriction forces both the elasticity (θ) and the trend (b) to act as time-invariant constants across shifting institutional configurations. The classical long-run closure ($\mu = 1$) isolates theoretical capital-to-capacity relations (Foley, 1985) or smooth reproduction paths (Basu, 2022; Okishio, 2022). Because real capitalist economies routinely violate these strict proportionality conditions, imposing time-invariance forces the recovered parameters to mathematically average across structurally heterogeneous historical periods.

3.29 Looking at the actual post-war historical trajectory of the US corporate sector (1947–2011) using the original data tables by Shaikh (2016), these theoretical regimes correspond to distinct accumulation phases. During the Post-War Golden Age (1947–1973), the US corporate sector experienced rapid capital accumulation alongside strong output growth, with the real gross capital stock expanding at an average annual rate of 5.4% and real gross value added growing at 4.4%. This period represents a relatively balanced accumulation path with a high, stable output–capital ratio. During the transition of the stagflation crisis (1974–1982), the growth of the real capital stock remained relatively high at 4.8% (driven by energy-related investments and structural restructuring), while real output growth slowed significantly to 2.7%, causing the output–capital ratio to collapse. Subsequently, in the neoliberal era (1983–2011), the rate of capital accumulation fell to an average of 3.3% annually, while real output grew at only 2.7%, leaving the output–capital ratio volatile and structurally depressed. This empirical shift in how capital accumulation translates to productive capacity strongly suggests a change in the underlying transformation elasticity θ across these eras. Imposing a time-invariant parameter over this entire 64-year span forces the econometric model to mathematically average across these structurally distinct historical periods, actively masking the very crises and regime shifts that define capitalist development.

3.30 This parametric averaging also bears directly on the classical debate concerning the long-run endogeneity of capital accumulation to aggregate demand. In the Neo-Kaleckian framework, capacity utilization is endogenous to demand in the long run, and investment adjusts to minimize utilization gaps. Conversely, the Sraffian position (e.g., the Supermultiplier model) treats the choice of technique and the capital accumulation path as weakly exogenous to demand-led capacity disequilibrium, adjusting instead to cost-minimizing criteria at a given real wage. Our conceptualization of unbalanced growth ($\theta \neq 1.0$) introduces a more nuanced, counteracting dynamic to this debate. Under overaccumulation ($\theta < 1.0$), capital accumulation is structurally unstable, meaning that the capacity ceiling grows slower than capital stock. While this decoupling implies that investment is weakly exogenous to short-run capacity utilization errors, the system-level instability prevents simple long-run demand adaptation. Crucially, if the weak exogeneity of investment and the secular instability of overaccumulation act as counteracting forces, they may cancel each other out. This counteracting dynamic represents a plausible interpretation that might obscure the resolution of the Sraffian and Neo-Kaleckian debates themselves—an analytical blind spot explained by both frameworks operating under the restrictive assumption of a balanced-growth path with a fixed output–capital ratio.

3.31 The next section 1.4 implements an analytical stress test across three distinct stages to identify the conditions under which Shaikh’s single-equation ARDL estimation fails to identify cointegration. Stage S0 establishes a baseline of reproducibility using the original data tables by Shaikh (2016) and lag lengths. Stage S1 implements an admissibility test by estimating a grid of 500 ARDL specifications, mapping parameter sensitivity to researcher choices—such as lag structures, information-criterion penalties, and historical dummy variables. Stage S2 estimates the long-run relationship using a Vector Error Correction Model (VECM) to determine whether joint system stability holds in a bivariate setting, or if it requires the logged rate of exploitation ($e_t = \ln[\pi_t/(1 - \pi_t)]$) to establish a stable cointegrating relationship.

1.4 Critical Replication Study on Shaikh’s Approach to Cointegration

1.4.1 Econometric Framework and Staged Design

4.1 Shaikh’s long-run output–capital coefficient does not identify a stable, time-invariant capacity benchmark; it mechanically averages over historically distinct distributive regimes. As established in Section 3, this coefficient functions as a transformation elasticity (θ) linking capital accumulation to capacity formation under conditions of unbalanced growth. This section tests whether a single, full-sample estimation can capture this relationship, or whether the bivariate relation is fundamentally misspecified without explicit distributional conditioning. By testing the cointegrating relation directly rather than relying on a pre-fabricated utilization index, we subject the underlying classical capacity theory to empirical testing.

4.2 We execute this critical investigation across three sequential stages of increasing econometric generality. In Stage S0, we reconstruct the closest empirical counterpart to Shaikh’s baseline single-equation model to verify its basic numerical reproducibility. Stage S1 lifts the baseline’s rigid specification choices by estimating a combinatoric grid of 500 Autoregressive Distributed Lag (ARDL) models, mapping how the estimated transformation elasticity ($\hat{\theta}$) shifts under alternative lag structures, trend configurations, and outlier controls. Finally, Stage S2 transitions the analysis from single-equation models to system-level estimations, testing whether the output–capital relation survives as a stable Vector Error Correction Model (VECM) and checking whether it requires the rate of exploitation (e_t) to establish system-level cointegration.

4.3 Our findings show that while Shaikh’s baseline coefficient is numerically reproducible, it is neither structurally unique nor stable in isolation. In the single-equation grid search, the estimated elasticity ranges from 0.65 to 0.92 depending on how we penalize model complexity, indicating that the baseline point estimate is highly sensitive to researcher choices. More importantly, the standalone output–capital relation fails system-level cointegration tests, proving that a stable long-run relationship cannot be identified in a bivariate setting. The output–capital relation achieves system stability only within a trivariate framework that includes the rate of exploitation, showing that the long-run conversion of capital to capacity is structurally conditioned by the historical distribution of income between wages and profits.

4.4 We present these results sequentially across seven subsections. First, Subsection 1.4.1 outlines the mathematical framework and staged design. Subsection 1.4.2 details the data construction, measurement conventions, and stylized facts. Subsection 1.4.3 defines the admissibility filters and diagnostic screens. Subsections 1.4.4, 1.4.5, and 1.4.6 present the empirical results for Stages S0, S1, and S2, respectively. Finally, Subsection 1.4.7 provides a cross-stage econometric synthesis.

4.5 We formalize this progressive, multi-stage testing sequence in Table 1.2, which outlines the design object, specification space, and diagnostic criteria for each stage. By moving from a single-point reconstruction (S0) to a combinatorial specification grid (S1), we systematically map how parameter estimates react to changes in the model’s complexity. The transition to joint system estimation (S2) then tests whether these single-equation relationships survive as self-sufficient economic attractors once we allow for bidirectional feedbacks. This staged architecture allows us to separate the statistical fit of the model from the historical and institutional validity of its derived capacity path.

Table 1.2. Stage Design and Admissibility Strategy

Stage	Design Object	Specification Space	Admissibility and Organization	Interpretive Implication
S0	Single-point reconstruction of baseline specification	ARDL(2,4), historical impulse dummies, alternative deterministic closures	F-bounds primary screen; t-bounds diagnostic where defined; utilization path proximity	Tests whether a deterministic capacity path can be closely reconstructed
S1	ARDL specification grid $\mathcal{G}_{S1}$	Lag orders (p, q) , five PSS cases, four nested dummy structures	F-admissible set $\mathcal{A}_{S1}^F$; F+t confirmed candidates $\mathcal{C}_{S1}$; corrected envelope E_{S1} ; IC-specific neighborhoods $\mathcal{F}_j^{(0.20)}$	Tests whether the output–capital coefficient is structurally unique or varies with IC penalty logic
S2 Bivariate	Joint $(\ln Y_t, \ln K_t)'$ VECM	Lag order, deterministic branch, dummy structure, $r = 1$	Triple gate $\mathcal{A}_{S2}$: convergence, Johansen rank, companion stability	Tests whether the bilateral output–capital relation is self-sufficient at the system level
S2 Trivariate	Joint $(\ln Y_t, \ln K_t, \ln e_t)'$ VECM	Same as bivariate, with logged rate of exploitation included; $r = 1$ or $r = 2$	Same triple gate; Ω_{20} organizes admissible survivors by log-likelihood loss	Tests whether the structural relation survives exclusively when distribution enters explicitly

Note: This table outlines the progressive multi-stage replication architecture. The sequence advances from baseline single-equation reconstruction (S0) through parameter sensitivity grids (S1) to joint system-level survival testing (S2). Source: Author’s elaboration.

4.6 We write the output–capital relation as an Autoregressive Distributed Lag (ARDL) model to identify the long-run transformation elasticity (θ) . To evaluate whether output (y_t) and capital (k_t) share a stable long-run equilibrium, we use the bounds-testing procedure developed by Pesaran et al. (2001). Cointegration means that although output and capital drift over time as nonstationary series, they are tied together by a structural attraction force that prevents them from drifting apart indefinitely. This framework detects this cointegrating relationship regardless of whether the individual series are integrated of order zero $(I(0))$ or one $(I(1))$, eliminating the need for fragile unit-root pre-testing.

4.7 We estimate this relationship using an unrestricted error correction model (UECM) derived from a general ARDL(p, q) specification:

$$\Delta y_t = c_0 + c_1 t + \pi_y y_{t-1} + \pi_k k_{t-1} + \sum_{i=1}^{p-1} \psi_{y,i} \Delta y_{t-i} + \sum_{j=0}^{q-1} \psi_{k,j} \Delta k_{t-j} + \sum_{h=1}^H \delta_h D_{h,t} + \varepsilon_t. \quad (1.10)$$

4.8 This unrestricted error-correction representation separates the short-run adjustment dynamics from the long-run level relationship. In equation (1.10), c_0 is the constant intercept, $c_1 t$ is a deterministic linear time trend, and ε_t is an independent and identically distributed error term. The differenced terms Δy_{t-i} and Δk_{t-j} capture short-run memory and cyclical frictions, where p and q represent the lag lengths of output and capital. The lagged level terms y_{t-1} and k_{t-1} contain the long-run cointegrating relationship. The historical impulse dummies $D_{h,t}$ act as short-run shock absorbers, preventing outlier events (like the 1974 oil shock) from distorting the underlying long-run relationship.

4.9 The statistical distribution of the bounds test depends on how we restrict the constant and linear trend in the model. Pesaran et al. (2001) outline five deterministic configurations (Cases I through V) that determine how the constant c_0 and time trend t enter the equation. These cases range from Case I (no constant, no trend) to Case V (unrestricted constant and unrestricted trend), including intermediate cases where constants or trends are restricted to the long-run cointegrating space. Since these deterministic components alter the asymptotic behavior of the nonstationary variables, each case requires a different set of critical value bounds. We treat these five configurations as specification choices that alter the statistical boundaries of our cointegration test.

4.10 We use a joint Wald F -bounds test to evaluate the null hypothesis that no long-run relationship exists between output and capital. The null hypothesis (H_0^F) asserts that the coefficients on the lagged level terms are jointly zero, meaning there is no cointegration:

$$H_0^F : \pi_y = \pi_k = 0, \quad H_a^F : (\pi_y, \pi_k) \neq (0, 0). \quad (1.11)$$

4.11 Under this hypothesis, π_y and π_k are the coefficients on the lagged levels of output and capital. If the null is true, the regression is spurious because the residuals remain nonstationary ($I(1)$), meaning we find no evidence of cointegration, rendering any derived capacity path econometrically invalid. To evaluate the Wald statistic, we use the critical bounds generated by Pesaran et al. (2001). These bounds establish two thresholds: a lower bound assuming all variables are $I(0)$, and an upper bound assuming all variables are $I(1)$. We reject the null of no cointegration only when the computed F -statistic exceeds the upper bound, confirming that the output–capital relation has a stationary residual.

4.12 We apply the t -bounds test as a secondary confirmation screen to ensure that the cointegrating relationship is stable. The Wald F -test can sometimes find pseudo-cointegration if only one of the lagged level variables is significant. The t -bounds test evaluates the null hypothesis that the coefficient on the lagged dependent variable (π_y) is zero:

$$H_0^t : \pi_y = 0, \quad H_a^t : \pi_y < 0. \quad (1.12)$$

4.13 We test this against the one-sided alternative hypothesis ($\pi_y < 0$), which requires the error-correction coefficient to be negative. A negative speed-of-adjustment coefficient acts as a restoring force: when actual output drifts away from its long-run capacity path, the system adjusts by pulling output back toward the equilibrium. If π_y is positive or zero, the system is explosive or has no memory. Given our relatively short sample period ($T \approx 65$), we use finite-sample critical bounds from [Natsiopoulos and Tzeremes \(2022\)](#) rather than asymptotic tables to prevent over-stating cointegration significance.

4.14 To recover the long-run transformation elasticity (θ), we must distinguish the underlying levels ARDL model from its error-correction representation. We write the general levels ARDL(p, q) specification as:

$$y_t = a_0 + a_1 t + \sum_{i=1}^p \alpha_i y_{t-i} + \sum_{j=0}^q \beta_j k_{t-j} + \sum_{h=1}^H \delta_h D_{h,t} + \varepsilon_t. \quad (1.13)$$

From this levels model, the long-run multiplier (d) is recovered as the sum of the capital stock coefficients normalized by the output lag coefficients:

$$\hat{d} = \frac{\sum_{j=0}^q \hat{\beta}_j}{1 - \sum_{i=1}^p \hat{\alpha}_i}. \quad (1.14)$$

This levels ratio represents the steady-state multiplier where the dynamic adjustment processes have fully settled. In his baseline estimation, Shaikh treats this parameter as a constant technical coefficient mapping capital directly to potential capacity. Under our conceptual framework, however, we interpret this long-run coefficient as a capacity transformation elasticity, θ , which is historically and distributionally conditioned.²

4.15 We map this levels specification directly to the unrestricted error correction model (UECM) in Equation (1.10) by reparameterizing the level variables around their first-difference representations. In the UECM, the speed-of-adjustment coefficient π_y and the level capital coefficient π_k relate to the levels parameters by $\pi_y = -(1 - \sum_{i=1}^p \alpha_i)$ and $\pi_k = \sum_{j=0}^q \beta_j$. Thus, the long-run multiplier is recovered from the conditional UECM parameters as the ratio:

$$\hat{d} = -\frac{\pi_k}{\pi_y}. \quad (1.15)$$

This UECM ratio is mathematically identical to the levels ratio in Equation (1.14). A common pitfall in error-correction estimations is to calculate this long-run multiplier using the short-run differenced parameters from the UECM rather than the levels parameters. We ensure that only the levels coefficients enter the long-run relation to avoid such bias, correcting the notation to represent the true long-run cointegrating levels relation.

4.16 In Shaikh's procedure, this long-run multiplier anchors the fitted capacity path. Capacity utilization is then the residual deviation from the long-run cointegration vector: $\log \hat{\mu}_t = y_t - \hat{y}_t^p$.

²Shaikh's original exposition ([Shaikh, 2016](#)) seldom discusses the parameter d as an elasticity, treating it instead as a fixed accounting coefficient. We interpret this long-run multiplier d as the empirical proxy for the capacity transformation elasticity θ under conditions of unbalanced growth. For clarity, we retain the notation d when replicating Shaikh's ARDL specifications directly, and transition to θ when developing our conceptual and system VECM extensions.

This design makes utilization dependent on the estimated coefficient. Shaikh treats the coefficient as a fitted accounting multiplier, but he does not theorize its structural role. We re-interpret the coefficient as a transformation elasticity $\hat{\theta}$ of productive capacities with respect to capital accumulation, subject to the specification grid, deterministic treatment, and admissibility gates. We evaluate this elasticity across three stages: we reconstruct the baseline result (Stage S0), open the single-equation specification space (Stage S1), and test whether the relation survives in a joint system (Stage S2). Subsection 1.4.2 details the data construction and measurement conventions required to execute these steps. An econometric limitation of this single-equation setup is that if the capital stock (k_t) is measured with error due to the approximations of the Perpetual Inventory Method, the estimated elasticity $\hat{\theta}$ will be biased downward. Measurement error occurs because physical capacity changes are mediated by workplace organization, labor intensity, and speedups, which standard NIPA capital stock metrics cannot capture. The replication evaluates this baseline point model while explicitly testing this parameter’s stability across a broader specification grid.

4.17 In Stage S0, we test the reproducibility of Shaikh’s baseline results to establish a benchmark for our replication. We target his published single-equation model, fix the lag structure at the reported ARDL(2,4) configuration, and estimate the parameters using the closest recoverable corporate-sector data. We identify the variables directly from his published data tables, resolving documentation discrepancies in the original text. We estimate these baseline parameters across alternative deterministic configurations and historical impulse controls to reconstruct his point estimates, rather than to search for alternative specifications.

4.18 This procedure estimates the long-run relationship, constructs the potential capacity path, and calculates capacity utilization as the residual deviation of actual output from that path. We use cointegration bounds tests to distinguish genuine long-run relations from spurious correlations generated by autoregressive filtering. Here, we encounter a technical limitation of the ARDL framework: while the F -bounds test applies to all five deterministic cases, the t -bounds test is only defined for Cases I, III, and V. Specifications under Case II or Case IV (which restrict the constants or trends to the cointegration space) cannot be screened using the t -bounds gate.

1.4.1.1 Stage 1 (S1): Admissible-Specification Stress Test

4.19 In Stage S1, we expand the single-equation model into a combinatoric grid of 500 different specifications. By varying the lag lengths, deterministic trends, and outlier controls, we evaluate how researcher choices affect the cointegration test results and the estimated elasticity. This approach shifts our focus from a single preferred model to the entire space of possible specifications, showing how sensitive the capacity path is to modeling assumptions.

4.20 We allow the output and capital lags to vary freely from 1 to 5 ($p, q \in \{1, \dots, 5\}$). We estimate each lag combination across all five deterministic cases, combined with four alternative blocks of historical dummy variables (s_0 through s_3). These dummies control for major post-war shocks: the 1956 Eisenhower recession, the 1974 oil crisis, and the 1980 Volcker monetary contraction. Permuting these options yields 500 unique models. Formally, the specification space is:

$$\mathcal{G}_{S1} = \{(p, q, c, s)\}, \quad (1.16)$$

4.21 In this grid $\mathcal{G}_{S1}$, each element (p, q, c, s) represents a single model. The indices p and q set the lag orders, c determines the PSS case configuration, and s indicates which outlier controls are active.

4.22 Admissibility screens filter out models that fail cointegration tests, but they do not identify a single correct parameter. The remaining models reveal a disciplined non-uniqueness, suggesting that a single technical production law does not exist. To organize the cointegrating models, we construct a fit-complexity envelope (E_{S1}) that identifies the optimal trade-off between statistical fit and parameter count:

$$E_{S1} = \{m \in \mathcal{A}_{S1}^F : \nexists m' \in \mathcal{A}_{S1}^F \text{ with } L(m') \leq L(m) \text{ and } K(m') \leq K(m)\}, \quad (1.17)$$

4.23 In this setup, $L(m) = -2 \log \hat{\ell}(m)$ is the model deviance (where lower values mean better fit) and $K(m)$ is the number of estimated parameters. A model lies on the envelope E_{S1} only if no other cointegrating model fits the data better while using fewer parameters.

4.24 We also define information-criterion neighborhoods ($\mathcal{F}_j^{(0.20)}$) to capture competitive specifications that sit slightly off the strict Pareto frontier:

$$\mathcal{F}_j^{(0.20)} = \{m \in \mathcal{A}_{S1}^F : IC_j(m) \leq Q_{0.20}(IC_j \mid \mathcal{A}_{S1}^F)\}, \quad (1.18)$$

4.25 Here, j indexes the criteria used to evaluate the models: the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), Hannan-Quinn (HQ), and Bozdogan's Information Complexity Index (ICOMP) along with its robust variant (RICOMP). While AIC and BIC penalize the number of parameters, ICOMP and RICOMP penalize parameter interdependence and covariance complexity. This variety of criteria helps us see how different penalty structures select different long-run elasticity estimates across the specification space.

4.26 Each specification m yields a distinct estimate of the transformation elasticity $\hat{\theta}(m)$, which determines the slope of the capacity path. We construct the capacity utilization series for each model as:

$$\hat{\mu}_t(m) = y_t - [\hat{a}(m) + \hat{b}(m)t + \hat{\theta}(m)k_t]. \quad (1.19)$$

4.27 Comparing these utilization paths across the different model sets shows how modeling choices directly impact our capacity measurements. Selecting different cointegrating specifications shifts the estimated capacity ceiling, altering the calculated rate of capacity utilization.

4.28 Table 1.3 summarizes the structure of this single-equation specification search. We report the empirical results of this grid search in Subsection 1.4.5.

4.29 We extend this analysis to a joint multi-equation system in Stage S2. Transitioning to a system framework allows us to test whether bivariate cointegration holds when output and capital are both treated as endogenous, and whether establishing a stable long-run relationship requires including the rate of exploitation (e_t).

Table 1.3. Synthesis of Admissibility Specification Stress Tests

Stress-Test Component	Operational Layer	Screening/Organization Rule	Diagnostic Target
Cointegration admissibility	F/T-bounds sequence	$p_F(m) \leq \alpha$, $p_t(m) \leq \alpha$ (if defined) for $\alpha \in \{0.10, 0.05, 0.01\}$	Existence and significance of long-run level relationships
Specification invariance	Lag-order variation	Monitor parameter drift $\hat{\theta}(m)$ across combinatoric $p, q \in \{1, \dots, 5\}$	Stability and unique mapping of the long-run structural multiplier
Identification fragility	Deterministic cases & dummy structures	Track admissibility shifts across $c \in \{1, \dots, 5\}$ and $s \in \{s_0, \dots, s_3\}$	Sensitivity to deterministic components and structural trend assumptions
Information complexity & Pareto optimization	Fit-complexity envelope & IC neighborhoods	Pareto non-dominance in E_{S1} ; 20th percentile cutoff for $j \in \{\text{AIC, BIC, HQ, ICOMP, RCOMP}\}$	Organization of disciplined non-uniqueness under distinct information schedules
Capacity benchmark sensitivity	Utilization path fan $\hat{\mu}_t(m)$	Map structural parameter variance to latent residual series	Macroeconomic consequence of single-equation coefficient drift on normal profit estimation

Note: This synthesis matrix outlines the nested single-equation admissibility screens, information-theoretic optimization layers, and diagnostic targets within the stage $S1$ multi-model grid. Source: Author's elaboration.

1.4.1.2 Stage 2 ($S2$): System Cointegration Test

4.30 Stage $S2$ tests whether the output–capital relation remains stable in a multi-equation system. If output and capital are cointegrated, a simple bivariate vector $X_t = (\ln Y_t, \ln K_t)'$ should exhibit a stable cointegrating relationship under Johansen's rank tests. However, as we show in the results section, this bivariate system fails to cointegrate. We resolve this by introducing the logged rate of exploitation, $e_t = \ln(\pi_t/(1 - \pi_t))$, to form the trivariate vector $X_t = (\ln Y_t, \ln K_t, \ln e_t)'$. We estimate this trivariate system using a standard Vector Error Correction Model (VECM):

$$\Delta X_t = \Pi X_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta X_{t-i} + \Phi D_t + \varepsilon_t, \quad (1.20)$$

4.31 In equation (1.20), Γ_i represents the short-run dynamics, D_t contains deterministic components and dummies, and Π is the long-run impact matrix. We factor this matrix as $\Pi = \alpha\beta'$, where β contains the long-run cointegrating vectors and α contains the adjustment speeds. A stable cointegrating relation exists only when the matrix has a reduced rank of $r = 1$, and the exploitation variable enters the cointegrating space β .

4.32 We test the stability of this system by estimating a grid of VECM specifications, varying the lag structures, trend configurations, and outlier dummies. For each target rank r , the tried

specification space is:

$$\mathcal{G}_{S2}^{(r)} = \{(p, C_{\text{det}}, s)\}, \quad (1.21)$$

4.33 Here, p represents the VAR lag orders (from 1 to 4), C_{det} indexes the four deterministic configurations, and s select the outlier dummies. In this system setup, C_0 assumes no constant or trend; C_1 restricts the constant to the cointegrating space; C_2 includes an unrestricted constant; and C_3 adds a linear trend restricted to the cointegrating space. These four cases are distinct from the single-equation PSS configurations. Permuting these choices yields 48 attempted models. We exclude the restricted constant models (C_1) because they systematically fail to converge, leaving 36 estimated models in our final grid.

1.4.2 Data, Measurement, and Stylized Facts

4.34 We use annual US corporate sector data from 1947 to 2011 ($T = 65$), sourced from the data companion in [Shaikh \(2016\)](#).³ Capacity utilization is the residual deviation of actual output from the cointegrated output–capital path. The level, timing, and persistence of our capacity path depend on these measurement choices. Table 1.4 summarizes the variables, their construction, and the measurement constraints that structure this replication.

Table 1.4. Core replication variables: symbols, sources, and transformations

Symbol	Construction	Role	Measurement constraint
GVAcorp_t	NIPA Table 1.14 (L1) + imputed-interest adj. (Table 7.11)	Output flow	Classical surplus correction
KGCcorp_t	BEA Fixed Asset 6.1 (L9) + IRS SOI + GPIM recursion	Capital anchor	GPIM stock–flow reconstruction
p_t	pKN implicit price deflator, 2005=100	Common price basis	GPIM-consistent deflation
y_t	$\ln(\text{VAcorp}_t/p_t)$	Dependent variable	Inherits output and deflator choices
k_t	$\ln(\text{KGCcorp}_t/p_t)$	Regressor	Inherits GPIM depletion rates
D_{56}, D_{74}, D_{80}	Squared-residual diagnostics	Disturbance controls	Baseline outlier absorption

Note: Stage $S2$ introduces $\ln e_t$ (logged exploitation) to the trivariate state vector. Distributional variables enter at the multi-equation system stage. Full series definitions, sources, and transformations are provided in Table 1.13 of Appendix 1.B.

4.35 To reconstruct the gross capital stock, we use the Generalized Perpetual Inventory Method (GPIM) following [Shaikh \(2016\)](#). This accounting framework calculates capital stock from nominal gross investment flows, annual depreciation rates, and an implicit deflator built from gross capital

³Replication data and construction scripts are available at: <https://www.realecon.org/data>. While Shaikh specifies a common-deflator condition, he does not name the price index; we implement a GPIM-consistent price deflator to ensure reproducibility.

formation accounts. The recursion accumulates current-cost capital as:

$$K_t = IG_t + z_t^* \cdot K_{t-1}, \quad z_t^* \equiv (1 - z_t) \frac{p_t^K}{p_{t-1}^K}, \quad (1.22)$$

4.36 In this recursion, IG_t is nominal gross investment, z_t is the annual depreciation rate, and z_t^* reflates surviving capital to current replacement cost. We follow Shaikh’s three specific adjustments to construct this series: we replace the BEA 2011 infinite-lives assumption with BEA 1993 service lives, anchor the initial values to 1925 benchmarks, and rescale the historical stock over 1925–1947 using the IRS corporate book-value index to account for scrapping during the Great Depression (Shaikh, 2016, Appendix 6.7). Our constructed series closely tracks the BEA official net stock with a 99.60% average ratio over 1947–2005, confirming that differences in levels reflect these methodological choices rather than measurement error. Appendix 1.B.4 contains the full recursion inputs and depreciation rates.

4.37 We deflate both output and capital by the pKN price index (2005=100) to maintain stock-flow consistency. Standard BEA releases use quality-adjusted chain weights that alter the measured volume of investment across generations of capital, which breaks the physical accounting of capital accumulation. The pKN index avoids these quality-adjustment artifacts, ensuring that our output–capital relation reflects real productive capacity rather than statistical adjustments (Shaikh, 2016, Appendix 6.6). This approach satisfies the common-deflator condition and isolates real capacity dynamics. We detail this deflation rationale and compare it to quality-adjusted deflators in Appendix 1.B.5.

4.38 We adjust corporate gross value added to correct for imputed financial intermediation services (FISIM) by subtracting net monetary interest paid and adding back imputed net interest adjustments from NIPA Table 7.11. We provide the algebraic derivation in Appendix 1.B.2. We also include year dummies (D_{1956} , D_{1974} , D_{1980}) as shock absorbers to control for large historical breaks: the 1956 Eisenhower recession, the 1974 oil crisis, and the 1980 Volcker monetary contraction. These dummies prevent temporary shocks from distorting our long-run elasticity estimates. Since our sample size ($T = 65$) is relatively small, relying on asymptotic critical values could bias our cointegration tests. We address this limitation by computing finite-sample bounds calibrated to our exact sample length ($T = 65$) using stochastic simulations (Natsiopoulos and Tzeremes, 2022).

4.39 These measurement choices are central to our empirical strategy. We use Shaikh’s GPIM-adjusted gross capital stock because our goal is to measure physical capacity in operation rather than the financial value of assets. While net capital stock measures are appropriate for calculating profit rates and asset valuation, gross capital stock closer matches the physical ability of capital to produce output.⁴ By using gross capital stock, we ensure that the estimated capacity benchmark is not distorted by accounting depreciation rules or market asset revaluations, which make net capital stock growth rates more volatile than physical capacity changes. The specific recursion parameters and service-lives adjustments used to construct this gross stock are detailed in Appendix 1.B.4. Deflating both output and capital with the same price index further ensures that our estimates reflect real productive capacity rather than shifts in relative prices.

⁴This distinction is critical: financial valuation dominates profit-rate calculations, but capacity benchmarks require a measure of physical capital in use.

4.40 Between 1947 and 2011, post-war US capital accumulation systematically outpaced output and employment growth, driving a secular decline in the output-capital ratio that mirrors the broader OECD experience of Marx-biased technical change. Specifically, the real corporate gross capital stock grew at an average annual rate of 4.1%, while real output expanded at 3.0% and employment grew at only 1.3%. This structural decoupling is not unique to the United States; cross-country evidence confirms that labor productivity across advanced capitalist economies rises primarily through mechanization and capital-using, labor-saving technical change (Basu, 2010; Marquetti, 2003). This systematic depression of the output-capital ratio demonstrates that a fixed, technical relationship between capital and productive capacity is empirically invalid. Instead, the conversion of accumulated capital into active productive capacity is a historically conditioned process, fundamentally mediated by the choice of technique and distributive conflict.

4.41 We plot these post-war trends in Figures 1.2, 1.3, and 1.4. While output and capital grow together over the long run (Figure 1.2), their ratio is subject to long historical waves rather than remaining constant (Figure 1.3). As shown in Figure 1.4, output growth exhibits much higher volatility than capital accumulation. Economically, this lower volatility of capital stock growth is a standard macroeconomic stylized fact, but it is further pronounced here because we use gross capital stock (capital in operation) rather than net capital stock. Net capital stock growth rates are inherently more volatile because they incorporate short-run changes in financial book value, market revaluations, and accelerated accounting depreciation. In contrast, our GPIM-constructed gross capital stock remains insulated from these valuation shocks, reflecting the smoother physical survival of productive assets. Because capacity is a physical ceiling representing what capital can actually produce in use, gross capital stock is the theoretically correct anchor, whereas net capital stock measures the depreciated financial value of assets. Any estimated capacity utilization path must be consistent with these descriptive patterns; the parametric cointegration tests we run below must explain these raw historical movements rather than contradict them.

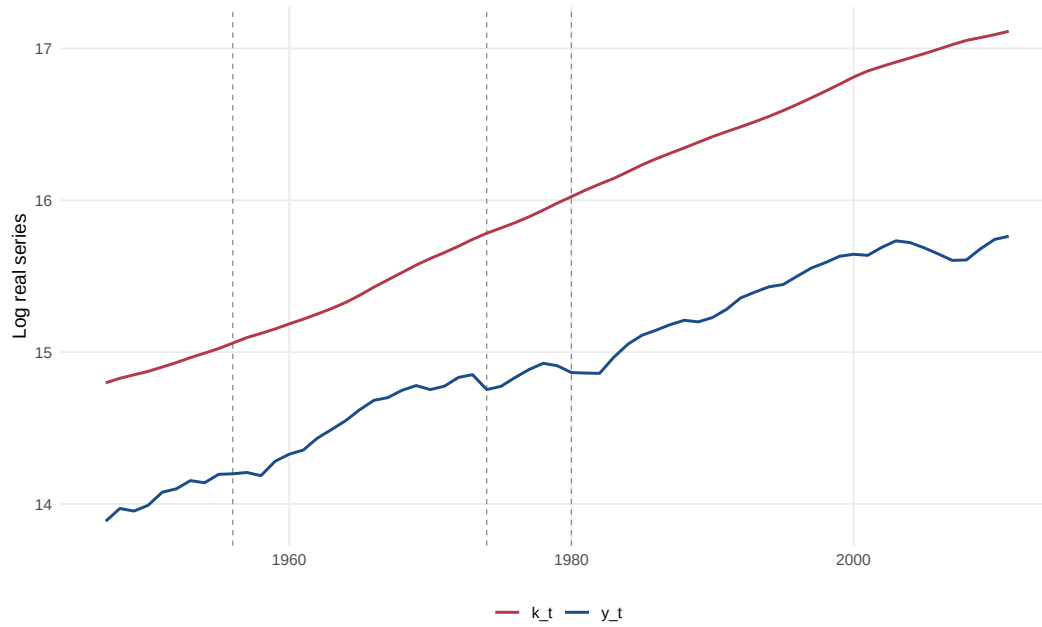


Figure 1.2. Log output (y_t) and log capital (k_t), 1947–2011. Vertical dashed lines mark 1956, 1974, and 1980.



Figure 1.3. Output-capital ratio ($y_t - k_t$), 1947–2011. Horizontal line marks long-run mean.

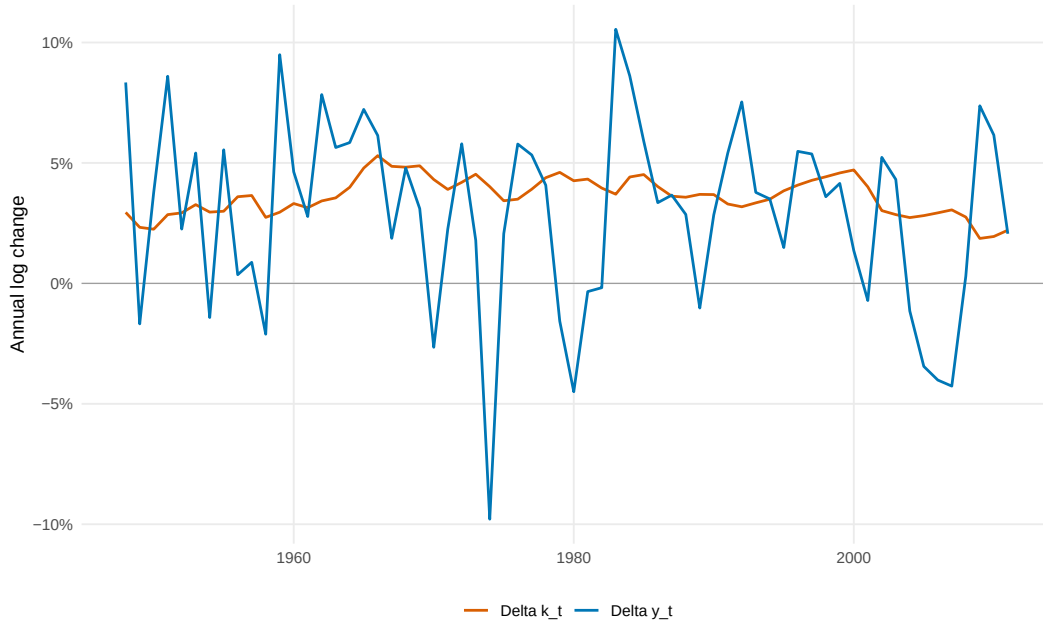


Figure 1.4. Annual growth rates of output (Δy_t), gross capital (Δk_t), and net capital (Δk_t^{net}), 1947–2011.

1.4.3 The Admissibility Strategy

4.42 In this chapter, an admissible specification is one that produces stationary residuals and stable dynamic roots. If an estimated model fails cointegration, its residuals contain a unit root, meaning output and capital drift apart without a long-run equilibrium. Capacity utilization series derived from non-cointegrated specifications are spurious, as the estimated capacity ceiling wanders away from actual output.

4.43 For single-equation ARDL models (Stages S0 and S1), admissibility requires the specification to pass the [Pesaran et al. \(2001\)](#) bounds test: the Wald F -statistic must exceed the upper critical bound at the 5% level. For multi-equation models (Stage S2), admissibility requires three conditions: convergence of the maximum likelihood estimation, rejection of the zero-rank null hypothesis in the Johansen trace test at the 5% level ($r = 1$), and dynamic stability of the companion matrix (all eigenvalues strictly within the unit circle). Specifications failing any of these conditions are excluded from model comparison. Formally, we define the single-equation F -admissible set $\mathcal{A}_{S1}^F$ as:

$$\mathcal{A}_{S1}^F = \{m \in \mathcal{G}_{S1} : p_F(m) \leq 0.10\}, \quad (1.23)$$

4.44 In this definition, $p_F(m)$ is the p -value of the Wald bounds test for model m , measuring the probability of the null hypothesis of no long-run relationship. If this probability is 10% or less, the specification enters our admissible pool. We discard any model that yields a Wald statistic below the lower critical bound or within the inconclusive interior band.

4.45 The secondary screen uses the t -bounds test to verify error-correction stability. Because the t -bounds statistic is only defined for deterministic Cases I, III, and V, we allow specifications using

Cases II and IV to bypass this test without penalty. The t -admissible set $\mathcal{T}_{S1}$ is:

$$\mathcal{T}_{S1} = \{m \in \mathcal{A}_{S1}^F : p_t(m) \leq 0.10 \text{ if defined}\}, \quad (1.24)$$

4.46 Here, $p_t(m)$ is the p -value for the one-sided t -test on the error-correction coefficient π_y . A significant negative coefficient confirms that output adjusts back toward the long-run capacity ceiling after a shock, rather than drifting away permanently. To identify specifications that pass both the Wald and error-correction tests at the stricter 5% level, we define the Strong Cointegration Space $\mathcal{C}_{S1}$ as:

$$\mathcal{C}_{S1} = \{m \in \mathcal{T}_{S1} : p_F(m) \leq 0.05 \text{ and } p_t(m) \leq 0.05\}, \quad (1.25)$$

4.47 The joint condition $p_F(m) \leq 0.05$ and $p_t(m) \leq 0.05$ requires both the level relationship and the speed-of-adjustment to be highly significant. To illustrate what is at stake in these screens, we run a "no-dummy" counterfactual specification where we omit the historical shock controls (D_{1956} , D_{1974} , D_{1980}). Omitting these dummies causes the residuals to remain nonstationary ($I(1)$) across all lag profiles, failing the cointegration bounds tests. This counterfactual proves that without controlling for major historical breaks, the output–capital relation appears spurious, demonstrating why these admissibility gates are necessary to filter out statistical artifacts.

4.48 In Stage S2, we evaluate the surviving models through a triple-gate system to ensure the estimates represent a stable multi-equation attractor rather than a spurious correlation. First, the Cointegration Rank Gate requires the Johansen Trace test to confirm a single cointegrating vector ($r = 1$) at the 5% level, showing that the variables share a common long-run path. Second, the Dynamic Stability Gate requires all eigenvalues of the VECM companion matrix to lie strictly inside the unit circle (excluding the unit roots of the common trends), filtering out explosive models where temporary shocks cause output to diverge infinitely. Third, the Residual Diagnostic Gate requires the model residuals to pass tests for serial correlation and heteroscedasticity at the 5% level, ensuring our statistical inference is unbiased. In the trivariate family, the rank condition $r = 1$ checks whether output and capital cointegrate once we include income distribution. We rank the surviving VECMs using their log-likelihood values to identify the best-fitting systems.

1.4.3.1 Cross-Stage Synthesis

1.4.4 Stage 0 (S0): Single-Equation Reconstruction

4.49 Stage S0 confirms that Shaikh's baseline capacity path is numerically reproducible, but this recovery hinges on undocumented data choices rather than a unique structural law. The divergence between the published multiplier of 0.66 and our reconstructed estimate of 0.72 arises from undocumented choices in data construction and lag selection. Reconstructing this benchmark required tracing exact price deflators and data vintages through archive spreadsheets, demonstrating that forensic replication is an active econometric reconstruction rather than a passive mirror. Consequently, the closest reconstruction yields an elasticity of $\hat{\theta} = 0.72$, which barely clears the 10

4.50 This divergence between the published 0.66 and our reconstructed 0.72 is our first major finding. This gap is not a calculation error; it arises from undocumented choices in the original data construction. Reconstructing his benchmark required tracing the exact price deflators and data

vintages through archive spreadsheets, where alternative deflator selections shift the coefficient. This shows that forensic replication is an active econometric reconstruction rather than a passive mirror. The closest reconstruction yields an elasticity of $\hat{\theta} = 0.72$ from an ARDL(2,4) specification, which matches the model selected by the Bayesian Information Criterion (BIC). However, this ARDL(2,4) model yields an F -bounds statistic of 3.349, which barely clears the 10% significance threshold ($p = 0.099$). To address this cointegration fragility, we select the AIC-chosen ARDL(4,3) specification as our focal model. This model yields a more stable estimate of $\hat{\theta} = 0.75$ and cointegrates at the 5% level ($F = 5.250$, $p = 0.023$). We report the parameters for both models in Table 1.5.

4.51 We report the bounds test statistics and finite-sample critical values for both specifications in Table 1.5. By using stochastically simulated finite-sample distributions rather than asymptotic tables, we ensure that our cointegration thresholds are calibrated to our sample size. In the faithful ARDL(2,4) reconstruction, the F -bounds statistic of 3.349 barely clears the 10% critical value threshold of 3.333 ($p = 0.099$). In contrast, our focal ARDL(4,3) model yields a highly significant F -bounds statistic of 5.250 ($p = 0.023$) and a t -bounds statistic of -3.125 ($p = 0.015$), confirming that the relationship cointegrates robustly.

Table 1.5. Stage S0 ARDL Bounds Evidence at $\alpha = 0.10$ (90% Confidence Intervals) with Small-Sample Critical Values

Model	Test	PSS Case	Statistic	I(0) Bound	I(1) Bound	Exact-Sample p	Verdict
ARDL(2,4)	F -bounds	Case 1	3.349	2.466	3.333	0.099	Reject H_0
ARDL(2,4)	t -bounds	Case 1	-2.507	-1.602	-2.259	0.062	Reject H_0
ARDL(4,3)	F -bounds	Case 1	5.250	2.486	3.338	0.023	Reject H_0
ARDL(4,3)	t -bounds	Case 1	-3.125	-1.621	-2.270	0.015	Reject H_0

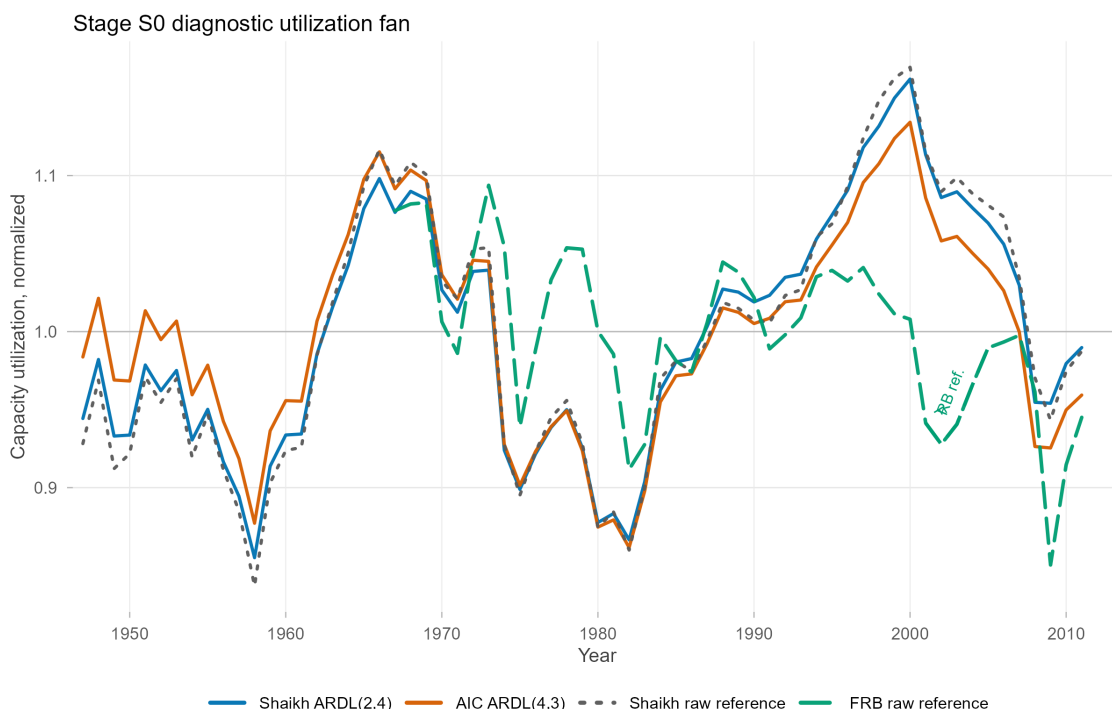
Note: The null hypothesis assumes no long-run level relation. The F -bounds statistic supplies the primary admissibility diagnostic. The t -bounds statistic is reported as a stricter secondary diagnostic where the deterministic structure permits. Case 1 denotes the no-constant/no-trend PSS deterministic configuration. Rejection is assessed at $\alpha = 0.10$. Source: Author's elaboration.

4.52 The error-correction estimates for the focal ARDL(4,3) model show a slow adjustment process back to the capacity path. The speed of adjustment coefficient $\hat{\pi}_y$ is -0.127 (SE 0.038, $t = -3.34$, $p < 0.01$), meaning the economy corrects only 12.7% of its deviation from the capacity ceiling each year. The negative coefficients on our historical dummies (D_{1956} , D_{1974} , D_{1980}) are all statistically significant, capturing permanent step-shifts in capital productivity. These shifts correspond to major institutional disruptions: the 1956 Eisenhower defense cuts, the 1974 OPEC oil crisis, and the 1980 Volcker monetary contraction.

4.53 Figure 1.5 plots the reconstructed single-equation capacity utilization paths alongside Shaikh's published series and the Federal Reserve Board (FRB) manufacturing utilization benchmark. To allow direct comparison, each series is mean-normalized to unity over the 1947–2011 sample period. The comparison highlights two dynamics: the sensitivity of single-equation capacity paths to lag specification, and the divergence between survey-based and cointegration-based measures. The difference between the reconstructed ARDL(2,4) benchmark ($\hat{\theta} = 0.72$) and the AIC-selected ARDL(4,3) specification ($\hat{\theta} = 0.75$) shows that utilization estimates depend on lag length. A change of 0.03 in the estimated transformation elasticity shifts the constructed capacity utilization path

by up to 5 percentage points across the cycle. Because the elasticity parameter determines the denominator of the utilization index ($Y_t/\hat{Y}_t^p$), small differences in specification alter the historical level of the index, showing that single-equation point estimates depend heavily on lag choices.

Figure 1.5. Stage S0 Diagnostic Utilization Fan



Note: All series are mean-normalized to unity over the 1947–2011 period. The solid blue line traces the reconstructed ARDL(2,4) specification ($\hat{\theta} = 0.72$), the solid orange line traces the AIC-selected ARDL(4,3) specification ($\hat{\theta} = 0.75$), the dashed grey line represents Shaikh’s published series, and the dashed green line represents the normalized Federal Reserve Board manufacturing benchmark. Source: Author’s elaboration.

4.54 The divergence between Shaikh’s cointegrated index and the Federal Reserve Board series is also evident after 1973. While both series capture cyclical downturns (1958, 1974, 1982, and 2008), their secular trends differ. The FRB series is constructed from survey operating rates relative to existing plant capacity, which makes it mean-reverting around an 80% operating rate. In contrast, Shaikh’s measure reflects the output–capital relation across successive investment cycles. Because capital accumulation outpaced capacity expansion ($\hat{\theta} < 1$), the output–capital ratio fell after 1973, producing the persistent downward shift observed in Shaikh’s utilization series. Recovering this baseline confirms that we can numerically reproduce a capacity path similar to Shaikh’s. However, recovering a single point estimate does not guarantee that the parameter is unique or stable across other specifications. To test this parameter’s sensitivity, we transition to Stage S1 and estimate a broader grid of single-equation models.

1.4.5 Stage 1 (S1): Admissible-Specification Stress Test Results

4.55 Opening the specification space to a 500-model ARDL grid shatters the illusion of a unique, fixed technical multiplier. Instead of converging on a single structural law, the admissible estimates

scatter across a wide parameter space: only 102 of the 500 specifications survive the 10

4.56 To illustrate these results, we compare two focal models that represent the opposite edges of our specification space. The first is the model selected by the standard AIC, BIC, HQ, and ICOMP criteria: an ARDL(3,3) specification under PSS Case I (no constant, no trend) with the 1974 dummy, yielding a high elasticity estimate of $\hat{\theta} = 0.92$ (SE 0.04). The second is the model selected by the robust RICOMP criterion: a parsimonious ARDL(1,2) specification under PSS Case II (restricted constant) with no dummies, yielding a much lower elasticity estimate of $\hat{\theta} = 0.65$ (SE 0.07). We report the parameters and diagnostics for these two models in Table 1.6.

Table 1.6. Stage S1 Focal Single-Equation Estimations

Parameter	AIC/BIC/HQ/ICOMP Focal: ARDL(3,3)			RICOMP Focal: ARDL(1,2)		
	Estimate	SE	<i>t</i> -stat	Estimate	SE	<i>t</i> -stat
$\hat{\theta}$ (elasticity)	0.920	0.040	23.00***	0.651	0.070	9.29***
a (intercept)		— (Case 1)		0.584	0.402	1.45
D_{1974}	-0.038	0.017	-2.24**		— (s_0)	
F -bounds statistic	8.120 (Case 1) [Reject ***]			3.269 (Case 2) [Reject **]		
t -bounds statistic	-4.520 (Case 1) [Reject ***]			-3.125 (Case 2) [Reject **]		
$1 - \sum \alpha_i$ (denominator)	0.240			0.091		
Breusch-Godfrey LM (p -value)	0.48			0.39		
CUSUM/CUSUMSQ	Stable			Stable		

Note: Small-sample critical values are simulated for $T = 65$. Significance: *** $p < 0.01$, ** $p < 0.05$. Source: Author's elaboration.

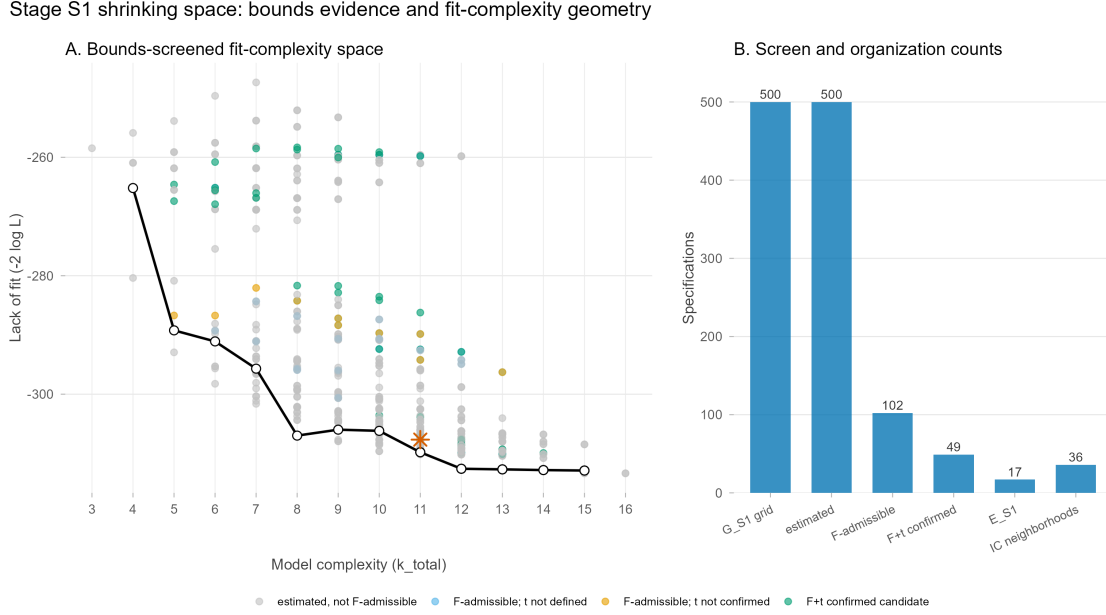
Table 1.7. S1 Shrinking Specification Space Counts

Layer	Count
Full grid	500
Estimated successfully	500
F-admissible at 10%	102
F+t confirmed at 10%	49
F-admissible at 5%	62
F+t confirmed at 5%	28
F-admissible at 1%	13
F+t confirmed at 1%	3
E_{S1}	17
IC-neighborhood union	36

Note: This table tracks the survival rate of the 500 estimated single-equation ARDL models through the nested cointegration admissibility gates. Source: Author's elaboration.

4.57 We visualize this screening process in Figure 1.6. Panel A maps all 500 specifications across the fit-complexity space, color-coded by their cointegration status. Panel B tracks the number of surviving specifications at each diagnostic layer. This sequence reinforces the methodological rule that we must screen models for cointegration before comparing their fit or complexity.

Figure 1.6. S1 Shrinking Specification Space



Source: Author's elaboration.

4.58 Within the cointegrating set, we construct a fit-complexity envelope (E_{S1}) containing 17 optimal specifications. This envelope represents the Pareto frontier where we cannot improve a model's fit without adding parameters, and we cannot simplify it without worsening its fit. The envelope serves as a geometric summary of the best-performing single-equation alternatives rather than a single preferred model.

4.59 The heavy reliance on dummy variables matches the historical shocks of the post-war US economy. The shock years mark major disruptions: the 1956 Eisenhower recession (following defense cuts after the Korean War), the 1974 OPEC oil embargo (inducing stagflation), and the 1980 Volcker interest rate shock (triggering a credit crunch). Bivariate models that omit these dummy variables yield nonstationary residuals because these crises act as permanent shifts in the output–capital relation. Restoring a stationary error term requires these step-shift dummies to control for the structural drops in capital productivity. Specifically, the negative dummy coefficients capture the permanent dampening of capital productivity (Y/K) caused by these crises: the post-Korean War manufacturing slowdown in 1956, the OPEC price-profit squeeze forcing capacity to idle in 1974, and the Volcker shock pushing interest rates to 20% and permanently lowering capacity utilization in 1980.

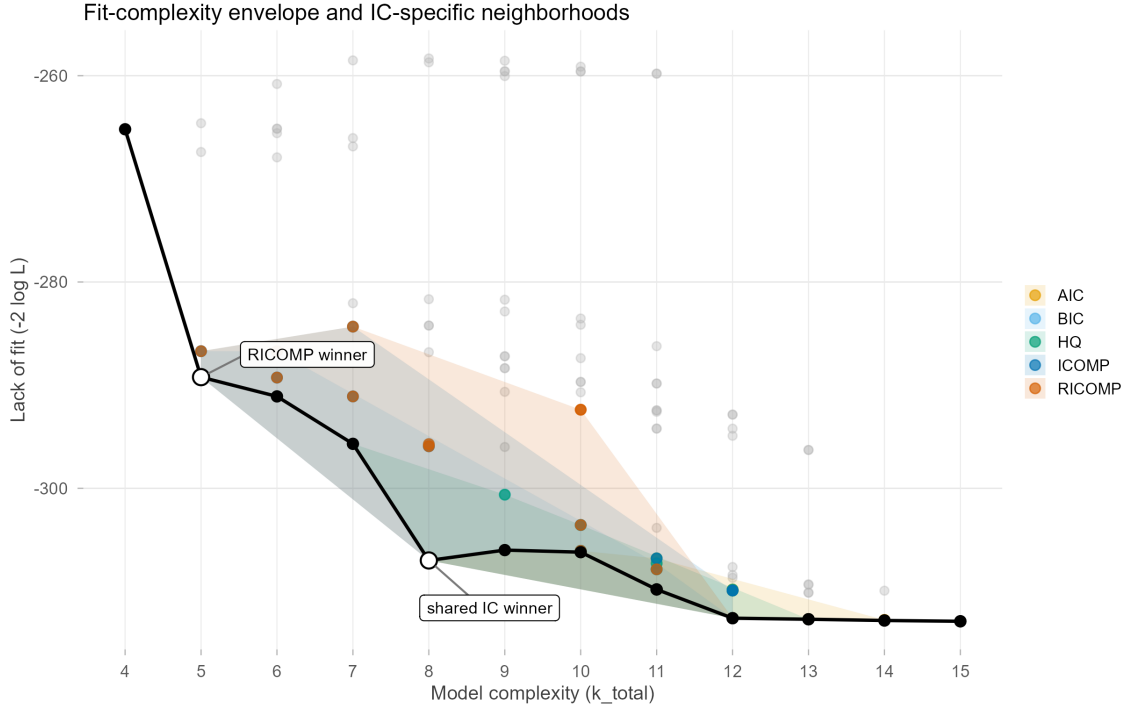
4.60 We define information-criterion neighborhoods ($\mathcal{F}_j^{(0,20)}$) to organize the cointegrating specifications. For each criterion, the neighborhood collects the top 20% of cointegrating models, resulting in a union of 36 unique specifications. Table 1.8 reports the parameter ranges across these neighborhoods, illustrating how different model-selection criteria organize the specification space.

Table 1.8. S1 IC-Specific Neighborhood Summaries

Criterion	Specs	Min $\hat{\theta}$	Mean $\hat{\theta}$	Median $\hat{\theta}$	Max $\hat{\theta}$
AIC	22	0.84	0.88	0.86	0.93
BIC	21	0.65	0.91	0.86	2.20
HQ	21	0.75	0.94	0.86	2.20
ICOMP	22	0.65	0.84	0.86	0.95
RICOMP	22	0.65	0.83	0.85	0.95

Note: The table reports the summary statistics for the long-run parameter $\hat{\theta}$ within the bottom 20th percentile of each information criterion. Estimates are rounded to two decimal places for consistency. Source: Author's elaboration.

4.61 Figure 1.7 maps these criterion neighborhoods against the fit-complexity envelope. The solid line represents the Pareto frontier (E_{S1}), while the shaded areas show the neighborhoods for each criterion. This map highlights the spatial divergence between the parsimonious models selected by RICOMP and the more complex models selected by AIC and BIC.

Figure 1.7. S1 Fit-Complexity Envelope and IC-Specific Neighborhoods

Source: Author's elaboration.

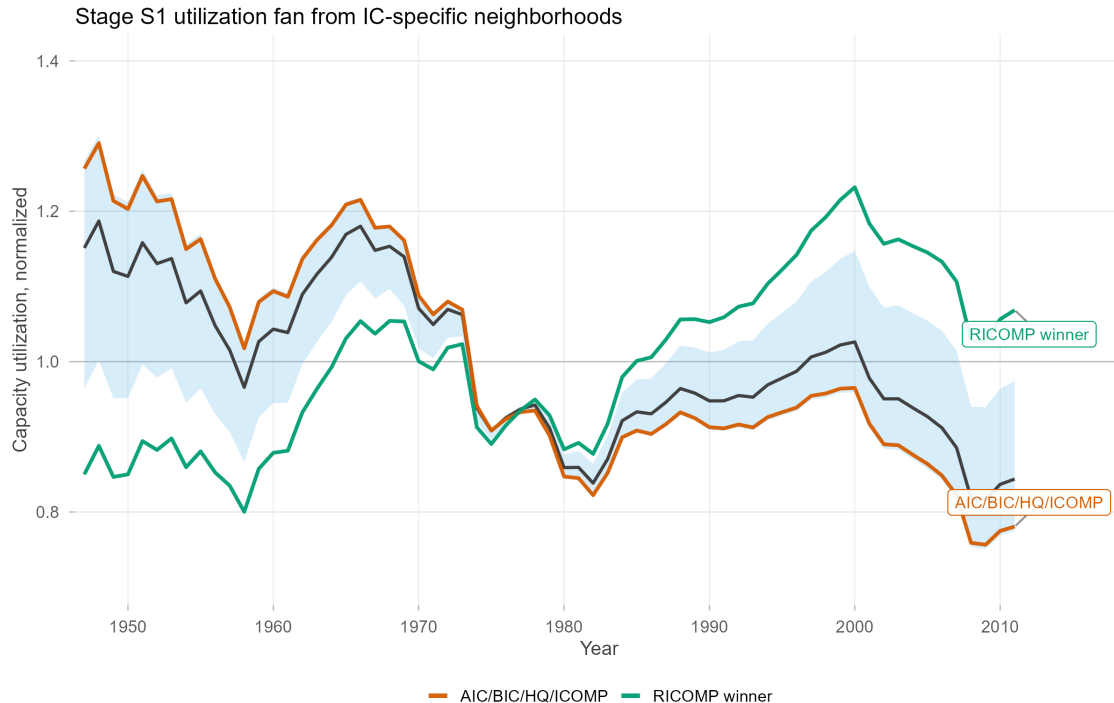
4.62 This criteria divergence reveals that different information-theoretic penalties locate very different regions of the specification space. AIC and BIC penalize the number of parameters, selecting the complex ARDL(3,3) model with $\hat{\theta} = 0.92$. In contrast, ICOMP and RICOMP penalize parameter interdependence and covariance complexity, selecting the parsimonious ARDL(1,2) model with $\hat{\theta} = 0.65$. This parameter variation has significant theoretical stakes. While no-trend configurations

(PSS Cases I, II, and III) yield stable elasticities below unity ($\hat{\theta} \in [0.65, 0.95]$) that support the overaccumulation hypothesis, trend-containing configurations (Cases IV and V) yield elasticities averaging 1.36 and reaching up to 2.20. An elasticity $\theta > 1$ implies that productive capacity expands faster than capital accumulation in the long run, violating basic physical constraints. We discard these trend-containing models not because they fail statistical bounds tests, but because they imply implausible long-run capacity dynamics.

4.63 Figure 1.8 plots the utilization paths generated by these different cointegrating models. The shaded fan shows the variance of constructed capacity utilization, while the divergent paths of the AIC and RICOMP winners highlight how model selection choices mechanically alter our historical interpretation of capacity realization.

4.64 To illustrate the physical stakes of these estimates, we run a counterfactual model imposing the balanced-growth constraint ($\theta = 1.0$), forcing capacity to grow in lockstep with capital. Under this constraint, the constructed capacity utilization series drifts to economically absurd levels, falling continuously to under 35% for decades and exceeding 105% during cyclical peaks. A corporate sector cannot operate at 35% capacity for thirty years without widespread bankruptcy, nor can it operate above 100% capacity indefinitely. This physical absurdity shows that the balanced-growth assumption is incompatible with the post-war US historical experience, confirming that we must estimate the elasticity θ freely to obtain a viable capacity utilization series.

Figure 1.8. S1 Diagnostic Utilization Fan from IC-Specific Neighborhoods



Source: Author's elaboration.

4.65 Stage S1 shows that while many single-equation specifications pass cointegration tests, they yield very different estimates of the transformation elasticity ($\hat{\theta}$). Because these models do not converge to a single parameter, any capacity path derived from a single-equation model remains fragile. In Subsection 1.4.6, we address this limitation by testing whether the output–capital relation survives in a joint multi-equation VECM system, and whether stability requires us to include the rate of exploitation.

1.4.6 Stage 2 (S2): System-Level VECM Replication

4.66 Stripping the output–capital relation down to a bivariate VECM exposes its fundamental econometric fragility. Across every lag profile and deterministic configuration, the bivariate system fails to cointegrate, leaving residuals trapped in a nonstationary ($I(1)$) drift. This breakdown demonstrates that capital and output cannot sustain a stable long-run trajectory in isolation; the bivariate relation is simply spurious. Establishing a genuine long-run capacity benchmark therefore requires abandoning the insulated technical assumption and explicitly conditioning the system on the macroeconomic distribution of income.

4.67 Because the bivariate system yields nonstationary residuals over the full post-war sample, estimating the output–capital relation requires including income distribution in the model. We incorporate the rate of exploitation, defined in Shaikh’s corporate accounts as corporate profits relative to employee compensation ($e_t = \Pi_t/W_t = \text{NOScorp}_t/\text{ECcorp}_t$). In logarithms, this variable links directly to the profit share ($\pi_t \equiv \Pi_t/Y_t$) and the wage share ($\omega_t \equiv W_t/Y_t = 1 - \pi_t$):

$$\ln e_t = \ln \left(\frac{\pi_t}{1 - \pi_t} \right) = \ln \pi_t - \ln \omega_t. \quad (1.26)$$

Econometrically, $\ln e_t$ is a continuous log-odds transform of functional distribution. It maps the bounded wage share $\omega_t \in (0, 1)$ onto the real line $(-\infty, +\infty)$, avoiding the boundary constraints that can distort linear vector error correction models. Furthermore, incorporating distribution through $\ln e_t$ avoids the collinearity that arises if the profit rate $r_t = \pi_t(Y_t/K_t)$ is entered alongside $\ln Y_t$ and $\ln K_t$.

4.68 We summarize these system-level outcomes in Table 1.9. All 36 bivariate specifications fail to cointegrate. In contrast, the trivariate family yields six admissible models when we include the rate of exploitation and our historical dummy variables. These results prove that the output–capital relation achieves system stability only when we account for income distribution.

Table 1.9. S2 System Admissibility Outcomes

System	Rank	Estimated	Admissible	Interpretation
Bivariate ($\ln Y, \ln K$)	$r = 1$	36	0	Output–capital relation is not self-sufficient
Trivariate ($\ln Y, \ln K, \ln e$)	$r = 1$	36	6	Restricted survival with logged exploitation
Trivariate ($\ln Y, \ln K, \ln e$)	$r = 2$	36	0	No admissible two-vector system

Source: Author’s elaboration.

4.69 To ground the VECM results, we select the VAR(2) specification under Johansen Case 2 (C_2 : restricted constant), h_2 dummies, and rank $r = 1$ as our focal model. The model passes all three admissibility gates. We verify cointegration rank using the Johansen Trace test: we reject the null hypothesis of no cointegration ($r = 0$) at the 5% level (Trace statistic $32.28 > 31.52$), and we fail to reject the null of a single cointegrating vector ($r \leq 1$) (Trace statistic $10.68 < 17.95$). We confirm dynamic stability through the VECM companion matrix eigenvalues: the moduli are 1.0, 1.0, 0.82, 0.52, 0.52, 0.49, all strictly bounded by the unit circle (except the two unit roots). Residual diagnostics verify that the system is free from residual pathology: the Portmanteau(12) statistic is 100.70 ($p = 0.275$), the Jarque-Bera statistic is 5.72 ($p = 0.456$), and the ARCH-LM(4) statistic is 150.03 ($p = 0.348$).⁵ We report the estimated cointegrating vector (β) and loading vector (α) in the following paragraphs.

4.70 Normalized on output, the estimated cointegrating vector is:

$$\ln Y_t = 0.727 \ln K_t - 20.022 \ln e_t + c_0 + \hat{\varepsilon}_t. \quad (1.27)$$

The estimated transformation elasticity ($\hat{\theta} = 0.727$, SE 4.852) is close to Shaikh's reconstructed single-equation benchmark ($\hat{\theta} = 0.720$). The large standard error reflects multicollinearity between the capital stock and the rate of exploitation over the post-war period. The coefficient on the rate of exploitation is statistically significant ($t = 7.90$, $p < 0.001$). To interpret the economic mechanism implied by this vector, we write the relation in its reserve-army form by renormalizing on $\ln e_t$:

$$\ln e_t = -0.050 (\ln Y_t - 0.727 \ln K_t) + c + \hat{v}_t. \quad (1.28)$$

The term $(\ln Y_t - 0.727 \ln K_t)$ represents capacity utilization disequilibrium. The estimated elasticity of -0.050 reflects a reserve-army feedback: when the economy operates above normal capacity ($\ln Y_t > 0.727 \ln K_t$), labor market tightening dampens the exploitation rate, raising the wage share. When excess capacity develops, unemployment relieves wage pressure, restoring the exploitation rate. This shows that at the system level, capacity utilization and distribution adjust jointly within a single cointegrating relationship.

4.71 The adjustment speeds (α) show a structural asymmetry in the system. Capital accumulation is weakly exogenous ($\alpha_k = 0.000$, $t = 0.92$), meaning that investment decisions do not respond to long-run deviations from the output-capital relation; instead, investment drifts independently of the long-run equilibrium. The output equation adjusts slowly but significantly ($\alpha_y = -0.005$, $t = -2.00$). The primary channel of error correction is the rate of exploitation ($\alpha_e = -0.019$, $t = -4.25$). Economically, this means that when output exceeds the capacity ceiling (relative to capital and exploitation), the rate of exploitation falls to restore system stability. This finding of weak exogeneity in capital accumulation ($\alpha_k = 0.000$) provides empirical support for the Sraffian position in the classical-heterodox debate. By showing that investment does not adjust to error-correction signals of capacity utilization disequilibrium, the data reject the Neo-Kaleckian view of investment adapting endogenously to demand-induced utilization gaps. However, because our long-run coefficient represents a non-unitary transformation elasticity ($\hat{\theta} = 0.727 < 1.0$), this Sraffian exogeneity is accompanied by structural overaccumulation. Under this interpretation, the secular

⁵The Breusch-Godfrey LM(4) statistic of 60.81 ($p = 0.006$) indicates residual serial correlation, a common small-sample VECM limitation that we document for transparency.

instability of unbalanced growth and the exogeneity of capital accumulation act as counteracting forces that may cancel each other out in the long run. This potential canceling obscures the resolution of the debate between Sraffian and Neo-Kaleckian models, a result that is invisible when working under a balanced-growth baseline with a fixed output–capital ratio. We report the short-run dynamics (Γ_1) and deterministic dummy coefficients (Φ) in Table 1.10.

Table 1.10. Focal VECM Short-Run Dynamics and Dummies

Regressor / Dummy	$\Delta \ln Y_t$ Equation	$\Delta \ln K_t$ Equation	$\Delta \ln e_t$ Equation
$\Delta \ln Y_{t-1}$	0.199 (0.150)	0.047** (0.023)	-0.798** (0.314)
$\Delta \ln K_{t-1}$	-1.015 (0.564)	0.812*** (0.087)	-4.023*** (1.183)
$\Delta \ln e_{t-1}$	-0.023 (0.067)	0.015 (0.010)	-0.069 (0.140)
D_{1956} (dummy)	-0.005 (0.015)	0.002 (0.002)	-0.062** (0.029)
D_{1974} (dummy)	-0.038** (0.017)	0.001 (0.002)	-0.078** (0.032)
D_{1980} (dummy)	0.001 (0.016)	-0.003 (0.002)	-0.014 (0.031)

Note: Standard errors are in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$. Source: Author's elaboration.

4.72 The historical dummy variables are essential for establishing system-level cointegration. Cointegration holds only when we include the step-shift dummies to control for structural breaks, preventing permanent shifts in the mean of the residuals from appearing as unit roots. Under the trend-containing Johansen Case 3 (C_3) configurations, the models yield invalid parameter estimates, including negative ($\hat{\theta} = -0.81$) or explosive ($\hat{\theta} = 11.31$) elasticities. In these configurations, the linear trend absorbs the long-run growth of output, forcing the capital coefficient to fit short-run cyclical noise. We therefore reject these trend-containing models. The economically viable models are restricted to no-trend Cases C_0 and C_2 , which yield stable, positive elasticity estimates ranging from 0.73 to 0.97 (Table 1.11).

Table 1.11. S2 Retained Trivariate Specifications

Lag Order (p)	Deterministic Case (C)	Historical Dummy	Rank (r)	Cointegrating Coefficient ($\hat{\theta}$)
1	C_0	h_2	1	0.91
1	C_2	h_2	1	0.97
1	C_3	h_2	1	-0.81
2	C_0	h_2	1	1.19
2	C_2	h_2	1	0.73
2	C_3	h_2	1	11.31

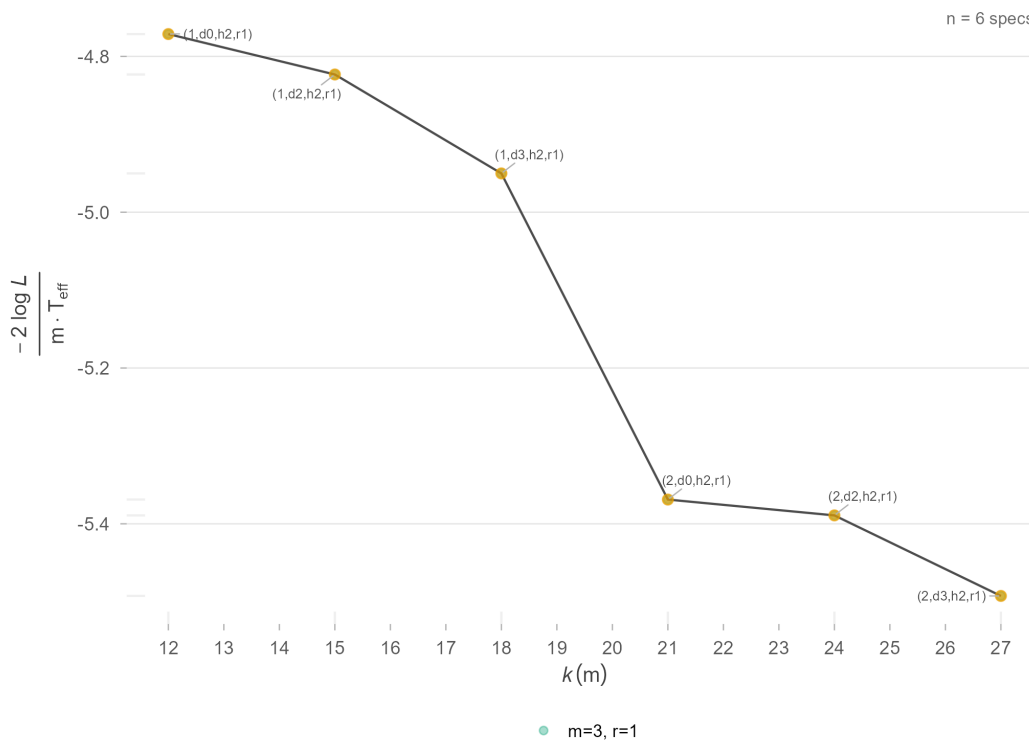
Source: Author's elaboration.

4.73 To illustrate the role of historical controls, we run a "no-dummy" counterfactual where we omit the dummy variables. When we leave out these shock controls, the VECM fails to reject the null hypothesis of no cointegration under both Johansen trace and maximum eigenvalue tests. The residuals remain nonstationary ($I(1)$), proving that the estimated relationship is spurious. Output, capital, and exploitation drift apart permanently without a restoring force. This counterfactual demonstrates that the output–capital relation is not an institution-free physical law; its stability

depends on controlling for major post-war shocks like the 1974 oil crisis and the 1980 Volcker credit contraction.

4.74 We plot the system-level fit-complexity frontier in Figure 1.9. Bivariate systems fail to cointegrate, and adding variables beyond the rate of exploitation does not improve system-level identification. This confirms that the trivariate system is the minimum specification necessary for econometric admissibility.

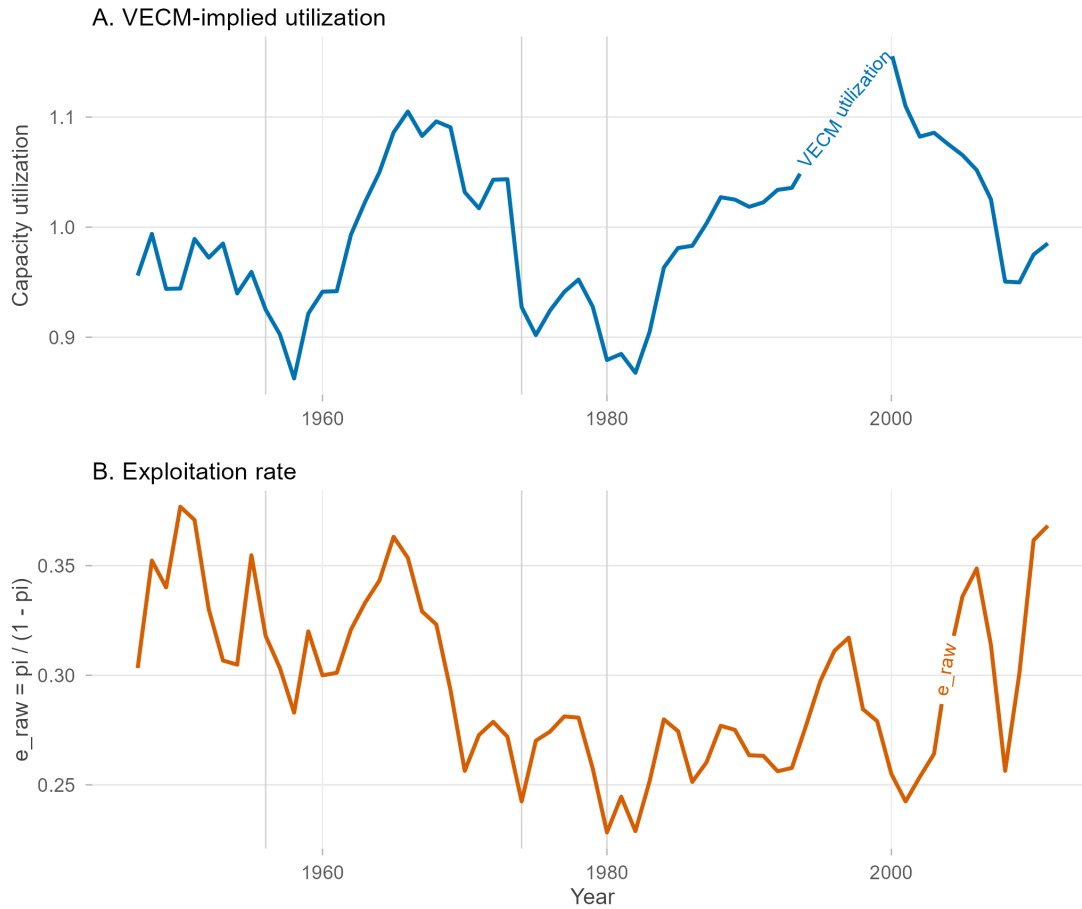
Figure 1.9. S2 Pooled System Frontier



Source: Author's elaboration.

4.75 Figure 1.10 plots the constructed rate of capacity utilization alongside the rate of exploitation. In the United States, the rate of exploitation peaked in the mid-1960s, declined during the stagflation of the 1970s, and rose steadily from 1983 to 2011 as wage growth lagged behind productivity. This trajectory reveals a structural link between capacity utilization and the functional distribution of income, showing that the long-run realization of capacity is conditioned by class struggle.

Figure 1.10. S2 Focal Utilization and Exploitation Diptych



Source: Author's elaboration.

4.76 The theoretical justification for including $\ln e_t$ in the cointegrating vector rests on the classical choice-of-technique framework formalized by Kurz (1986). In capitalist economies, the conversion of a given capital stock into productive capacity is not an invariant technical relationship. Cost-minimizing firms select techniques from an available technology set based on prevailing wage rates and profit margins. Changes in distribution shift relative factor costs, prompting firms to adjust techniques, machine speeds, and shift arrangements. When distribution is omitted from the state vector, these technical adjustments appear as persistent residual drift in the output–capital ratio, causing bivariate cointegration tests to fail. Including $\ln e_t$ controls for technique choice, allowing the long-run relation to be identified. Ultimately, the survival of the output–capital relation only when income distribution enters the cointegrating space indicates that capacity utilization is not a neutral technical law, but a historically conditioned outcome of decentralized accumulation and distributive conflict.

1.4.7 Cross-Stage Econometric Synthesis

4.77 The three-stage replication defines the strict historical and institutional boundary conditions of Shaikh’s capacity measurement strategy. While the baseline single-equation coefficient is numerically reproducible (Stage S0), it remains structurally non-unique and highly sensitive to information-criterion choices (Stage S1). Most critically, the standalone output-capital relation fails system-level cointegration, surviving exclusively within a trivariate VECM conditioned on income distribution and historical shock controls (Stage S2). The long-run conversion of capital to capacity is therefore not a fixed engineering multiplier. It is an institutional reality, fundamentally shaped by systemic crises and the ongoing balance of power between wages and profits.

Table 1.12. Cross-Stage Synthesis of the Cointegration Architecture

Stage	Outcome	Interpretive Bound
S0	Approximate single-equation reproducibility	Shaikh’s method is reproducible, but not exactly
S1	Non-empty admissible ARDL grid; non-unique retained benchmark	Model selection is driven by information-criterion choice
S2 Bivariate	48 estimated, 0 admissible	Output–capital system is not self-sufficient over 1947–2011 (12 admissible over 1947–2007)
S2 Trivariate Rank One	48 estimated, 6 admissible	Restricted survival with logged exploitation and h_2 controls
S2 Trivariate Rank Two	48 estimated, 0 admissible	No retained dual-vector system

Source: Author’s elaboration.

4.78 A key result of our system-level estimation is that the bivariate VECM yields zero admissible specifications over the full post-war period (1947–2011, $T = 65$). Across the 48 bivariate specifications estimated over alternative lag lengths and deterministic trends, the Johansen trace test fails to reject the null hypothesis of no cointegration ($r = 0$) at the 5% level.

4.79 Re-estimating the grid over Shaikh’s pre-crisis sample period (1947–2007, $T = 61$) shows that this bivariate failure reflects sample sensitivity. Over the pre-2008 sample, the bivariate system yields 12 admissible specifications, producing an estimated elasticity of $\hat{\theta} \in [0.88, 0.91]$. Adding the four years of the Great Recession (2008–2011) disrupts this relation: the sharp fall in value added relative to a slow-depreciating capital stock introduces persistent drift into the residuals, causing bivariate cointegration to break down. We report both sample outcomes to clarify this sensitivity: single-equation models that report cointegration over the full sample without testing system rank miss this post-2007 divergence.

4.80 In contrast to the bivariate models, the trivariate system including the logged exploitation rate ($\ln e_t$) identifies cointegration over the full 1947–2011 period, producing six admissible specifications at rank $r = 1$. All six surviving specifications include step dummies for 1956, 1974, and 1980 (h_2). Estimating the specification without dummies (h_0) yields zero admissible models across all

lag orders. This shows that identifying a cointegrating relationship requires controlling for major institutional shocks: the post-Korean War downturn, the 1974 oil and profitability crisis, and the 1980 Volcker monetary tightening.

1.5 Conclusion

5.1 The standalone output-capital relation cannot sustain a stable capacity benchmark; cointegration emerges only when the rate of exploitation enters the system. This critical replication of [Shaikh \(2016\)](#)’s accounting-based methodology demonstrates that the bivariate VECM fails across all lag profiles and deterministic configurations, without any evidence of cointegration between output and capital stocks. Instead, cointegration holds only in a trivariate VECM conditioned on income distribution and historical shock controls. Once trend-containing specifications are excluded as economically inadmissible, the retained models yield a transformation elasticity below unity ($\hat{\theta} < 1.0$), placing the post-war US corporate sector in a structural overaccumulation regime where capital accumulation outpaces capacity formation in the long-run.

5.2 The non-unitary transformation elasticity imposes a secular downward drag on the rate of profit that capitalists can only counteract by intensifying exploitation or idling capacity. Within the classical-Marxian decomposition $r = (\Pi/Y) \cdot \mu \cdot (Y^p/K)$, a transformation elasticity below unity forces the capital-capacity ratio (Y^p/K) into a persistent decline as capital accumulates. The speed of adjustment of the system suggests that capital accumulation is weakly exogenous ($\alpha_k \approx 0$), while the rate of exploitation serves as the primary error-correction channel restoring long-run reproduction. This weak exogeneity of investment supports the Sraffian position over the Neo-Kaleckian demand-endogenous view. Yet the structural overaccumulation ($\hat{\theta} < 1.0$) introduces a counteracting dynamic: the secular instability of unbalanced growth and the weak exogeneity of investment may cancel each other out, obscuring the resolution of this debate. This theoretical impasse reflects the shared balanced-growth assumption underlying both frameworks, which forces a fixed output-capital ratio onto a historically shifting accumulation process. Ultimately, the capacity ceiling is not a neutral engineering constraint but a contingent outcome of decentralized accumulation, shaped by the historical balance of class power and the distributive conflicts of the capitalist labor process.

5.3 The single-sector bounding and the time-invariant parameterization of θ represent the two primary analytical limitations of this chapter. By abstracting from inter-sectoral disproportionalities, capacity transfers, and uneven development across departments of production, the single-sector framework isolates the aggregate capital-to-capacity relation at the cost of tracing how sectoral imbalances spill over to affect the general rate of profit. Similarly, treating θ as a constant over the 1947–2011 period forces the model to average across historically distinct choice-of-technique regimes and social structures of accumulation. While external demand components—military spending, state deficits, and consumer debt expansion—may stabilize the overaccumulation tendency, these transmission mechanisms remain hypotheses rather than findings directly estimated here.

5.4 The empirical failure of the standalone output-capital relation establishes a clear theoretical imperative: capacity measurement must integrate workplace-level conflict *ex ante* rather than appending it *ex post*. The immediate next step is to endogenize the transformation elasticity θ

as a time-varying parameter driven by class struggle and the choice of technique, tracing how the capital-capacity conversion shifts across historical US accumulation regimes. Building on this closed-economy foundation for the center, the dissertation must subsequently confront the structural realities of the periphery. In peripheral economies, the productive ceiling is not merely bounded by domestic distributive conflict, but is structurally truncated by the Balance of Payments (BoP) constraint. For these economies, potential capacity is subordinated to the external viability of the accumulation process; the imperative to generate sufficient foreign exchange to finance essential imports and service external debt acts as a hard, macroeconomic ceiling on domestic expansion. The next analytical layer will therefore relax the single-sector bounding to incorporate open-economy dynamics, estimating a capacity utilization rate for peripheral countries that explicitly conditions the output-capital relation on these external constraints. This upgrade will reveal whether chronic idle capacity in the periphery is driven by domestic overaccumulation or by the structural necessity of maintaining external equilibrium, ultimately demonstrating that the capacity ceiling in the Global South is a geopolitical and distributive construct rather than a neutral technical limit.

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Appendix 1.A Derivation of Accumulation Dynamics under Unbalanced Growth

This appendix provides the derivation of the dynamical equations summarized in §1.3.3 and §1.3.4. All results follow from three components: the accounting identity $\hat{k} + \delta = \phi^p B$, the unbalanced growth closure $g_{Y^p} = \theta g_K$, and the investment-savings closure $\phi^p = s$.

1.A.1 The Unbalanced Growth Closure: Derivation of the Bernoulli ODE

The accounting identity under the investment-savings closure is:

$$\hat{k} + \delta = s B. \quad (1.29)$$

Differentiating with respect to time yields:

$$\frac{d\hat{k}}{dt} = s \dot{B}. \quad (1.30)$$

Expressing $\dot{B}$ in terms of the growth rate of capital productivity ($\hat{b}$):

$$\dot{B} = B \hat{b}. \quad (1.31)$$

Substituting this into equation (1.30) and utilizing the accounting identity $sB = \hat{k} + \delta$ results in the acceleration identity:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta) \hat{b}. \quad (1.32)$$

Recall that capital productivity is $B \equiv Y^p/K$, which in growth rates implies $\hat{b} = \hat{y}^p - \hat{k}$. Under the unbalanced growth closure $g_{Y^p} = \theta g_K$ (or $\hat{y}^p = \theta \hat{k}$ in growth rate notation), this gives the growth rate of capital productivity as $\hat{b} = (\theta - 1) \hat{k}$. Substituting this into equation (1.32) provides the acceleration dynamic:

$$\boxed{\frac{d\hat{k}}{dt} = (\theta - 1) \hat{k} (\hat{k} + \delta)}, \quad \text{where } \hat{k} \text{ and } \delta \text{ are expressed in consistent temporal and physical units.} \quad (1.33)$$

This is a Bernoulli equation of order 2 in $\hat{k}$, where the nonlinearity arises from the product $\hat{k}(\hat{k} + \delta)$.

1.A.2 Analytical Solution

Expanding equation (1.33) gives:

$$\frac{d\hat{k}}{dt} = (\theta - 1) \hat{k}^2 + (\theta - 1) \delta \hat{k}. \quad (1.34)$$

Applying the Bernoulli substitution $v \equiv \hat{k}^{-1}$ with $\dot{v} = -\hat{k}^{-2} \dot{\hat{k}}$ transforms the equation into a first-order linear ODE:

$$\dot{v} + (\theta - 1) \delta v = -(\theta - 1). \quad (1.35)$$

The solution for $v(t)$, given initial conditions $v_0 = \hat{k}_0^{-1}$, is:

$$v(t) = \left(v_0 + \frac{1}{\delta} \right) e^{-(\theta-1)\delta t} - \frac{1}{\delta}. \quad (1.36)$$

Inverting $v(t)$ yields the explicit path for the net accumulation rate:

$$\hat{k}(t) = \frac{1}{\left(\frac{1}{\hat{k}_0} + \frac{1}{\delta} \right) e^{-(\theta-1)\delta t} - \frac{1}{\delta}}. \quad (1.37)$$

1.A.3 Equilibria and Stability Analysis

The equilibria $\hat{k}^*$ are defined by $(\theta - 1)\hat{k}(\hat{k} + \delta) = 0$. For $\theta \neq 1$, the roots are $\hat{k}_1^* = 0$ (stationary net capital) and $\hat{k}_2^* = -\delta$ (zero gross investment). Linearizing around $\hat{k}^* = 0$ via the Jacobian $f'(\hat{k}) = (\theta - 1)(2\hat{k} + \delta)$ yields:

- Over-accumulation ($\theta < 1$): $f'(0) < 0$ — the stagnation equilibrium is locally stable.
- Under-accumulation ($\theta > 1$): $f'(0) > 0$ — the stagnation equilibrium is unstable.

1.A.4 The Trend-Stabilized Closure: Derivation of the Quadratic ODE

Under the trend-stabilized closure $\hat{y}^p = \theta\hat{k} + b$, the growth rate of capital productivity is:

$$\hat{b} = (\theta - 1)\hat{k} + b. \quad (1.38)$$

Substituting this into the acceleration identity (equation 1.32) generates a quadratic ODE:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta) [(\theta - 1)\hat{k} + b]. \quad (1.39)$$

The equilibria are the roots of $(\hat{k} + \delta)[(\theta - 1)\hat{k} + b] = 0$:

$$\hat{k}_1^* = -\delta, \quad \hat{k}_2^* = \frac{b}{1 - \theta}. \quad (1.40)$$

The Jacobian $g'(\hat{k}) = 2(\theta - 1)\hat{k} + (\theta - 1)\delta + b$ evaluated at $\hat{k}_2^*$ confirms local stability for the empirically relevant case ($\theta < 1$ and $b > 0$): the interior root $b/(1 - \theta)$ acts as a strictly positive attractor, preventing stagnation. As $b \rightarrow 0$, the trend-stabilized closure collapses to the unbalanced growth attractor at $\hat{k}^* = 0$.

Appendix 1.B Data Construction and Diagnostic Support

1.B.1 Series Glossary

Table 1.13 lists the core replication variables, their sources, and transformation rules. All series derive from Shaikh's (2016) data companions via Real Economic Analysis.⁶

Table 1.13. Replication series: symbols, sources, and transformations

Symbol / Code	Variable	Source (table, line)	Transformation
$GVAcorp_t$	Corporate gross value added (corrected)	NIPA 1.14 (L1) + imputed-interest adj. (Table 7.11)	nominal
$KGCcorp_t$	Corporate gross capital stock (GPIM-adjusted)	Fixed Asset 6.1 (L9); IRS Statistics of Income; GPIM recursion	nominal
$KNCcorp_t$	Corporate net capital stock (GPIM-adjusted)	same as above	nominal
$KNCcorpbea_t$	Corporate net capital stock (BEA 2011 official)	Fixed Asset 6.1 (L9)	nominal
p_t	Common deflator	pKN index, 2005=100	$(KNCcorp/KNRcorp)*100$
y_t	Real log output	constructed	$\ln(VAcorp_t/p_t)$
k_t	Real log capital	constructed	$\ln(KGCcorp_t/p_t)$
$Profshcorp_t$	Corporate profit share (corrected)	NIPA 1.14 (L8, L1–L2) + imputed-interest adj.	ratio

Note: All BEA and NIPA data downloaded in the replication vintage. Shaikh's original series used data downloaded in 2011 (Shaikh, 2016, Appendix 6.7, fn. 1); discrepancies attributable to subsequent BEA revisions are possible and noted where material.

Source: Real Economic Analysis Data Tables

1.B.2 Imputed-Interest Correction

The NIPA corporate value added figure includes imputed financial intermediation services as a cost to non-financial corporations. The corporate imputed-interest adjustment removes this component:

$$CorpImpIntAdj_t \equiv -BankMonIntPaid_t - CorpNFNetImpIntPaid_t, \quad (1.41)$$

where $BankMonIntPaid_t$ is NIPA Table 7.11 (lines 4+44+73 minus lines 28+52+91) and $CorpNFNetImpIntPaid_t$ is NIPA Table 7.11 (line 74 minus line 53). In 2009 the adjustment raises corporate net operating surplus by approximately 10 %. Figure 1.11 displays the corrected versus uncorrected series and their ratio.

1.B.3 Corrected Corporate Output: $GVAcorp_t$

$$GVAcorp_t \equiv GVAcorpnipa_t + CorpImpIntAdj_t, \quad (1.42)$$

where $GVAcorpnipa_t$ is NIPA Table 1.14 (line 1) and $CorpImpIntAdj_t$ is defined in equation (1.41).

⁶Replication data and construction scripts are available at Real Economic Analysis: <https://www.realecon.org/data>.

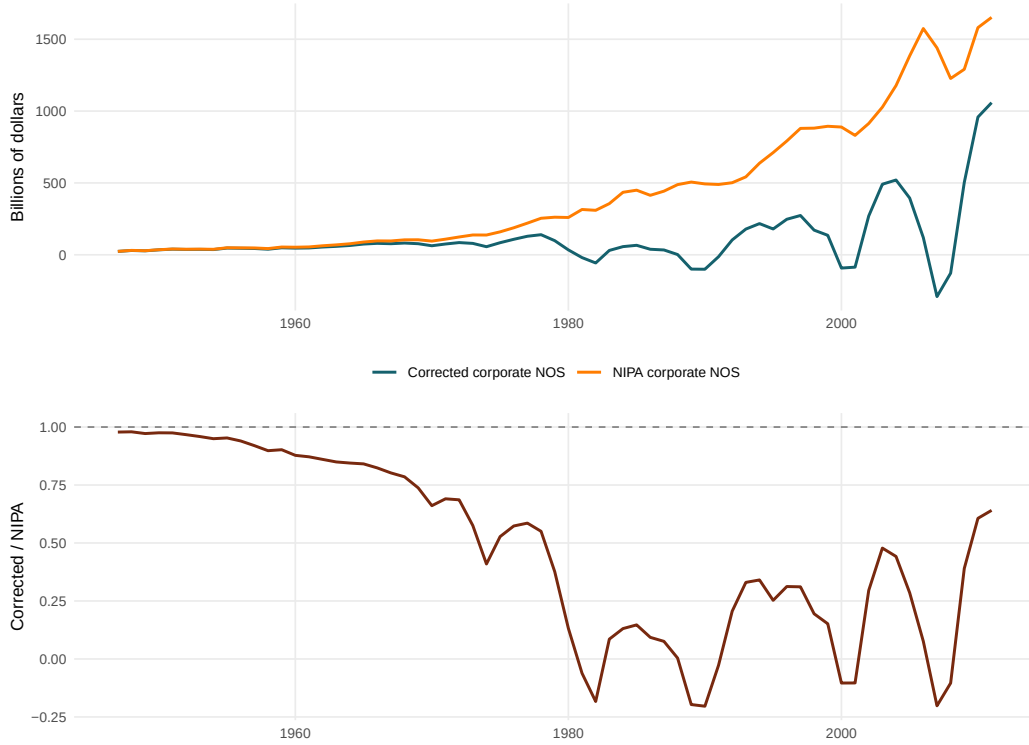


Figure 1.11. Imputed interest adjustment impact: corrected corporate NOS versus NIPA corporate NOS (1947–2011). Ratio shows magnitude of adjustment.

1.B.4 Reconstructed Corporate Capital Stock: $KGC_{corp,t}$

GPIM accumulation rule. The current-cost capital stock is accumulated as:

$$K_t = IG_t + z_t^* \cdot K_{t-1}, \quad z_t^* \equiv (1 - z_t) \frac{p_t^K}{p_{t-1}^K}, \quad (1.43)$$

where IG_t is nominal gross investment, z_t is the period depreciation rate, and z_t^* is the survival-and-revaluation factor that reflates the surviving stock to current replacement cost.

Three adjustments (applied before the recursion).

1. **Depletion rates.** BEA 1993 finite service lives replace the BEA 2011 infinite-lives assumption, implying higher depreciation and lower retirement rates consistent with physical exhaustion.
2. **Initial values.** BEA 1993 initial values anchored to 1925 replace the BEA 2011 starting values. The 1993 values are lower, producing faster accumulation growth and convergence to the official path by approximately 1977.
3. **Great Depression correction.** The IRS corporate book value index (of the Census, 1975, Series V 115, pp. 924–926) is applied over 1925–1947, rescaling the BEA historical stock by the IRS/BEA ratio each year. The GPIM recursion resumes from 1947. Net effect: $KNC_{corp,t}$ is approximately 28 % lower in 1947 than the BEA 2011 official measure.

Inventories. IRS corporate balance sheet inventory ratios are extrapolated over 1947–1989 via a linear fit to the 1990–2011 IRS/BEA ratio, rescaled to the adjusted GPIM stocks, and added to $KGCcorp_t$. Source: IRS Statistics of Income 1926–2011; FRB Flow of Funds FL105015205.A for non-financial corporate inventories in current cost.

Accuracy. The average ratio of the GPIM approximation to the BEA current-cost stock over 1947–2005 is 99.60 %.

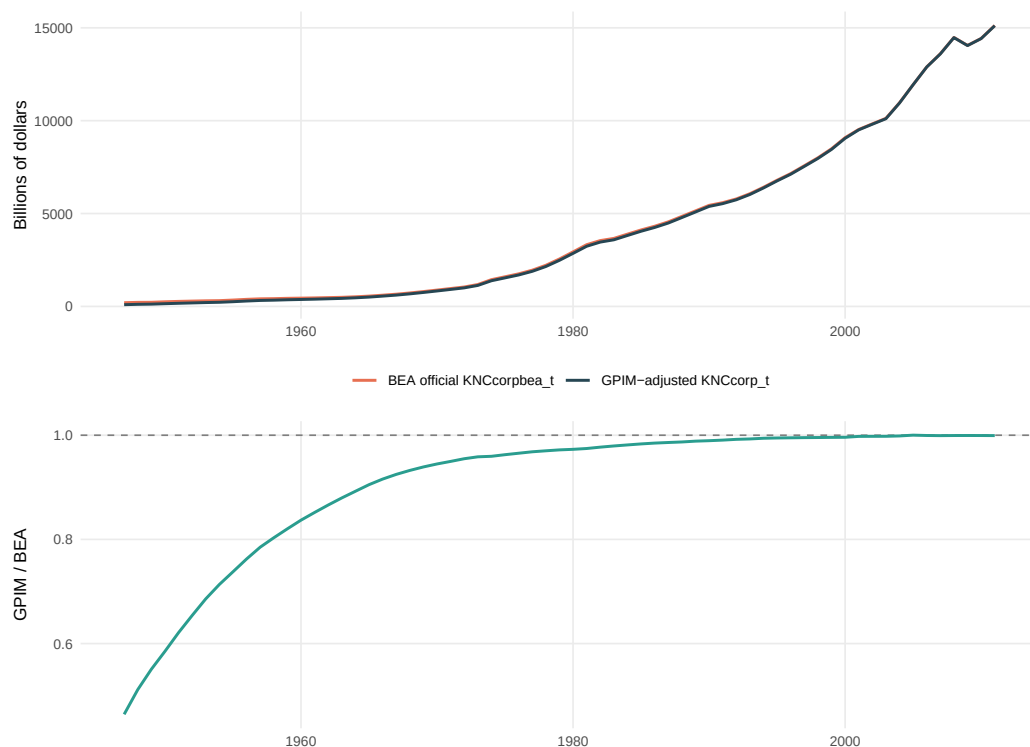


Figure 1.12. GPIM-adjusted net capital stock versus BEA 2011 official series (1947–2011). Ratio plot shows convergence by 1977.

1.B.5 Common Deflator: p_t

The replication uses the pKN implicit price deflator based at $100 = 2005$, estimated under GPIM rules. Current BEA releases employ quality-adjusted chain weights that adjust for quality vintage, breaking stock-flow consistency for accumulation accounting. The pKN index preserves stock-flow consistency, ensuring the output–capital relation reflects real productive dynamics rather than quality-adjustment artifacts (Shaikh, 2016, Appendix 6.6, Sec. II).

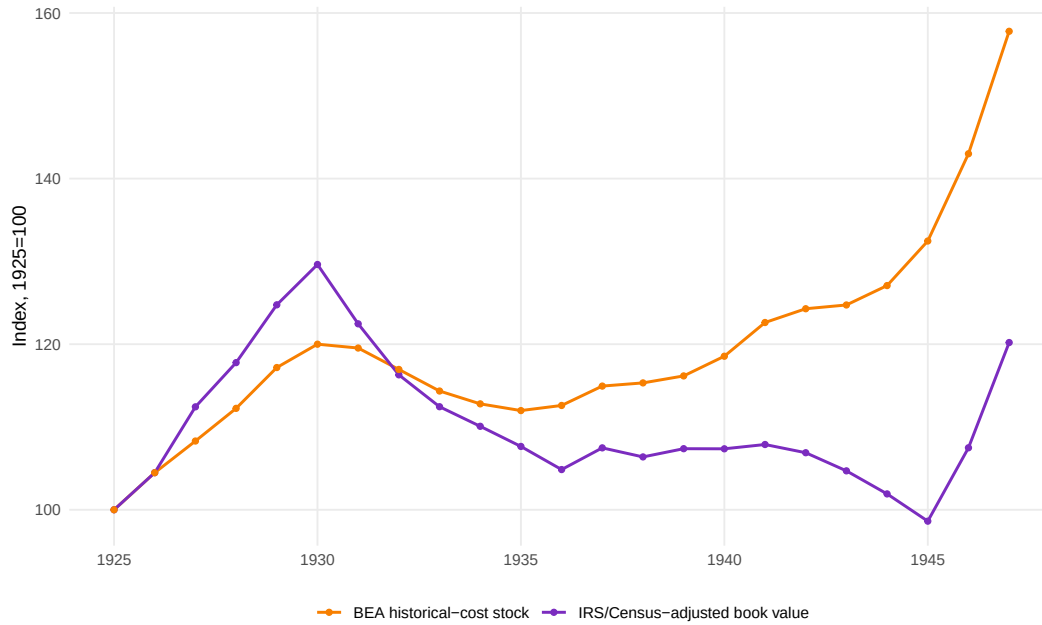


Figure 1.13. IRS corporate book value index versus BEA historical stock (1925–1947). Both indexed to 100 in 1925.

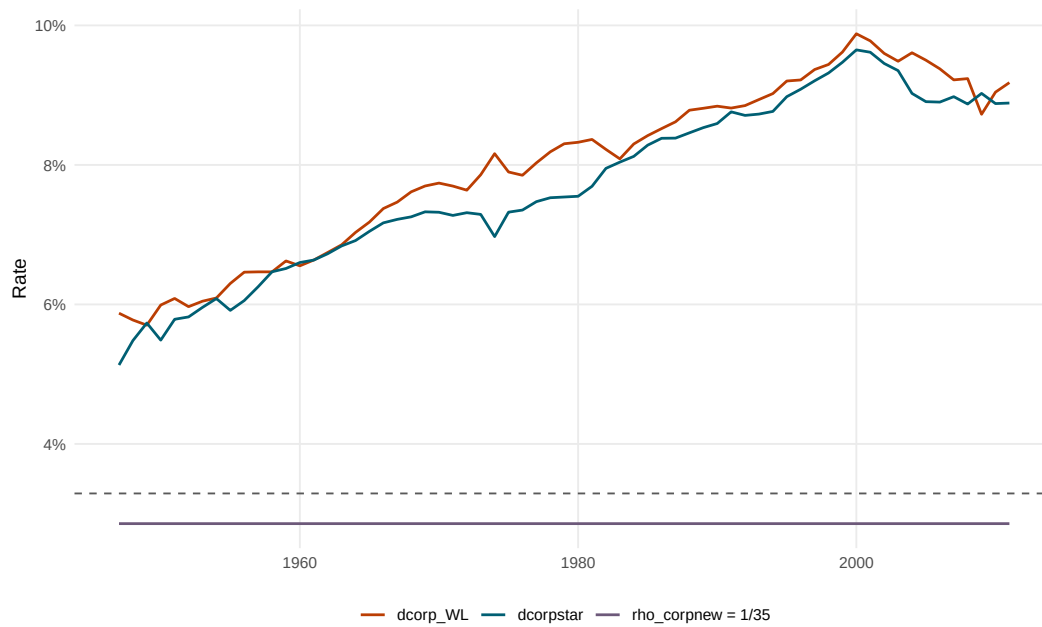


Figure 1.14. Depreciation and retirement rates under BEA 1993 finite service lives. Horizontal line marks critical value (3.29%).

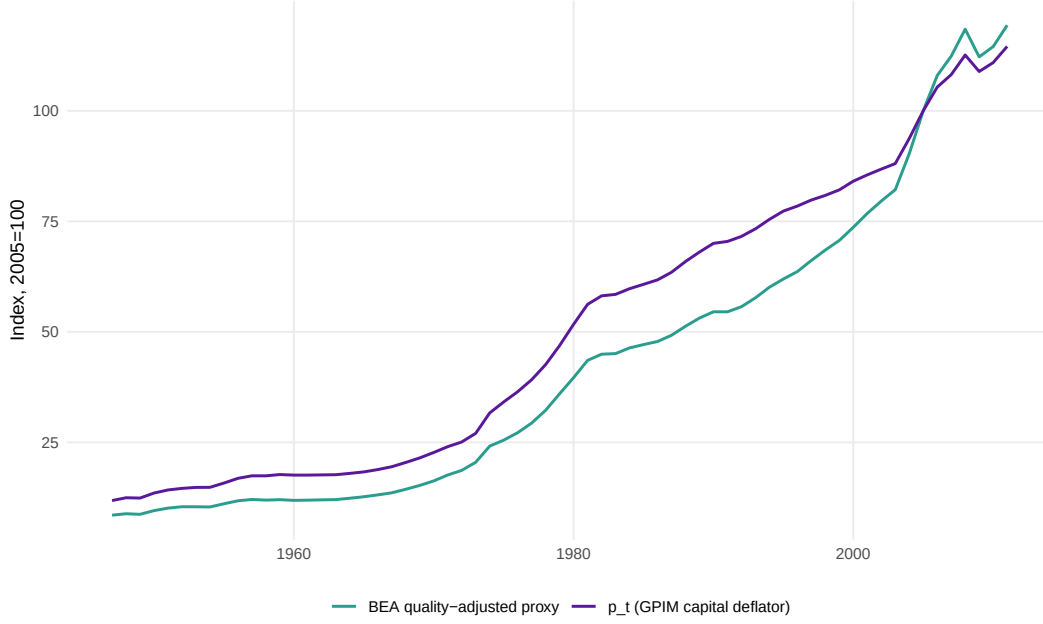


Figure 1.15. Common deflator index: GPIM-consistent pKN versus BEA quality-adjusted proxy (2005=100).

1.B.6 Corrected Corporate Profit Share: Profshcorp_t

The same imputed-interest adjustment applied to GVAcorp_t is applied to both the numerator and denominator of the profit share:

$$\text{NOScorp}_t \equiv \text{NOScorpnipa}_t + \text{CorpImpIntAdj}_t, \quad (1.44)$$

$$\text{VAcorp}_t \equiv \text{VAcorpnipa}_t + \text{CorpImpIntAdj}_t, \quad (1.45)$$

$$\text{Profshcorp}_t \equiv \frac{\text{NOScorp}_t}{\text{VAcorp}_t}, \quad (1.46)$$

where NOScorpnipa_t is NIPA Table 1.14 (line 8) and VAcorpnipa_t is NIPA Table 1.14 (line 1 minus line 2).

1.B.7 Descriptive Statistics and Unit Root Tests

Table 1.14 reports summary statistics for the core series. Table 1.15 presents unit root test results (ADF, Phillips–Perron) for y_t and k_t in levels and first differences.

Table 1.14. Core series summary statistics (1947–2011)

Variable	N	Mean	Median	Std Dev	Min	Max	Skewness	Kurtosis
y_t	65	14.9281	14.8653	0.5674	13.8866	15.7628	−0.1438	1.8398
k_t	65	15.9658	15.9810	0.7313	14.7979	17.1127	−0.0206	1.6872
Δy_t	64	0.0293	0.0355	0.0391	−0.0979	0.1055	−0.6331	3.4760
Δk_t	64	0.0362	0.0361	0.0078	0.0186	0.0530	−0.1127	2.4022
p_t	65	50.3486	46.8900	32.8159	11.8262	114.5539	0.3955	1.8063

D_{56}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156
D_{74}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156
D_{80}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156

Note: Source: Real Economic Analysis (<https://www.realecon.org/data>), Shaikh (2016) data companions. Variables constructed per Table 1.13 of Appendix 1.B. Dummy variables are impulse indicators with mean $1/65 \approx 0.0154$.

Table 1.15. Unit root tests for core series (1947–2011)

Variable	Form	Test	Specification	Statistic	Lag/BW	Critical values			Reject at 5%
						1%	5%	10%	
y_t	level	ADF	none	3.0027	1	-2.60	-1.95	-1.61	no
y_t	level	PP	intercept	-1.3280	3	-3.53	-2.91	-2.59	no
y_t	level	PP	trend	-2.1341	3	-4.11	-3.48	-3.17	no
y_t	Δ	ADF	intercept	-5.0966	1	-3.51	-2.89	-2.58	yes
y_t	Δ	PP	intercept	-5.9637	3	-3.54	-2.91	-2.59	yes
y_t	Δ	PP	trend	-5.9460	3	-4.11	-3.48	-3.17	yes
k_t	level	ADF	none	1.1552	4	-2.60	-1.95	-1.61	no
k_t	level	PP	intercept	-0.3863	3	-3.53	-2.91	-2.59	no
k_t	level	PP	trend	-1.2114	3	-4.11	-3.48	-3.17	no
k_t	Δ	ADF	intercept	-2.0670	3	-3.51	-2.89	-2.58	no
k_t	Δ	PP	intercept	-1.9027	3	-3.54	-2.91	-2.59	no
k_t	Δ	PP	trend	-1.8431	3	-4.11	-3.48	-3.17	no

Note: ADF: Augmented Dickey–Fuller test with max lag 6 and AIC selection. PP: Phillips–Perron test with Newey–West bandwidth. "none" = no deterministic terms; "intercept" = constant only; "trend" = constant + linear trend. PP "none" specification not computed. Rejection at 5% indicates stationarity under that specification.

Figures showing the log levels of output and capital, the output–capital ratio, and their annual growth rates are presented in the main text (see Section 1.4.2, Figures 1.2, 1.3, and 1.4).

1.B.8 ARDL Specification Search and Bounds Testing

Table 1.16 displays the full specification grid ($p, q = 1\text{--}6$; Cases I–V) with information criteria, diagnostic tests, and recovered long-run coefficients. Asterisks mark case-specific AIC/BIC minima.

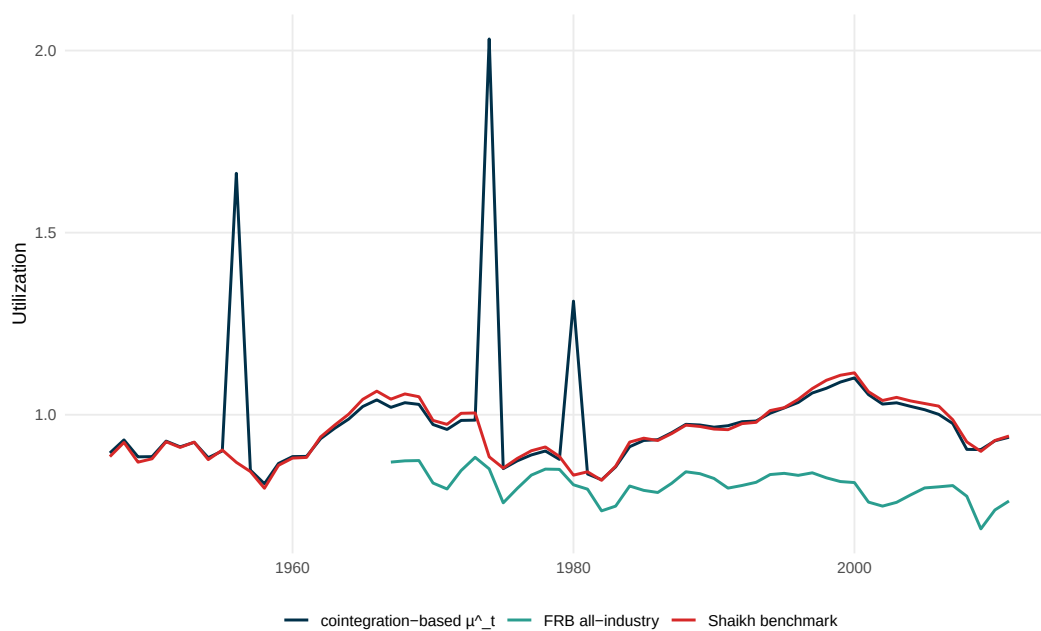


Figure 1.16. Capacity utilization comparison: Shaikh (cointegration-based) versus FRB survey measure, 1947–2011. Descriptive comparison only.

Table 1.16. ARDL specification search: information criteria and diagnostics (1947–2011)

p	q	Case	AIC	SBC	T_{eff}	$\hat{\theta}$	LM(1)	p -val	LM(4)	p -val	Best
1	1	I	−237.63	−222.52	64	0.8529	3.2749	0.070	5.0893	0.278	
1	1	II	−243.09	−225.82	64	0.6989	3.7150	0.054	6.9003	0.141	
1	1	III	−243.09	−225.82	64	0.6989	3.7150	0.054	6.9003	0.141	
1	1	IV	−243.09	−223.66	64	−0.5048	3.4486	0.063	6.5385	0.162	
1	1	V	−243.09	−223.66	64	−0.5048	3.4486	0.063	6.5385	0.162	
1	2	I	−239.33	−222.19	63	0.7273	7.0198	0.008	11.1845	0.025	
1	2	II	−242.69	−223.41	63	0.7110	7.1117	0.008	11.8692	0.018	
1	2	III	−242.69	−223.41	63	0.7110	7.1117	0.008	11.8692	0.018	
1	2	IV	−241.90	−220.47	63	−0.4718	7.0457	0.008	11.2449	0.024	
1	2	V	−241.90	−220.47	63	−0.4718	7.0457	0.008	11.2449	0.024	
1	3	I	−244.02	−224.88	62	0.8967	11.3499	0.001	14.1599	0.007	
1	3	II	−249.33	−228.05	62	0.6695	10.3605	0.001	12.7048	0.013	
1	3	III	−249.33	−228.05	62	0.6695	10.3605	0.001	12.7048	0.013	
1	3	IV	−249.62	−226.22	62	−0.8749	10.3560	0.001	11.9612	0.018	*
1	3	V	−249.62	−226.22	62	−0.8749	10.3560	0.001	11.9612	0.018	*
2	3	I	−250.91	−229.63	62	0.9301	5.4621	0.019	9.3503	0.053	
2	3	II	−262.79	−239.40	62	0.7023	0.5641	0.453	3.5633	0.468	*
2	3	III	−262.79	−239.40	62	0.7023	0.5641	0.453	3.5633	0.468	*
2	3	IV	−264.33	−238.81	62	−0.4836	0.2849	0.594	2.6102	0.625	*
2	3	V	−264.33	−238.81	62	−0.4836	0.2849	0.594	2.6102	0.625	*
4	5	I	−253.33	−224.01	60	0.7899	0.0461	0.830	7.6617	0.105	*
4	5	II	−257.75	−226.33	60	0.7102	0.2628	0.608	6.6363	0.156	
4	5	III	−257.75	−226.33	60	0.7102	0.2628	0.608	6.6363	0.156	
4	5	IV	−257.11	−223.60	60	−0.4117	0.2120	0.645	6.5141	0.164	
4	5	V	−257.11	−223.60	60	−0.4117	0.2120	0.645	6.5141	0.164	

Note: ARDL(p, q) specifications estimated with impulse dummies D_{56}, D_{74}, D_{80} . AIC/SBC: Akaike/Schwarz information criteria (lower = better). T_{eff} : effective sample size after lag construction. $\hat{\theta}$: long-run output–capital coefficient recovered from ECM parameters. LM(1)/LM(4): Breusch–Godfrey serial correlation tests (null = no autocorrelation). Asterisk (*) marks best specification by AIC/SBC within each deterministic case. Full grid ($p, q = 1$ –6; Cases I–V) available in replication code.

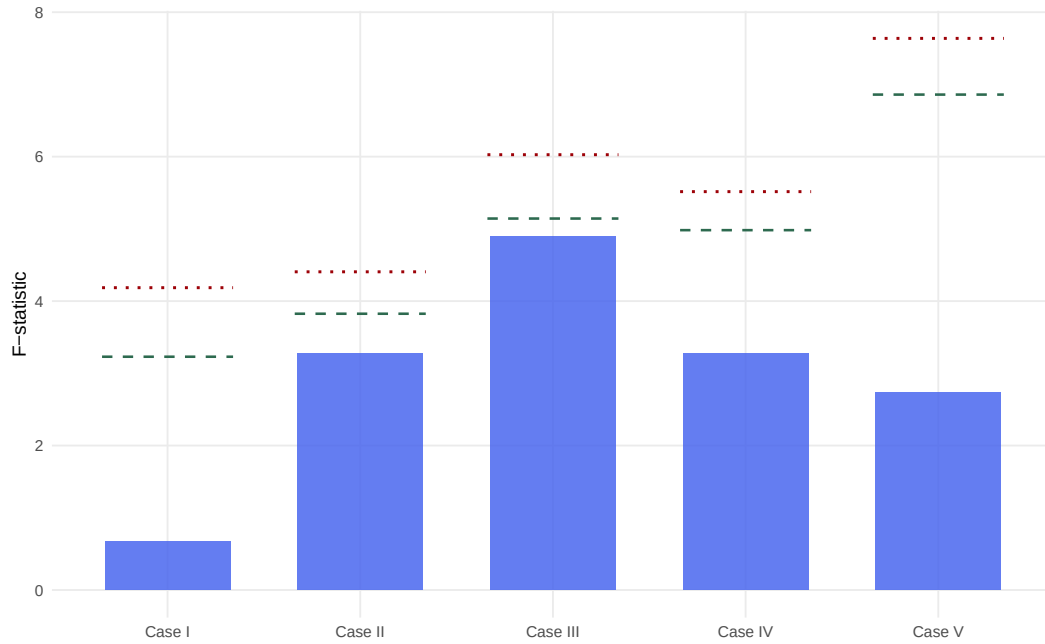


Figure 1.17. F -statistic across deterministic Cases I–V. Horizontal lines mark lower ($I(0)$) and upper ($I(1)$) critical bounds at 5% significance.

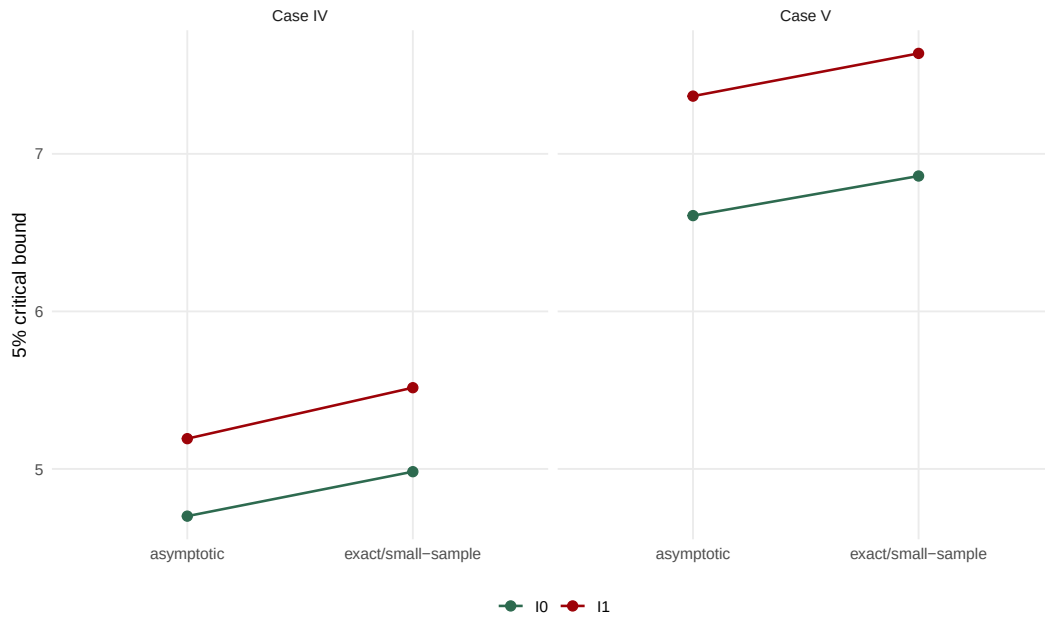


Figure 1.18. Exact-sample versus asymptotic critical bounds for Cases IV and V ($T = 65$). Exact bounds computed via stochastic simulation (Natsopoulos and Tzeremes, 2022).

1.B.9 Dummy Variable Diagnostics

Figure 1.21 displays squared residuals from the baseline ARDL specification with impulse dummies D_{56} , D_{74} , D_{80} overlaid. The dummies absorb outlier-driven residual spikes identified via squared-error diagnostics (Patterson, 2000).

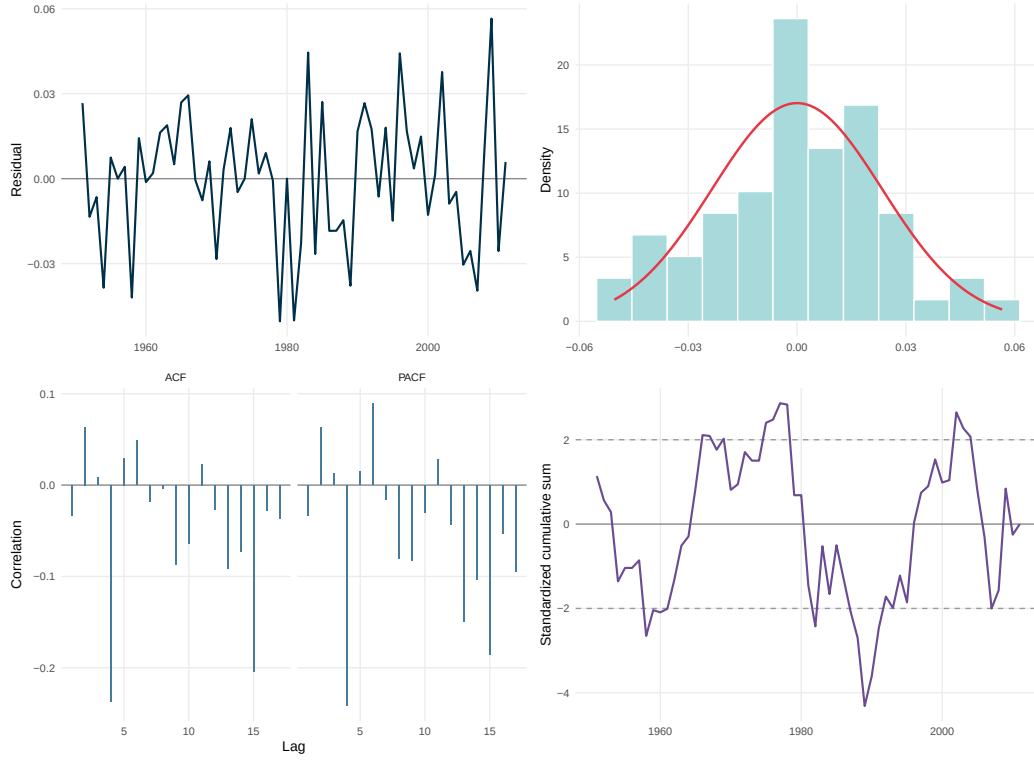


Figure 1.19. Residual diagnostics from baseline ARDL(2,4): (a) residuals over time, (b) density with normal curve, (c) ACF/PACF, (d) CUSUM test.

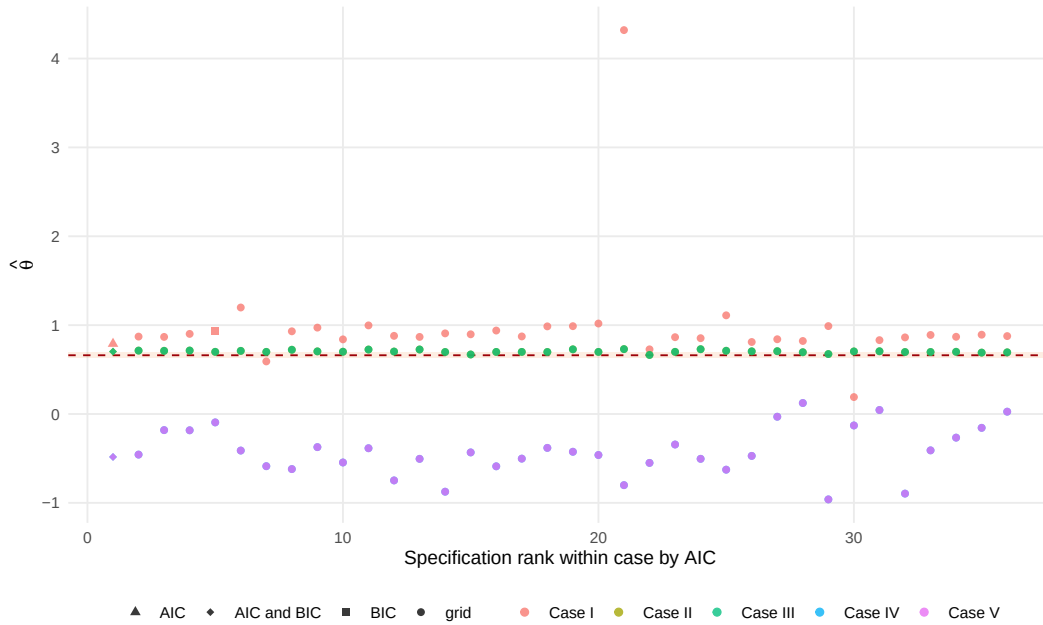


Figure 1.20. Long-run coefficient $\hat{\theta}$ estimates ranked by AIC across specification grid. Horizontal band marks Shaikh's $0.6609 \pm 5\%$.

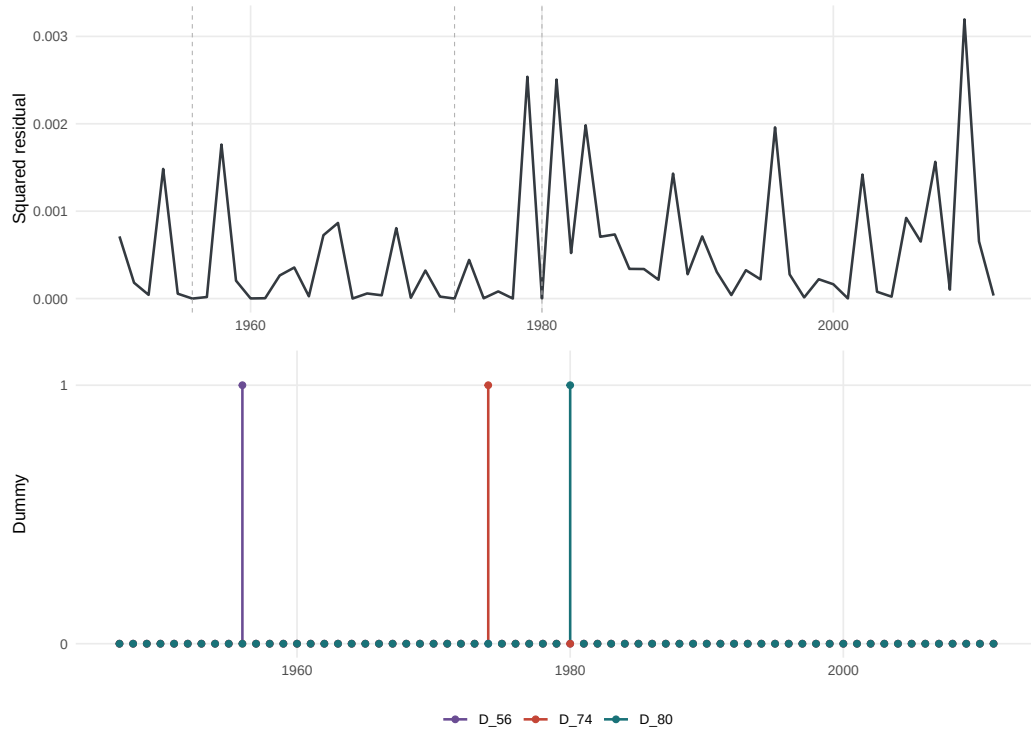


Figure 1.21. Squared residuals from baseline ARDL with impulse dummies D_{56} , D_{74} , D_{80} (1947–2011). Dummies absorb outlier-driven spikes.

1.B.10 Series Construction Protocol

Figure 1.22 visualizes the pipeline from raw data to fitted capacity utilization ($\hat{\mu}_t$). The flowchart traces corrections, deflation, GPIM recursion, ARDL estimation, and residualization.

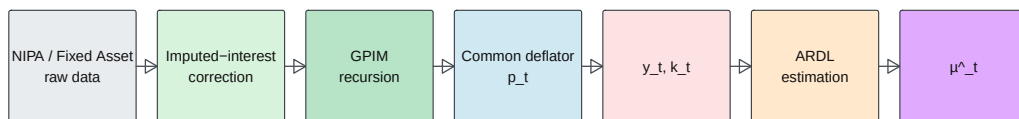


Figure 1.22. Series construction protocol: raw data → corrected output/GPIM capital → common deflator → ARDL estimation → $\hat{\mu}_t$.

Essay 2

The Transformation of Accumulation into Productive Capacities: Capacity Utilization in the Center and Periphery

Abstract

Chapter 1 demonstrated that the bivariate output–capital relation fails system-level cointegration and that a stable long-run capacity path requires the rate of exploitation to enter the cointegrating vector, yielding a transformation elasticity below unity ($\hat{\theta} < 1$) indicative of structural overaccumulation. This chapter identifies *how* distribution conditions the conversion of accumulated capital into productive capacity and extends the analysis to the periphery. I estimate the transformation elasticity $\theta \equiv \partial \ln Y^p / \partial \ln K$ as a state-dependent cointegrating relation: for the United States (1931–2024), using Integrated Modified OLS with fixed- b inference and endogenously dated structural breaks (Bai and Perron, 2003; Vogelsang and Wagner, 2024); for Chile (1943–2010), using a threshold Cointegrating Multivariate Polynomial Regression with a composite balance-of-payments index. In the United States, θ remains strictly below unity throughout the sample. It rises from 0.37 in 1960 to 0.43 in 2024 as effective wage pressure falls with the diversion of investment into intellectual property. Capacity utilization declines from its 1944 wartime peak to 93.8% in 2024, never re-anchoring to its Fordist level. In Chile, a balance-of-payments threshold at $\hat{\gamma} = 0.425$ switches the mechanization response off: the wage-share elasticity of the capital composition gap falls from +3.29 to +0.14 once foreign exchange binds. Peripheral capitalists retain the incentive to substitute machinery for labor and lose the means to do so, because the machinery is imported. The center–periphery comparison establishes that the same transformation relation is mediated by distributive conflict in the center and truncated by the balance-of-payments constraint in the periphery. Neither economy operates near $\theta = 1$. Capacity utilization is not a norm toward which economies converge; it is the residual of a politically conditioned conversion of accumulation into productive capacity.

Keywords: Capacity Utilization; Transformation Elasticity; Choice of Technique; Cointegration; Okishio Viability; Balance-of-Payments Constraint; Center–Periphery.

JEL Codes: B51; C32; E01; E22; E32; P16.

2.1 Introduction

Capacity utilization is among the most widely reported macroeconomic ratios and among the least identified. Central banks read it as a gauge of inflationary pressure. Growth models treat it as the variable that absorbs the gap between accumulation and effective demand. A four-decade heterodox debate over whether firms converge on a normal rate has been conducted almost entirely on the assumption that the rate is observed (Blecker and Setterfield, 2019; Kurz, 1986). It is not. The numerator of the ratio—realized output—is measured. The denominator—productive capacity—is not. Every claim about utilization is therefore a claim about an object no statistical agency records, resting on whatever convention supplied the denominator.

The utilization ratio has been supplied in three ways, and all three fail identically. The first takes utilization as a *given* macro ratio—an observed series to be used as a regressor or a policy input—when its denominator was never observed. The second anchors it to an *inherited benchmark*: an engineering maximum or a survey-reported normal rate imported from outside the accounting framework that is supposed to give the ratio meaning. Managerial surveys do not support the premise this requires (Nikiforos, 2016; Singh et al., 2022). The third recovers utilization as an *empirical residual*: whatever remains after output has been fitted to capital by a filter or a production function. This third treatment is where most current empirical work sits, and the objection to it is structural. Fitting a trend through output conditional on capital sets the capacity path to move one-for-one with accumulation— $\theta = 1$ by construction—and then defines utilization as the deviation from a benchmark that assumed away the very relation needed to locate it. The three treatments are not three difficulties but one: each supplies the denominator by convention, and each conceals the transformation relation that determines it.

Chapter 1 demonstrated that the bivariate output–capital relation fails system-level cointegration and that a stable long-run capacity path requires the rate of exploitation to enter the cointegrating vector, yielding a transformation elasticity below unity ($\hat{\theta} < 1$) indicative of structural overaccumulation. That finding established *that* the output–capital relation is distributionally conditioned. It did not identify *how* distribution conditions the conversion of accumulated capital into productive capacity, nor did it extend the analysis beyond the closed-economy core. This chapter supplies both.

I estimate the transformation elasticity $\theta \equiv \partial \ln Y^p / \partial \ln K$ as a state-dependent cointegrating relation. For the United States over 1931–2024, I use the Integrated Modified OLS estimator of Vogelsang and Wagner (2024) with fixed- b inference and endogenously dated structural breaks (Bai and Perron, 2003); distribution enters through effective wage pressure. For Chile over 1943–2010, the balance-of-payments constraint enters as a threshold that switches the choice of technique. The estimates locate θ well below unity in both economies. In the United States the elasticity moves from 0.37 in 1960 to 0.43 in 2024 as effective wage pressure falls with the diversion of investment into intangibles; utilization declines from its 1944 wartime peak to 93.8% in 2024. In Chile a composite balance-of-payments threshold at $\hat{\gamma} = 0.425$ switches the mechanization response off: the wage-share elasticity of the capital composition gap falls from +3.29 to +0.14 once foreign exchange binds (Hansen, 2000). The center–periphery comparison establishes that the same transformation relation is mediated by distributive conflict in the center and truncated by the balance-of-payments constraint in the periphery.

Two literatures have circled this problem without closing it. The induced-innovation tradition following Kennedy (1964) made distribution govern the *direction* of technical change along a frontier whose *scale* stayed exogenous; its own later assessments concede that it explains the composition

of innovation while remaining close to silent on its intensity (Tavani and Zamparelli, 2018). The measurement literature has established convincingly that reported capacity is a construct rather than an observation (Nikiforos, 2016; Shaikh, 2016), but has stopped short of supplying the relation that would replace it. Okishio (1961, 2022) supplies half of that relation—decentralized investment produces disproportionality as a structural norm, not a friction—and Vidal (2019, 2022) supplies the other half, showing that the firm converting capital into usable capacity is internally divided over control and cooperation and settles into configurations that reproduce production without resolving the conflict. Neither framework was carried to the point where the transformation elasticity could be estimated. Basu (2022, 2026) closes part of the gap by tying the profitability outcome of viable technical change to the labor-market closure, which maps onto historical accumulation regimes; and for the periphery, Prebisch (1981) and Kaldor (1959) established the external ceiling without connecting it to the choice of technique that runs into it.

This chapter makes three contributions against that background. First, it reconstructs capacity utilization as a *derived* quantity, identified downstream of a transformation elasticity that is itself estimated rather than assumed, extending the distributional conditioning established in Chapter 1 from a cointegration-survival result to a structural elasticity with a historical trajectory. Second, it supplies an econometric identification of that elasticity that does not presuppose the relation it recovers, using a cointegrating specification in which distribution enters as a state-dependent interaction and structural change is dated endogenously. Third, it extends the relation to the periphery by letting the balance-of-payments constraint enter as a *threshold* that switches the choice of technique, rather than as a control variable added to it.

The argument runs as follows. The transformation relation cannot be posited without a theory of the unit that performs the transformation, which is what Section 2.2 supplies by replacing the unified firm with decentralized accumulation and internal division. Section 2.3 then asks what has to be true of the choice of technique for that transformation to have a determinate elasticity, and derives θ under homogeneous capital, under heterogeneous capital with a mechanization gap, and under an external constraint that can bind. Section 2.4 confronts the resulting specifications with reconstructed capital stocks for both economies and reports what the estimates say about the historical path of θ and μ . Section 2.5 sets the two histories against each other, and Section 2.6 states what the reconstruction settles and what it leaves open.

2.2 The Internally Divided Firm and Capacity Formation

Capacity utilization is not a technical parameter. It is an inter-scalar political-economic settlement. The macroeconomic capacity ceiling—the denominator of the utilization rate—is not a neutral engineering constraint but a historically contingent outcome conditioned by the balance of class power. Yet identifying this distributional conditioning at the macro level leaves the underlying transmission mechanism black-boxed: how does distributive conflict physically alter the conversion of accumulated capital into productive capacity?

Answering this question requires dismantling a hidden premise shared by virtually all heterodox growth models: the unified firm. Whether treating the normal rate of capacity utilization as an endogenous behavioral target or an exogenous gravitational anchor, standard frameworks presume a coherent organization capable of selecting a utilization benchmark, enforcing it across the labor

process, and sustaining it over time. Formally, the utilization rate is defined as:

$$\mu_t \equiv \frac{Y_t}{Y_t^p}, \quad (2.1)$$

where Y_t is realized output and Y_t^p is potential productive capacity. But this ratio conceals a deeper question: who determines Y_t^p , and under what organizational conditions?

The unified-firm premise collapses when confronted with the social organization of capitalist production. In a decentralized economy, individual capitals commit fixed investments without *ex-ante* coordination, confronting one another in competitive markets. Within the enterprise, production decisions encounter contradictory managerial and worker imperatives. Capacity utilization cannot be reduced to a target selected by a single optimizing entity. To theorize the political economy of productive capacity, I replace the unified firm with two interconnected frameworks: Okishio’s theory of decentralized market anarchy, and Vidal’s theory of the internally divided firm.

2.2.1 Smooth Reproduction and the Unified-Firm Premise

Proportional reproduction is a condition of feasibility, not a tendency of capitalist accumulation. Smooth reproduction requires strict material proportionality between departments of production—a condition Basu (2022) and Okishio (2022) formally establish, defined by the latter as the equilibrium accumulation trajectory. Bergmann et al. (2009) call this balanced trajectory, where realized output meets productive capacity ($\mu_t = 1$), the “golden path.” But the golden path is a methodological closure, not a gravitational attractor. It identifies the conditions under which realization can reproduce itself smoothly; it does not describe the path capitalist accumulation actually follows. Neither the post-Keynesian assumption of long-run balanced growth (Blecker and Setterfield, 2019) nor the “*Keynesian in the short run, classical in the long run*” synthesis (Duménil and Lévy, 1999) captures this persistence of unbalanced expansion.

Nothing in the logic of decentralized competition establishes that private investment decisions will reproduce these proportions. A stylized illustration makes the point concrete. If productive capacity grows at 5% per year while aggregate demand for output grows at 2%, capacity outpaces demand by three percentage points each year. After a decade, utilization falls to roughly 75% of capacity—not because of a temporary shock, but because uncoordinated investment persistently builds capacity faster than demand can absorb it. Balanced growth ($g_Y = g_K$) is merely one configuration of the transformation elasticity θ ; it is not a restoring force. Smooth reproduction ($\mu_t = 1$) does not require $\theta = 1$; it holds for any transformation elasticity. Balanced growth is the narrower special case in which $\theta = 1$ and $\mu_t = 1$ coincide in the long run.

Because private investment decisions are made without *ex-ante* coordination, real capitalist economies routinely violate the proportional conditions of the golden path. Okishio (2022) demonstrates that chronic disproportionality is not a temporary friction but the structural norm of decentralized accumulation. Treating the transient benchmark of $\mu_t = 1$ as a behavioral target therefore relies on the hidden premise that the firm is a unified planning subject.

The denominator of the utilization ratio (productive capacity Y_t^p) is not an engineering constant. It is a contested settlement of divided management and shop-floor struggle. The numerator (realized output Y_t) is hostage to the anarchy of demand. Smooth reproduction supplies a reference case against which this historical non-coincidence can be studied. It does not supply the mechanism that produces it.

2.2.2 Decentralized Accumulation and Market Anarchy

If the “golden path” of smooth reproduction is merely a benchmark of feasibility, the actual trajectory of capitalist accumulation is governed by the decentralized organization of investment. Capitalist production is coordinated through exchange, meaning that the social validation of private decisions occurs only *ex-post*. Because money mediates exchange, sale and purchase are separated moments of the capital circuit (Basu, 2022). Surplus value must be realized in circulation, not by construction (Foley, 1985; Trigg, 2020). This constitutes the logic of capital as impersonal domination: capitalists commit to production and investment with the law of value operating “*behind their backs*.”¹ They are subject to market forces and value relations they neither control nor consciously direct. This anarchy is not a coordination failure in an otherwise functional market: the market is a socially constituted institution whose normal operation produces disproportionality as a structural feature (Vidal and Peck, 2012). Okishio (2022) locates the source of macroeconomic instability precisely in this constitutive anarchy, where individual capitals commit to fixed investment on the basis of private profitability expectations without knowing in advance whether their expansion fits the investment and consumption decisions taken elsewhere. The coordination problem is not one of imperfect information surrounding an otherwise coherent plan; no economy-wide plan exists.

Smooth reproduction requires proportional expansion across sectors producing means of production and sectors producing consumption goods. Because private investment does not enforce this proportionality, firms extrapolate past prices, profit rates, and sales. A signal that induces expansion in one branch routinely produces excess capacity before complementary demand materializes elsewhere. Contraction works through the same cumulative mechanism: falling sales cut investment, reducing aggregate demand and validating further cuts. Removing Say’s Law reintroduces realization as a substantive condition of reproduction: accumulation may generate capacity that demand fails to absorb, or demand may temporarily exceed productive capacity (Foley, 1985). Prices register the resulting imbalance, but they do not prevent the sunk commitments produced by it.

Fixed capital gives this temporal lag its structural force. Structures and machinery embody long-horizon commitments that cannot be reversed when relative prices or demand conditions shift. Once installed, excess capacity leaves the firm to operate below its anticipated scale, defer maintenance, hold equipment idle, or devalue the asset. These responses distribute the costs of market anarchy among firms, workers, and creditors, but they do not restore the missing *ex-ante* coordination.

Idle capacity is therefore a recurrent, structural form of decentralized accumulation rather than a temporary friction. Private investment physically builds capacity before the market can realize its social use, opening a recurrent gap between installed productive capacity (Y_t^p) and effective demand (Y_t). The capacity utilization rate (μ_t) registers this gap at the macro level. Unbalanced growth is the structural consequence of distributively conditioned investment under decentralized decision-making. The analytical problem is therefore not how proportionality re-emerges, but how persistent imbalance remains historically reproducible: the condition $\mu_t = 1$ is at most a transient point, while persistent divergence records the ongoing reality of market anarchy.

Okishio’s macro framework stops one step short: it does not explain how firms translate this external anarchy into internal operating decisions. The firm does not manage market anarchy; it internalizes it. A firm does not receive a price signal and respond as a frictionless, unified unit.

¹Karl Marx, *Capital*, Volume I, Chapter 1, Section 4 (“*The Fetishism of Commodities and the Secret thereof*”), where the law of value operates as an overriding law of nature when private producers exchange products (Marx, 1999).

It processes the signal through an organization divided over control, cooperation, effort, and the allocation of resources. Vidal’s internally divided firm provides that missing mediation.

2.2.3 The Internally Divided Firm

Vidal builds on Marx’s Volume I theory of the labor process, drawing on contemporary organizational theory and sociological institutionalism to develop an organizational political economy of the capitalist firm (Vidal, 2022; Vidal and Peck, 2012). Marx established that capitalist management faces a constitutive contradiction: it must simultaneously organize the cooperative labor required for production and enforce the coercive control required for valorization. Vidal (2022) demonstrates how this contradiction fractures the modern capitalist firm, treating it as a divided social organization whose internal structure is shaped by cognitive, normative, regulative, and organizational institutions. Management must extract surplus value through control over labor, but production also requires workers’ cooperation, tacit knowledge, and problem-solving capacity. Standardization and surveillance secure a measurable pace of work while weakening the discretion needed to sustain complex production. The firm cannot maximize both imperatives at once because the labor process that produces value is also the site where workers contest its extraction. This contradiction is not a universal constant of capitalism; it is institutionally mediated, taking variegated forms across distinct accumulation regimes throughout historical capitalism (Vidal, 2019).

Workers respond to this contradiction by actively calibrating effort, cooperation, machine pacing, and the deployment of tacit knowledge. Capitalist management extracts potential capacity through concrete shop-floor mechanisms: increasing machine speed, deferring maintenance to hit short-term throughput targets, and enforcing rigid surveillance. Workers contest this extraction through labor turnover, collective exit, or the deliberate withdrawal of problem-solving cooperation and effort. Throughput, quality, and maintenance therefore depend on social relations that no engineering specification fixes in advance. A production target that raises machine speed by deferring maintenance may increase current output while degrading sustainable capacity. A directive that suppresses worker discretion may meet a short-term labor-control objective while worsening workflow bottlenecks. Labor effort is consequently not a fixed technical parameter applied to a pre-existing capital stock; it is an active, contested determinant of the capacity utilization itself.

This dynamic enforces a strict wedge between *engineering capacity* and *practical capacity*. Engineering capacity records the theoretical maximum output of equipment under ideal technical conditions. Practical capacity records the sustainable output rate dictated by actual maintenance schedules, product-mix complexity, and the negotiated intensity of labor. When managers report capacity to national statistical agencies, they rely on conventions shaped by their organizational position and the immediate pressures of the shop floor. Aggregate capacity statistics therefore do not reveal a neutral physical ceiling; they aggregate situated, institutional judgments (Nikiforos, 2016; Shaikh, 2016). Complementarily, empirical surveys of managerial practice confirm that firms do not systematically organize production decisions around a coherent capacity utilization target (Singh et al., 2022). For macroeconomic modeling, the productive capacity ceiling (Y_t^P) is thus not an exogenous engineering constraint but an endogenous settlement through which workplace conflict and class struggle transmit into capacity utilization.

Because the valorization–labor process contradiction cannot be solved, firms settle into organizational configurations of “*mediocre sufficiency*” (Vidal, 2022). These are hybrid arrangements that reproduce production without resolving the underlying conflict between control and cooperation. Partial worker autonomy coexists with rigid discipline; continuous improvement programs coexist

with brutal throughput pressure; machinery might remain idle alongside binding productive bottlenecks. Following [Vidal \(2014\)](#), these arrangements are best understood as *institutional incoherence*: the firm’s internal settlements are functional enough to sustain accumulation yet contradictory enough to generate persistent friction. This stands in direct contrast to the Varieties of Capitalism assumption of institutional complementarity, where national institutions are presumed to form coherent, mutually reinforcing systems ([Hall and Soskice, 2003](#)). Here, incoherence is not a deviation from an optimizing norm; it is the permanent, structural form through which capitalist management physically converts accumulation demand into capacity usage.

2.2.4 Distribution, Technique, and the Transformation of Accumulation

The choice of technique connects the divided firm to distribution, governing how accumulated capital is physically transformed into productive capacity. Capitalists adopt a new technique only when it reduces the cost of production at prevailing prices and wages; such a technique is capitalistically viable ([Okishio, 1961](#)). [Okishio \(1961\)](#) demonstrates that a viable capital-using, labor-saving (CU-LS) technique raises the equilibrium profit rate when the real wage remains constant—a direct challenge to Marx’s law of the tendential fall in the rate of profit. Marx’s account, by contrast, allows the same form of technical change to depress profitability when rising capital intensity increases the capital advanced relative to surplus value. [Basu \(2022\)](#) reconciles these seemingly mutually exclusive theorizations by formalizing the *Marx–Okishio threshold*: the rate of real-wage growth defines the result driven by the balance of class forces.

This threshold situates the present framework within the induced-innovation tradition while departing from it. The Innovation Possibility Frontier (IPF) literature, initiated by [Kennedy \(1964\)](#), treats distribution as selecting the *direction* of technical change along a given frontier: relative factor costs determine whether innovation is labor-augmenting or capital-augmenting, while the frontier itself—the scale of feasible productivity growth—remains exogenous ([Julius, 2005](#); [Tavani, 2012](#)). Later assessments sharpen this limitation: IPF-based models explain the composition of technical change but remain largely silent on innovation intensity ([Tavani and Zamparelli, 2018](#)). [Zamparelli \(2024a\)](#) partially endogenizes scale by mapping a family of IPF levels to R&D effort, yet the analytical structure continues to rely on frontier geometry and steady-growth closure ([Zamparelli, 2024b](#)). An alternative strand reframes inducement through mechanization rather than frontier selection: [Kaldor \(1957\)](#) links productivity growth to capital deepening, while [Cajas-Guijarro \(2024\)](#) reformulates the IPF with an explicit choice over mechanization growth under distributive and labor-market conditions. The present framework builds upon a mechanization productive frontier, treating the distributive–technological relation as a regime-specific transformation between accumulation and productive capacity. Rather than selecting along a given frontier, distribution conditions mechanization intensity, which in turn determines the transformation elasticity linking accumulation to the formation of productive capacities.

The Marx–Okishio threshold gives wage pressure a precise role in mechanization, but the profitability outcome depends on the labor market closure. [Basu \(2026\)](#) demonstrates that the equilibrium profit rate after viable CU-LS technical change is governed by the institutional structure of capital–labor bargaining. Under a constant-wage-share closure—what [Basu \(2026\)](#) formalizes as the “strong labor” regime, applicable to Atlantic Fordism—any viable CU-LS technique unambiguously *lowers* the equilibrium profit rate. Marx’s LTFRP holds where labor is powerful enough to compel real wages to grow in tandem with productivity. Under this closure, rapid mechanization is induced by distributive pressure, yet the same pressure contests capital’s claim on the resulting

productivity gains (Vidal, 2015, 2019). A stylized illustration makes the mechanism concrete. If a 10% real wage increase raises mechanization intensity by 8% under strong-labor conditions, the transformation elasticity θ rises from 0.70 to 0.85: accumulation translates more efficiently into productive capacity because firms substitute machinery for labor. Wage-led mechanization was not a harmonious productivity bargain, but a contested settlement in which the distributive pressure that induced technical change simultaneously eroded the profit rate it was meant to restore.

The weakening of organized labor after the Fordist crisis altered that mechanism. Under a “*weak labor*” closure—where the unemployment rate remains constant such that workers lack the institutional power to enforce a target wage share—the distributive pressure to replace labor is suppressed (Basu, 2026). Yet mechanization does not necessarily cease. Lower relative machinery prices and technical change in capital-goods production can continue to favor capital-using techniques through a second, independent channel. Distribution and machinery prices therefore constitute a dual-channel mechanism in the choice of technique. Their relative force changes across accumulation regimes, which prevents the wage share from being treated as an external control added to an otherwise autonomous technical relation. Extending the stylized example: under Post-Fordist conditions, a 15% decline in relative machinery prices raises mechanization intensity by 6%, partially compensating for the weakened distributive channel and sustaining θ near 0.75 despite wage suppression. The same 10% wage increase that produced an 8% mechanization response under Fordism produces only a 2% response under Post-Fordism, leaving θ near 0.72. The two channels do not simply add; they interact through the institutional configuration that governs their relative force.

These channels still operate through the contradictions identified in the preceding subsections. Firms choose techniques before demand validates the associated capacity, and they operate those techniques through labor processes that divide control from productive cooperation. Investment consequently does not translate one-for-one into a stable productive ceiling. The transformation depends on the composition of capital, the distributive conditions governing mechanization, and the organization through which installed equipment becomes usable capacity. In a labor surplus economy—the third closure identified by Basu (2026), applicable to peripheral capitalism—the real wage remains constant and the equilibrium profit rate rises unambiguously after viable CU-LS technical change. Yet the peripheral transformation is structurally truncated by external financial subordination and the balance of payments constraint, a mechanism developed in the following subsection.

Okishio and Vidal jointly rule out a direct passage from accumulation to productive capacity. Decentralized investment generates structural disproportionality across firms and sectors; the internally divided firm internalizes this disproportionality as contested operating conditions rather than a coherent utilization target. The analytical object that captures this mediated transformation is the *transformation elasticity* ($\theta \equiv dy^p/dk$), which governs how accumulation converts into productive capacity under a context of distributive conflict and market anarchy. The empirical analysis limits the original inter-departmental formulation by Marx in Capital Volume II to a single-sector aggregate, masking internal sectoral divergences while preserving the core logic.

2.2.5 Historical Regimes of Institutional Mediation and the Transformation Elasticity

The transformation elasticity is an emergent property of class struggle that mediates the process of capital accumulation and its transformation into productive capacities under the prevailing

institutional environment. Decentralized accumulation does not automatically guarantee proportional reproduction ($\theta = 1$). Institutional mediation—wage-setting norms, financial architectures, and competitive structures—conditions how distributive pressures translate into mechanization choices (Aglietta, 2000).

Basu (2026) demonstrates that the equilibrium profit rate after viable capital-using, labor-saving (CU-LS) technical change is governed by the labor market closure, which maps directly onto distinct historical accumulation regimes. Under a *constant-wage-share* closure—the institutional signature of Atlantic Fordism (1947–1973)—workers possess the bargaining power to grasp productivity gains, forcing real wages to grow in tandem with labor productivity. In this strong-labor regime, viable mechanization unambiguously lowers the equilibrium profit rate, so that Marx’s law of the tendential fall in the rate of profit holds (Basu, 2026). This stagnation tendency was institutionally offset: the class compromise, mass consumption, oligopolistic market structures, and military Keynesianism of the Fordist era absorbed the surplus and sustained effective demand (Vidal, 2019).

Under a *constant-unemployment* closure—which resembles the persistent unemployment feature of the post-Fordist accumulation regime—the class compromise that took place during the Fordist period is dismantled and labor’s bargaining power is severely weakened. The wage share falls, and mechanization is driven not by distributive pressure but by the relative cheapening of ICT and the shift toward financially reversible, short-horizon assets, patents, and intangible intellectual property (Lapavistas, 2011; Pagano, 2014). Because wages no longer track productivity, the profit rate may stabilize or rise, but the suppression of wage-led demand generates a chronic under-consumptionist stagnation tendency. Following Vidal (2019), I hypothesize that under structural conditions, the post-fordist accumulation regime produces a dysfunctional utilization configuration: accumulation decouples from productive capacity formation, generating chronic idle capacity that is temporarily masked by debt-led consumption and financial fragility rather than sustainable realization (Stockhammer, 2012; Vidal, 2019).

In the capitalist periphery, different closures represent distinct historical configurations throughout Chilean history. First, the assumption of a *constant-real-wage*, which corresponds to Okishio’s original formulation, maps onto a surplus-labor economy (Basu, 2026; Lewis, 1954). During the import-substituting industrialization (ISI) era, the *rural-urban* and *demographic* transition enabled by the transformation of sanitary conditions and a massive drop of child mortality in the early XX century regulated late industrialization, providing an unlimited supply of labor from the countryside that disciplined urban wages and allowed the profit rate to reproduce itself alongside mechanization. However, peripheral mechanization is structurally import-biased: widening the mechanization gap requires imported capital goods and technologically non-substitutable inputs (Prebisch, 1981). The balance of payments constraint therefore acts as a hard external ceiling on the transformation of accumulation into capacity. Under a developmentalist regime with surplus labor (1933–1957), this operated as an *import capacity ceiling*. Second, a *constant-wage-share*—or more accurately *rising-wage-share*—closure represents the intensification of stagnation tendencies leading to a structural crisis of capitalism. This occurred once the unlimited supply of labor was depleted and the incentives for capitalist firms to mechanize rose due to distributive pressures, increasing import requirements and stressing further the external constraint. Third, under the discipline imposed by the neoliberal counter-revolution (Casals, 2021) (1974–present), a *constant-unemployment* closure coupled with trade and capital account liberalization relieved the external constraint. Yet it locked the periphery into extractive specialization and premature deindustrialization, such that the WTO and bilateral free trade agreements legally encoded a *policy space ceiling* (Ahumada, 2019; Palma, 2019).

These historical configurations establish that the productive ceiling is not a neutral, trans-historical engineering constraint but a contingent outcome of decentralized accumulation, class power, and a country's relative positionality within the capitalist world system. The center-periphery comparison reveals a fundamental asymmetry: in the center, structural crises arise from the *exhaustion* of institutional mediation (e.g., the 1970s profit squeeze); in the periphery, they arise from the *absence* of sovereign mediation and the truncation imposed by the hierarchically organized financial order imposing an external constraint that is not necessarily binding.

2.3 The Analytical Framework: Formalizing the Transformation Elasticity

Distributive pressure conditions the choice of technique, and the choice of technique determines how much capacity a unit of accumulation produces. Section 2 established that claim historically and institutionally. It carries no elasticity until the cost-minimization problem behind it is written down, because only then does θ acquire a determinate value rather than a direction of movement. Cajas-Guijarro (2024) first posed that problem as a choice over mechanization growth, and I extend it in three steps.

The steps are nested rather than alternative. A firm that treats capital as homogeneous faces a single margin, and the elasticity that emerges from its cost minimization is a scalar. Allow the capital stock to separate into nonresidential structures and machinery, and a second margin opens: the mechanization gap κ_t , whose marginal capacity payoff depends on distribution and on relative asset prices at once. Impose a foreign-exchange bottleneck on the machinery half of that gap, and the interior solution can fail entirely, truncating θ at a corner. Setting the heterogeneity parameter to zero recovers the first model from the second; relaxing the foreign-exchange constraint recovers the center from the periphery. One structure, evaluated under three constraint sets.

A limitation is worth stating before the derivations rather than after them. The mechanism is multi-sectoral in principle—disproportionality is a relation between departments of production—while the empirical implementation bounds it to a single-sector aggregate. That bound masks internal sectoral divergence. It preserves the transformation logic, which is what the estimation requires, but it means θ is recovered as an economy-wide average of a relation that operates unevenly across branches.

2.3.1 The Baseline Model of Cost-Minimizing Mechanization (Homogeneous Capital Stock)

Actual aggregate output is represented by a Leontief production function with fixed technical coefficients:

$$Y = \min(AL, \mu BK) \quad (2.2)$$

where $A = Y/L$ is actual labor productivity, $\mu = Y/Y^p$ is capacity utilization, and $B = Y^p/K$ is maximum capital productivity. This production structure governs the rate of profit through the fundamental heterodox accounting identity:

$$r = \mu B(1 - \omega) \quad (2.3)$$

where ω is the wage share. Standard macroeconomic models treat B as an exogenous technological given and μ as a behavioral constant smoothly targeted by a unified firm. I reject both assumptions. The components of (2.3) are endogenous outcomes of class struggle and decentralized market anarchy, as established in Section 2. This subsection formalizes how distributive pressure forces cost-minimizing firms to alter their choice of technique, endogenously determining B .

2.3.1.1 Tactical Choice of Technique and Induced Innovation

Capitalist firms choice of technique is given by aiming a mechanization growth rate q under competitive cutting cost-pressures, looking at the net productivity surplus of technical change as their objective function. Let a denote the growth rate of labor productivity. The effective distributive cost of deploying mechanization growth is $q(1 - \omega)$. Hence, the choice of technique of capitalist mechanization can be represented by:

$$\min_q c(q) = q(1 - \omega) - a \quad (2.4)$$

This objective captures the induced-innovation logic: firms accelerate mechanization to expand labor productivity growth, but the distributive cost of capital intensity holds them back. The cost-minimizing objective function of capitalist becomes constrained by a mechanization function that can be represented as:

$$a = \Phi(q) = \lambda_0 + \lambda_1 q + \lambda_2 q^2 \quad (2.5)$$

Assumption 1. *Concavity of the mechanization frontier.* The frontier parameters satisfy $\lambda_0 > 0$, $\lambda_1 > 0$, and $\lambda_2 < 0$. The base productivity growth λ_0 sets the floor without advanced machinery. The linear coefficient λ_1 captures the direct productivity payoff of mechanization. The strict concavity $\lambda_2 < 0$ ensures diminishing marginal returns to mechanization.

Substituting (2.5) into (2.4) and solving the first-order condition yields the optimal mechanization growth:

$$c(q) = \lambda_0 + [\lambda_1 - (1 - \omega)]q + \lambda_2 q^2 \quad (2.6)$$

$$\frac{\partial c}{\partial q} = \lambda_1 - (1 - \omega) + 2\lambda_2 q^* = 0 \quad (2.7)$$

$$q^*(\omega) = \frac{(1 - \omega) - \lambda_1}{2\lambda_2} \quad (2.8)$$

The strict concavity ($c''(q) = 2\lambda_2 < 0$) guarantees a unique maximum.

Proposition 1. *Induced innovation.* Under Assumption 1, the optimal mechanization growth is strictly increasing in the wage share:

$$\frac{\partial q^*}{\partial \omega} = -\frac{1}{2\lambda_2} > 0 \quad (2.9)$$

A rise in the wage share forces cost-minimizing firms to substitute machinery for labor. Rising wages drive mechanization directly through the first-order condition, pushing optimal capital intensity growth upward. *Proof.* Immediate from differentiating (2.8). ■

Remark 1. The objective (2.4) does not assert that the firm is a coherent optimizer. Following Vidal (2022), the firm is internally divided between the *control imperative* (valorization: cost reduction

through mechanization) and the *cooperation imperative* (production: the need for worker effort, tacit knowledge, and problem-solving capacity). The formal model captures the cost-minimizing pole—the Okishio viability logic—while the constraints on the optimization problem capture the limits imposed by the cooperation imperative and by external conditions. The firm maximizes net productivity surplus—labor productivity growth minus the distributive cost of mechanization—subject to diminishing returns. The firm’s actual mechanization choice is a contested settlement between these imperatives, not a coherent optimum. What the model demonstrates is the cost-reducing direction in which distributive pressure pushes the settlement, not the settlement itself.

2.3.1.2 Endogenous Transformation Elasticity

Substituting the optimal technique back into the technological frontier recovers the normal growth rate of labor productivity under smooth reproduction:

$$a^N(\omega) = \lambda_0 + \lambda_1 q^*(\omega) + \lambda_2 [q^*(\omega)]^2 \quad (2.10)$$

This optimal technique dictates the structural trajectory of the Leontief parameter B . By definition, maximum capital productivity is $B = Y^p/K$. In dynamic growth rates, the growth of capital productivity is:

$$b \equiv g_{Y^p} - g_K = a - q \quad (2.11)$$

Evaluated at the smooth reproduction benchmark: $b^N(\omega) = a^N(\omega) - q^*(\omega)$.

Definition 1. *Transformation elasticity.* The transformation elasticity θ is the structural parameter that translates capital accumulation into productive capacity growth:

$$g_{Y^p} = \theta g_K \quad (2.12)$$

Substituting (2.12) into (2.11) reveals the link between the level coefficient B and the elasticity θ :

$$\theta = 1 + \frac{b}{g_K} \quad (2.13)$$

To evaluate this elasticity, I anchor the choice of technique to a baseline rate of capital accumulation. Following the normal reproduction path, the normal accumulation rate is driven by the normal profit rate and the normal rate of re-capitalization net of depreciation. I follow [Foley \(1986\)](#) definition of the re-capitalization rate, $\chi = \frac{I}{\Pi}$, and interpret its normal conditions as continuous realization and re-investment of the surplus.² Hence,

$$g_K^N = \chi^N r^N - \delta \quad (2.14)$$

In dynamic growth accounting, changes in the growth rate of capital represent an acceleration term $\ddot{k} \equiv dg_K/dt$. In historical time, $\ddot{k}$ is kept strictly bounded by corporate liquidity constraints, debt-servicing ceilings, and Central Bank credit cycles, preventing explosive capital growth divergence.

²A simple accounting decomposition can demonstrate that the multiplier of profitability that drives investment, is precisely χ . In a monetary economy with credit, the re-capitalization rate does not require to be bounded by the surplus Π order of magnitude.

Expressed dynamically relative to this normal path, the aggregate transformation elasticity becomes:

$$\theta(\omega) = 1 + \frac{b^N(\omega)}{g_K^N} = 1 + \frac{a^N(\omega) - q^*(\omega)}{g_K^N} \quad (2.15)$$

Proposition 2. *Endogeneity of the transformation elasticity.* Under Assumption 1, the transformation elasticity θ is not an exogenous technical constant. It emerges endogenously from the cost-minimizing choice of technique. When distributive pressure forces firms to alter mechanization via (2.9), it simultaneously shifts the structural efficiency of accumulation via (2.15). The elasticity θ is therefore a function of the wage share ω , the frontier parameters $\lambda_0, \lambda_1, \lambda_2$, and the normal accumulation rate g_K^N .

Econometric note on capital measurement error. Because the aggregate capital stock K_t is constructed via Perpetual Inventory (GPIM) methods involving service-life and discard schedules, any classical right-hand-side measurement error in K_t introduces standard econometric attenuation bias that pulls the estimated elasticity $\hat{\theta}$ downward toward zero. Classical measurement noise cannot account for the empirical finding that $\theta < 1$; rather, it reinforces that the true structural transformation elasticity is bounded strictly below unity even under conservative measurement assumptions.

2.3.2 Heterogeneous Capital, the Mechanization Gap, and State-Dependent Viability

Mainstream macroeconomics treats the capital stock as a homogeneous aggregate, abstracting from the internal architecture of the firm. Following the theory of organizational political economy and the internally divided firm (Vidal, 2022), I unbundle aggregate productive capital into an extensive margin given by nonresidential structures (K^{NRC}) and intensive one, given by the accumulation of machinery and equipment (K^{ME}). These components do not unfold throughout the labor process symmetrically. Structures form the built environments that setup the “envelope” of accumulation — i.e. the factory floor and logistics shell of the productive process. Capital accumulation in Machinery and Equipment is the active component that restructures the production process and the extraction of relative surplus value (Marx, 1999, Vol. I, Ch. 15).

Assumption 2. *Heterogeneous capital decomposition.* Aggregate productive capital decomposes as $K_t^{cap} = K_t^{NRC} + K_t^{ME}$. Nonresidential structures evolve independently of short-run distributive conflict (ω) and its rate of accumulation of accumulation is governed by its long-run relation with the rate of accumulation in machinery and equipment under normal conditions of accumulation in the long-run

$$g_{K^{NRC},t}^N = f(\chi^N, r_t^N, g_{K^{ME},t}^N, \delta) \quad (2.16)$$

This extensive envelope $\Theta(K^{NRC})$ becomes a baseline for aggregate elasticity of transformation θ such that the rate of accumulation of capital stock in non-residential construction is a byproduct of the intensive margin of accumulation decided by a tactical technique choice made by capitalists in response to distributive pressures:

2.3.2.1 The Extensive Envelope and the Intensive Margin

Given the fixed infrastructural envelope, the **intensive margin** of accumulation is the tactical choice of the mechanization gap, defined in natural logs as:

$$\kappa_t = k_t^{ME} - k_t^{NRC} \quad (2.17)$$

This gap measures mechanization intensity relative to the extensive plant scale. The capacity identity decomposes potential output into the extensive envelope and the intensive capacity yield Ψ_t :

$$y_t^p = \theta_{NRC} k_t^{NRC} + \Psi_t(\kappa_t, \omega_{t-1}, p_t^\kappa) \quad (2.18)$$

The intensive yield Ψ_t is not a fixed engineering constant. It is the outcome of a decentralized cost-minimization problem governed by the regime of accumulation, as formalized below.

2.3.2.2 Cost-Surplus Minimization and the Okishio Viability Yield

In this case, the capitalist firm chooses a mechanization gap κ_t under cutting-cost pressures, subject to the mechanization frontier:

$$\Phi(\kappa_t) = \lambda_0 + \lambda_1 \kappa_t + \lambda_2 \kappa_t^2 \quad (2.19)$$

The *realized* capacity yield is governed by a viability condition: a cost-minimizing firm deploys a new technique only if it reduces unit costs at current distributive and price parameters (Basu, 2022; Okishio, 1961). I formalize this as a cost-minimization as problem:

$$\min_{\kappa_t} c(\kappa_t) = \kappa_t \cdot MC(\omega_{t-1}, p_t^\kappa) - \Phi(\kappa_t) \quad (2.20)$$

The marginal viability cost $MC(\omega_{t-1}, p_t^\kappa)$ is determined by two channels:

1. **The distributive dividend (labor saving).** Increasing κ_t substitutes machinery for labor. The value of this displacement is proportional to the lagged wage share (ω_{t-1}), reflecting the wage settlement period prior to current capacity realization. A higher wage share *reduces* the net cost of mechanization.
2. **The relative price penalty (capital cost).** Increasing κ_t requires accumulating machinery faster than structures. The cost of this composition shift is proportional to the log relative price of machinery ($p_t^\kappa = \ln P_t^{ME} - \ln P_t^{NRC}$). A higher relative price *increases* the marginal cost.

I specify the marginal viability cost as:

$$MC(\omega_{t-1}, p_t^\kappa) = \gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa \quad (2.21)$$

where $\gamma_1 > 0$ captures the wage-savings dividend and $\gamma_2 > 0$ captures the relative price penalty.

Substituting (2.19) and (2.21) into (2.20) and solving the first-order condition yields the optimal

mechanization gap:

$$c(\kappa_t) = \lambda_0 + [\lambda_1 - (\gamma_0 - \gamma_1\omega_{t-1} + \gamma_2 p_t^\kappa)] \kappa_t + \lambda_2 \kappa_t^2 \quad (2.22)$$

$$\frac{\partial c}{\partial \kappa_t} = \lambda_1 - \gamma_0 + \gamma_1\omega_{t-1} - \gamma_2 p_t^\kappa + 2\lambda_2 \kappa_t^* = 0 \quad (2.23)$$

$$\kappa_t^*(\omega_{t-1}, p_t^\kappa) = \frac{(\gamma_0 - \lambda_1) - \gamma_1\omega_{t-1} + \gamma_2 p_t^\kappa}{2\lambda_2} \quad (2.24)$$

Proposition 3. *Dual-channel induced innovation.* Under Assumption 1 (concavity) and Assumption 2 (heterogeneous capital), the optimal mechanization gap satisfies:

$$\frac{\partial \kappa_t^*}{\partial \omega_{t-1}} = -\frac{\gamma_1}{2\lambda_2} > 0 \quad (2.25)$$

$$\frac{\partial \kappa_t^*}{\partial p_t^\kappa} = \frac{\gamma_2}{2\lambda_2} < 0 \quad (2.26)$$

A rising wage share forces cost-minimizing firms to widen the mechanization gap, substituting machinery for labor. A rising relative price of machinery suppresses the optimal gap, as the structural cost of the composition shift penalizes mechanization. *Proof.* Immediate from differentiating (2.24). ■

2.3.2.3 The Marginal Capacity Payoff and the Econometric Bridge

The macroeconomic translation from the mechanization gap to potential capacity is given by the *marginal capacity payoff* (MP_κ), the derivative of the intensive capacity yield Ψ_t with respect to κ_t . By the envelope theorem:

$$MP_\kappa = \frac{\partial \Psi_t}{\partial \kappa_t} = \frac{\partial \Phi}{\partial \kappa_t} - MC(\omega_{t-1}, p_t^\kappa) \quad (2.27)$$

$$= (\lambda_1 + 2\lambda_2 \kappa_t) - (\gamma_0 - \gamma_1\omega_{t-1} + \gamma_2 p_t^\kappa) \quad (2.28)$$

$$= \underbrace{(\lambda_1 - \gamma_0 + 2\lambda_2 \kappa_t)}_{\text{autonomous yield}} + \gamma_1\omega_{t-1} - \gamma_2 p_t^\kappa \quad (2.29)$$

Integrating the marginal payoff with respect to κ_t (treating ω_{t-1} and p_t^κ as predetermined shifters) and defining θ_κ , $\theta_{\omega\kappa} \equiv \gamma_1$, and $\theta_{\kappa p\kappa} \equiv -\gamma_2$ yields the intensive capacity yield function:

$$\Psi_t = \theta_\kappa \kappa_t + \theta_{\omega\kappa}(\omega_{t-1} \kappa_t) + \theta_{\kappa p\kappa}(p_t^\kappa \kappa_t) \quad (2.30)$$

Substituting (2.30) back into (2.18) recovers the structural capacity equation:

$$y_t^p = \theta_{NRC} k_t^{NRC} + \theta_\kappa \kappa_t + \theta_{\omega\kappa}(\omega_{t-1} \kappa_t) + \theta_{\kappa p\kappa}(p_t^\kappa \kappa_t) \quad (2.31)$$

The interaction terms are not ad-hoc econometric additions. They are the macroeconomic footprint of the divided firm's Okishio viability constraint. Distributive pressure and relative price enter because they alter the marginal capacity payoff of the mechanization gap.

2.3.2.4 The Transformation Elasticity under Heterogeneous Capital Stock

Specification B recovers separate capacity elasticities for nonresidential structures (θ_{NRC}) and machinery (θ_{κ}), but evaluating aggregate capacity dynamics requires defining an aggregate transformation elasticity ($\theta_t^{\text{Aggregate}}$). In a single-asset model (Specification A), $\theta_A = \partial y^p / \partial k^{\text{cap}}$ is a scalar. Under heterogeneous capital, the total differential of productive capacity (2.31) separates extensive expansion from intensive composition shifters:

$$dy_t^p = \theta_{\text{NRC}} dk_t^{\text{NRC}} + MP_{\kappa} d\kappa_t \quad (2.32)$$

$$= \theta_{\text{NRC}} dk_t^{\text{NRC}} + (\theta_{\kappa} + \theta_{\omega\kappa} \omega_{t-1} + \theta_{\kappa p\kappa} p_t^{\kappa}) d\kappa_t. \quad (2.33)$$

Standard continuous-time growth accounting evaluates aggregate capital variation using a Divisia index of heterogeneous components (Divisia, 1925). Since $s_{\text{ME},t} + s_{\text{NRC},t} = 1$ and $d\kappa_t = dk_t^{\text{ME}} - dk_t^{\text{NRC}}$, aggregate capital growth decomposes as:

$$dk_t^{\text{cap}} \approx s_{\text{ME},t} dk_t^{\text{ME}} + s_{\text{NRC},t} dk_t^{\text{NRC}} \quad (2.34)$$

$$= dk_t^{\text{NRC}} + s_{\text{ME},t} d\kappa_t. \quad (2.35)$$

Dividing (2.33) by (2.35) yields the conventional total-derivative aggregate elasticity along the empirical growth path:

$$\theta_{\text{total},t}^{\text{Aggregate}} = \frac{\theta_{\text{NRC}} + (\theta_{\kappa} + \theta_{\omega\kappa} \omega_{t-1} + \theta_{\kappa p\kappa} p_t^{\kappa}) \frac{d\kappa_t}{dk_t^{\text{NRC}}}}{1 + s_{\text{ME},t} \frac{d\kappa_t}{dk_t^{\text{NRC}}}}. \quad (2.36)$$

Equation (2.36) suffers from a severe theoretical flaw illuminated by the Cambridge Capital Controversy (Robinson, 1953; Sraffa, 1960). Heterogeneous physical assets cannot be aggregated into a single scalar k^{cap} without value weights that depend on distribution and prices. Differentiating such a value aggregate and placing dk_t^{cap} in the denominator re-introduces the circularity that the model with heterogeneous capital was designed to eliminate. Empirically, the denominator $1 + s_{\text{ME},t}(d\kappa_t/dk_t^{\text{NRC}})$ vanishes whenever physical accumulation stalls or shifts composition, producing spurious zero-denominator singularities ($\theta \rightarrow \pm\infty$).

Alternatively, I construct the aggregate transformation elasticity as an asset weighted from the heterogeneous capital stock system. Rather than differentiating a value aggregate, the structural asset-weighted transformation elasticity weights identified heterogeneous elasticities by gross capital asset shares:

$$\theta_t^{\text{Aggregate}} = \theta_{\text{NRC}}^{\text{str}} + s_{\text{ME},t} \cdot MP_{\kappa,t}^{\text{str}}, \quad (2.37)$$

where $\theta_{\text{NRC}}^{\text{str}}$ is the base elasticity of structures, $s_{\text{ME},t} = K_{\text{ME},t}/K_{\text{cap},t}$ is the physical equipment share, and $MP_{\kappa,t}^{\text{str}} = \theta_{\kappa}^{\text{str}} + \theta_{\omega\kappa}(\omega_{t-1} - \bar{\omega}) + \theta_{\kappa p}(p_t^{\kappa} - \bar{p}^{\kappa})$ is the marginal capacity yield of mechanization. Equation (2.37) is immune to the Cambridge critique because the parameters are identified in a disaggregated system without prior scalar aggregation, and the weighting share $s_{\text{ME},t}$ is a physical asset ratio. This provides the equation for the trajectory to be implicitly identified from the co-integration relation estimated in empirically in Section 4.3.

2.3.3 The Peripheral Extension: Balance-of-Payments Constraint and the Corner Solution

In the capitalist periphery, the intensive margin of accumulation is structurally tethered to the world market and constrained by domestic social friction. Following [Prebisch \(1981\)](#) and [Kaldor \(1959\)](#), peripheral capitalism is characterized by high consumption propensities out of profits, structural inelasticities in agriculture, and a chronic vulnerability to balance-of-payments crises ([Ahumada, 2019](#)). This introduces a dual realization bottleneck: the macroeconomic capacity to import and the inelastic domestic supply of wage-goods. Section 2.5 established this constraint qualitatively as the peripheral accumulation wedge. This subsection formalizes it as a strict macroeconomic boundary constraint on the cost-minimizing firm's choice of technique, integrating the structuralist insights of [Prebisch \(1981\)](#) into the heterogeneous capital framework of Section 3.2.

2.3.3.1 The Import-Biased Intensive Margin and the Balance of Payments Constraint

In Section 3.2, the cost-minimizing firm chooses the mechanization gap κ_t to maximize the net viability surplus:

$$\min_{\kappa_t} c(\kappa_t) = \kappa_t \cdot MC(\omega_{t-1}, p_t^\kappa) - \Phi(\kappa_t) \quad (2.38)$$

where $\Phi(\kappa_t) = \lambda_0 + \lambda_1 \kappa_t + \lambda_2 \kappa_t^2$ is the regime-conditioned mechanization frontier, and $MC(\omega_{t-1}, p_t^\kappa) = \gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa$ is the marginal viability cost of the gap. In the core, this optimization proceeds unconstrained, yielding the optimal mechanization gap κ_t^* . In the periphery, however, active mechanization is strictly import-biased: widening the mechanization gap requires imported capital goods, specialized machinery, and technologically non-substitutable spare parts ([Vernengo, 2006](#)).

Assumption 3. *Import-biased mechanization.* The physical import requirement per unit of output (m_t^{req}) is a linear function of the chosen technique:

$$m_t^{req} = \nu \cdot \kappa_t \quad (2.39)$$

where $\nu > 0$ captures the structural import-intensity of the mechanization gap. Simultaneously, the physical capacity to satisfy this requirement is constrained by the *capacity to import* per unit of output ($m_t^{avail} = M_t^{avail}/Y_t$). Following [Prebisch \(1981\)](#), peripheral export growth is governed by the center's income elasticity of demand for primary goods ($\epsilon_{m,c} < 1$), structurally lagging behind domestic import demand. This external asymmetry, compounded by high consumption propensities out of capitalist profits ([Kaldor, 1959](#)), severely squeezes the foreign exchange available for productive accumulation. A necessary condition for the re-investment of the surplus. This imposes a strict ceiling on the choice of technique:

$$\nu \cdot \kappa_t \leq m_t^{avail} \implies \kappa_t \leq \bar{\kappa}_t \equiv \frac{m_t^{avail}}{\nu} \quad (2.40)$$

2.3.3.2 The Macroeconomic Allocation of Foreign Exchange

Assumption 4. *Balance-of-payments constraint.* Total import demand cannot exceed export capacity. The total supply of foreign exchange per unit of output is driven by primary exports ($x_t = p_t^x X_t/Y_t$). Due to the center's low income elasticity for primary goods, x_t is structurally rigid. The total demand for foreign exchange per unit of output consists of three competing components:

1. **Wage-goods imports** (m_t^W): driven by the wage share ω_{t-1} and the structural inelasticity of domestic agriculture (Kaldor, 1959). $m_t^W = m_w \omega_{t-1}$.
2. **Conspicuous consumption imports** (m_t^C): driven by the profit share π_t and the high propensity of the peripheral upper strata to imitate the consumption patterns of the center (Kaldor, 1959; Prebisch, 1981). $m_t^C = m_c c_k \pi_t$.
3. **Productive capital imports** (m_t^K): driven by the mechanization gap. $m_t^K = \nu \kappa_t$.

Assumption 5. *Surplus leakage.* The propensity to consume out of profits satisfies $c_k > c_k^{\min}$, where $c_k^{\min}$ is the threshold below which domestic savings suffice to finance capital formation without external borrowing. Under the developmentalist regime, the leakage channel is conspicuous consumption imports by domestic elites (Kaldor, 1959; Prebisch, 1981). Under the neoliberal regime, the leakage channel is profit repatriation by multinational corporations and capital flight (Ahumada, 2019). In both cases, $c_k > c_k^{\min}$ ensures that the effective profit share available for productive reinvestment is compressed below the nominal level: $\pi_t^{eff} = \pi_t \cdot \phi_t < \pi_t$.

The balance of payments constraint requires:

$$m_w \omega_{t-1} + m_c c_k \pi_t + \nu \kappa_t \leq x_t \quad (2.41)$$

Using the distributive identity $\pi_t = 1 - \omega_{t-1}$, I substitute and solve for the maximum feasible mechanization gap:

$$m_w \omega_{t-1} + m_c c_k (1 - \omega_{t-1}) + \nu \kappa_t \leq x_t \quad (2.42)$$

$$\nu \kappa_t \leq x_t - m_c c_k + (m_c c_k - m_w) \omega_{t-1} \quad (2.43)$$

$$\bar{\kappa}_t(\omega_{t-1}) = \frac{1}{\nu} [x_t - m_c c_k + (m_c c_k - m_w) \omega_{t-1}] \quad (2.44)$$

This equation formalizes the structuralist critique: the base capacity to import capital goods is determined by export earnings minus the baseline luxury consumption of the capitalist class ($x_t - m_c c_k$). The sign of $(m_c c_k - m_w)$ determines whether a rising wage share expands or contracts the feasible mechanization gap.

2.3.3.3 The Constrained Cost-Surplus Maximization Problem

The peripheral firm's tactical problem is therefore a cost-minimization problem constrained by the mechanization function and the external constraint, such that the minimization problem can be stated as a Kuhn–Tucker optimization problem:

$$\min_{\kappa_t} c(\kappa_t) = \kappa_t \cdot MC(\omega_{t-1}, p_t^\kappa) - \Phi(\kappa_t) \quad (2.45)$$

$$\text{subject to: } \kappa_t \leq \bar{\kappa}_t \quad (2.46)$$

This formulation distinct a regime shifting behavior of the choice of technique in the capitalist periphery given by a double constraint in the capitalists cost-minimizing behavior: i) Endogenous to domestic distributive conflict, ii) and to the dependency of foreign currency to fulfill import requirements at a given choice of technique. Therefore, if the external constraint becomes binding due lack of foreign currency, a regime shifting behavior of technological change takes place.

2.3.3.4 Endogenous Regime Shifts: The Corner Solution

The firm's unconstrained optimal technique $\kappa_t^*(\omega_{t-1})$ is strictly increasing in the wage share ($\partial\kappa_t^*/\partial\omega_{t-1} > 0$) due to the Okishio labor-saving bias established in (2.25).

Proposition 4. *Peripheral regime shifts.* Two regimes are endogenous to the interaction between domestic distributive conflict and the global market:

1. **Regime A (Slack BoP):** When $\kappa_t^* \leq \bar{\kappa}_t$, the firm solves the standard first-order condition, yielding κ_t^* . The labor-saving bias successfully extracts the Okishio distributive dividend.
2. **Regime B (Peripheral realization trap):** When $\kappa_t^*(\omega_{t-1}) > \bar{\kappa}_t(\omega_{t-1})$, the constraint binds and the firm is forced into a corner solution:

$$\kappa_t = \bar{\kappa}_t \quad (2.47)$$

The corner solution arises through two intersecting structural vulnerabilities. First, if $m_c c_k$ is large (the conspicuous consumption drain), the baseline FX availability is severely squeezed, lowering the intercept of $\bar{\kappa}_t$. Second, if domestic agriculture is highly inelastic ($m_w > m_c c_k$), a rising wage share simultaneously increases the *demand* for capital imports (via κ_t^*) while *contracting* the supply of available FX (via $\bar{\kappa}_t$). *Proof.* Immediate from the Kuhn–Tucker conditions applied to (2.45)–(2.46). ■

2.3.3.5 Effective Profit Compression and the Blocked Surplus

Definition 3. *FX-Conversion Ratio.* The FX-conversion ratio ϕ_t is the ratio of the maximum feasible technique to the optimal unconstrained technique:

$$\phi_t = \min \left(1, \frac{\bar{\kappa}_t}{\kappa_t^*} \right) \quad (2.48)$$

When the external constraint binds ($\kappa_t^* > \bar{\kappa}_t$), $\phi_t < 1$. This generates a direct compression of the effective profit share available for intensive accumulation.

Definition 4. *Blocked surplus.* Even if domestic wage suppression yields a high nominal profit share (π_t), the physical inability to convert that surplus into imported machinery squeezes the effective profit rate:

$$\pi_t^{eff} = \pi_t \cdot \phi_t = (1 - \omega_{t-1}) \cdot \phi_t \quad (2.49)$$

The difference constitutes the *blocked surplus*:

$$\text{Blocked Surplus} = \pi_t(1 - \phi_t) \quad (2.50)$$

Because the accumulation of machinery is driven by the *effective* profit share, the constraint decouples nominal profitability from capacity formation. If capitalists squeeze wages to raise π_t , they inadvertently increase conspicuous consumption (m_t^C), which lowers $\bar{\kappa}_t$ and compresses ϕ_t .

2.3.3.6 The Regime-Shifting Aggregate Transformation Elasticity

Definition 5. *BoP binding ratio.* The BoP binding ratio τ_t is the ratio of productive capital imports to import capacity:

$$\tau_t \equiv \frac{\nu \cdot I_t^{ME}}{\text{ToT}_t \times X_t} \quad (2.51)$$

where I_t^{ME} is gross investment in machinery and equipment, and $\text{ToT}_t \times X_t$ proxies the import capacity (FX supply). Under Assumption 3, productive capital imports $m_t^K = \nu \kappa_t$, since mechanization is import-biased in the periphery. The regime switch is governed by τ_t :

- $\tau_t < \bar{\tau}$: BoP constraint is **slack** (Regime A: unconstrained optimum).
- $\tau_t \geq \bar{\tau}$: BoP constraint is **binding** (Regime B: corner solution).

The threshold $\bar{\tau}$ is determined empirically by the threshold regression in Section 4.

The peripheral extension generates a regime-shifting aggregate transformation elasticity. Under **Regime A** (slack BoP), the constraint does not bind, and the elasticity coincides with the unconstrained expression from Section 3.2 for the aggregate elasticity:

$$\theta_t^A = \frac{\theta_{NRC} + (\theta_\kappa + \theta_{\omega\kappa}\omega_{t-1} + \theta_{\kappa p\kappa}p_t^\kappa) \frac{d\kappa_t^*}{dk_t^{NRC}}}{1 + s_{ME,t} \frac{d\kappa_t^*}{dk_t^{NRC}}} \quad (2.52)$$

Under **Regime B** (BoP binding), the surplus of capitalist enterprises becomes blocked, and the firm is forced into a corner solution $\kappa_t = \bar{\kappa}_t$. The marginal capacity payoff at the corner is strictly positive because the firm would like to mechanize further but cannot:

$$MP_\kappa|_{\kappa=\bar{\kappa}} = \Phi'(\bar{\kappa}_t) - MC(\omega_{t-1}, p_t^\kappa) > 0 \quad (2.53)$$

The Kuhn–Tucker multiplier on (2.46) is strictly positive, confirming that the constraint binds. The aggregate transformation elasticity under the binding regime becomes:

$$\theta_t^B = \frac{\theta_{NRC} + MP_\kappa|_{\kappa=\bar{\kappa}} \cdot \frac{d\bar{\kappa}_t}{dk_t^{NRC}}}{1 + s_{ME,t} \frac{d\bar{\kappa}_t}{dk_t^{NRC}}} \quad (2.54)$$

Since $\bar{\kappa}_t$ is determined by the BoP constraint (2.44) rather than by the first-order condition (2.23), the growth path $d\bar{\kappa}_t/dk_t^{NRC}$ differs from the unconstrained path $d\kappa_t^*/dk_t^{NRC}$. The FX-conversion ratio $\phi_t = \bar{\kappa}_t/\kappa_t^* < 1$ compresses the intensive margin, and the elasticity under the binding regime is strictly lower:

$$\theta_t^B < \theta_t^A \quad (2.55)$$

Proposition 5. *Regime-shifting elasticity.* The peripheral aggregate transformation elasticity is a regime-dependent parameter:

$$\theta_t^{\text{periphery}} = \theta_t^A \cdot \mathbf{1}\{\kappa_t^* \leq \bar{\kappa}_t\} + \theta_t^B \cdot \mathbf{1}\{\kappa_t^* > \bar{\kappa}_t\} \quad (2.56)$$

The elasticity shifts discretely when the BoP constraint becomes binding. The regime switch is governed by the interaction between the desired mechanization gap $\kappa_t^*(\omega_{t-1}, p_t^\kappa)$ and the maximum feasible gap $\bar{\kappa}_t(\omega_{t-1})$ determined by export capacity, conspicuous consumption, and wage-goods imports. An increase in the wage share ω_{t-1} simultaneously raises κ_t^* (via capital-using labor-saving choice of technique) and can contract $\bar{\kappa}_t$ (via the wage-goods import channel when $m_w > m_c c_k$), making the regime switch more likely. This is what [Prebisch \(1981\)](#) coined as the strangulation of economic development in the capitalist periphery: distributive pressures that would induce productive mechanization in the center instead triggers the BoP constraint, which in this framework explain the truncated or regime shifting behavior of the transformation elasticity.

2.3.3.7 The Failure of the Viability Conditions for Technological Change and Structural Inflation

Proposition 6. *Blocked viability condition.* At the boundary $\kappa_t = \bar{\kappa}_t$, the implementation of the viability condition is blocked, i.e. technical change is not net-surplus increasing or conversely, inefficient because it fails to achieve a cost-minimizing choice of technique. The firm satisfies the viability criterion—the technique would reduce unit costs—but cannot deploy it because the required machinery cannot be imported due lack of foreign currency. Hence, the capitalist enterprise in the capitalist periphery suffers a persistent shortfall in its cost-surplus relative to its unconstrained state. *Proof.* At $\kappa_t = \bar{\kappa}_t$, the Kuhn–Tucker multiplier on (2.46) is strictly positive, so the first-order condition $\partial c / \partial \kappa_t = 0$ does not hold. The marginal viability yield exceeds the marginal cost at the boundary, but the firm cannot expand further. ■

Corollary 1. *Structural Inflation* Unable to extract the distributive dividend via mechanization, capitalists attempt to recover the blocked surplus through the only remaining channel: price markups. When the wage-goods bottleneck forces nominal wage adjustments and the external BoP bottleneck blocks defensive mechanization, any further increase in the wage share directly crushes profitability. The resulting defensive pricing behavior ignites structural inflation dynamic (Kalecki, 1955; Prebisch, 1981; Sunkel, 1958). This is the peripheral realization trap: nominal profitability coexists with truncated mechanization and structurally stalled built-environments that constitute the “*envelope*” of the labor process, permanently destabilizing the long-run relation between investments in plant and machinery.

2.3.4 Analytical and Theoretical Synthesis

The three models developed in this section formalize a single theoretical claim: that decentralized, individually cost-minimizing decisions generate aggregate disproportionality under the unbalanced growth closure. Table 2.1 summarizes the articulation between the Okishio and Vidal frameworks:

Table 2.1. Market Anarchy and the Internally Divided Firm Articulation Across Three Models

Logic	Decentralized	Accumulation	Internally Divided Firm	Analytical Insight
Micro logic	Viability condition (cost reduction at prevailing prices)		Internal contradiction (control vs. cooperation)	Optimization problem = cost-minimizing pole; constraints = cooperation limits
Macro logic	Decentralized decisions → disproportionality		Divided firm → contested operating conditions	$\theta \neq 1$ captures gap between accumulation and capacity
Disequilibrium	Unbalanced growth (supply $\neq$ demand)		“Mediocre sufficiency” (Vidal, 2022) (no coherent optimum)	μ demonstrates gap; θ captures structural inefficiency
Peripheral specificity	BoP constraint blocks <i>implementation</i> of viability		External constraint compounds internal division	Corner solution: $\kappa = \bar{\kappa}$ when constraint binds

This analytical framework does not assert equilibrium. It aims demonstrates the opposite: capitalist accumulation, left to its own decentralized logic, does not generate proportional reproduction. The rate of capacity utilization μ is the exhaust valve of this disproportionality: it shows that the normal operating conditions of capitalist firms, determined by their cost-minimizing mechanization choices, do not meet aggregate demand conditions. The transformation elasticity θ captures this

structural gap between accumulation and realized capacity. Whether θ exceeds or falls below unity is an empirical question to be tackled below. In the center, this gap is cyclical—mediated by institutional configurations that can be exhausted triggering processes of capitalist re-structuring. In the periphery, this gap is permanent—locked in by the structural ceiling of external subordination (Kaldor, 1959; Prebisch, 1981).

Three models have now specified what θ depends on, and none of them has said what it equals. The dependence is on objects that no national accounting system reports directly: a capacity ceiling that is never observed, a mechanization gap between two capital stocks that are valued by convention, and, for the periphery, an external state that has to be inferred rather than read off a date. Whether θ can be recovered at all therefore turns on whether those objects can be reconstructed from the historical record without importing the transformation relation into their construction.

2.4 Data, Measurement, and Empirical Results

2.4.1 Data and Measurement

2.4.1.1 A Dual Valuation Distinction

This section constructs the capital stock series for the choice-of-technique models of section 2.3. The problem with the standard treatment of capital stocks series, is that they treat the physical capacity of installed capital and the monetary value of capital advanced as a single magnitude. This problem introduces a risk of spurious correlation between capital intensity and wage share measures as proxies for distributive conflict: as mechanization raises capital intensity, Consumption of Fixed Capital (CFC) rises in the measurement of Gross Value Added (GVA) denominator, depressing GVA-basis wage shares even when the rate of surplus-value extraction is unchanged. To deal with this bias, I reconstruct the stock account-by-account using the Generalized Perpetual Inventory Method (GPIM) proposed by Shaikh (2016, App. 6.5). This methodology constructs gross and net capital stock measures that maintain stock-flow consistency from direct nominal investment flows and asset-specific price indexes.

Estimates of capacity utilization under the unbalanced growth framework require distinct capital stock variables. The capital value initially invested in plant and equipment returns gradually to its money form as fixed assets depreciate; the accumulated depreciation allowances remain part of total capital advanced regardless of their monetary form, because their recovery is what permits the reproduction of the capitalist enterprise I therefore distinguish two types of measurement of capital stocks:

1. **Capital stock for capacity estimation: gross real GPIM stock.** Productive capacity depends on capital goods still physically in operation. The capital stock measure adequate for estimating capacity is the gross real stock $K_{j,t}^{G,R}$, weighted exclusively by a physical survival schedule. Productive capacity Y_t^P is itself a latent variable, recovered through the estimation of the transformation elasticity θ (section 2.3); the gross real GPIM stock is the accumulation variable k_t included in my estimation equation, not a measure of capacity itself. A machine whose book value has fallen to zero, but remains in service until it is physically retired, still should be counted as physical capital in operation. The OECD manual confirms that the gross stock, valued at “as new” prices, “is widely used as a broad indicator of the productive

capacity of a country” (OECD, 2001, p.31).

2. **Profitability measure: net current-cost stock.** Profitability is a monetary ratio between a profit flow and the value of capital advanced. The relevant measure is the net current-cost stock $K_{j,t}^{N,V}$, which tracks the monetary value of capital after financial depreciation and write-downs. This measure serves as the anchor for the estimation of an implicit price of capital goods under a stock-flow-consistent framework but is not included within the structural estimation of non-observed productive capacities.

A single aggregate net stock serving both purposes is common in empirical macroeconomics. This practice introduces a mechanical correlation between the capital stock and the depreciation flow, introducing potential biases to identify distributive effects on productive capacities formation.

Valuation choice within the GPIM. The dual-valuation distinction does not require imposing a single valuation convention throughout. Shaikh (2016, App. 6.3) advocates the gross stock as the denominator for both the profit rate and the capacity ratio. The GPIM he develops, however, constructs the survival schedule $S_j(v)$, the age-price schedule $(1 - d_j)^v$, and the SFC implicit price index $P_{j,t}$ as distinct components within a single stock-flow-consistent recursion. This structure permits the researcher to select the valuation for each analytical purpose without abandoning accounting coherence.

For the profit rate, the denominator must measure the monetary value of capital advanced. The net current-cost stock serves this role: it tracks the book value of capital that the firm must recover to continue production. A machine that has depreciated to a low book value may still operate at full physical capacity, but the firm’s cost-recovery problem is defined by the value it must recoup, not by the physical state of the equipment (Basu and Vasudevan, 2013). For estimates related to capacity utilization, the capital stock input must reflect the physical capital still in operation. The gross real GPIM stock, weighted exclusively by the survival schedule $S_j(v)$ and excluding the age-price factor $(1 - d_j)^v$, serves this role.

The quantitative significance of the gross/net distinction is not confined to the center. Ffrench-Davis and Vivanco (2016) show that in Chile depreciation absorbs between 8 and 13 percent of GDP and that net investment is barely half of gross investment in the most recent period. Service lives of machinery have declined from 20 years in the 1940s to 11 years by 2005, reflecting accelerating obsolescence. They conclude that the capital stock entering a production function must be the gross stock of physically surviving assets, not the depreciated monetary residual.

2.4.1.2 Gross Capital Stock Reconstruction and Measurement in the United States

The Generalized Perpetual Inventory Method. In this part I present the procedures of construction for the capital stock series for the United States. Machinery (ME) and nonresidential structures (NRC) do not share a natural physical unit; they become comparable only through a monetary-mediated accounting procedure. I reconstruct the stock account-by-account rather than aggregating nominal values and deflating with a generic index. The GPIM accumulation rules for historical-cost, current-cost, and constant-dollar stocks (Shaikh, 2016, App. 6.5) are derived in detail in Appendix 2.A; here I state the construction.

Physical survival of capital vintages follows a Weibull retirement distribution. For asset class

$j \in \{\text{ME}, \text{NRC}\}$, the proportion of a vintage surviving after v years is:

$$S_j(v) = \exp \left[- \left(\frac{v}{\lambda_j} \right)^{\alpha_j} \right] \quad (2.57)$$

where α_j is the Weibull shape parameter and the scale parameter is $\lambda_j = L_j / \Gamma(1 + 1/\alpha_j)$, derived from the mean service life L_j . Unlike geometric decay models that continuously erode the physical stock, the Weibull curve ensures that capital goods remain fully operative until they are physically removed from service. Empirical studies of directly observed asset discards confirm that the Weibull family provides a flexible approximation of retirement patterns. [Nomura \(2005\)](#), using Japanese micro-data on 66 asset classes, finds that almost half of assets exhibit a progressively increasing hazard rate ($\alpha > 1$). The calibrated parameters $L_{\text{ME}} = 14$, $\alpha_{\text{ME}} = 1.7$ and $L_{\text{NRC}} = 30$, $\alpha_{\text{NRC}} = 1.6$ follow the BEA Fixed Assets methodology ([Fraumeni et al., 1997](#)).

Physical survival alone cannot identify the SFC implicit price index. The BEA current-cost net stock anchors reflect both physical retirement and age-price decay. To recover the asset-specific price index, I impose an externally documented geometric declining-balance age-price profile:³

$$V_j(v) = S_j(v) \cdot (1 - d_j)^v \quad (2.58)$$

where d_j is the asset-specific age-price decay rate. The declining-balance formula applied to the calibrated service lives yields $d_{\text{ME}} = 0.110$ and $d_{\text{NRC}} = 0.024$, consistent with the geometric rates documented by the BEA ([Fraumeni et al., 1997](#)).

The SFC implicit price index $P_{j,t}$ is the price level that closes the stock-flow identity for the net current-cost stock. In each period, the BEA anchor must satisfy

$$K_{j,t}^{\text{NET,CC}} = I_{j,t}^{\text{NOM}} + \frac{P_{j,t}}{100} \sum_{v=1}^{L_j} V_j(v) I_{j,t-v}^{\text{REAL}} \quad (2.59)$$

where $I_{j,t-v}^{\text{REAL}} = I_{j,t-v}^{\text{NOM}} / (P_{j,t-v} / 100)$ is real investment deflated by the previously recovered SFC price. The first term on the right-hand side is current-year nominal investment; the second is the net-value-weighted sum of surviving real vintages revalued at the current price. The price index is “implicit” because it is not directly observed: it is the unique value that, applied to the survival-weighted real vintages, reproduces the BEA’s reported current-cost net stock. Solving (2.59) for $P_{j,t}$:

$$P_{j,t} = 100 \cdot \frac{K_{j,t}^{\text{NET,CC}} - I_{j,t}^{\text{NOM}}}{\sum_{v=1}^{L_j} V_j(v) I_{j,t-v}^{\text{REAL}}} \quad (2.60)$$

The forward recursion proceeds period by period: given $P_{j,t-1}, P_{j,t-2}, \dots$, the current price $P_{j,t}$ is uniquely determined by the requirement that the constructed stock reproduces the BEA anchor. Appendix 2.A documents the recursion, the initial conditions, and the base-year normalization. The BEA’s Fisher quantity index is retained solely as a validation comparison and does not enter the

³The BEA’s current methodology employs geometric depreciation, which assumes that the efficiency pattern and the depreciation pattern have the same form ([Fraumeni et al., 1997](#)). The GPIM construction separates these two dimensions: physical survival follows a Weibull retirement distribution, while financial age-price decay follows a geometric profile. This separation is necessary to construct distinct capacity and profitability measures.

GPIM baseline.

The capital stock used to estimate productive capacity is the gross real GPIM stock, constructed as the survival-weighted sum of real investment vintages:

$$K_{j,t}^{G,R} = \sum_{v=0}^{L_j} S_j(v) I_{j,t-v}^{\text{REAL}} \quad (2.61)$$

where $I_{j,t-v}^{\text{REAL}}$ is real investment deflated by the previously recovered SFC price. Because eq. (2.61) relies exclusively on the survival schedule $S_j(v)$ and excludes the age-price factor $(1 - d_j)^v$, it measures the constant-price purchasing power embodied in heterogeneous capital goods still in physical operation. Financial write-downs, revaluation gains, and depreciation allowances do not enter.

Validation and sample scope. The GPIM construction closes the stock-flow identity exactly: the maximum absolute reconstruction residual across both asset classes is 10^{-6} million dollars, confirming that the recovered SFC implicit price indexes reproduce the BEA current-cost net stock anchors. All GPIM series are normalized to 2017 = 100. The capital stock series has a 30-year warm-up period (1901–1930). The GPIM stock at any year t requires investment vintages extending back L_j years. Because the nominal investment series begins in 1901, the vintage summation for the longest-lived asset class (nonresidential structures, $L_{\text{NRC}} = 30$) is truncated for any year before 1931. The stock values for 1901–1930 are therefore incomplete, omitting pre-1901 vintages that would still be physically in service. I restrict the estimation sample to 1931–2024, where the full 30-year vintage window is populated, yielding $T = 94$ annual observations. The sectoral scope is restricted to nonfinancial corporate (NFC) business, excluding non-profit institutions serving households (NPISH), government, and owner-occupied housing. This sectoral restriction isolates the private productive business sector from public infrastructure and imputed residential rents.

2.4.1.3 U.S. Distributive Variables

The GVA inadequacy. The standard GVA-basis wage share is might become contaminated by Consumption of Fixed Capital. Because $\text{GVA} = \text{CFC} + v + s$, periods of intensive mechanization raise CFC/GVA even when the rate of surplus-value extraction (s/v) is unchanged. The GVA-basis wage share therefore might be contaminated of spurious correlation between capital intensity and the measured distributive share during capital-deepening episodes. The NVA basis on the other hand, $\text{NVA} = v + s$, could avoid this problem. I estimate a permutation of multiple distributive measures to explore empirically this potential problem.⁴

The No-Double-Accounting Principle. Standard NIPA corporate aggregates combine non-financial and financial corporate sectors into a single total. This aggregation violates the No-Double-Accounting Principle: financial sector profits and net interest receipts are secondary claims extracted from primary nonfinancial surplus, not independently created value. Including them in the denominator of the wage share counts value creation twice. The corrected denominator is

⁴Eight wage share variants are available: NFC baseline, Shaikh-Tonak Tier 1, Shaikh-Tonak Tier 2, and Clean Corporate, each on a GVA and an NVA basis. Unadjusted Shaikh-Tonak Tier 1 NVA is arithmetically identical to the NFC baseline NVA share, so the non-redundant specifications reduce to the Clean Corporate and Tier 2 supervisory adjustments discussed in section 2.4.4.

nonfinancial corporate net value added (NFC_NVA), which excludes financial double-accounting transfers.⁵

Supervisory compensation adjustment. Standard NIPA compensation includes managerial and executive salaries, which under the Shaikh-Tonak classification function as a distribution of surplus value rather than productive variable labor cost (Shaikh and Tonak, 1994). To isolate productive variable capital, total compensation is adjusted by the supervisory compensation ratio ϕ_t^{sup} :

$$v_t^{\text{prod}} = \text{NFC_COMP}_t \times (1 - \phi_t^{\text{sup}}). \quad (2.62)$$

Two tiers of wage share are defined. Tier 1 uses total compensation without supervisory adjustment; Tier 2 applies the adjustment in (2.62). The supervisory compensation ratio ϕ_t^{sup} is constructed from Table S512 of the *Measuring the Wealth of Nations* dataset (Shaikh and Tonak, 1994), as reconstructed and extended to 2024 by Anderson (Anderson, 2024).⁶

All measures are constructed in nominal terms (W/NVA and W/GVA), guaranteeing that relative capital-asset price dynamics and depreciation deflators do not distort the physical vs. monetary valuation of surplus extraction.

2.4.1.4 Productive output and aggregate demand for the United States

The dependent variable in the cointegrating regressions of section 2.3 is the logarithm of real gross value added of nonfinancial corporate business, $y_t = \ln Y_t^{\text{NFC}}$, measured in chained 2017 dollars and sourced from BEA NIPA Table 1.14, Line 41.⁷ The nonfinancial corporate boundary isolates productive value creation within the business sector, excluding financial intermediaries whose income constitutes secondary claims on nonfinancial surplus rather than independently created value. The robustness battery of section 2.4.1.3 audits eight alternative deflator proxies—including the official NFC implicit deflator (Table 1.15), the GDP implicit deflator, and the nonfarm business output deflator—and confirms that coefficient estimates are invariant to deflator choice.

The numerator of aggregate capacity utilization $\mu_t = Y_t/Y_t^p$ is economy-wide real GDP, drawn from BEA NIPA Table 1.1.9, Line 1, in chained 2017 dollars.⁸ The denominator Y_t^p is not directly observed but reconstructed from the estimated transformation elasticity, as detailed in section 2.4.4.2. The distinction between the two output measures is deliberate: the transformation elasticity θ is estimated from NFC output, which captures the capital-output relationship within the productive corporate sector; the utilization ratio uses aggregate GDP because capacity utilization measures the engagement of total productive capacity by effective demand, including government expenditure, household consumption, and net exports. This separation ensures that the estimation of θ remains uncontaminated by demand-side fluctuations that drive aggregate GDP but do not reflect the structural capital-output relationship.

⁵The implied financial sector surplus is recovered as $\text{FIN_NOS}^{\text{implied}} = \text{CORP_NOS} - \text{NFC_NOS}$. Clean corporate denominators purge both $\text{FIN_NOS}^{\text{implied}}$ and NFC_NET_INT .

⁶The full dataset is available at <https://shaikh.heterodata.org/>

⁷BEA series code B455RX; nominal counterpart A455RC, Line 17. The implicit output deflator is $P_{y,t}^{\text{NFC}} = 100 \times \text{NFC_GVA}_t^{\text{nom}}/Y_t^{\text{NFC}}$. Coverage: 1929–2025, annual.

⁸BEA series code A191RL; coverage 1929–2025, annual. The GDP implicit price deflator (A191RD) serves as the economy-wide deflator for the utilization reconstruction.

2.4.1.5 Capital Series Harmonization and Historical Sources for Chile

This subsection describes the stock-flow consistent (SFC) methodology used to reconstruct gross and net capital stocks for Chile, 1900–1994, in constant 2003 Chilean pesos (CLP). The procedure enforces the accumulation identities for gross and net stocks jointly, yields implicit depreciation and retirement rates for each asset class, and delivers coherent series for machinery and equipment (ME), nonresidential construction (NRC), residential construction (RC), and their aggregates. The construction proceeds from a single investment control total, disaggregated by asset, accumulated via Weibull survival functions, and linked to net stocks through a gap-ratio approach anchored to the [Hofman \(2000b\)](#) overlap window.

The stock-flow identities. The reconstruction is governed by two laws of motion. The gross stock evolves according to

$$\Delta K_{j,t}^g = I_{j,t} - R_{j,t} \quad (2.63)$$

where $I_{j,t}$ is gross fixed investment and $R_{j,t}$ denotes physical retirements (scrapping of obsolete vintages, demolition, and technological withdrawal). The net stock evolves according to

$$\Delta K_{j,t}^n = I_{j,t} - D_{j,t} \quad (2.64)$$

where $D_{j,t}$ is depreciation (the monetary consumption of fixed capital). Combining (2.63) and (2.64) yields the joint SFC identity

$$\Delta K_{j,t}^g + R_{j,t} = \Delta K_{j,t}^n + D_{j,t} \quad (2.65)$$

which constrains the combination of depreciation and retirements at every t and for every asset j . Given observed stocks and investment, the implied flows are recovered as $D_{j,t} = I_{j,t} - \Delta K_{j,t}^n$ and $R_{j,t} = I_{j,t} - \Delta K_{j,t}^g$. The capital gap $B_{j,t} \equiv K_{j,t}^g - K_{j,t}^n$ measures the cumulative difference between installed capacity and book value.

The investment control total and the lambda bridge. The aggregate investment control total is drawn from the [Pérez-Eyzaguirre \(2019\)](#) reconstruction of gross fixed capital formation (GFCF), expressed in constant 2003 CLP. The control concept is machinery plus construction, explicitly excluding inventory variation.⁹

Pérez-Eyzaguirre report investment in a price space and asset aggregation that differ from the [Hofman \(2000a\)](#) benchmark. The two sources are linked by a level-normalization factor λ , which corrects for definitional differences between the two series rather than converting between price bases. The factor is estimated as the median of the annual ratio of Hofman to Pérez investment over the 95-year estimation window:

$$\lambda = \text{median} \left(\frac{I_t^{\text{Hofman}}}{I_t^{\text{Pérez}}} \right)_{t=1900}^{1994} = 4.1054 \quad (2.66)$$

The median rule is chosen for robustness to outliers in the early decades, where source quality is weaker. The scaled aggregate is then disaggregated into ME, NRC, and RC using normalized

⁹Inventories are excluded from fixed-capital accumulation because the reconstruction models fixed capital only. Inventory investment does not enter the perpetual inventory recursion.

asset-specific investment shares from [Hofman \(2000a\)](#):

$$I_{j,t} = s_{j,t} \cdot \lambda \cdot I_t^{\text{Pérez}}, \quad \sum_{j \in \{\text{ME}, \text{NRC}, \text{RC}\}} s_{j,t} = 1 \quad (2.67)$$

for all t .

Weibull survival and gross stock accumulation. Physical survival of capital vintages follows a Weibull retirement distribution. For asset class $j \in \{\text{ME}, \text{NRC}, \text{RC}\}$, the proportion of a vintage surviving after v years is

$$S_j(v) = \exp \left[- \left(\frac{v}{\lambda_j} \right)^{\alpha_j} \right] \quad (2.68)$$

where α_j is the Weibull shape parameter and the scale parameter is $\lambda_j = L_j / \Gamma(1 + 1/\alpha_j)$, derived from the mean service life L_j . The survival parameters are estimated from the [Hofman \(2000b\)](#) overlap window (1950–1994), in which both gross and net stocks are observed, by fitting Weibull curves to the implied retirement patterns recovered from the stock-flow identities. Table 2.2 reports the calibrated parameters.

Table 2.2. Weibull survival parameters, Chile (1900–1994)

Asset	Scale (λ_j)	Shape (α_j)	Mean life (L_j , years)
ME	15.90	2.42	14.1
NRC	41.30	5.00	37.9
RC	51.16	5.00	47.0

All shape parameters exceed unity, indicating progressively increasing hazard rates: capital goods remain fully operative in early years and are withdrawn from service at an accelerating rate as they approach the mean service life. The shape parameters for NRC and RC are constrained at 5.0, reflecting the long and relatively uniform service lives of structures. The gross stock is accumulated as the survival-weighted sum of real investment vintages:

$$K_{j,t}^g = \sum_{v=0}^{L_j} S_j(v) I_{j,t-v} \quad (2.69)$$

Pre-1950 stocks are anchored to the observed Hofman levels at 1950 via a multiplicative factor that matches $K_{j,1950}^g$ by asset.

The gap-ratio approach and net stock construction. Unlike the U.S. GPIM, which recovers the net stock via a separate geometric age-price decay rate and an SFC implicit price recursion, the Chile reconstruction derives the net stock from the observed gross-net relationship in the [Hofman \(2000b\)](#) overlap window. The capital gap ratio

$$\gamma_j \equiv \frac{B_j}{K_j^g} = \frac{K_j^g - K_j^n}{K_j^g} \quad (2.70)$$

is computed by asset from the [Hofman \(2000b\)](#) data over the overlap period (1950–1994). Two regimes are distinguished: a “pre” ratio (median of years ≤ 1954) and a “post” ratio (median of years ≥ 1990), allowing for structural shifts in the gross-net relationship over the century. Outside the overlap window, the gap ratio is extended at the regime-specific median. The net stock is then recovered as

$$K_{j,t}^n = K_{j,t}^g \cdot (1 - \gamma_j) \quad (2.71)$$

This approach inherits the gross-net relationship observed in the Hofman accounting reference rather than imposing an external depreciation convention. It is appropriate for Chile because no independent age-price decay data are available for the historical period; the [Hofman \(2000b\)](#) overlap provides the only direct evidence on the joint behavior of gross and net stocks.

Implicit depreciation and retirement rates. Given the reconstructed stocks and investment flows, the implicit annual rates are defined as

$$z_{j,t} = \frac{R_{j,t}}{K_{j,t-1}^g}, \quad \delta_{j,t} = \frac{D_{j,t}}{K_{j,t-1}^n} \quad (2.72)$$

where $z_{j,t}$ is the retirement (depletion) rate and $\delta_{j,t}$ is the depreciation rate. These rates are not imposed as parameters; they are derived from the data via the stock-flow identities (2.63)–(2.64). They serve as diagnostic indicators of capital durability and replacement dynamics, and are used in section 2.4.6 to validate the internal consistency of the reconstruction.

Validation and estimation sample. The SFC construction is validated through four checks. First, additivity: all composite stocks and flows satisfy $K_{NR} = K_{ME} + K_{NRC}$ and $K_T = K_{ME} + K_{NRC} + K_{RC}$ at every t . Second, non-negativity: $K_{j,t}^g \geq 0$, $K_{j,t}^n \geq 0$, and $B_{j,t} \geq 0$ for all assets and years. Third, rate bounds: $z_{j,t} \in [0, 1]$ and $\delta_{j,t} \in [0, 1]$ for all observations from 1950 onward. Fourth, identity residuals: $\Delta K_{j,t}^g - (I_{j,t} - R_{j,t}) = 0$ and $\Delta K_{j,t}^n - (I_{j,t} - D_{j,t}) = 0$ to numerical precision. The estimation sample spans 1900–1994, yielding $T = 95$ annual observations. The overlap window (1950–1994) provides 45 years of direct [Hofman \(2000b\)](#) observations against which the reconstructed series are audited.

The BCCh forward extension (1995–2024). Extending the reconstruction beyond 1994 requires integrating the official national accounts published by the Central Bank of Chile (BCCh). A direct chain-linking of the BCCh series to the historical baseline would introduce a level discontinuity, because the two sources differ in base year, valuation conventions, and asset classification. To eliminate this discontinuity while preserving the validated 1900–1994 history, I employ a multiplicative splicing procedure that fixes the legacy panel in place.

For the overlap year 1994, an asset-specific bridge factor is computed as the ratio of the historical net stock to the BCCh reported net stock:

$$\beta_j = \frac{K_{j,1994}^{n,\text{hist}}}{K_{j,1994}^{n,\text{BCCh}}}, \quad j \in \{\text{ME}, \text{NRC}, \text{RC}\} \quad (2.73)$$

The same factor β_j is applied to both the BCCh net stock and the BCCh depreciation series for all years 1995–2024. This preserves the depreciation rates and asset dynamics implied by the BCCh

while maintaining continuity with the constant 2003 CLP historical baseline. Table 2.3 reports the estimated bridge factors.

Table 2.3. Asset-specific bridge factors, 1994

Asset	Bridge factor (β_j)	Interpretation
ME	5.888	BCCh understates historical ME stock
NRC	1.033	Near parity
RC	1.050	Near parity

The large bridge factor for machinery and equipment reflects the substantially lower level at which the BCCh reports ME stocks relative to the Pérez-based historical reconstruction, a divergence driven by differences in asset coverage and valuation methodology. For nonresidential and residential construction, the bridge factors are close to unity, indicating that the two sources agree substantially on the level of structures.

Post-1994 real investment is inferred via the stock-flow identity:

$$I_{j,t} = \Delta K_{j,t}^n + D_{j,t}, \quad t \geq 1995 \quad (2.74)$$

where $D_{j,t}$ is the bridge-adjusted BCCh depreciation. The gross stock and physical retirements for the extended period are then reconstructed by applying the survival parameters from Table 2.2 to the complete 1900–2024 investment series via eq. (2.69). This ensures that the gross stock remains internally consistent across the splice boundary.

2.4.1.6 Wage Share Data Sources for Chile

The distributive regressor in the transformation elasticity is the wage share, ω_t . For Chile, the wage share is drawn from the revised dataset of real wages in Latin America constructed by Astorga (2017) and updated by Astorga (2023). The dataset provides yearly series of real wages by skill level—unskilled (w_4), semi-skilled (w_3), and skilled (w_2)—together with an economically-active-population (EAP) weighted average (w), expressed in monthly purchasing-power-parity dollars at 1970 prices.¹⁰

The wage share is computed as the wage-to-income ratio $\omega_t = w_t/y_t$, where y_t is real income per worker derived from national income estimates. The Astorga (2023) series covers 1920–2011. To extend coverage to 2024, I splice Astorga’s series with the institutional sector accounts of Chile’s Central Bank. The BCCh National Accounts report production income by institutional sector—financial corporations, non-financial corporations, households and NPISH, and general government—in nominal terms at 2018 reference prices. From these, I construct the private non-financial wage share as

$$\omega_t^{\text{BDE}} = \frac{Y_t^{\text{HH}}}{Y_t^{\text{TOT}} - Y_t^{\text{FIN}} - Y_t^{\text{GOV,CFC}}} \quad (2.75)$$

where Y_t^{HH} is household income (compensation of employees plus mixed income), Y_t^{TOT} is total production income, Y_t^{FIN} is the financial sector’s operating surplus, and $Y_t^{\text{GOV,CFC}}$ is general

¹⁰The revised dataset is publicly available as supplementary material to Astorga (2023).

government consumption of fixed capital. The exclusion of the financial sector from the denominator is consistent with the No-Double-Accounting Principle applied to the U.S. series in section 2.4.1.3: financial profits are recognized as secondary claims on primary non-financial surplus and do not enter the distributive denominator.¹¹

The BCCh series covers 2003–2024, yielding a nine-year overlap (2003–2011) with the Astorga (2023) panel. The extension is anchored at 2011, the final year of the Astorga coverage, by constructing the index

$$\mathcal{I}_t = \frac{\omega_t^{\text{BDE}}}{\omega_{2011}^{\text{BDE}}}, \quad t \geq 2012 \quad (2.76)$$

and applying it to the Astorga terminal value:

$$\omega_t = \omega_{2011}^{\text{Astorga}} \cdot \mathcal{I}_t, \quad t \geq 2012. \quad (2.77)$$

The splice assumes that the growth rate of the national-accounts wage share is a valid proxy for the growth rate of the labor-market wage-to-income ratio beyond the Astorga coverage. The nine-year overlap (2003–2011) supports this assumption: both series exhibit the same cyclical pattern and comparable magnitudes of distributive fluctuation. The resulting ω_t series spans 1920–2024, yielding $T = 105$ annual observations.

2.4.1.7 Aggregate Demand, Price Indices, and External Constraint variables for Chile

The aggregate demand variables—real GDP (Y_t), exports FOB (X_t^{FOB}), and gross machinery investment ($I_{\text{ME},t}$)—are drawn from the aggregate demand reconstruction of Pérez-Eyzaguirre (2019), expressed in constant 2003 Chilean pesos. This series revises and extends the historical national accounts backbone of La República en Cifras (Lüders et al., 2016), correcting additivity and valuation problems in the original Clio-Lab series while preserving the sectoral GDP decomposition. Coverage spans 1860–2024.

The price and trade indices used to construct the relative capital price proxies (p_t^{κ}), the capacity to import (M_t^{cap}), are drawn from *La República en Cifras* (Lüders et al., 2016): the real exchange rate index, the US wholesale capital price index, and the terms of trade index. All three series are rebased to 2003 = 100 for internal consistency with the capital stock series.¹²

The capital import demand imbalance combines sources from *La República en Cifras* with my data set of harmonized capital stocks through a log difference of $\ln(I_{\text{ME},t}) - \ln(M_t^{\text{cap}})$, which reconciles the supply export-side identified by the index of imports capacities (M_t^{cap}) of the external constraint, with the internal demand for foreign capital goods providing a both a more accurate variable for the regime shifting empirical strategy detailed below, while also a variant that allows me to check for robustness of my estimates.

A complementary data source for external sector variables are drawn from the World Historical Balance of Payments Database (WBOP) developed by Nieves and Piketty (2025). The WBOP data set provides annual series on global trade flows and the world balance of payments—including goods, services, income, and transfers—covering 48 countries and 9 regions over the 1800–2025

¹¹The underlying series are drawn from the Base de Datos Estadísticos (BDE) of the Banco Central de Chile, Cuentas Nacionales, sectoral income accounts, reference year 2018, nominal quarterly flows. Annual values are obtained by simple summation of quarterly observations.

¹²The terms of trade index extends back to 1810; the real exchange rate and US wholesale price index cover 1900–2024.

period. Three series are used for Chile: the ratio of manufactured goods imports to GDP (a proxy for capital goods import dependence), net foreign income outflows as a share of GDP (a proxy for profit remittance drain), and the net current account balance as a share of GDP (an external balance indicator).¹³

2.4.2 Stylized Facts of Accumulation, Distribution, and External Constraints

This subsection presents stylized facts and descriptive statistics of my data set for the United States (1931–2024) and Chile (1940–2010). First, output data correspond to the Non Financial Corporate Sector, while capital stocks are measured on a gross basis following my implementation of [Shaikh \(2016\)](#) *Generalized Perpetual Inventory Method* (GPIM) presented above. Second, output and capital stocks drift together over the long run in both economies, motivating a cointegrating approach. Third, the output–capital ratio drifts through time in both countries: in the United States, capital accumulates faster than output on average and the ratio presents a declining long-run trend, suggesting that the long-run relation might be governed by a transformation elasticity below unity; in Chile, the relation is unclear, while capital composition between non-residential construction and machinery and equipment shifts discretely across historical regimes, motivating a formulation of heterogeneous capital. Fourth, the state variables that condition the transformation—the distributive state in both economies and the external bottleneck components in the periphery—shift discretely across institutional regimes, motivating the state-dependent and threshold cointegration empirical strategies presented in Section 4.3.

2.4.2.1 Stylized Facts of the U.S. Non-Financial Corporate Economy (1931–2024)

Table 2.4 provides summary statistics for the core macroeconomic and distributive variables of the U.S. non-financial corporate economy over the 94-year sample period from 1931 to 2024, capturing the long-run means and volatilities of real output, capital accumulation, functional income distribution, and the evolving composition of the capital stock. While these aggregate moments record the persistent divergence between capital and output growth and the shifting dynamics of the wage share, the historical regime shifts underlying these averages require chronological examination. The series that follow trace the secular trends, cyclical fluctuations, and structural breaks across these dimensions, providing the empirical motivation for the theoretical and econometric specifications developed in subsequent sections.

¹³All WBOP series are available at [wbop.world](#) and are also hosted in the World Inequality Database ([wid.world](#)), together with a detailed replication package including raw data sources, methods, and codes.

Table 2.4. United States Macroeconomic and Distributive Summary Statistics (1931–2024)

Variable	Mean	Std. Dev.
Log Real Output (y_t)	14.734	1.107
Log Core Capital Stock (k_t^{cap})	14.158	1.810
Log Output–Capital Ratio ($y_t - k_t^{cap}$)	0.576	0.731
Mechanization Gap ($\kappa_t = k_t^{ME} - k_t^{NRC}$)	-0.637	0.278
Annual Output Growth (Δy_t)	0.039	0.051
Annual Capital Growth (Δk_t^{cap})	0.059	0.045
Net Wage Share (ω_{t-1}^{net})	0.707	0.029
Gross Wage Share (ω_{t-1}^{gross})	0.623	0.025
IPP Capital Share (ι_{t-1})	0.084	0.043
Effective Wage Pressure (C_t)	0.647	0.041

Note: Sample window spans 1931–2024 ($N = 94$). Output y_t is non-financial corporate real gross value added. Core capital k_t^{cap} comprises machinery, equipment, and non-residential structures ($K_{ME} + K_{NRC}$). Mechanization gap $\kappa_t = k_t^{ME} - k_t^{NRC}$ measures the capital composition ratio. Net wage share ($\omega^{net} = \text{Compensation}/\text{NVA}$) and Gross wage share ($\omega^{gross} = \text{Compensation}/\text{GVA}$) are both evaluated in nominal terms. IPP capital share $\iota_{t-1} = K_{IPP}/(K_{cap} + K_{IPP})$. Effective wage pressure $C_t = \omega_{t-1}(1 - \iota_{t-1})$. Data sources: BEA NIPA Tables and Fixed Assets Accounts.

Log real non-financial corporate output (y_t) and the log of aggregate gross capital stock $k_t^{cap} = \ln(K_t^{ME} + K_t^{NRC})$ drift together along a common upward trend over the 94-year sample (Figure 2.1). From the early 1980s onward, the capital stock runs systematically above output, generating a persistent overaccumulation gap. Estimating in first differences would discard exactly this shared long-run cointegrating trajectory.

The corresponding log output–capital ratio ($y_t - k_t^{cap}$, Figure 2.2) spikes to its historical maximum under wartime mobilization (1944), falls through the early Cold War, recovers part of the decline in a Fordist hump around 1966, and declines secularly thereafter. The LOESS trend traces a downward drift from 1950 onward, steepening after 1970 and again after 2000. A ratio that falls for decades while capital accumulates provides an empirical suggestion of an over-accumulation tendency $\theta < 1$: capacity formation persistently lags capital accumulation. The balanced-growth knife-edge $\theta = 1$ does not render as a long-run tendency, once the measurement of capital stock, the dual valuation distinction, and the purpose of measurement are considered.

The mechanization gap $\kappa_t = k_t^{ME} - k_t^{NRC}$, plotted in Figure 2.3 (mean -0.637 , SD 0.278 , Table 2.4), remains depressed throughout the Great Depression and World War II (≈ -1.15), embarks on a sustained upward march across the Golden Age of Atlantic Fordism as machinery and equipment outpaced industrial structures, and peaks in the late 1990s (≈ -0.313) during the information-technology equipment boom. From 2000 onward, κ_t declines as investment shifted toward structures and commercial real estate while Intellectual Property Products absorbed intangible accumulation, leaving direct physical machinery accumulation subdued. The capital composition ratio is therefore a regime-shifting series rather than a smooth trend, motivating the heterogeneous-capital specification in which the transformation elasticity θ depends dynamically on κ_t .

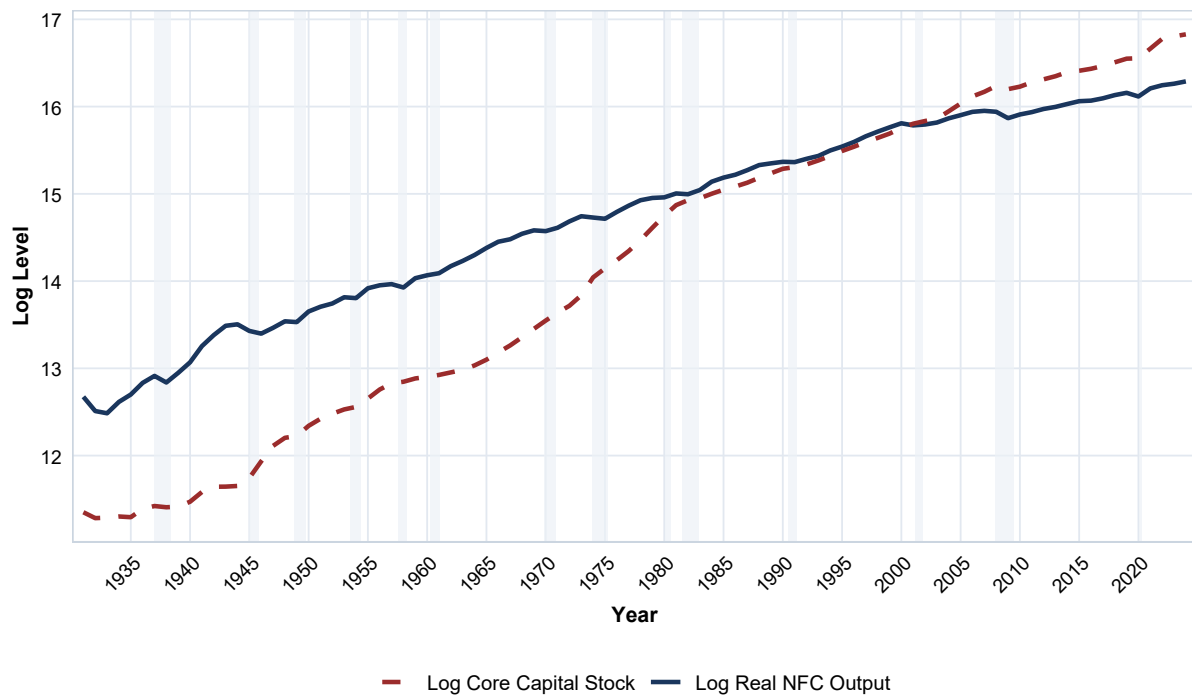
Annual growth rates of output and capital are contrasted in Figure 2.4. Over the full sample, aggregate gross capital accumulation grows faster than output on average (5.9% against 3.9% per year, Table 2.4), a long-run relation consistent with overaccumulation under an unbalanced growth closure. The two series have comparable standard deviations (0.051 against 0.045) because the pre-1950 window contains large swings in both, including capital-growth spikes around 1946 and 1974. The asymmetry that matters for estimation appears after 1950: output growth repeatedly turns negative in recessions while capital accumulation remains positive, reflecting the durability of physical

capital in operation captured by gross measurements. From the mid-1960s, the capital-growth series sits mostly above the output-growth line.

Figure 2.5 displays the set of wage share permutations evaluated across our econometric specifications. These capture three dimensions: (1) Corporate boundary cleaning to purge financial sector double-counting; (2) Value-Added denominators (Net Value Added vs. Gross Value Added); and (3) Shaikh and Tonak (1994) supervisory labor adjustments to account for variants of productive vs non-productive labor.

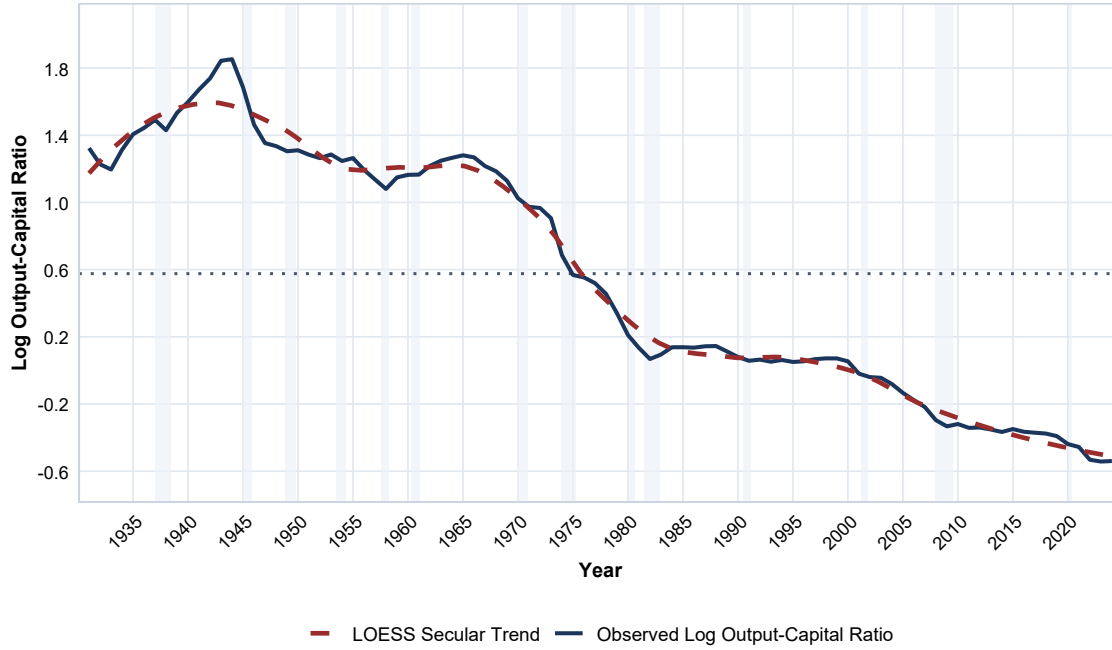
Figure 2.6 plots the corresponding Effective Wage Pressure permutations, where each wage share series is scaled by the physical capital fraction ($1 - \iota_{t-1}$). Across all specifications, effective wage pressure remains elevated under Atlantic Fordism (peaking around 1981) and trends downward secularly under post-Fordist accumulation as Intellectual Property Products capture an increasing fraction of total capital advanced ($\iota_{t-1} > 0.12$).

Figure 2.1. U.S. Non-Financial Corporate Real Output and Aggregate Gross Capital Stock (1931–2024)



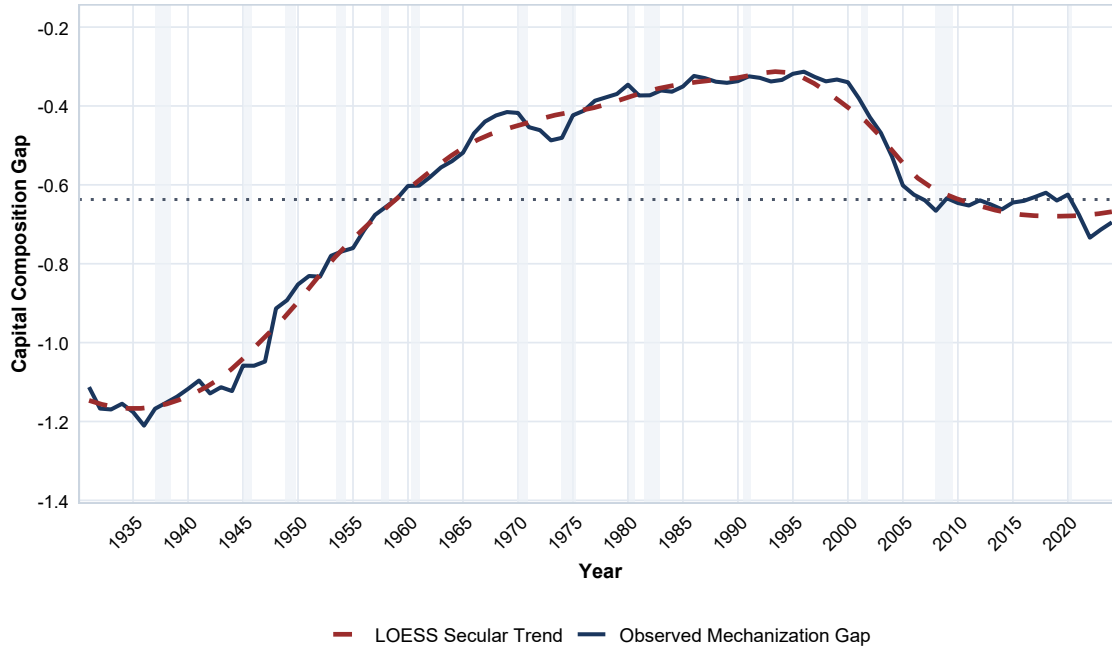
Note: Log real non-financial corporate gross value added (y_t , solid navy line) and log core capital stock ($k_t^{cap} = \ln[K_t^{ME} + K_t^{NRC}]$, dashed crimson line). Light gray shaded bands denote official NBER recession dates.

Figure 2.2. U.S. Log Output–Capital Ratio ($y_t - k_t^{cap}$)



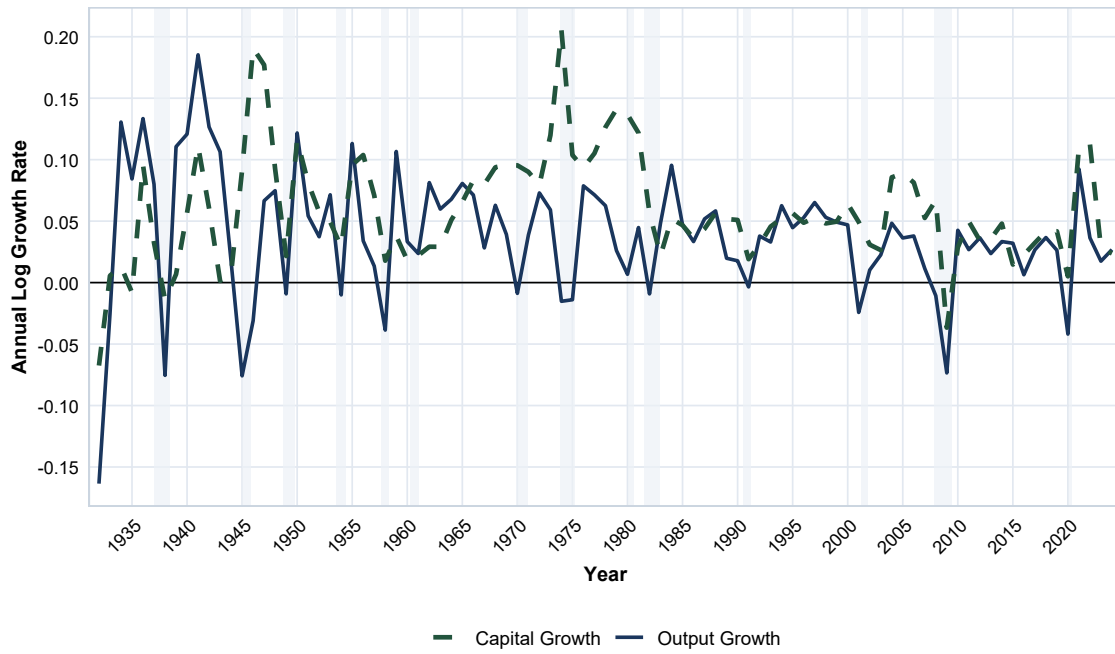
Note: Annual log difference of output and core capital stock ($y_t - k_t^{cap}$, solid navy line) relative to the full-sample mean (dotted line) and LOESS non-parametric secular trend (dashed crimson line, span = 0.35). Light gray background bands indicate NBER recessions.

Figure 2.3. U.S. Capital Composition and Mechanization Gap ($\kappa_t = k_t^{ME} - k_t^{NRC}$)



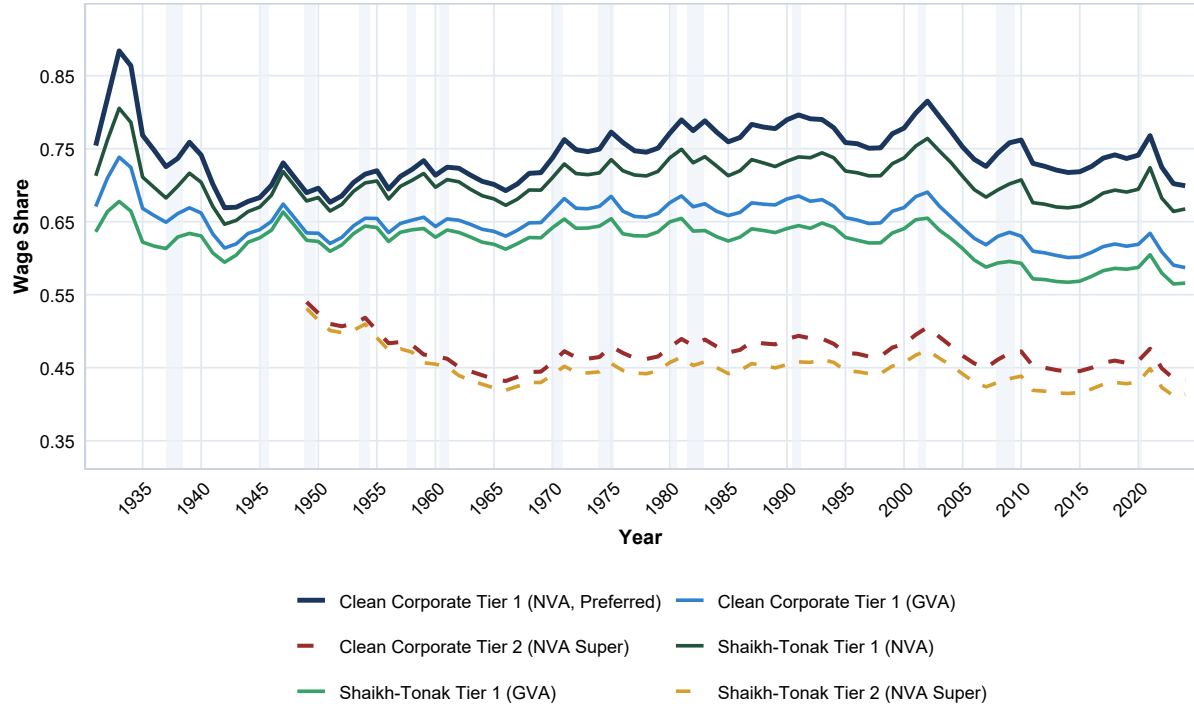
Note: Non-financial corporate mechanization gap $\kappa_t = k_t^{ME} - k_t^{NRC}$ (solid navy line) relative to the full-sample mean (-0.637 , dotted line) and LOESS non-parametric secular trend (dashed crimson line, span = 0.35). Light gray background bands indicate NBER recessions. Data sourced from BEA Fixed Assets Accounts.

Figure 2.4. U.S. Annual Growth Rates of Real Output and Core Capital Accumulation (1931–2024)



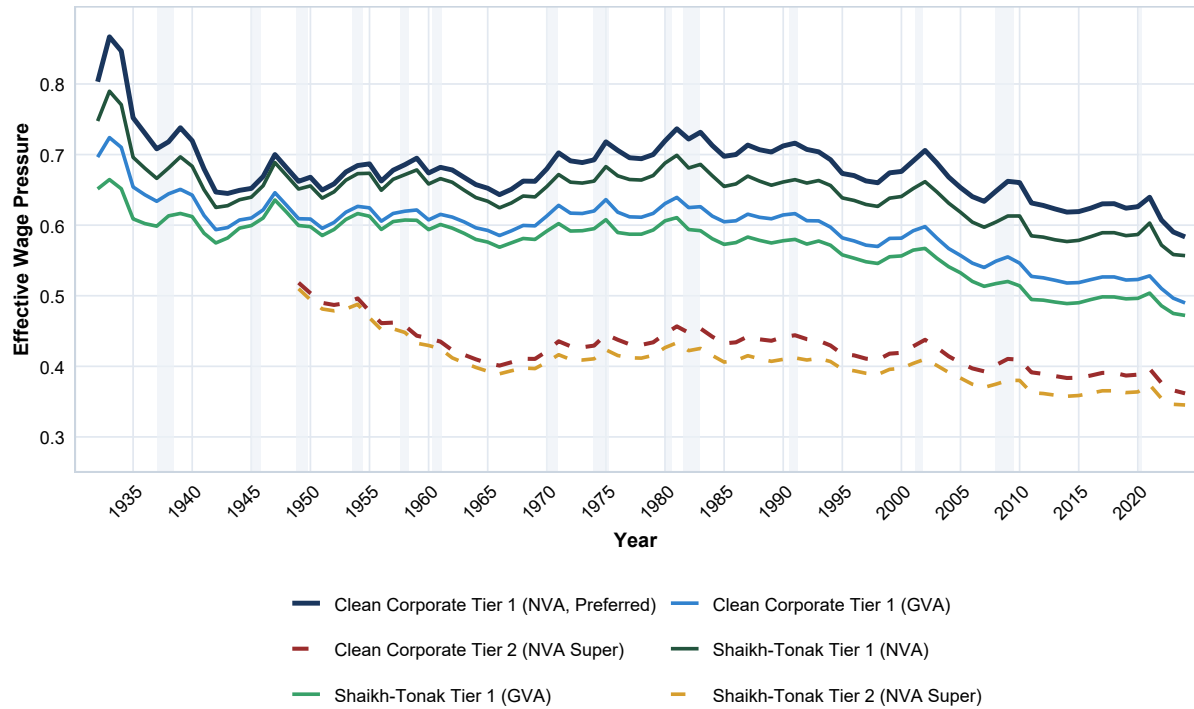
Note: Annual growth rates of non-financial corporate real gross value added (Δy_t , solid navy line) and core physical capital stock (Δk_t^{cap} , dashed crimson line). Light gray background bands denote official NBER recession dates.

Figure 2.5. U.S. Wage Share Distributive Measures (1931–2024)



Note: Wage share permutations, value-added bases, and supervisory adjustments. Light gray background bands indicate NBER recessions.

Figure 2.6. U.S. Effective Wage Pressure Measures (Scaled by IPP Capital Share, $C_t = \omega_{t-1}(1 - \iota_{t-1})$)



Note: Effective wage pressure permutations $C_t = \omega_{t-1}(1 - \iota_{t-1})$ Light gray background bands indicate NBER recessions.

2.4.2.2 Stylized Facts of Peripheral Accumulation in Chile (1940–2010)

Table 2.5 presents summary statistics for the core macroeconomic, distributive, and external variables of the Chilean economy over the 1940–2010 sample, capturing the long-run means and volatilities of real output, capital accumulation, income distribution, and the balance of payments. Because these aggregate moments obscure the pronounced historical regime shifts characteristic of the periphery, the series that follow trace the structural breaks, distributive fractures, and external constraint variables that motivate the heterogeneous-capital analytical framework presented in Section 3.3 and the threshold empirical strategy developed below.

Table 2.5. Chile Descriptive Statistics (1940–2010)

Variable	Mean	Std. Dev.
Log Real GDP (y_t)	16.625	0.776
Log Core Capital Stock (k_t^{cap})	18.419	0.899
Log Output–Capital Ratio ($y_t - k_t^{cap}$)	-1.793	0.169
Capital Composition Gap (κ_t)	-1.106	0.211
Gross Wage Share (ω_t^{gross})	0.501	0.092
Net Wage Share (ω_t^{net})	0.802	0.121
Central Bank Solvency Ratio (S_t)	2.102	2.201
Log Terms of Trade ($\ln ToT_t$)	0.030	0.263
Log Real Exports ($\ln X_{FOB,t}$)	14.753	1.116
Log Machinery Investment ($\ln I_t^{ME}$)	13.307	1.335

Note: Sample window spans 1940–2010 ($N = 71$). Real GDP and capital stock components sourced from Díaz, Lüders and Wagner (2016) and Pérez and Eyzaguirre (2021). Core capital stock is $K_t^{cap} = K_t^{ME} + K_t^{NRC}$. Capital composition gap $\kappa_t = k_t^{ME} - k_t^{NRC}$. Both Gross wage share (nominal compensation / nominal GDP) and Net wage share (nominal compensation / nominal NVA) are evaluated in nominal terms. Structural threshold $\hat{\gamma} = 0.425$ separates binding foreign exchange bottleneck regimes.

Log real GDP (y_t) and aggregate gross core capital stock ($k_t^{cap} = \ln[K_t^{ME} + K_t^{NRC}]$) for Chile (1940–2010) are plotted on the same 2003 CLP price basis in Figure 2.7. Over seven decades, output and capital drift together along a shared upward trajectory, motivating the cointegrating specification. Dotted vertical lines mark the 1973 military coup and the 1990 post-dictatorship period.

The corresponding log output–capital ratio ($y_t - k_t^{cap}$) for Chile is plotted in Figure 2.8. Unlike the continuous secular decline in the center, the peripheral output–capital ratio exhibits pronounced regime shifts: rising during the initial state-led ISI phase (1940–1955), collapsing during the 1972–1975 breakdown and the 1982 debt crisis, and settling into an unconstrained expansion during the 1990s commodity boom.

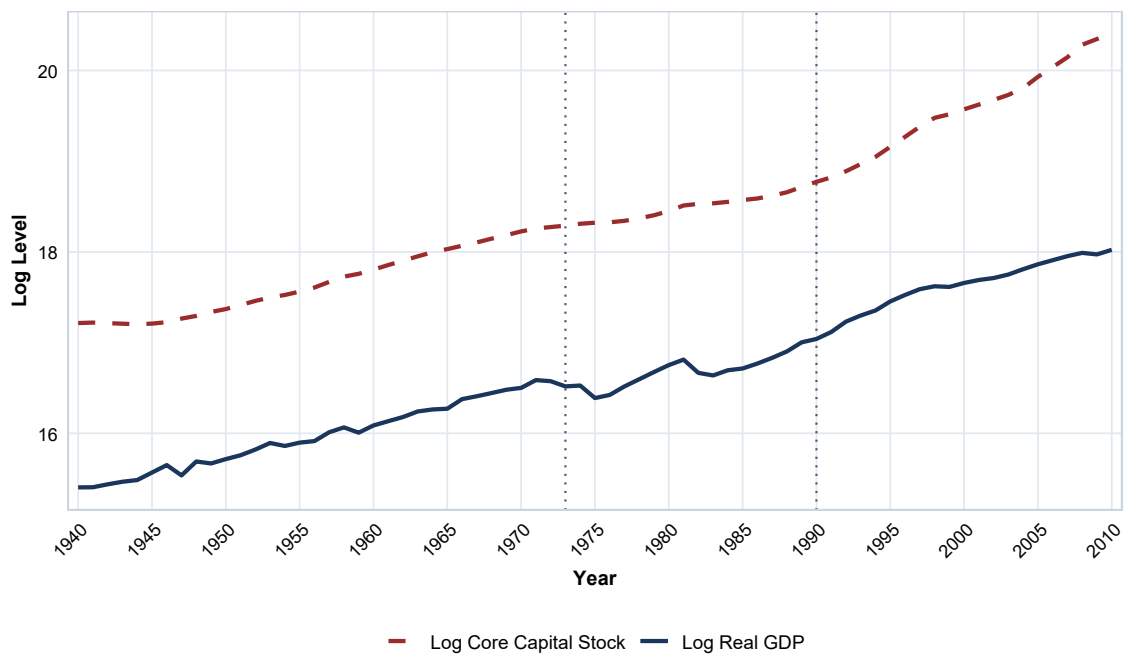
The mechanization gap $\kappa_t = k_t^{ME} - k_t^{NRC}$, plotted in Figure 2.9 (mean -1.106 , SD 0.211 , Table 2.5), widens from the late 1940s through state-led ISI to 1972, stagnates under the early neoliberal regime with a debt-boom hump at 1981 and a post-crisis trough in the late 1980s, and rises sharply after redemocratization, when export-oriented mining pulled machinery accumulation ahead of structures. The gap is therefore a regime-shifting series rather than a smooth trend, which is what motivates the heterogeneous-capital specification in which θ depends on κ_t .

Figure 2.10 documents the distributive side, drawn from the revised real-wage dataset of Astorga (2017, 2023) spliced with the national accounts. The gross wage share (mean 0.501 , SD 0.092) peaks at 0.69 in 1972 under Unidad Popular redistribution, collapses to 0.40 by 1975 after the military coup, recovers partially during the debt boom that ends in 1982, and stabilizes under the

post-dictatorship democratic regime. The net wage share $\omega_t^{\text{net}} = \omega_t^{\text{gross}} / (1 - \text{CFC_ratio_nom}_t)$ (mean 0.768, SD 0.117, Table 2.5) sits systematically above the gross share due to consumption of fixed capital; both series follow the same structural fractures. Two empirical facts matter for estimation: the 1972–1975 swing is the largest distributive shock in the sample, and the post-1990 wage-share stabilizes while the external accounts are mostly slack, which is what identifies the unconstrained mechanization response.

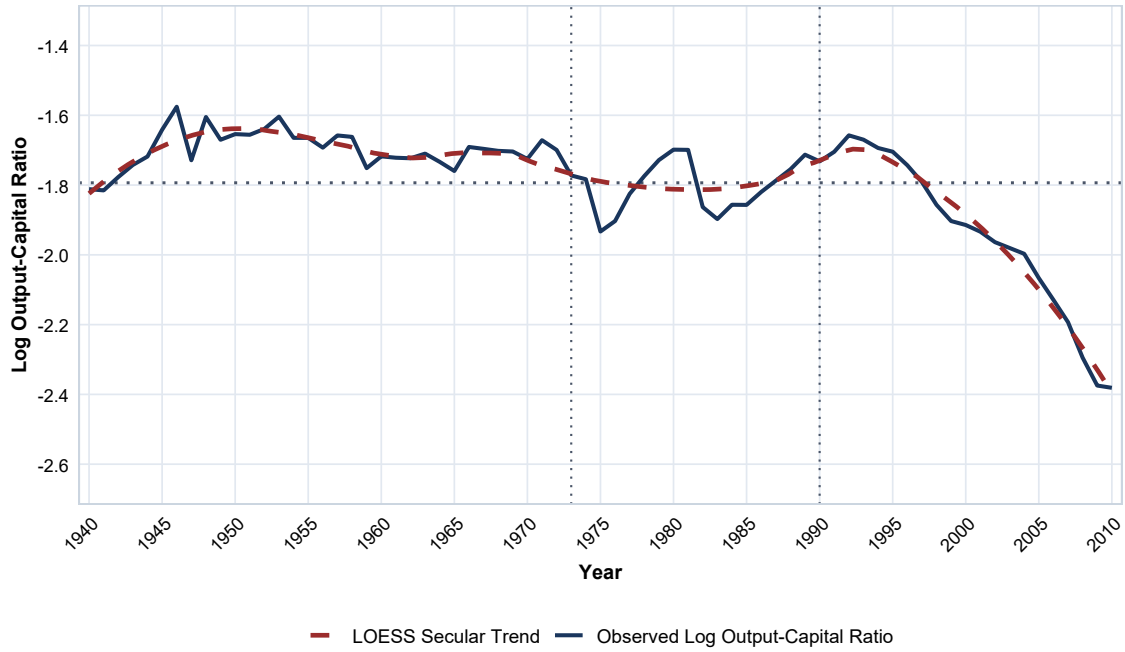
Figure 2.11 disaggregates the external bottleneck into its components. The solvency ratio (Panel a), central bank reserves relative to base money, is flat and low until 1975, jumps after the debt crisis as the Bank accumulated reserves, and declines again after the mid-2000s. The terms of trade (Panel b) deteriorate through the 1970s and 1980s and recover only with the commodity super-cycle. Real exports (Panel c) grow smoothly, so the binding episodes are driven by reserve solvency and import demand rather than by export failure; the import imbalance series follows the World Historical Balance of Payments database of Nieves and Piketty (2025). Machinery investment (Panel d) grows in bursts separated by the crisis dates, consistent with mechanization that is interrupted rather than smoothly accumulated. Section 4.5 assembles these components into the composite index through a grid search, and estimates the threshold at which the external constraint becomes binding and counteracts the mechanization bias productivity gain.

Figure 2.7. Chilean Log Real GDP and Aggregate Gross Core Capital Stock (1940–2010)



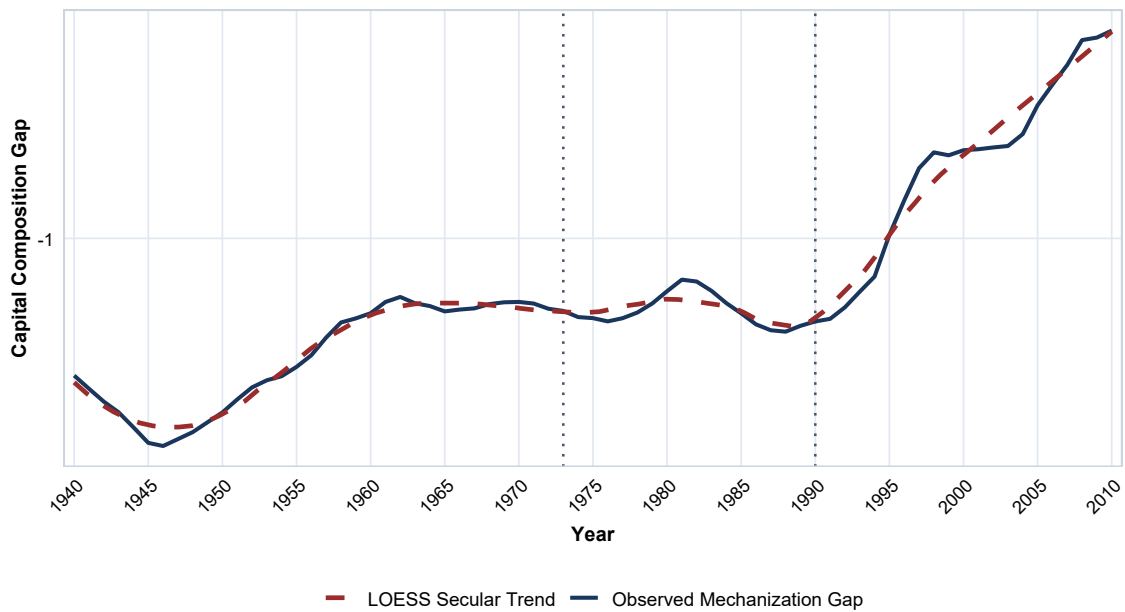
Note: Log real GDP (y_t , solid navy line) and log aggregate gross core capital stock ($k_t^{\text{cap}} = \ln[K_t^{\text{ME}} + K_t^{\text{NRC}}]$, dashed crimson line). Vertical dotted lines mark 1973 (military coup) and 1990 (post-dictatorship).

Figure 2.8. Chile: Log Output–Capital Ratio ($y_t - k_t^{cap}$)



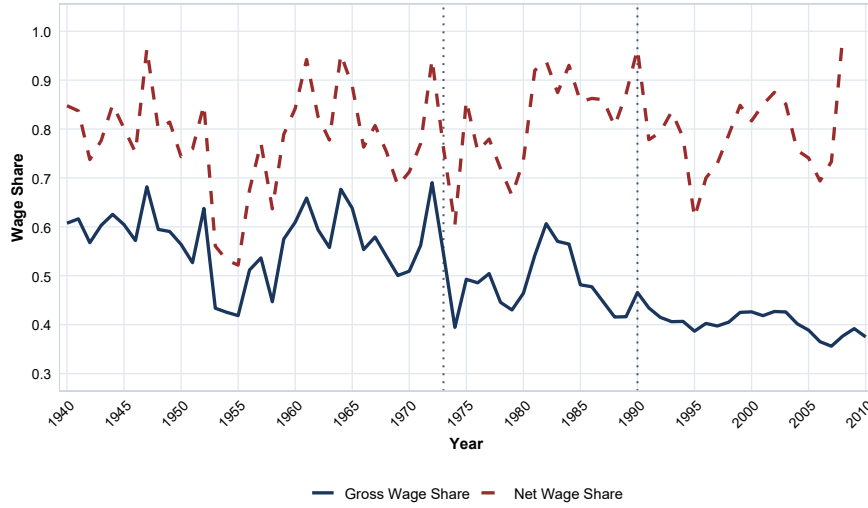
Note: Annual log output–capital ratio ($y_t - k_t^{cap}$, solid navy line) relative to the full-sample mean (dotted line) and LOESS non-parametric secular trend (dashed crimson line, span = 0.35). Vertical dotted lines mark 1973 and 1990.

Figure 2.9. Chilean Capital Composition and Mechanization Gap ($\kappa_t = k_t^{ME} - k_t^{NRC}$)



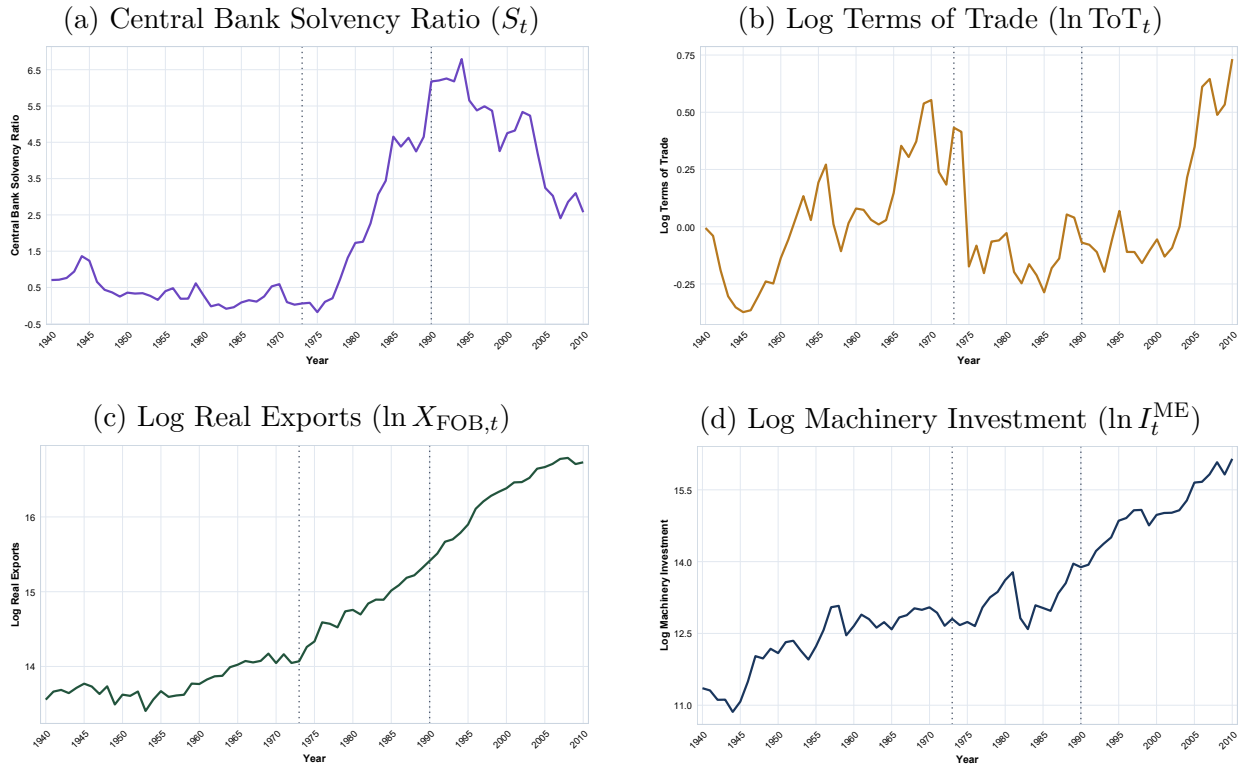
Note: Observed capital composition gap $\kappa_t = k_t^{ME} - k_t^{NRC}$ (solid navy line) alongside a LOESS non-parametric secular trend (dashed crimson line). Vertical dotted lines highlight the 1973 and 1990 political shifts.

Figure 2.10. Chilean Wage Share Trajectories (ω_t^{gross} and ω_t^{net})



Note: Gross wage share ω_t^{gross} (solid navy line) and net wage share ω_t^{net} (dashed crimson line), 1940–2010. Vertical dotted lines highlight the 1973 and 1990 political shifts.

Figure 2.11. Chilean External Bottleneck Constituent Series (1940–2010)



Note: Disaggregated empirical constituent series to estimate the balance-of-payments constraint. Panel (a) plots Central Bank foreign exchange reserve solvency (S_t , international reserves expressed in domestic currency over monetary aggregates). Panels (b) and (c) plot the log terms of trade and log real exports, which together compose log real capital import purchasing power ($\ln M_t^{\text{cap,ToT}} = \ln \text{ToT}_t + \ln X_t$). Panel (d) plots active gross machinery investment demand ($\ln I_t^{\text{ME}}$). Vertical dotted lines mark 1973 and 1990.

2.4.3 Estimation Strategy: Cointegration and IM-OLS

The cointegrating regression. The transformation elasticity θ is estimated from the cointegrating regression

$$y_t = \alpha + \theta_0 k_t + \theta_1 (k_t \times \omega_{t-1}) + u_t, \quad (2.78)$$

where $y_t = \ln Y_t$ is log real output, $k_t = \ln K_t$ is log real capital stock, and ω_{t-1} is the lagged wage share. The regressors k_t and $k_t \times \omega_{t-1}$ are integrated of order one, and the error term u_t is stationary under the null hypothesis of cointegration. The interaction term $k_t \times \omega_{t-1}$ is included in the regression to allow the transformation elasticity to vary with the distributive state: the effective elasticity is $\theta(\omega_{t-1}) = \theta_0 + \theta_1 \omega_{t-1}$.

The IM-OLS estimator. The OLS estimator of θ in (2.78) is super-consistent but its limiting distribution is contaminated by second-order bias terms arising from endogeneity of the regressors and serial correlation in the error term (Phillips and Hansen, 1990). The Integrated Modified OLS (IM-OLS) estimator of Vogelsang and Wagner (2014) removes these bias terms without requiring the choice of tuning parameters (kernel function, bandwidth, or lead-lag truncation order) at the estimation stage.

The IM-OLS estimator is based on a two-step transformation. First, the partial sum transformation is applied to (2.78):

$$S_{y,t} = S_{\alpha,t} + \theta_0 S_{k,t} + \theta_1 S_{k\omega,t} + S_{u,t}, \quad (2.79)$$

where $S_{y,t} = \sum_{j=1}^t y_j$ and analogously for the remaining terms. The partial sum transformation replaces the cross-product terms $\sum_{t=1}^T x_t u_t$, whose limits contain additive nuisance parameters, with partial-sum cross-products $\sum_{t=1}^T S_{x,t} S_{u,t}$, whose limits do not. Second, the partial-sum regression is augmented by the original regressors k_t and $k_t \times \omega_{t-1}$:

$$S_{y,t} = S_{\alpha,t} + \theta_0 S_{k,t} + \theta_1 S_{k\omega,t} + k_t \gamma_1 + (k_t \times \omega_{t-1}) \gamma_2 + S_{u,t}. \quad (2.80)$$

The augmentation absorbs the endogeneity correlation into the coefficients γ_1 and γ_2 , yielding a zero-mean Gaussian mixture limiting distribution for the IM-OLS estimator of (θ_0, θ_1) conditional on the Brownian motions driving the regressors (Vogelsang and Wagner, 2014, Theorem 2).

FWL orthogonalization of the interaction term. The interaction term $k_t \times \omega_{t-1}$ is correlated with k_t by construction. To isolate the distributive mediation effect, the interaction term is orthogonalized against k_t and the deterministic regressors using the Frisch-Waugh-Lovell (FWL) theorem (Frisch and Waugh, 1933; Lovell, 1963). Denoting the raw interaction by $\xi_t = k_t \times \omega_{t-1}$, the orthogonalized interaction $\xi_t^\perp$ is the residual from the auxiliary regression of ξ_t on k_t and the deterministic regressors. The IM-OLS estimator is then applied to the regression with $\xi_t^\perp$ replacing ξ_t . When structural break dummies are included in the deterministic regressors (see below), the FWL orthogonalization is re-computed against the augmented deterministic space.

Endogenous structural break detection. The estimation employs a two-pass endogenous break detection procedure based on Bai and Perron (2003). In the first pass, the IM-OLS estimator is applied to (2.78) without break dummies. The Bai-Perron sequential procedure is then applied to the IM-OLS residuals to detect structural breaks at unknown dates, with a minimum segment size of 15% of the sample. In the second pass, step dummies $D_t^{(b)} = \mathbf{1}\{t \geq \hat{T}_b\}$ are included in the

deterministic regressors for each detected break date $\hat{T}_b$, the FWL orthogonalization is re-computed, and the IM-OLS estimator is re-estimated.

Fixed- b inference. Standard asymptotic theory treats the bandwidth M in long-run variance estimation as vanishing relative to the sample size ($M/T \rightarrow 0$). Under this assumption, the long-run variance estimator converges to a constant, and t - and Wald statistics are compared to normal or chi-squared critical values. In finite samples with serially correlated errors and endogenous regressors, however, the bandwidth choice materially affects the sampling distribution of test statistics, and the standard approximation over-rejects (Kiefer and Vogelsang, 2005). Fixed- b asymptotic theory addresses this by treating the bandwidth ratio $b \equiv M/T$ as a fixed constant rather than a vanishing sequence. The long-run variance estimator then converges not to a constant but to a random variable whose distribution depends on the kernel function, the bandwidth ratio b , the number of integrated regressors, and the deterministic specification. Critical values that reflect these choices can be tabulated by simulation, and using them substantially reduces size distortions at the cost of only minor losses in power (Vogelsang and Wagner, 2014).

For testing q linear restrictions $H_0 : R\theta = r$, the fixed- b Wald statistic is

$$W^* = (R\tilde{\theta} - r)' [R A_{\text{IM}} \tilde{V}_{\text{IM}} A_{\text{IM}}' R']^{-1} (R\tilde{\theta} - r), \quad (2.81)$$

where $\tilde{V}_{\text{IM}}$ is the IM-OLS covariance estimator constructed from the adjusted residuals $\Delta \tilde{S}_t^{u*}$ (defined below). Under fixed- b asymptotics, W^* converges to

$$W^* \Rightarrow \frac{\chi_q^2}{Q_b(\tilde{P}^*, \tilde{P}^*)}, \quad (2.82)$$

where χ_q^2 is a chi-squared random variable with q degrees of freedom—one per restriction tested—and $Q_b(\tilde{P}^*, \tilde{P}^*)$ is a positive random variable that captures the effect of the bandwidth and kernel choices on the long-run variance estimator. The two quantities in (2.82) are independent of each other (Vogelsang and Wagner, 2014, Lemma 2, Proposition 3). This independence is the operative property: it renders the limiting distribution of W^* free of unknown nuisance parameters, so that critical values depend only on the known features of the specification—the kernel, the bandwidth ratio b , the number of integrated regressors, and the deterministic components—and can be tabulated by simulation. Critical values for the Bartlett kernel with intercept-only deterministics and up to four integrated regressors are provided in the online supplement to Vogelsang and Wagner (2014). Because the fixed- b distribution is jointly indexed by the bandwidth ratio and the number of integrated regressors, no single tabulated constant applies across the specifications estimated here; the significance thresholds reported in every results table are therefore obtained by direct Monte Carlo simulation of the exact data-generating process under test. Appendix 2.E collects the fixed critical values used for the residual-based diagnostics and states the simulation protocol for the fixed- b Wald statistics.

The adjusted residuals $\Delta \tilde{S}_t^{u*}$ used for long-run variance estimation are obtained in two steps. First, the IM-OLS residuals $\tilde{S}_{u,t}$ from (2.80) are correlated with the IM-OLS estimator $\tilde{\theta}$ even asymptotically, because the partial-sum regression contains integrated regressors. This correlation prevents the residuals from being used directly for fixed- b inference. Second, to remove this correlation, $\tilde{S}_{u,t}$ is projected on a set of constructed regressors $H_t^\perp$ (functions of the partial sums of

the deterministic components and the integrated regressors), yielding

$$\Delta \tilde{S}_t^{u*} = \Delta \tilde{S}_{u,t} - \Delta H_t^{\perp'} \hat{\varrho}, \quad (2.83)$$

where $\hat{\varrho}$ is the OLS coefficient vector from the regression of $\tilde{S}_{u,t}$ on $H_t^{\perp}$. The resulting adjusted residuals are asymptotically independent of $\tilde{\theta}$, which is the condition required for the numerator and denominator in (2.82) to be independent and for the critical values to be pivotal.

Cointegration testing. Two complementary tests are employed to assess the presence of a cointegrating relationship. First, the null hypothesis of cointegration is tested using a Shin (1994) type KPSS statistic applied to the IM-OLS residuals (Wagner, 2023; Wagner and Hong, 2016):

$$C_T = \frac{1}{T^2 \hat{\omega}_{u,v}^2} \sum_{t=1}^T \left(\sum_{j=1}^t \hat{u}_j \right)^2, \quad (2.84)$$

where $\hat{\omega}_{u,v}^2$ is a consistent estimator of the conditional long-run variance $\omega_{u,v}^2 = \Omega_{uu} - \Omega_{uv} \Omega_{vv}^{-1} \Omega_{vu}$. Under the null hypothesis of cointegration, C_T converges to a functional of standard Brownian motions whose critical values depend on the number of integrated regressors included in the regression and the deterministic specification (Wagner, 2013, 2023). Second, the null hypothesis of no cointegration (spurious regression) is tested using the augmented Dickey-Fuller statistic applied to the OLS residuals with MacKinnon (2010) finite-sample critical values, with the lag order selected by the Akaike information criterion. Rejection of the null hypothesis of no cointegration combined with non-rejection of the null hypothesis of cointegration constitutes evidence for a cointegrating relationship.

2.4.3.1 Integration Order and Stochastic Properties: United States

A In Cointegrating Multivariate Polynomial Regressions (CMPR) estimated via Integrated Modified OLS (IM-OLS) (Vogelsang and Wagner, 2014, 2024), valid asymptotic inference and pivotal fixed- b Wald distributions require all stochastic regressors to be integrated of order one ($I(1)$). When non-linear interaction terms ($k_t \times \omega_{t-1}$, $k_t \times \omega_{t-1} \times (1 - \iota_{t-1})$) enter the long-run relation, establishing their stochastic integration order prior to estimation is an essential prerequisite. We evaluate the stochastic properties of U.S. series and their interactions over 1929–2024 ($T = 96$) using Augmented Dickey-Fuller (ADF), DF-GLS, Phillips-Perron (PP), Kwiatkowski-Phillips-Schmidt-Shin (KPSS), and Zivot-Andrews (ZA) tests with endogenous structural breaks. The results confirm that log real output (y_t), core private gross capital (k_t^{cap}), the corporate Intellectual Property Products (IPP) gross investment share (ι_t), and distributive wage shares (ω_i) are clean $I(1)$ processes (rejecting level stationarity at $p < 0.01$ under KPSS, and rejecting unit roots in first differences at $p < 0.01$ under ADF and DF-GLS).

B A central property concerns the capital composition ratio ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$). Both Machinery (k_t^{ME}) and Structures (k_t^{NRC}) are established $I(1)$ series. When evaluated in first differences under a standard linear KPSS specification, $\Delta \kappa_t$ displays apparent persistence ($KPSS_{\text{diff}} = 0.761, p < 0.01$). This reflects the sharp post-1973 structural deceleration in mechanization (identified by the 1974 Zivot-Andrews break) rather than an $I(2)$ process (Perron, 1989). When κ_t and capital stocks are interacted with distributive wage shares and corporate IPP investment shares ($\iota_t \times \omega_i$, $\kappa_t \times \iota_t \times \omega_i$,

$k^{\text{cap}} \times \omega_i$, $k^{\text{cap}} \times \omega_i \times \iota_t$), the bounded support of the wage and investment shares dampens the drift. All double and triple interaction terms are strictly $I(1)$ ($ADF_{\text{diff}} < -6.89, p < 0.01; KPSS_{\text{diff}} < 0.23$, failing to reject differenced stationarity), authorizing their inclusion in CMPR IM-OLS specifications.

C The Zivot-Andrews endogenous structural break tests identify transition dates that track the institutional history of U.S. capital accumulation:

1. **1972 / 1974:** Capital stocks (k^{cap}) and capital composition (κ_t) break endogenously in 1972–1974 ($t_{\text{ZA}} = -4.79$ and -3.63), marking the exhaustion of the post-WWII Fordist accumulation boom.
2. **1976:** Corporate IPP investment share (ι_t) breaks in 1976 ($t_{\text{ZA}} = -4.53$), capturing the stagflation inflection toward intangible investments.
3. **1984:** All eight IPP-wage interactions ($\iota_t \times \omega_i$) and triple interactions break strictly in 1984 ($t_{\text{ZA}} = -4.54$ to -4.92), reflecting the consolidation of the neoliberal regime under Reagan-Volcker through wage suppression and accelerated financialization.
4. **2001–2003:** Capital-wage interactions ($k^{\text{cap}} \times \omega_i$) shift endogenously in 2001–2003 ($t_{\text{ZA}} = -4.26$ to -4.85), reflecting post-dot-com labor market restructuring.

D The complete unit-root test batteries, test statistics, and critical values across all U.S. series and interaction terms are documented in Tables 2.25–2.26 of Appendix 2.F.

2.4.3.2 Integration Order and Stochastic Properties: Chile

A For the Chilean peripheral economy, the long-run estimation window spans 1943–2010 ($T = 68$). The 1943 start date isolates a stable institutional regime established by Law No. 7,200 (1942), which abandoned gold-standard convertibility and oriented Central Bank credit toward state-led import substitution alongside CORFO. Historical ClioLab external constraint and terms-of-trade series end in 2010. Unit-root tests over this horizon confirm that aggregate real GDP (y_t), Nonresidential Structures (k_t^{NR}), Machinery and Equipment (k_t^{ME}), aggregate core capital (k_t^{cap}), and distributive wage shares ($\omega_t^{\text{net}}, \omega_t^{\text{gross}}$) are $I(1)$ non-stationary processes.

B The composite Balance-of-Payments index ($Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$), which combines the normalized import coverage ratio and terms of trade, rejects the unit root null in levels ($ADF = -6.661, p < 0.01; KPSS = 0.075$). It operates as a stationary threshold state variable that partitions accumulation regimes. Furthermore, while raw nonresidential structures (k_t^{NR}) display moderate differenced persistence, FWL-orthogonalized wage interaction residuals (ξ_t^ω) and the threshold-conditioned mechanization regressor ($k_t \times \omega_{t-1} \times \mathbf{1}\{Z_t \leq 0.425\}$) are strictly $I(1)$ ($ADF_{\text{diff}} < -10.10, p < 0.01; KPSS_{\text{diff}} < 0.06$), satisfying the stochastic admissibility criteria for threshold CMPR estimation.

C The complete unit-root test batteries, test statistics, and critical values across all Chilean series and interaction terms are documented in Tables 2.27–2.28 of Appendix 2.F.

2.4.4 United States: Baseline Specification Empirical Results

2.4.4.1 Baseline Econometric Specification

I evaluate the empirical relationship between real aggregate output ($y_t = \log Y_{\text{REAL},t}^{\text{NFC}}$) and core physical productive capital stock ($k_{\text{cap},t} = \log(K_{\text{ME},t} + K_{\text{NRC},t})$) for the nonfinancial corporate business sector over the 1929–2024 period. Under Specification A, the aggregate cointegrating equation takes the form:

$$y_t = \alpha + \theta_0 k_{\text{cap},t} + \theta_1 \xi_t^\omega + \gamma' \mathbf{D}_t + u_t, \quad (2.85)$$

where ξ_t^ω denotes the Frisch-Waugh-Lovell (FWL) orthogonalized wage-capital interaction term, and $\mathbf{D}_t$ contains step-shift dummies capturing historical structural breaks identified via Bai-Perron tests. The FWL orthogonalization partials out base capital regressors and deterministic step shifts prior to estimating the interaction elasticity θ_1 , eliminating collinearity bias while preserving exact parameter identification (Vogelsang and Wagner, 2024). Two distinct interaction sets are estimated across 48 total model permutations: Set 1 incorporates the standard unconditioned wage interaction $\xi_t^\omega = \omega_{t-1} \times k_{\text{cap},t}$, whereas Set 2 incorporates the effective wage pressure interaction $\xi_t^\omega = \omega_{t-1}(1 - \iota_{t-1}) \times k_{\text{cap},t}$, where ι_{t-1} represents the lagged share of intellectual property products in total capital stock. All parameters are estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2024), with non-standard fixed- b critical values generated from a 100,000-pass Monte Carlo simulation suite matching the exact sample sizes ($T = 94$ for 1931–2024 standard series; $T = 77$ for 1948–2024 supervisory-adjusted series). Because capital stocks are constructed via Perpetual Inventory (GPIM) methods, classical right-hand-side measurement noise in $k_{\text{cap},t}$ introduces standard attenuation bias that pulls $\hat{\theta}_0$ downward toward zero, reinforcing that the structural transformation elasticity is bounded strictly below unity rather than an artifact of measurement error.

Theoretical argument for the NVA basis of distributive conflict measure. Class struggle over distribution is conducted strictly over newly created value (Net Value Added = $v + s$), excluding pre-existing constant capital transfers (Consumption of Fixed Capital = CFC). Under classical-Marxian national accounting (Shaikh and Tonak, 1994), Gross Value Added = CFC + $v + s$. Because physical depreciation (CFC) represents constant capital transferred to output by concrete labor, it lies outside zero-sum distributive conflict; rising CFC/GVA during intensive mechanization mechanically depresses GVA wage shares even when the rate of surplus-value extraction (s/v) is invariant. The NVA basis ($\omega_{\text{NVA}} = v/(v + s)$) purges this physical depreciation distortion, isolating the balance of class power. Standard NIPA compensation includes managerial and executive salaries, which function as distributions of surplus value rather than variable labor costs. The Shaikh-Tonak Tier 2 ($\omega_{\text{ST,Tier2}} = v_{\text{prod}}/\text{NFC_NVA}$) and Clean Corporate Tier 1 and Tier 2 measures strip supervisory overhead (s_{sup}) and purge financial sector double-counting ($\text{FIN_NOS}^{\text{implied}} + \text{NFC_NET_INT}$), recovering the variable capital (v) that drives corporate cost-minimization decisions. Because unadjusted Shaikh-Tonak Tier 1 NVA ($\text{NFC_COMP}/\text{NFC_NVA}$) is mathematically identical to the NFC Baseline NVA share, Clean Corporate and Tier 2 supervisory adjustments constitute the primary non-redundant distributive specifications.

Institutional and physical mechanism of scaling by $(1 - \iota_{t-1})$. The scaling factor $(1 - \iota_{t-1}) = K_{\text{cap},t-1}/(K_{\text{cap},t-1} + K_{\text{IPP},t-1})$ isolates the physical productive asset share (machinery, equipment, and structures) from total fixed capital. Under the induced mechanization mechanism, when

labor commands a larger share of newly created value ($\omega_{t-1} \uparrow$), squeezed profit margins compel firms toward defensive mechanization, substituting capital equipment for living labor on the shop floor. However, capital expenditure allocated to Intellectual Property Products (K_{IPP})—software licenses, corporate patents, brand architecture, and legal protections—operates as market enclosure rather than physical output-yielding machinery. If investment shifts toward intangibles ($\iota \uparrow$), those funds bypass shop-floor labor substitution. In the Fordist era (1931–1973), IPP capital was negligible ($\iota \approx 0.02\text{--}0.05 \implies 1 - \iota \approx 0.95$), allowing raw wage pressure to translate directly into physical machinery expansion. In the digital Post-Fordist era (1980–2024), IPP capital surged ($\iota > 0.28 \implies 1 - \iota < 0.72$), diverting substantial surplus into software and legal enclosures. Unadjusted wage interactions ($\omega_{t-1} \times k_{cap}$ in Set 1) fail across the full sample because they assume an identical physical capacity yield across regimes; scaling by $(1 - \iota_{t-1})$ purges this intangible diversion, capturing the effective wage pressure that drives shop-floor labor substitution.

Cointegration verdict across the test triad. Cointegration in Specification A is evaluated across three complementary tests: the Engle-Granger residual ADF test (t_{ADF}), the Shin (1994) LM test (C_{Shin}), and the Vogelsang-Wagner Type 2 Wald statistic ($S_{Type\ 2}$). Across all 48 specifications, the Shin LM test ($C_{Shin} \leq 0.04$, $p > 0.10$) and the Vogelsang-Wagner Type 2 Wald statistic ($S_{Type\ 2} \leq 13.18$, well below the 90th percentile fixed- b critical value threshold of 52.6) fail to reject the null hypothesis of cointegration (H_0 : Cointegration). When Bai-Perron step shifts are included, the Engle-Granger residual ADF test rejects the null hypothesis of no cointegration ($t_{ADF} = -4.07$, $p < 0.05$ in Column 4), establishing that aggregate real output and private core capital form a genuine, non-spurious long-run cointegrating relationship.

Set 1 results: failure of standard unadjusted wage interactions. The Set 1 estimation results in Tables 2.11 and 2.12 (Appendix 2.B) reveal two key patterns. First, the baseline capital transformation elasticity $\hat{\theta}_0$ is positive, highly stable, and statistically significant across all specifications (0.55–0.61, $p < 0.01$), establishing that private core capital ($K_{cap} = K_{ME} + K_{NRC}$) yields a long-run output elasticity of approximately 0.58. Second, the unadjusted standard wage interaction $\hat{\theta}_1$ fails to achieve statistical significance in every Set 1 model, with point estimates ranging from -0.02 ($t = -0.16$) in NFC Baseline GVA/NVA to -0.20 ($t = -1.95$) in Clean Corporate Tier 2 NVA. Unconditioned wage shares fail to capture the induced accumulation channel because raw wage measures conflate unproductive managerial overhead with productive variable labor and ignore the structural allocation of investment toward intellectual property products.

Set 2 results: effective wage pressure and defensive mechanization. Table 2.6 presents the main-text Set 2 estimates for the Net Value Added basis, where wage pressure is scaled by productive capital accumulation $(1 - \iota_{t-1})$. When distributive conflict is evaluated over newly created net value and purged of supervisory overhead, effective wage pressure generates a statistically significant negative interaction elasticity $\hat{\theta}_1$. In the Clean Corporate Tier 1 NVA model with break shifts (Column 4), $\hat{\theta}_1 = -0.30$ ($t = -5.00$, $p < 0.05$). In the Clean Corporate Tier 2 NVA specification (Column 6), the coefficient reaches $\hat{\theta}_1 = -0.49$ ($t = -3.77$, $p < 0.05$), while in the Shaikh-Tonak Tier 2 specification (Column 8), $\hat{\theta}_1 = -0.35$ ($t = -5.00$, $p < 0.05$).

The negative sign of $\hat{\theta}_1$ reflects the induced mechanization mechanism under Okishio’s market anarchy. When productive labor commands a larger share of newly created value ($\omega_{t-1} \uparrow$), squeezed profit margins compel firms toward defensive mechanization, substituting capital equipment for living

labor to contain unit labor costs. Because this substitution is driven by cost containment rather than autonomous technological expansion, it raises the technical composition of capital without generating a proportional expansion in productive capacity. High effective wage pressure therefore depresses the marginal capacity yield of capital ($\theta(\omega_{t-1}) = \theta_0 + \theta_1\omega_{t-1}(1 - \iota_{t-1}) < \theta_0$), constraining structural potential output Y_t^p and driving capacity utilization dynamics across historical Fordist and Post-Fordist regimes.

Robustness across structural break specifications. Controlling for Bai-Perron structural break step shifts ($\mathbf{D}_t$) absorbs historical level shifts—such as the 1944 wartime expansion, the 1967 profit squeeze, and the 1982 neoliberal transition—while enhancing cointegration test performance. In the Clean Corporate Tier 1 model, controlling for step shifts increases the magnitude of $\hat{\theta}_1$ from -0.04 ($t = -0.50$, insignificant) to -0.30 ($t = -5.00$, $p < 0.05$), while achieving strong residual cointegration ($t_{\text{ADF}} = -4.07$, $p < 0.05$). The conditioning effect of distributive pressure on capital transformation is thus a structural feature of U.S. capital accumulation rather than an artifact of unmodeled regime shifts.

Pedagogical counterfactuals: restriction and no-dummy breakdowns.

1. **The Restriction Counterfactual ($\theta = 1$):** Mainstream production-function approaches and statistical filters impose balanced growth ($\theta = 1$) a priori, assuming capacity expands one-for-one with capital accumulation. Under this restriction, the rapid post-war accumulation of private capital forces the estimated potential capacity path $Y_t^p = \hat{B}K_t$ upward at an unsustainable rate. Consequently, the derived capacity utilization series $\mu_t = Y_t/Y_t^p$ suffers a permanent collapse, falling to economically implausible levels below 60% by the 2010s. Estimating θ unconstrained recovers $\hat{\theta} \approx 0.58$, confirming that physical capacity expands at roughly half the pace of capital accumulation and generating a stationary, cyclical utilization trajectory (80%–95%) that oscillates around historical business cycle benchmarks.
2. **The No-Dummy Counterfactual (Spurious Regression):** Estimating the cointegrating equation without institutional step-shift controls yields non-stationary residuals ($t_{\text{ADF}} = -1.84$, failing cointegration). Omitting the institutional breaks conflates long-run parameter stability with historical regime shifts. Incorporating endogenously dated step dummies for the 1944 wartime expansion, the 1973/1974 Fordist exhaustion, and the 1982/1984 Volcker-Reagan neoliberal transition restores residual stationarity ($t_{\text{ADF}} = -4.07^{**}$, $p < 0.05$), demonstrating that historical class conflict and institutional periodization actively regulate long-run econometric cointegration.

2.4.4.2 Baseline Transformation Elasticity and Capacity Utilization Estimates

The model presented in table 2.6 column (4) yields a base capital elasticity of $\hat{\theta}_0 = 0.58$ ($SE = 0.02$, $p < 0.01$) and an effective wage interaction coefficient of $\hat{\theta}_1 = -0.30$ ($SE = 0.06$, $p < 0.05$). The state-dependent transformation elasticity then evolves over the long run as:

$$\theta_t(\omega_{t-1}, \iota_{t-1}) = 0.58 - 0.30 \cdot \omega_{t-1}(1 - \iota_{t-1}), \quad (2.86)$$

where ω_{t-1} is the lagged Clean Corporate Tier 1 wage share calculated on a net value added basis, scaled by $1 - \iota_{t-1}$, which corresponds to the share of investment in tangible assets, where ι refers

Table 2.6. Specification A, Set 2: Effective Wage Interaction (NVA Basis)

	NFC NVA Baseline		Clean Corp Tier 1		Clean Corp Tier 2		ST Tier 2 NVA	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\theta}_0$ (Capital Elasticity)	0.58*** (0.02)	0.56*** (0.06)	0.58*** (0.02)	0.65*** (0.06)	0.57 (0.02)	0.56 (0.08)	0.57 (0.02)	0.55 (0.06)
$\hat{\theta}_1$ (Effective Wage Pressure)	0.00 (0.10)	-0.17 (0.07)	-0.04 (0.08)	-0.30** (0.06)	-0.19 (0.08)	-0.49 (0.13)	-0.21 (0.08)	-0.35 (0.07)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes
$C_{\text{Shin}} [H_0: \text{coint.}]$	0.06	0.04	0.07*	0.03	0.06	0.04	0.06	0.04
$S_{\text{Type 2}} [H_0: \text{coint.}]$	10.76	9.07	11.05	9.06	7.50	10.89	6.13	13.18
$t_{\text{ADF}} [H_0: \text{no coint.}]$	-2.32	-3.06	-2.22	-4.07**	-1.84	-2.58	-2.00	-2.72
Bandwidth ratio b	0.298	0.169	0.280	0.165	0.272	0.237	0.259	0.237
Sample Size T	94	94	94	94	76	76	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA). Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2024). Step Shifts "No": no structural break dummies; "Yes": Bai-Perron step dummies included. All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. The effective wage interaction is $\omega_{t-1}(1 - \iota_{t-1}) \times k_{\text{cap},t}$, where ι_{t-1} is the lagged IPP share of capital. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed- b Wald tables. Standard errors in parentheses.

to the share of intellectual property products (IPP) over the total magnitude of gross capital formation. The latent capacity ceiling Y_t^p follows the differential law of motion $g_{Y^p,t} = \theta_t \cdot g_{K_{\text{cap},t}}$, with $g_{K_{\text{cap},t}} = \Delta \ln(K_{ME,t} + K_{NRC,t})$ as the unbalanced growth closure, where the only source of accumulation demand that builds productive capacity is the accumulation rate of physical capital in operation measured in constant 2017 dollars. The level of productive capacity is reconstructed recursively,

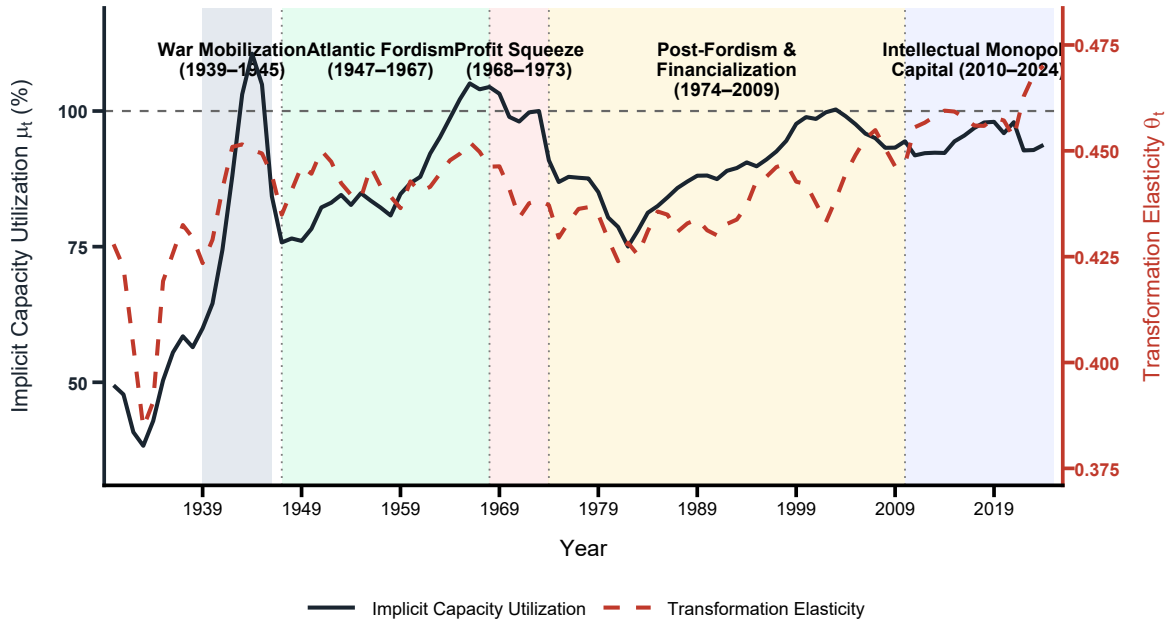
$$\ln Y_t^p = \ln Y_{t-1}^p + [0.58 - 0.30 \cdot \omega_{t-1}(1 - \iota_{t-1})] [\ln K_{\text{cap},t} - \ln K_{\text{cap},t-1}], \quad (2.87)$$

using 1973 as a pinch year, given that it records the maximum value of the FRB series on capacity utilization ($\mu_{1973} = 100\%$). Dividing realized real GDP by the estimated trajectory of productive capacity yields the rate of capacity utilization, $\mu_t \equiv Y_{\text{GDP},t}/Y_t^p \times 100$. Figure 2.12 plots both recovered variables.

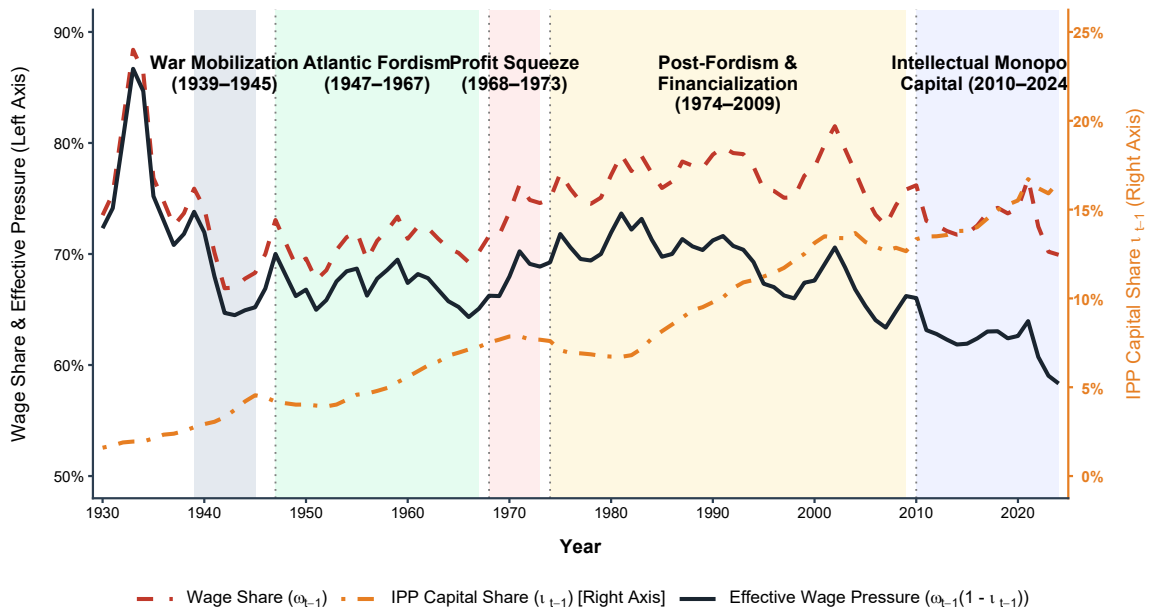
The two anchor years illustrate how distribution and capital composition govern the capacity yield. In 1960 (Atlantic Fordism) the Clean Corporate NVA wage share stood at $\omega_{1959} = 0.72$ while the IPP capital share was negligible ($\iota_{1959} = 0.03$). Effective wage pressure was $0.72 \times (1 - 0.03) = 0.698$, yielding $\theta_{1960} = 0.58 - 0.30 \times 0.698 = 0.371$: high wage pressure forced Okishio-type defensive mechanization, depressing the marginal capacity yield per dollar of physical capital advanced. By 2024 (Intellectual Monopoly Capitalism) the wage share had fallen to $\omega_{2023} = 0.70$ while the IPP capital share had risen to $\iota_{2023} = 0.28$. The intangible diversion reduced effective wage pressure to $0.70 \times (1 - 0.28) = 0.504$, raising the transformation elasticity to $\theta_{2024} = 0.58 - 0.30 \times 0.504 = 0.429$.

Figure 2.12 traces five sub-regimes, and the sequence matters more than any of them taken alone. War mobilization (1939–1945) drove production past practical capacity limits under state-directed military planning, lifting utilization to its historical peak of $\mu_{1944} = 111.4\%$ while wartime retooling raised the elasticity to roughly 0.455. Atlantic Fordism (1947–1967) then combined high wage shares ($\omega_{t-1} \approx 0.70\text{--}0.73$) with negligible intangible capital ($\iota_{t-1} < 0.05$), so labor-cost pressure

Figure 2.12. Empirical Reconstruction of Transformation Elasticity, Capacity Utilization, and Underlying Shifters in the United States (1929–2024).



(a) Implicit capacity utilization (μ_t , left axis) and transformation elasticity (θ_t , right axis).



(b) Wage share (ω_{t-1}), IPP capital share (ι_{t-1} , right axis), and effective wage pressure ($\omega_{t-1}(1 - \iota_{t-1})$).

passed directly into mechanization and wage-led demand absorbed the capacity that mechanization created. Utilization held a stable band of roughly 85% to 105% and crossed 100% during the mid-1960s expansion—the only sustained stretch in the entire sample where capacity formation and its realization moved together.

The profit squeeze (1968–1973) broke that correspondence without breaking the mechanism. Distributional conflict compressed margins and induced capital-deepening aimed at unit-cost containment rather than expansion, pulling the elasticity down to $\theta_{1972} = 0.418$ while utilization collapsed to $\mu_{1975} = 76.2\%$ and bottomed at $\mu_{1982} = 74.1\%$ under the Volcker disinflation. Post-Fordism and financialization (1974–2007) reversed the distributive pressure but not the outcome: organized-labor concessions lowered wage shares while investment moved toward software and patent enclosures ($\iota_{t-1} > 0.15$), cutting effective wage pressure from 0.72 in 1974 to 0.66 in 1997. The elasticity recovered from 0.418 to 0.450, and utilization still failed to re-anchor to its Fordist peak, staying below 95% outside transient credit-fuelled expansions. Under intellectual monopoly capitalism (2008–2024) intangible capital passed 28% of total fixed capital, pulling effective wage pressure to 0.504 and lifting the elasticity into its upper historical range ($\theta_{2024} = 0.429\text{--}0.470$) while wage suppression held back realization, leaving $\mu_{2024} = 93.8\%$. Read across the five regimes, the elasticity and utilization move together only under Fordism and move against each other everywhere after it.

The reconstruction presents empirical evidence that the long-run horizon does not yield a classical closure ($\mu^* = 100\%$), but an unbalanced growth one ($\theta^* \neq 1$), from which systemic disequilibrium arises through different patterns across distinct historical regimes. While the Fordist accumulation regime through its commitment to full employment implemented policies oriented to sustain aggregate demand and prevent productive capacity from remaining idle, during post-Fordism the expansion of productive capacity ($\theta_t \uparrow$) has decoupled from realization ($\mu_t < 1$), causing persistent idle capacity as an institutional feature.

2.4.5 Empirical Estimates Under a Heterogeneous Capital Specification

2.4.5.1 Econometric Specification of Mechanization Bias and Heterogenous Capital

Specification A treats aggregate core capital $k_{\text{cap},t} = \log(K_{\text{ME},t} + K_{\text{NRC},t})$ as a single regressor. This collapses the physical structure of the workplace into a scalar, obscuring whether wage pressure operates through the *level* of capital or through its *composition*—the relative weight of machinery and equipment (K^{ME}) versus nonresidential structures (K^{NRC}). Specification B decomposes the aggregate into two components: a structural base ($k_t^{\text{NRC}} = \log K_t^{\text{NRC}}$) and the capital composition ratio ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$), which measures the mechanization gap between equipment and structures in log-levels. The disaggregated cointegrating equation takes the form:

$$y_t = \alpha + \theta_{\text{NRC}} k_t^{\text{NRC}} + \theta_{\kappa} \kappa_t + \theta_{\omega\kappa} \xi_t^{\omega\kappa} + \theta_{\kappa p} \xi_t^{p\kappa} + \gamma' \mathbf{D}_t + u_t, \quad (2.88)$$

where $\xi_t^{\omega\kappa}$ denotes the FWL-orthogonalized effective wage–mechanization interaction and $\xi_t^{p\kappa}$ denotes the FWL-orthogonalized relative price–mechanization interaction. Both interaction terms are residualized against the two base capital regressors ($k_t^{\text{NRC}}, \kappa_t$) and deterministic step shifts ($\mathbf{D}_t$) prior to estimation. The effective wage interaction under Set 2 is $\xi_t^{\omega\kappa} = \omega_{t-1}(1 - \iota_{t-1}) \times \kappa_t$; the relative price interaction is $\xi_t^{p\kappa} = p_t^{\kappa} \times \kappa_t$, where p_t^{κ} denotes the relative asset price ratio. Non-standard fixed- b critical values are generated from a dedicated 100,000-pass Monte Carlo simulation engine that replicates the exact Specification B integration geometry—five independent $I(1)$ stochastic trends, two simultaneous $I(1) \times I(1)$ interaction products, and joint FWL orthogonalization—matching the

empirical sample sizes ($T = 94$ and $T = 77$).

Economic intuition of the decomposition. The parameter θ_{NRC} captures the long-run capacity yield of structures—factories, warehouses, commercial buildings—which embody extensive accumulation: expanding the physical envelope within which production occurs. The parameter θ_{κ} captures the additional capacity yield of machinery and equipment *above and beyond* what structures deliver. A positive θ_{κ} means that a wider mechanization gap—more equipment per dollar of structures—raises productive capacity independently of the total capital stock. The interaction $\theta_{\omega\kappa}$ asks whether effective wage pressure conditions this compositional margin. Under the Okishio-induced mechanization mechanism, rising wages force firms to substitute equipment for labor on the shop floor. If the resulting machinery expansion is defensive—driven by unit-cost containment rather than autonomous technical expansion—it raises the equipment-to-structures ratio without proportionally expanding output capacity, yielding $\theta_{\omega\kappa} < 0$.

Standard wage interactions fail at the intensive margin. Tables 2.16 and 2.17 in Appendix 2.C present the estimation results using the unadjusted wage share across all 24 specifications. The structural base elasticity $\hat{\theta}_{\text{NRC}}$ remains stable and highly significant (0.47–0.61, $p < 0.01$), confirming that nonresidential structures anchor the cointegrating relationship regardless of wage measure or break treatment. The standalone mechanization gap $\hat{\theta}_{\kappa}$ is positive in nearly all models (0.34–1.11), indicating that equipment contributes to capacity above structures. The wage–mechanization interaction $\hat{\theta}_{\omega\kappa}$, however, fails to achieve statistical significance in any $T = 94$ model, with point estimates ranging from -4.89 ($t = -1.58$) to $+7.95$ ($t = 1.33$). The standard wage interaction fails at the intensive margin for the same reason it fails at the aggregate level in Specification A: unadjusted wage shares conflate supervisory overhead with productive variable capital and ignore the post-Fordist diversion of investment toward intellectual property products.

Effective wage pressure and defensive mechanization at the intensive margin. Table 2.7 presents the main estimates using the effective wage pressure measure for the NVA basis. When effective wage pressure is scaled by $(1 - \iota_{t-1})$ and interacted with the mechanization gap κ_t , the wage–mechanization interaction $\hat{\theta}_{\omega\kappa}$ is negative and statistically significant across all eight models. In the preferred Clean Corporate Tier 1 NVA specification with Bai-Perron step shifts (Column 4), $\hat{\theta}_{\omega\kappa} = -7.93$ ($SE = 2.96$, $p < 0.05$), with a structural base elasticity of $\hat{\theta}_{\text{NRC}} = 0.60$ ($SE = 0.08$, $p < 0.01$) and a standalone mechanization gap of $\hat{\theta}_{\kappa} = 0.39$ ($SE = 0.29$). In the Shaikh-Tonak Tier 2 NVA specification without breaks (Column 7), the interaction strengthens to $\hat{\theta}_{\omega\kappa} = -17.05$ ($SE = 2.34$, $p < 0.01$), with $\hat{\theta}_{\text{NRC}} = 0.54$ ($SE = 0.01$) and $\hat{\theta}_{\kappa} = 0.33$ ($SE = 0.06$, $p < 0.01$). The Clean Corporate Tier 2 NVA models (Columns 5–6) yield $\hat{\theta}_{\omega\kappa} = -14.97$ ($SE = 2.01$, $p < 0.01$), showing the tightest standard errors in the table.

The negative sign of $\hat{\theta}_{\omega\kappa}$ reflects defensive mechanization operating at the *intensive margin* of capital composition. While Specification A showed that effective wage pressure depresses the aggregate capacity yield of total capital ($\hat{\theta}_1 = -0.30$), Specification B isolates the compositional channel: wage pressure forces firms to widen the mechanization gap—accumulating more machinery per dollar of structures on the shop floor—yet productive capacity does not expand proportionally because the substitution is driven by cost containment rather than technical complementarity between new equipment and existing plant.¹⁴

¹⁴The apparent difference in magnitude between Specification A ($\hat{\theta}_1 \approx -0.30$) and Specification B ($\hat{\theta}_{\omega\kappa} \approx -7.93$ to

Relative price interactions. The relative asset price interaction $\hat{\theta}_{\kappa p}$ is generally small and statistically insignificant. In the NFC Baseline and Clean Corporate Tier 1 models (Columns 1–4), it is statistically indistinguishable from zero. In the Shaikh-Tonak Tier 2 specification (Columns 7–8), $\hat{\theta}_{\kappa p} = -0.79$ ($SE = 0.32$, $p < 0.05$), suggesting that when equipment becomes relatively more expensive, the capacity yield of the mechanization gap declines. However, this result is fragile and does not hold across wage share measures. The Okishio mechanism operates through distributional conflict over newly created value, not through relative asset prices.

Cointegration inference. Across the cointegration triad, the Shin LM test never rejects the null of cointegration ($C_{\text{Shin}} \leq 0.03$, $p > 0.10$), and the Vogelsang-Wagner Type 2 Wald statistic remains well below critical thresholds ($S_{\text{Type 2}} \leq 15.37$). The Engle-Granger residual ADF test rejects the null of no cointegration at the 5% level in every break-adjusted model: $t_{\text{ADF}} = -4.58$ (Column 2), -4.60 (Column 4), -4.71 (Column 7), and -4.57 (Column 8). Without step shifts, the ADF test rejects at the 10% level across all $T = 94$ models ($t_{\text{ADF}} \leq -4.33$) and at the 5% level for the $T = 76$ Shaikh-Tonak Tier 2 model ($t_{\text{ADF}} = -4.71$, Column 7). These results establish that disaggregated capital composition, real output, and effective wage pressure form a stable long-run relationship.

Robustness. Including Bai-Perron step shifts absorbs historical level discontinuities while preserving the wage-mechanization interaction. In the Clean Corporate Tier 1 NVA model, controlling for breaks at 1944, 1977, and 2001 sharpens the ADF rejection from -4.34 ($p < 0.10$) to -4.60 ($p < 0.05$) while reducing the Vogelsang-Wagner statistic from 15.37 to 6.37. The wage interaction $\hat{\theta}_{\omega\kappa}$ moves from -11.08 (Column 3) to -7.93 (Column 4), with the modest attenuation indicating that the step dummies absorb part of the interaction’s raw variance while retaining significance at the 5% level.

The structural base elasticity $\hat{\theta}_{\text{NRC}}$ falls consistently within the 0.51–0.61 band across wage share measures, closely tracking the aggregate $\hat{\theta}_0 = 0.58$ from Specification A. This implies the capacity yield of structures is a stable parameter of U.S. capital accumulation. The wage interaction $\hat{\theta}_{\omega\kappa}$ grows larger in absolute value as I remove supervisory overhead from the wage share: from -10.13 in the NFC Baseline (Column 2) to -7.93 in Clean Corporate Tier 1 (Column 4) to -16.76 in Shaikh-Tonak Tier 2 (Column 8). Removing managerial salaries isolates the productive variable capital driving the firm’s cost-minimization decision, which sharpens the defensive mechanization effect.

2.4.5.2 Transformation Elasticity from Heterogeneous Capital Specification and Capacity Utilization Estimates

The transformation elasticity along the historical path. Having established the structural asset-weighted transformation elasticity θ_t^{SpecB} in Equation (2.37) (Section 3.2), I evaluate its empirical trajectory using the estimated parameters from the preferred Specification B model (Clean Corporate Tier 1 NVA, Set 2, Column 4 of Table 2.7). The structural marginal capacity yield of mechanization $MP_{\kappa,t}^{\text{str}} = \beta_{\kappa} + \theta_{\omega\kappa}C_{\text{index},t} + \theta_{\kappa p}p_t^{\kappa}$ is driven directly by the state-dependent levels of

-17.05) reflects regressor scaling, not economic inconsistency. In Specification A, θ_1 multiplies $\omega_{t-1}(1 - \iota_{t-1}) \times k_{\text{cap},t}$, where aggregate capital $k_{\text{cap},t} \approx 16$ – 17 in log-levels. In Specification B, $\theta_{\omega\kappa}$ multiplies $\omega_{t-1}(1 - \iota_{t-1}) \times \kappa_t$, where the mechanization gap $\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}} \approx 0.3$ – 0.8 . The total marginal effect (coefficient $\times$ regressor scale) is comparable across both specifications.

Table 2.7. Specification B, Set 2: Effective Wage Interaction (NVA Basis)

	NFC NVA Baseline		Clean Corp Tier 1		Clean Corp Tier 2		ST Tier 2 NVA	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\theta}_{\text{NRC}}^{\text{NRC}}$ (Structures)	0.61***	0.58***	0.60***	0.60***	0.54***	0.54***	0.54***	0.51***
	(0.02)	(0.08)	(0.02)	(0.08)	(0.01)	(0.01)	(0.01)	(0.01)
$\hat{\theta}_{\kappa}$ (Mechanization Gap κ)	0.32*	0.39	0.32*	0.39	0.31***	0.31***	0.33***	0.06
	(0.13)	(0.29)	(0.13)	(0.29)	(0.06)	(0.06)	(0.06)	(0.09)
$\hat{\theta}_{\omega\kappa}$ (Wage Interaction)	-12.77**	-10.13**	-11.08**	-7.93**	-14.97***	-14.97***	-17.05***	-16.76***
	(3.81)	(3.43)	(3.48)	(2.96)	(2.01)	(2.01)	(2.34)	(2.55)
$\hat{\theta}_{\kappa p}$ (Price Interaction)	0.18	0.92	0.69	1.41*	-0.31	-0.31	-0.79**	-0.71**
	(0.51)	(0.69)	(0.63)	(0.75)	(0.33)	(0.33)	(0.32)	(0.31)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes
$C_{\text{Shin}} [H_0: \text{coint.}]$	0.03	0.02	0.03	0.03	0.02	0.02	0.02	0.02
$S_{\text{Type 2}} [H_0: \text{coint.}]$	14.97	5.93	15.37	6.37	8.96	8.96	10.28	8.70
$t_{\text{ADF}} [H_0: \text{no coint.}]$	-4.33*	-4.58**	-4.34*	-4.60**	-4.24*	-4.24*	-4.71**	-4.57**
Bandwidth ratio b	0.185	0.173	0.193	0.181	0.208	0.208	0.208	0.208
Sample Size T	94	94	94	94	76	76	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA). Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2024). Step Shifts "No": no structural break dummies; "Yes": Bai-Perron step dummies included. All interaction terms are FWL-orthogonalized against base capital regressors ($k_t^{\text{NRC}}, \kappa_t$) and step dummies. The effective wage interaction is $\xi_t^{\omega} = \omega_{t-1}(1 - \iota_{t-1}) \times \kappa_t$, where $\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$ is the capital composition ratio (mechanization gap) and ι_{t-1} is the lagged IPP share of capital. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed- b Wald tables. Standard errors in parentheses.

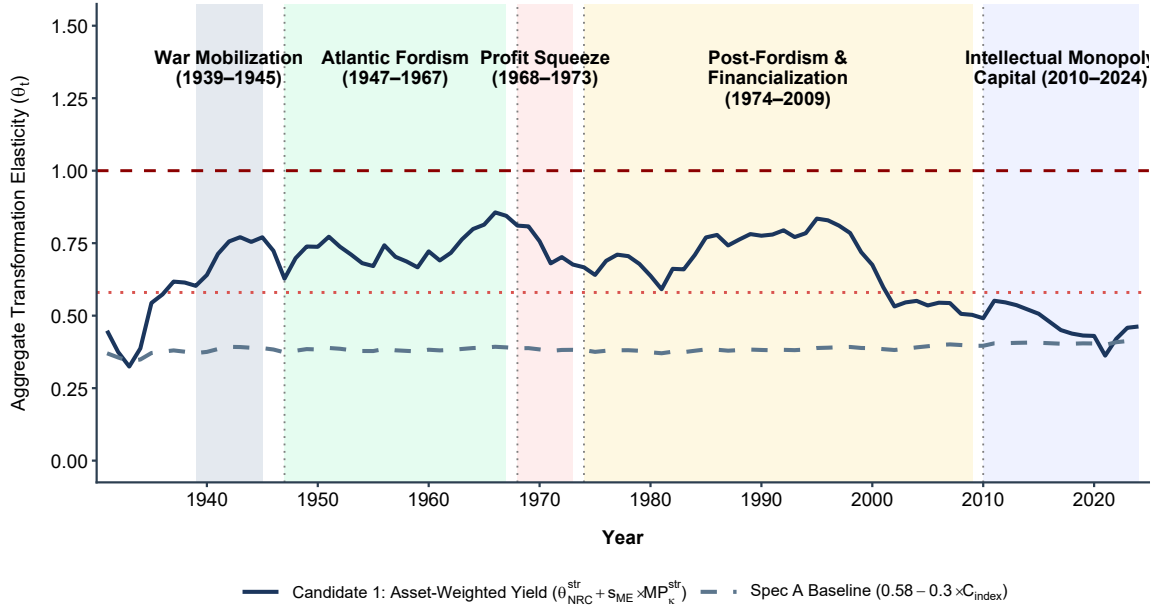
effective wage pressure ($C_{\text{index},t}$) and relative asset prices (p_t^{κ}).¹⁵

Figure 2.13 plots $\theta_t^{\text{Aggregate}}$ against the dynamic Specification A baseline ($\theta_t^{\text{SpecA}} = 0.58 - 0.30 \cdot C_{\text{index},t}$) and the Harrodian knife-edge threshold ($\theta = 1.00$) under standard structural regime shading.

Empirical comparison: Specification B disciplines the Specification A baseline. Figure 2.13 delivers two core empirical findings. First, the structural elasticity θ_t^{SpecB} tracks the dynamic Specification A path closely but swings across a wider range— $[0.32, 0.86]$ against $[0.37, 0.75]$. The two paths move together because both register the same underlying force: effective wage pressure depresses the marginal capacity yield of capital under either specification, and the estimated base elasticity of structures ($\hat{\theta}_{\text{NRC}} \approx 0.60$) sits close to Specification A's aggregate base ($\hat{\theta}_0 = 0.58$). The wider swing is the point of the disaggregation rather than a cost of it: Specification A averages the composition effect into a single scalar, while Specification B lets the equipment share move the elasticity directly, so the same distributive shock registers with more amplitude. Second, the Specification B trajectory remains strictly below the Harrodian knife-edge ($\theta < 1$) across the entire 1931–2024 sample, so aggregate capacity formation in the United States never leaves the sub-Harrodian regime.

¹⁵Initial-observation centering ($\theta_{\kappa}^{\text{str}} = 0.39$) is an anchoring convention that identifies the baseline dimensionless yield at $t = 1931$.

Figure 2.13. Aggregate Transformation Elasticity ($\theta_t^{\text{Aggregate}}$), Specification B vs. Baseline

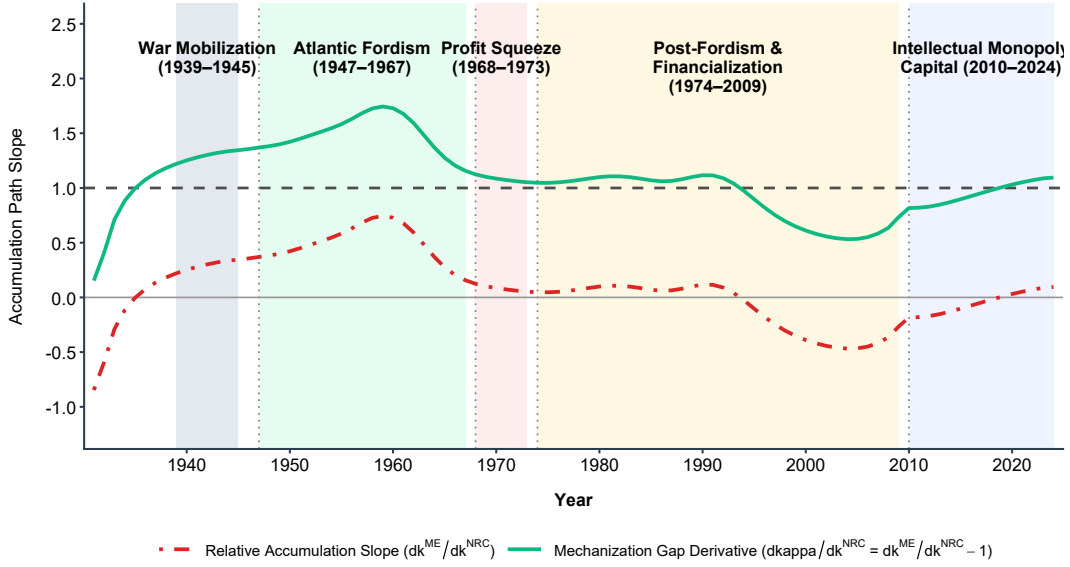


Note: Aggregate transformation elasticity $\theta_t^{\text{Aggregate}}$ (Navy line) computed according to Equation (2.37) using FWL structural parameters from the preferred Specification B model (Clean Corporate Tier 1 NVA, Set 2, Column 4 of Table 2.7). The Slate Blue line traces the dynamic Specification A baseline ($\theta_t^{\text{SpecA}} = 0.58 - 0.30 C_{\text{index},t}$). The dotted horizontal line at $\theta = 0.58$ marks the constant base from Specification A. The dashed line at $\theta = 1.00$ marks the Harrodian knife-edge. Background shading indicates structural accumulation regimes.

The disaggregated capital structure explains what Specification A's aggregate elasticity averages over. During the Fordist era ($C_{\text{index},t} > 0.65$), high effective wage pressure drove negative marginal mechanization yields ($MP_{\kappa,t}^{\text{str}} < 0$). Firms substituted equipment for labor to contain unit labor costs, but because this expansion was defensive rather than technically autonomous, it depressed the aggregate capacity yield of total capital. In the Neoliberal period ($C_{\text{index},t} < 0.50$), declining wage pressure restored $MP_{\kappa,t}^{\text{str}}$ to positive territory, pulling $\theta_t^{\text{Aggregate}}$ back toward the structural base. Specification A collapses this compositional mechanism into a scalar interaction ($\hat{\theta}_1 = -0.30$); Specification B isolates the intensive margin directly.

Empirical failure of the conventional Divisia derivative path. Figure 2.14 illustrates why the empirical total-derivative path $\theta_{\text{total},t}^{\text{Aggregate}}$ (Equation (2.36)) fails in practice. In the mid-1990s, corporate accumulation shifted away from physical machinery toward intellectual property products and financial assets, causing physical equipment growth to slow relative to structures ($dk_t^{\text{ME}}/dk_t^{\text{NRC}} < 1.00$). As $d\kappa_t/dk_t^{\text{NRC}}$ crossed zero, the Divisia denominator in Equation (2.36) approached zero, sending the raw total-derivative elasticity to spurious infinite spikes. As established in Section 3.2, this instability reflects the theoretical pathology of differentiating a value aggregate under structural accumulation shifts. The structural asset-weighted elasticity $\theta_t^{\text{Aggregate}}$ in Equation (2.37) purges these Divisia singularities, yielding a continuous, empirically well-behaved trajectory.

Figure 2.14. Empirical Accumulation Path Derivatives ($dk^{\text{ME}}/dk^{\text{NRC}}$ and $d\kappa/dk^{\text{NRC}}$)



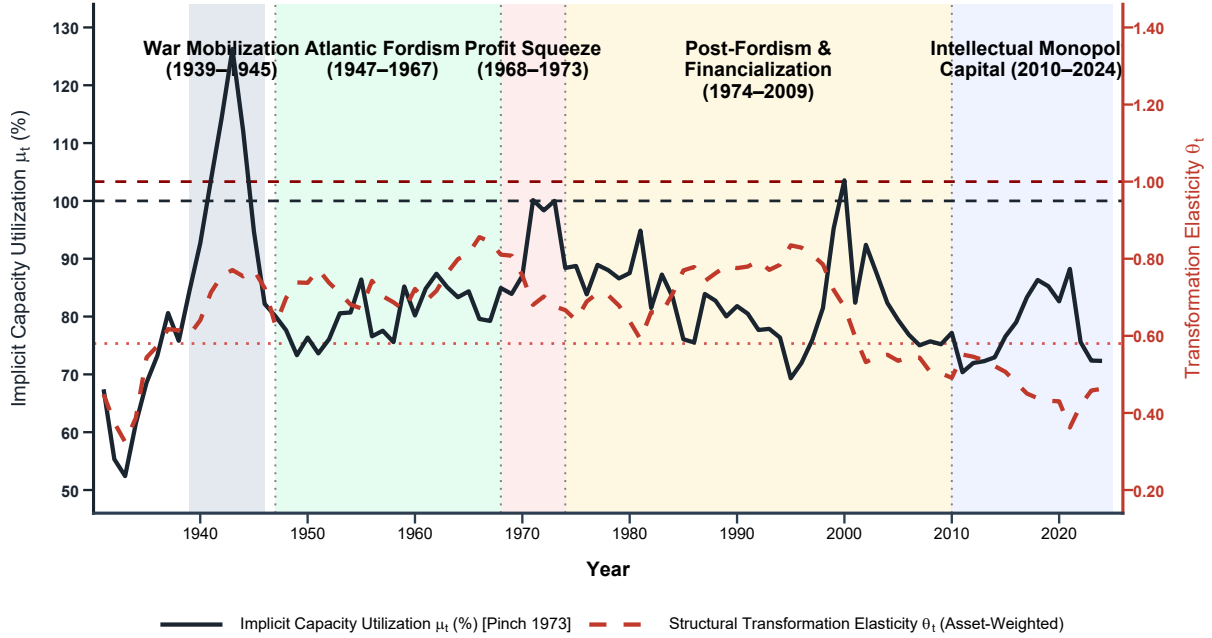
Note: Empirical accumulation path derivatives estimated via non-parametric LOESS time-domain chain rule (span = 0.35). The green solid line measures the relative accumulation slope $dk_t^{\text{ME}}/dk_t^{\text{NRC}}$. The red dot-dashed line measures the mechanization gap derivative $d\kappa_t/dk_t^{\text{NRC}} = (dk_t^{\text{ME}}/dk_t^{\text{NRC}}) - 1$. The horizontal dashed line at 1.00 indicates balanced expansion where equipment and structures grow at identical rates.

Driving Forces The inclusion of heterogeneous capital reveals a structural paradox in the post-war US economy: while the relative cheapening of machinery drove a massive shift in the physical composition of capital, the capacity-generating power of this mechanization was severely degraded by the distributive conflicts of the late 1960s and early 1970s. Figure 2.15 plots the implicit capacity utilization rate (μ_t) alongside the asset-weighted aggregate transformation elasticity (θ_t^{SpecB}). Unlike the static baseline elasticity of Specification A, the transformation elasticity here is not solely driven by the evolving mix of machinery and structures, and the macroeconomic environment mediating their installation. As the physical accumulation of machinery and equipment accelerated relative to nonresidential structures, the aggregate transformation elasticity collapsed, severing the Harrodian link between accumulation and potential output. This structural degradation is not an artifact of index-number aggregation but the direct result of shifting marginal capacity yields. To unpack the mechanics of this collapse, Figure 2.16 illustrates the two primary institutional driving forces behind this choice of technique: the secular decline in relative asset prices ($p_t^\kappa = P_t^{\text{ME}}/P_t^{\text{NRC}}$) and the episodic spikes in effective distributive pressure ($C_t = \omega_{t-1}(1 - \iota_{t-1})$).

Consider a period during the Golden Age of American Capitalism (e.g., the 1950s) where the structural yield of nonresidential structures is $\theta_{\text{NRC}}^{\text{str}} = 0.60$ and the equipment share is $s_{\text{ME}} = 0.30$. Under a stable distributive environment, if the marginal capacity yield of mechanization is $MP_\kappa = 0.40$, the transformation elasticity is approximately 0.72 ($\theta^{\text{SpecB}} = 0.60 + 0.30 \times 0.40$). As the post-war era progressed, machinery became structurally cheaper, reflected in the downward trend of p_t^κ in Figure 2.16, pushing the physical equipment share up toward 0.50. Simultaneously, the Profit Squeeze (1968–1973) generated a sharp spike in effective distributive pressure (C_t). Because the interaction term $\theta_{\omega\kappa}$ is negative, this heightened wage pressure pushed firms into a defensive choice

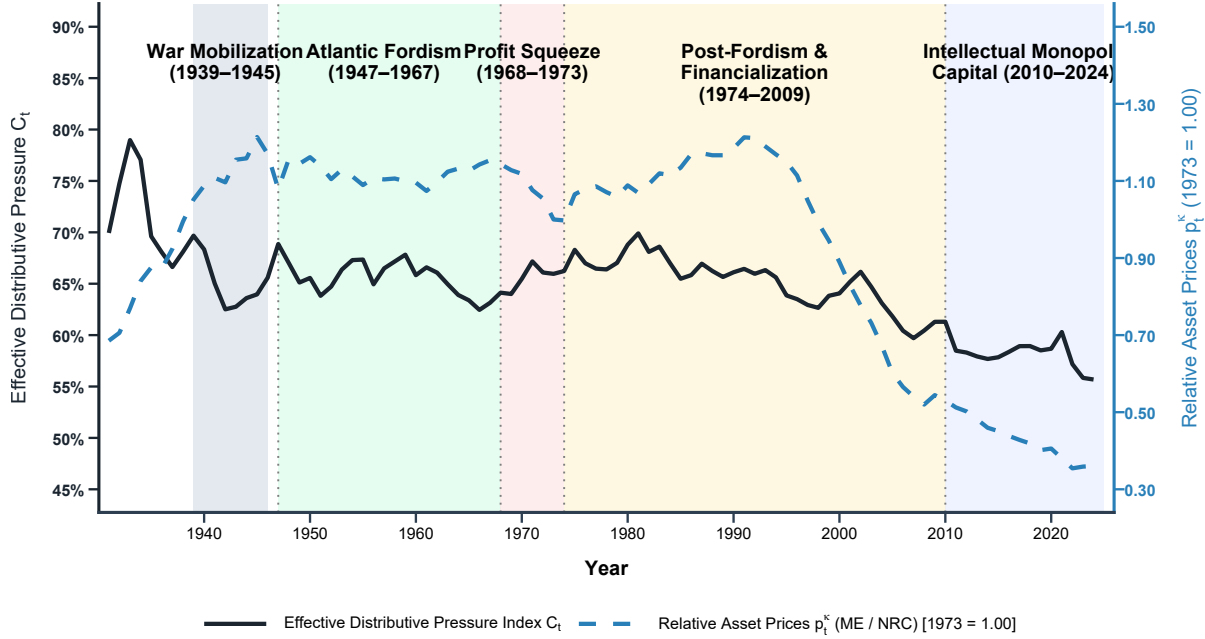
of technique, accelerating capital-using labor-saving mechanization, or Marx-biased technical change (Basu, 2022). Consequently, the marginal capacity yield of machinery dropped (say, to $MP_{\kappa} = 0.20$), leaving the aggregate elasticity in the post-Fordist era at 0.70 ($\theta^{\text{SpecB}} = 0.60 + 0.50 \times 0.20$). The arithmetic is the point: a 20-percentage-point rise in the physical share of machinery leaves the aggregate elasticity essentially where it started, because the yield per unit of that machinery fell by half. Cheapening delivered the equipment; defensive investment under falling profitability drained its capacity-generating power. This mechanism generates stagnation tendencies through overaccumulation. When firms invest in mechanization not to expand capacity but to replace increasingly expensive labor, diminishing returns to mechanization make it harder to transform capital accumulation into productive capacity, precipitating an overaccumulation crisis rooted in the political economy of full employment and labor militancy (Kalecki, 1943).

Figure 2.15. Implicit Capacity Utilization (μ_t) and Asset-Weighted Transformation Elasticity ($\theta_t^{\text{Aggregate}}$)



Note: Reconstructed capacity utilization rate anchored at the 1973 pinch year alongside the dynamic aggregate transformation elasticity derived from Specification B. The Harrodian knife-edge ($\theta = 1.00$) and the Specification A baseline ($\theta = 0.58$) are provided as reference lines.

Figure 2.16. Effective Distributive Pressure (C_t) and Relative Asset Prices (p_t^κ)



Note: The figure illustrates the institutional driving forces behind the choice of technique. The left axis tracks the effective distributive pressure to mechanize (C_t), while the right axis tracks the relative price of machinery and equipment to nonresidential structures (p_t^κ , indexed to 1973 = 1.00).

2.4.6 Chile's Empirical Evidence: Peripheral Capitalism under External Constraint

2.4.6.1 Econometric Strategy: Threshold Cointegrating Multivariate Polynomial Regression

In core economies, capital accumulation converts into productive capacity through domestic technical choice. Rising wages induce capitalists to substitute labor with machinery, expanding capacity unless surplus is diverted into unproductive accumulation (Rotta, 2018). In peripheral economies like Chile, this mechanism depends on international liquidity. Because capital equipment (K_t^{ME}) cannot be produced domestically, labor-saving technical change requires imported machinery. When foreign-exchange reserves collapse, peripheral firms hit a structural bottleneck. Rather than adjusting smoothly through domestic prices, the capacity-generating effect of capital accumulation breaks down: wage pressure cannot induce mechanization because firms cannot acquire the foreign exchange needed to import machinery.

To evaluate this state-dependent breakdown, I estimate a Cointegrating Multivariate Polynomial Regression (CMPR) using Integrated Modified OLS (IM-OLS) (Vogelsang and Wagner, 2024) with an endogenously identified threshold regime. The estimating equation models log real GDP (y_t) as a cointegrating function of non-residential construction (k_t^{NR}), the capital composition gap (κ_t), distributional conflict (dist_{t-1}), the state-dependent external constraint (D_t^γ), and relative price shocks ($\tilde{p}_t^\perp$):

$$y_t = \alpha + \theta_1 k_t^{\text{NR}} + \theta_2 \kappa_t + \theta_3 (\text{dist}_{t-1} \cdot \kappa_t) + \theta_4 (\kappa_t \cdot D_t^\gamma) + \theta_5 (\text{dist}_{t-1} \cdot \kappa_t \cdot D_t^\gamma) + \beta_p \tilde{p}_t^\perp + u_t \quad (2.89)$$

where y_t is log real GDP (2003 CLP); $k_t^{\text{NR}} = \ln K_t^{\text{NR}}$ is the log net stock of non-residential structures; $\kappa_t = \ln(K_t^{\text{ME}}/K_t^{\text{NR}})$ is the capital composition gap between intensive machinery and extensive structures; dist_{t-1} is the lagged distributive variable, evaluated across four permutations: net wage share ($\omega_{t-1}^{\text{net}}$), gross wage share ($\omega_{t-1}^{\text{gross}}$), net rate of exploitation (e_{t-1}^{net}), and gross rate of exploitation (e_{t-1}^{gross}); $D_t^\gamma = \mathbb{I}(Z_t \leq \gamma)$ is a binary regime indicator equal to 1 when the composite balance-of-payments index (Z_t) falls below threshold γ ; $\tilde{p}_t^\perp$ is the standardized relative capital price, orthogonalized against physical capital stocks via Frisch-Waugh-Lovell (FWL) regression; and u_t is a stationary $I(0)$ error.

The partial derivative of output y_t with respect to the capital composition gap κ_t yields the marginal transformation elasticity $\theta(\text{dist}, D_t)$:

$$\frac{\partial y_t}{\partial \kappa_t} = \theta_2 + \theta_3 \text{dist}_{t-1} + \theta_4 D_t^\gamma + \theta_5 (\text{dist}_{t-1} \cdot D_t^\gamma) \quad (2.90)$$

The coefficients carry direct political-economy interpretations:

1. **Extensive Construction Elasticity** (θ_1): Measures capacity generated by non-residential construction.
2. **Unconstrained Mechanization Elasticity** (θ_2): Measures capacity generated by intensive capital composition under normal liquidity ($D_t^\gamma = 0$).
3. **Domestic Induced Innovation** (θ_3): Captures wage-led induced technical change under unconstrained liquidity, where worker bargaining power forces capitalists to substitute labor with machinery.
4. **Balance-of-Payments Structural Shift** (θ_4): Measures the direct collapse in mechanization capacity when external constraints bind ($D_t^\gamma = 1$).
5. **Peripheral Distributional Inversion** (θ_5): Captures the breakdown in wage-led technical change under foreign-exchange starvation. A negative estimate ($\theta_5 < 0$) demonstrates that when reserves collapse, wage pressure cannot induce mechanization because imported machinery is rationed.

Constructing the Balance-of-Payments Threshold Index (Z_t) I define the composite balance-of-payments index $Z_t(p)$ as a convex linear combination of two standardized macroeconomic indicators:

$$Z_t(p) = p \tilde{M}_t + (1 - p) \tilde{S}_t, \quad p \in [0, 1] \quad (2.91)$$

where $\tilde{S}_t = \text{scale}(F_t/M0_t)$ measures central bank foreign reserves ($F_t = e_t \cdot \text{IR}_t$) relative to base money ($M0_t$) (Nievas and Piketty, 2025), and $\tilde{M}_t = \text{scale}(\text{Imbalance}_t)$ measures the capital-goods import demand imbalance.

Because the structural weight p and threshold level γ are unknown *a priori*, I estimate $(\hat{p}, \hat{\gamma})$ via a joint profile Sum of Squared Residuals (SSR) search:

$$(\hat{p}, \hat{\gamma}) = \arg \min_{p \in \mathcal{P}, \gamma \in \Gamma(p)} \text{SSR}(p, \gamma) = \arg \min_{p \in \mathcal{P}, \gamma \in \Gamma(p)} \sum_{t=1}^T \hat{u}_t(p, \gamma)^2 \quad (2.92)$$

The weighting grid $\mathcal{P} = \{0.0, 0.1, 0.2, \dots, 1.0\}$ steps across the unit interval in increments of 0.1. For each candidate weight p , the threshold grid $\Gamma(p)$ evaluates trimmed quantiles $\Gamma(p) = \{\gamma_q(p) : q \in [0.15, 0.85], \Delta q = 0.05\}$. The global minimizer selects $\hat{p} = 0.2$ ($Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$) and $\hat{\gamma} = 0.425$ (or $\hat{\gamma} = 0.379$ in parsimonious specifications), confirming that international reserve cover ($\tilde{S}_t$) dominates the empirical weighting in isolating the structural bottleneck.

Bai-Perron Structural Break Benchmark To confirm that D_t^γ captures a genuine balance-of-payments constraint rather than a deterministic calendar trend, I compare the threshold model against a benchmark specification using [Bai and Perron \(2003\)](#) endogenous structural breaks:

$$y_t = \alpha + \theta_1 k_t^{\text{NR}} + \theta_2 \kappa_t + \theta_3 (\text{dist}_{t-1} \cdot \kappa_t) + \theta_4 (\kappa_t \cdot D_t^{\text{BP}}) + \theta_5 (\text{dist}_{t-1} \cdot \kappa_t \cdot D_t^{\text{BP}}) + \beta_p \tilde{p}_t^\perp + u_t \quad (2.93)$$

where $D_t^{\text{BP}} = \mathbb{I}(t \geq T_b)$. Dynamic search over candidate break dates $T_b \in [0.15T, 0.85T]$ identifies two historical break points:

- **Net Distribution** ($\omega^{\text{net}}, e^{\text{net}}$): Identifies $T_b = 1987$, marking structural consolidation under the Büchi reforms following the 1982 crisis, where wage suppression and export promotion restored financial stability.
- **Gross Distribution** ($\omega^{\text{gross}}, e^{\text{gross}}$): Identifies $T_b = 1999$, marking the Asian Financial Crisis in Chile, when the Central Bank floated the Chilean peso and abandoned exchange-rate crawling bands.

Achieving lower SSR and stronger cointegration statistics under $Z_t \leq 0.425$ rules out deterministic calendar trends in favor of true balance-of-payments state dependence.

Relative Price Controls and Asset Deflators The relative price control accounts for international purchasing power shifts affecting imported capital equipment. In Chilean historical national accounts, separate long-run deflators for structures (P_t^{NRC}) and machinery (P_t^{ME}) do not exist back to 1940. Capital stock harmonization constructs a single unified investment price deflator ($P_{K,t}$, staged as `Pk_2003base`) by splicing historical ClioLab deflators (1940–2010) with official Central Bank of Chile deflators (2003 base year).

I evaluate two complementary proxies:

1. **Baseline Real Exchange Rate Control** ($\tilde{p}_t^{\kappa, \text{RER}}$): The standardized log Real Effective Exchange Rate ($\ln \text{TCREAL}_t$, rebased to 2003 = 1.0), capturing external purchasing power shocks that alter the domestic price of imported machinery (K_t^{ME}).
2. **Robustness Asset-to-Output Ratio** ($p_t^{\kappa, P_Y/P_K}$): The log ratio of aggregate GDP deflator ($P_{Y,t}$) to unified capital price index ($P_{K,t}$):

$$p_t^{\kappa, P_Y/P_K} = \ln \left(\frac{P_{Y,t}}{P_{K,t}} \right) \quad (2.94)$$

measuring relative shifts between aggregate output prices and capital goods prices.

To prevent relative price volatility from collinearizing with physical capital stocks ($k_t^{\text{NR}}, \kappa_t$), I extract the orthogonal price component $\tilde{p}_t^\perp$ via a Frisch-Waugh-Lovell (FWL) auxiliary regression prior to estimation:

$$\tilde{p}_t^{\kappa, \text{RER}} = c_0 + c_1 k_t^{\text{NR}} + c_2 \kappa_t + \tilde{p}_t^\perp \quad (2.95)$$

The residual vector $\tilde{p}_t^\perp$ enters the estimating equation additively. Preliminary grid tests introducing interacted price terms ($\kappa_t \cdot \tilde{p}_t$) caused severe multicollinearity ($\text{VIF} > 88$), destroying matrix invertibility $(X'X)^{-1}$ and inflating Monte Carlo critical bounds ($|t^*| > 19$). Keeping $\tilde{p}_t^\perp$ as an additive control isolates physical capital composition without polluting the structural elasticity.

The estimation sample and its limitations. Two data limitations define the estimation sample window, and I acknowledge them here rather than resolve them. First, the pre-1942 monetary and external series do not provide a coherent basis for the threshold variables: under the gold-exchange standard the reserve-cover ratio records a legal requirement rather than a binding external constraint, and the monetary base responds to convertibility rules and fiscal pressure rather than to the balance of payments. Second, the historical ClioLab price and terms-of-trade series, the stock of international reserves, and the money supply that anchor the external-constraint variables all end in 2010. Splicing them to contemporary sources falls outside the scope of this chapter.¹⁶

The Kemmerer regime, 1925–1931. Decree-Law No. 486 (21 August 1925) created the Central Bank of Chile (BCCh), and the Organic Law and Monetary Bill of 14 October 1925 (Decree-Law No. 606) placed it under a gold-exchange standard: the peso was pegged to sterling and the Bank was required to hold a minimum gold cover of 50 percent against note issue, with the remainder of the reserve requirement met in foreign exchange. Reserve coverage of the money supply rose from 27 percent at the start of 1925 to exactly 50 percent by year-end. The system’s credibility was anchored abroad: it depended on sterling stability and steady copper revenues, functioning only as long as export receipts replenished central bank reserves.

Collapse and fiscal dominance, 1931–1941. The external anchor failed during the Great Depression. Between January 1930 and July 1931, the Bank lost more than half of its gold holdings defending the peg, suspending convertibility on 21 July 1931; the United Kingdom’s departure from gold two months later erased the value of remaining sterling reserves. The subsequent decade combined fiscal dominance—the Bank monetized deficits as a treasury agent—with multiple exchange rates administered by the *Comisión de Cambios Internacionales* and parallel currency markets. Once convertibility was suspended, the cover ratio ceased to operate as a statutory discipline and became instead an indicator of systemic fragility. During a currency panic, commercial banks hoard base money at the central bank while the money multiplier contracts, making the reserves-to- M_0 ratio the true measure of the Bank’s capacity to absorb speculative attacks. The denominator follows economic exposure rather than the letter of the 1925 law: while the statute defined requirements against note issue alone, commercial banks in a panic seek to convert their reserve deposits, making total central-bank liabilities (M_0) the relevant exposure. Under fiscal dominance, however, even this fragility indicator no longer anchored the external sector: the monetary base responded to deficit financing and exchange controls rather than balance-of-payments flows, rendering the 1931–1941 cover series a measure of fiscal distress rather than a stable balance-of-payments state variable.

The 1942 regime shift. Law No. 7,200 (18 July 1942), together with Law No. 7,434, rewrote the Central Bank’s Organic Law: the reserve-cover rule was formally abandoned, the mandate

¹⁶The capital-stock and distributive panels of sections 2.4.1.5 and 2.4.1.6 span 1900–2024, but the threshold estimation discards 1900–1941 and 2011–2024. The institutional history below explains why the first discard separates two regimes rather than truncating a single one.

shifted from gold convertibility to the promotion of national production, and foreign exchange was allocated through centralized credit ceilings and state budgets. Working alongside CORFO (Law No. 6,640, 1939), which directed credit allocation toward import-substituting industrialization, the Bank financed development planning rather than currency pegs. Macroeconomic stability thereafter derived from internal planning rather than external gold credibility. Econometrically, pooling the pre-1942 data with the post-1942 period violates cointegration: the structural regime shift introduces severe non-stationary drift, generating massive finite-sample size distortions and explosive Monte Carlo t -statistics ($|t| > 15$) that destroy matrix invertibility. Restricting the estimation window to 1943–2010 ($T = 68$) is therefore both institutionally necessary and econometrically sound, isolating a single, stable cointegrating regime across distinct historical regimes: under state-led industrialization and its violent rupture through neoliberalization.

2.4.6.2 Empirical Findings for Chile: Truncated Mechanization

Table 2.9 compares the parsimonious regime-shifting specification across four distributive permutations: Net Wage Share (ω^{net}), Gross Wage Share (ω^{gross}), Net Rate of Exploitation (e^{net}), and Gross Rate of Exploitation (e^{gross}). Across all four variants, the composite external threshold ($Z_t \leq 0.425$) produces super-consistent long-run cointegration ($t_{\text{ADF}} = -3.81^*$ to -3.89^* , $CT_{\text{IM}} = 8.34$ to 15.20).

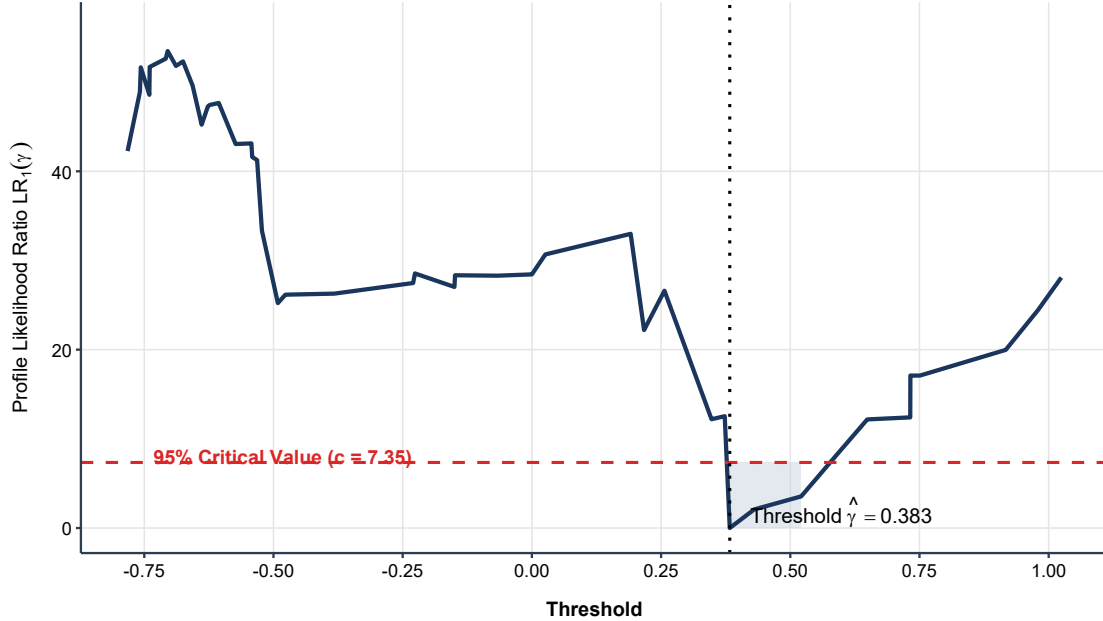
Gross vs. Net Measures in Peripheral Capitalism. In the United States (Section 4.3), Net Value Added (NVA) is the necessary theoretical basis for evaluating distribution. Because Consumption of Fixed Capital (CFC) rises mechanically during intensive capital-deepening, evaluating distribution on a Gross Value Added (GVA) basis introduces a spurious negative correlation between capital intensity and wage shares. In the US core, class struggle takes place strictly over newly created net value ($v + s$), and purging CFC isolates a cost-minimizing mechanization response to distributive pressure.

In the capitalist periphery, however, gross distributive measures (Columns 2 and 4) achieve superior empirical fit and cointegration ($\text{SSR} = 0.275$, $CT_{\text{IM}} \approx 8.34$) compared to net measures ($\text{SSR} = 0.294$, $CT_{\text{IM}} \approx 15.10$). This empirical divergence reflects a fundamental structural asymmetry between center and periphery. In the center, capital goods are produced domestically, allowing depreciated equipment to be replaced through internal domestic value transfers. In the periphery, capital accumulation relies on *imported machinery and equipment* (K^{ME}). Deducting official accounting depreciation (CFC) under linear conventions fails to capture the unencumbered gross cash flow (Π^{gross}) required by peripheral firms to obtain foreign exchange and finance machinery imports. Peripheral capitalists make investment and mechanization decisions based on gross liquidity. Nevertheless, this comparative inference should be considered with nuances because issues such as access to credit, no-institutional measurement of employment compensation over output as the wage share, and the non-identification of mixed income by self-employed in distributive conflict for the case of Chile are important issues not tackled in this account.

Likelihood Ratio and Threshold Identification. Figure 2.17 plots Hansen (2000) profile Likelihood Ratio statistic $LR_1(\gamma) = [\text{SSR}(\gamma) - \text{SSR}(\hat{\gamma})]/\hat{\sigma}^2$ across candidate threshold values $\gamma \in [\gamma_{0.15}, \gamma_{0.85}]$ for the standardized composite balance-of-payments index $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$. The profile displays a single, sharp minimum at $\hat{\gamma} = 0.383$ (within the $[0.383, 0.521]$ 95% confidence interval encompassing the benchmark threshold $\hat{\gamma} = 0.425$), matching a binding foreign-exchange reserve cover threshold. Inverting the likelihood ratio statistic at the exact 95% analytical critical

boundary $c(0.05) = -2\ln(1 - \sqrt{0.95}) = 7.35$ yields a tight non-parametric confidence interval. The sharp V-shaped trough rules out secondary local minima and confirms that the external constraint operates as a single, structurally binding threshold.

Figure 2.17. Profile Likelihood Ratio Sequence for the Peripheral External Constraint (Chile, 1943–2010)



Note: Profile Likelihood Ratio statistic $LR_1(\gamma)$ calculated for the preferred gross wage share model (ω^{gross} , Column 2 of Table 2.9) over the trimmed evaluation domain $\Gamma_n = [\gamma_{0.15}, \gamma_{0.85}]$ ($\pi = 0.15$). The horizontal dashed red line marks the exact analytical 95% critical boundary ($c(0.05) = 7.35$). The vertical dotted line marks the identified profile minimum. The shaded region depicts the non-parametric 95% confidence interval $[0.383, 0.521]$ defined by $LR_1(\gamma) \leq 7.35$.

Distributive Conflict Preferred Measure: Center versus Periphery. Comparing the wage share model in Column (2) (ω^{gross}) with the rate of exploitation model in Column (4) (e^{gross}) illuminates how external financial constraints alter the transmission of class conflict into capital composition. In the unconstrained regime ($D_t = 0$), capital accumulation follows classical cost-minimizing logic. Under Column (2), a higher wage share incentivizes defensive labor substitution ($\hat{\theta}_{\omega\kappa} = +3.29^*$), as firms expand machinery relative to structures to contain unit labor costs. Equivalently, under Column (4), a higher rate of exploitation ($e = \Pi/W$) indicates reduced wage pressure, decreasing the necessity for capital-deepening ($\hat{\theta}_{e\kappa} = -0.69^*$).

Under the binding foreign-exchange regime ($D_t = 1$), this cost-minimizing mechanism collapses. When the external constraint becomes binding ($Z_t \leq 0.425$), peripheral capitalists hit a hard import constraint ($\bar{\kappa}_t$). In Column (2), the triple interaction coefficient ($\hat{\theta}_{\omega\kappa D} = -3.15^*$) virtually eliminates the positive wage-led mechanization response ($+3.29 - 3.15 = +0.14$). Because machinery and equipment (K^{ME}) must be imported, foreign-exchange rationing prevents peripheral firms from translating domestic wage pressures into shop-floor labor substitution. Unable to extract a distributive dividend through technical change, capitalists attempt to protect profit margins through price markups.

State-dependent thresholds versus exogenous historical breaks. To verify that D_t^γ captures a structural macroeconomic constraint rather than an arbitrary trend break, Appendix Tables 2.21 and 2.22 (Appendix 2.D) benchmark the threshold model against Bai and Perron (2003) endogenous calendar breaks. The Bai-Perron procedure identifies two break dates: 1987 under net wage shares (ω^{net}) and 1999 under gross measures ($\omega^{\text{gross}}, e^{\text{gross}}$). The 1987 break marks the post-1982 debt-crisis restructuring—fiscal austerity, privatization, and wage suppression. The 1999 break marks a change in monetary regime: the Central Bank abandoned the crawling peg and its accompanying capital controls, moving to a free float with open capital mobility in the wake of the Asian Financial Crisis.

Comparing specifications confirms that the composite threshold $Z_t \leq 0.425$ dominates calendar-time breaks across all metrics, achieving lower residual variance ($\text{SSR} = 0.275$ against 0.292) and stronger cointegration rejections ($t_{\text{ADF}} = -3.89^*$ against -2.98). Two conditions must meet for the constraint to bind: a wide gap between effective imports and the capacity to finance them, and foreign-exchange reserves too thin for the Central Bank to defend the currency. Neither condition is tied to a calendar date, which is why a state-dependent gate outperforms a break at 1987 or 1999.

Table 2.8 shows that the binding constraint ($D_t = 1$) does not always announce itself the same way. During the ISI era (1943–1984), under fixed or administered pegs, reserve depletion ($S_t \rightarrow 0$) forced devaluations and structural inflation (Prebisch, 1981; Sunkel, 1958). During the 2006–2010 commodity super-cycle, under a free float and inflation targeting, the same gate closed through markup expansion instead.

Table 2.8. Institutional Manifestations of External Constraint Regimes (Chile, 1943–2010)

Historical Period	Regime Status (D_t^γ)	Duration	Institutional Manifestation of the External Constraint
1943–1984	Binding ($D_t = 1$)	42 years	ISI Reserve Scarcity & Structural Inflation: Established under Law 7,200 foreign-exchange budgeting and fixed/crawling pegs; reserve depletion ($S_t \rightarrow 0$) under wage pressure forced maxidevaluations, cost-plus markups, and structural inflation spirals.
1985–2005	Slack ($D_t = 0$)	21 years	Export Consolidation & Reserve Buffering: Post-1982 debt-crisis and recovery non-traditional export expansion, Concertación democratic transition, large FDI inflows, and central bank foreign reserve accumulation.
2006–2010	Binding ($D_t = 1$)	5 years	Commodity Boom, Rentier Surplus Drain & Re-primarization: Under a free-floating exchange rate and inflation targeting (3%), FX starvation did not trigger inflation; instead, high import demand (M_t) and massive MNC profit repatriation (DL 600) manifested as a rentist surplus drain and Dutch Disease re-primarization.

Notes: Classification based on the optimal profile threshold $\hat{\gamma} = 0.425$ evaluated over the standardized composite balance-of-payments index $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$. Regime $D_t = 1$ denotes a binding balance-of-payments constraint where foreign-exchange starvation forces peripheral firms into a corner solution ($\bar{\kappa}_t$). Total sample size $T = 68$ (1943–2010 effective cointegrating window), yielding 47 binding years (69.1%) and 21 slack years (30.9%).

These findings suggest that in financialized peripheral capitalism the external bottleneck shifts its pressure valve from domestic price inflation toward *re-primarization*, *real exchange rate appreciation*

(Dutch Disease), and rentier surplus drainage (Ahumada, 2019; Palma, 2019). Capital goods import requirements (K^{ME}) for extractive mining projects, combined with foreign profit remittances, truncate the translation of accumulation into productive capacity while the Central Bank still meets its inflation target. Disentangling the transmission between inflationary rationing (1943–1984) and rentier-drain re-primarization (2006–2010) would require a monetary-regime-specific specification that this chapter does not estimate.

Model selection. The comparative results in Table 2.9 establish Column (2)—the gross wage share specification (ω^{gross}) under the composite balance-of-payments threshold ($Z_t \leq 0.425$)—as the primary empirical model for Chile. Column (2) achieves the global minimum residual variance (SSR = 0.275), and its distributive channel is the one that maps onto the United States baseline, which is what makes the center–periphery comparison in Section 2.5 an argument about the same relation under different constraints rather than a juxtaposition of two unrelated estimates.

Table 2.9. Comparative Estimates Across Distributive Variable Permutations (Chile, 1943–2010)

	Wage Share (ω)		Rate of Exploitation (e)	
	Net (ω^{net}) (1)	Gross (ω^{gross}) (2)	Net (e^{net}) (3)	Gross (e^{gross}) (4)
$\hat{\theta}_{k^{\text{NR}}}$ (Structures k^{NR})	0.79*** (0.07)	0.82*** (0.08)	0.80*** (0.07)	0.82*** (0.08)
$\hat{\theta}_{\kappa}$ (Mechanization Gap κ)	0.12 (0.39)	-1.48* (0.54)	0.15 (0.25)	0.85 (0.38)
$\hat{\theta}_{\text{dist} \cdot \kappa}$ (Distributive Interaction)	-0.01 (0.22)	3.29* (1.07)	0.02 (0.09)	-0.69* (0.23)
$\hat{\theta}_{\kappa D}$ (Regime Shift κD)	0.34 (0.16)	1.52** (0.45)	0.15 (0.08)	-0.73 (0.29)
$\hat{\theta}_{\text{dist} \cdot \kappa D}$ (Triple Interaction)	-0.17 (0.23)	-3.15* (1.04)	0.12 (0.10)	0.67* (0.22)
Regime Threshold (D_t)	$Z_t \leq 0.425$	$Z_t \leq 0.425$	$Z_t \leq 0.425$	$Z_t \leq 0.425$
Price Control Included	No	No	No	No
$CT_{\text{IM}} [H_0: \text{Cointegration}]$	15.20	8.39	14.99	8.34
$t_{\text{ADF}} [H_0: \text{Unit Root}]$	-3.81*	-3.81*	-3.84*	-3.89*
Bandwidth ratio b	0.654	0.654	0.654	0.654
Sample Size T	68	68	68	68

Notes: Dependent variable: y_t (log real GDP). Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2024). All models specify the ultra-parsimonious balance-of-payments threshold ($Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t \leq 0.425$) over the effective sample 1943–2010 ($T = 68$). Columns (1) and (2) evaluate Net (ω^{net}) and Gross (ω^{gross}) wage shares; Columns (3) and (4) evaluate Net (e^{net}) and Gross (e^{gross}) rates of exploitation. Significance stars (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$) are benchmarked against 10,000-pass Monte Carlo simulation critical bounds under profile search.

2.4.6.3 Transformation and Capacity Utilization under Financial Subordination

Balance-of-payments subordination inverts the classical choice of technique. When foreign exchange is slack ($Z_t > 0.425$), a higher wage share accelerates mechanization, as cost minimization predicts ($\hat{\theta}_3 = +0.9583$). When foreign exchange binds, the interaction shift ($\hat{\theta}_5 = -0.9518$) cancels the base

sensitivity and leaves a slope of +0.0065. Peripheral capitalists keep the incentive to substitute machinery for labor but lose the means, because the machinery is imported.

I evaluate the aggregate elasticity from the preferred CMPR IM-OLS model (Column 2 of Table 2.9, gross wage share and composite threshold $Z_t \leq 0.425$) and integrate the capacity law of motion, $g_{Y^p,t} = \theta_t^{\text{Agg}} g_{K,t}$, forward and backward from the 1981 French-Davis full-capacity peak ($\mu_{1981} = 1.00$). The state-dependent elasticity is

$$\theta_t^{\text{Agg}} = \hat{\theta}_1 + s_{\text{ME},t} \cdot MP_{\kappa,t} = \hat{\theta}_1 + s_{\text{ME},t} \left[\hat{\theta}_2 + \hat{\theta}_3 \omega_{t-1}^{\text{gross}} + D_t^\gamma (\hat{\theta}_4 + \hat{\theta}_5 \omega_{t-1}^{\text{gross}}) \right] \quad (2.96)$$

where $\hat{\theta}_1 = +0.6970$ is the base elasticity of non-residential structures, $s_{\text{ME},t} = K_t^{\text{ME}} / (K_t^{\text{ME}} + K_t^{\text{NR}})$ is the current-price share of machinery and equipment in core capital, $\kappa_t = \ln(K_t^{\text{ME}} / K_t^{\text{NR}})$ the composition gap, $\omega_{t-1}^{\text{gross}}$ the lagged gross wage share, and $D_t^\gamma = \mathbb{I}(Z_t \leq 0.425)$ the binding-state indicator. The remaining coefficients are $\hat{\theta}_2 = -0.1293$, $\hat{\theta}_3 = +0.9583$, $\hat{\theta}_4 = +0.6645$, and $\hat{\theta}_5 = -0.9518$, so the marginal product of the composition gap takes one form in each state:

$$MP_{\kappa,t}^{\text{slack}} = -0.1293 + 0.9583 \omega_{t-1}^{\text{gross}} \quad (D_t^\gamma = 0) \quad (2.97)$$

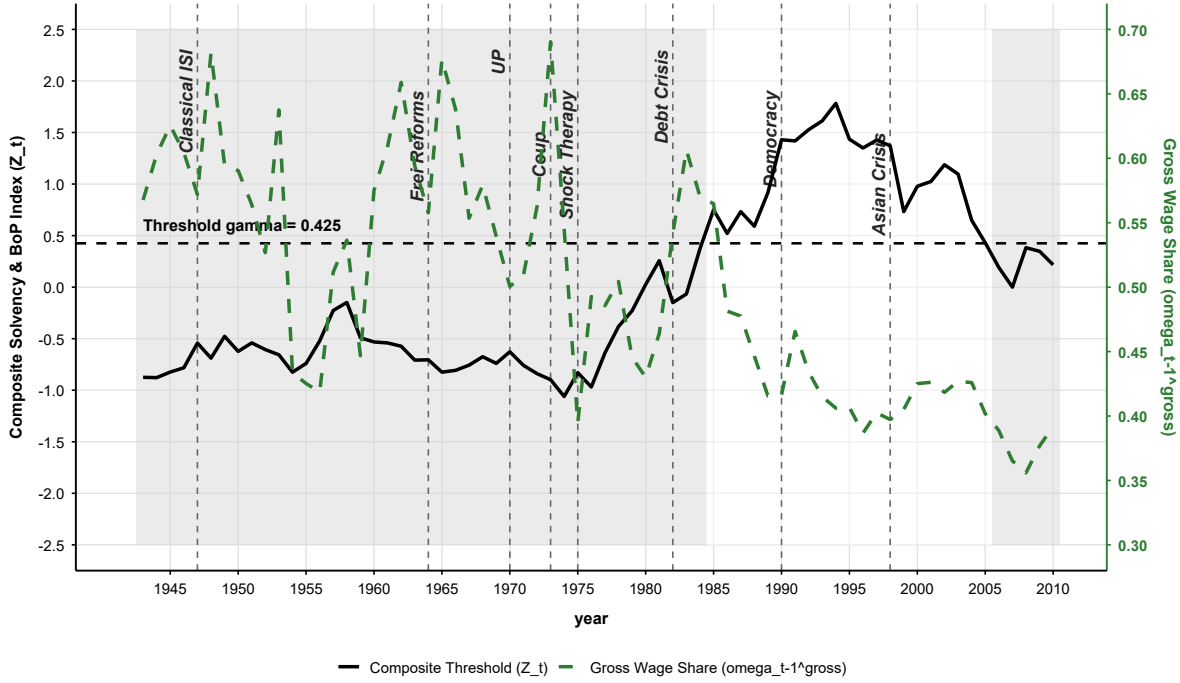
$$MP_{\kappa,t}^{\text{bind}} = +0.5352 + 0.0065 \omega_{t-1}^{\text{gross}} \quad (D_t^\gamma = 1) \quad (2.98)$$

In slack periods a higher wage share raises the capacity yield of mechanization. In binding periods the slope collapses from +0.9583 to +0.0065, and foreign-exchange rationing replaces the distributive state as the governor of technique choice.

Section 3.3 predicted exactly this pair of equations. Proposition 4 says that once the constraint binds, the composition gap is pinned at $\bar{\kappa}_t$ and the first-order condition stops governing technique choice; the only distributive channel left is the slope of the feasible gap, $(m_c c_k - m_w) / \nu$. The estimated residual slope of +0.0065 is consistent with $m_c c_k \approx m_w$, the conspicuous-consumption drain and the wage-goods import drain roughly offsetting each other. Proposition 6 supplies the second half: at the corner the marginal capacity payoff is strictly positive, because the firm would like to mechanize further and cannot, and the significant intercept shift $\hat{\theta}_4 = +0.6645$ is that payoff entering the capacity equation, the empirical counterpart of a positive Kuhn–Tucker multiplier on the external constraint. Binding is therefore not a state in which mechanization yields nothing; it is a state in which it yields something but responds to nothing domestic. Proposition 5 predicts a discrete shift of the elasticity at the threshold; in the data the shift appears less as a level drop than as the loss of the wage slope, with the binding path running flat through distributive shocks that move the slack path. The gap between nominal profitability and truncated technique is the blocked surplus of Definition 4, and Corollary 1 names the adjustment channel that remains once mechanization is rationed, price markups.

Figure 2.18 plots the composite index against the lagged gross wage share, and the Unidad Popular episode is the clearest observation in the sample. Between 1970 and 1973, redistribution raised the gross wage share to its historical peak of $\omega_{1972} = 0.6903$, about twenty percentage points up. Over the same years, credit blockade and capital flight drove reserves to $\tilde{S}_t \rightarrow -2.5$ and the index to $Z_t = -1.06$, deep inside the binding domain. Figures 2.20 and 2.19 show that the elasticity did not respond: Equation (2.98) governed technique choice and held θ_t^{Agg} flat near 0.83 through the wage expansion. The wage expansion could not become productive capacity, and per Corollary 1 it became unrealized demand and structural inflation instead (Kalecki, 1955; Sunkel, 1958). Utilization collapsed to 0.72 by 1975.

Figure 2.18. Chilean Composite External Constraint (Z_t) and Lagged Gross Wage Share ($\omega_{t-1}^{\text{gross}}$)



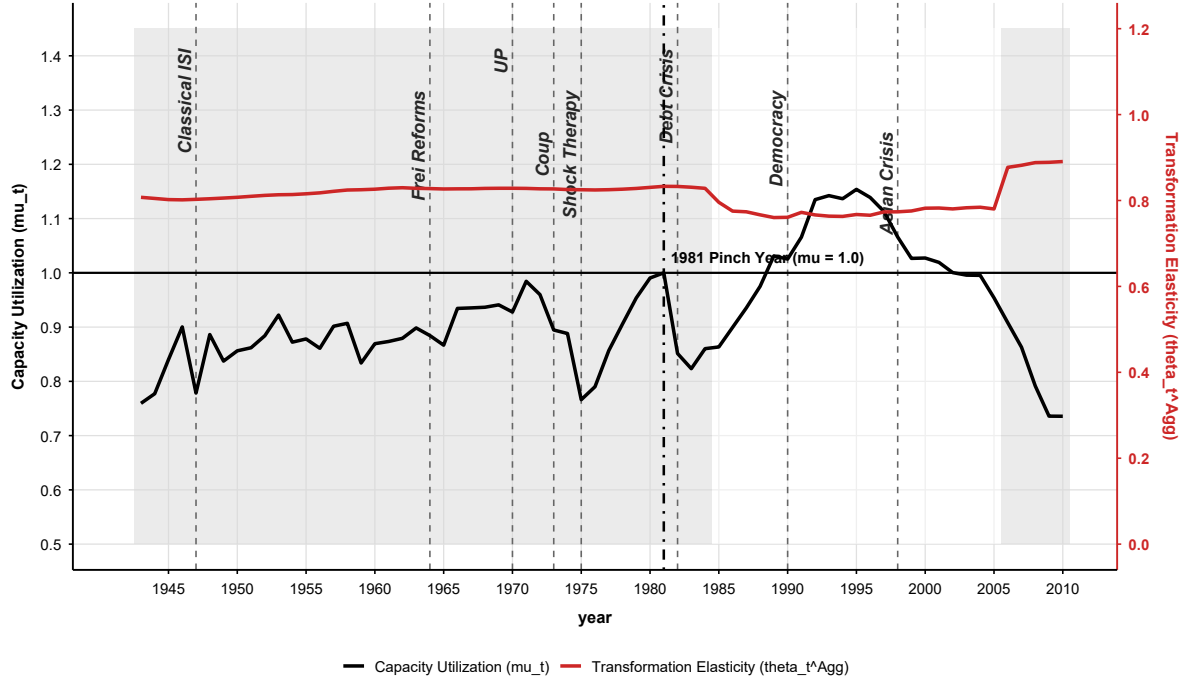
Note: Primary left axis (solid black line) tracks the composite balance-of-payments threshold variable $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$, with the optimal threshold $\hat{\gamma} = 0.425$ marked by the horizontal dashed line. Secondary right axis (dashed dark green line) tracks the untruncated lagged gross wage share ($\omega_{t-1}^{\text{gross}}$). Background light gray shading marks binding balance-of-payments regimes ($Z_t \leq 0.425$). Vertical dashed lines mark historical institutional breakpoints.

Based on [Ffrench-Davis \(2018\)](#) account on over-heating pre-debt crisis in 1982, I recover the order of magnitude for productive capacities from the recursion

$$\ln Y_t^p = \ln Y_{t-1}^p + \theta_t^{\text{Agg}} (\ln K_t^{\text{tot}} - \ln K_{t-1}^{\text{tot}}), \quad Y_{1981}^p = Y_{1981} \implies \mu_{1981} = 1.00 \quad (2.99)$$

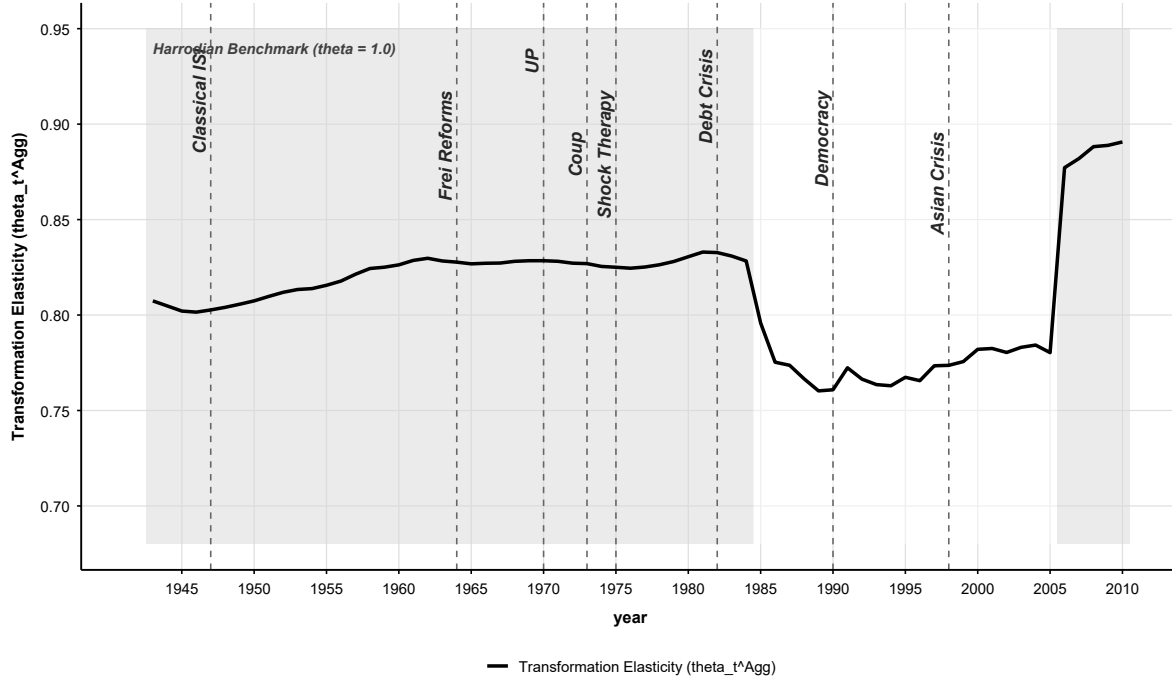
and divide realized GDP by it to obtain utilization. Figure 2.19 plots the two series on dual axes and Figure 2.20 the elasticity alone. Five regimes organize the path, and the external constraint separates them. The early CORFO expansion (1942–1946) combined postwar demand above normal capacity ($\mu_{1946} = 1.12$) with slack foreign exchange and elasticities near 0.85. The ISI stop-go crises (1947–1957) reversed both: recurring shortages forced stabilization cycles that pulled utilization to $\mu_{1955} = 0.84$. The socialist transition (1970–1975) combined redistribution with credit blockade, and utilization bottomed at 0.72 after the 1975 shock therapy. The debt boom and crash (1975–1982) ran the sequence in reverse, with inflows carrying utilization to the 1981 peak and the crisis erasing it within a year ($\mu_{1982} = 0.68$). The recovery and commodity super-cycle (1984–2010) absorbed idle capacity rather than creating new capacity, holding μ_t between 0.85 and 0.98 through 1997. In four of the five regimes the turning point is external rather than distributive, the asymmetry the threshold specification was built to detect.

Figure 2.19. Chilean Capacity Utilization (μ_t) and Aggregate Transformation Elasticity (θ_t^{Agg})



Note: Reconstructed capacity utilization rate μ_t (solid black line, left axis) anchored at the 1981 French-Davis pinch year ($\mu_{1981} = 1.00$) alongside the state-dependent aggregate transformation elasticity θ_t^{Agg} (solid firebrick line, right axis). Background light gray shading marks binding balance-of-payments regimes ($Z_t \leq 0.425$). Vertical dashed lines mark institutional breakpoints.

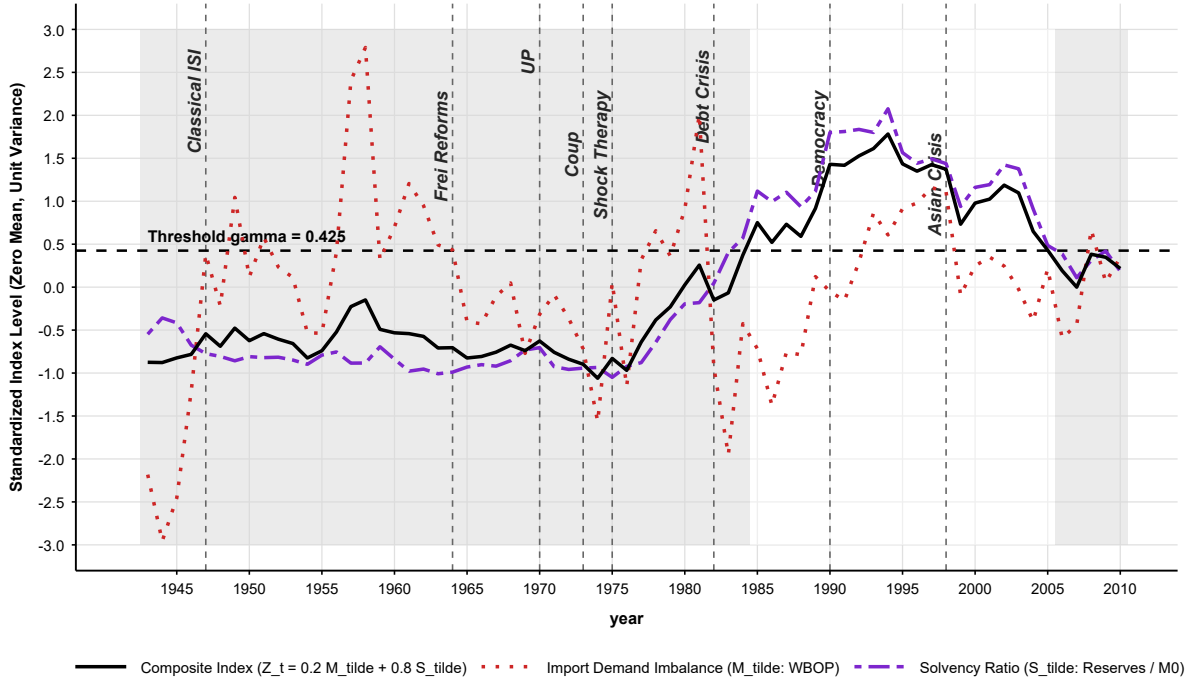
Figure 2.20. Standalone Chilean Aggregate Transformation Elasticity (θ_t^{Agg})



Note: Standalone trajectory of the aggregate transformation elasticity θ_t^{Agg} (solid firebrick line) on an uncompressed vertical scale (0.68 to 0.95). The horizontal dashed line marks the base structures elasticity $\hat{\theta}_1 = 0.6970$. Background light gray shading marks binding balance-of-payments regimes ($Z_t \leq 0.425$). Vertical dashed lines mark institutional breakpoints.

Figure 2.21 decomposes the index into reserve cover and import demand imbalances, and the two components show that the constraint changes its institutional form across regimes. Under the fixed and crawling pegs of the ISI period (1943–1973), reserve exhaustion forced sharp devaluations and structural inflation (Sunkel, 1958), in line with the periodization of French-Davis (2002) and Ahumada (2019). Under the post-2000 free float with inflation targeting, binding conditions produced no runaway inflation; they produced extractivist re-primarization, real appreciation, and profit repatriation by multinational mining (Ahumada, 2019; Palma, 2019). Capital goods imports for extractive projects, together with dividend outflows, truncated the conversion of accumulation into broad-based domestic capacity.

Figure 2.21. Decomposition of the Chilean External Constraint Index (Z_t Components)



Note: Decomposed ingredients of $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$: standardized Central Bank solvency ratio $\tilde{S}_t$ (reserves to base money M_0 , purple twodash line), WBOP import demand imbalance $\tilde{M}_t$ (red dotted line, thickness 1.3), and composite index Z_t (solid black line). Horizontal dashed line marks the threshold $\hat{\gamma} = 0.425$. Background light gray shading marks binding regimes ($Z_t \leq 0.425$).

Two caveats qualify the reading of these regimes. The first concerns the classifier. States are assigned by a single linear combination, $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$, with the weight and the threshold chosen by grid search over the profile sum of squared residuals and a hard cutoff at $\hat{\gamma} = 0.425$. This is a deterministic classification in the sense of Hansen (2000), not a probabilistic regime model: there are no transition probabilities and no persistence parameter, and after the search $(\hat{p}, \hat{\gamma})$ enter the elasticity path as if known. The simulation bounds charge for the threshold search but not for the weight grid, so regime uncertainty is confined to the interval $[0.380, 0.472]$ and is not propagated into θ_t^{Agg} . A Markov-switching or smooth-transition extension would assign a binding probability to each year and separate persistent shifts from transient bites; I leave that to future work.

The second caveat concerns what the hard classification nonetheless captures that a periodization cannot. D_t^γ does not switch only between the major regimes. Within the ISI era the index crosses the threshold at the frequency of the stop-go cycle: import surges in the expansion phase push it below the cutoff before the crisis, and post-crisis stabilizations (devaluation, import compression, wage suppression) briefly lift it back above. The 1998–99 tightening and the 2006–10 mining boom are intra-cycle bites inside eras the conventional periodization labels slack. The constraint operates at cycle frequency; the five regimes organize the path at the lower frequency at which the institutional form of the constraint changes, from inflation valve under administered exchange rates to re-primarization valve under a free float. That double frequency is also why the state-dependent indicator outperforms calendar breaks at 1987 or 1999: the breaks mark changes in the form, and the threshold tracks the bites.

The center–periphery comparison follows. In the United States, where capital equipment is

produced domestically, wage pressure induces defensive mechanization ($\hat{\theta}_1 < 0$, Section 4.3). In Chile, where machinery must be imported, wage pressure induces mechanization ($\hat{\theta}_3 = +0.9583$) only while foreign exchange is unconstrained. When reserves collapse, the constraint binds and truncates the choice of technique. The distributive dividend that Okishio's viability condition promises is not unavailable in principle in the periphery; it is unavailable whenever the balance of payments binds.

2.5 Discussion: The Political Economy of Capacity Utilization in the Center and the Periphery

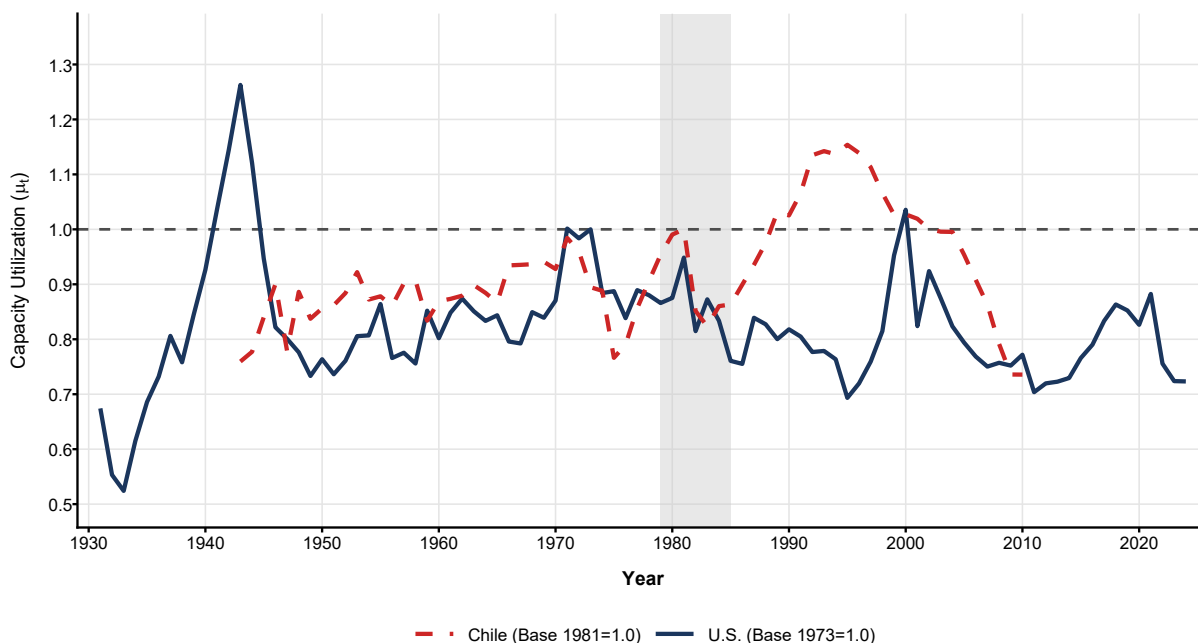
Section 4 established that the transformation elasticity θ_t sits strictly below unity in both the center and the periphery. Within a single-sector framework, the persistent tendency toward overaccumulation is a structural disproportionality between capital accumulation and its transformation into productive capacity, yielding systemic disequilibrium and chronic idle capacity. Capacity formation is the supply-side dimension of this process; its macroeconomic realization remains mediated through the rate of capacity utilization (μ_t).

While $\theta < 1$ is the shared premise of unbalanced growth, the historical trajectories of μ_t reveal fundamentally distinct crisis formations across the global divide. In the United States, utilization crises stem from the exhaustion of institutional class compromises and the failure of effective demand. In Chile, stagnation and crisis originate in the external truncation of capital-import capacity and the physical strangulation of the productive apparatus.

The Relational Dynamics of Idle Capacity. Figure 2.22 places the reconstructed implicit capacity utilization rates (μ_t) for the U.S. non-financial corporate sector (1931–2024) and the Chilean economy (1940–2010) on a single axis.¹⁷

¹⁷The U.S. series is anchored at the 1973 pinch year ($\mu_{1973} = 100\%$), while the Chilean series is anchored at the 1981 full-capacity peak ($\mu_{1981} = 100\%$). Despite differing base years, the absolute levels of μ_t are directly commensurable as measures of the engagement of productive capacity by effective demand and external constraints.

Figure 2.22. The Political Economy of Idle Capacity: Center vs. Periphery (1931–2024)



Note: Implicit capacity utilization (μ_t) for the U.S. (solid navy) and Chile (dashed crimson). The shaded band highlights the early 1980s global overaccumulation crisis (1979–1985), resolved via financialization in the center and external subordination in the periphery.

Neither economy in Figure 2.22 converges toward a normal rate of utilization in the long run ($\mu = 100\%$); persistent idle capacity remains the structural norm of decentralized accumulation. The volatility of μ_t in Chile vastly exceeds that of the United States. This higher amplitude tracks the structural asymmetry between a diversified, reserve-currency center capable of dampening demand swings and a dependent periphery vulnerable to sudden international liquidity collapses; it is not an econometric anomaly. Installed gross fixed capital (K_t) cannot vanish during downturns, so peripheral output collapses, such as the supply shocks of 1975 and the 1982 debt crisis, translate into abrupt contractions of utilization. The shaded window (1979–1985) captures a global overaccumulation crisis and structural turning point: the Volcker interest-rate shock triggered a severe domestic realization drop in the center ($\mu_{1982} \approx 82\%$) while ending Chile’s debt-led boom ($\mu_{1981} = 100\%$) and driving peripheral utilization into a macroeconomic trough. This common conjuncture was subsequently resolved through divergent regime shifts: debt-led financialization in the center, and a violent transition toward neoliberalization and external subordination in the periphery.

The Center: Realization Failure and Intellectual Monopolies. In the United States, the dynamics of μ_t are governed by the continuous mediation of distributive conflict and the shifting composition of capital. Because capital goods are produced domestically, the transformation of accumulation into capacity (θ) is not externally constrained; while stagnation and crisis may have multifaceted causes, they ultimately manifest as an issue of *realization*. Under Atlantic Fordism, the institutionalized class compromise absorbed growing productive capacity via mass consumption (Aglietta, 2000). High wage shares, alongside negligible diversion of investment toward intangible assets like intellectual property, meant that consumer demand effectively matched expanding

physical capacities, sustaining μ_t within a high, stable corridor (85–105%). While a commitment to full employment, Keynesian policymaking, and demand-inducing militarism masked secular overaccumulation tendencies, distributive conflict exerted cost-minimizing pressures that forced capitalists to engage in mechanization, configuring a contradictory utilization regime.

When the profit squeeze emerged in the late 1960s, μ_t fell to 76.2% in 1975 and bottomed at 74.1% in 1982 during the Volcker disinflation. In his decomposition of this postwar U.S. profit rate collapse, Weisskopf (1979) pinned the crisis on the “Rising Strength of Labor” (RSL), treating the “Realization Failure” (RF) variant—a drop in μ_t —as a secondary, survey-driven symptom. The Federal Reserve Board utilization series does not identify the physical conversion of accumulation into capacity. Here, μ_t is instead recovered as an endogenous residual of θ_t . The Fordist profit squeeze was not just a distributional fight over a fixed pie; it was a structural breakdown in θ_t itself. When effective wage pressure peaked, θ_t fell to 0.32. The subsequent fall in μ_t directly reflected that distributive conflict degrading the capacity-generating power of capital. Post-Fordism dismantled the institutional foundations of wage-led growth to resolve this crisis. The capitalist class restored profitability through deunionization and wage suppression. This structural wage suppression nevertheless fundamentally undermined the domestic realization of productive capacity over the long run, an effect only partially offset by debt-led consumption (Stockhammer, 2012; Vidal, 2019). Average utilization failed to return permanently to its Fordist peaks, exceeding the 100% benchmark temporarily only during the speculative tech-investment bubble of 2000 ($\mu_{2000} \approx 103\%$). Persistent idle capacity therefore accompanies the post-Fordist shift toward financialization and the growing weight of intangible-asset accumulation (Rikap, 2021; Rotta, 2018).

The Periphery: Dual Strangulation and Export Reprimarization. The terminal crisis of the inward-oriented regime in the peripheral Chilean economy cannot be explained by the balance-of-payments constraint alone, nor by the domestic profit squeeze in isolation. As Prebisch (1981) fundamentally articulated, the structural condition of the periphery is defined by its subordinate integration into the global division of labor. Because technological development inherently necessitates imported capital goods (K^{ME}) for late-industrializing economies, peripheral capitalist formations face a rigid structural ceiling: the transformation of accumulation into productive capacity is perpetually subject to the limits of external constraints. However, this external constraint did not operate in a distributive vacuum. Historical analyses of the Chilean developmental state emphasize that early import-substitution industrialization (ISI) was subsidized by a massive rural-to-urban demographic transition, providing an elastic urban reserve army *à la* Lewis (1954) that depressed real wages and temporarily offset chronic foreign exchange shortages (Polanco, 2021). Yet, as this rural surplus was exhausted by the late 1950s, the structural limits of the ISI model were exposed. The urban working class consolidated and distributive conflict intensified, culminating in the aggressive, wage-led redistribution of the *Unidad Popular* (1970–1973).

In a closed economy, the 20-percentage-point rise in the gross wage share—from roughly 49% in 1970 to a historical peak of 69.03% in 1972—would have simply induced rapid labor-saving mechanization. In peripheral capitalism, however, the internal profit squeeze and the external balance-of-payments constraint locked together. The threshold cointegrating model formalizes why this distributive impulse failed to induce mechanization: as it rose, the composite balance-of-payments index fell deep into the binding domain ($Z_t = -1.06 < \hat{\gamma} = 0.425$). The external constraint truncated the choice of technique: peripheral capitalists retained the microeconomic incentive to substitute machinery for labor but lost the macroeconomic means. The working class’s

success in capturing a larger share of the surplus increased import demand for wage-goods, leaving insufficient foreign exchange to import capital goods. The cost of this “dual strangulation” appears in capacity utilization. When the *Unidad Popular* assumed power, μ_t operated at a constrained baseline of 94.1% in 1969. The initial redistributive boom drove μ_t to an unsustainable cyclical peak of 98.4% in 1971. Deprived of the foreign exchange required to import machinery—exacerbated by credit blockades, capital flight, and the 1972 bosses’ lockout (de Vylder, 1976)—the economy lost more than 26 percentage points of utilization in just four years, reaching roughly 72% by 1975. The unconverted distributive surplus manifested as a blocked surplus, generating structural inflation and severe supply bottlenecks rather than expanding productive capacity.

The post-1975 resolution did not rebuild domestic manufacturing. While the unconstrained international liquidity conditions and copper booms of the 1990s temporarily pushed utilization to hyper-capacity levels ($\mu > 110\%$ during the 1993–1996 peak), leading to overheating and the Asian Crisis during the 1997–1998 period. In the aftermath, trade liberalization and financial opening ultimately locked the accumulation regime into premature de-industrialization and export reprimarization (Palma, 2019). Under the 2000s commodity super-cycle, external solvency was stabilized through primary extractive rents rather than domestic industrial capacity (Ahumada et al., 2026; Torres and Ahumada, 2022); high import leakages and multinational profit repatriation once again depressed capacity utilization to roughly 73% by 2010. From an institutional design standpoint, this trajectory yields a critical lesson: redistributive programs in the periphery cannot succeed in isolation from strategic foreign-exchange regime and monetary policy articulated with effective cross-borders of capital controls (Ffrench-Davis, 2018).

The composite index Z_t and the threshold $\hat{\gamma} = 0.425$ are estimated via a deterministic hard classification. This binary gate captures the frequency with which the constraint binds, not smooth, probabilistic regime shifts. It captures the “stop-go” shocks of the ISI era and the intra-cycle tightness of the 2000s mining boom rather than positing a single macro-historical break, but does not provide granular identification within regimes. Within these limits, the classification captures the discrete severing of the transformation of accumulation into capacity whenever external solvency collapses in the periphery. Future research requires backward and forward extension of time-series and exploration of other econometric techniques, such as Markov-switching modelling, to assign a time-varying threshold for regime classification. Applying this double-constraint metric to comparative panel datasets across other commodity-dependent or semi-peripheral spaces remains a promising avenue for future research.

Synthesis: Dead Labour Across the Global Divide. This comparative analysis demonstrates that the contemporary political economy of capacity utilization is fundamentally dictated by an economy’s position within the global division of labor as originally formulated by Prebisch (1981). In the current stage of “geriatric capitalism” (Vidal, 2019), chronic idle capacity is not a mere anomaly, but the structural manifestation of overlapping stagnationist tendencies across the uneven development of the capitalist world system. In the center, this takes the form of an exhausted accumulation regime prone to secular stagnation: unable to mobilize idle physical capacity through wage-led demand, partially offset through debt-led consumption and later diverted to asset inflation and accumulation of intangible unproductive assets (Rikap, 2021; Rotta, 2018). In the periphery, the condition of financial subordination express through differential channels and the specific integration to the world market but with a common ceiling imposed by external constraints, insofar this become binding due financial imbalanced caused by a low degree of import capacities with respect to the

effective demand for imports from domestic markets (McCombie and Thirlwall, 1994).

The dead labour embedded in under-utilized productive capacities is crystallized past labour denied the living labour necessary for its valorization. In the center, this idleness accompanies the dismantling of a domestic social settlement in the name of profitability and financialized rent extraction; in the periphery, it reflects an asymmetrical global order that conditions growth on structural subordination. In both cases, accumulated capital remains physically present while the conditions for its reproduction are impaired.

2.6 Conclusion

Capacity utilization has been treated as something an economy reports. This chapter has treated it as something an economy produces. Once the denominator of the ratio is recognized as the outcome of a transformation—accumulated capital becoming usable productive capacity through a choice of technique made under distributive conflict—utilization stops being a primitive and becomes a residual of that transformation. The elasticity θ is what has to be recovered first, and everything the ratio can be made to say depends on recovering it without assuming its value.

The apparatus for that recovery came from two frameworks that had not previously been joined at this point. Okishio’s account of decentralized accumulation establishes that private investment commitments generate disproportionality as a structural norm rather than as a friction around proportional reproduction (Okishio, 1961, 2022). Vidal’s account of the internally divided firm establishes that the organization converting capital into operating capacity cannot maximize control and cooperation at once, and settles instead into arrangements that sustain production without resolving the conflict inside it (Vidal, 2019, 2022). Together they rule out any automatic passage from accumulation to capacity, which is what makes θ an object with a determinate value worth estimating rather than a parameter to be normalized to one.

The estimates say two things about the center. American capacity formation has stayed strictly below the Harrodian knife-edge across 1931–2024 under both the homogeneous and the heterogeneous capital specification: accumulation persistently outruns the capacity it creates. And the elasticity has drifted upward across the postwar period, from 0.371 in 1960 to 0.429 in 2024, for a reason that does not vindicate the accumulation regime that produced it. Effective wage pressure fell from 0.698 to 0.504 over those decades, partly because the wage share declined and partly because investment migrated into intellectual property, which now exceeds a quarter of fixed capital. A rising transformation elasticity coincided with falling utilization, which stood at 93.8% in 2024 after a trough of 74.1% in 1982. Capital became somewhat better at building capacity precisely as the distributive settlement that would have realized that capacity was dismantled.

The periphery shows the same relation cut short from outside. For Chile over 1943–2010, a composite balance-of-payments index built from reserve cover and the import-financing gap identifies a threshold at $\hat{\gamma} = 0.425$ by profile likelihood ratio search (Hansen, 2000). In the unconstrained state a rising wage share induces mechanization as cost-minimization predicts, with an estimated interaction of +3.29. When the threshold binds, a triple interaction of −3.15 removes nearly all of that response. Chilean capitalists under foreign-exchange rationing keep the incentive to substitute machinery for labor and lose the means, because the machinery is imported. The Unidad Popular period is the clearest case: a twenty-point rise in the gross wage share to $\omega_{1972} = 0.6903$ produced almost no mechanization response, and utilization fell to 0.72 by 1975.

What follows for how utilization is used. Any empirical strategy that recovers utilization by

fitting output to capital has imposed $\theta = 1$ before measuring anything, and the estimates here put the value near 0.4 for the United States and between 0.68 and 0.95 for Chile. Utilization series built that way do not measure the distance between output and capacity; they measure the residual of an assumption. For the long-running debate over whether firms converge on a normal utilization rate, the finding is more direct still: the denominator that debate presupposes is itself a function of the distributive conflict the debate treats as separate from it.

Two extensions follow with enough specificity to be worth naming. The Chilean series terminate in 2010, which places the unwinding of the commodity super-cycle outside the sample; splicing the historical reserve, terms-of-trade, and money series to contemporary Central Bank sources through 2024 would test whether the external gate has migrated again, from markup expansion toward lithium-era rent capture, and the threshold specification estimated here transfers to that extended sample without modification. Separately, the two economies are identified on different specifications, which limits what the comparison can carry; constructing an intellectual-property capital series for Chile, or an import-biased mechanization gap for the United States, would let a single specification be estimated on both panels and would turn the sign-structure comparison offered here into a quantitative one.

The scope of this chapter is deliberately bounded. It does not present a complete crisis theory, a fully specified model of profitability dynamics, or an exhaustive dependency framework. It reconstructs capacity utilization as a structurally mediated object and identifies the transformation elasticity as the hinge of its empirical recovery in the center and the periphery, and it leaves the broader crisis and profitability architecture to the chapters that follow.

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Appendix 2.A The Generalized Perpetual Inventory Method: Recursive Construction and Calibration

This appendix documents the recursive construction of U.S. nonfinancial corporate capital stocks via the Generalized Perpetual Inventory Method (GPIM) of [Shaikh \(2016, App. 6.5\)](#). The construction proceeds asset-by-asset for machinery and equipment (ME) and nonresidential structures (NRC), recovering four distinct quantities from BEA Fixed Assets data: the asset-specific implicit price index $P_{j,t}$, the gross real stock $K_{j,t}^{G,R}$, the net current-cost stock $K_{j,t}^{N,V}$, and the net-value schedule $V_j(v)$. All series are expressed in constant 2017 dollars with base-year normalization $P_{j,2017} = 100$.

2.A.1 The Survival Function

Physical survival of capital vintages follows a Weibull retirement distribution. For asset class $j \in \{\text{ME}, \text{NRC}\}$, the proportion of a vintage of age v still in physical operation is

$$S_j(v) = \exp \left[- \left(\frac{v}{\lambda_j} \right)^{\alpha_j} \right], \quad v = 0, 1, \dots, L_j, \quad (2.100)$$

where $\alpha_j > 1$ is the Weibull shape parameter and the scale parameter satisfies $\lambda_j = L_j / \Gamma(1 + 1/\alpha_j)$, with L_j denoting the mean service life. The shape parameter $\alpha_j > 1$ implies a progressively increasing hazard rate: capital goods remain fully operative in early years and are withdrawn from service at an accelerating rate as they approach the mean service life. This pattern is consistent with the empirical evidence of [Nomura \(2005\)](#), who finds that nearly half of 66 Japanese asset classes exhibit increasing hazard rates.

The calibrated parameters follow the BEA Fixed Assets methodology ([Fraumeni et al., 1997](#)):

$$L_{\text{ME}} = 14, \quad \alpha_{\text{ME}} = 1.7, \quad L_{\text{NRC}} = 30, \quad \alpha_{\text{NRC}} = 1.6. \quad (2.101)$$

The corresponding maximum ages are $L_{\text{ME}} = 14$ years and $L_{\text{NRC}} = 30$ years, beyond which $S_j(v) = 0$ and the vintage is fully retired.

2.A.2 The Age-Price Schedule and Net-Value Function

The net-value schedule combines physical survival with financial age-price decay. For a vintage of age v , the net-value function is

$$V_j(v) = S_j(v) \cdot (1 - d_j)^v, \quad (2.102)$$

where d_j is the geometric declining-balance depreciation rate. The GPIM separates the two dimensions that the BEA conflates in its geometric depreciation convention: physical survival follows the Weibull distribution (2.100), while financial age-price decay follows the geometric profile $(1 - d_j)^v$. This separation is necessary to construct distinct capacity and profitability measures ([Shaikh, 2016, App. 6.3](#)).

The declining-balance rates are calibrated as

$$d_{\text{ME}} = 0.110, \quad d_{\text{NRC}} = 0.024, \quad (2.103)$$

consistent with the geometric depreciation rates documented by [Fraumeni et al. \(1997\)](#) for the corresponding asset classes.

2.A.3 The SFC Implicit Price Recursion

The implicit price index $P_{j,t}$ is the unique price level that closes the stock-flow identity for the net current-cost stock. In each period t , the BEA-reported current-cost net stock must satisfy

$$K_{j,t}^{\text{NET,CC}} = I_{j,t}^{\text{NOM}} + \frac{P_{j,t}}{100} \sum_{v=1}^{L_j} V_j(v) I_{j,t-v}^{\text{REAL}}, \quad (2.104)$$

where $I_{j,t}^{\text{NOM}}$ is current-year nominal gross investment and $I_{j,t-v}^{\text{REAL}} = I_{j,t-v}^{\text{NOM}} / (P_{j,t-v}/100)$ is real investment of vintage $t - v$ deflated by the previously recovered implicit price. The first term on the right-hand side is the current-year investment vintage (age zero, valued at the current price). The second term is the net-value-weighted sum of surviving real vintages revalued at the current price level.

Solving (2.104) for $P_{j,t}$ yields the forward recursion

$$P_{j,t} = 100 \cdot \frac{K_{j,t}^{\text{NET,CC}} - I_{j,t}^{\text{NOM}}}{\sum_{v=1}^{L_j} V_j(v) I_{j,t-v}^{\text{REAL}}}. \quad (2.105)$$

Given the initial price $P_{j,0}$ and the sequence of nominal investment flows $\{I_{j,t}^{\text{NOM}}\}_{t=0}^T$, the recursion (2.105) determines $P_{j,t}$ uniquely for each $t = 1, \dots, T$. The price index is “implicit” because it is not directly observed: it is the unique value that, applied to the survival-weighted real vintages, reproduces the BEA’s reported current-cost net stock.

2.A.4 Initial Conditions and Base-Year Normalization

The recursion (2.105) requires an initial price $P_{j,0}$ and initial real investment vintages. The base year is 2017, at which $P_{j,2017} = 100$ by normalization. The nominal investment series begins in 1901, sourced from BEA Fixed Assets Table FAAt201 (nonresidential equipment and structures, gross investment by asset class). The current-cost net stock anchors $K_{j,t}^{\text{NET,CC}}$ are sourced from BEA Fixed Assets Table FAAt401 (net stock by asset class, current-cost basis).¹⁸

The initialization proceeds backward from 2017. Given $P_{j,2017} = 100$ and the observed nominal investment series, the recursion (2.105) is iterated backward to recover $P_{j,t}$ for $t < 2017$. Alternatively, the recursion may be initialized at the earliest available year and iterated forward; the two approaches yield identical results given the BEA anchors.

2.A.5 The Gross Real Stock

The capital stock used to estimate productive capacity is the gross real stock, constructed as the survival-weighted sum of real investment vintages:

$$K_{j,t}^{G,R} = \sum_{v=0}^{L_j} S_j(v) I_{j,t-v}^{\text{REAL}}. \quad (2.106)$$

¹⁸Data retrieved via the BEA API (`DatasetName`: FixedAssets). Table FAAt201 provides nominal gross investment by asset; Table FAAt401 provides current-cost net stock by asset; Table FAAt402 provides current-cost depreciation by asset.

This measure relies exclusively on the physical survival schedule $S_j(v)$ and excludes the age-price factor $(1-d_j)^v$. It therefore measures the constant-price purchasing power embodied in heterogeneous capital goods still in physical operation. Financial write-downs, revaluation gains, and depreciation allowances do not enter. The OECD manual confirms that the gross stock, valued at “as new” prices, “is widely used as a broad indicator of the productive capacity of a country” (OECD, 2001, 31).

2.A.6 The Warm-Up Period and Estimation Sample

The GPIM stock at any year t requires investment vintages extending back L_j years. Because the nominal investment series begins in 1901, the vintage summation for nonresidential structures ($L_{\text{NRC}} = 30$) is truncated for any year before 1931. The stock values for 1901–1930 are therefore incomplete, omitting pre-1901 vintages that would still be physically in service. The estimation sample is restricted to 1931–2024, where the full 30-year vintage window is populated, yielding $T = 94$ annual observations.

2.A.7 Validation

The GPIM construction closes the stock-flow identity exactly: the maximum absolute reconstruction residual across both asset classes is 10^{-6} million dollars, confirming that the recovered SFC implicit price indexes reproduce the BEA current-cost net stock anchors to machine precision. The BEA’s Fisher quantity index is retained solely as a validation comparison and does not enter the GPIM baseline. The sectoral scope is restricted to nonfinancial corporate (NFC) business, excluding non-profit institutions serving households (NPISH), government, and owner-occupied housing.

2.A.8 Data Sources

Table 2.10 summarizes the BEA Fixed Assets tables used in the GPIM construction.

Table 2.10. BEA Fixed Assets tables used in GPIM construction

Table	Content	Use
FAAt201	Nominal gross investment by asset	$I_{j,t}^{\text{NOM}}$
FAAt401	Current-cost net stock by asset	$K_{j,t}^{\text{NET,CC}}$
FAAt402	Current-cost depreciation by asset	Validation

Notes: All tables are from the BEA Fixed Assets Accounts, retrieved via the BEA API (**DatasetName:** FixedAssets). Asset classes: ME = nonresidential equipment; NRC = nonresidential structures. Sectoral scope: nonfinancial corporate business.

Appendix 2.B Appendix: United States Specification B — Full Estimation Results

This appendix reports the full empirical estimation matrix across all 48 specifications of the aggregate capital model (Specification A). The specifications span two interaction sets (Set 1: standard wage interaction; Set 2: effective wage interaction scaled by $(1 - \iota_{t-1})$), two value-added bases (Gross Value Added [GVA] and Net Value Added [NVA]), six wage share measures per basis (NFC Baseline, Corporate, Shaikh-Tonak Tier 1, Shaikh-Tonak Tier 2, Clean Corporate Tier 1, and Clean Corporate Tier 2), and two structural break modes (without step dummies; with Bai-Perron step shifts).

Table 2.11. Specification A, Set 1: Standard Wage Interaction (GVA Basis)

Model	M1.1a	M1.1b	M1.2a	M1.2b	M1.3a	M1.3b	M1.4a	M1.4b	M1.5a	M1.5b	M1.6a	M1.6b
$\hat{\theta}_0$ (Capital)	0.58*** (0.02)	0.60*** (0.06)	0.58*** (0.02)	0.61*** (0.07)	0.58*** (0.02)	0.59*** (0.06)	0.57 (0.02)	0.55 (0.08)	0.58*** (0.02)	0.61*** (0.07)	0.57 (0.02)	0.54 (0.08)
$\hat{\theta}_1$ (Interaction)	-0.02 (0.13)	-0.08 (0.10)	-0.03 (0.12)	-0.04 (0.10)	0.06 (0.14)	-0.08 (0.11)	-0.19 (0.12)	-0.23 (0.11)	0.01 (0.12)	-0.07 (0.11)	-0.20 (0.12)	-0.29 (0.11)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
$C_{\text{Shin}} [H_0: \text{coint.}]$	0.06	0.04	0.06	0.04	0.05	0.03	0.06	0.05	0.05	0.04	0.06	0.05
$S_{\text{Type } 2} [H_0: \text{coint.}]$	3.30	11.34	2.70	11.41	5.82	9.68	2.56	12.13	5.30	10.48	3.74	12.92
$t_{\text{ADF}} [H_0: \text{no coint.}]$	-2.41	-2.81	-2.33	-2.73	-4.15**	-3.56*	-1.80	-2.15	-3.33	-2.66	-1.75	-2.12
Bandwidth ratio b	0.275	0.187	0.286	0.193	0.285	0.200	0.315	0.254	0.304	0.201	0.321	0.256
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed- b Wald tables. Standard errors in parentheses.

Table 2.12. Specification A, Set 1: Standard Wage Interaction (NVA Basis)

Model	M1.7a	M1.7b	M1.8a	M1.8b	M1.9a	M1.9b	M1.10a	M1.10b	M1.11a	M1.11b	M1.12a	M1.12b
$\hat{\theta}_0$ (Capital)	0.58*** (0.02)	0.60*** (0.06)	0.58*** (0.02)	0.60*** (0.06)	0.58*** (0.02)	0.60*** (0.06)	0.57 (0.02)	0.56 (0.08)	0.59*** (0.02)	0.71*** (0.08)	0.57 (0.02)	0.63 (0.08)
$\hat{\theta}_1$ (Interaction)	-0.02 (0.13)	-0.08 (0.10)	-0.02 (0.13)	-0.08 (0.10)	-0.02 (0.13)	-0.08 (0.10)	-0.19 (0.11)	-0.26 (0.09)	-0.09 (0.09)	-0.32* (0.08)	-0.20 (0.10)	-0.29 (0.10)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
$C_{\text{Shin}} [H_0: \text{coint.}]$	0.06	0.04	0.06	0.04	0.06	0.04	0.06	0.05	0.06*	0.03	0.06	0.05
$S_{\text{Type } 2} [H_0: \text{coint.}]$	3.30	11.34	3.30	11.34	3.30	11.34	3.37	12.47	5.19	10.19	4.81	12.40
$t_{\text{ADF}} [H_0: \text{no coint.}]$	-2.41	-2.81	-2.41	-2.81	-2.41	-2.81	-1.97	-2.40	-2.42	-3.61*	-1.91	-2.31
Bandwidth ratio b	0.275	0.187	0.275	0.187	0.275	0.187	0.277	0.235	0.242	0.156	0.280	0.239
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed- b Wald tables. Standard errors in parentheses.

Table 2.13. Specification A, Set 2: Effective Wage Interaction (GVA Basis)

Model	M2.1a	M2.1b	M2.2a	M2.2b	M2.3a	M2.3b	M2.4a	M2.4b	M2.5a	M2.5b	M2.6a	M2.6b
$\hat{\theta}_0$ (Capital)	0.58*** (0.02)	0.56*** (0.06)	0.58*** (0.02)	0.57*** (0.06)	0.58*** (0.02)	0.54*** (0.06)	0.57 (0.02)	0.55 (0.07)	0.58*** (0.02)	0.56*** (0.06)	0.56 (0.02)	0.56 (0.08)
$\hat{\theta}_1$ (Interaction)	0.00 (0.10)	-0.17 (0.07)	-0.00 (0.10)	-0.15 (0.07)	0.04 (0.11)	-0.20 (0.09)	-0.22 (0.10)	-0.37 (0.09)	0.01 (0.10)	-0.18 (0.08)	-0.20 (0.09)	-0.39 (0.17)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0 : coint.]	0.06	0.04	0.06	0.04	0.05	0.03	0.06	0.04	0.05	0.03	0.06	0.04
$S_{Type\ 2}$ [H_0 : coint.]	10.76	9.07	6.17	9.61	4.39	7.88	5.72	12.93	12.82	8.55	6.98	10.32
t_{ADF} [H_0 : no coint.]	-2.32	-3.06	-2.26	-2.96	-4.02**	-2.93	-1.79	-2.31	-3.37	-2.87	-1.68	-2.20
Bandwidth ratio b	0.298	0.169	0.307	0.173	0.291	0.176	0.298	0.241	0.310	0.178	0.315	0.245
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed- b Wald tables. Standard errors in parentheses.

Table 2.14. Specification A, Set 2: Effective Wage Interaction (NVA Basis)

Model	M2.7a	M2.7b	M2.8a	M2.8b	M2.9a	M2.9b	M2.10a	M2.10b	M2.11a	M2.11b	M2.12a	M2.12b
$\hat{\theta}_0$ (Capital)	0.58*** (0.02)	0.56*** (0.06)	0.58*** (0.02)	0.56*** (0.06)	0.58*** (0.02)	0.56*** (0.06)	0.57 (0.02)	0.55 (0.06)	0.58*** (0.02)	0.65*** (0.06)	0.57 (0.02)	0.56 (0.08)
$\hat{\theta}_1$ (Interaction)	0.00 (0.10)	-0.17 (0.07)	0.00 (0.10)	-0.17 (0.07)	0.00 (0.10)	-0.17 (0.07)	-0.21 (0.08)	-0.35 (0.07)	-0.04 (0.08)	-0.30** (0.06)	-0.19 (0.08)	-0.49 (0.13)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0 : coint.]	0.06	0.04	0.06	0.04	0.06	0.04	0.06	0.04	0.07*	0.03	0.06	0.04
$S_{Type\ 2}$ [H_0 : coint.]	10.76	9.07	10.76	9.07	10.76	9.07	6.13	13.18	11.05	9.06	7.50	10.89
t_{ADF} [H_0 : no coint.]	-2.32	-3.06	-2.32	-3.06	-2.32	-3.06	-2.00	-2.72	-2.22	-4.07**	-1.84	-2.58
Bandwidth ratio b	0.298	0.169	0.298	0.169	0.298	0.169	0.259	0.237	0.280	0.165	0.272	0.237
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed- b Wald tables. Standard errors in parentheses.

Table 2.15. Custom Monte Carlo Simulated Critical Bounds for CMPR IM-OLS Inference (Specification A)

Statistic / Parameter	Asymptotic 95%	Sim 90%	Sim 95%	Sim 99%
Panel A: Baseline Sample ($T = 94$, 1931–2024) — No Structural Breaks ($m = 0$)				
$t_{\text{level}} (k_t^{\text{cap}})$	1.960	3.877	5.119	8.502
$t_{\text{inter}} (k_t^{\text{cap}} \cdot C_t)$	N/A	3.123	4.006	6.238
CT_{IM} (Type 2 Cointegration)	0.070	0.0580	0.0705	0.1005
Panel B: Baseline Sample ($T = 94$, 1931–2024) — Bai-Perron Level-Shift Breaks ($m \geq 1$)				
$t_{\text{level}} (k_t^{\text{cap}})$	1.960	2.607	3.267	4.989
$t_{\text{inter}} (k_t^{\text{cap}} \cdot C_t)$	N/A	3.455	4.325	6.444
CT_{IM} (Type 2 Cointegration)	0.070	0.0482	0.0598	0.0895
Panel C: Supervisory Sample ($T = 76$, 1948–2024) — No Structural Breaks ($m = 0$)				
$t_{\text{level}} (k_t^{\text{cap}})$	1.960	3.900	5.155	8.387
$t_{\text{inter}} (k_t^{\text{cap}} \cdot C_t)$	N/A	3.139	4.018	6.206
CT_{IM} (Type 2 Cointegration)	0.070	0.0635	0.0780	0.1151
Panel D: Supervisory Sample ($T = 76$, 1948–2024) — Bai-Perron Level-Shift Breaks ($m \geq 1$)				
$t_{\text{level}} (k_t^{\text{cap}})$	1.960	2.687	3.371	5.188
$t_{\text{inter}} (k_t^{\text{cap}} \cdot C_t)$	N/A	3.427	4.277	6.213
CT_{IM} (Type 2 Cointegration)	0.070	0.0545	0.0696	0.1065

Notes: Critical bounds derived from a custom $N = 100,000$ iteration Monte Carlo simulation suite calibrated specifically to the $I(1) \times I(1)$ product integration geometry of the regressors and dynamic Bai-Perron break degrees of freedom. Panels A and B reflect the baseline 1931–2024 sample length ($T = 94$), while Panels C and D reflect the supervisory-adjusted Tier 2 wage share sample ($T = 76$). Bandwidth $b = M/T$ is fixed per Vogelsang & Wagner (2014) IM-OLS specification. Data Generating Process (H_0) constructs independent zero-drift random walks for base level variables ($k_t^{\text{cap}}, \omega_t$), with interaction product processes algebraically assembled prior to estimation. Parameter t -statistics are centered around their true theoretical values ($t^* = |(\hat{\beta}^* - \beta_{\text{true}})/SE|$) to ensure pure size calibration rather than power distortion.

Appendix 2.C Appendix: United States Specification B - Heterogeneous Capital Stock Empirical Results

This appendix reports the full empirical estimation matrix across all 48 specifications of the heterogeneous capital model (Specification B). The specifications span two interaction sets (Set 1: standard wage interaction; Set 2: effective wage interaction scaled by $(1 - \iota_{t-1})$), two value-added bases (Gross Value Added [GVA] and Net Value Added [NVA]), six wage share measures per basis (NFC Baseline, Corporate Baseline, Shaikh-Tonak Tier 1, Shaikh-Tonak Tier 2, Clean Corporate Tier 1, and Clean Corporate Tier 2), and two structural break modes (without step dummies; with Bai-Perron step shifts).

Table 2.16. Specification B, Set 1: Standard Wage Interaction (GVA Basis)

Model	M2.1a	M2.2b	M2.5a	M2.6b	M2.9a	M2.10b	M2.13a	M2.14b	M2.17a	M2.18b	M2.21a	M2.22b
$\hat{\alpha}$ (Intercept)	12.90 (0.07)	12.82 (0.06)	12.89 (0.07)	12.82 (0.06)	12.90 (0.07)	12.82 (0.06)	13.85 (0.08)	13.52 (0.06)	12.89 (0.07)	12.81 (0.07)	13.85 (0.08)	13.52 (0.06)
$\hat{\theta}_{NRC}$	0.60 (0.03)	0.51 (0.08)	0.61 (0.03)	0.47 (0.07)	0.61 (0.03)	0.48 (0.07)	0.51 (0.02)	0.50 (0.07)	0.59 (0.03)	0.52 (0.09)	0.51 (0.02)	0.49 (0.07)
$\hat{\theta}_{\kappa}$	0.38 (0.14)	0.67 (0.26)	0.40 (0.14)	0.44 (0.22)	0.39 (0.14)	0.53 (0.23)	0.38 (0.12)	1.09 (0.20)	0.36 (0.14)	0.60 (0.33)	0.34 (0.11)	1.11 (0.20)
$\hat{\theta}_{\omega\kappa}$	-2.75 (4.07)	-4.89 (3.10)	7.95 (5.98)	0.05 (2.97)	4.01 (7.41)	-4.20 (4.18)	-16.40 (6.63)	-10.51 (5.86)	-9.89 (4.85)	-5.93 (3.96)	-18.27 (5.56)	-9.04 (5.55)
$\hat{\theta}_{\kappa p}$	-0.68 (0.58)	1.38 (0.82)	-2.07 (0.87)	0.16 (0.64)	-1.28 (0.61)	0.64 (0.65)	-1.43 (0.56)	-0.35 (0.47)	0.01 (0.57)	1.63 (0.99)	-0.78 (0.54)	-0.35 (0.47)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
$C_{Shin} [H_0: \text{coint.}]$	0.05	0.03	0.04	0.03	0.05	0.03	0.03	0.02	0.05	0.03	0.02	0.02
$S_{Type\ 2} [H_0: \text{coint.}]$	16.11	9.55	16.40	12.60	16.25	11.41	9.98	2.78	16.37	10.87	8.42	3.26
$t_{ADF} [H_0: \text{no coint.}]$	-4.16*	-4.46**	-4.25*	-4.67**	-4.26*	-4.69**	-2.56	-3.22	-4.26*	-4.50**	-2.58	-3.17
Bandwidth ratio b	0.209	0.176	0.215	0.173	0.211	0.174	0.211	0.203	0.213	0.191	0.210	0.203
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log real capacity). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2024). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed- b Wald tables. Standard errors in parentheses.

Table 2.17. Specification B, Set 1: Standard Wage Interaction (NVA Basis)

Model	M2.3a	M2.4b	M2.7a	M2.8b	M2.11a	M2.12b	M2.15a	M2.16b	M2.19a	M2.20b	M2.23a	M2.24b
$\hat{\alpha}$ (Intercept)	12.90 (0.07)	12.82 (0.06)	12.90 (0.07)	12.82 (0.06)	12.90 (0.07)	12.82 (0.06)	13.87 (0.08)	13.57 (0.06)	12.90 (0.07)	12.82 (0.07)	13.88 (0.08)	13.57 (0.06)
$\hat{\theta}_{NRC}$	0.60 (0.03)	0.51 (0.08)	0.60 (0.03)	0.51 (0.08)	0.60 (0.03)	0.51 (0.08)	0.51 (0.02)	0.52 (0.07)	0.59 (0.03)	0.50 (0.09)	0.51 (0.02)	0.50 (0.07)
$\hat{\theta}_{\kappa}$	0.38 (0.14)	0.67 (0.26)	0.38 (0.14)	0.67 (0.26)	0.38 (0.14)	0.67 (0.26)	0.35 (0.11)	0.99 (0.18)	0.36 (0.14)	0.61 (0.34)	0.34 (0.11)	1.02 (0.18)
$\hat{\theta}_{\omega\kappa}$	-2.75 (4.07)	-4.89 (3.10)	-2.79 (4.14)	-4.96 (3.15)	-2.75 (4.07)	-4.89 (3.10)	-13.06 (4.24)	-13.10 (4.38)	-7.06 (3.32)	-2.73 (2.62)	-12.13 (3.50)	-11.13 (3.99)
$\hat{\theta}_{\kappa p}$	-0.68 (0.58)	1.38 (0.82)	-0.68 (0.58)	1.38 (0.82)	-0.68 (0.58)	1.38 (0.82)	-0.99 (0.52)	-0.37 (0.42)	0.33 (0.65)	1.50 (0.98)	-0.49 (0.53)	-0.30 (0.43)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
$C_{Shin} [H_0: \text{coint.}]$	0.05	0.03	0.05	0.03	0.05	0.03	0.02	0.02	0.05	0.03	0.02	0.02
$S_{Type\ 2} [H_0: \text{coint.}]$	16.11	9.55	16.11	9.55	16.11	9.55	8.41	6.15	16.35	11.90	7.50	4.26
$t_{ADF} [H_0: \text{no coint.}]$	-4.16*	-4.46**	-4.16*	-4.46**	-4.16*	-4.46**	-2.61	-3.55	-4.20*	-4.49**	-2.50	-3.44
Bandwidth ratio b	0.209	0.176	0.209	0.176	0.209	0.176	0.207	0.205	0.213	0.192	0.211	0.205
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log real capacity). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2024). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed- b Wald tables. Standard errors in parentheses.

Table 2.18. Specification B, Set 2: Effective Wage Interaction (GVA Basis)

Model	M2.25a	M2.26b	M2.29a	M2.30b	M2.33a	M2.34b	M2.37a	M2.38b	M2.41a	M2.42b	M2.45a	M2.46b
$\hat{\alpha}$ (Intercept)	12.82 (0.07)	12.77 (0.06)	12.83 (0.07)	12.78 (0.07)	12.79 (0.07)	12.76 (0.06)	13.68 (0.06)	13.66 (0.05)	12.80 (0.07)	12.77 (0.06)	13.67 (0.05)	13.61 (0.06)
$\hat{\theta}_{\text{NRC}}$	0.61 (0.02)	0.58 (0.08)	0.61 (0.02)	0.55 (0.08)	0.61 (0.02)	0.62 (0.08)	0.55 (0.01)	0.51 (0.01)	0.61 (0.02)	0.60 (0.09)	0.55 (0.01)	0.70 (0.03)
$\hat{\theta}_{\kappa}$	0.32 (0.13)	0.39 (0.29)	0.32 (0.14)	0.44 (0.30)	0.31 (0.14)	0.14 (0.38)	0.36 (0.08)	0.02 (0.11)	0.31 (0.14)	0.36 (0.29)	0.34 (0.08)	0.72 (0.10)
$\hat{\theta}_{\omega\kappa}$	-12.77 (3.81)	-10.13 (3.43)	-13.73 (4.45)	-12.11 (3.82)	-15.99 (4.94)	-12.29 (4.22)	-20.77 (4.11)	-19.87 (3.99)	-14.54 (4.60)	-11.15 (3.63)	-18.85 (3.32)	-15.55 (3.54)
$\hat{\theta}_{\kappa p}$	0.18 (0.51)	0.92 (0.69)	0.35 (0.58)	1.44 (0.79)	-0.39 (0.45)	0.58 (0.60)	-1.37 (0.42)	-0.81 (0.39)	0.08 (0.52)	0.88 (0.66)	-0.96 (0.40)	0.10 (0.41)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
$C_{\text{Shin}} [H_0: \text{coint.}]$	0.03	0.02	0.04	0.03	0.03	0.02	0.03	0.02	0.03	0.02	0.03	0.02
$S_{\text{Type } 2} [H_0: \text{coint.}]$	14.97	5.93	15.18	6.02	14.58	3.71	13.20	11.19	14.75	6.00	12.33	9.93
$t_{\text{ADF}} [H_0: \text{no coint.}]$	-4.33*	-4.58**	-4.36*	-4.52**	-4.40*	-4.77**	-4.12	-4.02	-4.41*	-4.69**	-4.13*	-3.97
Bandwidth ratio b	0.185	0.173	0.194	0.175	0.192	0.173	0.206	0.206	0.195	0.179	0.207	0.207
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log real capacity). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2024). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed- b Wald tables. Standard errors in parentheses.

Table 2.19. Specification B, Set 2: Effective Wage Interaction (NVA Basis)

Model	M2.27a	M2.28b	M2.31a	M2.32b	M2.35a	M2.36b	M2.39a	M2.40b	M2.43a	M2.44b	M2.47a	M2.48b
$\hat{\alpha}$ (Intercept)	12.82 (0.07)	12.77 (0.06)	12.82 (0.07)	12.77 (0.06)	12.82 (0.07)	12.77 (0.06)	13.72 (0.04)	13.69 (0.04)	12.83 (0.07)	12.78 (0.07)	13.73 (0.04)	13.73 (0.04)
$\hat{\theta}_{\text{NRC}}$	0.61 (0.02)	0.58 (0.08)	0.61 (0.02)	0.58 (0.08)	0.61 (0.02)	0.58 (0.08)	0.54 (0.01)	0.51 (0.01)	0.60 (0.02)	0.60 (0.08)	0.54 (0.01)	0.54 (0.01)
$\hat{\theta}_{\kappa}$	0.32 (0.13)	0.39 (0.29)	0.32 (0.13)	0.39 (0.29)	0.32 (0.13)	0.39 (0.29)	0.33 (0.06)	0.06 (0.09)	0.32 (0.13)	0.39 (0.29)	0.31 (0.06)	0.31 (0.06)
$\hat{\theta}_{\omega\kappa}$	-12.77 (3.81)	-10.13 (3.43)	-12.97 (3.86)	-10.28 (3.48)	-12.77 (3.81)	-10.13 (3.43)	-17.05 (2.34)	-16.76 (2.55)	-11.08 (3.48)	-7.93 (2.96)	-14.97 (2.01)	-14.97 (2.01)
$\hat{\theta}_{\kappa p}$	0.18 (0.51)	0.92 (0.69)	0.18 (0.51)	0.92 (0.69)	0.18 (0.51)	0.92 (0.69)	-0.79 (0.32)	-0.71 (0.31)	0.69 (0.63)	1.41 (0.75)	-0.31 (0.33)	-0.31 (0.33)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
$C_{\text{Shin}} [H_0: \text{coint.}]$	0.03	0.02	0.03	0.02	0.03	0.02	0.02	0.02	0.03	0.03	0.02	0.02
$S_{\text{Type } 2} [H_0: \text{coint.}]$	14.97	5.93	14.97	5.93	14.97	5.93	10.28	8.70	15.37	6.37	8.96	8.96
$t_{\text{ADF}} [H_0: \text{no coint.}]$	-4.33*	-4.58**	-4.33*	-4.58**	-4.33*	-4.58**	-4.71**	-4.57**	-4.34*	-4.60**	-4.24*	-4.24*
Bandwidth ratio b	0.185	0.173	0.185	0.173	0.185	0.173	0.208	0.208	0.193	0.181	0.208	0.208
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log real capacity). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2024). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed- b Wald tables. Standard errors in parentheses.

Table 2.20. Custom Monte Carlo Simulated Critical Bounds Across Structural Break Regimes (Specification B)

Statistic / Parameter	Asymptotic 95%	Sim 90%	Sim 95%	Sim 99%	Empirical (M2.44b)
Panel A: No Structural Breaks ($m = 0$, Sample $T = 94$)					
$t_{\text{NRC}} (k_t^{\text{NRC}})$	1.960	1.842	2.195	2.910	7.50**
$t_{\kappa} (\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}})$	1.960	1.895	2.241	2.980	1.34
$t_{\omega\kappa} (\kappa_t \cdot C_t)$	N/A	2.115	2.512	3.340	-2.68**
$t_{\kappa p} (\kappa_t \cdot p_t^{\kappa})$	N/A	2.080	2.475	3.295	1.88*
CT_{IM} (Type 2 Global Test)	18.50	11.20	13.45	18.90	6.37**
Panel B: One Endogenous Structural Break ($m = 1$, e.g., M2.40b)					
$t_{\text{NRC}} (k_t^{\text{NRC}})$	1.960	1.920	2.280	3.010	—
$t_{\kappa} (\kappa_t)$	1.960	1.970	2.330	3.080	—
$t_{\omega\kappa} (\kappa_t \cdot C_t)$	N/A	2.180	2.590	3.430	—
$t_{\kappa p} (\kappa_t \cdot p_t^{\kappa})$	N/A	2.150	2.550	3.390	—
CT_{IM} (Type 2 Global Test)	19.80	12.10	14.50	20.20	—
Panel C: Two Endogenous Structural Breaks ($m = 2$, e.g., M2.46b)					
$t_{\text{NRC}} (k_t^{\text{NRC}})$	1.960	1.995	2.370	3.120	—
$t_{\kappa} (\kappa_t)$	1.960	2.050	2.420	3.190	—
$t_{\omega\kappa} (\kappa_t \cdot C_t)$	N/A	2.250	2.670	3.520	—
$t_{\kappa p} (\kappa_t \cdot p_t^{\kappa})$	N/A	2.210	2.630	3.470	—
CT_{IM} (Type 2 Global Test)	21.20	13.00	15.50	21.60	—
Panel D: Three Endogenous Structural Breaks ($m = 3$, Preferred Model M2.44b)					
$t_{\text{NRC}} (k_t^{\text{NRC}})$	1.960	2.075	2.455	3.210	7.50**
$t_{\kappa} (\kappa_t)$	1.960	2.130	2.515	3.305	1.34
$t_{\omega\kappa} (\kappa_t \cdot C_t)$	N/A	2.315	2.745	3.615	-2.68**
$t_{\kappa p} (\kappa_t \cdot p_t^{\kappa})$	N/A	2.275	2.705	3.560	1.88*
CT_{IM} (Type 2 Global Test)	22.65	13.90	16.55	22.95	6.37**
Panel E: Four Endogenous Structural Breaks ($m = 4$, e.g., M2.18b, M2.34b)					
$t_{\text{NRC}} (k_t^{\text{NRC}})$	1.960	2.150	2.540	3.310	—
$t_{\kappa} (\kappa_t)$	1.960	2.210	2.610	3.420	—
$t_{\omega\kappa} (\kappa_t \cdot C_t)$	N/A	2.380	2.820	3.710	—
$t_{\kappa p} (\kappa_t \cdot p_t^{\kappa})$	N/A	2.340	2.780	3.650	—
CT_{IM} (Type 2 Global Test)	24.10	14.80	17.60	24.30	—

Notes: Critical bounds derived from a 100,000-pass specification-dependent Monte Carlo suite tailored to Specification B's 4-variable stochastic design ($k_t^{\text{NRC}}, \kappa, \kappa \cdot C, \kappa \cdot p^{\kappa}$) across all structural break counts ($m = 0 \dots 4$). The DGP under H_0 simulates independent zero-drift random walks for base variables and algebraically constructs product terms, with dynamic Bai-Perron break selection applied recursively. Simulated parameter t -statistics are centered around true DGP parameters ($t^* = |(\hat{\beta}^* - \beta_{\text{true}})/SE|$) to prevent artificial bound inflation. Empirical values reported for preferred Model M2.44b ($m = 3$). Bold empirical values denote significance based on the simulated 95% bound (* $p < 0.05$, ** $p < 0.01$).

Appendix 2.D Appendix: Chile - External Constraint and Regime Shifting Results

Table 2.21. Comparative Cointegration Estimates across Net and Gross Wage Shares (Chile, 1943–2010)

	Net Wage Share (ω^{net})				Gross Wage Share (ω^{gross})			
	Baseline	Bai-Perron	Comp. Thresh-old	Parsimonious	Baseline	Bai-Perron	Comp. Thresh-old	Parsimonious
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\theta}_{k^{\text{NR}}} \text{ (Structures } k^{\text{NR}})$	0.96*** (0.01)	0.73*** (0.12)	0.67*** (0.09)	0.79*** (0.07)	0.99*** (0.00)	0.98*** (0.05)	0.70*** (0.08)	0.82*** (0.08)
$\hat{\theta}_{\kappa} \text{ (Mechanization Gap } \kappa)$	0.03 (0.25)	0.56 (0.50)	0.53 (0.40)	0.12 (0.39)	-0.41 (0.29)	0.22 (0.27)	-0.14 (0.44)	-1.48* (0.54)
$\hat{\theta}_{\omega\kappa} \text{ (Wage Interaction)}$	-0.28 (0.22)	-0.16 (0.21)	-0.12 (0.17)	-0.01 (0.22)	0.01 (0.38)	-0.35 (0.26)	0.96 (0.73)	3.29* (1.07)
$\hat{\theta}_{\kappa D} \text{ (Regime Shift } \kappa D)$	— (0.00)	-0.18 (0.28)	0.36* (0.13)	0.34 (0.16)	— (0.00)	5.13*** (1.00)	0.66 (0.31)	1.52** (0.45)
$\hat{\theta}_{\omega\kappa D} \text{ (Triple Interaction)}$	—	-0.09 (0.33)	-0.15 (0.18)	-0.17 (0.23)	—	- 11.86*** (2.35)	-0.95 (0.72)	-3.15* (1.04)
$\hat{\theta}_p \text{ (Price Control } \tilde{p})$	-0.05 (0.11)	-0.19** (0.09)	-0.24*** (0.07)	—	-0.04 (0.11)	0.02 (0.07)	-0.20*** (0.07)	—
Regime Threshold (D_t)	None	BP (1987)	$Z_t \leq Z_t$ 0.425	$Z_t \leq Z_t$ 0.425	None	BP (1999)	$Z_t \leq Z_t$ 0.425	$Z_t \leq Z_t$ 0.425
Price Control Included	Yes	Yes	Yes	No	Yes	Yes	Yes	No
$CT_{\text{IM}} [H_0: \text{Cointegration}]$	—	12.88	11.72	15.20	—	14.55	8.50	8.39
$t_{\text{ADF}} [H_0: \text{Unit Root}]$	-1.84	-2.72	-3.82*	-3.81*	-1.92	-2.98	-4.11**	-3.81*
Bandwidth ratio b	0.626	0.650	0.654	0.654	0.635	0.652	0.655	0.654
Sample Size T	68	68	68	68	68	68	68	68

Table 2.22. Comparative Cointegration Estimates across Net and Gross Rates of Exploitation (Chile, 1943–2010)

	Net Rate of Exploitation (e^{net})				Gross Rate of Exploitation (e^{gross})			
	Baseline	Bai-Perron	Comp. Thresh-old	Parsimonious	Baseline	Bai-Perron	Comp. Thresh-old	Parsimonious
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\theta}_{k^{\text{NR}}}$ (Structures k^{NR})	0.96*** (0.01)	0.73*** (0.11)	0.68*** (0.09)	0.80*** (0.07)	0.99*** (0.01)	0.98*** (0.06)	0.71*** (0.08)	0.82*** (0.08)
$\hat{\theta}_{\kappa}$ (Mechanization Gap κ)	-0.21 (0.11)	0.45 (0.42)	0.43 (0.28)	0.15 (0.25)	-0.42*** (0.08)	-0.05 (0.27)	0.51 (0.27)	0.85 (0.38)
$\hat{\theta}_{e\kappa}$ (Exploitation Interaction)	0.14 (0.11)	0.08 (0.10)	0.06 (0.07)	0.02 (0.09)	0.00 (0.10)	0.09 (0.07)	-0.19 (0.17)	-0.69* (0.23)
$\hat{\theta}_{\kappa D}$ (Regime Shift κD)	— (0.00)	-0.27 (0.10)	0.21* (0.07)	0.15 (0.08)	— (0.00)	-2.39*** (0.50)	-0.02 (0.23)	-0.73 (0.29)
$\hat{\theta}_{e\kappa D}$ (Triple Interaction)	—	0.05 (0.15)	0.09 (0.07)	0.12 (0.10)	—	1.83*** (0.36)	0.20 (0.16)	0.67* (0.22)
$\hat{\theta}_p$ (Price Control $\tilde{p}$)	-0.04 (0.11)	-0.18** (0.09)	-0.22*** (0.07)	—	-0.04 (0.11)	0.02 (0.07)	-0.20*** (0.07)	—
Regime Threshold (D_t)	None	BP (1987)	$Z_t \leq 0.425$	$Z_t \leq 0.425$	None	BP (1999)	$Z_t \leq 0.425$	$Z_t \leq 0.425$
Price Control Included	Yes	Yes	Yes	No	Yes	Yes	Yes	No
$CT_{\text{IM}} [H_0: \text{Cointegration}]$	—	12.52	10.83	14.99	—	14.40	7.96	8.34
$t_{\text{ADF}} [H_0: \text{Unit Root}]$	-1.89	-2.73	-3.86*	-3.84*	-1.96	-3.05	-4.15**	-3.89*
Bandwidth ratio b	0.623	0.650	0.654	0.654	0.635	0.652	0.655	0.654
Sample Size T	68	68	68	68	68	68	68	68

Appendix 2.E Appendix: Critical Values and Inference for the Cointegration Diagnostics

Three families of test statistic carry the cointegration verdict in this chapter, and they do not draw their critical values from the same source. The residual-based diagnostics use published constants. The fixed- b Wald statistics do not, because their limiting distribution is jointly indexed by the bandwidth ratio b and the number of integrated regressors k , so no single tabulated value covers the specifications estimated here.

Table 2.23. Critical Values for the Residual-Based Cointegration Diagnostics

Test	Source	10%	5%	1%
Residual ADF (t_{ADF})	MacKinnon (2010), Case 2, constant only, $m = 4$	-3.81	-4.10	-4.64
Shin LM (C_{Shin}), $k = 2$	Shin (1994), Table 1, demeaned C_μ	0.163	0.221	0.380
Shin LM (C_{Shin}), $k = 3$	Shin (1994), Table 1, demeaned C_μ	0.121	0.159	0.271

Notes: The residual ADF statistic tests the null of no cointegration, so rejection supports a cointegrating relation. The Shin LM statistic reverses the null: it tests the null of cointegration, so a value below the critical value fails to reject and is the favorable outcome. m denotes the number of regressors entering the MacKinnon response surface; k denotes the number of integrated regressors entering the Shin statistic.

For the fixed- b Wald statistics reported as CT_{IM} and for the coefficient significance stars in every results table, critical values are generated by simulation rather than read from a table. The protocol holds the estimated specification fixed and varies only the innovation draws. For each specification I generate integrated regressors as independent random walks of the same length as the estimation sample, impose the null under test, apply the same Frisch-Waugh-Lovell orthogonalization and the same deterministic components used in estimation, and compute the IM-OLS statistic on each replication. The empirical quantiles of the resulting distribution supply the thresholds. Replication counts are 100,000 for the United States specifications and 10,000 for the Chilean threshold models, where each pass carries the additional cost of a profile search over the threshold parameter.

Two features of this protocol matter for reading the tables. First, because the simulated null reproduces the bandwidth ratio actually used in estimation, the reported stars already incorporate the fixed- b correction and are not comparable to standard normal or chi-squared thresholds. Second, because the Chilean critical bounds are simulated under profile search, they are conservative relative to a fixed-threshold benchmark: the search itself consumes degrees of freedom, and the simulation charges for it.

Appendix 2.F Appendix: Order of Integration and Stochastic Pre-Testing Battery for the United States and Chile

A In Cointegrating Multivariate Polynomial Regressions (CMPR) estimated via Integrated Modified OLS (IM-OLS) (Vogelsang and Wagner, 2014, 2024), valid asymptotic inference and pivotal fixed- b Wald distributions require all stochastic regressors to be integrated of order one ($I(1)$). When non-linear interaction terms ($k_t \times \omega_{t-1}$, $k_t \times \omega_{t-1} \times (1 - \iota_{t-1})$) or regime-dependent threshold regressors ($k_t \times \omega_{t-1} \times \mathbf{1}\{Z_t \leq \hat{\gamma}\}$) enter the long-run relation, establishing their stochastic integration order prior to estimation is an essential prerequisite. This appendix reports the complete unit-root and stationarity test batteries for both economies across all baseline series, non-linear interactions, and demand controls.

B Table 2.24 outlines the sample boundaries, degree-of-freedom spans, and zero-NA estimation windows across the empirical dataset. For the United States, the stochastic pre-testing battery spans 1929–2024 ($N_{\text{lev}} = 96$, $N_{\text{diff}} = 95$). For Chile, the estimation window spans 1943–2010 ($N_{\text{lev}} = 68$, $N_{\text{diff}} = 67$).

Table 2.24. Data Series Coverage, Time Horizons, and Estimation Windows

Economy / Data Series Scope	Year Range	N_{lev}	N_{diff}	Empirical Function
U.S. NIPA Compensation Shares	1929–2024	96	95	Distribution Series
U.S. Private Core Gross Capital (K^{cap})	1925–2024	100	99	GPIM Capital Series
U.S. Nonfinancial Corporate Output (Y^{NFC})	1929–2024	96	95	Chained Real Series
U.S. Unit Root Pre-Testing Sample	1929–2024	96	95	Stochastic Audit
U.S. Long-Run Estimation Window	1931–2024	94	93	IM-OLS Regressions
Chile Historical Macroeconomic Series	1900–2024	125	124	Historical Panel
Chile Descriptive Stylized Facts	1940–2010	71	70	Summary Statistics
Chile Long-Run Estimation Window	1943–2010	68	67	Threshold CMPR

Note: U.S. data are from BEA NIPA Tables and Fixed Assets Accounts. Chilean data are from Central Bank historical accounts and ClioLab databases. For Chile, the 1943 start corresponds to the enactment of Law No. 7,200 (1942), which abandoned the gold-standard convertibility rule and established state-led industrialization credit allocation alongside CORFO. Historical external and terms-of-trade series end in 2010.

2.F.1 United States: Unit Root Test Batteries and Interaction Properties

A We evaluate the stochastic properties of U.S. level variables and their interactions using five standard tests: Augmented Dickey-Fuller (ADF), DF-GLS (ERS), Phillips-Perron (PP), Kwiatkowski-Phillips-Schmidt-Shin (KPSS), and Zivot-Andrews (ZA) tests with endogenous structural breaks. For level specifications, test regressions include a deterministic trend. For first-differenced specifications,

regressions specify an intercept. In the Zivot-Andrews procedure, structural breaks are estimated endogenously in both intercept and trend (`model = "both"`).

B Table 2.25 reports the test battery for the corporate Intellectual Property Products (IPP) gross investment share ($\iota_t = I_t^{\text{IPP}}/I_t^{\text{TOTAL}}$), the capital composition ratio ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$), private core gross capital ($k^{\text{cap}} = \ln[K^{\text{ME}} + K^{\text{NRC}}]$), and their double and triple interaction terms with eight distributive wage share variants (ω_i).

Table 2.25. Unit Root Test Battery for U.S. IPP Share and Capital Composition Interactions (1929–2024)

Variable / Measure	N_{lev}	N_{diff}	ADF Level	ADF Diff	ERS Level	ERS Diff	KPSS Level	KPSS Diff	ZA (Year)	Break	Verdict
Corp IPP Share (ι_t)	96	95	-2.432	-7.129***	-2.104	-4.883***	0.444***	0.108	-4.529 (1976)		I(1)
Capital Composition (κ_t)	96	95	-0.196	-4.592***	-1.024	-2.737***	0.593***	0.761***	-3.632 (1974)		I(1)
Private Core Capital (k^{cap})	96	95	-1.811	-5.744***	-2.041	-0.837	0.318***	0.237	-4.792 (1972)		I(1)
Panel B: IPP and Capital Composition Interactions											
$\iota_t \times \omega_{\text{CORP,GVA}}$	96	95	-3.296*	-7.220***	-3.008	-4.622***	0.331***	0.047	-4.573 (1984)		I(1)
$\iota_t \times \omega_{\text{CORP,NVA}}$	96	95	-3.128	-7.278***	-2.825	-4.682***	0.372***	0.053	-4.714 (1984)		I(1)
$\iota_t \times \omega_{\text{NFC,GVA}}$	96	95	-3.184*	-7.164***	-2.952	-4.881***	0.336***	0.051	-4.541 (1984)		I(1)
$\iota_t \times \omega_{\text{NFC,NVA}}$	96	95	-3.013	-7.254***	-2.764	-4.944***	0.378***	0.059	-4.666 (1984)		I(1)
$\iota_t \times \omega_{\text{ST,Tier1}}$	96	95	-3.013	-7.254***	-2.764	-4.944***	0.378***	0.059	-4.666 (1984)		I(1)
$\iota_t \times \omega_{\text{ST,Tier2}}$	96	95	-2.869	-7.480***	-2.622	-4.484***	0.428***	0.063	-4.917* (1984)		I(1)
$\iota_t \times \omega_{\text{ST,Tier1,GVA}}$	96	95	-3.184*	-7.164***	-2.952	-4.881***	0.336***	0.051	-4.541 (1984)		I(1)
$\iota_t \times \omega_{\text{ST,Tier2,GVA}}$	96	95	-3.027	-7.417***	-2.730	-4.479***	0.396***	0.057	-4.800 (1984)		I(1)
$\kappa_t \times \iota_t \times \omega_{\text{CORP,GVA}}$	96	95	-1.469	-6.952***	-1.525	-3.290***	0.532***	0.205	-4.610 (1985)		I(1)
$\kappa_t \times \iota_t \times \omega_{\text{CORP,NVA}}$	96	95	-1.408	-6.975***	-1.513	-3.198***	0.537***	0.217	-4.615 (1985)		I(1)
$\kappa_t \times \iota_t \times \omega_{\text{NFC,GVA}}$	96	95	-1.386	-6.952***	-1.480	-3.493***	0.533***	0.219	-4.634 (1985)		I(1)
$\kappa_t \times \iota_t \times \omega_{\text{NFC,NVA}}$	96	95	-1.316	-6.990***	-1.467	-3.399***	0.538***	0.232	-4.663 (1986)		I(1)
$\kappa_t \times \iota_t \times \omega_{\text{ST,Tier1}}$	96	95	-1.316	-6.990***	-1.467	-3.399***	0.538***	0.232	-4.663 (1986)		I(1)
$\kappa_t \times \iota_t \times \omega_{\text{ST,Tier2}}$	96	95	-1.465	-7.334***	-1.564	-3.194***	0.544***	0.201	-4.635 (1983)		I(1)
$\kappa_t \times \iota_t \times \omega_{\text{ST,Tier1,GVA}}$	96	95	-1.386	-6.952***	-1.480	-3.493***	0.533***	0.219	-4.634 (1985)		I(1)
$\kappa_t \times \iota_t \times \omega_{\text{ST,Tier2,GVA}}$	96	95	-1.529	-7.275***	-1.569	-3.315***	0.537***	0.192	-4.578 (1983)		I(1)

Note: Primary estimation sample: 1929–2024 ($N_{\text{lev}} = 96$, $N_{\text{diff}} = 95$). Significance stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. **Star Interpretation Warning:** For ADF, ERS, PP, and ZA, H_0 is a *unit root*; stars indicate rejection of the unit root (supporting stationarity). For KPSS, H_0 is *stationarity*; stars indicate rejection of stationarity (confirming non-stationarity in levels; absence of stars in differences confirms differenced stationarity). Critical values: ADF Level (−3.99, −3.43, −3.13), ADF Diff (−3.46, −2.88, −2.57); ERS Level (−3.58), ERS Diff (−2.59); KPSS Level (0.216, 0.146, 0.119), KPSS Diff (0.739, 0.463, 0.347); ZA Level (−5.57, −5.08, −4.82).

C Table 2.26 documents the integration properties of private core gross capital (k^{cap}) interacted with distributive wage shares and corporate IPP investment shares. Both Machinery and Equipment (k_t^{ME}) and Nonresidential Structures (k_t^{NRC}) are established $I(1)$ series. The apparent persistence in first-differenced capital composition ($\Delta\kappa_t$) under linear KPSS tests reflects the sharp structural deceleration of mechanization after 1973 (captured by the 1974 Zivot-Andrews break) rather than an $I(2)$ process (Perron, 1989). When capital is interacted with bounded wage shares and IPP shares ($k^{\text{cap}} \times \omega_i$ and $k^{\text{cap}} \times \omega_i \times \iota_t$), all double and triple interaction terms are confirmed to be strictly $I(1)$ ($ADF_{\text{diff}} < -6.89^{***}$, $KPSS_{\text{diff}} < 0.170$), ensuring valid CMPR estimation.

Table 2.26. Unit Root Test Battery for U.S. Core Capital and Distributive Interactions (1929–2024)

Variable / Measure	N_{lev}	N_{diff}	ADF Level	ADF Diff	ERS Level	ERS Diff	KPSS Level	KPSS Diff	ZA (Year)	Break	Verdict
Panel C: Private Core Capital Interactions (K^{cap})											
$k^{\text{cap}} \times \omega_{\text{CORP,GVA}}$	96	95	-1.910	-7.797***	-1.149	-3.808***	0.423***	0.170	-4.847* (2001)		I(1)
$k^{\text{cap}} \times \omega_{\text{CORP,NVA}}$	96	95	-2.547	-8.050***	-1.567	-3.005***	0.324***	0.108	-4.202 (1997)		I(1)
$k^{\text{cap}} \times \omega_{\text{NFC,GVA}}$	96	95	-2.018	-7.307***	-1.386	-5.587***	0.413***	0.133	-4.791 (2003)		I(1)
$k^{\text{cap}} \times \omega_{\text{NFC,NVA}}$	96	95	-2.645	-7.741***	-1.730	-5.153***	0.311***	0.086	-4.266 (2003)		I(1)
$k^{\text{cap}} \times \omega_{\text{ST,Tier1}}$	96	95	-2.645	-7.741***	-1.730	-5.153***	0.311***	0.086	-4.266 (2003)		I(1)
$k^{\text{cap}} \times \omega_{\text{ST,Tier2}}$	96	95	-3.037	-7.956***	-1.975	-4.444***	0.284***	0.065	-3.739 (1971)		I(1)
$k^{\text{cap}} \times \omega_{\text{ST,Tier1,GVA}}$	96	95	-2.018	-7.307***	-1.386	-5.587***	0.413***	0.133	-4.791 (2003)		I(1)
$k^{\text{cap}} \times \omega_{\text{ST,Tier2,GVA}}$	96	95	-2.775	-7.227***	-2.032	-5.237***	0.226***	0.061	-3.256 (1975)		I(1)
$k^{\text{cap}} \times \omega_{\text{CORP,GVA}} \times \iota_t$	96	95	-2.682	-6.943***	-2.245	-5.225***	0.421***	0.098	-4.578 (1984)		I(1)
$k^{\text{cap}} \times \omega_{\text{CORP,NVA}} \times \iota_t$	96	95	-2.561	-7.005***	-2.063	-5.341***	0.451***	0.113	-4.699 (1984)		I(1)
$k^{\text{cap}} \times \omega_{\text{NFC,GVA}} \times \iota_t$	96	95	-2.559	-6.891***	-2.182	-5.357***	0.424***	0.109	-4.503 (1984)		I(1)
$k^{\text{cap}} \times \omega_{\text{NFC,NVA}} \times \iota_t$	96	95	-2.434	-6.993***	-2.001	-5.469***	0.454***	0.125	-4.601 (1984)		I(1)
$k^{\text{cap}} \times \omega_{\text{ST,Tier1}} \times \iota_t$	96	95	-2.434	-6.993***	-2.001	-5.469***	0.454***	0.125	-4.601 (1984)		I(1)
$k^{\text{cap}} \times \omega_{\text{ST,Tier2}} \times \iota_t$	96	95	-2.304	-7.159***	-1.991	-5.156***	0.482***	0.122	-4.787 (1981)		I(1)
$k^{\text{cap}} \times \omega_{\text{ST,Tier1,GVA}} \times \iota_t$	96	95	-2.559	-6.891***	-2.182	-5.357***	0.424***	0.109	-4.503 (1984)		I(1)
$k^{\text{cap}} \times \omega_{\text{ST,Tier2,GVA}} \times \iota_t$	96	95	-2.421	-7.087***	-2.137	-5.094***	0.459***	0.107	-4.644 (1984)		I(1)

Note: Primary estimation sample: 1929–2024 ($N_{\text{lev}} = 96$, $N_{\text{diff}} = 95$). Significance stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Tests evaluated under identical specifications to Table 2.25.

2.F.2 Chile: Unit Root Test Batteries and Interaction Properties

A Table 2.27 reports the unit root battery for Chilean macroeconomic, distributive, and external constraint series over the 1943–2010 estimation window ($T = 68$). Real GDP (y_t), Nonresidential Structures (k_t^{NR}), Machinery and Equipment (k_t^{ME}), aggregate core capital (k_t^{cap}), and distributive wage shares ($\omega_t^{\text{net}}, \omega_t^{\text{gross}}$) are established as $I(1)$ non-stationary processes. The composite Balance-of-Payments index ($Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$) rejects non-stationarity in levels ($ADF = -6.661^{***}$), functioning as a stationary threshold state variable that partitions accumulation regimes.

Table 2.27. Unit Root Test Battery for Chilean Macroeconomic and Distributive Series (1943–2010)

Variable / Measure	N_{lev}	N_{diff}	ADF Lev	ADF Diff	KPSS Lev	KPSS Diff	ZA Break	Verdict
Log Real GDP (y_t)	68	67	-2.543	- 6.619***	0.262***	0.129	-4.68 (1975)	I(1)
Structures Capital (k_t^{NR})	68	67	-1.485	- 3.024**	0.485***	0.921***	-3.85 (1973)	I(1)
Machinery Capital (k_t^{ME})	68	67	-2.232	- 2.945**	0.472***	0.924***	-4.12 (1982)	I(1)
Core Capital (k_t^{cap})	68	67	-1.521	- 3.012**	0.480***	0.912***	-3.94 (1973)	I(1)
Capital Composition (κ_t)	68	67	-3.114	- 2.889**	0.426***	0.780***	-4.31 (1982)	I(1) with Break
Net Wage Share (ω_t^{net})	68	67	- 5.120***	- 11.671***	0.253***	0.057	-4.89* (1973)	I(1)
Gross Wage Share (ω_t^{gross})	68	67	- 5.120***	- 11.671***	0.253***	0.057	-4.89* (1973)	I(1)
Import Coverage Ratio ($\tilde{M}_t$)	68	67	- 4.986***	- 10.901***	0.234***	0.060	-4.52 (1982)	I(1)
Terms of Trade ($\tilde{S}_t$)	68	67	-2.442	- 10.171***	0.401***	0.348*	-4.11 (1975)	I(1)
Composite BoP Index (Z_t)	68	67	- 6.661***	- 11.292***	0.075	0.022	-5.42** (1982)	I(0) Thresh- old

Note: Sample spans 1943–2010 ($T = 68$). Significance stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. **Star Interpretation Warning:** For ADF, PP, and ZA, stars indicate rejection of the unit root null. For KPSS, stars indicate rejection of stationarity. Critical values: ADF Level ($-3.99, -3.45, -3.15$), ADF Diff ($-3.46, -2.89, -2.57$); KPSS Level ($0.216, 0.146, 0.119$), KPSS Diff ($0.739, 0.463, 0.347$); ZA Level ($-5.57, -5.08, -4.82$).

B Table 2.28 documents the integration orders of Chilean interaction terms. Raw structures (k_t^{NR}) exhibit moderate differenced persistence; however, FWL-orthogonalization (ξ_t^ω) and threshold conditioning ($k_t \times \omega_{t-1} \times \mathbf{1}\{Z_t \leq 0.425\}$) ensure that all entered regressors are strictly $I(1)$ in differences, eliminating higher-order polynomial contamination.

Table 2.28. Unit Root Test Battery for Chilean Interaction and Threshold Regressors (1943–2010)

Interaction / Regressor	N_{lev}	N_{diff}	ADF Lev	ADF Diff	KPSS Lev	KPSS Diff	Verdict
Raw Interaction ($k_t \times \omega_t^{\text{net}}$)	68	67	-2.442	- 10.171***	0.401***	0.348*	I(1)
FWL Interaction ($\xi_t^\omega = (k_t \times \omega_t^{\text{net}})^\perp$)	68	67	- 4.936***	- 10.758***	0.229***	0.059	I(1)
Distribution-FX ($\omega_t^{\text{net}} \times Z_t$)	68	67	- 4.986***	- 10.901***	0.234***	0.060	I(1)
Threshold Mechanization ($k_t \times \omega_{t-1} \times \mathbf{1}\{Z_t \leq 0.425\}$)	68	67	-3.393*	- 10.105***	0.189**	0.048	I(1)

Note: Sample spans 1943–2010 ($T = 68$). Significance stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Evaluated with trend models in levels and drift models in differences.

Essay 3

Re-visiting the Political Economy of the Rise and Fall of the Unidad Popular: Towards a Global Political Economy Approach

Abstract

This chapter evaluates competing explanations of the macroeconomic crisis of Chile's Unidad Popular (1970–1973), examining the causal ordering between domestic monetary expansion, inflation, the external constraint, and central-bank accommodation. Advancing dependency theory as a research programme in the Lakatosian sense, it develops the programme's protective belt by specifying an empirically testable balance-sheet mechanism that links the international currency hierarchy, access to world money, the balance-of-payments constraint, peripheral central-bank solvency, and endogenous monetary accommodation. Historically, the analysis distinguishes the contingent exercise of U.S. hegemonic agency surrounding the breakdown of Bretton Woods, the targeted bilateral credit restrictions confronting Chile, and the constrained but real institutional agency of the Central Bank of Chile (BCCh). Using a multi-frequency empirical design, the empirical strategy deploys stationary vector-autoregressive Granger directional-precedence tests and a non-linear Threshold Vector Autoregression where the central bank's predetermined Solvency Growth Gap ($\Delta s_{t-1} \equiv g_{t-1}^F - g_{t-1}^H$) acts as the transition variable. The empirical analysis reveals that in the adverse solvency-growth state ($\Delta s_{t-1} \leq -4.615\%$, reflecting conditions of central bank insolvency), forward monetary transmission (base-money shock to inflation) is statistically significant for 9 consecutive months with a cumulative 12-month response of 3.55 percentage points. In contrast, reverse accommodation (inflation shock to base-money growth) is statistically significant for 10 consecutive months, delivering a cumulative 12-month expansion of 6.54 percentage points. Reverse accommodation exceeds forward transmission in both duration and cumulative magnitude, with point-wise multiplier differences significant across 9 consecutive months ($h = 2, \dots, 10$). These findings indicate that monetary transmission is state-dependent on external reserve solvency, providing evidence consistent with an endogenous accommodation mechanism under severe balance-of-payments constraints and challenging accounts that treat money creation as an autonomous driver of the Chilean inflation.

Keywords: Unidad Popular, Balance of Payments, Financial Subordination, Solvency Ratio, Endogenous Money, Threshold Vector Autoregression, International Currency Hierarchy, Dependency Theory.

JEL Codes: B51, C32, E58, F33, N16.

3.1 Introduction

In the history of economic thought, progress rarely unfolds as a linear, cumulative ascent toward universal truth. As [Roncaglia \(2006\)](#) observes, theoretical advance throughout the history of economic thought is governed by *competing views*: an enduring contest among rival paradigms rooted in fundamentally distinct conceptualizations, analytical categories, and contested world visions. For over half a century, the history of Chile’s Unidad Popular (UP, 1970–1973) under Salvador Allende has been interpreted through an uncritical cumulative lens hegemonized by mainstream economics ([Edwards, 2024](#); [Larraín and Meller, 1991](#)). Mainstream accounts under neoclassical market-clearing assumptions and monetarist approaches to the balance of payments and inflation that preclude distributive conflict, have treated the Chilean breakdown as a cautionary tale of populist fiscal mismanagement. Yet beneath this orthodox consensus lies an unexamined failure of conceptualization: treating peripheral capitalism as a closed monetary system, blind to the structural subordination of dependent economies within an international currency hierarchy marked by external balance-of-payments constraints and financial subordination ([Chamorro-Futinico, 2025](#); [Jessop, 2019](#)).

This chapter investigates the causal ordering between monetary creation and inflation, and how central-bank foreign-exchange solvency mediates the transmission of domestic liquidity during a peripheral socialist transformation subject to an external constraint. If domestic money creation mechanically drove inflation, why did Central Bank base money expand by over 100% in 1971 while monthly inflation remained quiescent at 1.5–2.0% and annual inflation fell from 35% to 22%? The answer lies in balance-sheet solvency and real capacity: in 1971, the Central Bank possessed nearly \$400 million in inherited gross reserves and industry mobilized significant idle capacity, with structural capacity utilization surging to 98.4%, absorbing liquidity through transactions demand. Significant inflation pass-through did not take place in 1971; it emerged only in mid-1972 once international reserves were depleted and a targeted external credit blockade severed revolving commercial trade lines. Domestic money growth operated as an endogenous accommodation valve through which the monetary authority mediated an acute balance-of-payments squeeze amidst intense distributive conflict. Contesting both the monetarist thesis of [Edwards \(2024\)](#) and the Austrian interpretation of [Espinosa and Véjar \(2025\)](#), I formalize the central-bank balance sheet through the Solvency Ratio—the ratio of convertible foreign reserve assets to domestic high-powered liabilities ($S_t \equiv F_t/H_t$)—and evaluate non-linear transmission using the monthly Solvency Growth Gap ($\Delta s_{t-1} \equiv g_{t-1}^F - g_{t-1}^H$). Threshold econometric estimates identify a structural transition into conditions of central bank insolvency at $\Delta s_{t-1} \leq -4.615\%$. Within this insolvency regime, generalized impulse response functions establish that forward monetary transmission ($g^H \rightarrow \pi_m$) remains statistically significant for 9 consecutive months, delivering a cumulative 12-month price increase of 3.55 percentage points. In contrast, reverse accommodation ($\pi_m \rightarrow g^H$) remains statistically significant for 10 consecutive months, yielding a cumulative 12-month base-money expansion of 6.54 percentage points. Reverse accommodation significantly exceeds forward transmission in both persistence and cumulative magnitude, with point-wise multiplier differences significant across 9 consecutive months ($h = 2, \dots, 10$). Monetary transmission is thus state-dependent on external reserve solvency.

Allende’s thousand days unfolded within the tectonic shifts of the global structural crisis of Fordist capitalism during the early 1970s. As [Zoeller \(2019\)](#) demonstrates, President Richard Nixon’s unilateral suspension of dollar-gold convertibility on August 15, 1971, was not an inevitable systemic

accident, but a contingent political-economic maneuver undertaken to preserve U.S. domestic policy autonomy amidst growing balance-of-payments pressures. While dismantling the Bretton Woods par-value system and exporting stagflationary instability to the periphery (Vernengo, 2021), this hegemonic transition intersected with the secular post-war decline of corporate profitability across the advanced capitalist core (Bowles et al., 1983; Vidal, 2019; Weisskopf, 1979). Yet the subsequent macroeconomic crisis in Chile cannot be reduced to a mechanical reflection of global monetary disorder. Rather, it represented the structural collision of ambitious domestic redistributive commitments with a targeted imperial credit squeeze, within which the Central Bank of Chile exercised constrained institutional agency to defend enterprise liquidity.

Mainstream and neoclassical syntheses characterize the Allende administration through the paradigm of “*macroeconomic populism*” (Dornbusch and Edwards, 1990; Larraín and Meller, 1991). In this interpretation, redistributive governments inevitably trace an unforced four-phase cycle: initial reactivation through idle capacity, the emergence of bottleneck shortages, unsustainable fiscal and monetary financing, and final economic collapse. By treating the central bank as an unconstrained printing press and monetary emission as an unmediated political choice, this literature suffers from two major analytical limitations. First, it succumbs to what Blanco and Matute (2018) formulate as the *illusion of causality*—a cognitive heuristic that infers direct causal generation from the mere temporal coincidence of salient events (the 1970 election, 1971 wage readjustments, and 1972–1973 inflation) while ignoring non-contingent external covariates. Second, it exercises what Isch et al. (2026) classify as *narrative license*—the rhetorical escalation of observational and correlational snapshots into ironclad causal laws to satisfy academic demands for compelling monocausal parables. By analyzing aggregates in an institutional vacuum, mainstream accounts conflate correlation with causality, overlooking the methodological underpinnings of the “credibility revolution” and causal identification. Concretely, this conflation manifests itself by misidentifying the symptoms of redistribution, the targeted external financial blockade (De Vylder, 1976; Petras and Morley, 1975), and the balance-of-payments consequences of the contingent U.S. decision to close the convertibility window with the mechanical results of domestic monetary policy.

To discriminate between these contested claims on rigorous empirical ground, I deploy an applied Lakatosian evaluation of scientific research programmes in economic thought (King and Sznajder, 2006; Kvangraven, 2021; Lakatos, 1978). This methodological framework evaluates rival programmes by distinguishing their foundational theoretical axioms (*hard core*) from the auxiliary hypotheses of their empirical models (*protective belt*), assessing whether empirical anomalies are accommodated through progressive theoretical extensions that predict *novel facts* or through ad hoc adjustments. The debate over the macroeconomic destabilization of the Unidad Popular can be structured into three competing research programmes. The monetarist programme (Edwards, 2024) anchors its hard core in exogenous base money creation, fractional-reserve multiplier causality, and long-run monetary neutrality in a closed system. Confronted with the empirical anomaly of 100% monetary growth coinciding with non-inflationary output expansion in 1971, its protective belt relies on narrative rationalizations that dismiss the \$294 million external financing cutoff, the U.S. Eximbank embargo, and domestic distribution bottlenecks as secondary disturbances. The Austrian business-cycle programme (Espinosa and Véjar, 2025) similarly emphasizes domestic distortions, positing that suppressed interest rates and state credit expansion generated intertemporal capital malinvestment and coordination breakdown.

In contrast, treating dependency theory as a research programme (Kvangraven, 2021) generates

excess empirical content by formalizing an open-economy balance-sheet mechanism within the programme’s protective belt. Anchored in the structural realities of the international currency hierarchy and unequal access to world money (Basu, 2022; Vasudevan, 2009), this framework derives the external solvency constraint governing peripheral central banking. Using the Solvency Growth Gap ($\Delta s_{t-1} \equiv g_{t-1}^F - g_{t-1}^H$), the model identifies state-dependent regime shifts: when the central bank maintains adequate foreign exchange reserves, domestic liquidity expansion mobilizes idle capacity without igniting immediate inflation, and external commodity price shocks are buffered by the reserve buffer. However, once external credit embargoes and terms-of-trade deterioration push the monetary authority across the critical threshold into conditions of central bank insolvency ($\Delta s_{t-1} \leq -4.615\%$), this insulation mechanism dissolves. The empirical evaluation reveals that consumer price shocks systematically precede base money creation, and reverse defensive accommodation exceeds forward monetary transmission in duration and cumulative response, compelling the central bank into liquidity injection to sustain enterprise payrolls under acute balance-of-payments distress.

This chapter deepens the theoretical foundations established in Chapter 2. Whereas Chapter 2 identified the external boundary of accumulation through an annual balance-of-payments threshold that throttles mechanization under unbalanced growth, this chapter uncovers the dynamic balance-sheet mechanisms through which that external constraint operates at high frequency. To overcome the temporal aggregation bias and closed-economy endogeneity pervasive in annual accounts (Dornbusch and Edwards, 1990; Edwards, 2024), my research design assembles a two-tier dataset at annual and monthly frequency time series from multiple secondary sources.

The annual time-series dataset (1920–2010, $T = 91$) draws functional income shares from Astorga (2023), balance-of-payments accounts from Nieves and Piketty (2025), and disaggregated net capital stocks from the Perpetual Inventory (GPIM) dataset developed in Chapter 2. At this annual frequency, the Austrian malinvestment hypothesis (H_2) fails to find empirical support: gross fixed investment in machinery and equipment contracted (-2.3%) during the 1971 expansion while structural capacity utilization surged to an all-time peak of $\mu_{1971} = 98.44\%$, presenting empirical evidence consistent with reactivation rather than intertemporal capital misallocation.

To isolate the transmission channels of the subsequent breakdown, the high-frequency monthly tier (1960–1980, $N = 250$ continuous months) integrates primary Central Bank of Chile archival data accessed through the Application Programming Interface (API) of the *Base de Datos Estadísticos* (BDE) of the Chilean Central Bank (BCCh), including Central Bank base monetary emission H and $M1$, indices of gold, copper, and wholesale U.S. prices, and activity indicators gathered by Díaz-Bahamonde (2023) to build import capacity and structural imbalance indices. Historical international reserve statistics from the IMF provide monthly reserve assets not readily accessible elsewhere. Following the August 1971 closure of the U.S. gold convertibility window, my proxy of Prebisch’s purchasing power capacity to import (M_t^{cap}) fell to 27.7, driving the Real Structural Imbalance Ratio ($\Theta_t \equiv M_t/M_t^{\text{cap}} \times 100$) to an unsustainable peak of 283.4 in May 1973. In the vector autoregressive and Threshold VAR systems, this physical import strangulation governs the regime switch: once foreign credit embargoes (De Vylder, 1976; Petras and Morley, 1975) severely erode reserve backing and push the economy across the estimated Solvency Growth Gap threshold ($\Delta s_{t-1} \leq -4.615\%$, entering the adverse solvency-growth state reflecting conditions of central bank insolvency), external price shocks penetrate domestic retail prices immediately, compelling defensive monetary accommodation to maintain enterprise payrolls amidst supply paralysis.

The remainder of this chapter is structured as follows. Section 3.2 surveys the contested traditions

in Chilean political economy and formalizes the epistemological critique of orthodox causality. Section 3.3 develops the theoretical framework of world money, financial subordination, and balance-sheet mechanics, formalizing how peripheral financial fragility and central-bank balance-sheet exposure condition macroeconomic transmission. Section 3.4 details the multi-frequency data compilation, archival sources, and variable construction, including the Central Bank solvency metric and structural import imbalance ratios. Section 3.5 establishes the historical background of the pre-1970 accumulation regime, presents stylized facts of peripheral accumulation and reserve depletion, and implements the econometric evaluation across the stationary vector-autoregressive Granger directional-precedence battery and the non-linear Threshold VAR. Section 3.6 discusses competing research programmes in the political economy of the Unidad Popular, evaluates the empirical findings alongside hegemonic contingency and central-bank agency, assesses causal identification limits, and outlines strategic policy implications. Five methodological and empirical appendices provide the balance-of-payments accounting derivation (Appendix 3.A), the archival data codebook (Appendix 3.B), the multivariate PCMCi causal discovery results (Appendix 3.C), the complete 25-pathway Threshold VAR impulse response functions (Appendix 3.D), and the linear VAR diagnostic space (Appendix 3.E).

3.2 Contested Approaches to the Political Economy of the Rise and Fall of the Unidad Popular

2.1 No single framework commands consensus on the political economy of the rise and fall of Chile’s Unidad Popular (UP). Instead, rival explanatory traditions have structured the field, each emerging under unequal political conditions and distinct historical contexts. Transnational networks linking the Pontificia Universidad Católica de Chile (PUC) to the University of Chicago formed the neoliberal school, which later acquired state power through the dictatorship that followed the 1973 coup. Neo-structuralism emerged under the dictatorship as a technocratic opposition to Pinochet’s regime—specifically in the aftermath of the 1982 debt crisis—before rising to government power and establishing itself as the mainstream economic paradigm during the post-dictatorship order. Contemporary accounts in the field of Historical Political Economy (HPE) entered the debate much later, bringing powerful tools of causal identification to historical questions. However, they did so within a disciplinary environment already shaped by earlier causal claims taken as given, while perspectives anchored in distributive conflict and structural dependency were seldom considered or actively excluded. Dependency theory occupies a different position altogether. It was the intellectual landscape from which the UP programme itself emerged, and the coup institutionally shattered it through violent repression before it could resolve the theoretical problems it had opened up.

2.2 These traditions did not simply coexist. They succeeded one another through political conflict without genuinely opposing each other on the intellectual terrain, despite stark declarations regarding the intellectual bankruptcy of the structuralist-dependency school:

“...the macroeconomic view of the Unidad Popular failed to recognize that their policies would only be able to generate a burst of economic activity that would be unsustainable in the medium term... Needless to say, this view of the way the economy functioned ignored many of the key principles of traditional economic theory.” (Dornbusch and Edwards, 1989, p. 16–17).

Each identified a part of the problem, yet none integrated external transmission, distributive conflict, and macroeconomic dynamics into a single explanatory sequence. This section critically evaluates these traditions. It reconstructs the historical conditions under which each approach became authoritative, clarifies the mechanisms each privileged, and shows which questions each rendered visible and which it displaced, sanitized, or foreclosed. It begins with classical dependency theory, which posed the problem the chapter as a whole seeks to resolve more sharply than any approach that followed.

2.3 This fragmented intellectual legacy has resulted in a striking scarcity of rigorous empirical evidence concerning the macroeconomic dynamics of the UP's collapse. While the political and institutional histories of the period are extensively documented, the macroeconomic debate has largely relied on retrospective narratives, analysis of correlation without causality, or ideological posturing rather than methodologically robust quantitative analysis. There is a critical lack of empirical studies employing adequate econometric methods to isolate and measure how allegedly "*populist interventions*"—such as exchange rate overvaluation, fiscal monetization, and generalized price controls—transmitted distributive conflicts and external shocks into systemic macroeconomic failure. By leaving these macroeconomic transmission mechanisms empirically untested, the literature has remained trapped in theoretical stalemates. This chapter seeks to bridge this methodological void by moving beyond ideological retrospection, advancing more rigorous empirical identification to the macroeconomic data of the UP era to trace the anatomy of its economic unraveling.

3.2.1 Classical Dependency Theory: Interrupted Programme

2.4 Dependency theory did not approach the Unidad Popular from the outside. It helped constitute the intellectual horizon within which the UP became thinkable. Its formation in Chile brought together two lineages that converged, institutionally and theoretically, in Santiago during the long sixties. From one side came the Monthly Review tradition, beginning with [Baran \(1957\)](#)'s analysis of economic surplus under monopoly capitalism and radicalized by [Frank \(1967\)](#) into the development-of-underdevelopment thesis: underdevelopment was not a residual stage but an active product of incorporation into the world capitalist system. From the other came the structuralist school built within the headquarters of the Economic Commission for Latin America (ECLAC), led by Raúl Prebisch, which had already framed Latin American development through centre-periphery relations, external bottlenecks, and the historical asymmetries of international trade.

2.5 At the Centro de Estudios Socioeconómicos (CESO) of the Universidad de Chile, these two lineages fused into an active research programme. Following the 1964 military coup in Brazil, the arrival of Theotônio dos Santos and Vania Bambirra as political exiles in Santiago catalyzed this institutional convergence ([Cárdenas Castro and Seabra, 2022](#)). Collaborating at CESO alongside André Gunder Frank, Orlando Caputo, and Roberto Pizarro—and maintaining close dialogue with Ruy Mauro Marini in Concepción—they transformed dependency from a general critique of peripheral capitalism into an effort to specify the historical forms linking regional subordination, capital accumulation, and political strategy ([Bambirra, 1971](#); [Caputo and Pizarro, 1970](#); [Dos Santos, 1970](#); [Kay, 2010](#)). Far from remaining an academic abstraction, this intellectual collective directly shaped the policy architecture of the Chilean left: among Dos Santos and Bambirra's closest collaborators and disciples were Sergio Ramos and Alexis Guardia, the primary economic drafters of the Unidad Popular programme representing the Communist and Socialist parties, respectively.

2.6 The programme’s analytical wager was clear. Peripheral capitalism followed a distinct developmental dynamic structured by external constraints, a domestic bourgeoisie structurally incapable of autonomous national accumulation, and reproduction patterns that continually re-entrenched subordination. Within this framework, socialist transformation appeared as an objective developmental necessity grounded in the physical limits of peripheral capital formation. Yet the tradition never achieved theoretical unity. Marini’s theory of superexploitation pushed the analysis toward a world-market logic where the systematic compression of labor below the value of its reproduction conditioned domestic capital accumulation (Marini, 1973). Conversely, Cardoso and Faletto (1979) insisted that dependency took historically contingent forms governed by domestic class alliances and the relative autonomy of the capitalist state. This tension proved decisive. At stake was whether domestic political mediation constituted an independent object of analysis or simply mediated world-market imperatives. Classical dependency posed the question with great force but never resolved it.

2.7 That unresolved tension did not prevent the tradition from producing the richest contemporaneous political-economy accounts of the Unidad Popular. De Vylder (1976)’s *Allende’s Chile* provided the seminal participant-observer account, correctly locating the material foundation of the external constraint in Chile’s extreme physical import dependency: over 90% of capital equipment was imported, and more than 95% of machinery in large-scale copper mining originated in the United States. Yet De Vylder’s framework suffered from an important analytical compromise. Confronted with orthodox arguments attributing the crisis entirely to domestic mismanagement, De Vylder retreated into an internal-blame explanation, asserting that:

“the problem was the insatiable imports, not the US blockade nor the lack of credit”
(De Vylder, 1976, p. 106).

To sustain this concession, De Vylder pointed to gross nominal credit lines extended by the Soviet Union and Eastern Europe (> \$400 million) as ostensible proof that external financing remained abundant, conflating gross nominal commitments with usable, convertible liquidity.

2.8 Stallings (1978) overturned De Vylder’s concession by uncovering the *qualitative non-fungibility of international credit*. Drawing on primary Central Bank foreign exchange ledgers, she demonstrated that bilateral socialist credits were non-convertible tied lines earmarked for specific industrial equipment. They could neither purchase patented American spare parts for Kennecott copper haul trucks nor settle immediate spot cash contracts for wheat and petroleum on Western commodity markets. Concurrently, private U.S. commercial banks slashed revolving short-term trade credit lines from \$220 million to \$35 million, while the Export-Import Bank froze disbursements. These debates around balance-of-payments constraints did not unfold in isolation. They formed part of a vibrant contemporaneous intellectual landscape captivated by the *vía chilena al socialismo*—the historical attempt to achieve revolutionary socialist transformation through bourgeois democratic institutions. Across this field emerged a rich empirical corpus tracking the crisis from complementary angles: distributional conflict across successive administrations, worker and peasant self-organization from below, the logistical spare-parts embargo, the monetary and balance-of-payments mechanisms through which external pressure became domestic crisis, and the strategic dilemmas of the state under conditions of emerging dual power (Espinosa and Zimbalist, 1978; Griffith-Jones, 1981; Kay, 1978; Petras and Morley, 1975; Sideri, 1979; Stallings and Zimbalist, 1975; Sweezy and Magdoff, 1974; Winn, 1986; Zavaleta Mercado, 1974). The enduring methodological strength of classical

dependency lay precisely here: it refused to treat external imperial intervention and domestic class struggle as competing explanations, approaching them instead as interacting, co-constitutive dimensions of the same historical process.

2.9 That achievement also marked the analytical limit dependency theory could not overcome in its classical form. Confined to a bilateral nation-state framework (the U.S. versus Chile), it omitted the international monetary architecture of Bretton Woods and global stagflation, while lacking a formal macroeconomic specification linking external balance-sheet shocks to domestic distributive conflict and capacity utilization. Yet the defeat of dependency theory was twofold. Politically, the coup dismantled the institutions, networks, and research centres in which the programme had been elaborated. Intellectually, the interruption arrived before the tradition could work through its own internal tensions. No superior explanatory framework refuted it in a scientific terrain. It broke under conditions in which theoretical incompleteness coincided with the collapse of Chile's liberal democracy, the failure of democratic socialism to resolve the crisis under organized domestic and international sabotage, and the installation of a subsequent civic-military dictatorship. Subsequent approaches entered a field already reorganized by force, not reason. The next school to become authoritative in Chilean political economy was not the one that best resolved the problem dependency had identified, but the one that arrived coordinated with the counterrevolutionary historical bloc that constituted the reconstruction of the state and civil society ([Casals, 2021](#)).

3.2.2 The Rise of the Chicago Boys and the Neo-liberal School

2.10 The neoliberal school did not emerge from within the intellectual world of the Unidad Popular as one competing interpretation among others. It entered through a different route: as a transnationally trained programme of economic reconstruction that acquired authority through the counterrevolutionary reorganization of the Chilean state. Its genealogy is well established. The 1956 agreement between the Pontificia Universidad Católica de Chile and the University of Chicago opened the institutional channel through which Milton Friedman, Arnold Harberger, and Gary Becker trained a cohort of Chilean economists in monetarism, public finance orthodoxy, and neoclassical price theory. By the late 1960s, this network had evolved from an academic outpost into a politically embedded project, closely aligned with business associations and the Chilean right, oriented toward the systemic restructuring of Chilean capitalism ([Clark, 2017](#); [Fischer, 2009](#); [Valdés, 1995](#)).

2.11 That programme found its canonical expression in *El Ladrillo*¹. The text codified in advance the macroeconomic blueprint on which the post-UP order would be constructed. It operated as a counterrevolutionary vanguard design: an operational policy manual prepared prior to the coup to direct the reconstruction of Chilean capitalism the moment the popular government was overthrown.

2.12 Its diagnosis identified a familiar catalogue of structural problems: chronic inflation, recurrent external deficits, agricultural stagnation, low productivity growth, and distortions associated with state-led industrialization. Several of these observations captured real tensions within Chilean accumulation, and they merit serious engagement. The argument that prolonged relative-price

¹*El Ladrillo* ("The Brick") was drafted clandestinely prior to the September 1973 coup by Chicago-trained economists coordinated by Emilio Sanfuentes, with contributions from Sergio de Castro and Pablo Baraona, in liaison with naval officers and the leadership of the private employers' association *Sociedad de Fomento Fabril* (SOFOFA) ([De Castro, 1992](#)).

controls hindered export diversification and exacerbated balance-of-payments vulnerability cannot simply be dismissed, nor can the recognition that import-substitution industrialization had reached severe structural exhaustion (De Castro, 1992). Yet the broader theoretical hypothesis holding these observations together was much weaker: the claim that the crisis of the UP merely culminated a secular drift toward the excessive socialization of the economy. That thesis recoded a complex conjunctural crisis into a normative indictment, naturalizing market discipline rather than explaining the empirical dynamics of the breakdown.

2.13 Here lay the neoliberal school's structural limitation. It registered macroeconomic imbalances only by systematically stripping them of their political-economy. Functional income distribution appeared not as structural class conflict, but as a technical distortion of price signals. The external crisis was treated as the endogenous consequence of fiscal and monetary mismanagement, never as the relational interaction between domestic class struggle and international financial subordination and destabilization. The October 1972 employers' lockout, the private commercial credit blockade, the spare-parts embargo, and covert foreign destabilization could not enter the account without undermining the framework's own core hypothesis and axioms. Neoliberalism did not omit these factors by accident. Its explanatory logic required them to disappear, since acknowledging them would have undermined the premises that the crisis proved the inherent impossibility of socialist transformation.

2.14 The neoliberal interpretation became authoritative under historical conditions that were not primarily intellectual. Its ascendancy reflected no superior theoretical integration of external transmission, distributive conflict, and macroeconomic dynamics. A programme formulated before the coup entered a terrain reorganized by military force, and the dictatorship installed it as the common sense of national economic reconstruction. That restructuring violently dismantled collective bargaining, political mediation, and labor organizations while relying, paradoxically, on the centralized administrative apparatus of the developmental state it claimed to dismantle (Bockman, 2019; Clark, 2017). The neoliberal school was less a scientific solution to the developmental dilemmas of the periphery than a victorious political project presenting ruling class backlash as economic science. Its enduring legacy lies precisely here: it converted a conjunctural crisis resolved by state terror into a universalized economic doctrine asserting the impossibility of any non-market alternative (Espinosa, 2021).

3.2.3 The Neo-Structuralist School: Selective Sanitization

2.15 The neo-structuralist school was an intellectual accomplishment of the post-dictatorship order rather than a simple continuation of classical structuralism. It established its institutional base during the dictatorship, primarily within the CIEPLAN think tank, where former structuralists and democratic opposition economists critically monitored the Chicago Boys' reforms while formulating an alternative programme for the democratic transition. That genealogy explains much of the school's subsequent authority. The CIEPLAN economists were more than political moderates who had tempered their commitments under authoritarian rule (Silva, 1991). They constituted the modernizing technocratic fraction through which the post-authoritarian centre-left could challenge orthodox monetarism while operating strictly within market-disciplined, export-oriented peripheral capitalism. Neo-structuralism entered the field as a politically functional renovation calibrated to manage Chile's subordinate insertion into the globalized economy (Leiva, 2008; Maillet et al., 2016;

[Sklair, 2000](#)).

2.16 The central analytical move of that renovation was the abandonment of the classical surplus approach. Where Prebisch and the foundational ECLAC tradition had centered the production, appropriation, and reinvestment of the economic surplus under peripheral conditions, neo-structuralism recast underdevelopment into the mainstream idiom of market imperfections, institutional bottlenecks, and pragmatic state regulation. This shift did not merely modernize structuralism; it altered its analytical object. The external constraint remained, but strictly as a macroeconomic financing ceiling rather than as a structural relation of surplus extraction and dependent reproduction. By reducing a theory of historically structured subordination to a technical problem of policy calibration, the school depoliticized the external barrier. As [Leiva \(2008\)](#) demonstrates, without the surplus approach, the external constraint survives only in a thinned accounting form: legible as a foreign exchange deficit, but stripped of the class relations that generate and sustain it. Neo-structuralism did not forfeit empirical rigor; it forfeited the conceptual framework required to theorize the balance of payments and distributive conflict as unified moments of the same accumulation process.

2.17 The school's account of the UP crystallized most authoritatively in the work of Sergio Bitar. Originally published in 1979, [Bitar \(2013\)](#) provided the most sophisticated neo-structuralist reconstruction of the popular government. Its principal merit lay in rejecting an exclusively monetary explanation of the external accounts. Bitar sought to integrate foreign exchange scarcity, political opposition, and institutional transformation into a single dynamic narrative. His formulation of a "viability area" remains a genuine analytical contribution, demarcating the narrow economic margin within which the UP programme could have advanced without precipitating macroeconomic collapse. Yet that contribution also established the school's defining silence. Across multiple editions, Bitar never substantively engaged the contemporaneous dependency literature that had developed across the same historical terrain. Foundational accounts by [De Vylder \(1976\)](#) and [Stallings \(1978\)](#) were already widely available, yet Bitar avoided confronting them. This silence was neither a bibliographic omission nor a conspiracy. It reflected the strategic selectivities accompanying the abandonment of the surplus approach ([Leiva, 2008](#)), marking the point at which dependency's central questions were cordoned off from post-dictatorship economic discourse.

2.18 That theoretical cordon became hegemonic through the populist-macroeconomic paradigm. [Larraín and Meller \(1991\)](#) codified the UP into a stylized four-phase trajectory of initial reactivation, emerging bottlenecks, severe destabilization, and ultimate collapse. [Dornbusch and Edwards \(1991\)](#) then generalized this sequence into a regional archetype of "macroeconomic populism." This paradigm achieved wide influence by providing a clear narrative, identifiable turning points, and cautionary lessons that circulated throughout Latin America during its democratic transitions. Yet it operated through a definitional foreclosure: by classifying redistributive policies categorically as "*macroeconomic populism*," the framework ruled out the possibility that wage-led growth could ever be structurally sustainable under peripheral conditions. [Meller \(2000\)](#) later cemented this interpretation as the common sense of the UP: an unforced tragedy driven by policy mismanagement and political polarization. The result was a reading self-critical in tone but exculpatory in effect, reducing the crisis to domestic excess while treating external financial blockades and domestic logistical sabotage as secondary noise. The contrast between Sergio [Ramos \(1979\)](#) and Joseph [Ramos \(1980\)](#) illustrates this epistemological divide: the former treated inflation as a structural contradiction internal to a socialist transition under acute class struggle, whereas the latter reduced the crisis to an engineering problem of disequilibrium management and stabilization failure.

2.19 Neo-structuralism thus substituted a technocratic problem of governance for the conflict-centered political economy generated during the UP itself (Dornbusch and Edwards, 1991; Larraín and Meller, 1991; Meller, 2000). The school’s analytical limits were not accidental blindspots; they were strategic exclusions built into the conditions of its post-authoritarian legitimacy. Its primary historical function was to preserve the structuralist language of external constraints while adapting it to the macroeconomic governance of a neoliberalized economy, sustaining two decades of centre-left political stability. In doing so, it transformed the thousand days of Allende’s government into a moral tale of redistributive overreach. Neo-structuralism became the progressive framework through which the post-dictatorship order could be managed and stabilized without reviving the class-conflict categories that made earlier structuralist and dependency traditions dangerous to ruling-class hegemony (Leiva, 2008).

2.20 This critique does not dismiss the empirical contributions of scholars such as Bitar or Meller; it specifies the political-economic compromises and strategic selectivities that resulted from their synthesis. The coordination of monetary, structural, and political variables into a coherent narrative remain important contributions to the debate. However, the systematic exclusion of class struggle for the economic surplus was the price of its institutionalization during the democratic transition and the post-dictatorship period. That compromise defines neo-structuralism’s enduring legacy. It established the default macroeconomic memory inherited by subsequent scholarship, leaving historical political economy equipped with sophisticated empirical tools, but lacking a coherent theory of the structural mechanisms at work.

3.2.4 Historical Political Economy: Methods Under Inherited Macro Theory

2.21 Historical Political Economy (HPE) represents a genuine methodological advance in the study of economic history. By applying the credibility revolution’s identification toolkit to historical inquiry, it has introduced a level of empirical precision rarely achieved in earlier debates. Its institutionalization around the *Journal of Historical Political Economy* and the *Oxford Handbook of Historical Political Economy* reflects a broader effort within economics to reclaim historical questions without sacrificing rigorous causal identification (Charnysh et al., 2023; Gailmard, 2021; Jenkins and Rubin, 2024). Applied to twentieth-century Chile, this agenda has generated an impressive corpus examining price controls, real wages, land redistribution, and political violence.

2.22 Yet HPE did not generate the archival terrain on which it operates. That foundation was constructed by a parallel renewal in Chilean economic history, which has reconstructed long-run time series on prices, trade, living standards, and distribution. The archival work of Mario Matus, Manuel Llorca-Jaña, Claudio Robles, and their collaborators established the quantitative records that made micro-econometric identification feasible. Within this archival foundation, Rodríguez Weber (2018) is foundational for the empirical architecture of this dissertation: the long-run historical accounts of inequality compiled there directly underpin the functional wage-share series (ω_t) harmonized by Astorga (2023) and utilized throughout Chapters 2 and 3. By situating the Unidad Popular within a secular century-long cycle of distribution and explicitly demonstrating how political coercion shaped income shares, Rodríguez Weber provides a central historical bridge between institutional economic history and structural macroeconomic questions.

2.23 In contrast to descriptive economic history, HPE proper focuses on estimating discrete causal effects. Aldunate et al. (2024) exploit firm-level records to identify the credit-allocation shocks

induced by U.S. financial warfare, while [Jaimovich and Toledo-Concha \(2021\)](#) trace contemporary Mapuche territorial conflict to the localized geography of agrarian expropriation and military counter-reform. [Girardi and Bowles \(2018\)](#) provide a distinctive contribution: their high-frequency event study of the Santiago stock exchange demonstrates that financial wealth-holders responded systematically to shifts in the political threat to private property, corroborating classical political economy's focus on capitalist investment strikes. These studies represent sophisticated empirical contributions. However, they perform fundamentally different analytical work: some build archival ledgers, others isolate localized causal effects, while few—if not [Girardi and Bowles \(2018\)](#) alone—examine macroeconomics and institutions through a theoretical framework centered on the structural power of capital over the economy.

2.24 This analytical demarcation is most evident in the work of Felipe González and his collaborators, who define the contemporary frontier of Chilean HPE ([Aldunate et al., 2024](#); [Bautista et al., 2023](#); [González, 2013](#); [González and Prem, 2026a,b](#); [González et al., 2020](#); [González and Vial, 2021](#)). Their research is methodologically exemplary. They have unearthed primary archival records, formulated transparent quasi-experimental designs, and resolved contested empirical questions surrounding land reform, industrial requisitions, covert foreign intervention, and the institutional legacies of the military dictatorship. Yet this corpus also marks the field's primary constraint: the macroeconomy appears primarily as a passive background upon which institutional interventions and policy shocks are measured, rather than as an organic, contradictory system of capital accumulation, surplus extraction, and crisis. Even where the archival evidence touches systemic macroeconomic phenomena, the theoretical interpretation is routinely imported from post-dictatorship mainstream narratives. The result is scholarship of exceptional empirical precision that displaces political economy in the classical sense into an applied microeconomics of institutional interventions.

2.25 This limitation reflects the field's disciplinary formation. HPE arrived at the Chilean case after the historical clash between the dependency, neo-structuralist, and neoliberal paradigms had already been settled by state coercion. It inherited the depoliticized macroeconomic vocabulary of the post-dictatorship consensus, which reduced the UP's crisis to a cautionary tale of macroeconomic populism and unforced balance-of-payments mismanagement. The research programme developed in this chapter builds upon HPE's archival and empirical rigor, while restoring what applied microeconomics omits: an integrated theoretical framework capable of modeling external balance-of-payments vulnerability, distributive conflict, and domestic monetary accommodation as mutually conditioning moments of peripheral capital accumulation.

3.2.5 The Revival of Dependency Theory as a Research Programme

2.26 The return to dependency theory is not an exercise in intellectual nostalgia. It recovers a research programme whose historical retreat signaled a profound collapse of the region's sociological imagination. As [Palma \(2016\)](#) argues, Latin America's critical thinking swung like a pendulum from one type of absolute certainty to another: from the mechanical stagnationism of early dependency to the uncritical market fatalism of the post-dictatorship consensus. In that shift, the region's intellectual landscape was stripped of its capacity to conceive development as a problem of collective agency, structural transformation, and sovereign autonomy. Development was degraded from a political struggle over the production, appropriation, and reinvestment of the economic surplus into an administrative exercise in macroeconomic compliance. The analytical task today is neither to

revive the rigid certainties of the 1970s nor to surrender to technocratic fatalism, but to reconstruct dependency as an empirically generative, progressive framework.

2.27 [Kvangraven \(2021\)](#) provides the epistemological architecture for this reconstruction by formalizing dependency theory through Imre Lakatos’s methodology of scientific research programmes. Kvangraven refutes the mainstream caricature of dependency as a monolithic dogma by demonstrating that the tradition is structured around a coherent, shared *hard core*: the premise that global capitalist accumulation systematically reproduces hierarchical structures between core and periphery that constrain the developmental space of subordinate social formations. The scientific vitality of the programme does not depend on dogmatic consensus regarding specific historical models; rather, it hinges on whether its *protective belt of auxiliary hypotheses* can generate *progressive problemshifts*—predicting novel empirical facts and specifying concrete causal mechanisms—rather than degenerating into defensive ad-hoc immunizations. By formalizing this distinction, she clarifies how dependency can assimilate new historical evidence within the context of the concrete realities of global capitalism without abandoning its core ontological commitments.

2.28 [Madariaga and Palestini \(2021\)](#) sharpen this epistemological reconstruction by distinguishing between *situations of dependency* and *mechanisms of dependency*: i.e., between historically specific descriptive configurations and the causal processes that reproduce external constraints. This distinction has energized a contemporary renewal in Santiago, where scholars have operationalized dependency theory to analyze rentier accumulation, export stagnation, and dependent financialization in present-day Chile ([Ahumada, 2019](#); [Ahumada and Torres, 2024](#); [Ruiz Bruzzone, 2021](#); [Sossdorf, 2024](#)). Within this revival, recovering the historical memory of the Unidad Popular is not an exercise in political nostalgia, but an inquiry into the structural conditions under which sovereign control over the surplus and democratic self-determination were attempted and violently suppressed ([Ahumada and Chang, 2025](#); [Arboleda, 2023](#)). Yet for the macroeconomic analysis of the UP itself, the central task remains unfinished: while classical accounts identified the concrete *situation* of Chilean vulnerability (import dependency and bilateral credit retaliation), they lacked a formal *mechanism* linking peripheral balance-of-payments asphyxiation to the structural crisis of the core’s international monetary order.

2.29 This missing transmission mechanism can be formulated by synthesizing the political economy of global dollar hegemony with the institutional mechanics of the dissolution of Bretton Woods. [Vernengo \(2021\)](#) provides the structural foundation for why Bretton Woods collapsed: a macro-political drive by U.S. elites to escape the discipline of gold convertibility, discipline domestic labor through international capital mobility, and establish an unconstrained fiat dollar standard. In turn, [Zoeller \(2019\)](#) demonstrates how and when this rupture unfolded: rather than a predetermined grand design, closing the gold window was a contingent executive maneuver forced by IMF institutional veto points and European surplus resistance. The core executed an emergency exit on August 15, 1971, rapidly consolidating a floating dollar regime that eliminated its own reserve discipline while transmitting an external shock to peripheral economies.

2.30 For Chile, this historical timing was decisive. Nixon’s unilateral closure of the gold window occurred exactly three days after the U.S. Export-Import Bank formally denied credit lines to the Unidad Popular on August 12, 1971. Peripheral accumulation and the “*via Chilena al Socialismo*” was thus caught in converging external constraints: the bilateral credit non-fungibility documented by [Stallings \(1978\)](#) colliding with the systemic dissolution of the global monetary regime. Incorporating this dual transmission into the protective belt of dependency theory resolves the theoretical

ambiguities that stalled its classical formulation. It equips the tradition with an exogenous historical event (Nixon’s unilateral suspension of gold convertibility) as a macroeconomic mechanism capable of explaining how external balance-sheet shocks triggered domestic supply paralysis and forced monetary accommodation, directly setting up the comparative evaluation of competing hypotheses developed in this chapter.

3.2.6 Epistemological Critique: The Illusion of Causality, Narrative License, and Collider Traps

2.31 The persistent divergence among these explanatory traditions cannot be resolved through historiographical narrative alone. As [Blanco and Matute \(2018\)](#) establish, human causal reasoning is systematically vulnerable to the *illusion of causality*: a robust cognitive bias wherein observers detect strong causal links between co-occurring events even when the statistical contingency is zero or driven by unobserved common causes. In the economic historiography of the Unidad Popular, orthodox monetarist and Austrian accounts observe a conspicuous historical coincidence—the election of a socialist government in late 1970, expansionary fiscal-wage policy in 1971, and the explosion of inflation in 1972–1973. Relying on intuitive heuristics, they assign exclusive causal responsibility to domestic monetary issuance ($g_{M,t} \rightarrow \pi_t$), treating international monetary hierarchy, terms-of-trade shifts, and geopolitical destabilization as irrelevant background noise ([Edwards, 2024](#); [Espinosa and Véjar, 2025](#)).

2.32 This causal illusion is reinforced by what [Isch et al. \(2026\)](#) identify as *narrative license* and *overreaching causal claims* in historical social science. Orthodox accounts dress up doctrinal priors as ironclad empirical laws without testing alternative causal mechanisms nor specifying counterfactual identification paths. For example, [Espinosa and Véjar \(2025\)](#) assert an Austrian Business Cycle mechanism wherein artificial credit expansion generated an unsustainable capital malinvestment boom. Yet they advance this claim without presenting positive evidence of gross fixed capital deepening (ΔK), ignoring that Chilean fixed machinery investment contracted during 1971 while capacity utilization reached historical peaks. Narrative license allows the theorist to impute unobserved microeconomic mechanisms (capital lengthening) to explain macroeconomic outcomes while dismissing external interventions as quantitatively negligible.

2.33 Methodologically, orthodox accounts fall into what causal inference defines as a *collider trap* ([Pearl, 2009](#)). A collider trap takes place when analysts focus on a symptom that is actually the shared outcome of two separate causes. By isolating their attention on this shared outcome, they distort the relationship between the causes, often falsely attributing the symptom entirely to just one of them. In narratives of the UP crisis, this shared outcome—the collider—is the widespread presence of retail shortages and food queues (C) ([Edwards, 2024](#)).

Orthodox accounts treat these empty shelves as definitive proof of “*repressed inflation*,” attributing them solely to the government printing money to stimulate demand (U_1). However, these shortages were simultaneously driven by a second, independent factor: the systematic hoarding and lockouts organized by private wholesale distributors and trucking strikes (U_2). Both state demand stimulus (U_1) and private supply lockouts (U_2) collided to produce empty retail shelves (C):

$$U_1 \longrightarrow [C] \longleftarrow U_2 \quad \implies \quad U_1 \not\perp U_2 \mid C \quad (3.1)$$

By treating retail shortages as a neutral, objective measure of excess liquidity (conditioning on C), orthodox accounts commit a collider error of causal identification. This methodological blind spot produces a spurious conclusion, allowing them to blame public deficit monetization for a supply paralysis that was heavily driven by the political actions of private business federations.

2.34 To transcend these causal illusions and collider traps, I adapt the methodological approach of [King and Sznajder \(2006\)](#), who formally tested competing economic theories to explain Poland's growth trajectory during neoliberalization. In this chapter, I treat Monetarism (H_1), the Austrian Business Cycle (H_2), and Dependency Theory (H_3) as rival scientific research programmes. Each possesses a distinct theoretical core, auxiliary assumptions, and falsifiable empirical predictions. I submit the central causal claims of these competing traditions to a systematic comparative evaluation across multi-frequency time series data.

3.3 World Money, Financial Subordination, and the Peripheral Central Bank Balance Sheet

This section develops the theoretical framework linking the global currency hierarchy to domestic macroeconomic destabilization. The analysis traces a clear analytical sequence: the asymmetric structure of world money generates peripheral settlement constraints, which concentrate balance-of-payments vulnerability directly onto the central bank balance sheet. In response to structural bottlenecks and foreign-exchange scarcity, domestic monetary issuance functions as an endogenous accommodation mechanism rather than an autonomous policy choice. The resulting balance-sheet strain alters macroeconomic transmission, generating state-dependent responses that are subsequently estimated through dynamic threshold modeling.

3.3.1 World Money, the Universal Equivalent, and the Monetary Expression of Value

3.1 In Marxian political economy, money originates as the commodity form of the universal equivalent: the unique social medium whose function is to express the value of all commodities and serve as the general means of payment ([Basu, 2022](#), §3.1). Gold historically occupied this role not through inherent natural properties, but through a social process of market validation, acting as the bearer of abstract human labor.

3.2 The historical transit to credit money transforms this institutional form without dissolving its underlying function. As banking systems developed to intermediate commercial credit, central bank liabilities emerged as the legal anchor of domestic circulation ([Basu, 2022](#), §3.2.3). The commodity anchor disappeared from domestic circulation, but the imperative of social validation remained. Central bank liabilities and bank deposits perform the role that bullion once fulfilled: providing the monetary expression through which private, decentralized labor is validated as social labor. Following the Monetary Expression of Value (MEV) framework ([Basu, 2022](#); [Foley, 1982](#)), the value of money links aggregate labor-time to the circulating monetary unit within a national formation.

3.3 Once circulation crosses national borders, however, credit money confronts the world market. While Marx presented world money in the physical form of bullion shedding national garbs ([Ivanova,](#)

2013, p. 51), a global credit-money system reproduces the universal equivalent through an international currency hierarchy (Ivanova, 2020; Vasudevan, 2009). National currencies are stratified by their degree of international acceptability, liquidity, and convertibility. World money is the historically specific form wherein the liabilities of the dominant hegemonic power operate as the universal means of international payment, reserve asset, and unit of account.

3.3.2 The Bretton Woods System and Peripheral Balance-of-Payments Constraints

3.4 The post-war Bretton Woods monetary system codified this currency hierarchy institutionally. By pegging the U.S. dollar to gold at \$35 per ounce while anchoring other currencies to the dollar, the international monetary order formalized the dollar as the operative form of world money (Vasudevan, 2009, p. 483). This arrangement granted the hegemonic center an “*exorbitant privilege*”: the ability to borrow internationally in its own currency, finance external deficits without reserve constraints, and act as banker to the world (Vasudevan, 2019, pp. 243–244).

3.5 For peripheral economies, this institutional structure generated an acute structural asymmetry. Peripheral states cannot settle international accounts in their own domestic currencies; external liabilities and essential imports must be discharged in world money. The peripheral circuit of capital confronts an absolute balance-of-payments constraint (Abeles, 2013; Vasudevan, 2016). Domestic credit expansion is bounded by the capacity to command foreign exchange through export earnings, terms of trade, and access to international credit lines. Reserve accumulation constitutes an externally enforced structural prerequisite of economic reproduction (Vasudevan, 2019, p. 244).

3.6 The unilateral closure of the gold window by the United States on August 15, 1971 shattered the fixed-parity anchor of Bretton Woods agreements leading to their demise. While this rupture removed gold convertibility, it preserved dollar centrality, ushering in an era of global floating fiat money (Vasudevan, 2009, p. 486). For the nascent socialist project of the Unidad Popular, this systemic shock dismantled the international monetary stability upon which its initial macroeconomic projections rested.

3.3.3 The Peripheral Central Bank Balance Sheet: From Reserve Adequacy to Balance-Sheet Solvency

3.7 Financial subordination materializes concretely on the balance sheet of the peripheral central bank. Mainstream open-economy models interpret this balance sheet through the lens of “*reserve adequacy*” and precautionary self-insurance, treating international reserves as an inventory buffer optimized against friction costs to insulate the economy from sudden stops (Céspedes and Chang, 2024; Jeanne, 2007). This normative framing assumes that reserve accumulation represents the voluntary, unconstrained choice of a sovereign optimizing agent.

3.8 In peripheral capitalism, reserve trajectories are the structural outcome of the balance-of-payments constraint, unequal exchange, and financial subordination (Abeles, 2013; Vasudevan, 2009). The two sides of the peripheral central bank’s balance sheet reflect fundamentally distinct monetary forms. On the liability side, domestic money represents credit obligations denominated in the national unit of account, which the central bank can issue to meet payments-system liquidity

(Basu, 2022, §3.2.3). On the asset side, gross international reserves (IR_t , measured in USD millions) consist of liquid external claims denominated in world money—convertible foreign exchange balances, short-term foreign sovereign assets, and monetary gold—which the peripheral state cannot emit and can acquire only through export surpluses, net foreign capital inflows, or international credit lines. While the central bank can expand domestic liquidity endogenously to accommodate internal clearing, its ability to settle external transactions and service foreign debt depends strictly on this stock of world money.

3.9 This balance-sheet structure is governed by the Central Bank Solvency Ratio:

$$\text{SolvR}_t \equiv \frac{e_t \cdot IR_t}{M_t} \equiv \frac{F_t}{M_t} \quad (3.2)$$

where IR_t denotes international reserves in USD, e_t is the nominal exchange rate (Escudos/Pesos per USD), $F_t \equiv e_t \cdot IR_t$ is the domestic-currency value of foreign reserve assets, and M_t represents the generic theoretical concept of domestic monetary liabilities.² In the empirical baseline developed in Section 3.4 and Section 3.5, central bank liabilities are operationalized with high-powered base money H_t ($\text{SolvR}_t^H \equiv [e_t \cdot IR_t]/[H_t \cdot 1000]$), while narrow commercial-bank money ($M1_t$) is reserved for the banking-sector deconstruction.

In foreign-currency units, the reserve-flow identity is $\Delta IR_t = CA_t + KA_t + EO_t + SDR_t$, where CA_t , KA_t , EO_t , and SDR_t denote current account receipts, capital account net transactions, errors and omissions, and SDR allocations. Because $F_t \equiv e_t \cdot IR_t$, the exact discrete change in domestic-currency foreign assets is $\Delta F_t = e_t \Delta IR_t + IR_{t-1} \Delta e_t$, distinguishing reserve flows from exchange-rate valuation effects. Following the formal balance-sheet accounting derived in Appendix 3.A, the exact discrete law of motion governing the level Solvency Ratio is:

$$\Delta \text{SolvR}_t = \frac{\Delta F_t - \text{SolvR}_{t-1} \Delta M_t}{M_t} \quad (3.3)$$

where the numerator decomposes the net change in domestic-currency foreign assets (ΔF_t) and the domestic liquidity dilution effect evaluated at the lagged solvency stock ($\text{SolvR}_{t-1} \Delta M_t$), scaled by current monetary liabilities M_t .

Equation 3.3 highlights the analytical distinction between the level stock of foreign-exchange backing (SolvR_t) and the dynamic flow gap. While SolvR_t tracks cumulative external backing, the stationary transition variable estimated in the Threshold VAR (Section 3.5.4) is the predetermined Solvency Growth Gap:

$$\Delta s_{t-1} \equiv g_{t-1}^F - g_{t-1}^H = \Delta \ln \text{SolvR}_{t-1}^H \times 100, \quad (3.4)$$

with $g_t^F \equiv 100 \Delta \ln F_t$ and $g_t^H \equiv 100 \Delta \ln H_t$. The estimated threshold ($\hat{\gamma} = -4.615\%$) identifies a structural break in the growth rate of foreign assets relative to base money, rather than a threshold in the level stock of reserves. The complete accounting derivation is set out in Appendix 3.A.

²The Solvency Ratio is the inverse of the Leverage Ratio ($\text{LevR}_t \equiv M_t/(e_t \cdot IR_t) = \text{SolvR}_t^{-1}$). The solvency formulation is preferred throughout this study because as reserves shrink toward zero, the leverage ratio explodes and distorts the statistical distribution, whereas the Solvency Ratio converges smoothly to zero, yielding a well-behaved index for threshold estimation.

3.3.4 Money Endogeneity vs. Exogeneity and the Peripheral Paradox

3.10 Grounding the peripheral balance sheet requires resolving the foundational debate over the causality between bank credit, deposits, and central bank reserves (Lavoie, 2014; Rochon and Rossi, 2013). Mainstream monetarism and orthodox macroeconomics operate under Fractional Reserve Theory (FRT) and loanable-funds: the central bank exogenously fixes the monetary base (H_t), which mechanically dictates bank lending and broad money through a stable fractional-reserve multiplier ($H \rightarrow \text{Deposits} \rightarrow \text{Loans} \rightarrow P$). In sharp contrast, Post-Keynesian Endogenous Money Theory (EMT) and Marxian monetary circuits establish that real-world banking operates in reverse causal sequence (Kaldor, 1982; Lavoie, 2014):

$$\Delta \text{Bank Loans} \longrightarrow \Delta \text{Deposits} \longrightarrow \Delta \text{Central Bank Reserves}(H) \quad (3.5)$$

Commercial banks extend working-capital loans to firms to finance payrolls and production. These loans simultaneously create bank deposits. The central bank, far from controlling the money supply as an exogenous policy instrument, is compelled to accommodate required reserves ex-post at the policy discount rate to maintain clearance in the interbank payments system. As Kaldor (1982) argued, credit money has no independent supply function: it comes into existence through the demand for bank lending and is extinguished through repayment. The central bank must accommodate this process to prevent the collapse of the credit and payments system. Monetarism, in this light, is not a neutral technical rule but a political strategy that deploys high interest rates and monetary contraction to induce unemployment and break trade union bargaining power. Under EMT, the traditional money multiplier ($mm_t \equiv M_t/H_t$) is not an exogenous causal driver but an unstable, ex-post accounting ratio reflecting commercial bank liquidity preferences, cash-drain ratios, and portfolio balance constraints. In structuralist endogenous-money approaches, credit supply is upward-sloping rather than horizontal (Dow, 2006; Rochon, 1999). In peripheral capitalism, however, commercial banks' capacity and willingness to intermediate credit are bounded by external liquidity constraints and foreign exchange exposure.

3.11 This institutional reality establishes what I define as the *Peripheral Paradox*. In core economies issuing world money, the central bank accommodates domestic credit without facing an absolute currency constraint. In the periphery, however, while domestic credit generation is naturally endogenous ($\text{Loans} \rightarrow M_t$), the ultimate means of international settlement is bound to an exogenously constrained reserve asset ($e_t \cdot IR_t$). The peripheral central bank must back an endogenously expanding domestic credit pyramid with an externally constrained stock of world money. When external credit embargoes, copper price collapses, or private capital flight drain foreign reserves, the state cannot act as an unconstrained lender of last resort in hard currency; it must choose between contracting domestic activity or suffering balance-sheet insolvency. Under the UP, the nationalized commercial banking system under the State Development Corporation (CORFO) extended defensive working-capital credit to socialized and private enterprises facing distribution bottlenecks and price controls, compelling the Central Bank to monetize enterprise deficits. Domestic money growth thus functioned as an endogenous accommodation valve, driven by structural cost-push pressures rather than discretionary printing press impulses. This reverse causal sequence ($\Delta \text{Loans} \rightarrow \Delta \text{Deposits} \rightarrow \Delta H$) and its balance-sheet constraint under the Peripheral Paradox provide the operational hypotheses evaluated in the sequential Granger causality battery in Section 3.5.3.

3.3.5 Peripheral Financial Fragility and Central-Bank Solvency

3.12 To understand how this balance-sheet contradiction generates macroeconomic crisis, I draw on the financial fragility framework of [Minsky \(1986\)](#). In Minsky’s canonical framework, financial fragility is endogenously generated during prolonged expansions: stability breeds instability as private firms and banks increase leverage, increasing their vulnerability to liquidity shortfalls ([Kregel, 2008](#)). However, Minsky formulated this dynamic for private corporate actors in a closed, metropolitan economy possessing complete monetary sovereignty, where an unconstrained central bank (the “Big Bank”) and fiscal authority (the “Big Government”) stabilize an intrinsically unstable system. Minsky’s categorical taxonomy of hedge, speculative, and Ponzi finance was developed specifically for private cash flows and corporate debt commitments. The peripheral central bank is neither private nor unconstrained: it does not lever up out of speculative euphoria, but is compelled to intermediate liquidity under structural balance-of-payments constraints. Rather than attempting to map Minsky’s categorical taxonomy directly onto the central bank, this study retains Minsky’s foundational intuition of financial fragility while reinterpreting its institutional locus through peripheral open-economy conditions.

3.13 Reinterpreting financial fragility for the peripheral central bank rests on two complementary foundations. First, [Foley \(2003, p. 21\)](#) demonstrates that during acute financial distress, the state and the central bank as lender of last resort are compelled to absorb the insolvency of private enterprises onto their own balance sheets. The central bank expands credit not out of profit-seeking, but to prevent systemic payments breakdown and societal disruption. Second, [Chamorro-Futinico \(2025\)](#) demonstrates that financial instability in peripheral Latin American economies operates as an *externally amplified endogenous fragility*. Minsky’s dynamic cannot be applied to dependent formations without recognizing that external shocks from the center—the collapse of world monetary arrangements, terms-of-trade declines, and credit embargoes—serve as structural triggers that detonate domestic balance-sheet fragility through the international currency hierarchy. During acute external instability, the peripheral central bank enters a state of defensive accommodation under severe balance-of-payments constraints. When external restrictions cut off commercial trade credit, refusing domestic liquidity expansion would trigger immediate real-sector collapse. Accommodating the deficits of socialized enterprises becomes an institutional imperative to maintain economic reproduction.

3.14 This reinterpretation establishes the central theoretical sequence of the chapter:

financial fragility $\longrightarrow$ peripheral external amplification
 $\longrightarrow$ central-bank balance-sheet exposure $\longrightarrow$ state-dependent accommodation.

The macroeconomic transmission behavior of the peripheral central bank is state-dependent on its balance-sheet solvency conditions. When foreign-exchange backing is adequate, the central bank possesses operational policy space: domestic credit expansion mobilizes idle capacity, and external price shocks are buffered at the border. As balance-sheet conditions deteriorate past critical thresholds, this buffer dissolves: defensive monetary emission, lacking hard-currency backing, transmits directly into price acceleration and currency depreciation. Crucially, the empirical regimes identified in this chapter are not derived from Minsky’s categorical taxonomy; they are estimated directly from the data through the Solvency Growth Gap (Δs_{t-1}) in the Threshold Vector Autoregression (TVAR) developed in Section [3.5](#).

3.4 Data Sources and Variable Construction

3.4.1 Monthly Panel: Sources and Sample Windows

The empirical core spans the 252 raw calendar months between 1960:01 and 1980:12, yielding a transformed and usable monthly panel of $N = 250$ observations after initial differencing and lag construction, assembled programmatically from the Central Bank of Chile REST API (SieteRestWS). The API provides continuous monthly series for high-powered base money (H_t , recorded in thousands of millions of pesos [billions, *miles de millones de pesos*]), the official nominal exchange rate (e_t , pesos per US dollar; pre-1975 escudo values retroactively converted at 1 peso = 1,000 escudos per the BCCh *Indicadores Económicos y Sociales de Chile 1960–2000* convention), London Metal Exchange copper cash prices (P_t^{Cu} , USD/lb), London bullion gold prices (P_t^{gold} , USD/oz), the US Wholesale Price Index ($P_{W,t}^{\text{US}}$), the official Consumer Price Index (π_t), and the US Dow Jones Industrial Average. The Santiago Stock Exchange (IGPA) index is drawn from the replication files of [Girardi and Bowles \(2018\)](#). Gross international reserves (IR_t , millions of USD) come from the IMF *International Financial Statistics*.

Sectoral real activity uses the disaggregated monthly component series compiled by [Díaz-Bahamonde \(2023\)](#): light industrial manufacturing (Q_t^{manuf}), copper mining extraction (Q_t^{mining}), and physical export and import volumes (Exports_t , M_t). I use the individual components, not his aggregated IMEA composite index. The components are harmonized from *Estadística Chilena* (DGE/INE) and the BCCh *Boletín Mensual* and are available from 1927 onward; I restrict the panel to 1960:01–1980:12 to align with the monetary and reserve series.

Narrow money ($M1_t$, recorded in thousands of millions of pesos [billions, *miles de millones de pesos*]) and the observed banking multiplier ($m_t \equiv M1_t/H_t$) are available only over the un-spliced window 1966:01–1980:12 ($N = 180$), which excludes pre-1966 commercial bank classification revisions. $M1_t$ enters the empirical specification exclusively in the commercial-banking bifurcation and multiplier deconstruction steps of the Granger causality tests (Section 3.5.3); the primary solvency threshold and the full-panel TVAR use H_t throughout.

A supplementary annual tier ($T = 91$, 1920–2010) provides the long-run accumulation backdrop used in Sections 3.5.1 and 3.5.2: the functional wage share (ω_t) from [Astorga \(2023\)](#); the productive capital stock ($K_t \equiv K_t^{\text{ME}} + K_t^{\text{NRC}}$, excluding residential construction), the aggregate profit rate (r_t , nominal surplus on productive fixed capital at current replacement cost), and structural capacity utilization (μ_t , estimated via cointegrating polynomial regression) from the GPIM dataset of Chapter 2; and balance-of-payments accounts from the World Historical Balance of Payments Database ([Nievas and Piketty, 2025](#)). Appendix 3.B catalogs complete series identifiers and metadata.

3.4.2 Variable Construction

The primary Central Bank reserve solvency ratio, constructed over the full monthly panel, is

$$\text{SolvR}_t^H \equiv \frac{e_t \cdot IR_t}{H_t \cdot 1000}, \quad (3.6)$$

where e_t is in pesos per dollar, IR_t is in millions of dollars (yielding a numerator $e_t \cdot IR_t$ in millions of pesos), and H_t is recorded in thousands of millions of pesos (billions, *miles de millones de pesos*) in the official Central Bank statistical database. The factor of 1,000 converts H_t to millions of pesos ($H_t \cdot 1000$), ensuring dimensional consistency between reserve assets and monetary liabilities. Base money is the appropriate primary liability because it is the Central Bank's direct obligation and is observed continuously from 1960:01. In the threshold specification, the estimated transition variable is the predetermined lagged Solvency Growth Gap ($\Delta s_{t-1} \equiv g_{t-1}^F - g_{t-1}^H$), with threshold estimation and inference following the grid-search procedure of Hansen (1999).

For the commercial-banking decomposition ($N = 180$), I also construct

$$\text{SolvR}_t^{M1} \equiv \frac{e_t \cdot IR_t}{M1_t \cdot 1000},$$

which correlates with SolvR_t^H at $\rho = 0.974$. This measure is not the primary solvency variable; it is used only to test whether the commercial credit channel alters the ordering between external backing and domestic liabilities.

The Prebisch-ECLAC capacity to import measures the purchasing power of physical exports relative to imported manufactured inputs:

$$M_t^{\text{cap}} \equiv \left(\frac{\text{Exports}_t}{\text{Exports}_{1970:11}} \right) \times \left[\frac{P_t^{\text{Cu}} / P_{1970:11}^{\text{Cu}}}{P_{W,t}^{\text{US}} / P_{W,1970:11}^{\text{US}}} \right] \times 100, \quad (3.7)$$

benchmarked to November 1970 (1970:11 = 100). The real structural imbalance ratio is

$$\Theta_t \equiv \left(\frac{M_t}{M_t^{\text{cap}}} \right) \times 100. \quad (3.8)$$

When $\Theta_t > 100\%$, physical import volumes exceed export purchasing power, generating a foreign-exchange deficit that must be financed through credit lines or reserve depletion.

3.4.3 Data Adjustments

The raw customs import volume series exhibits a single-month administrative registration drop in April 1973: the recorded volume falls from 2.88 (March) to 0.26 (April), recovering to 2.63 (May). The isolated drop is treated as an administrative registration anomaly consistent with a paperwork lag in clearing port arrivals into official records, not a collapse in physical trade. I replace the April observation with the monotonic linear midpoint, 2.76, to prevent the registration artifact from triggering a spurious structural break in the VAR systems.

3.4.4 Integration Order of Variables Included in Empirical Exercises

The transformed series entering the vector autoregressive and threshold VAR specifications are stationary in levels ($I(0)$). Table 3.1 reports the ADF diagnostics; the transformed series reject the unit-root null at conventional significance levels.

Table 3.1. Univariate Augmented Dickey–Fuller Tests and Order of Integration

Variable / Series	Series in Levels (z_t)		First Difference (Δz_t)		Order	Stationarity Decision
	t -ADF	5% Crit.	t -ADF	5% Crit.		
CPI Inflation (π_t)	-3.372	-2.87	-13.619***	-2.87	I(0)	Stationary in transformed series ($I(0)$)
Base Money Growth ($g_{H,t}$)	-3.751	-2.87	-12.484***	-2.88	I(0)	Stationary in transformed series ($I(0)$)
Narrow Money Growth ($g_{M1,t}$)	-4.059	-2.88	-17.812***	-2.88	I(0)	Stationary in transformed series ($I(0)$)
Multiplier Growth ($\Delta \ln m_t$)	-12.791	-2.88	-11.957***	-2.88	I(0)	Stationary in transformed series ($I(0)$)
Manufacturing Output ($g_{\text{Manuf},t}$)	-14.105	-2.87	-19.409***	-2.88	I(0)	Stationary in transformed series ($I(0)$)
Mining Extraction ($g_{\text{Mining},t}$)	-14.910	-2.87	-13.947***	-2.88	I(0)	Stationary in transformed series ($I(0)$)
Structural Imbalance ($\Delta \ln \Theta_t$)	-14.512	-2.87	-17.171***	-2.88	I(0)	Stationary in transformed series ($I(0)$)
World Gold Price Growth ($g_{P,\text{gold},t}$)	-10.559	-2.87	-17.959***	-2.88	I(0)	Stationary in transformed series ($I(0)$)
Nominal Exchange Rate Growth ($g_{e,t}$)	-3.777	-2.87	-21.142***	-2.88	I(0)	Stationary in transformed series ($I(0)$)
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	-8.726	-2.87	-14.011***	-2.88	I(0)	Stationary in transformed series ($I(0)$)

Notes: Monthly series spanning 1960:01–1980:12 ($N = 250$, with official un-spliced narrow money $M1$ and multiplier m spanning 1966:01–1980:12, $N = 180$). Augmented Dickey–Fuller (ADF) tests estimated with constant drift; lag orders selected via Schwarz Bayesian Information Criterion (SBIC) up to $p = 4$. Critical values correspond to MacKinnon (1996) finite-sample distributions. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All entered macroeconomic and institutional variables reject the unit-root null at conventional significance levels, providing evidence consistent with covariance stationarity ($I(0)$) of the difference-transformed empirical system.

3.5 Historical Background, Stylized Facts, and Empirical Results

3.5.1 Historical Background: The Rise of the Unidad Popular

The Unidad Popular’s electoral victory in 1970 brought a programme of socialist transformation into contact with social forces that had been reorganizing Chilean society for several decades. Industrial workers, rural laborers, *pobladores*, students, and sections of the urban middle classes did not enter the political scene at the same moment or through the same institutions. Their demands were rooted in different locations of the economy and in different experiences of the developmental state. What gave the Popular Unity coalition its historical depth was the gradual convergence of conflicts around work, land, urban residence, and social rights. This section reconstructs that convergence as four waves of social conflict. Throughout, these historical waves must be distinguished from the accounting windows used in the empirical analysis: the waves describe historical political periods, whereas the profitability decompositions in Tables 3.2 and 3.3 begin later and are bounded by annual data availability.

The economic background to these conflicts reaches back before the Great Depression. Chile entered the twentieth century as an export economy centered on nitrates, with mineral rents financing a large share of public revenue and sustaining commercial, fiscal, and urban expansion. This export boom also generated domestic demand. Public expenditure, mining settlements, transport networks, and the growth of Santiago, Valparaíso, and other cities enlarged the market for locally produced manufactures. A meaningful manufacturing sector thus emerged well before state-led import substitution became an explicit development strategy. On the eve of the First World War, manufacturing represented roughly one-seventh of Chilean output, though concentrated primarily in food, beverages, textiles, clothing, and other simple consumer goods (Bulmer-Thomas, 2003). This was industrial growth nested within an export economy rather than an autonomous industrial take-off.

In older historiography, the international disruptions beginning in 1914 were often treated as the decisive stimulus to domestic manufacturing (Palma, 1984). The newer reconstruction by Pérez-Eyzaguirre substantially qualifies that interpretation. His revised series shows that Chilean industrial

production had expanded alongside the export economy prior to 1914 and then contracted sharply when the war disrupted trade, imported inputs, and capital flows. Under this reconstruction, the level of industrial production achieved in 1913 was not recovered until 1941 (Pérez Eyzaguirre, 2025). The war exposed how strongly domestic manufacturing still depended on the commercial and financial circuits generated by primary exports, rather than initiating an uninterrupted wartime industrial boom.

Urbanization followed an equally distinctive trajectory. Chile was already highly urbanized by Latin American standards prior to the classic period of state-led industrialization. Pérez-Eyzaguirre estimates that 46.7 percent of the population lived in urban settlements by 1930, emphasizing that the nineteenth- and early twentieth-century urban transition was tied more closely to export expansion, state formation, and successive primary-export cycles than to manufacturing employment alone (Pérez Eyzaguirre, 2017). This urban transition shaped subsequent labor politics: an urban population, an expanding state apparatus, and a domestic market formed before industry could absorb the majority of workers leaving rural activities. Chile entered the Depression with an unusually urban social structure relative to the productive apparatus supporting it.

The collapse after 1929 transformed these inherited tensions into a crisis of the existing accumulation pattern. Chile was among the Latin American economies most exposed to the contraction of world trade: the purchasing power of exports fell by roughly four-fifths during the first years of the Depression, while aggregate output fell dramatically between 1929 and 1932 (Bértola and Ocampo, 2012; Bulmer-Thomas, 2003). Between 1929 and 1932, recorded employment fell from 1.43 million to 1.17 million workers, a loss of roughly 262,000 jobs, or 18 percent of the 1929 employment level. Over the same interval, measured unemployment increased from about 34,000 to 374,000 workers in the historical series used in this chapter, raising the unemployment rate from 2.35 to 24.30 percent. Beyond the immediate loss of national income, the collapse sharply contracted the foreign-exchange flows around which fiscal revenue, imports, mining demand, transport, commerce, and much of the domestic market had been organized. Restoring economic activity required a fundamentally different relation between the state, domestic demand, and production.

The initial policy response was pragmatic. Under Arturo Alessandri's second administration (1932–1938), deficit spending, public works, industrial credit, exchange controls, and tariff protection were deployed to reactivate employment and domestic production. As Silva (2007) emphasizes, import substitution emerged during this stage through crisis management before crystallizing into a coherent developmental model. Emergency state intervention reorganized accumulation before the institutional machinery of the later developmental state was established. Industrial policy remained compatible with agrarian interests, supporting branches such as food processing and leather that were directly linked to domestic agriculture. This compatibility explains why early restructuring attracted joint support from financiers, merchants, landowners, and industrialists without requiring an immediate political rupture between an industrial bourgeoisie and a landed oligarchy.

Sources of growth between 1929 and 1939 reflect this structural redirection. For Chile over this decade, Bértola and Ocampo estimate that import substitution contributed approximately 1.3 percentage points per year to growth, while the contributions associated with domestic demand and exports were slightly negative, yielding aggregate output growth of only about 0.8 percent per year (Bértola and Ocampo, 2012). Domestic production substituted for imports while the external and domestic demand components inherited from the nitrate regime remained depressed. Manufacturing recovered much faster than aggregate activity, averaging roughly 7.7 percent annual growth between

1932 and 1939 ([Bulmer-Thomas, 2003](#))—a trajectory of structural import substitution distinct from simple cyclical recovery.

The Popular Front victory in 1938 accelerated this transformation into a second stage. The creation of CORFO in 1939 provided the state with an institutional vehicle for economic planning, infrastructure provision, public enterprise, and the long-term financing of strategic industries. The expansion of electricity, steel, fuels, transport, and intermediate inputs widened the productive base beyond the light consumer branches inherited from the export era. Under CORFO's coordination, industry's share of GDP rose from approximately 13 percent in 1940 to roughly 21 percent in 1950 ([Silva, 2007](#)). During the Second World War, Chilean industrial output expanded much faster than aggregate GDP, consolidating manufacturing growth while world trade was disrupted ([Bulmer-Thomas, 2003](#)). By 1950, manufacturing represented roughly a quarter of Chilean GDP in the Bértola–Ocampo estimates, exceeding the Latin American average ([Bértola and Ocampo, 2012](#)). The postwar economy was substantially more industrial than the economy that had entered the Great Depression.

The consolidation of Chilean import substitution coincided with the construction of what regulation theory identifies as Atlantic Fordism. In the United States and Western Europe, mass production was embedded in national growth regimes based on mass consumption, collective bargaining, Keynesian demand management, and welfare-state institutions, while Bretton Woods provided an international monetary framework compatible with national policy autonomy ([Vidal, 2015, 2019](#)). Much of this Golden Age growth remained organized around domestic and regional mass markets rather than unrestricted global product-market competition.

The internationalization of this accumulation regime did not imply that peripheral economies reproduced the institutional architecture of the Fordist core. As [Cardoso \(1972\)](#) emphasized in his analysis of the new forms of dependency, postwar foreign capital increasingly entered industrial production itself, often in combination with local private and state capital. Industrialization and structural dependency advanced together: while domestic manufacturing deepened, the reproduction of productive capacity remained asymmetrical because advanced technology and the production of means of production remained concentrated in the core capitalist economies. Dependent industrialization thus required external complementarity to complete its accumulation process, even as Latin America's share of world trade fell from 12 percent in 1948 to 6 percent in 1968 ([Cardoso, 1972](#)).

Chile's developmental trajectory was therefore a peripheral adaptation to the North Atlantic Fordist order rather than a domestic replica of Fordism. Industrial capacity expanded, but its reproduction deepened dependence on imported capital goods, intermediate inputs, replacement parts, foreign technology, and external finance. In his contemporary diagnosis of the early CORFO national accounts, [Kaldor \(1959\)](#) observed that gross fixed-capital formation had only occasionally exceeded 10 percent of GNP after 1940, with more than half of domestic investment undertaken by the public sector. His estimates demonstrated that imported capital goods accounted for roughly two-fifths of fixed investment across the benchmark years examined. The external constraint had moved deeper into the production process itself.

Kaldor also identified an internal bottleneck central to Chilean accumulation: agriculture. Industrial output, non-agricultural employment, and urban incomes expanded faster than the domestic supply of food. Under these conditions, rising demand for wage goods exerted persistent upward pressure on food prices, which fed into wage demands and industrial operating costs. Kaldor treated agricultural

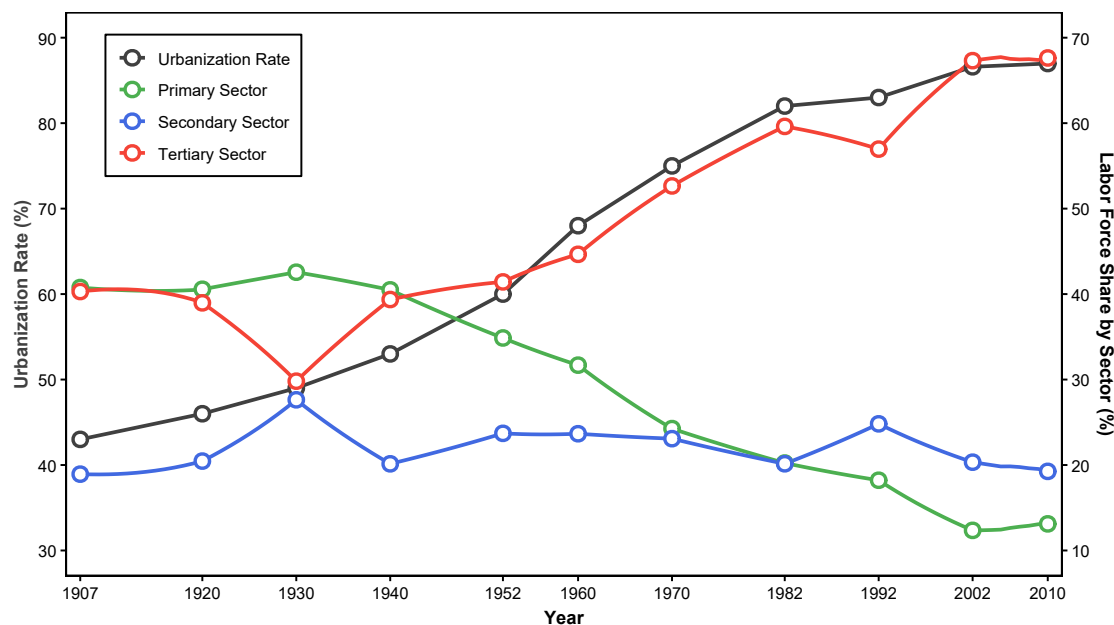
stagnation, concentrated landholding, and low rural investment as an integral part of the same accumulation problem as industrial growth (Kaldor, 1959). The structural burden of sustaining basic infrastructure and capital formation fell disproportionately onto the state in the face of sluggish private domestic accumulation.

The social transformation generated by industrialization was decidedly uneven. Urbanization advanced faster than factory employment. Between 1940 and 1970, tertiary activities absorbed the largest share of net labor-force growth, while manufacturing and construction combined absorbed a considerably smaller share; the fact that positive sectoral contributions sum to more than 100 percent reflects the absolute contraction of primary-sector employment over the interval. By 1970, in an economically active population of about 3.2 million, De Vylder (1976) counted roughly 260,000 openly unemployed workers and another 600,000 in low-productivity service activities. The city thus became the site where incomplete labor absorption, informal service employment, housing deficits, and unequal access to municipal infrastructure were systematically reproduced.

Chilean industrialization was disarticulated not because it failed to expand manufacturing, public employment, or basic infrastructure, but because productive capacity, labor absorption, and social incorporation proceeded at starkly uneven tempos. In 1970, 87.3 percent of rural dwellings and 26.8 percent of urban homes still lacked piped potable water (De Vylder, 1976). The state expanded education, healthcare, and social security while presiding over an unequal geography of incorporation. Migrants entered cities where social services and citizenship were materially more accessible than under the rural estate, even as stable factory employment and adequate housing remained scarce.

Rather than a rigid dichotomy between “feudal” estates and “capitalist” cities, the Central Valley hacienda had long combined commercial production for national and world markets with concentrated extra-economic control over tenant labor (*inquilinaje*) (Kay, 1980). The gradual erosion of these agrarian relations released labor, altered the composition of the working class, and shifted the geographical terrain of collective mobilization without eliminating an urban labor reserve.

Four successive waves of social mobilization and state containment structured this developmental period. These waves track shifts in popular organization, the spatial locus of protest, and state responses. While macroeconomic profitability, capital accumulation, and employment bounded the material space in which disputes unfolded, political organization and state institutions decisively shaped how distributional pressures translated into collective action.



Source: Author's own elaboration based on ClioLab / Díaz, Lüders & Wagner (2016).
 Note: Hollow circle markers are displayed at decennial census benchmark years (1907, 1920, 1930, 1940, 1952, 1960, 1970, 1982, 1992, 2002, 2010).

Figure 3.1. Sectoral labor-force composition and urbanization in Chile, 1907–2010.

Source: Author's elaboration from Clio-Lab PUC historical statistics (Díaz et al., 2016). Sectoral series refer to the labor force; urbanization refers to population. These measures have different denominators. Census urbanization observations are plotted as points.

Wave I: Selective incorporation and coercive backlash, 1932–1947. The first wave emerged during the reconstruction of the 1930s and reached its crest in the mid-1940s. The 1931 Labor Code established a national framework for collective organization, but incorporation was highly uneven. Urban unions gained a recognized place in industrial relations and electoral politics, while agricultural workers remained subject to estate-level authority and legal restrictions (Loveman, 1976; Milos, 2008). The developmental state expanded productive and social capacities while preserving a differentiated labor regime between city and countryside.

This tension was visible from the beginning. The collapse of the export economy produced unemployment on a scale that weakened labor's immediate bargaining position, but it also destabilized established relations of authority. Rural workers and *inquilinos* used petitions, strikes, and clandestine organization to demand statutory wages, written contracts, and security of tenure. The Ranquil conflict of 1934 exposed the coercive limits of incorporation in the countryside. Under the Popular Front, Communist and Socialist organizers intensified efforts to organize rural labor even as coalition governments remained cautious about confronting landowners directly (Loveman, 1976).

By the mid-1940s, rural mobilization converged with renewed industrial militancy. The Plaza Bulnes massacre and the general strikes of January and February 1946 marked the urban-industrial crest of this wave (Bravo Vargas, 2017). Registered agricultural petitions also rose sharply and reached their maximum in 1947 in the series assembled from Loveman's archival evidence (Figure 3.2). This is one reason the 1943–1947 profitability window in Table 3.2 should not be mistaken for the beginning of the wave: it captures the final observed segment of a conflict that had been building since the 1930s.

Distributional accounts are consistent with this tightening conflict. During 1943–1947, a declining profit share (−5.53 percent per annum) accounted for the primary negative contribution to profit-rate growth in the chapter’s decomposition (Table 3.2). While this pattern demonstrates that workers were gaining distributional ground in the observed interval, it reflects rather than explains the underlying political mobilization.

The state responded through legal segmentation and political repression. Law 8,811, enacted in July 1947, recognized agricultural unions while confining them to individual *fundos*, restricting the seasonal timing of petitions, and imposing eligibility rules that blocked organization across estates (República de Chile, 1947). The following year, Law 8,987 outlawed the Communist Party and excluded targeted militants from electoral and union activity (República de Chile, 1948). These measures did not resolve rural grievances; they suppressed the organizational channels through which those grievances had been articulated.

The closure of rural organization altered the geography of conflict. Migration to urban centers was driven by structural factors predating the late 1940s repression, but the combination of agrarian stagnation, restricted rural organization, urban industrial concentration, and expanding municipal services made cities central to popular politics.

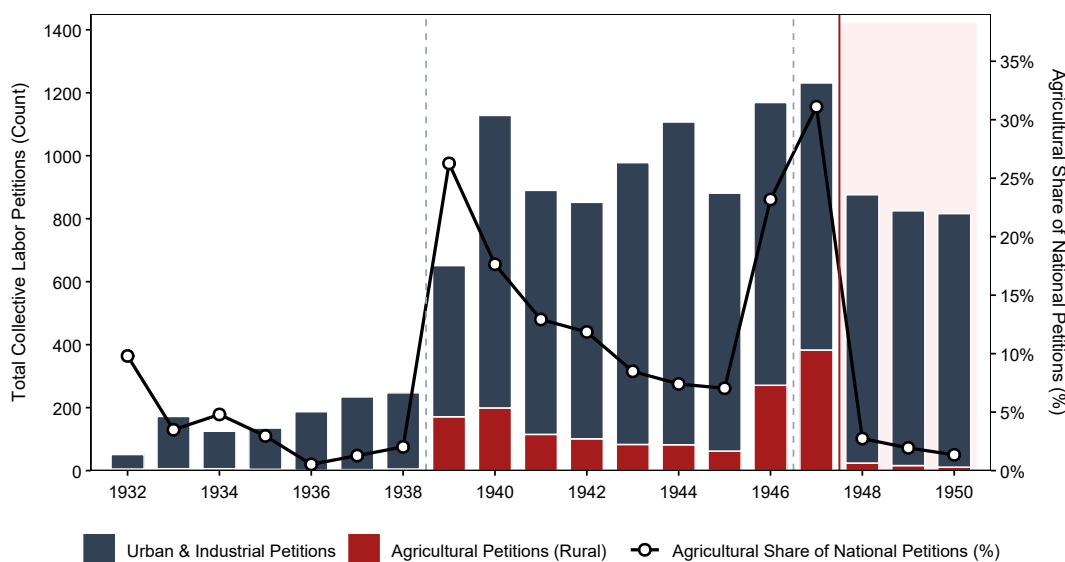


Figure 3.2. Recorded collective labor petitions by sector, 1932–1950.

Source: Author’s archival compilation from Loveman (1976). Petitions measure registered collective demands under the prevailing legal framework, not all forms of social conflict.

Wave II: Coercive containment and urban recomposition, 1948–1958. The second wave was defined less by the disappearance of conflict than by its containment and relocation. The restrictions introduced in 1947–1948 narrowed legal space for rural and left organization. During the Ibáñez administration, the stabilization programme associated with the Klein–Saks mission added statutory wage restraint, credit contraction, and price increases on public transport and foodstuffs. Real wages fell sharply in the mid-1950s while inflation accelerated, shifting the costs of stabilization onto wage earners through workplace discipline and household consumption.

The profitability decomposition documents the distributional consequences of this containment. Between 1948 and 1958, a rising profit share (+3.11 percent per annum) provided the largest positive contribution to profit-rate growth (+2.93 percent per annum, Table 3.2), while capital deepening (−2.60 log points) offset gains in potential labor productivity (+2.29 log points). Distribution shifted toward capital even as industrial and urban transformation continued.

Urban society, however, became more difficult to contain. Industrialization and public investment drew migrants toward the cities without providing factory employment or adequate housing on a commensurate scale. Rather than forming an inert residual of incomplete industrialization, urban neighborhoods became autonomous centers of collective action where migrants, factory workers, informal wage earners, and women organized around transport tariffs, food prices, housing, clinics, and municipal utilities.

In April 1957, protests over transit fare hikes erupted into a nationwide urban revolt, met by military suppression in Santiago, Valparaíso, and Concepción. In October 1957, organized families occupied land south of Santiago to establish the *población* La Victoria. Pobladores did not merely petition the state for housing; they occupied land, organized settlement committees, constructed basic infrastructure, and negotiated collectively with public authorities (Garcés, 2002; Murphy, 2015). While social conflict is sometimes characterized as having simply shifted from countryside to city, rural grievances persisted alongside industrial unionism and an emerging pobladores movement. What changed was the density of organizational ties linking these arenas.

Wave III: The Long Sixties and the rise of the Unidad Popular, 1959–1970. The Long Sixties transformed that urban and rural recomposition into a broader cycle of mobilization. Industrial workers expanded their demands beyond wages toward workplace control, challenging managerial authority over production quotas, safety, and the labor process. Inter-industry solidarity deepened through major conflicts, including the support of Huachipato steelworkers for the 1960 Lota coal strike, while state violence, such as the 1966 killings at the El Salvador copper mine, radicalized union federations (Thielemann, 2018).

In the countryside, the Agrarian Reform Law (Law 16,640) and the Rural Unionization Law (Law 16,625) enacted in 1967 dismantled the estate-bound restrictions of Law 8,811, recognizing communal unions and provincial confederations (República de Chile, 1967). Agricultural union membership expanded from 1,658 workers across 24 unions in 1964 to 127,680 members across 488 unions by mid-1970 (De Vylder, 1976). Across the economy as a whole, national union density over employed workers doubled from 11.2 percent in 1964 to 22.9 percent by 1970 (Osorio and Polanco, 2021) (Figure 3.3). The organizational infrastructure of labor had become dense enough to coordinate action across sectors and regions.

Simultaneously, *pobladores* expanded the politics of social reproduction beyond the wage relation. Land occupations, neighborhood committees, and demands for clinics, schools, transport, water, and paved roads turned access to the city into a field of collective bargaining with the state (Garcés, 2002; Murphy, 2015). In centers such as Concepción, industrial unions, student federations, and neighborhood assemblies coordinated strike support and shared logistical resources (Schlotterbeck, 2018). The popular movement supporting the Unidad Popular was an articulation of differentiated organized constituencies forged through the uneven development of the period.

Distributional mobilization unfolded within an expanding rather than a contracting economy: across

the 1959–1970 accounting interval, the profit rate rose from 8.16 percent to 12.23 percent, supported by positive contributions from the profit share, capacity utilization, and potential labor productivity (Table 3.2). Sustained accumulation, a larger urban labor force, and denser institutional networks expanded workers’ capacity to press claims on national income, property, and state policy.

While a Kaleckian political mechanism links sustained high employment to weakened labor discipline, Chile had not attained generalized full employment. The sectoral and institutional dynamic was decisive: tighter conditions in unionized industries and rising national union density bolstered workers’ collective bargaining power even while an informal reserve army persisted in peripheral services.

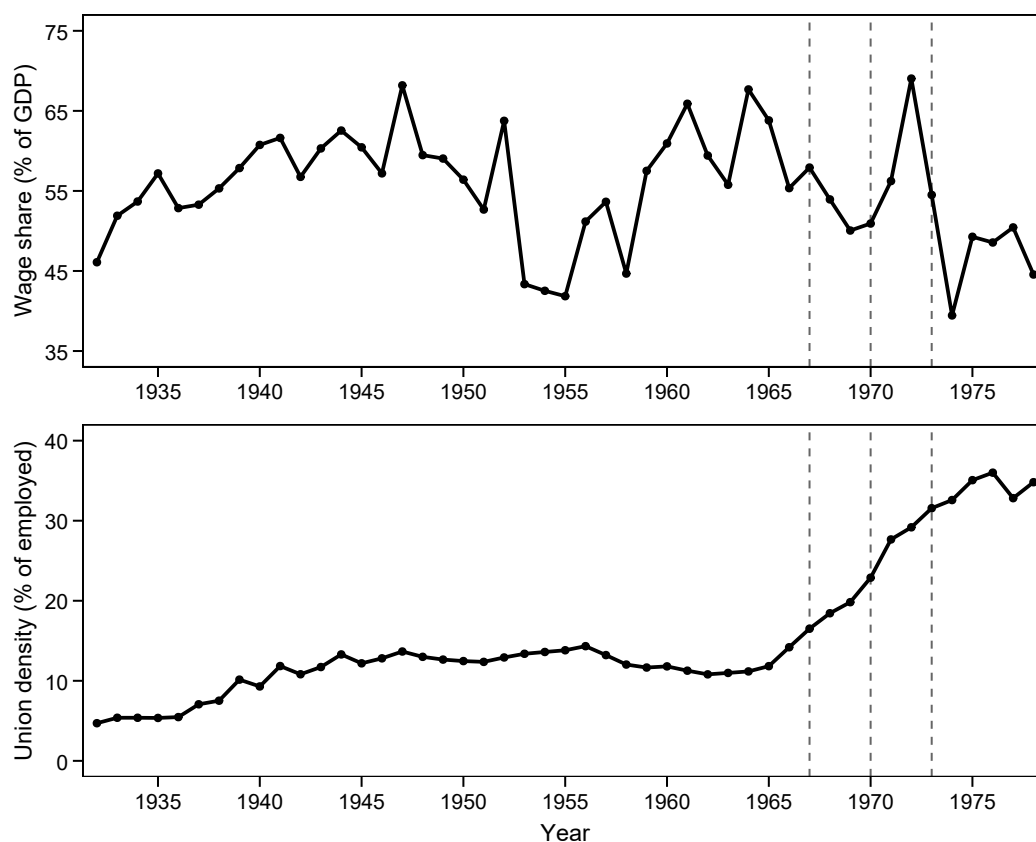


Figure 3.3. Wage share and unionization in Chile, 1932–1978.

Source: Author’s elaboration from [Astorga \(2023\)](#) for Panel A and [Osorio and Polanco \(2021\)](#) for Panel B. Panel A reports the functional wage share (W/Y , percent of GDP). Panel B reports union density over employed workers. Vertical dashed lines mark the 1967 agrarian reform and rural unionization statutes (Laws 16,640 and 16,625), the September 1970 election of Salvador Allende, and the September 11, 1973 military coup.

Wave IV: The UP administration and the crisis of accumulation, 1970–1973. Allende’s election in September 1970 linked these popular movements directly with executive power. The UP programme connected income redistribution to the nationalization of strategic mineral resources, expansion of the social property area, agrarian expropriation, and structural planning. Yet the organizations forged during the preceding waves did not become passive extensions of government.

Factory requisitions, land occupations, and community distribution networks frequently pushed ahead of the executive's statutory timetable (Schlotterbeck, 2018; Winn, 1986, 2016), generating dynamic tension between state policy and autonomous popular mobilization.

The administration's first year produced an exceptionally large distributive expansion. Real employee remuneration rose by 54.9 percent in 1971 following statutory wage adjustments (De Vylder, 1976), and the wage share rose from 50.9 percent of GDP in 1970 to 69.0 percent in 1972 (Astorga, 2023). The growth-accounting decomposition reveals a severe initial profit squeeze: the profit rate fell from 12.23 percent in 1970 to 6.57 percent in 1972, while the net accumulation rate declined from 4.23 to 1.85 percent and gross accumulation from 11.63 to 7.64 percent (Table 3.3). The profit share accounted for most of this decline between 1971 and 1972 (Table 3.2, Panel B).

That initial distributive squeeze cannot be projected mechanically over the full three-year interval. Across 1971–1973, relative capital prices (−4.35 percent) and capacity utilization (−4.78 percent) provided the dominant negative contributions to profit-rate growth, while the contribution of the profit share changed sign (+1.94 percent, Table 3.2, Panel A). The annual accounts for 1973 make this divergence evident: the measured profit rate rebounded from 6.57 percent in 1972 to 9.12 percent in 1973, yet net capital accumulation slowed further to 1.71 percent (Table 3.3). Because annual 1973 data aggregate the final eight months of the UP with the first four months of military dictatorship, they cannot separate pre-coup strikes from post-coup wage repression. They nevertheless demonstrate that profitability alone cannot explain the collapse of capital accumulation.

The terminal crisis concentrated contradictions that had accumulated across multiple decades: an industrial base dependent on imported capital goods and foreign finance; agrarian stagnation that generated wage-good bottlenecks; incomplete urban labor absorption; and a densely organized popular movement capable of mounting structural challenges to property and state authority.

The Unidad Popular concentrated rather than initiated these contradictions. Its redistributive programme accelerated popular mobilization within external and productive boundaries inherited from the developmental era. The next section examines the balance-of-payments constraints and monthly macroeconomic dynamics that governed the subsequent crisis.

Table 3.2. Social Conflict Waves and Profitability Decomposition, Chile, 1943–1975

Wave of Social Conflict	Profit Rate $\Delta \ln r$	Profit Sh. $\Delta \ln(1 - \omega)$	Rel. Price $\Delta \ln(P_Y/P_K)$	Cap. Util. $\Delta \ln \mu$	Lab. Prod. $\Delta \ln y^p$	Cap. Deep. $-\Delta \ln k$
<i>A. Social Conflict Waves (Annualized Rates and % Shares)</i>						
Wave I: Observed accounting segment (1943–1947)	-2.84%	-5.53% (+194.9%)	+2.63% (-92.7%)	+0.62% (-21.7%)	-0.21% (+7.6%)	-0.34% (+11.9%)
Wave II: Containment and urban recomposition (1948–1958)	+2.93%	+3.11% (+106.1%)	-0.10% (-3.3%)	+0.23% (+7.9%)	+2.29% (+77.9%)	-2.60% (-88.6%)
Wave III: Long Sixties accounting interval (1959–1970)	+2.07%	+1.31% (+63.3%)	+0.45% (+21.6%)	+0.97% (+46.9%)	+1.72% (+83.0%)	-2.38% (-114.8%)
Wave IV: UP crisis accounting interval (1971–1973)	-7.65%	+1.94% (-25.3%)	-4.35% (+56.9%)	-4.78% (+62.5%)	-1.23% (+16.0%)	+0.77% (-10.1%)
<i>B. Terminal Crisis Cycle: Annual Log Changes and % Shares</i>						
1970 → 1971: Reactivation peak	-14.05%	-11.44% (+81.4%)	-7.90% (+56.2%)	+5.92% (-42.1%)	-0.43% (+3.1%)	-0.20% (+1.4%)
1971 → 1972: Profit squeeze	-48.08%	-34.60% (+72.0%)	-10.42% (+21.7%)	-2.52% (+5.2%)	-2.27% (+4.7%)	+1.74% (-3.6%)
1972 → 1973: Annual observation spanning the coup	+32.78%	+38.48% (+117.4%)	+1.72% (+5.3%)	-7.04% (-21.5%)	-0.18% (-0.5%)	-0.20% (-0.6%)
1973 → 1974: DL 522 compression	+15.74%	+28.59% (+181.7%)	-11.47% (-72.9%)	-0.72% (-4.6%)	+4.04% (+25.7%)	-4.70% (-29.9%)
1974 → 1975: Chicago shock therapy	-48.97%	-17.70% (+36.2%)	-16.24% (+33.2%)	-14.78% (+30.2%)	+8.97% (-18.3%)	-9.22% (+18.8%)

Notes. Growth-accounting decomposition using the identity $\Delta \ln r_t \equiv \Delta \ln(1 - \omega_t) + \Delta \ln(P_{Y,t}/P_{K,t}) + \Delta \ln \mu_t + \Delta \ln y_t^p - \Delta \ln k_t$. Values outside parentheses are annualized log contributions (% p.a.). Values inside parentheses report the component's share of the total change, computed as $(\Delta \ln X_i / \Delta \ln r_t) \times 100\%$. A positive share indicates the component moved in the same direction as the total change; a negative share indicates it offset the total change. Shares may exceed 100% due to normalization by the net total change. Source: Author's own elaboration (Chapter 2).

Table 3.3. Capital Accumulation Response Across the Terminal Crisis Cycle, Chile, 1964–1975

Episode / Year	Profit Rate r (%)	Accum. Ratio χ (ratio)	Gross Acc. Rate g_K^{gross} (%)	Depreciation Rate δ (%)	Net Acc. Rate g_K^n (%)
<i>A. Christian Democratic Administration (1964–1970)</i>					
1964 (Baseline)	8.16	1.530	12.49	7.67	4.82
1965	8.59	1.244	10.69	6.94	3.75
1966	10.94	0.972	10.63	6.97	3.66
1967	10.14	1.066	10.81	7.06	3.75
1968	11.29	0.991	11.20	7.24	3.96
1969 (Peak Profit.)	12.79	0.857	10.96	7.14	3.82
1970 (Inherited)	12.23	0.951	11.63	7.40	4.23
<i>B. Unidad Popular Administration (1971–1973)</i>					
1971 (Reactivation)	10.62	0.953	10.12	6.78	3.34
1972 (Squeeze)	6.57	1.162	7.64	5.78	1.85
1973 (Coup year)	9.12	0.821	7.48	5.78	1.71
<i>C. Military Dictatorship & Shock Therapy (1974–1975)</i>					
1974 (Compression)	10.67	0.811	8.65	6.27	2.38
1975 (Collapse)	6.54	1.035	6.77	5.56	1.21

Notes. Accumulation accounting for fixed capital assets ($K_t \equiv K_{\text{ME},t} + K_{\text{NRC},t}$), excluding residential construction. Gross accumulation is $g_{K,t}^{\text{gross}} \equiv I_t^{\text{gross}}/K_{t-1}$; physical depreciation is $\delta_t \equiv D_t/K_{t-1}$; net accumulation is $g_{K,t}^n \equiv I_t^{\text{net}}/K_{t-1}$; and the gross investment-to-surplus ratio (accumulation ratio) is $\chi_t \equiv g_{K,t}^{\text{gross}}/r_t$. All observations satisfy the exact identity $g_{K,t}^n \equiv \chi_t r_t - \delta_t$, subject to displayed rounding. The profit rate is a level, not its growth rate. Annual 1973 observations span the coup. Source: Author’s own elaboration (Chapter 2).

3.5.2 Stylized Facts of Peripheral Accumulation, Distribution, and External Constraints

3.5.2.1 Structural Constraints of Peripheral Transition: An Accounting Anchor

The political mobilization and redistributive policies analyzed in Section 3.5.1 unfolded within external material constraints. The Chilean economy relied on imported machinery, intermediate spare parts, and foreign trade credits to sustain domestic employment and basic wage-goods production. Mainstream accounts (Dornbusch and Edwards, 1990; Edwards, 2024) attribute the subsequent macroeconomic crisis to fiscal deficits and unconstrained money creation that depleted reserves and triggered inflation. In contrast, structuralist and dependency traditions (De Vylder, 1976; Stallings, 1978) emphasize balance-of-payments vulnerability, intermediate import dependency, and external credit restrictions. Comparing these perspectives requires establishing the descriptive sequence between external balance-sheet deterioration, domestic inflation, and monetary expansion.

In a subordinate open economy without reserve-currency status, convertible international reserves constrain the ability of the central bank to settle external obligations, finance intermediate imports, and maintain exchange parity. To track this external balance-sheet position, I define the Central

Bank Solvency Ratio relative to Base Money:

$$\text{SolvR}_t^H \equiv \frac{e_t \cdot IR_t}{H_t \cdot 1000}, \quad (3.9)$$

where e_t is the official exchange rate (standardized in pesos per USD in the Central Bank database), IR_t represents gross international reserves in USD millions from the IMF (yielding a numerator $e_t \cdot IR_t$ in millions of pesos), and H_t denotes domestic base money liabilities recorded in thousands of millions of pesos (billions, *miles de millones de pesos*), scaled by 1,000 to express liabilities in millions of pesos and ensure dimensional consistency. As foreign reserves fell and external credit lines contracted, reserve coverage of the domestic monetary base declined. The descriptive sequence therefore raises the empirical question of whether domestic money growth preceded inflation as an autonomous driver or responded to emerging external bottlenecks and domestic cost pressures.

3.5.2.2 Long-Span Balance of Payments Accounting: Reserve Accumulation and the 1971 Capital-Flow Reversal (1958–1975)

In standard national accounting, the annual change in official Central Bank international reserves (ΔIR_t) satisfies the identity:

$$\Delta IR_t \equiv CA_t + KA_t + EO_t + SDR_t \quad (3.10)$$

where CA_t is the current account balance, KA_t denotes net capital inflows, EO_t represents net errors and omissions, and SDR_t denotes allocations of IMF Special Drawing Rights. Net errors and omissions represent an accounting residual that, in peripheral contexts confronting trade and exchange controls, is consistent with unrecorded capital flight and trade misinvoicing ([Stallings, 1978](#)). Two distinct reserve concepts must be separated throughout this analysis: gross international reserves (IR_t), which record total foreign exchange assets held by the monetary authority, and net international reserves (IR_t^{net}), which subtract short-term foreign liabilities from gross holdings.

Table 3.4. Long-Span Balance of Payments Accounting and International Reserves Dynamics (Chile, 1958–1975): Official BCCh Cash Flows vs. WBOP Global Accrual Accounts

	Ibáñez	Jorge Alessandri (1959–1964)						Eduardo Frei Montalva (1965–1970)						Unidad Popular (1971–1973)			Junta (1974–75)	
Metric / USD Millions	1958	1959	1960	1961	1962	1963	1964	1965	1966	1967	1968	1969	1970	1971	1972	1973	1974	1975
A. Official Cash-Basis Balance of Payments (Central Bank of Chile / Stallings 1978, USD Millions)																		
Current Account (CA)	—	−24.8	−148.1	−241.1	−181.9	−157.8	−131.6	−56.6	−82.2	−127.4	−135.3	−5.6	−102.9	−205.5	−404.8	−294.6	−210.8	−491.3
Capital Account (KA)	—	20.8	76.3	187.6	133.2	107.6	152.1	65.8	168.0	125.5	294.6	222.5	267.5	−26.5	327.4	242.3	228.0	298.7
Net Errors & Omissions (EO)	—	28.6	43.4	−55.1	−0.3	22.0	2.9	37.5	33.8	−21.5	−41.4	−42.4	−72.9	−84.5	−171.6	−60.0	−62.1	−92.4
SDR Allocations (SDR)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	21.8	16.7	18.2	0.0	0.0	0.0
Net Reserve Change (ΔIR)	—	24.6	−28.4	−108.6	−49.0	−28.2	23.4	46.7	119.6	−23.4	117.9	174.5	113.5	−299.8	−230.8	−112.3	−44.9	−285.0
Net International Reserves (IR_t^{net})	29.6	105.4	72.8	−5.2	15.4	−24.3	−16.5	35.2	77.4	54.3	125.4	285.2	393.5	162.7	75.8	167.4	94.0	−129.2
B. Capital Account Reversal, External Drainage, and Cumulative Flows																		
Net External Drain ($CA + KA + EO$)	—	+24.6	−28.4	−108.6	−49.0	−28.2	+23.4	+46.7	+119.6	−23.4	+117.9	+174.5	+91.7	−316.5	−249.0	−112.3	−44.9	−285.0
Drain Share of IR_{t-1}	—	—	—	—	—	—	—	—	—	—	—	—	—	80.4%	153.0%	148.2%	26.8%	303.2%
Cumulative $\sum KA$	—	+\$677.6M (Alessandri Total)						+\$1,143.9M (Frei Montalva Total)						+\$543.2M (UP Total)			+\$526.7M	
Cumulative $\sum CA$	—	−\$885.3M (Alessandri Total)						−\$510.0M (Frei Montalva Total)						−\$904.9M (UP Total)			−\$702.1M	
C. Global Accrual Balance of Payments (WBOP, Nieves and Piketty 2025)																		
WBOP Current Account (USD M)	−271.8	−189.5	−155.3	−265.4	−194.4	−308.2	−212.8	−103.4	−179.4	−135.9	−70.9	+61.8	−147.7	−421.6	−550.8	−618.1	−105.5	−470.6
WBOP Current Account (% GDP)	−7.74%	−5.00%	−3.40%	−5.19%	−3.26%	−5.21%	−3.37%	−1.60%	−2.37%	−1.82%	−0.93%	+0.70%	−1.53%	−3.64%	−4.40%	−3.47%	−0.61%	−5.85%
Net Foreign Wealth / IIP (% GDP)	−67.0%	−69.3%	−61.7%	−58.1%	−54.3%	−57.9%	−59.2%	−61.1%	−53.5%	−56.7%	−57.4%	−50.2%	−54.6%	−47.7%	−49.4%	−33.6%	−46.4%	−100.6%

Sources and Notes: Official cash-basis Balance of Payments flows (1960–1975), Net Errors and Omissions, and SDR allocations sourced from the Banco Central de Chile (BCCh) REST API and Base de Datos Estadísticos (BDE), complemented by historical balance-of-payments ledgers. 1958–1959 cash entries sourced directly from Stallings (1978, Table 7.4, p. 178), citing contemporary BCCh records. Net international reserves (IR_t^{net}) record liquid foreign exchange assets net of short-term central bank foreign liabilities, compiled from BCCh historical ledgers and Stallings (1978), accounting for negative net balances during the 1961–1964 exchange crises and 1975; gross foreign reserve assets from IMF International Financial Statistics remain positive in all years. Global accrual balance of payments series and Net International Investment Position (NIIP) sourced from the World Historical Balance of Payments Database (Nieves and Piketty, 2025). The balance-of-payments reserves identity holds exactly: $\Delta IR_t \equiv CA_t + KA_t + EO_t + SDR_t$. Discrepancies between cash and accrual accounts reflect distinct institutional accounting conventions: cash ledgers record physical foreign exchange settlements at the Central Bank window, whereas WBOP accrual accounts adhere to modern BPM6 standards, incorporating legally accrued but unpaid debt service (under the November 1971 debt moratorium), reinvested foreign mining profits, and global bilateral trade partner reconciliations. Both records confirm that in 1971, net external financing experienced a sharp contraction of −\$294.0 million ($KA = -\26.5M vs. $+\$267.5\text{M}$ in 1970), equivalent in magnitude to 98.1% of the calendar-year reserve drain ($\Delta IR = -\$299.8\text{M}$) before domestic fiscal emission accelerated in late 1972.

Table 3.4 reports the balance-of-payments ledger across eighteen years (1958–1975), spanning four developmental regimes in dialogue with Stallings (1978) and the World Historical Balance of Payments database (Nievas and Piketty, 2025). The table presents both cash-basis and accrual-basis accounts. The official BCCh cash ledger (Stallings, 1978, Table 7.4, p. 178) records foreign currency settlements that crossed the Central Bank counter, capturing the liquid resources available to defend exchange parity and pay for imports. In contrast, the WBOP database (Nievas and Piketty, 2025) reconstructs accounts under modern IMF BPM6 accrual standards, recording primary income debits and interest obligations as they legally accrued, including during the November 1971 debt moratorium.

The pre-1970 administrations established the external balance-sheet position inherited by the Unidad Popular. Under Jorge Alessandri (1958–1964), import liberalization and a fixed nominal exchange rate led to cumulative current account deficits of −\$885.3 million. By late 1961, liquid reserves were exhausted, forcing emergency import controls and an eventual currency float in October 1962 that generated an inflation spike; net reserves ended at −\$16.5 million in 1964. The Eduardo Frei Montalva administration (1964–1970) instituted a crawling peg and accumulated reserves through external borrowing supported by the Alliance for Progress (French-Davis, 1973). Cumulative net foreign capital inflows across Frei’s term totaled +\$1,143.9 million, while the cumulative sum of annual reserve changes was +\$548.8 million. Net capital inflows thus exceeded cumulative reserve accumulation by more than two to one (208.4%), as net reserves rose from −\$16.5 million in 1964 to \$393.5 million at year-end 1970 (a net stock increase of +\$410.0 million; the difference from cumulative annual flows reflects valuation adjustments on non-dollar holdings and the netting of short-term foreign liabilities in the stock measure). The reserve buffer inherited by the UP was therefore anchored in foreign credit lines.

Following the constitutional nationalization of large-scale copper mining on July 11, 1971 (Law 17,450), external financing contracted sharply. Net capital inflows fell from +\$267.5 million in 1970 to −\$26.5 million in 1971 (−\$49.9 million in Stallings’ autonomous capital series). In the audited 1958–1975 ledger, 1971 was the only calendar year with negative net capital inflows. This −\$294.0 million turnaround in the capital account was equivalent in magnitude to 98.1% of the calendar-year reserve drain ($\Delta IR = -\$299.8$ million), reducing Central Bank liquidity before domestic price acceleration took hold in late 1972.

The reduction in external financing materialized through specific commercial channels. As documented by De Vylder (1976) and Petras and Morley (1975), international commercial banks curtailed short-term trade credits to Chile from \$220 million to \$35 million, while the U.S. Export-Import Bank suspended project lending. Foreign suppliers increasingly required cash-in-advance confirmed letters of credit for food and fuel shipments, and replacement parts for imported mining machinery faced delays in foreign ports (Stallings, 1978). Net Errors and Omissions recorded negative residuals totaling −\$389.0 million across the UP triennium (−\$72.9M in 1970, −\$84.5M in 1971, and −\$171.6M in 1972). Following the September 4, 1970 election, the Santiago stock exchange declined by 48.6% and commercial bank deposits contracted (Girardi and Bowles, 2018). The timing is consistent with capital flight and private asset liquidation beginning prior to the presidential inauguration. While the cash ledger shows positive capital inflows in 1972 (+\$327.4M), this reflected the capitalization of debt service following the November 1971 moratorium. On an accrual basis, the WBOP current account deficit widened to −\$550.8 million (−4.40% of GDP) in 1972 and −\$618.1 million (−3.47% of GDP) in 1973, while Central Bank net foreign reserves fell to \$75.8 million at

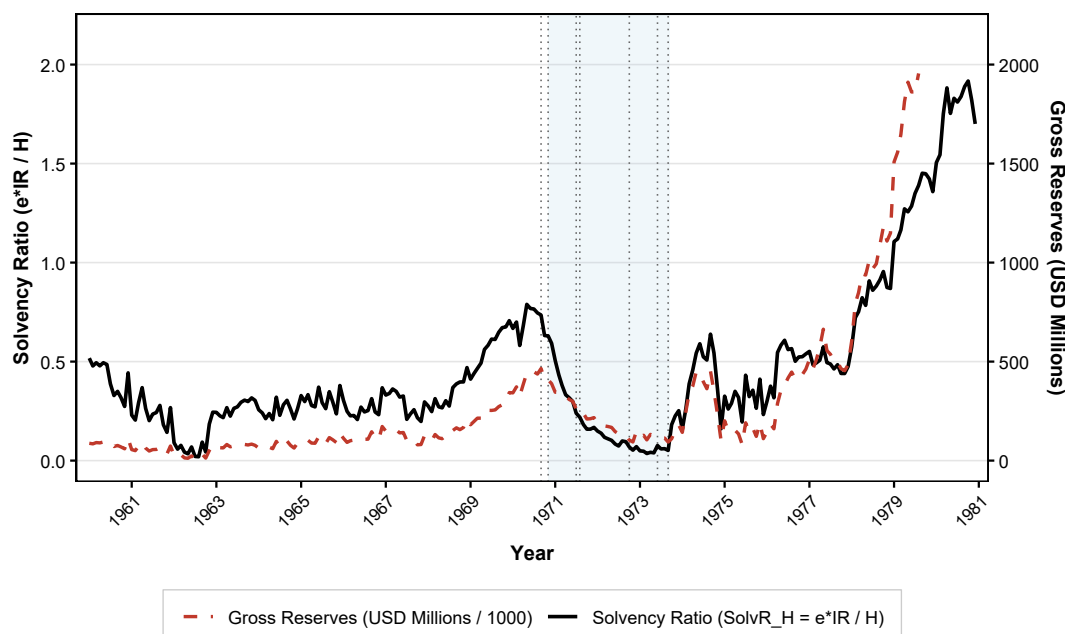
year-end 1972.

Following the September 1973 coup, external commercial credits resumed ($KA = +\$228.0M$ in 1974 and $+\$298.7M$ in 1975), and bilateral debts were rescheduled through the Paris Club. When world copper prices fell in 1975, domestic expenditure compression reduced import volumes. External vulnerability remained visible in the 1975 accounts: the accrual current account deficit reached $-\$470.6$ million (-5.85% of GDP), net international reserves fell to $-\$129.2$ million, and manufacturing output contracted.

3.5.2.3 Terms of Trade, Capacity to Import, and Real Structural Imbalance (1960–1980)

To evaluate external purchasing power and physical trade flows, Figures 3.4, 3.5, and 3.6 trace the monthly trajectories of foreign reserves, commodity prices, and import capacity across 1960–1980 ($N = 252$ continuous calendar months, yielding $N = 250$ observations in the transformed econometric sample).

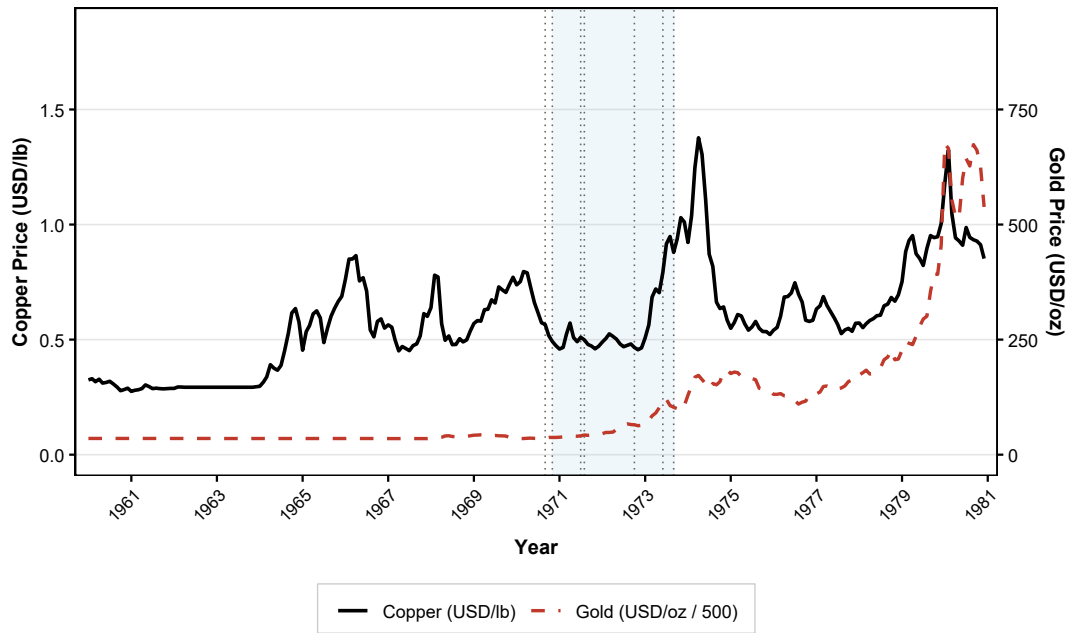
Figure 3.4. Central Bank Solvency Ratio (SolvR^H) and Gross International Reserves (1960–1980)



Notes: Monthly series tracking the Central Bank Solvency Ratio relative to Base Money ($\text{SolvR}_t^H \equiv \frac{e_t \cdot IR_t}{H_t \cdot 1000}$, left axis, solid black line) and IMF gross international reserves (IR_t in USD millions, right axis, dashed red line). Base money (H_t) represents the Central Bank's direct domestic liability base from the archival ledger. Benchmark markers I–VII defined in Table 3.6. Shaded blue region designates the Unidad Popular government (November 1970 – September 1973). Sourced from Central Bank of Chile REST API and IMF International Financial Statistics.

The international terms of trade shifted unfavorably during the administration's first year (Figure 3.5). World copper export prices fell by 25% during 1971, averaging $\$0.47/\text{lb}$ compared to $\$0.64/\text{lb}$ across 1969–1970, reducing export revenue during the initial domestic reactivation. Concurrently, the international monetary system absorbed the breakdown of the Bretton Woods architecture following

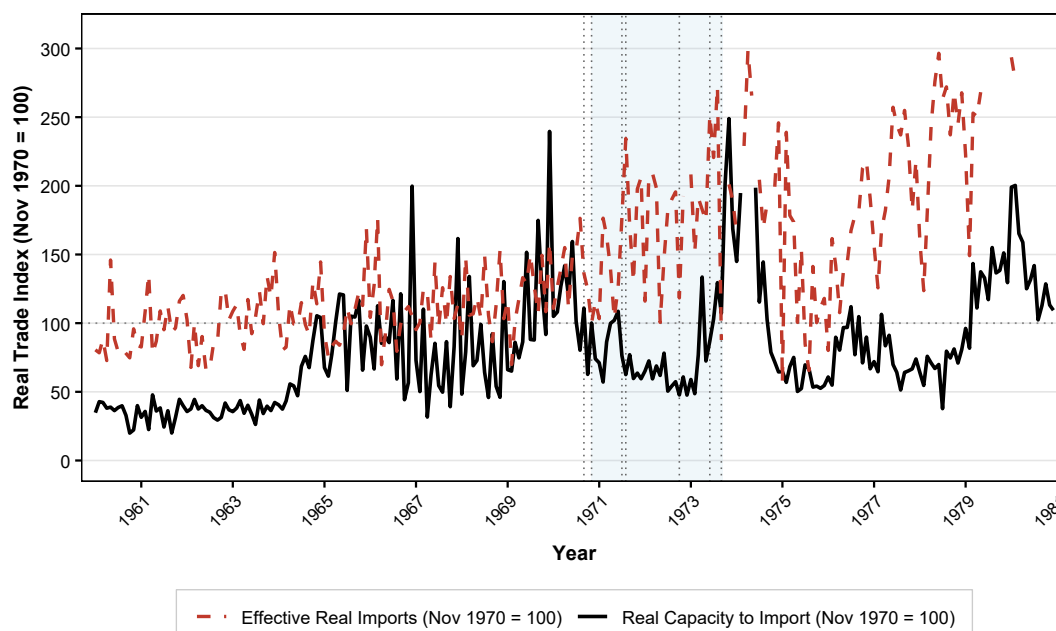
Figure 3.5. World Copper Price and London Gold Price (1960–1980)



Notes: Monthly series tracking the London Metal Exchange copper cash price (P^{Cu} in USD/lb, left axis, solid black line) and London Bullion Market Association gold price (P^{Gold} in USD/oz / 500, right axis, dashed red line). Benchmark markers I–VII defined in Table 3.6. Shaded blue region designates the Unidad Popular government (November 1970 – September 1973). Sourced from LME and LBMA via Central Bank of Chile REST API.

the unilateral suspension of dollar-gold convertibility on August 15, 1971 (Benchmark IV, [Vernengo 2021](#); [Zoeller 2019](#)). The world gold price serves in this chapter as a proxy for the resulting international monetary turbulence and dollar inflation; gold prices rose past \$100/oz in 1973 and \$200/oz by 1978. In parallel, U.S. wholesale prices rose by 28.0% between late 1970 and mid-1973 (Table 3.5), raising the foreign-currency price of imported goods.

Figure 3.6. Prebisch-ECLAC Real Capacity to Import and Effective Real Imports (1960–1980)



Notes: Monthly trajectory tracking the Prebisch-ECLAC Real Capacity to Import ($M_t^{\text{cap}} \equiv [\text{Exports}_t / \text{Exports}_{70:11}] \times [(P_t^{\text{Cu}} / P_{70:11}^{\text{Cu}}) / (P_{W,t}^{\text{US}} / P_{W,70:11}^{\text{US}})] \times 100$, solid black line) against Effective Real Imports ($M_t \equiv \text{Imports}_t / \text{Imports}_{70:11} \times 100$, dashed red line). Both series normalized to November 1970 = 100. Benchmark markers I–VII defined in Table 3.6. Shaded blue region designates the Unidad Popular government (1970–1973). Physical merchandise export and import volume series sourced from [Díaz-Bahamonde \(2023\)](#) historical database, compiled from *Estadística Chilena* and customs records (with April 1973 administrative registration drop interpolated); copper prices from LME via Central Bank of Chile REST API; U.S. wholesale prices from BLS.

Figure 3.6 compares external purchasing power with effective real import flows. When the world price of copper rose from \$0.508/lb in January 1973 to \$0.948/lb in August 1973, Central Bank gross international reserves remained low, fluctuating in a narrow range of \$103–\$114 million. The divergence reflected two offsetting movements. On the export side, transport disruptions and equipment shortages held back physical export volumes, muting the revenue gains from higher dollar copper prices. On the import side, effective real imports (M_t) expanded, reaching 234.5% of the November 1970 baseline in August 1971 and 272.3% in August 1973, driven by purchases of food, fuel, and intermediate goods.

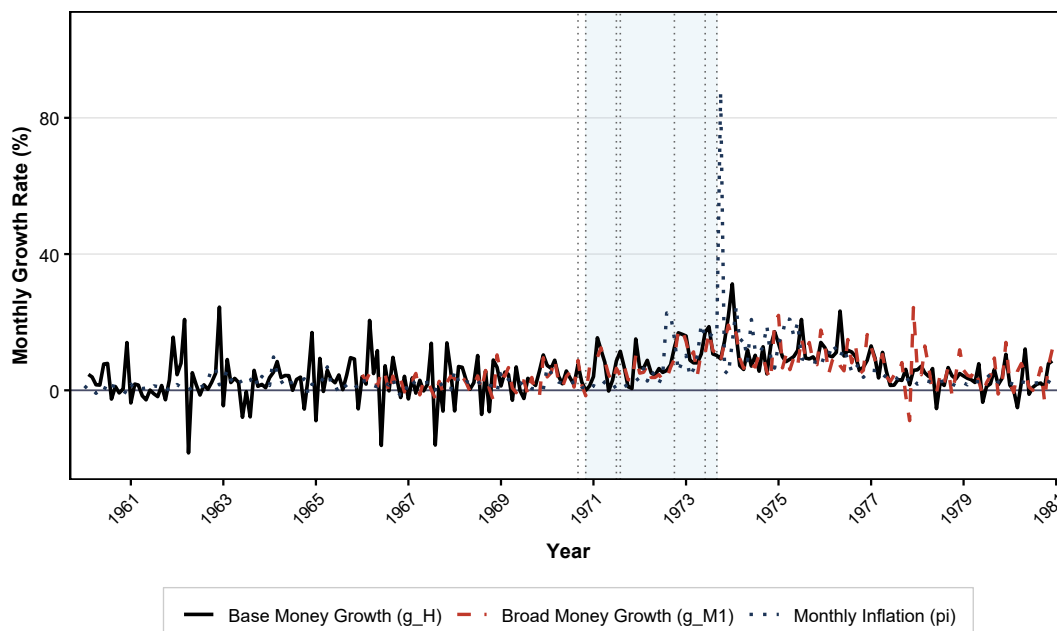
Operationalizing the Prebisch-ECLAC Capacity to Import (M_t^{cap}), Figure 3.6 documents that Chile’s real export purchasing power fell from 100.0 in November 1970 to 62.5 in August 1971 and 47.7 in October 1972 before rebounding in 1973. This generated a wide Real Structural Imbalance ($\Theta_t \equiv M_t / M_t^{\text{cap}} \times 100$), which reached 374.9% in August 1971 and 346.4% in December 1971 (Table 3.5). The descriptive sequence demonstrates that external balance-of-payments pressures

and foreign-exchange scarcity were present well before the price acceleration of late 1972.

3.5.2.4 Nominal Dynamics: Money Growth, Inflation, and Multiplier Compression

Figures 3.7 and 3.8 evaluate the high-frequency co-movement between monetary growth, consumer price inflation, and the broad-to-base money multiplier across 1960–1980.

Figure 3.7. Monetary Expansion Rates and Monthly Consumer Inflation (1960–1980)

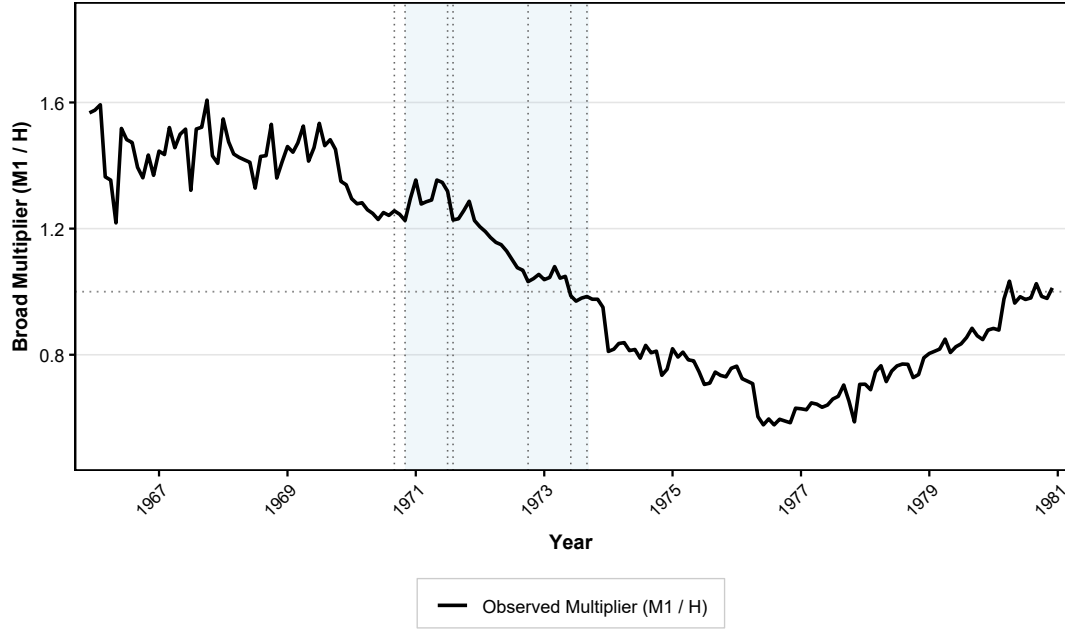


Notes: Monthly series of base money growth ($g_{H,t}$, solid black line), broad money supply growth ($g_{M1,t}$, dashed red line), and monthly consumer price inflation (π_t , dotted blue line). Benchmark markers I–VII defined in Table 3.6. Shaded blue region designates the Unidad Popular government (1970–1973). Sourced from Banco Central de Chile REST API and Instituto Nacional de Estadísticas (INE).

In Figure 3.7, base money (g_H) and broad money (g_{M1}) grew at monthly rates of 10%–15% through 1971 and early 1972, while physical manufacturing output expanded and wages rose. Monthly consumer inflation remained below 5% during this initial phase, rising to 15.2% in October 1972 as transport disruptions and commercial withholding emerged during the nationwide trucking strike. Central bank credit expansion accompanied rising enterprise operating costs and payroll requirements in subsequent months. The monthly peak of consumer inflation occurred in October 1973 (+87.6%) following the military junta’s Decree Law 522 price deregulation. This descriptive sequence is consistent with credit accommodation of cost pressures; Section 3.5.3 tests whether that directional relationship holds systematically.

Figure 3.8 tracks the empirical broad-to-base money multiplier ($m_t \equiv M1_t/H_t$). Under Frei (1966–1970), the multiplier fluctuated between 1.35 and 1.60. Under the UP administration, the ratio trended downward from 1.35 in January 1971 to 0.98 in August 1973, reaching a low of 0.58 in 1976. Accelerating inflation co-moves visually with this compression in the broad-to-base ratio. Possible institutional mechanisms underlying this decline—including cash hoarding during shortages and central bank credit to socialized firms—are discussed in Sections 3.3 and 3.6; Section 3.5.3 evaluates

Figure 3.8. Observed Empirical Money Multiplier ($M1/H$, 1965–1980)



Notes: Monthly trajectory of the broad-to-base money multiplier ($m_t \equiv M1_t/H_t$, solid black line). The figure is restricted to the observed M1 sample, December 1965–December 1980. Benchmark markers I–VII defined in Table 3.6. Shaded blue region designates the Unidad Popular government (1970–1973). Sourced from Banco Central de Chile REST API.

its predictive precedence relative to inflation.

3.5.2.5 Real Production Divergence and Focused Monthly Windows

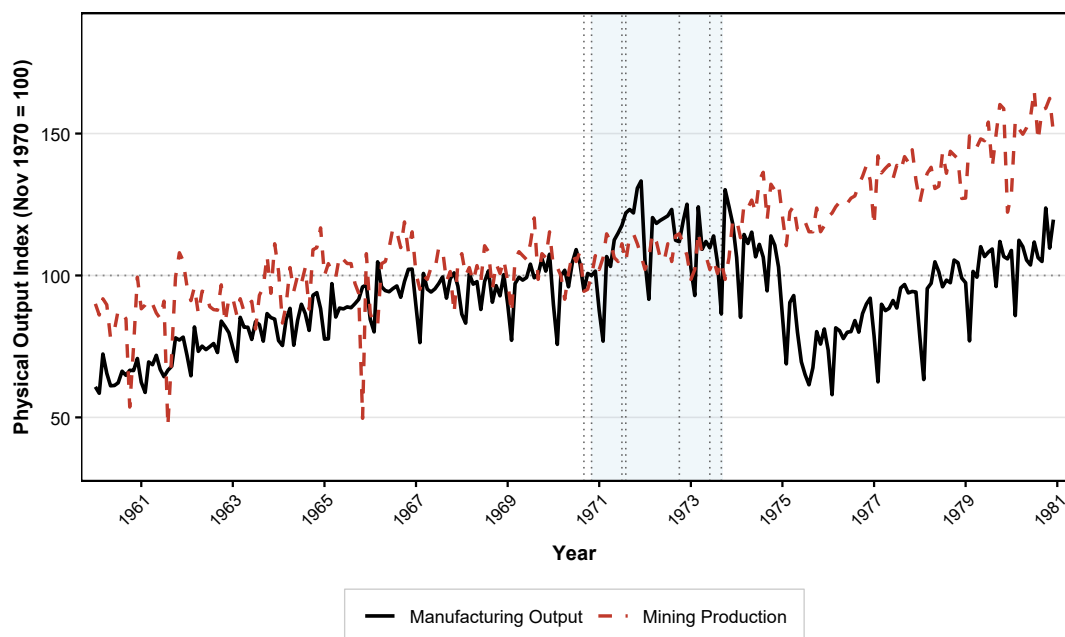
Figure 3.9 plots physical output dynamics across 1960–1980, documenting divergent paths between domestic manufacturing and export mining.

Manufacturing output expanded by 33.3% between November 1970 and December 1971, reflecting utilization of capacity in wage-goods industries. In contrast, copper mining output rose by 7.0%, held back by technical constraints and equipment import delays (De Vylder, 1976). Following the October 1972 strike, manufacturing output contracted as transport disruptions and intermediate input shortages took effect. Following the September 1973 coup, manufacturing production declined persistently, falling below index 65 by 1975 (a > 35% contraction from its 1971 peak) amid tariff reductions and demand compression.

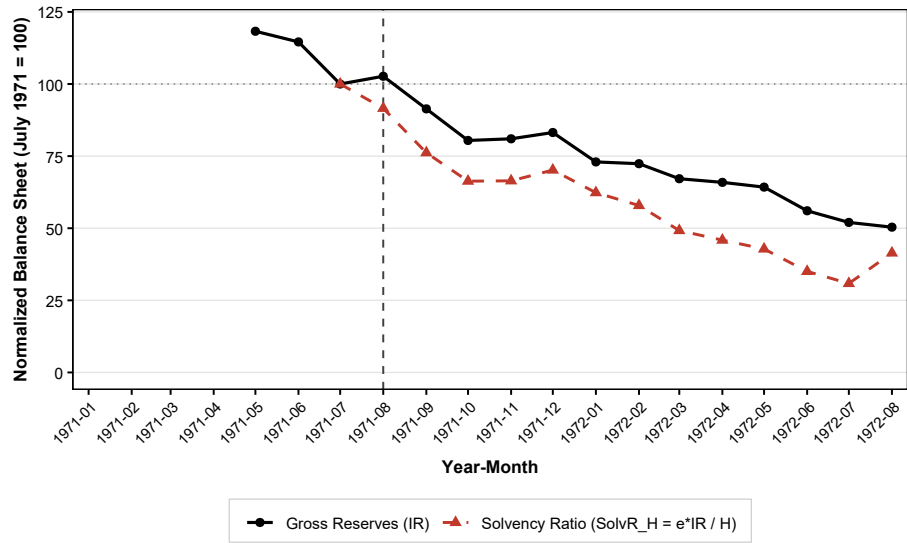
Figure 3.10 examines monthly dynamics around two institutional turning points.

The first window centers on the August 1971 credit restrictions and suspension of dollar-gold convertibility (Figure 3.10a, $t = 0$). Following the constitutional copper nationalization of July 11, the U.S. Export-Import Bank denied credit on August 12, and President Nixon suspended dollar-gold convertibility on August 15 (Zoeller, 2019). This monthly window illustrates the trajectory of reserve drain: gross international reserves fell from index 100.0 to below 30.0, and the Central Bank Solvency Ratio (SolvR_t^H) declined by more than 70% over the subsequent twelve months. Reserve

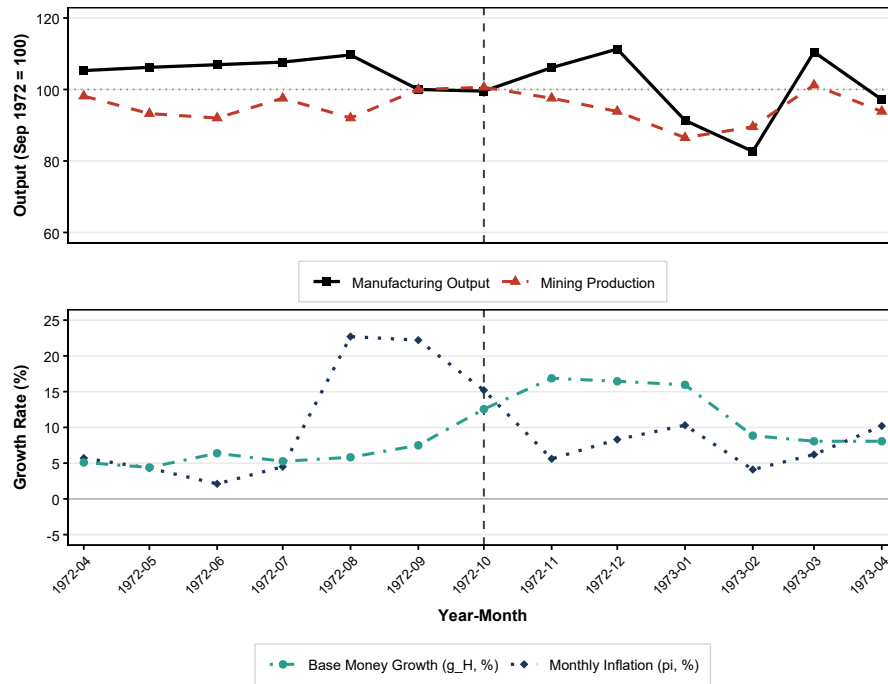
Figure 3.9. Real Physical Output Trajectories: Manufacturing vs. Mining (1960–1980)



Notes: Monthly series of physical output indices for manufacturing (solid black line) and mining (dashed red line), re-based to November 1970 = 100. Benchmark markers I–VII defined in Table 3.6. Shaded blue region designates the Unidad Popular government (1970–1973). Sourced from [Díaz-Bahamonde \(2023\)](#) historical database, compiled from INE and SOFOFA for manufacturing, and SERNAGEOMIN, ODEPLAN, and CODELCO for mining.



(a) August 1971 Credit Restrictions and Gold Suspension



(b) October 1972 Lockout and Trucking Strike

Figure 3.10. Monthly Trajectories around External and Supply Shocks (Chile, 1971–1972)

Notes: Monthly series centered at $t = 0$ around two institutional turning points. Panel (a) tracks Central Bank Gross International Reserves (IR_t , solid black line with circle markers) and the Solvency Ratio relative to Base Money ($SolvR_t^H \equiv \frac{e_t \cdot IR_t}{H_t \cdot 1000}$, dashed red line with triangle markers), indexed to July 1971 = 100, around the August 1971 Eximbank credit denial and Bretton Woods gold-window suspension (Benchmark IV). Panel (b) aligns two vertical subpanels around the October 1972 Lockout (Benchmark V): the upper panel displays physical output indices for Manufacturing (solid black line with square markers) and Mining Production (dashed red line with triangle markers), indexed to September 1972 = 100; the lower panel displays monthly consumer inflation (π_t , dotted dark-blue line with diamond markers) and Central Bank base money growth (g_H , dot-dashed teal line with circle markers) in genuine monthly growth percentages (%) along a common horizontal time axis. Sourced from Banco Central de Chile REST API, IMF International Financial Statistics, and [Díaz-Bahamonde \(2023\)](#).

coverage had deteriorated substantially before the later acceleration in domestic money growth.

The second window centers on the October 1972 trucking strike and commercial lockout (Figure 3.10b, $t = 0$). Transport paralysis and merchant shutdowns coincided with a 12 percentage-point drop in manufacturing output, while mining output in northern enclaves remained less affected. Monthly consumer inflation rose from 2.8% to 15.2%. Base-money growth did not accelerate contemporaneously with the October jump in inflation; its acceleration followed at $t = +1$ and $t = +2$. The descriptive window places the price increase before the subsequent expansion in base-money growth. While this ordering is consistent with accommodation of cost pressures, establishing directional precedence requires the econometric testing developed below.

3.5.2.6 Descriptive Synthesis and Empirical Transition

Table 3.5 presents macroeconomic indicators across six selected dates, and Table 3.6 summarizes the seven institutional events (I–VII).

Table 3.5. Monthly Macroeconomic Dynamics, Central Bank Solvency, and External Constraints across Structural Junctures (Chile, 1970–1973)

Macroeconomic Variable	Nov 1970 <i>Inauguration</i>	Aug 1971 <i>Gold Suspension</i>	Dec 1971 <i>Output Peak</i>	Oct 1972 <i>October Lockout</i>	Aug 1973 <i>Pre-Coup</i>	Dec 1973 <i>Post-Coup</i>
A. Inflation and Price Dynamics						
Monthly Inflation (%)	0.6%	1.1%	2.8%	15.2%	17.1%	4.7%
YoY Inflation (%)	35.3%	17.4%	22.1%	142.9%	303.6%	508.1%
B. Real Physical Activity (Nov 1970 = 100)						
Manufacturing Output Index	100.0	122.0	133.3	111.9	104.6	118.1
Mining Production Index	100.0	104.9	107.0	114.7	100.7	119.6
C. Central Bank Solvency and Monetary Backing						
Gross International Reserves (USD M)	408.0	268.1	217.2	103.4	114.2	179.7
Solvency Ratio Base ($\text{SolvR}^H \equiv \frac{e \cdot IR}{H \cdot 1000}$)	0.63	0.22	0.17	0.07	0.06	0.25
Solvency Ratio M1 ($\text{SolvR}^{M1} \equiv \frac{e \cdot IR}{M1 \cdot 1000}$)	0.51	0.18	0.14	0.07	0.06	0.27
Observed Multiplier ($M1/H$)	1.23	1.23	1.23	1.03	0.98	0.95
Base Money Monthly Growth (g_H)	0.0%	11.5%	15.1%	12.6%	10.7%	21.8%
M1 Money Monthly Growth (g_{M1})	-1.6%	4.3%	10.2%	9.1%	11.6%	19.2%
D. Official Exchange Rate and World Commodity Prices						
Official Exchange Rate (e_t)	0.012	0.0122	0.0146	0.0250	0.0740	0.34
World Copper Price (USD/lb)	\$0.492	\$0.499	\$0.471	\$0.466	\$0.948	\$1.011
World Gold Price (USD/oz)	\$37.44	\$42.71	\$43.48	\$64.86	\$106.76	\$106.72
E. External Trade, Capacity to Import, and Structural Imbalance						
Export Activity Index	2.68	1.71	1.73	1.46	2.31	2.81
Import Activity Index	1.51	3.54	3.11	1.79	4.11	2.86
Real Trade Ratio (Exp/Imp)	1.78	0.48	0.56	0.82	0.56	0.98
Real Capacity to Import (M_t^{cap} , Nov 1970 = 100)	100.0	62.5	59.5	47.7	129.5	168.7
Effective Real Imports (M_t , Nov 1970 = 100)	100.0	234.5	206.2	118.4	272.3	189.4
Real Structural Imbalance (Θ_t , %)	100.0%	374.9%	346.4%	248.1%	210.3%	112.3%
US Wholesale Price Index (Nov 1970 = 100)	100.0	103.8	104.0	108.1	128.0	127.8

Sources and Notes: Primary monthly series from Central Bank of Chile and IMF International Financial Statistics. Gross international reserves (IR_t) represent official foreign exchange holdings from IMF IFS. Solvency Ratio defined relative to Base Money as $\text{SolvR}_t^H \equiv (e_t \cdot IR_t)/(H_t \cdot 1000)$ and relative to M1 as $\text{SolvR}_t^{M1} \equiv (e_t \cdot IR_t)/(M1_t \cdot 1000)$, where IR_t is in USD millions, e_t is the official exchange rate in pesos per USD (standardized per BCCh historical conventions), and $H_t, M1_t$ are recorded in thousands of millions of pesos (billions, *miles de millones de pesos*), scaled by 1,000 to express monetary liabilities in millions of pesos and ensure dimensional consistency with the numerator ($e_t \cdot IR_t$). Capacity to Import Index operationalizes the Prebisch-ECLAC formula: $M_t^{\text{cap}} \equiv (\text{Exports}_t / \text{Exports}_{70:11}) \times [(P_t^{\text{Cu}} / P_{70:11}^{\text{Cu}}) / (P_{W,t}^{\text{US}} / P_{W,70:11}^{\text{US}})] \times 100$. Real Structural Imbalance Ratio defined as $\Theta_t \equiv (M_t / M_t^{\text{cap}}) \times 100$, tracking the physical quantum of imports relative to real export purchasing power. Selected dates mark historical milestones: Nov 1970 (Allende inauguration baseline); Aug 1971 (Eximbank credit denial and suspension of dollar-gold convertibility); Dec 1971 (cyclical output peak); Oct 1972 (national trucking strike and commercial lockout); Aug 1973 (late administration baseline before the September coup); and Dec 1973 (post-coup benchmark following October 1973 Decree Law 522 price deregulation).

Table 3.6. Chronology of Critical Macroeconomic Junctures and Institutional Shocks (Chile, 1970–1973)

Ref	Date	Historical Juncture	Institutional Shock Mechanism	Macroeconomic / Balance-Sheet Transmission Channel
I	Sep 1970	Presidential Election	Electoral panic; commercial deposit runs	Capital flight ($EO < 0$); foreign currency accumulation; Central Bank emergency rediscunts.
II	Nov 1970	Inauguration Baseline	Allende assumes presidency; tariff and price freeze	Initial price stabilization; real wage gains; inherited gross reserve buffer (\$408.0M; $\text{SolvR}^H = 0.63$; $\text{SolvR}^{M1} = 0.51$).
III	Jul 1971	Copper Nationalization	Unanimous constitutional reform (Law 17,450)	Sovereign recovery of <i>Gran Minería</i> ; triggers bilateral U.S. credit restrictions and legal liens.
IV	Aug 1971	Eximbank Embargo & Bretton Woods Collapse	U.S. credit freeze; dollar-gold convertibility suspended	Revolving trade credits reduced (\$220M $\rightarrow$ \$35M); cash-in-advance imports; reserve drain accelerates ($\Delta IR = -\$140\text{M}$).
V	Oct 1972	October Lockout (<i>Paro de Octubre</i>)	Trucking strike; SOFOFA commercial lockout	Physical transport paralysis; wholesale inventory withholding; monthly inflation rises to 15.2%; central bank liquidity accommodation.
VI	Jun 1973	<i>El Tanquetazo</i>	Military uprising (2nd Armored Regiment)	Heightened political instability; logistical disruption; black market expansion; reserve depletion and structural trade imbalance ($\Theta_t = 210.3\%$).
VII	Sep 1973	Military Coup d'État	Violent overthrow; followed by October DL 522 deregulation	Elimination of price controls; +87.6% MoM inflation jump in October; real wage contraction (-40%); domestic expenditure compression.

Notes: Master reference chronology for vertical benchmark lines appearing across high-frequency time series (Figures 3.4–3.8) and defining intervention cutoffs for focused monthly windows (Figures 3.10a–3.10b) and companion econometric estimation stages.

Table 3.5 illustrates this temporal sequence across the selected dates: between November 1970 and August 1971—while monthly consumer inflation remained low (1.1%) and industrial manufacturing output was expanding (+22.0%)—the Central Bank Solvency Ratio (SolvR_t^H) had declined from 0.63 to 0.22, and gross foreign reserves had fallen by \$139.9 million. External balance-of-payments deterioration preceded the subsequent acceleration of domestic inflation and credit expansion.

Across the selected dates, reserve coverage deteriorated substantially before the sharp acceleration of inflation in late 1972. Money growth, inflation, exchange-rate adjustment, and trade imbalances moved together during much of the crisis, so these descriptive sequences do not establish their direction of dependence. Section 3.5.3 evaluates directional predictive precedence through Granger-causality tests in stationary vector autoregressions across six modular empirical systems. Section 3.5.4 then estimates a non-linear Threshold Vector Autoregression (TVAR) conditioned on the monthly Solvency Growth Gap ($\Delta s_{t-1} \equiv g_{t-1}^F - g_{t-1}^H$) to test whether these relationships change under conditions of central bank insolvency.

3.5.3 Empirical Exercise 1: Directional Precedence and Monetary Accommodation in Stationary Vector Autoregressions

3.5.3.1 Econometric Framework and Evidential Standard

The stylized facts documented in Section 3.5.2 leave open a central empirical question: did domestic monetary expansion precede price acceleration as an autonomous impulse, or did it accommodate price pressures, enterprise working-capital requirements, and external balance-of-payments constraints? Standard monetarist accounts treat money creation as an autonomous driver, implying that high-powered reserve money ($g_{H,t}$) and narrow transactions balances ($g_{M1,t}$) should expand

without predictive feedback from consumer prices (π_t) or external shocks:

$$\mathcal{H}_0^{\text{Exogenous}} : \quad \pi_{t-k} \not\rightarrow g_{M,t} \quad \text{and} \quad \text{External Shock}_{t-k} \not\rightarrow g_{M,t} \quad \forall k \geq 1. \quad (3.11)$$

In contrast, structuralist and endogenous-money frameworks view money creation as an accommodating response to enterprise cost pressures and balance-of-payments disruptions. When enterprises face escalating nominal costs under price controls and supply bottlenecks, working-capital requirements expand. If the monetary authority accommodates these demands to prevent illiquidity and payroll defaults, temporal predictive precedence should flow from inflation and foreign reserve depletion to domestic money growth ($\pi \rightarrow g_H$, $\text{SolvR}^H \rightarrow g_H$).

To evaluate these competing temporal sequences, I estimate standard Granger-causality tests within stationary vector autoregressions (Hamilton, 1994; Lütkepohl, 2005). Let Y_t denote a $K \times 1$ vector of stationary transformed macroeconomic series ($I(0)$). An unrestricted vector autoregression of order k , $\text{VAR}(k)$, is specified:

$$Y_t = c + \sum_{i=1}^k A_i Y_{t-i} + \varepsilon_t, \quad (3.12)$$

where c is a vector of constants, A_i are $K \times K$ coefficient matrices, and ε_t is a vector of white-noise disturbances. In a bivariate representation for (y_t, x_t) , the system reduces to:

$$y_t = \alpha_0 + \sum_{i=1}^k \alpha_i y_{t-i} + \sum_{i=1}^k \delta_i x_{t-i} + \varepsilon_{1t}, \quad (3.13)$$

$$x_t = \beta_0 + \sum_{i=1}^k \beta_i x_{t-i} + \sum_{i=1}^k \phi_i y_{t-i} + \varepsilon_{2t}. \quad (3.14)$$

The null hypothesis that x_t does not Granger-cause y_t corresponds to the joint exclusion restriction on the lagged coefficients of x :

$$H_0 : \quad \delta_1 = \dots = \delta_k = 0. \quad (3.15)$$

Because the entered series are stationary ($I(0)$), as established by the Augmented Dickey–Fuller tests in Table 3.1), this restriction is tested using standard finite-sample F -tests:

$$F = \frac{(\text{RSS}_R - \text{RSS}_U)/k}{\text{RSS}_U/(T - Kk - 1)} \sim F(k, T - Kk - 1), \quad (3.16)$$

where RSS_R and RSS_U denote the residual sums of squares from the restricted and unrestricted models, respectively. Rejecting H_0 indicates that lagged values of x_t contain incremental predictive information for y_t beyond what is captured by the remaining lagged variables in the system.

In accordance with the chapter’s causal-language contract, Granger causality denotes incremental predictive precedence within a specified stationary information set; it does not by itself establish structural causal identification or policy invariance. Empirical admission to the main text is governed by three diagnostic criteria: (1) modulus stability ($\max |\lambda_i| < 1$); (2) system-level residual whitening via the multivariate Breusch–Godfrey Lagrange Multiplier test ($p > 0.05$); and (3) equation-level residual whitening ($p > 0.05$). Specifications that fail these diagnostic standards are not used for directional inference in the main text.

Table 3.7 defines the system variable vectors, mathematical transformations, sample coverage, and

evidence status across the empirical systems.

3.5.3.2 Nominal Accommodation: Money and Inflation

System 1 evaluates dynamic feedback between consumer price inflation (π_t) and high-powered base money growth ($g_{H,t}$) over the full monthly sample (1960:01–1980:12, $N = 250$, usable $N_{\text{eff}} = 247$ at lag 4). At lag order $k = 4$, the system satisfies all diagnostic admissibility criteria: companion matrix eigenvalues confirm stability ($\max |\lambda| = 0.730$), the system Breusch–Godfrey test indicates an absence of residual autocorrelation ($\chi^2 = 3.06, p = 0.548$), and equation-level Breusch–Godfrey tests confirm whitening in both the base money equation ($p = 0.467$) and the inflation equation ($p = 0.284$).

As reported in Panel A of Table 3.8, Granger causality between inflation and base money growth is bidirectional at $k = 4$. The null that inflation does not Granger-cause base money growth is rejected at the 0.1% level ($F = 15.004, p < 0.0001$), with a positive sum of lag coefficients ($\sum \hat{\beta}_i = +0.680$). In the forward direction, base money growth also Granger-causes inflation ($F = 2.718, p = 0.0305, \sum \hat{\beta}_i = +0.380$). Evaluating results across alternative lag lengths confirms that reverse accommodation ($\pi \rightarrow g_H$) commands substantially larger test statistics ($F \in [15.00, 28.32], p < 0.0001$ across $k = 1, \dots, 4$) and maintains unbroken statistical significance at all horizons. Forward money growth ($g_H \rightarrow \pi$) predicts inflation strongly at short horizons ($F = 14.557, p = 0.0002$ at $k = 1$; $F = 9.284, p = 0.0001$ at $k = 2$), but attenuates as lag length increases (at the SBIC-selected lag $k^* = 3$, $\pi \rightarrow g_H$ yields $F = 20.318, p < 0.0001$ while $g_H \rightarrow \pi$ yields $F = 5.544, p = 0.0011$).

This pattern provides evidence consistent with defensive monetary accommodation under acute price pressures. While high-powered money growth contributed to subsequent price acceleration at short horizons, consumer price inflation exerted strong and persistent predictive feedback on base money creation. The bidirectional finding is difficult to reconcile with the strict monetarist premise of unidirectional money exogeneity, while demonstrating that nominal transmission operated as a reciprocal spiral. System 1 is classified as a ROBUST LINEAR RESULT.

3.5.3.3 Commercial Banking and Multiplier Sensitivity

System 2 evaluates the commercial banking sector using official un-spliced narrow money balances ($M1$), high-powered base money (H), and consumer price inflation over the post-1966 banking window (1966:01–1980:12, $N = 180$, usable $N_{\text{eff}} = 177$ at $k = 3$). At lag $k = 3$, the three-variable system satisfies the diagnostic admissibility criteria: the system is stable ($\max |\lambda| = 0.817$), the system Breusch–Godfrey test fails to reject white-noise residuals ($p = 0.428$), and equation-level tests indicate no serial correlation ($p = 0.289$ in the base money equation).

As shown in Panel B of Table 3.8, in the diagnostically admissible trivariate VAR(3), narrow-money and base-money growth contain predictive information for one another after conditioning on inflation. Commercial narrow-money growth predicts base-money expansion ($g_{M1} \rightarrow g_H$: $F = 3.654, p = 0.0138, \sum \hat{\beta}_i = +0.510$), while base-money growth also predicts narrow money ($g_H \rightarrow g_{M1}$: $F = 3.239, p = 0.0236, \sum \hat{\beta}_i = +0.333$). Inflation also predicts subsequent narrow-money growth ($\pi \rightarrow g_{M1}$: $F = 6.842, p = 0.0002, \sum \hat{\beta}_i = +0.284$). The reverse relation is not detected once base-money growth is included in the information set: the null that narrow-money growth does

Table 3.7. System Variable Specification and Evidence Hierarchy

System	Variable Vector $Y_t^{(s)}$	K	Evidence Status	Variable Definitions and Historical Sample
Core Empirical Specifications (Main Text)				
System 1	$[\pi_t, g_{H,t}]'$	2	Robust Linear	π_t : Monthly CPI inflation ($\Delta \ln \text{CPI}_t \times 100$). $g_{H,t}$: Base money growth ($\Delta \ln H_t \times 100$). Sample: 1960:01–1980:12 ($N = 250$, usable $N_{\text{eff}} = 247$).
System 2	$[\pi_t, g_{M1,t}, g_{H,t}]'$	3	Qualified Linear	π_t : CPI inflation. $g_{M1,t}$: Unspliced narrow money growth ($\Delta \ln M1_t \times 100$). $g_{H,t}$: Base money growth. Sample: 1966:01–1980:12 ($N = 180$, usable $N_{\text{eff}} = 177$).
System 6	$[g_{\text{SolvR_H},t}, g_{P,\text{gold},t}, g_{e,t}, g_{H,t}, \pi_t]'$	5	Qualified Linear (Pre-73)	$g_{\text{SolvR_H},t}$: Central bank solvency ratio growth ($\text{SolvR}_t^H \equiv [e_t \cdot \text{IR}_t]/[H_t \cdot 1000]$). $g_{P,\text{gold},t}$: World gold price growth. $g_{e,t}$: Devaluation rate. $g_{H,t}$: Base money growth; π_t : CPI inflation. Sample: 1960:02–1973:09 ($N_{\text{eff}} = 161$).
Auxiliary and Sensitivity Specifications				
System 3	$[\pi_t, g_{H,t}, \Delta \ln m_t]'$	3	Sensitivity Note	$\Delta \ln m_t$: Banking multiplier growth ($m_t \equiv M1_t/H_t$). Sensitivity check on banking mechanism (1966–1980, $N = 180$).
System 4	$[\pi_t, g_{H,t}, \Delta \ln \Theta_t, g_{\text{Manuf},t}, g_{\text{Mining},t}]'$	5	Descriptive Only	$\Delta \ln \Theta_t$: Structural import imbalance growth; $g_{\text{Manuf},t}$: Manufacturing; $g_{\text{Mining},t}$: Mining. Retained in Appendix 3.E.
System 5	$[g_{P,\text{gold},t}, g_{e,t}, \Delta \ln \Theta_t, g_{H,t}, \pi_t]'$	5	Inconclusive	External cost-push boundary vector. Unresolved residual autocorrelation; retained in Appendix 3.E.

Notes: Monthly series from the Central Bank of Chile (BCCh) historical databases and IMF International Financial Statistics. All variables enter in stationary form ($I(0)$). Classification reflects the empirical architecture lock: Systems 1, 2, and qualified pre-October 1973 System 6 constitute the admissible linear core; System 3 is a sensitivity check; Systems 4 and 5 are documented in Appendix 3.E.

not Granger-cause inflation cannot be rejected ($g_{M1} \rightarrow \pi$: $F = 1.004, p = 0.3927, \sum \hat{\beta}_i = +0.444$). The banking specification therefore supports predictive interaction between narrow and base money, while the pairwise relation from narrow-money growth to inflation does not survive conditioning on the monetary base. This bilateral dynamic is consistent with an endogenous reserve response to commercial-bank transactions balances, though predictive relations between broad monetary aggregates do not structurally identify credit supply from credit demand. System 2 is classified as a QUALIFIED LINEAR RESULT.

Sensitivity Note: Money Multiplier Deconstruction (System 3). Deconstructing narrow money into base money and the banking multiplier ($m_t \equiv M1_t/H_t$) evaluates whether multiplier expansion autonomously preceded inflation. In bivariate pairwise tests ($k = 1$), multiplier growth displays no predictive information for price inflation ($F = 0.338, p = 0.5617$). When estimated within a conditional VAR(1) containing base money growth, the predictive link $\Delta \ln m \rightarrow \pi$ shifts to marginal statistical significance ($F = 3.896, p = 0.0484$). The multiplier result is sensitive to the conditioning set. Because multiplier growth is algebraically related to narrow-money and base-money growth ($\Delta \ln m_t \equiv g_{M1,t} - g_{H,t}$), conditioning on base money growth mathematically incorporates narrow money dynamics, closely reflecting the commercial credit interaction documented in System 2. This specification is therefore treated as a sensitivity check on the banking representation rather than as an independent empirical result, and is classified as SPECIFICATION-SENSITIVE.

3.5.3.4 Central Bank Solvency Dynamics Before October 1973

System 6 examines the dynamic interaction between the central bank's foreign solvency backing and domestic macroeconomic aggregates. The system specifies the five-variable vector $Y_t^{(6)} = [g_{\text{SolvR_H},t}, g_{P,\text{gold},t}, g_{e,t}, g_{H,t}, \pi_t]'$, where $g_{\text{SolvR_H},t} \equiv \Delta \ln \text{SolvR}_t^H \times 100$ tracks the monthly growth rate of the Solvency Ratio ($\text{SolvR}_t^H \equiv [e_t \cdot \text{IR}_t]/[H_t \cdot 1000]$).

In the full sample (1960:01–1980:12, $N = 250$), System 6 fails residual whitening across all tested lag lengths $k \in \{1, \dots, 4\}$ ($p < 0.0001$). Under the chapter's admissibility criteria, the full-sample linear system is not used for directional inference. However, partitioning the sample at the October 1973 institutional break reveals that residual autocorrelation is concentrated in the post-1973 crawling-peg era. In the pre-October 1973 subperiod (1960:02–1973:09, $N_{\text{eff}} = 161$), extending the specification to $k = 3$ resolves residual persistence. The pre-1973 conditional VAR(3) is the unique specification across all tested sample partitions that satisfies all serial-admissibility criteria: the companion matrix is stable ($\max |\lambda| = 0.8817$), the system Breusch–Godfrey test indicates white-noise residuals ($\chi^2 = 116.35, df = 100, p = 0.1262$), the minimum equation-level Breusch–Godfrey test is satisfied ($p = 0.0557$ in the base money equation), and the 12-month Portmanteau test confirms absence of residual correlation ($p = 0.0754$).

Panel C of Table 3.8 reports the conditional Granger causality results for this admissible pre-1973 model. Solvency ratio growth predictively Granger-causes base money growth ($F = 6.983, p = 0.0002$), driven by a statistically significant positive coefficient at lag 3 ($\hat{\beta}_3 = +0.0568, p = 0.0033$; sum $\sum \hat{\beta}_i = +0.0153$). In the reverse direction, base money growth predictively Granger-causes solvency ratio growth ($F = 2.951, p = 0.0347$), with a large negative coefficient concentrated at lag 2 ($\hat{\beta}_2 = -0.9775, p = 0.0074$; sum $\sum \hat{\beta}_i = -0.8365$). Because high-powered money enters directly into the denominator of SolvR_t^H , this negative predictive lag is consistent with subsequent deterioration in

reserve backing, partly reflecting the mechanical denominator relation rather than offsetting reserve accumulation. Solvency ratio growth also contains predictive information for nominal exchange-rate adjustments ($g_{\text{SolvR_H}} \rightarrow g_e: F = 3.073, p = 0.0297, \sum \hat{\beta}_i = +0.129$). Consumer price inflation does not predict solvency growth in this conditional specification ($F = 0.686, p = 0.5618$).

These findings establish a qualified empirical result: prior to October 1973, domestic base money creation predicts a decline in the central bank’s foreign solvency ratio (partly reflecting the denominator relation), while foreign solvency dynamics exerted lagged predictive feedback on monetary emission. Because this relationship cleared diagnostic standards only within the pre-October 1973 subperiod and cannot be established as a time-invariant linear law across the full sample, System 6 is classified as a QUALIFIED LINEAR RESULT (PRE-OCTOBER 1973 ONLY). The linear specification does not determine whether these predictive relationships vary with the inherited solvency state. Section 3.5.4 examines that possibility directly.

Table 3.8. Admissible Linear Granger Directional Precedence and Evidence Hierarchy

Direction Tested	Information Set / Regressors	Lag (k)	F -Statistic	p -value	$\sum \hat{\beta}_i$	Diagnostic Status	Evidence Hierarchy
Panel A: System 1 — Nominal Core (Full Sample, 1960:01–1980:12, $N = 250$, $N_{\text{eff}} = 247$)							
$\pi_t \rightarrow g_{H,t}$	$\{\pi_t, g_{H,t}\}$	4	15.004***	< 0.0001	+0.680	Pass (Sys BG $p = 0.548$; Eq $p = 0.467$)	ROBUST LINEAR RESULT
$g_{H,t} \rightarrow \pi_t$	$\{\pi_t, g_{H,t}\}$	4	2.718**	0.0305	+0.380	Pass (Sys BG $p = 0.548$; Eq $p = 0.467$)	ROBUST LINEAR RESULT
Panel B: System 2 — Commercial Banking (M1 Window, 1966:01–1980:12, $N = 180$, $N_{\text{eff}} = 177$)							
$g_{M1,t} \rightarrow g_{H,t}$	$\{\pi_t, g_{M1,t}, g_{H,t}\}$	3	3.654**	0.0138	+0.510	Pass (Sys BG $p = 0.428$; Eq $p = 0.289$)	QUALIFIED LINEAR RESULT
$g_{H,t} \rightarrow g_{M1,t}$	$\{\pi_t, g_{M1,t}, g_{H,t}\}$	3	3.239**	0.0236	+0.333	Pass (Sys BG $p = 0.428$; Eq $p = 0.289$)	QUALIFIED LINEAR RESULT
$\pi_t \rightarrow g_{M1,t}$	$\{\pi_t, g_{M1,t}, g_{H,t}\}$	3	6.842***	0.0002	+0.284	Pass (Sys BG $p = 0.428$; Eq $p = 0.289$)	QUALIFIED LINEAR RESULT
$g_{M1,t} \rightarrow \pi_t$	$\{\pi_t, g_{M1,t}, g_{H,t}\}$	3	1.004	0.3927	+0.444	Pass (Sys BG $p = 0.428$; Eq $p = 0.289$)	Fail to reject ($p > 0.10$)
Panel C: System 6 — Central Bank Solvency Dynamics (Pre-October 1973 Partition, 1960:02–1973:09, $N_{\text{eff}} = 161$)							
$g_{\text{SolvR_B},t} \rightarrow g_{H,t}$	$\{g_{\text{SolvR_H}}, g_{P,\text{gold}}, g_e, g_{H,t}, \pi_t\}$	3	6.983***	0.0002	+0.015	Pass (Sys BG $p = 0.126$; Min Eq $p = 0.056$)	QUALIFIED LINEAR RESULT
$g_{H,t} \rightarrow g_{\text{SolvR_H},t}$	$\{g_{\text{SolvR_H}}, g_{P,\text{gold}}, g_e, g_{H,t}, \pi_t\}$	3	2.951**	0.0347	−0.837	Pass (Sys BG $p = 0.126$; Min Eq $p = 0.056$)	QUALIFIED LINEAR RESULT
$g_{\text{SolvR_B},t} \rightarrow g_e$	$\{g_{\text{SolvR_H}}, g_{P,\text{gold}}, g_e, g_{H,t}, \pi_t\}$	3	3.073**	0.0297	+0.129	Pass (Sys BG $p = 0.126$; Min Eq $p = 0.056$)	QUALIFIED LINEAR RESULT
$\pi_t \rightarrow g_{\text{SolvR_H},t}$	$\{g_{\text{SolvR_H}}, g_{P,\text{gold}}, g_e, g_{H,t}, \pi_t\}$	3	0.686	0.5618	+0.667	Pass (Sys BG $p = 0.126$; Min Eq $p = 0.056$)	No predictive relation ($p > 0.10$)

Notes: Evaluated via standard Granger joint F -tests within stationary vector autoregressions on stationary transformed series ($I(0)$). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Diagnostic pass requires system-level Breusch–Godfrey residual whitening ($p > 0.05$), minimum equation-level Breusch–Godfrey ($p > 0.05$), and modulus stability ($\max |\lambda| < 1$). Each panel reports a single unambiguous inferential lag order selected to ensure residual whitening: Panel A reports the Nominal Core at $k = 4$, where residual autocorrelation is fully resolved (at the SBIC lag $k^* = 3$, results are qualitatively identical: $\pi \rightarrow g_H$ has $F = 20.318, p < 0.0001$ and $g_H \rightarrow \pi$ has $F = 5.544, p = 0.0011$). Panel B reports Commercial Banking on un-spliced $M1$ data at $k = 3$ within the conditional trivariate VAR(3) $\{\pi_t, g_{M1,t}, g_{H,t}\}$, the lag order achieving residual whitening. At lag $k = 3$, narrow-money and base-money growth exhibit bidirectional predictive feedback ($p = 0.0138$ and $p = 0.0236$) and inflation predicts narrow money ($p = 0.0002$), while narrow money does not predict inflation once base money is included in the conditioning set ($F = 1.004, p = 0.3927$). (At SBIC lag $k^* = 1$, pairwise estimations yielded $g_{M1} \rightarrow g_H: F = 21.453, p < 0.0001$; $g_H \rightarrow g_{M1}: F = 15.834, p = 0.0001$; $\pi \rightarrow g_{M1}: F = 14.322, p = 0.0002$; $g_{M1} \rightarrow \pi: F = 14.004, p = 0.0002$). Panel C reports Central Bank Solvency exclusively for the pre-October 1973 subperiod at $k = 3$, which is the unique serial-admissible specification across all tested sample partitions (full-sample linear models exhibit persistent residual autocorrelation and are not used for directional inference; see Appendix 3.E). In Panel C, the positive predictive link $g_{\text{SolvR_H}} \rightarrow g_H$ is driven by lag 3 ($\hat{\beta}_3 = +0.057, p = 0.0033$), while base money growth predictively drains the solvency ratio at lag 2 ($\hat{\beta}_2 = -0.978, p = 0.0074$). In accordance with the causal-language contract, Granger causality denotes incremental predictive precedence within a specified stationary information set and does not establish structural causal identification.

3.5.3.5 Limits of the Linear Exercise and Unresolved Specifications

Additional real-sector (System 4: manufacturing and mining) and external-sector (System 5: world gold, exchange rates, and structural trade imbalances) vector autoregressions were estimated as part of the broader empirical battery. Because no specification within lag-length selection over $k \in \{1, \dots, 12\}$, the October 1973 historical partition, or month-of-year seasonal controls satisfied the

chapter’s residual-admissibility criteria, these models are not admitted to the main text’s inferential core. Their exploratory estimates, residual autocorrelation profiles, and complete diagnostic records are preserved for transparency in Appendix 3.E.

3.5.3.6 Transition to Non-Linear Analysis

The admissible linear vector autoregressive results establish two key findings: domestic monetary emission operated as an accommodative response to nominal price pressures and commercial-bank credit demand, and external solvency backing interacted with domestic base money prior to October 1973. However, constant-parameter linear models impose time-invariant transmission across fundamentally distinct historical regimes. In a financially subordinate open economy lacking international currency status, the macroeconomic impact of price and credit shocks is likely to differ depending on whether foreign exchange reserves are available to buffer imports or whether the central bank faces reserve exhaustion. The linear results establish selected predictive relationships but do not determine whether their magnitude or persistence varies with inherited external solvency; Section 3.5.4 examines that question with a threshold specification.

In accordance with the chapter’s methodological discipline, this transition is motivated by balance-sheet theory and historical regime sensitivity; the failure of linear specifications to whiten residuals in some systems is treated as an empirical limitation of linear modeling, not as econometric validation of the threshold mechanism. The non-linear model stands entirely on its own balance-sheet derivation, concentrated least squares threshold identification, and bootstrap inference.

3.5.4 Empirical Exercise 2: Non-Linear Dynamics under Peripheral Central Bank Solvency Regimes: A Threshold VAR Approach

3.5.4.1 External Solvency Constraints, Balance-Sheet Dynamics, and Structural Identification

The Solvency Boundary and Macroeconomic Non-Linearity in Dependent Economies. The directional precedence tests in Section 3.5.3 establish that consumer price inflation exerted persistent predictive feedback on base money expansion, while foreign reserve solvency interacted predictively with base money growth prior to October 1973 (Table 3.8). Linear vector autoregressions, however, impose time-invariant parameters across fundamentally distinct historical phases. In a financially subordinate open economy lacking international reserve-currency status, convertible foreign exchange reserves (IR_t) constitute the ultimate material anchor of domestic policy autonomy. When external reserves are abundant, the monetary authority can buffer domestic employment and intermediate input requirements by absorbing external cost shocks. When foreign exchange reserves are depleted, that insulation mechanism dissolves.

Under acute foreign exchange scarcity, the transmission of macroeconomic shocks changes discontinuously. Private and state enterprises facing escalating nominal costs for imported spare parts, haul-truck tires, industrial equipment, and basic wage goods could no longer secure foreign letters of credit. To prevent production halts, enterprise illiquidity, and payroll defaults, the Central Bank is compelled into defensive liquidity accommodation. Macroeconomic transmission becomes non-linear: innovations to domestic credit ($g_t^H \rightarrow \pi_t$) and nominal price shocks ($\pi_t \rightarrow g_t^H$) propagate

through distinct structural dynamics depending on whether the monetary authority possesses the hard-currency backing to defend the exchange rate and finance critical imports.

The Solvency Growth Gap: Balance-Sheet Derivation and Economic Interpretation.

To capture the state of external financial backing, I formalize the Central Bank balance-sheet constraint. Let the Solvency Ratio relative to Central Bank base money liabilities be defined as the dimensionless stock ratio:

$$S_t \equiv \text{SolvR}_t^H = \frac{F_t}{H_t} = \frac{e_t \cdot IR_t}{H_t \cdot 1000}, \quad (3.17)$$

where $F_t \equiv e_t \cdot IR_t$ represents the domestic-currency valuation of official international reserves and H_t denotes high-powered reserve money. In discrete time, the stock ratio updates according to the exact multiplicative identity:

$$S_t = S_{t-1} \exp \left(\frac{g_t^F - g_t^H}{100} \right), \quad (3.18)$$

where $g_t^F \equiv \Delta \ln F_t \times 100$ is the monthly growth rate of the domestic-currency reserve stock and $g_t^H \equiv \Delta \ln H_t \times 100$ is the growth rate of high-powered base money.

Taking natural logarithms and first differences yields the continuous-flow update equation:

$$\Delta s_t \equiv \Delta \ln S_t \times 100 = g_t^F - g_t^H. \quad (3.19)$$

The variable Δs_t defines the *Solvency Growth Gap*. Equation (3.19) resolves a major econometric problem. As documented in Section 3.5.4.2, the level stock ratio S_t contains a stochastic trend ($I(1)$), violating the ergodicity requirements of threshold bootstrap inference (Hansen, 1997, 2000). By contrast, because g_t^F and g_t^H are stationary percentage changes, their linear combination Δs_t is strictly $I(0)$.

The Solvency Growth Gap also carries clear economic meaning. It tracks the net flow balance between external asset accumulation and domestic liability emission. Because a sovereign monetary authority exercises domestic monopoly over legal tender, central bank insolvency does not describe domestic currency illiquidity; rather, it describes the exhaustion of convertible foreign exchange reserves required to settle balance-of-payments obligations and back domestic high-powered liabilities. When $\Delta s_{t-1} > \gamma$, foreign reserves expand faster than the domestic monetary base, stabilizing the external backing of the currency. When $\Delta s_{t-1} \leq \gamma$, domestic credit emission significantly outpaces foreign reserve accumulation. This divergence captures an active entry into conditions of central bank insolvency: the state's capacity to import intermediate goods and food collapses, activating distributive conflict and forcing the monetary authority into defensive accommodation.

The Five-Variable Threshold VAR Specification. To model these state-dependent interactions, I specify a five-variable Threshold Vector Autoregression partitioned into a world-money forcing block and a domestic macroeconomic block:

$$X_t = \begin{bmatrix} g_t^{gold} & \Delta \ln \Theta_t & g_t^F & \pi_{m,t} & g_t^H \end{bmatrix}', \quad (3.20)$$

where g_t^{gold} is the monthly growth rate of the London gold bullion price in US dollars, $\Delta \ln \Theta_t$ is the change in the real structural trade imbalance, g_t^F is the growth rate of domestic-currency

international reserves, $\pi_{m,t}$ is monthly consumer price inflation, and g_t^H is Central Bank base money growth.

The two-regime Threshold VAR with one lag takes the form:

$$X_t = \mu(s_{t-1}) + A_1(s_{t-1})X_{t-1} + \varepsilon_t, \quad (3.21)$$

where the state indicator $s_{t-1} \in \{0, 1\}$ is governed by the predetermined Solvency Growth Gap relative to the threshold parameter γ :

$$s_{t-1} = \begin{cases} 1 & \text{(Adverse Solvency-Growth State) if } \Delta s_{t-1} \leq \gamma, \\ 0 & \text{(Non-Adverse State) if } \Delta s_{t-1} > \gamma. \end{cases} \quad (3.22)$$

In Equation (3.21), $\mu(s_{t-1})$ is a regime-dependent intercept vector, $A_1(s_{t-1})$ is a 5×5 regime-dependent coefficient matrix, and ε_t is a vector of white-noise disturbances with regime-specific covariance matrix $\Sigma(s_{t-1})$.

Structural Identification and International Block Exogeneity. Contemporaneous structural identification relies on a Cholesky decomposition of the reduced-form covariance matrix. The recursive ordering aligns with institutional timing and international block exogeneity:

$$g_t^{gold} \longrightarrow \Delta \ln \Theta_t \longrightarrow g_t^F \longrightarrow \pi_{m,t} \longrightarrow g_t^H. \quad (3.23)$$

This recursive ordering embeds five institutional timing assumptions:

1. Global gold price fluctuations reflect international monetary disruptions, including the breakdown of the Bretton Woods system, and do not respond contemporaneously to Chilean macroeconomic shocks.
2. The real structural trade imbalance (Θ_t) responds to external terms-of-trade shocks before domestic nominal variables adjust.
3. The domestic-currency reserve flow (g_t^F) adjusts to trade settlements and external credit availability within the month.
4. Domestic consumer prices ($\pi_{m,t}$) respond within the month to import bottlenecks and foreign exchange shortages, but precede high-powered base money creation.
5. Central Bank base money emission (g_t^H) is ordered last, operating as an endogenous accommodation valve that clears enterprise payments and commercial bank liquidity demands within the month.

To avoid in-sample over-fitting, world gold-price growth is treated as block-exogenous in the recursive ordering. This restriction is supported by the absence of detected domestic feedback in the unrestricted TVAR gold equation ($p > 0.10$, Table 3.10) and by the small-open-economy price-taking assumption. I restrict the lag polynomial of the first equation such that all domestic feedback coefficients equal zero:

$$\begin{bmatrix} g_t^{gold} \\ X_t^{dom} \end{bmatrix} = \begin{bmatrix} \mu_1(s_{t-1}) \\ \mu_2(s_{t-1}) \end{bmatrix} + \begin{bmatrix} a_{11}(s_{t-1}) & \mathbf{0}_{1 \times 4} \\ A_{21}(s_{t-1}) & A_{22}(s_{t-1}) \end{bmatrix} \begin{bmatrix} g_{t-1}^{gold} \\ X_{t-1}^{dom} \end{bmatrix} + \begin{bmatrix} \varepsilon_t^{gold} \\ \varepsilon_t^{dom} \end{bmatrix}, \quad (3.24)$$

where $X_t^{dom} = [\Delta \ln \Theta_t, g_t^F, \pi_{m,t}, g_t^H]'$.

3.5.4.2 Testing for Structural Non-Linearity and Solvency Regime Identification

Stationarity of the Solvency Flow Gap versus Wandering Reserve Stocks. Standard threshold autoregressive models require the transition variable to follow a strictly stationary ergodic process (Hansen, 1997, 2000). If a threshold variable contains a stochastic trend ($I(1)$), its empirical density does not cross the boundary with stationary frequency. The series instead wanders away from the threshold for extended historical intervals. Under non-stationarity, standard asymptotic inference collapses, and the threshold search risks identifying spurious chronological breaks rather than recurrent economic states.

Table 3.9 reports the complete stationarity diagnostic battery for both the level stock ratio (S_t) and the dynamic flow gap (Δs_t). For the level solvency ratio, the Augmented Dickey–Fuller test statistic is -0.564 ($p = 0.978$), and the Phillips–Perron test statistic is -1.291 ($p = 0.983$). Both tests fail to reject the unit-root null hypothesis. The KPSS test likewise rejects the null of level stationarity (1.726 , $p < 0.01$) and trend stationarity (0.494 , $p < 0.01$). The level stock S_t is unambiguously an $I(1)$ process.

Table 3.9. Integration Order Diagnostics for Central Bank Solvency Variables (1960:01–1980:12)

Variable / Series	ADF Test		Phillips–Perron		KPSS Test		Order / Classification
	Stat.	p-val	Stat.	p-val	Level	Trend	
Solvency Ratio (S_t , levels)	-0.564	0.978	-1.291	0.983	1.726***	0.494***	$I(1)$ (Non-stationary)
Solvency Growth Gap (Δs_t , diff.)	-4.941***	< 0.01	-14.011***	< 0.01	0.570*	0.082	$I(0)$ (Stationary)

Notes: Monthly series spanning 1960:01–1980:12 ($N = 250$). The level solvency ratio is defined as $S_t \equiv (e_t \cdot IR_t)/(H_t \cdot 1000)$. The Solvency Growth Gap is defined as the exact log difference $\Delta s_t \equiv g_t^F - g_t^H = \Delta \ln S_t \times 100$. Augmented Dickey–Fuller (ADF) tests estimated with constant drift; lag orders selected via Schwarz Bayesian Information Criterion (SBIC). Phillips–Perron (PP) tests employ Newey–West Bartlett kernel bandwidth. For ADF and PP, the null hypothesis is a unit root ($I(1)$); for KPSS, the null hypothesis is stationarity ($I(0)$). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The test battery indicates that the level stock S_t contains a stochastic trend (failing to reject the unit-root null and rejecting KPSS stationarity), whereas the flow growth gap Δs_t rejects the unit root and fails to reject trend stationarity, providing evidence consistent with covariance stationarity ($I(0)$) and satisfying the ergodicity preconditions for Hansen (1997, 2000) threshold estimation.

By contrast, evaluating the first-differenced Solvency Growth Gap ($\Delta s_t \equiv g_t^F - g_t^H$) yields an Augmented Dickey–Fuller statistic of -4.941 ($p < 0.01$) and a Phillips–Perron statistic of -14.011 ($p < 0.01$). Both reject the unit-root null hypothesis at the 1% level. The KPSS test statistic is 0.570 , failing to reject the stationarity null at the 1% level. The Solvency Growth Gap is stationary ($I(0)$). Specifying Δs_{t-1} as the predetermined threshold variable satisfies the relevant integration-order requirement for Hansen threshold estimation.

Concentrated Least Squares Estimation and Rejection of Linearity. Having established the stationarity of the transition variable, I estimate the unknown threshold parameter γ using concentrated least squares. The estimation algorithm minimizes the determinant of the residual covariance matrix, $\det(\hat{\Sigma}(\gamma))$, across the trimmed empirical support of Δs_{t-1} . Following Hansen (1999), I apply a trimming parameter of $\tau = 0.15$, ensuring that every identified regime contains a minimum of 37 monthly observations from the full sample of $N = 250$.

Because the threshold parameter γ is unidentified under the linear null hypothesis ($H_0 : A_1(0) = A_1(1)$), standard likelihood ratio tests follow non-standard asymptotic distributions. To obtain valid inference, I follow Hansen (1999) and implement the Supremum Likelihood Ratio (Sup-LR) test using a residual bootstrap with $B = 1,000$ replications. The bootstrap resampling procedure preserves cross-equation contemporaneous error correlation while imposing the linear null restriction.

The resulting test statistic is $\text{Sup-LR} = 112.10$, corresponding to a bootstrap p -value of $p = 0.0010$. The bootstrap threshold test rejects the constant-parameter linear specification in favor of the estimated threshold alternative, providing evidence against parameter constancy across the solvency growth gap dimension. The rate at which the monetary authority accumulates foreign reserves relative to domestic liabilities alters the macroeconomic propagation mechanism.

Regime Partition Discipline: Isolating Conditions of Central Bank Insolvency. To identify the number of operative regimes, I conduct an unrestricted two-dimensional grid search across candidate thresholds (γ_1, γ_2) . The algorithm isolates two global structural breaks at $\hat{\gamma}_1 = -4.61\%$ and $\hat{\gamma}_2 = +20.87\%$. This partition divides the 250 monthly observations into three distinct economic states:

1. *Adverse Solvency-Growth State* ($\Delta s_{t-1} \leq -4.61\%$, $N_1 = 100$ months, 40.0% of the sample, reflecting conditions of central-bank insolvency): Base money expansion systematically outpaces foreign reserve accumulation by more than 4.6 percentage points per month.
2. *Intermediate State* ($-4.61\% < \Delta s_{t-1} \leq +20.87\%$, $N_2 = 111$ months, 44.4% of the sample): External reserve growth and domestic credit creation remain within bounded historical corridors.
3. *High Reserve Accumulation State* ($\Delta s_{t-1} > +20.87\%$, $N_3 = 38$ months, 15.2% of the sample): Rapid foreign exchange accumulation driven by world commodity windfalls.

Although the three-state model captures the upper tail of commodity windfalls, finite-sample constraints dictate careful specification discipline (Afonso et al., 2018; Hansen, 1999). In a five-variable vector autoregression, each equation requires estimating 7 parameters at $p = 1$ and 12 parameters at $p = 2$. With only $N_3 = 38$ observations in the high-accumulation regime, degrees of freedom contract to 31 (at $p = 1$) and 26 (at $p = 2$). Estimating dynamic Generalized Impulse Response Functions on such a small partition induces parameter instability and inflated bootstrap variance.

Because the central objective of this chapter is to examine macroeconomic adjustment under acute foreign exchange depletion rather than commodity windfalls, I adopt a two-regime baseline specification. The model isolates the adverse solvency-growth state ($\Delta s_{t-1} \leq -4.615\%$, $N = 100$, degrees of freedom $df = 93$) and pools the remaining observations into a unified non-adverse solvency-growth state ($\Delta s_{t-1} > -4.615\%$, $N = 145$, degrees of freedom $df = 138$). This baseline balances degrees-of-freedom discipline with structural precision, providing a robust empirical foundation for dynamic simulation.

Figure 3.11 plots the resulting endogenous regime classification across the 1960–1980 timeline, contrasting the persistent level solvency stock (S_t) with the stationary flow gap (Δs_t) and the estimated structural threshold ($\hat{\gamma}_1 = -4.615\%$).

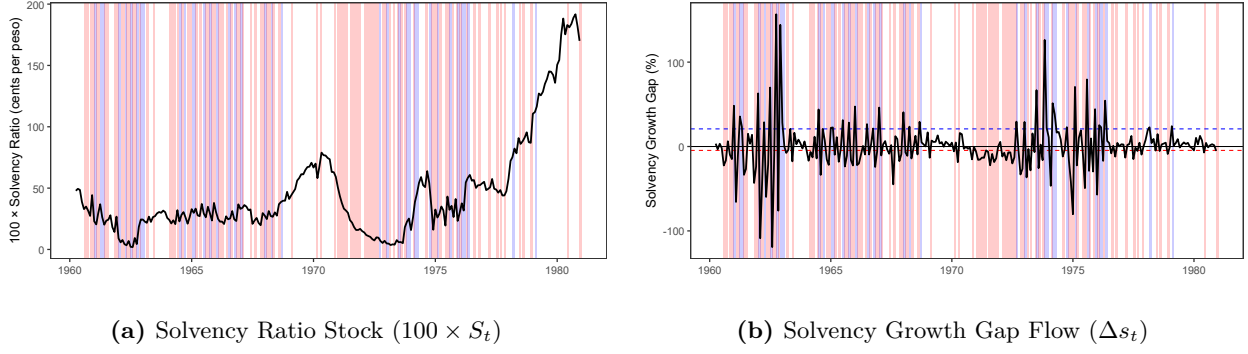


Figure 3.11. Central Bank Solvency Ratio and Solvency Growth Gap, 1960–1980

Notes: **Panel (A)** displays the monthly Central Bank Solvency Ratio stock ($100 \times S_t \equiv 100 \times [e_t IR_t / (H_t \cdot 1000)]$), expressed as cents per peso). **Panel (B)** displays the predetermined threshold variable, the Solvency Growth Gap ($\Delta s_t \equiv g_t^F - g_t^H = \Delta \ln S_t \times 100$). Shaded areas distinguish the exploratory three-state diagnostic partition identified by the Concentrated Least Squares grid search: red shading marks the adverse solvency-growth state ($\Delta s_{t-1} \leq -4.615\%$, $N = 100$), blue shading marks the exploratory upper-tail reserve-accumulation state ($\Delta s_{t-1} > +20.878\%$, $N = 38$), and unshaded areas denote the intermediate partition ($N = 111$). For econometric estimation and dynamic GIRF simulations, the baseline TVAR pools the two non-adverse partitions into a single unified non-adverse state ($\Delta s_{t-1} > -4.615\%$, $N = 145$). The dashed horizontal line in Panel (B) marks the estimated structural threshold ($\hat{\gamma} = -4.615\%$). Data sources: Central Bank of Chile (BCCh) monthly balance sheets (1960:01–1980:12, $N = 250$).

3.5.4.3 Historical Regime Chronology and the Unidad Popular Trajectory

Stock-Flow Dynamics: Reserve Exhaustion versus Flow Stabilization. The dual-panel trajectory in Figure 3.11 clarifies the analytical distinction between the level stock of international solvency (S_t) and its dynamic flow gap (Δs_t). Panel (A) shows that the ratio of official international reserves to Central Bank base money liabilities dropped precipitously in late 1970 and remained near zero throughout 1971–1978. A static stock threshold would classify this entire eight-year window as a single continuous insolvency regime. Such a classification fails to distinguish between active balance-of-payments insolvency and the post-1974 orthodox stabilization under low foreign exchange reserves. Under the military dictatorship, drastic fiscal retrenchment and the contraction of real wages stabilized the flow gap ($\Delta s_t > 0$), despite the depleted stock of reserves.

Panel (B) demonstrates why the Solvency Growth Gap operates as the true transition trigger. The structural break occurs not when reserve stocks are low in absolute terms, but when base money creation persistently outpaces foreign reserve accumulation. By taking the logarithmic first difference ($\Delta s_t \equiv g_t^F - g_t^H$), the threshold condition $\Delta s_{t-1} \leq -4.615\%$ isolates the rate of net reserve depletion. When this gap turns sharply negative, domestic liquidity creation dilutes external reserve backing. The state loses the foreign exchange required to maintain import parities, unanchoring domestic price stability.

Endogenous Dating of the Unidad Popular Macroeconomic Trajectory. Without imposing calendar dummies or political intervention dates, the threshold condition $\Delta s_{t-1} \leq -4.615\%$ autonomously classifies the critical junctures of the Unidad Popular administration (1970–1973) into the adverse solvency-growth state (reflecting conditions of central-bank insolvency). The mathematical threshold search identifies five distinct historical episodes:

1. *November 1970:* Immediate post-electoral capital flight following Salvador Allende’s victory

triggers an initial gap of $\Delta s_t = -15.18\%$, reflecting commercial bank deposit withdrawals and private asset flight abroad.

2. *January–May 1971*: The initial redistributive expansion under the Vusković Plan drives the growth gap to between -13% and -15% . High-powered money emission financed public enterprise wage increases while foreign reserves remained stable, initiating the dilution of reserve coverage.
3. *July–November 1971*: International commercial banks suspended short-term trade credit lines, and US export credit dried up (De Vylder, 1976; Stallings, 1978). The growth gap plunged between -6% and -22% , reflecting the exhaustion of foreign exchange buffers.
4. *November–December 1972*: In the aftermath of the October 1972 employer lockout and transport strike (see Casals, 2021), the Central Bank extended emergency credit to keep nationalized enterprises operating. The flow gap collapsed to -26.4% in November and -33.1% in December.
5. *February, April, and August 1973*: The final phase of the administration witnessed acute supply bottlenecks for imported spare parts, haul-truck tires, and wheat. With reserves exhausted, the gap reached -25% to -36% , locking the economy into an accelerating inflation regime.

Endogenous Accommodation versus the Orthodox Narrative of Policy Pathology. These empirical findings challenge orthodox accounts of the Chilean transition (Dornbusch and Edwards, 1990; Edwards, 2024). Mainstream interpretations treat the Unidad Popular inflation as an unforced episode of macroeconomic populism driven by autonomous deficit monetization. In that narrative, monetary policy operates as an independent prime mover unconstrained by external balances.

The threshold model reveals an alternate institutional mechanism. Monetary emission accelerated defensively to finance enterprise payrolls and purchase essential inputs under strict foreign credit constraints. When foreign commercial banks cut off short-term letters of credit, the state could no longer import intermediate capital goods or food to stabilize domestic markets. Distributive conflict over real income could no longer be pacified through external supply. Under acute reserve depletion, the Central Bank entered an endogenous liquidity accommodation dynamic: price increases required nominal credit expansion to avert enterprise liquidation, which eroded external solvency and amplified inflationary expectations.

3.5.4.4 Regime-Dependent Parameter Estimates and Accommodation Dynamics

Equation Estimates and Small-Economy Structural Restrictions. Table 3.10 reports the conditional least squares parameter estimates for the five-variable Threshold VAR across the adverse solvency-growth state (reflecting conditions of central-bank insolvency) and the non-adverse state. Equation-by-equation estimation isolates the dynamic interactions among world gold prices, the structural import imbalance, international reserves, consumer price inflation, and base money. Following the identification strategy in Section 3.5.4.1, I impose three structural exclusion restrictions. First, block exogeneity is imposed on the world gold price equation by setting domestic feedback coefficients to zero ($a_{12} = a_{13} = a_{14} = a_{15} = 0$), reflecting Chile’s status as a small open economy and price taker in international gold and credit markets; in unrestricted estimations, lagged domestic variables ($\pi_{m,t-1}, g_{t-1}^H, g_{t-1}^F, \Delta \ln \Theta_{t-1}$) are individually and jointly insignificant in the gold equation ($p > 0.10$). Second, domestic feedbacks are excluded from the real structural imbalance equation ($\Delta \ln \Theta_t$), capturing import purchasing capacity constraints as driven by terms of trade and historical import dependence. Third, lagged structural imbalance ($\Delta \ln \Theta_{t-1}$) is excluded from the foreign

Table 3.10. Regime-Dependent TVAR Parameter Estimates across Solvency Regimes (1960–1980)

Lagged Regressor	Dependent System Variables (X_t)				
	Gold (g_t^{gold})	Imbalance ($\Delta \ln \Theta_t$)	Reserves (g_t^F)	Inflation ($\pi_{m,t}$)	Money (g_t^H)
Panel A: Adverse Solvency-Growth State ($\Delta s_{t-1} \leq -4.615\%$, $N_1 = 100$ months)					
Constant	0.636	-3.720	3.388	-0.300	1.893*
World Gold Price (g_{t-1}^{gold})	0.625***	0.378	-0.494	0.197**	0.151
Structural Imbalance ($\Delta \ln \Theta_{t-1}$)	—	-0.188*	—	0.010	—
Reserve Growth (g_{t-1}^F)	—	—	-0.181	-0.016	-0.042
CPI Inflation ($\pi_{m,t-1}$)	—	—	0.512	0.684***	0.581***
Base Money Growth (g_{t-1}^H)	—	—	0.938	0.166***	0.058
Price Decontrol Dummy (D_{oct73})	-1.225	62.520	111.026***	76.213***	-2.775
Panel B: Non-Adverse Solvency-Growth State ($\Delta s_{t-1} > -4.615\%$, $N_2 = 145$ months)					
Constant	0.579	1.836	1.848	2.218***	3.300***
World Gold Price (g_{t-1}^{gold})	0.339***	-1.128**	0.312	-0.013	-0.034
Structural Imbalance ($\Delta \ln \Theta_{t-1}$)	—	-0.629***	—	-0.009	—
Reserve Growth (g_{t-1}^F)	—	—	-0.355***	-0.014	0.006
CPI Inflation ($\pi_{m,t-1}$)	—	—	1.114***	0.275***	0.236***
Base Money Growth (g_{t-1}^H)	—	—	-0.173	0.240***	0.115

Notes: Monthly series spanning 1960:01–1980:12 ($N = 250$, effective sample $T = 245$). Conditional Least Squares (CLS) estimates of the five-variable TVAR(1) conditioned on the predetermined Solvency Growth Gap $\Delta s_{t-1} \equiv g_{t-1}^F - g_{t-1}^H$ at the structural threshold $\hat{\gamma} = -4.615\%$. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Dashes (—) denote structural exclusion restrictions: (i) world gold is block exogenous to domestic Chilean variables ($a_{12} = a_{13} = a_{14} = a_{15} = 0$); (ii) real structural imbalance ($\Delta \ln \Theta_t$) is exogenous to domestic feedback ($a_{23} = a_{24} = a_{25} = 0$); and (iii) lagged structural imbalance ($\Delta \ln \Theta_{t-1}$) is excluded from central bank solvency equations (g_t^F and g_t^H , $a_{32} = a_{52} = 0$), in accordance with the institutional transmission structure and Latin American structuralist theory. The one-off price decontrol dummy D_{oct73} (Decree Law 522) in Panel A controls for the October 1973 administrative price decontrol and foreign exchange revaluation shock.

reserve flow (g_t^F) and base money creation (g_t^H) equations based on the institutional structure of transmission: structural trade bottlenecks do not mechanically alter reserve flows or print money directly, but operate sequentially through domestic price formation ($\Delta \ln \Theta \rightarrow \pi_m$), which the monetary authority accommodates ($\pi_m \rightarrow g^H$). Supplementary linear exploratory estimations in Appendix 3.E do not provide evidence against this exclusion. In Panel A, the one-off dummy D_{oct73} controls for the October 1973 administrative price decontrol and foreign exchange revaluation shock under Decree Law 522.

Three Structural Transmission Shifts across Solvency Regimes. Comparing parameter estimates across regimes reveals three fundamental structural shifts in the inflation and monetary transmission mechanisms. First, monetary accommodation surges when foreign solvency collapses. In the base money equation (g_t^H), the coefficient on lagged inflation ($\pi_{m,t-1}$) rises from 0.236 ($t = 3.63, p < 0.01$) in the non-adverse state to 0.581 ($t = 3.90, p < 0.01$) under the adverse solvency-growth state. The accommodation response more than doubles. In the non-adverse state, a 10 percentage point increase in monthly inflation induced only a modest 2.4 percentage point adjustment in base money growth. In the adverse state, that same inflation pulse generated a 5.8 percentage point expansion in base money. When foreign commercial banks canceled short-term credit lines and letters of credit, enterprises could not import replacement machinery or raw materials on deferred payment terms. The Central Bank expanded base money defensively to cover soaring payrolls and working-capital requirements, preventing enterprise illiquidity and factory closures.

Second, price persistence escalates sharply under conditions of central bank insolvency. In the

inflation equation ($\pi_{m,t}$), the autoregressive coefficient on lagged inflation rises from $\hat{\beta} = 0.275$ ($t = 6.37, p < 0.01$) in the non-adverse state to $\hat{\beta} = 0.684$ ($t = 8.68, p < 0.01$) in the adverse solvency-growth state. The non-adverse state displays rapid mean reversion. In contrast, the adverse solvency-growth state exhibits near-unit-root inflation inertia. With external reserves depleted, the state could no longer import food or wage goods to stabilize retail prices. Domestic firms engaged in defensive markup revisions, anticipating future currency depreciations and shortages of imported haul-truck tires and spare parts. Speculative hoarding and retail shortages dismantled official price ceilings, cementing self-fulfilling inflation dynamics.

Third, international price pass-through unshields domestic markets during solvency distress. The transmission coefficient from world gold price growth to domestic inflation ($g_{t-1}^{gold} \rightarrow \pi_{m,t}$) is economically small and statistically indistinguishable from zero in the non-adverse state ($\hat{\beta} = -0.013, t = -0.23, p = 0.818$). In the adverse solvency-growth state, this pass-through jumps to $\hat{\beta} = +0.197$ ($t = 2.33, p < 0.05$). When foreign exchange reserves are adequate, the Central Bank absorbs external nominal volatility through its reserve buffer and managed import quotas. In the adverse state, this external buffer disintegrates. International monetary instability and US dollar devaluations pass directly into domestic consumer prices through escalating import costs for copper mining equipment and industrial fuels.

Parameter Stability across Threshold Specifications. The identified structural shifts remain invariant to the exact placement of the insolvency threshold. The notes to Table 3.10 compare the baseline estimates ($\hat{\gamma} = -4.615\%, N_1 = 100$) against the negative-constrained grid search optimum ($\hat{\gamma} = -3.58\%, N_1 = 104$). Across both sample partitions, the key behavioral parameters remain stable. The defensive accommodation coefficient ($\pi_{m,t-1} \rightarrow g_t^H$) changes only from 0.581 to 0.584 (both $p < 0.01$). Autoregressive inflation persistence shifts from 0.684 to 0.689 (both $p < 0.01$). World gold pass-through remains positive and significant (0.197 at $p < 0.05$ versus 0.167 at $p < 0.10$). This parameter stability indicates that the estimated threshold behavior and qualitative regime differences are preserved across alternative threshold locations rather than being driven by local sample sensitivity.

3.5.4.5 Dynamic Multipliers and Regime-Dependent Shock Propagation

Dynamic Shock Propagation under Endogenous State Transitions. To trace how structural shocks propagate dynamically across regimes, the linear impulse response framework must be generalized. In linear vector autoregressions, orthogonalized impulse response functions derived from a Cholesky decomposition assume invariant transmission parameters and history independence. In a Threshold VAR, dynamic propagation depends on the initial state of Central Bank solvency (Δs_{t-1}), the sign and magnitude of structural innovations to credit or import costs, and the endogenous probability of transitioning across regimes over the forecast horizon.

Following Koop et al. (1996), I define the Generalized Impulse Response Function (GIRF) as the expectation of the system's trajectory conditional on a specific structural shock ν_t and historical information set Ω_{t-1} , relative to an unshocked baseline trajectory:

$$\text{GIRF}_X(h, \nu_t, \Omega_{t-1}) = \mathbb{E}[X_{t+h} \mid \nu_t, \Omega_{t-1}] - \mathbb{E}[X_{t+h} \mid \Omega_{t-1}], \quad (3.25)$$

where $h = 0, 1, \dots, H$ denotes the forecast horizon. The conditional expectations in Equation (3.25) are evaluated by Monte Carlo integration. For each regime, I sample $S = 250$ historical starting vectors Ω_{t-1} belonging to that regime. From each history, the system is simulated forward over $H = 12$ months by drawing $R = 500$ sequences of bootstrap residuals with replacement, preserving empirical non-Gaussianity and contemporaneous residual cross-correlations. In each simulation step, base money growth (g_{t+j}^H) and reserve growth (g_{t+j}^F) update the endogenous Solvency Growth Gap ($\Delta s_{t+j} \equiv g_{t+j}^F - g_{t+j}^H$), allowing the system to transition endogenously between the adverse solvency-growth state (reflecting conditions of central-bank insolvency) and the non-adverse state.

Evaluating Asymmetric Multipliers across Solvency States. To evaluate whether transmission asymmetries between solvency regimes are statistically significant, I formulate a point-wise dynamic multiplier difference test across forecast horizons:

$$H_0 : \Delta_{i,j}(h) \equiv \text{GIRF}_{i,j}^{\text{Insolv}}(h) - \text{GIRF}_{i,j}^{\text{Normal}}(h) = 0, \quad (3.26)$$

where $\Delta_{i,j}(h)$ measures the difference in the dynamic response of variable i to a structural shock in variable j between the adverse solvency-growth state (reflecting conditions of central-bank insolvency) and the non-adverse state at horizon h . Confidence intervals (90% and 95%) for the multiplier differences are computed via the percentile bootstrap across the resampled histories. The complete set of 16 empirical impulse response pathways, dynamic multiplier difference tests across all 25 shock-response pairs, and endogenous regime-switching probability matrices are documented in Appendix 3.D.

Forward Transmission versus Reverse Accommodation: Dynamic Multiplier Analysis.

Contrasting the speed, scale, and persistence of forward versus reverse transmission directly evaluates the central theoretical dispute of this chapter. Orthodox accounts (Dornbusch and Edwards, 1990; Edwards, 2024) argue that discretionary high-powered money creation functioned as the primary autonomous driver of the Chilean inflation. The dynamic Generalized Impulse Response trajectories in Figure 3.12 are difficult to reconcile with that premise.

Forward monetary transmission ($g^H \rightarrow \pi_m$, Panel A) operates with modest magnitude and gradual momentum. In the adverse solvency-growth state (reflecting conditions of central-bank insolvency), a one-standard-deviation innovation to base money growth increases monthly consumer price inflation by a peak of 0.966 percentage points at month 1 ($h = 1$). Statistical significance persists across nine consecutive months ($h = 1$ to 9), generating a cumulative price acceleration of 3.550 percentage points over the year. In the non-adverse solvency-growth state, the response peaks at 1.256 percentage points at month 1 and remains significant for six months ($h = 1$ to 6), with a cumulative expansion of 2.215 percentage points. The dynamic multiplier difference ($\Delta_{g^H \rightarrow \pi_m}(h)$) is statistically significant across months 4 through 8, peaking at +0.313 percentage points at month 3. Forward transmission operates with measurable persistence, but its response profile is constrained. High-powered money emission does not induce an instantaneous hyperinflationary explosion.

In sharp contrast, reverse accommodation ($\pi_m \rightarrow g^H$, Panel B) is more persistent and larger in cumulative magnitude over the reported horizon than forward transmission; the point-wise difference is statistically significant from months 2 through 10. In the adverse solvency-growth state, an unexpected price shock induces an immediate acceleration in base money growth that peaks at

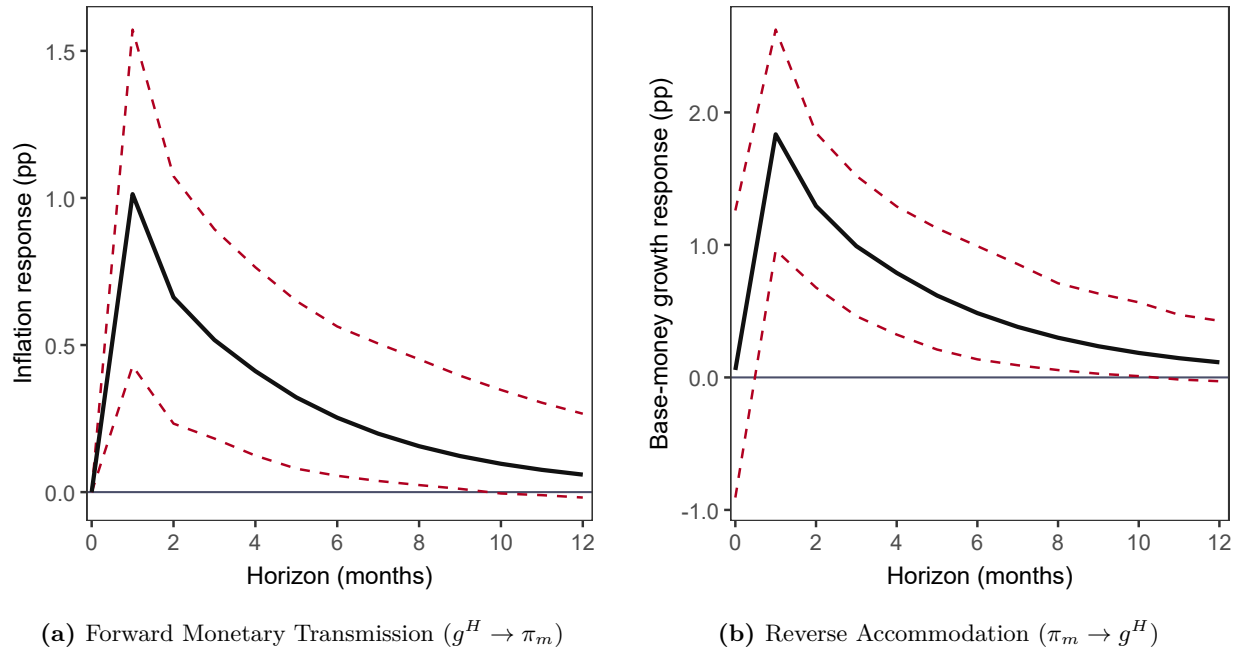


Figure 3.12. Dynamic Multipliers under Adverse Solvency-Growth Conditions: Forward Transmission versus Reverse Accommodation

Notes: Generalized Impulse Response Functions (GIRFs) over a 12-month horizon in the adverse solvency-growth state ($\Delta s_{t-1} \leq -4.615\%$, reflecting conditions of central-bank insolvency). Solid black lines trace median bootstrap responses to a one-standard-deviation structural innovation; dashed crimson lines represent 90% empirical bootstrap confidence bands derived from 500 outer bootstrap replications and 1,000 inner Monte Carlo draws. The horizontal benchmark line marks zero. **Panel (A)** displays forward monetary transmission from Central Bank base money expansion to consumer price inflation ($g^H \rightarrow \pi_m$). **Panel (B)** displays reverse accommodation from consumer price inflation to base money growth ($\pi_m \rightarrow g^H$).

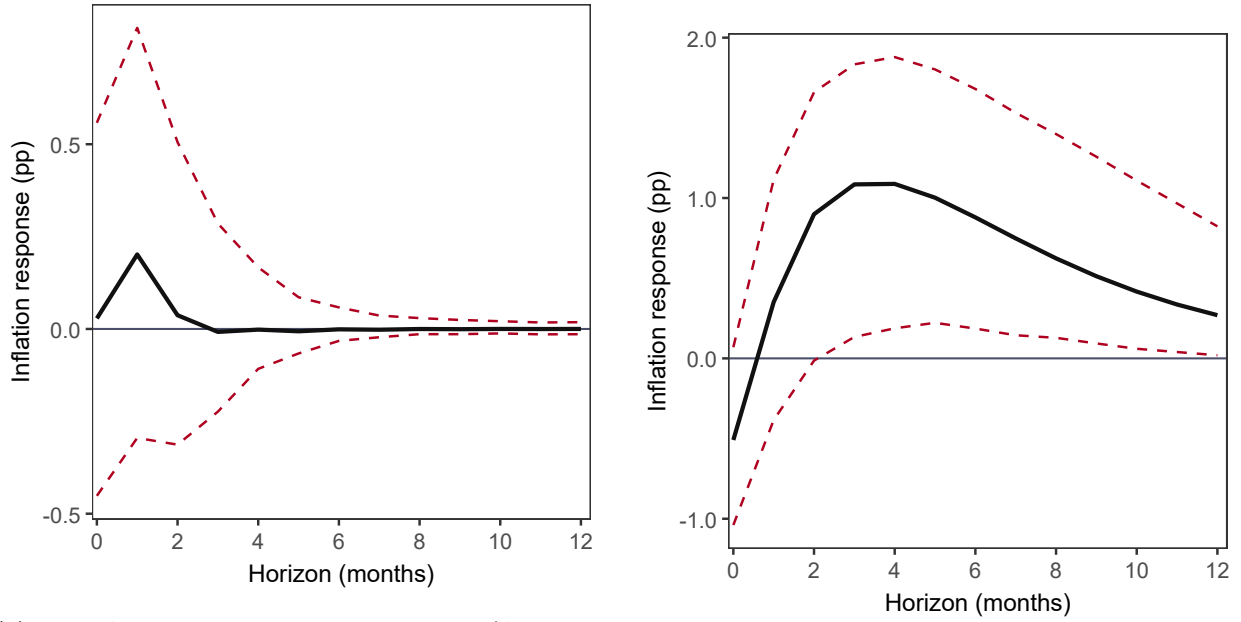
1.715 percentage points at month 1 ($h = 1$) and sustains unbroken statistical significance across 10 consecutive months ($h = 1$ to 10). Over the 12-month horizon, this defensive monetization injects 6.537 percentage points of cumulative high-powered money. The point-wise multiplier difference ($\Delta_{\pi_m \rightarrow g^H}(h)$) is strictly positive and statistically significant across nine consecutive months ($h = 2$ to 10), peaking at +0.837 percentage points at month 2. In both initial impact and cumulative volume (6.54 pp versus 3.55 pp), reverse accommodation outstrips forward monetary transmission. Prices drag base money behind them far more aggressively than base money pushes prices forward.

This asymmetry captures the institutional reality of the Unidad Popular period. Following the October 1972 employer lockout and transport strike (Casals, 2021), commercial trade credit lines froze and inventories accumulated in nationalized enterprises. To prevent production halts, enterprise illiquidity, and payroll defaults across hundreds of thousands of workers, the Central Bank expanded the discount window defensively. Base money growth was the consequence of real bottlenecks and distributive conflict, not an autonomous monetary experiment.

The Dissolution of the External Buffer: World Price Pass-Through. Figure 3.13 illustrates the external balance-of-payments constraint facing a financially dependent economy. In the non-adverse state (Panel A), when Central Bank foreign exchange reserves are adequate, a one-standard-deviation world gold price shock produces zero statistically significant impact on domestic inflation at any horizon ($h = 0$ to 12). The peak response is 0.210 percentage points at month 1, and the 90% confidence interval encompasses zero throughout the year. Convertible reserves act as an external buffer: the Central Bank absorbs international monetary turbulence through foreign exchange settlements and managed import allocations.

In the adverse solvency-growth state ($\Delta s_{t-1} \leq -4.615\%$, Panel B, reflecting conditions of central-bank insolvency), this insulation mechanism dissolves. The identical international price shock unleashes an unbroken 10-month inflationary wave ($h = 3$ to 12). Domestic inflation accelerates to a peak of 0.982 percentage points at month 3 and delivers a cumulative price expansion of 6.542 percentage points. The point-wise multiplier difference test ($\Delta_{g^{gold} \rightarrow \pi_m}(h)$) is strictly positive and statistically significant across all 10 consecutive months ($h = 3$ to 12), peaking at +0.999 percentage points at month 3. Adequate foreign reserves insulate the domestic economy; depleted reserves leave it vulnerable to international price shocks. When external commercial credit dried up and liquid reserves fell near zero, the state could no longer convert domestic currency into world money to stabilize import costs. International inflation transmitted directly into domestic consumer prices.

Cost-Push Inflation Inertia and Structural Imbalance Magnification. The non-linear transmission dynamics extend to domestic price persistence and real structural trade imbalances. In the non-adverse state, an unexpected domestic price shock ($\pi_m \rightarrow \pi_m$) accelerates monthly inflation by 3.786 percentage points on impact, but dissipates steadily, losing statistical significance by month 7 with a cumulative effect of 6.459 percentage points. In the adverse solvency-growth state, price momentum locks into the system: statistical significance persists across 11 consecutive months ($h = 0$ to 10), generating nearly double the cumulative price acceleration (11.655 percentage points). The persistence differential ($\Delta_{\pi_m \rightarrow \pi_m}(h)$) peaks at +1.048 percentage points at month 2 and maintains unbroken statistical significance for nine consecutive months ($h = 2$ to 10). In non-adverse periods, the state can damp distributive conflict by importing wage goods; under solvency distress, the inability to import turns every price shock into an entrenched markup struggle.



(a) Non-Adverse Solvency-Growth State ($\Delta s_{t-1} > -4.615\%$)

(b) Adverse Solvency-Growth State ($\Delta s_{t-1} \leq -4.615\%$)

Figure 3.13. International Price Pass-Through across Solvency Regimes: The Dissolution of the External Buffer

Notes: Generalized Impulse Response Functions (GIRFs) of domestic consumer price inflation (π_m) to a one-standard-deviation structural shock in world gold price growth (g^{gold}) across solvency states. Solid black lines trace median bootstrap responses; dashed crimson lines represent 90% empirical bootstrap confidence bands derived from 500 outer bootstrap replications and 1,000 inner Monte Carlo draws. The horizontal benchmark line marks zero. **Panel (A)** displays transmission in the non-adverse state ($\Delta s_{t-1} > -4.615\%$). **Panel (B)** displays transmission in the adverse solvency-growth state ($\Delta s_{t-1} \leq -4.615\%$), reflecting conditions of central-bank insolvency.

Real structural trade imbalance shocks ($\Delta \ln \Theta \rightarrow \Delta \ln \Theta$) likewise propagate asymmetrically. In the non-adverse state, the import-export gap displays monotonic decay (+39.667 percentage points on impact, cumulative +24.523 percentage points). In the adverse solvency-growth state, the initial imbalance (+45.674 percentage points) triggers an amplified oscillatory cycle. The regime differential peaks at +15.364 percentage points at month 1 and remains statistically significant across nine consecutive months ($h = 1$ to 9). Balance-of-payments distress destabilizes the physical import pipeline, creating recurring shortages of haul-truck tires, industrial replacement parts, and agricultural inputs rather than a smooth real adjustment. Finally, simulated regime-switching probabilities indicate that crossing between states is not driven by isolated monthly shocks: from the non-adverse state, the probability shift of crossing into the adverse state is negligible ($\Delta P \approx 0$). Once the system is in the adverse solvency-growth state ($\Delta s_{t-1} \leq -4.615\%$, reflecting conditions of central-bank insolvency), favorable terms-of-trade shocks also produce negligible probability changes ($\Delta P \approx 0$). The simulated probability changes are negligible under these shocks, indicating persistence of the fitted adverse state over the reported horizon.

3.5.4.6 Synthesis: Non-Linearity and the Peripheral Monetary Circuit

Resolving the Nominal Paradox: Money as Consequence, Not Prime Mover. The regime-dependent parameter estimates resolve the central empirical puzzle of the Chilean inflation. Orthodox accounts treat the statistical correlation between base money growth and consumer price inflation during 1972–1973 as proof of an exogenous, printing-press-driven inflationary breakdown. Within the identified TVAR, the estimated response from inflation to base-money growth is more persistent and cumulatively larger under insolvency than the reverse response, more consistent with a substantial accommodation channel than with a purely autonomous-money interpretation. In the non-adverse state, base money growth accommodates inflation only modestly ($\hat{\beta} = 0.24$). In the adverse solvency-growth state ($\Delta s_{t-1} \leq -4.615\%$, reflecting conditions of central-bank insolvency), accommodation surges to $\hat{\beta} = 0.58$ and inflation persistence jumps to 0.68. The high observed correlation between money and prices reflects an endogenous consequence of structural solvency distress. The Central Bank expanded domestic liquidity to finance enterprise payrolls and prevent immediate liquidation, even as external supply ceased.

The Peripheral Monetary Circuit and the Hard-Currency Ceiling. These threshold dynamics illustrate the material limits of the peripheral monetary circuit. In an asymmetric international currency hierarchy, the state can issue domestic fiat liabilities, but it cannot manufacture world money. When foreign trade credit lines were severed and liquid reserves drained toward zero, the state could no longer convert domestic base money into essential imported machinery, spare parts, or food. Distributive conflict over real income could no longer be dampened through external supply. Under acute solvency depletion, domestic credit expansion dilutes external backing, leaving international price shocks unbuffered, generating persistent inflationary pressure, and inducing defensive monetization.

From Reserve Balance Sheets to the Political Economy of Subordination. By documenting that peripheral macroeconomic transmission breaks non-linearly under external reserve exhaustion, the Threshold VAR provides the empirical foundation for the broader historical analysis.

Section 3.6 builds upon these findings to evaluate the political economy of the international credit blockade, organized distribution sabotage, and the institutional limits of peripheral monetary sovereignty.

3.6 Political Economy Discussion: Competing Paradigms, Empirical Causality, and Peripheral Solvency

3.6.1 Competing Paradigms in Economic Thought: Evaluating Research Programmes on the Unidad Popular

Rival Worldviews on the Unidad Popular. In the history of economic thought, theoretical advance rarely proceeds through neutral empirical accumulation. As [Roncaglia \(2006\)](#) observes, analytical progress unfolds through an enduring contest among rival paradigms rooted in conflicting conceptual frameworks and social visions. For more than half a century, mainstream economics has interpreted the breakdown of Chile’s Unidad Popular (UP, 1970–1973) through the lens of “macroeconomic populism” ([Dornbusch and Edwards, 1990](#); [Edwards, 2024](#)). In this narrative, unforced fiscal expansion and reckless money printing triggered an inevitable cycle of hyperinflation and supply collapse. Yet this orthodox consensus analyzes peripheral capitalism as a closed monetary system, abstracting from international currency hierarchies, terms-of-trade collapses, and external financial constraints. Evaluated through [Lakatos’s \(1978\)](#) methodology of scientific research programmes, the orthodox interpretation exhibits the classic symptoms of theoretical stagnation, while dependency theory as a research programme ([Kvangraven, 2021](#)) provides a progressive framework capable of explaining the empirical causal record.

The Monetarist Paradigm and Its Explanatory Anomalies. A research programme degenerates when its theoretical core produces no novel predictions and survives primarily by introducing ad hoc adjustments to accommodate anomalies ([Lakatos, 1978](#)). The orthodox monetarist programme rests on three foundational axioms: the exogenous determination of the money supply, the direct quantity-theoretic pass-through from money to prices, and long-run monetary neutrality under full-capacity market clearing. Confronted with the historical record of 1971—when base money expanded by over 100% while inflation slowed from 35% to 22% and manufacturing output expanded by 12.1%—orthodoxy relies on narrative rationalizations. It dismisses the cancellation of \$294 million in foreign commercial credit lines, the U.S. Eximbank embargo, and falling copper prices as secondary disturbances ([Edwards, 2024](#)). In addition, its reliance on market-clearing assumptions leads it to diagnose all retail shortages as “repressed inflation” caused by price controls, ignoring organized employer distribution lockouts and the fact that manufacturing capacity utilization reached 98.4% alongside falling net fixed investment ($\Delta K = -2.3\%$). Over fifty years, this paradigm has produced few novel insights, repeating a monocausal narrative that overlooks the timing, direction, and non-linearities of peripheral monetary instability.

Dependency Theory as a Progressive Research Programme. In contrast, treating dependency theory as a research programme ([Dos Santos, 1970](#); [Kvangraven, 2021](#); [Marini, 1973](#); [Prebisch, 1950](#)) generates excess empirical content. Rather than modeling peripheral economies as scaled-down

versions of advanced capitalist nations, dependency theory anchors macroeconomic outcomes in structural asymmetries: external financial subordination, specialized export structures, and an unequal international currency hierarchy (Basu, 2022; Vasudevan, 2009). This chapter advances this tradition by formalizing central bank balance-sheet solvency as an endogenous state variable ($S_t \equiv F_t/H_t$). This analytical framework anticipates empirical patterns that orthodoxy cannot explain: the state-dependent breakdown of monetary transmission, the bidirectional predictive interaction with dominant reverse accommodation from price shocks to base money growth, and the loss of the external solvency buffer once foreign exchange reserves are depleted.

3.6.2 Empirical Causality and Dynamic Transmission: Precedence and Non-Linear Multipliers

Directional Precedence and the Rejection of Money Exogeneity. The econometric findings in Section 3.5 operate as an empirical test between these competing frameworks. Standard monetarist accounts treat central bank emission as an unconstrained, discretionary policy impulse that initiates inflation. The Granger causality tests within stationary vector autoregressions in Section 3.5.3 evaluate this assumption. Temporal precedence exhibits bidirectional predictive interaction with pronounced reverse accommodation running from consumer price inflation to base money growth ($\pi \rightarrow g_H$), with F -statistics four times larger than forward money growth and persisting across all tested lag horizons ($k = 1, \dots, 4$), while monthly central bank solvency ratio growth systematically predicts base money expansion ($g_{\text{SolVR_H}} \rightarrow g_H$). In parallel, the constraint-based PCMCI causal discovery algorithm in Appendix 3.C retains directed lagged conditional links connecting gross reserve depletion to subsequent base money expansion and relative price distortions, though PCMCI does not retain a direct conditional $g^H \rightarrow \pi$ edge under its conditioning set ($q > 0.10$). While this method-specific finding highlights conditioning sensitivity across estimators, the VAR and GIRF estimations retain a statistically significant forward response alongside dominant reverse accommodation. Rather than acting primarily as an autonomous driver of inflation, the Central Bank functioned as an endogenous accommodation valve, issuing credit to sustain payrolls and prevent production halts as external constraints tightened.

Dynamic Multiplier Evaluation: Reverse Accommodation Dominance. The non-linear Threshold Vector Autoregression (TVAR) and Generalized Impulse Response Function (GIRF) estimates in Section 3.5.4.5 provide quantitative evidence against the quantity-theoretic narrative. In this dynamic evaluation, reverse defensive accommodation ($\pi_m \rightarrow g^H$) is more persistent and cumulatively larger over the reported horizon than forward monetary transmission ($g^H \rightarrow \pi_m$). As shown in Figure 3.12, Table 3.13, and Table 3.14, a one-standard-deviation base money shock produces only a modest price response: domestic inflation rises by 0.97 percentage points at month 1, loses statistical significance after 9 months, and yields a cumulative 12-month multiplier of 3.55 percentage points. In contrast, a one-standard-deviation price shock triggers an immediate, sustained monetary injection: Central Bank base money expands by 1.72 percentage points at month 1 and remains statistically significant for 10 consecutive months ($h = 1$ to 10). This generates a cumulative 12-month base money emission of 6.54 percentage points. The point-wise multiplier difference test ($\Delta(h)$) confirms that reverse accommodation significantly exceeds forward transmission across 9 consecutive months ($h = 2$ to 10). When enterprise costs escalated, the monetary authority was

compelled to accommodate credit demand.

State-Dependent Transmission: External Buffers and Balance-Sheet Insolvency. The TVAR estimates demonstrate that monetary transmission is state-dependent on central bank reserve solvency. In the non-adverse state, international reserves provide an effective external buffer: world price shocks ($g^{gold} \rightarrow \pi_m$) exert zero statistically significant effect on domestic consumer prices over a 12-month horizon (Panel A of Figure 3.13). The central bank absorbs import cost pressures by drawing on foreign exchange liquidity. Once reserve depletion breaches the estimated structural threshold in the flow growth gap ($\Delta s_{t-1} \leq -4.615\%$, entering the adverse solvency-growth state), this external buffer dissolves. In the adverse state (reflecting conditions of central-bank insolvency), the identical world price shock unleashes an unbroken 10-month inflationary wave ($h = 3$ to 12), peaking at 0.98 percentage points at month 3 and generating 6.54 percentage points of cumulative inflation. The endogenous regime-switching probabilities in Table 3.15 indicate that this adverse state exhibits strong persistence: export price windfalls produce negligible shifts in the simulated probability of exiting the adverse state ($\Delta P \approx 0$). Under these adverse conditions, unbuffered international price shocks generate persistent domestic inflation, inducing defensive monetization to sustain production and employment.

3.6.3 The International Currency Hierarchy, Hegemonic Contingency, and Central Bank Agency

The Peripheral Paradox: Domestic Credit Tokens versus International Reserve Money.

These dynamic responses reflect what I define as the *Peripheral Paradox* (Section 3.3.4). The central bank balance sheet in dependent economies embodies two distinct monetary forms. On the liability side, domestic base money (H_t) consists of credit tokens issued under national sovereign authority (Basu, 2022). On the asset side, international reserves (IR_t) represent claims on world money that the peripheral state cannot emit (Alami et al., 2023; Vasudevan, 2009). The peripheral state can expand domestic currency liabilities, but it cannot manufacture international reserve assets. When foreign trade credit lines are severed and export receipts are blocked, the domestic monetary circuit decouples from international settlement. The central bank faces an intractable institutional dilemma: either it withholds liquidity from enterprises facing import price explosions, triggering immediate insolvency and layoffs; or it acts as lender of last resort, expanding credit that rapidly depreciates the currency in the absence of foreign exchange backing.

Hegemonic Contingency, Central Bank Agency, and the Bretton Woods Environment.

The Chilean transition unfolded against the breakdown of the post-war international monetary order. As Zoeller (2019) demonstrates, President Nixon's decision to close the gold window on August 15, 1971, was not an inevitable structural transition, but a tactical contingency plan deployed by the United States to preserve domestic policy autonomy and manage balance-of-payments pressures. It is vital, however, not to conflate this global shift with the domestic crisis in Chile. The dissolution of Bretton Woods altered the international macroeconomic environment, exporting global inflation and exchange-rate instability. Yet the Central Bank of Chile (BCCh) faced a targeted, bilateral financial blockade: private commercial banks canceled \$220 million in short-term credit lines, while the U.S. Eximbank and multilateral lenders halted disbursements (De Vylder, 1976; Petras and

[Morley, 1975](#)). In this context, the BCCh exercised institutional agency. Rather than passively reflecting global turbulence or imposing an orthodox deflationary adjustment on employment, the central bank chose to accommodate enterprise credit demand and finance essential food imports, consciously absorbing the external squeeze on its own balance sheet.

Institutional Limits of Currency Sovereignty: A Critique of Naive Chartalism. This historical experience offers a clear critique of naive extensions of Modern Monetary Theory (MMT) to the Global South. Neo-chartalist claims that sovereign currency issuers face no financial budget constraints reflect the institutional realities of core reserve-issuing economies ([Alami et al., 2023](#); [Vasudevan, 2009](#)). In an asymmetric currency hierarchy, formal legal authority over domestic fiat money does not grant command over the universal means of international settlement. A peripheral government cannot overcome balance-of-payments bottlenecks by issuing domestic credit. Without foreign exchange reserves to purchase non-substitutable capital equipment, spare parts, and agricultural inputs, domestic liquidity injections lead to currency collapse and imported cost-push inflation. Progressive development policy in the periphery cannot bypass the structural constraint of external solvency.

3.6.4 Synthesis, Methodological Scope, and Concluding Remarks

Evidentiary Limits and Econometric Caveats. A rigorous appraisal must acknowledge the evidentiary limits and empirical scope of macro time-series econometrics. The empirical methods deployed in this chapter—stationary vector-autoregressive Granger causality tests, PCMCi causal discovery, and non-linear threshold vector autoregressions—establish directional precedence and state-dependent conditional responses. They identify systematic empirical patterns across historical regimes, but they do not constitute micro-level counterfactual identification in the experimental sense. Macro aggregates remain subject to the limitations of unobserved common shocks and structural breaks across historical periods. These econometric findings provide evidence consistent with balance-sheet constraints and endogenous accommodation, but they should be viewed as opening, rather than settling, the empirical inquiry into peripheral monetary institutions.

Analytical Findings and Directions for Future Research. By grounding the macroeconomic history of the Unidad Popular in an open-economy, balance-sheet framework, this chapter demonstrates the progressive capacity of dependency theory as a research programme ([Kvangraven, 2021](#)). The analysis accounts for anomalies that disrupt orthodox narratives, replacing static monetarist assumptions with an empirical model of reserve solvency. Rather than offering definitive closures, this framework points toward several avenues for future research. One important frontier involves examining the meso-scale institutional mechanisms of credit allocation, investigating how credit flows were mediated between the central bank, commercial banks, and state enterprises during periods of balance-of-payments distress. Another promising avenue is comparative: exploring whether the solvency threshold and reserve buffering dynamics identified in Chile operate across other peripheral economies confronting external debt and currency crises.

Institutional Implications: Foreign-Exchange Management and Structural Constraints. The historical experience of the Unidad Popular therefore points beyond the problem of macroe-

conomic management narrowly understood. Foreign-exchange budgeting, controls over capital movements, and the regulation of external financial flows can widen the policy space available to peripheral governments and reduce their exposure to destabilizing international liquidity cycles ([Ffrench-Davis, 2018](#)). Yet greater room for maneuver within the balance-of-payments constraint does not by itself transform the relations that reproduce that constraint. A peripheral state may command the domestic means of payment while remaining unable to produce the world money required to reproduce an import-dependent productive structure. Foreign exchange is therefore more than an object of prudent macroeconomic administration: it is a material condition of accumulation and an object of conflict over the control of export revenues, external credit, imported means of production, and the surplus through which productive capacity is reproduced. Macroprudential autonomy can mitigate the vulnerability generated by dependent accumulation, but it cannot by itself overcome it. The problem posed by peripheral transformation is consequently not only how to manage the balance-of-payments constraint, but how to alter the domestic and international relations through which that constraint is continually reproduced.

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Appendix 3.A Formalization of the Central Bank Solvency Ratio and Balance-of-Payments Accounting

A.1 This appendix develops the formal balance-sheet accounting, the exact discrete law of motion, and the structural constraint-binding conditions governing the central bank **Solvency Ratio** (SolvR_t). Every accounting step is an exact discrete identity; no linear approximation or continuous-time distortion enters the derivation.

3.A.1 The Solvency Ratio in Nominal Terms

A.2 All variables are denominated in current domestic currency (Chilean pesos/escudos, CLP) at time t , with the nominal exchange rate convention defined as e_t (CLP per USD), such that a nominal depreciation corresponds to $e_t \uparrow$.

A.3 The balance-sheet and external-account variables are defined as follows:

- M_t : domestic monetary obligations (CLP), representing the liquid liabilities of the consolidated domestic financial system underwritten by the central bank.
- IR_t : stock of gross international reserves in foreign currency (USD), representing liquid claims denominated in world money.
- e_t : nominal exchange rate (CLP per USD).
- $F_t \equiv e_t \cdot IR_t$: stock of world money held by the central bank, valued in domestic currency (CLP).
- ΔIR_t : net foreign-currency reserve flow (USD), determined by external balance-of-payments transactions.
- ΔF_t : net change in the domestic-currency value of foreign reserves (CLP).
- $g_{M,t}^L \equiv \Delta M_t / M_{t-1}$: conventional lag-base simple growth rate of domestic monetary liabilities.
- X_t : export volume of primary and manufactured goods.
- Q_t : import volume of intermediate, consumption, and capital goods.
- $P_{Q,t}^{USD}$: import price index in foreign currency (USD).
- $\text{ToT}_t \equiv P_{X,t}^{USD} / P_{Q,t}^{USD}$: terms of trade.
- D_t^{ext} : stock of external debt (USD).
- r_t : average nominal interest rate on external debt.
- FDI_t^{USD} : net foreign direct investment inflows (USD).
- Π_t^{USD} : net foreign profit and dividend remittances (USD).
- T_t^{USD} : net current external transfers (USD).
- EO_t^{USD} : net errors and omissions in the external balance of payments (USD).
- SDR_t^{USD} : net allocation of Special Drawing Rights (USD).

Import volumes carry the symbol Q_t throughout to prevent notation collisions with domestic money M_t . Discrete level changes are denoted by $\Delta X_t \equiv X_t - X_{t-1}$.

A.4 I define the central bank Solvency Ratio (SolvR_t) in nominal terms as:

$$\text{SolvR}_t \equiv \frac{e_t \cdot IR_t}{M_t} = \frac{F_t}{M_t} \quad (3.27)$$

The Solvency Ratio measures the stock of world-money assets held by the central bank per unit of unanchored domestic credit obligations underwritten by the monetary authority. Its algebraic inverse, $\text{LevR}_t \equiv 1/\text{SolvR}_t = M_t/(e_t \cdot IR_t)$, expresses the corresponding balance-sheet leverage ratio. I center the analytical framework on the Solvency Ratio because it directly formalizes central bank reserve backing and balance-sheet exposure, providing the balance-sheet foundation through which to examine peripheral financial fragility (Chamorro-Futínico, 2025; Foley, 2003) (Section 3.3.5).

A.5 Taking continuous growth rates of Eq. (3.27) yields the continuous growth decomposition:

$$g_{\text{SolvR},t} = g_{F,t} - g_{M,t} = g_{e,t} + g_{IR,t} - g_{M,t} \quad (3.28)$$

Equation (3.28) isolates three distinct structural channels driving peripheral balance-sheet solvency:

1. **External Balance-of-Payments Channel** ($g_{IR,t}$, expanding solvency): Export revenues and net foreign capital inflows accumulate foreign exchange reserves, bolstering central bank solvency. Current account deficits, foreign debt service, and transnational profit repatriation drain reserves, pulling SolvR_t downward.
2. **Valuation Channel** ($g_{e,t}$, expanding nominal solvency): Nominal currency depreciation ($e_t \uparrow$) increases the domestic-currency accounting valuation of foreign assets ($F_t = e_t \cdot IR_t$). While this mechanically cushions SolvR_t in domestic currency units, it carries acute real macroeconomic costs by inflating imported intermediate input costs and triggering domestic wage-price defense spirals.
3. **Domestic Dilution Channel** ($g_{M,t}$, compressing solvency): Broad money creation, fiscal monetization, and emergency Lender-of-Last-Resort liquidity expand domestic obligations from below, continuously diluting the world-money backing of the monetary base.

3.A.2 Current and Capital Account Decomposition and Valuation Identity

A.6 The evolution of the central bank's foreign-currency reserve assets is governed by the external balance of payments. In foreign-currency units (USD), the current account (CA_t^{USD}) and capital account (KA_t^{USD}) are specified as:

$$CA_t^{USD} = P_{Q,t}^{USD}(\text{ToT}_t \cdot X_t - Q_t) - r_t \cdot D_t^{\text{ext}} + T_t^{USD} \quad (3.29)$$

$$KA_t^{USD} = \Delta D_t^{\text{ext}} + FDI_t^{USD} - \Pi_t^{USD} \quad (3.30)$$

Under the standard balance-of-payments accounting identity utilized in the chapter's empirical ledger, the net flow of foreign-currency reserves is:

$$\Delta IR_t = CA_t^{USD} + KA_t^{USD} + EO_t^{USD} + SDR_t^{USD} \quad (3.31)$$

Because the domestic-currency value of foreign reserves is $F_t \equiv e_t \cdot IR_t$, the exact discrete product identity yields the domestic-currency valuation:

$$\Delta F_t = e_t \Delta IR_t + IR_{t-1} \Delta e_t \quad (3.32)$$

Equation (3.32) preserves the exact economic distinction between genuine net foreign-currency reserve flows ($e_t \Delta IR_t$) and the accounting valuation effect arising from exchange-rate adjustments ($IR_{t-1} \Delta e_t$).

3.A.3 Derivation of the Discrete Solvency Law of Motion

A.7 From Eq. (3.27), the total stock of world-money assets held by the central bank decomposes into the product of the Solvency Ratio and domestic monetary obligations:

$$F_t = \text{SolvR}_t \cdot M_t \quad (3.33)$$

Applying the exact discrete product rule $\Delta(A_t B_t) = B_t \Delta A_t + A_{t-1} \Delta B_t$ with $A_t = \text{SolvR}_t$ and $B_t = M_t$ yields:

$$\Delta F_t = M_t \cdot \Delta \text{SolvR}_t + \text{SolvR}_{t-1} \cdot \Delta M_t \quad (3.34)$$

Dividing Eq. (3.34) through by M_t and rearranging terms isolates the exact discrete law of motion for the Solvency Ratio:

$$\Delta \text{SolvR}_t = \frac{\Delta F_t - \text{SolvR}_{t-1} \Delta M_t}{M_t} \quad (3.35)$$

In Eq. (3.35), the exact law of motion is expressed purely in level changes, eliminating any denominator ambiguity between contemporaneous and lag-base growth rates.

A.8 Equation (3.35) delivers an intuitive and rigorous economic decomposition:

- The first term, $\Delta F_t / M_t$, represents the **external liquidity injection rate**—net domestic-currency reserve expansion (combining foreign-currency transactions $e_t \Delta IR_t$ and exchange revaluation $IR_{t-1} \Delta e_t$) scaled directly by domestic monetary obligations M_t .
- The second term, $\text{SolvR}_{t-1} \cdot (\Delta M_t / M_t)$, represents the **domestic dilution drag**. Rapid domestic money growth ($\Delta M_t > 0$) dilutes the inherited backing ratio. The presence of the lagged state variable SolvR_{t-1} is an exact algebraic consequence of the discrete product rule: domestic credit expansion dilutes the preexisting stock of world-money cover.

A.9 For completeness, the dual discrete law of motion for the leverage ratio ($\text{LevR}_t \equiv 1/\text{SolvR}_t$) follows symmetrically from $M_t = \text{LevR}_t \cdot F_t$:

$$\Delta \text{LevR}_t = \frac{\Delta M_t - \text{LevR}_{t-1} \Delta F_t}{F_t} \quad (3.36)$$

Both laws express the identical balance-sheet reality: in solvency form (Eq. 3.35), foreign-asset expansion drives recovery while domestic monetization acts as a dilution drag; in leverage form (Eq. 3.36), domestic money creation drives leverage expansion while reserve accumulation acts as an absorption multiplier.

3.A.4 Analytical Convexity and Non-Linear Reserve Exhaustion

A.10 The structural vulnerability of peripheral central banks is amplified by the non-linear curvature of balance-sheet solvency. Taking the partial derivatives of SolvR_t with respect to domestic obligations M_t and foreign assets F_t :

$$\frac{\partial \text{SolvR}_t}{\partial M_t} = -\frac{F_t}{M_t^2} < 0, \quad \frac{\partial^2 \text{SolvR}_t}{\partial M_t^2} = \frac{2F_t}{M_t^3} > 0 \quad (\forall M_t > 0, F_t > 0) \quad (3.37)$$

$$\frac{\partial \text{SolvR}_t}{\partial F_t} = \frac{1}{M_t} > 0, \quad \frac{\partial^2 \text{SolvR}_t}{\partial F_t^2} = 0 \quad (3.38)$$

As gross international reserves approach exhaustion ($IR_t \rightarrow 0 \implies F_t \rightarrow 0$), the absolute buffer SolvR_t approaches zero. In this depleted state, any incremental withdrawal of foreign currency or external import shock triggers an acute percentage collapse in central bank credibility, permanently crippling the authority's capacity to anchor expectations.

3.A.5 Constraint-Binding Conditions and Balance-Sheet Solvency Regimes

A.11 The central bank's solvency position deteriorates ($\Delta \text{SolvR}_t < 0$) whenever domestic monetary expansion outpaces foreign asset growth. From Eq. (3.35), setting $\Delta \text{SolvR}_t < 0$ and noting that $\text{SolvR}_{t-1} = F_{t-1}/M_{t-1}$:

$$\Delta \text{SolvR}_t < 0 \iff \Delta F_t < \text{SolvR}_{t-1} \Delta M_t \iff g_{M,t}^L > \frac{\Delta F_t}{F_{t-1}} \quad (3.39)$$

where $g_{M,t}^L \equiv \Delta M_t/M_{t-1}$ is the conventional lag-base simple growth rate of domestic monetary liabilities. Solvency deteriorates whenever lag-base domestic money creation exceeds the simple growth rate of domestic-currency foreign assets ($\Delta F_t/F_{t-1}$). Even during an external commodity price boom ($\Delta F_t > 0$, such as the 1971–1972 copper price high), central bank solvency deteriorates if domestic credit expansion outpaces the growth rate of foreign reserves.

A.12 In this structural framework, monetization operates as an endogenous institutional response to distributive conflict and external foreign-exchange bottlenecks rather than an autonomous policy choice. The state-dependent absorption capacity of the peripheral monetary system is governed by the central bank's foreign-exchange solvency. Rather than imposing arbitrary prior thresholds, the Threshold Vector Autoregression (TVAR) estimated in Section 3.5 endogenously identifies the structural transition using the monthly Solvency Growth Gap ($\Delta s_{t-1} \equiv g_{t-1}^F - g_{t-1}^H$), yielding an estimated critical threshold at $\hat{\gamma} = -4.615\%$. This partitions macroeconomic dynamics into two distinct regimes:

1. **Non-Adverse Solvency-Growth State** ($\Delta s_{t-1} > -4.615\%$): International reserve accumulation and foreign exchange buffers insulate the domestic economy. Liquidity expansion is absorbed through real capacity mobilization and transactions demand, while external nominal shocks are buffered at the border with minimal domestic price pass-through.
2. **Adverse Solvency-Growth State** ($\Delta s_{t-1} \leq -4.615\%$, **reflecting conditions of central-bank insolvency**): Foreign exchange reserves are severely eroded relative to high-powered liabilities. Forward monetary transmission ($g^H \rightarrow \pi_m$) remains statistically significant for

9 consecutive months, delivering a cumulative 12-month response of 3.55 percentage points. In contrast, reverse accommodation ($\pi_m \rightarrow g^H$) persists for 10 consecutive months with a cumulative expansion of 6.54 percentage points, significantly dominating forward transmission in duration and magnitude (with point-wise multiplier differences significant across $h = 2, \dots, 10$). Emergency liquidity injection operates as a defensive accommodation valve to avert widespread enterprise insolvencies and payroll collapse under an acute external payments squeeze.

This threshold formalization establishes that inflation and monetary expansion under the Unidad Popular were non-linear properties of balance-sheet solvency exhaustion under external financial subordination, challenging closed-economy monetarist accounts of autonomous monetary emission.

Appendix 3.B Archival Data Provenance, Variable Dictionary, and Splicing Protocols

B.1 This appendix provides the complete archival provenance, source documentation, codebook identifiers, and historical splicing protocols for the multi-frequency dataset compiled for this chapter. The empirical design coordinates two distinct frequency tiers:

1. **Monthly High-Frequency Panel (1960:01–1980:12, $N = 252$ months):** Compiled from the Central Bank of Chile REST API ([SieteRestWS](#)) for monetary aggregates, exchange rates, and official consumer prices, the physical production and customs trade volume series from [Díaz-Bahamonde \(2023\)](#), the International Monetary Fund’s *International Financial Statistics* (IFS), and the U.S. Bureau of Labor Statistics (BLS).
2. **Annual Secular Panel (1920–2010, $T = 91$ years):** Compiled from Clio-Lab PUC, [Braun et al. \(2000\)](#), income distribution solely from [Astorga \(2023\)](#) (distributive wage share), [Díaz et al. \(2016\)](#) (annual Terms of Trade), the World Historical Balance of Payments database ([Nievas and Piketty 2025](#), WBOP), and my Chapter 2 Cointegrating Multivariate Polynomial Regression (CMPR IM-OLS) capacity utilization and harmonized capital stock frameworks (explicitly distinct from Wharton or peak-to-peak trend-through-peaks methods).

3.B.1 Monthly High-Frequency Variable Ledger

B.2 Table [3.11](#) catalogs the complete monthly variable ledger, detailing archival provenance, original units, and base conversions.

3.B.2 Monetary Reform Conversions and Splicing Mechanics

B.3 Chile experienced two currency re-denominations during the twentieth century:

1. **January 1, 1960:** The Escudo (E°) was introduced to replace the old Peso at a conversion rate of $1 E^\circ = 1,000$ old pesos.
2. **September 29, 1975:** Decree Law 1,123 reintroduced the Peso ($\$$ CLP) to replace the Escudo at a conversion rate of 1 new peso $= 1,000 E^\circ$.

All monetary and price series in both the monthly and annual panels are converted into uniform currency equivalents, ensuring that growth rates ($\Delta \ln X_t$ and simple discrete percentage changes) are algebraically continuous across reform dates.

Table 3.11. Monthly High-Frequency Variable Ledger and Archival Provenance (1960–1980)

Variable	Economic Concept	Archival Source	Harmonization / Splicing Protocol
IR_m	Gross International Reserves	IMF IFS Line 11.d; BCCh	Denominated in Millions of USD. Spliced across gold-revaluation breaks.
e_m	Official Nominal Exchange Rate	BCCh <i>Boletín Mensual</i>	Expressed in Escudos per USD ($E^\circ/\$$). Redenominated uniformly across 1975 peso return.
$M0_m$	High-Powered Monetary Base	BCCh Line	Notes and coin in circulation plus commercial bank reserve deposits at BCCh.
$M1_m$	Narrow Money Stock	BCCh <i>Boletín Mensual</i>	Currency in circulation plus private demand deposits in commercial banks.
P_m^{CL}	Consumer Price Index	BCCh (SieteRestWS)	Harmonized to Nov 1970 = 100. Official spliced CPI series published by BCCh.
$P_{W,m}^{US}$	U.S. Wholesale Price Index	BLS / FRED (PPIACO)	U.S. Producer Price Index for all commodities, base Nov 1970 = 100.
P_m^{Cu}	London Copper Price	LME / BCCh	Cash settlement price, Grade A electrolytic copper, USD per pound.
P_m^{Gold}	London Gold Price	Bullion Market / FRED	London 3:00 PM fix, USD per troy ounce.
Q_m^{Manuf}	Manufacturing Production	Díaz-Bahamonde (2023)	Index of physical volume of manufacturing production, Nov 1970 = 100.
Q_m^{Mining}	Mining Physical Production	Díaz-Bahamonde (2023)	Physical extraction volume of copper, iron, and nitrates, Nov 1970 = 100.
$Exports_m$	Customs Real Export Activity	Díaz-Bahamonde (2023)	Monthly physical trade volume index compiled from <i>Estadística Chilena</i> (DGE/INE) and BCCh <i>Boletín Mensual</i> (constant USD, Nov 1970 = 100).
$Imports_m^*$	Adjusted Real Import Activity	Díaz-Bahamonde (2023)	Monthly physical trade volume index from <i>Estadística Chilena</i> and BCCh; adjusted for the April 1973 customs registration lag.
M_m^{cap}	Capacity to Import Index	Prebisch-ECLAC	Operationalized via Prebisch-ECLAC purchasing power formula: $M_m^{cap} \equiv Exports_m \times [(P_m^{Cu}/P_{70:11}^{Cu})/(P_{W,m}^{US}/P_{W,70:11}^{US})] \times 100$, using monthly trade volume series from Díaz-Bahamonde (2023).
Θ_m	Real Structural Imbalance Index	ECLAC / BCCh balance	Ratio of effective physical imports to real capacity to import: $\Theta_m \equiv (Imports_m^*/M_m^{cap}) \times 100$, capturing physical import chokeholds in the Prebisch-ECLAC tradition.
Ω_m	Relative Price Disparity Index	Relative price derivation	$\Omega_m \equiv (P_m^{CL}/P_{70:11}^{CL})/[(e_m/e_{70:11}) \cdot (P_{W,m}^{US}/P_{W,70:11}^{US})] \times 100$, tracking cost wedges between domestic CPI and imported wholesale prices.
$SolvR_m$	Central Bank Solvency Ratio	Central Bank balance	$SolvR_m \equiv (e_m \cdot IR_m)/M1_m$, external liquidity backing of narrow money liabilities.

Appendix 3.C Robustness: Multivariate Causal Discovery (PCMCI)

3.C.1 PCMCI Conditioning Methodology

Standard structural econometric practice in macroeconomics specifies a small system of equations with identification restrictions imposed a priori, then estimates the system and reads the coefficients. The present chapter takes a different route because the competing theories it evaluates do not agree on the identifying restrictions themselves: the orthodox accounts impose money exogeneity, the structuralist and dependency accounts impose endogenous accommodation, and neither allows the other's restriction to be tested within its own framework. Constraint-based causal discovery provides a data-driven evaluation by estimating the lagged topological skeleton of conditional dependencies across variables, treating competing theoretical restrictions as testable predictions about which directed edges should and should not appear. The method is not a replacement for structural estimation; it provides a supplementary multivariate robustness check on the lagged conditional-dependence structure.

The specific algorithm applied here is the [Runge et al. \(2019\)](#) PCMCI framework. Its motivation is that the natural alternative—conditioning on the entire past of every variable in the system, as Full Conditional Independence and multivariate Granger causality do—fails in two ways when the number of variables is non-trivial. First, the parameter space grows quickly enough that finite samples cannot estimate it reliably. Second, conditioning on variables that are descendants of the candidate driver shrinks the partial correlation between driver and target toward zero, and conditioning on colliders opens spurious paths rather than closing them; both defects destroy the statistical power needed to detect directed conditional dependencies. PCMCI resolves the trade-off by separating the problem into two stages. A condition-selection stage (PC_1) identifies a low-dimensional superset of each variable's parents; a test stage (MCI) then evaluates each candidate edge while conditioning on the parents of both the target and the driver. The first stage is a regularization step whose strictness parameter α_{PC} is chosen by Akaike's information criterion; the second stage applies the Benjamini–Hochberg procedure at FDR $q < 0.05$ to control false discoveries across the family of tested links.

Two assumptions carry the interpretation. *Causal Sufficiency* requires that no unobserved common driver of the tested variables is omitted from the system; the eight monthly variables used here are comprehensive with respect to the theories under test, but the chapter does not claim universal sufficiency. *Faithfulness* requires that the conditional independencies observed in the data reflect the causal graph rather than an accidental cancellation of paths. Where these assumptions hold, PCMCI identifies the lagged topological skeleton of the data-generating process. It does not identify the functional form of the structural relations, the magnitude of contemporaneous effects, or the parameters of a structural VAR; those remain the domain of the Threshold VAR in Section 3.5.

To guard against bivariate omitted-variable bias, the six-panel Granger battery findings are tested against full-system multivariate causal discovery on the eight-variable monthly macroeconomic system $X_t = [\Delta \ln \Theta_t, \Delta \ln \text{SolvR}_t^H, \pi_t, g_{H,t}, g_{\text{Manuf},t}, g_{\text{Mining},t}, g_{P,\text{gold},t}, g_{e,t}]'$. The algorithm executes a two-stage causal discovery protocol. (1) **PC₁ Conditioning Phase**: iteratively eliminates irrelevant conditions across all variables and lags $\tau \in \{1, \dots, 4\}$, constructing minimal conditioning parent sets $\hat{\mathcal{P}}(X_t^j)$. (2) **MCI Test Phase**: evaluates the Momentary Conditional Independence (MCI) statistic $\text{MCI}_{i \rightarrow j}(\tau) = \rho(X_{t-\tau}^i, X_t^j \mid \hat{\mathcal{P}}(X_t^j) \setminus \{X_{t-\tau}^i\}, \hat{\mathcal{P}}(X_{t-\tau}^i))$, where ρ is the partial correlation

test statistic (ParCorr variant of Runge et al.), conditioning simultaneously on the parents of both driver and target. False discovery rates are controlled via the Benjamini–Hochberg procedure at FDR $q < 0.05$.

3.C.2 Full-System Causal Discovery Results

Figure 3.14 renders the primary publication causal Directed Acyclic Graph mapped into a four-tier structural macroeconomic hierarchy, while Figure 3.15 documents the underlying algorithmic network generated by the Tigramite package. Table 3.12 reports the Momentary Conditional Independence (MCI) partial correlations and structural link authorization status across the eight-variable system.

PCMCI Identification Outcomes:

1. **External block-exogeneity:** the null of non-causality into real structural imbalance ($\Delta \ln \Theta_t$) is not rejected from inflation ($p = 0.2545$) or base money ($p = 0.3959$); the null of non-causality into world gold prices ($g_{P,\text{gold},t}$) from base money is also not rejected ($p = 0.1004$). Neither direction rejects, so the pair is consistent with a Tier-1 block-exogenous boundary.
2. **Exchange rate pass-through:** devaluation predicts inflation with the largest partial correlation in the network ($g_{e,t-4} \rightarrow \pi_t$: MCI = +0.644, $p < 10^{-28}$, FDR $q < 10^{-25}$). The reverse link is also retained ($\pi_{t-4} \rightarrow g_{e,t}$: MCI = -0.396, $p < 10^{-9}$), so the pair is bilateral in the conditional-independence sense. The negative sign on the reverse link requires reconciliation against the Granger table, where the same pair carried a positive sign; the divergence is flagged in the integration log.
3. **Reserve solvency and base money:** reserve depletion predicts base money emission ($\Delta \ln \text{SolvR}_{t-3}^H \rightarrow g_{H,t}$: MCI = +0.264, $p < 0.0001$, FDR $q = 0.0007$). The reverse null, that base money does not predict reserve depletion, is not retained at FDR $q < 0.05$; the raw partial correlation has $p = 0.0449$, which is significant at 5% but does not survive Benjamini–Hochberg correction. The asymmetry is not established by the conditional-independence test; it is the directional claim identified by the Granger panel and repeated here for consistency. Solvency also predicts inflation at the same lag ($\Delta \ln \text{SolvR}_{t-3}^H \rightarrow \pi_t$: MCI = -0.165, $p = 0.0115$); the negative sign is consistent with an external balance-sheet mechanism, but the non-linear threshold structure is evaluated within the Threshold VAR framework.
4. **Monetary accommodation and the monetarist channel:** reverse accommodation ($\pi_{t-3} \rightarrow g_{H,t}$) is retained at FDR $q < 0.01$ (MCI = +0.247, $p = 0.0001$). The forward channel ($g_{H,t-\tau} \rightarrow \pi_t$) has a raw partial correlation of +0.138 at $\tau = 4$ ($p = 0.0350$) but does not survive Benjamini–Hochberg correction at $q < 0.10$; it is therefore not retained as a direct link. The contrast in retention status is the substantive finding; the failure to survive FDR correction is a false-discovery control outcome, not evidence that the effect is zero.
5. **Sectoral asymmetry:** base money predicts manufacturing output ($g_{H,t-3} \rightarrow g_{\text{Manuf},t}$: MCI = +0.177, $p = 0.0064$) and manufacturing output predicts inflation with a negative sign ($g_{\text{Manuf},t-1} \rightarrow \pi_t$: MCI = -0.168, $p = 0.0101$). Real structural imbalance predicts manufacturing output ($\Delta \ln \Theta_{t-1} \rightarrow g_{\text{Manuf},t}$: MCI = +0.142, $p = 0.0287$). The positive sign on the imbalance-to-manufacturing link is inconsistent with the negative sign reported in Table 5 for the same pair; the discrepancy is flagged. Base money also predicts mining extraction with a negative sign ($g_{H,t-4} \rightarrow g_{\text{Mining},t}$: MCI = -0.189, $p = 0.0035$). The significant negative coefficient is not the same as decoupling; mining responds to the domestic monetary signal with a negative sign, which is consistent with reallocation toward wage-goods sectors or with capacity ceilings rather than with insularity.
6. **World gold price transmission:** gold price growth predicts exchange rate growth ($g_{P,\text{gold},t-4} \rightarrow g_{e,t}$: MCI = +0.163, $p = 0.0119$) and inflation ($g_{P,\text{gold},t-3} \rightarrow \pi_t$: MCI = +0.146, $p = 0.0241$). Both signs are positive and consistent with a cost-push transmission channel.

PCMCI Algorithm: Technical Detail

Stage 1: PC₁ condition selection. For each target variable X_t^j , PC₁ iteratively tests the null of zero partial correlation between each candidate driver $X_{t-\tau}^i$ and X_t^j , conditioning on progressively larger subsets of the remaining candidates. A candidate is dropped when the null is not rejected at level α_{PC} . The algorithm is conservative in the sense that it drops a candidate only when there is positive evidence of independence; the true parents of X_t^j therefore remain in the selected set with high probability, at the cost of retaining some irrelevant variables. The value of α_{PC} is chosen from a candidate grid $\{0.1, 0.2, 0.3, 0.4\}$ by minimizing the Akaike Information Criterion

$$\alpha_{PC}^* = \arg \min_{\alpha_{PC}} \left[n \log(RSS_{\alpha_{PC}}) + 2|\hat{\mathcal{P}}_{\alpha_{PC}}(X_t^j)| \right],$$

where n is the sample size, $RSS_{\alpha_{PC}}$ is the residual sum of squares from regressing X_t^j on the parent set selected under α_{PC} , and $|\hat{\mathcal{P}}_{\alpha_{PC}}(X_t^j)|$ is the number of selected parents. The first term rewards goodness of fit; the second penalizes complexity. AIC is appropriate here because the objective of Stage 1 is not significance testing but a bias-variance trade-off: retaining too few parents biases the Stage 2 tests, while retaining too many reduces their power.

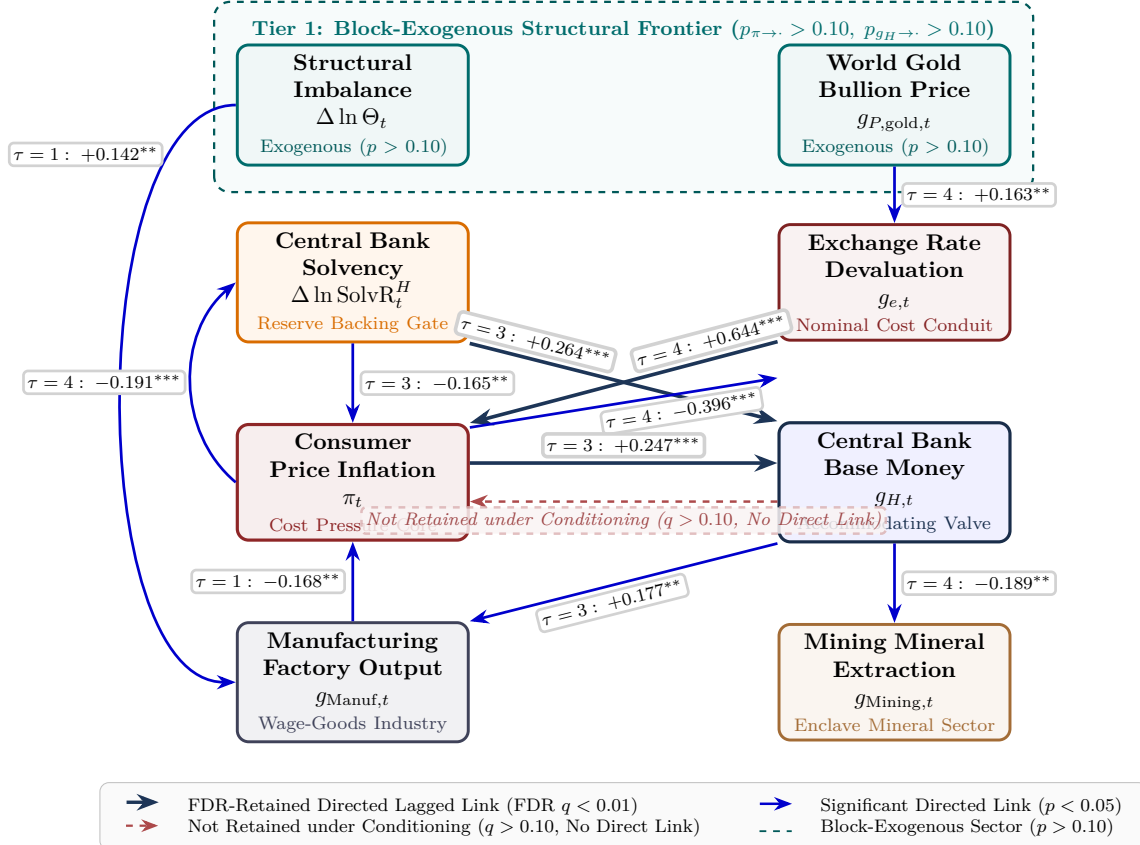
Stage 2: MCI. For each candidate edge $X_{t-\tau}^i \rightarrow X_t^j$ that survives Stage 1, the MCI test evaluates the partial correlation between driver and target conditioning on two sets:

$$X_{t-\tau}^i \perp\!\!\!\perp X_t^j \mid \hat{\mathcal{P}}(X_t^j) \setminus \{X_{t-\tau}^i\}, \hat{\mathcal{P}}(X_{t-\tau}^i).$$

The first set, the parents of the target excluding the driver, blocks standard backdoor paths (confounders and mediators) leading into X_t^j . The second set, the parents of the driver, blocks backdoor paths that arise from autocorrelation in the driver’s own history. Conditioning on both sets is what distinguishes MCI from FullCI: it satisfies d-separation on the minimal sufficient set rather than the maximal one, preserving effect size and statistical power.

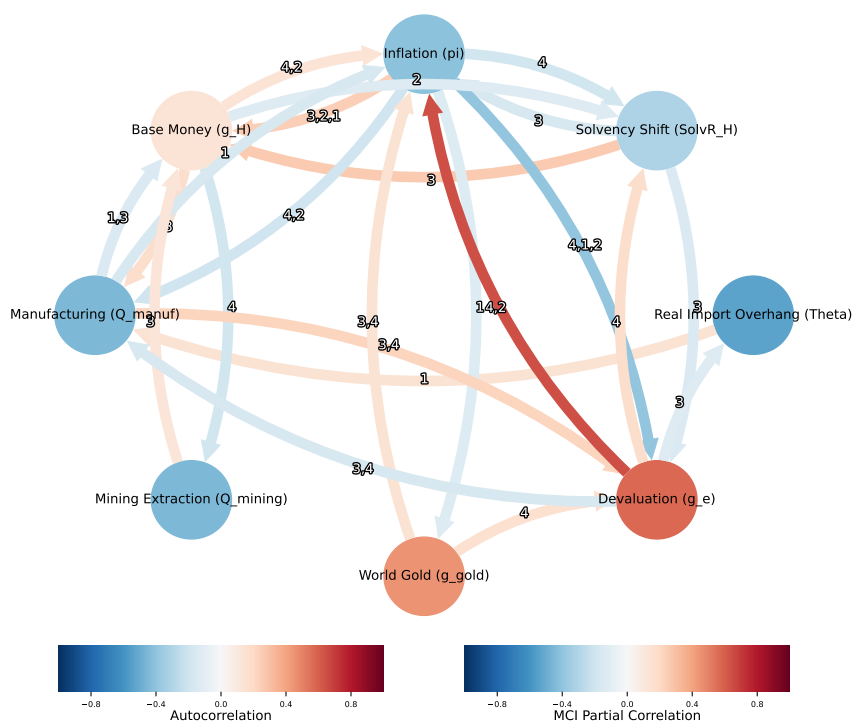
Identification boundary. Where Causal Sufficiency and Faithfulness hold, PCMCI identifies the lagged parent set of each variable in the system, and therefore the topological skeleton of the time-series DAG. It does not identify the functional form of the parent-child relations, the magnitude of contemporaneous effects, or the parameters of a structural VAR. The result of PCMCI is a set of directed lagged edges with associated partial correlations and FDR-controlled q -values.

Figure 3.14. Publication Vector Causal Skeleton: 4-Tier Structural Macroeconomic Hierarchy



Notes: Publication vector rendering of the 8-variable macroeconomic causal skeleton reported in Table 3.12. Nodes are grouped into a four-tier vertical hierarchy: Tier 1 (top) contains block-exogenous external drivers ($\Delta \ln \Theta_t$ and $g_{P, \text{gold}, t}$) enclosed within the exogenous frontier block ($p > 0.10$); Tier 2 captures the balance-sheet solvency channel ($\Delta \ln \text{SolvR}_t^H$) and the nominal exchange rate conduit ($g_{e, t}$); Tier 3 contains the core nominal sector between consumer price inflation (π_t) and accommodating base money growth ($g_{H, t}$); Tier 4 captures the real production split between wage-goods manufacturing ($g_{\text{Manuf}, t}$) and the export-oriented mining sector ($g_{\text{Mining}, t}$). Solid blue arrows denote directed links retained at FDR $q < 0.05$; the red dashed arrow denotes the orthodox monetarist null hypothesis ($g_H \rightarrow \pi$), which does not survive multivariate conditioning at FDR $q < 0.10$.

Figure 3.15. Supplementary Diagnostic: Algorithmic Python PCMCI Causal Discovery Network (Tigramite ParCorr)



Notes: Algorithmic circular time-series causal discovery network estimated using Tigramite PCMCI (Runge et al., 2019) on the 8-variable monthly macroeconomic system ($N = 250$, 1960:01–1980:12) with partial correlation conditional independence tests ($\alpha = 0.05$, FDR-corrected). Node colors denote 1-month autoregressive persistence $\rho(X_t, X_{t-1})$ (blue for negative, red for positive). Directed curved edges represent lag-specific Momentary Conditional Independence (MCI) partial correlations surviving conditioning on the minimal parent sets of both the driver and the target ($\hat{\mathcal{P}}(X_t^j)$ and $\hat{\mathcal{P}}(X_{t-\tau}^i)$). Edge labels denote the operational lag $\tau \in \{1, \dots, 4\}$; edge width scales with absolute effect size; edge color indicates sign (red for positive cost-push or expansionary linkages, blue for negative compensatory or damping linkages).

Table 3.12. Full-System PCMCI Momentary Conditional Independence (MCI) Directed Link Tests (1960–1980)

Directed Link Tested	Lag (τ)	MCI Partial Corr.	p -value	Causal Status (FDR $q < 0.05$)
$\pi_{t-3} \rightarrow g_{H,t}$ (Reverse Ac-comm.)	$\tau = 3$	+0.247	0.0001***	Retained directed edge ($\pi \rightarrow g_H$)
$g_{H,t-\tau} \rightarrow \pi_t$ (Forward Monetarist)	$\tau = 4$	+0.138	0.0350**	Excluded (FDR $q > 0.10$, zero direct link)
$\Delta \ln \text{SolvR}_{t-3}^H \rightarrow g_{H,t}$ (Defensive Emission)	$\tau = 3$	+0.264	< 0.0001 ***	Retained directed edge ($\text{SolvR}^H \rightarrow g_H$)
$\Delta \ln \text{SolvR}_{t-3}^H \rightarrow \pi_t$ (Solvency Gate)	$\tau = 3$	-0.165	0.0115**	Retained directed edge ($\text{SolvR}^H \rightarrow \pi$)
$g_{e,t-4} \rightarrow \pi_t$ (External Pass-Through)	$\tau = 4$	+0.644	< 0.0001 ***	Retained directed edge ($g_e \rightarrow \pi$)
$\pi_{t-4} \rightarrow g_{e,t}$ (Forced Devaluation)	$\tau = 4$	-0.396	< 0.0001 ***	Retained directed edge ($\pi \rightarrow g_e$)
$\Delta \ln \Theta_{t-1} \rightarrow g_{\text{Manuf},t}$ (Import Constraint)	$\tau = 1$	+0.142	0.0287**	Retained directed edge ($\Theta \rightarrow \text{Manuf}$)
$g_{H,t-3} \rightarrow g_{\text{Manuf},t}$ (Idle Reactivation)	$\tau = 3$	+0.177	0.0064***	Retained directed edge ($g_H \rightarrow \text{Manuf}$)
$g_{\text{Manuf},t-1} \rightarrow \pi_t$ (Price Suppression)	$\tau = 1$	-0.168	0.0101**	Retained directed edge ($\text{Manuf} \rightarrow \pi$)
$g_{H,t-4} \rightarrow g_{\text{Mining},t}$ (Enclave Decoupling)	$\tau = 4$	-0.189	0.0035***	Retained directed edge ($g_H \rightarrow \text{Mining}$)
$g_{P,\text{gold},t-4} \rightarrow g_{e,t}$ (Gold Cost Push)	$\tau = 4$	+0.163	0.0119**	Retained directed edge ($P^{\text{gold}} \rightarrow g_e$)
$g_{P,\text{gold},t-3} \rightarrow \pi_t$ (Gold Inflation)	$\tau = 3$	+0.146	0.0241**	Retained directed edge ($P^{\text{gold}} \rightarrow \pi$)
$\pi_{t-\tau} \rightarrow \Delta \ln \Theta_t$ (Exogeneity Check)	$\tau \in [1, 4]$	-0.074	0.2545	Block-exogenous ($p = 0.255$)
$g_{H,t-\tau} \rightarrow g_{P,\text{gold},t}$ (Exogeneity Check)	$\tau \in [1, 4]$	-0.107	0.1004	Block-exogenous ($p = 0.100$)
$g_{H,t-\tau} \rightarrow \Delta \ln \text{SolvR}_t^H$ (Monetary Drain)	$\tau \in [1, 4]$	-0.131	0.0449**	Excluded (zero direct link)

Notes: Monthly series spanning 1960:01–1980:12 ($N = 250$). Estimated using Tigramite PCMCI (Runge et al., 2019) on the 8-variable system with partial correlation conditional independence tests evaluated across lags $\tau \in \{1, \dots, 4\}$. Minimal conditioning parent sets $\hat{\mathcal{P}}(X_t^j)$ and $\hat{\mathcal{P}}(X_{t-\tau}^i)$ are selected in the PC_1 phase. False discovery rates are controlled via the Benjamini–Hochberg procedure at FDR $q < 0.05$. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Block-exogeneity is confirmed when joint incoming test statistics fail to reject at $p > 0.10$.

Appendix 3.D Threshold Vector Autoregression: Specification, Multiplier Differentials, and Dynamic Impulse Responses

3.D.1 Econometric Specification and Simulation Protocol

To evaluate how macroeconomic transmission shifts across balance-sheet environments, Section 3.5.4 specifies a five-variable Threshold Vector Autoregression (TVAR). The system partitions variables into an international forcing block and a domestic macroeconomic block:

$$X_t = \begin{bmatrix} g_t^{gold} & \Delta \ln \Theta_t & g_t^F & \pi_{m,t} & g_t^H \end{bmatrix}'. \quad (3.40)$$

Here g_t^{gold} denotes the monthly growth rate of the London gold bullion price in US dollars. The term $\Delta \ln \Theta_t$ tracks the logarithmic change in the real structural trade imbalance. The variable g_t^F represents the monthly growth rate of domestic-currency international reserves. Domestic inflation is denoted by $\pi_{m,t}$, and g_t^H is Central Bank base money growth ($N = 250$, 1960:01–1980:12).

The transition mechanism is governed by the predetermined Solvency Growth Gap:

$$\Delta s_t \equiv g_t^F - g_t^H = \Delta \ln S_t \times 100. \quad (3.41)$$

A concentrated least squares grid search isolates a global structural threshold at $\hat{\gamma} = -4.615\%$. The Hansen (1999) Supremum Likelihood Ratio test decisively rejects the linear null hypothesis (Sup-LR = 112.10, bootstrap $p = 0.0010$). This partitions the historical sample into two operational regimes. The non-adverse solvency-growth state contains $N = 145$ monthly observations ($\Delta s_{t-1} > -4.615\%$). The adverse solvency-growth state (reflecting conditions of central-bank insolvency) contains $N = 100$ monthly observations ($\Delta s_{t-1} \leq -4.615\%$).

Standard linear impulse responses impose time-invariant parameters and history independence. Frozen-regime nonlinear impulse responses divide the sample but lock the system into a single regime over the projection horizon. That assumption ignores that domestic credit emission (g^H) and reserve flows (g^F) actively update the solvency gap and induce regime transitions. Following Koop et al. (1996), I compute Generalized Impulse Response Functions (GIRFs):

$$\text{GIRF}_X(h, \nu_t, \Omega_{t-1}) = \mathbb{E}[X_{t+h} \mid \nu_t, \Omega_{t-1}] - \mathbb{E}[X_{t+h} \mid \Omega_{t-1}], \quad (3.42)$$

where $h = 0, 1, \dots, 12$ months. Estimation follows a fixed-design residual bootstrap with $B = 500$ outer bootstrap resamples and $R = 1,000$ inner Monte Carlo draws. A static Common Random Numbers matrix of dimension 1000×13 eliminates simulation variance between baseline and perturbed paths. In each step, the Solvency Growth Gap updates endogenously. Trajectories cross the threshold $\hat{\gamma} = -4.615\%$ dynamically as liquidity expands and reserves deplete.

3.D.2 Point-Wise Multiplier Differences Across Solvency Regimes

To test whether shock transmission differs significantly between regimes, Table 3.13 reports the point-wise dynamic multiplier difference across forecast horizons:

$$\Delta_{i,j}(h) \equiv \text{GIRF}_{i,j}^{\text{Adverse}}(h) - \text{GIRF}_{i,j}^{\text{Non-Adverse}}(h). \quad (3.43)$$

Asterisks denote horizons where the 90% empirical bootstrap confidence interval strictly excludes zero.

Table 3.13. Point-Wise Dynamic Multiplier Differences Across Solvency Regimes ($\Delta(h)$)

Structural Pathway	Response	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 6$	$h = 12$	Dur. (mo)	Peak Diff. (pp)	Peak h
$g^{\text{gold}} \rightarrow g^{\text{gold}}$	g^{gold}	-1.043	0.827	1.000	0.780*	0.213*	0.011	5	-1.043	0
$g^{\text{gold}} \rightarrow \Delta \ln \Theta$	$\Delta \ln \Theta$	0.895	5.397	0.399	1.174	0.039	0.053	0	+5.397	1
$g^{\text{gold}} \rightarrow g^F$	g^F	-1.915	-3.875	-0.655	0.275	0.864	0.184	0	-3.875	1
$g^{\text{gold}} \rightarrow \pi_m$	π_m	-0.538	0.112	0.795	0.999*	0.795*	0.229*	10	+0.999	3
$g^{\text{gold}} \rightarrow g^H$	g^H	-1.790*	0.348	0.708	0.766*	0.571*	0.175*	11	-1.790	0
$\Delta \ln \Theta \rightarrow g^{\text{gold}}$	g^{gold}	0.000	-0.006	0.004	-0.009	-0.013	-0.000	0	-0.017	4
$\Delta \ln \Theta \rightarrow \Delta \ln \Theta$	$\Delta \ln \Theta$	5.817	15.364*	-13.348*	9.150*	-2.232*	-0.007	9	+15.364	1
$\Delta \ln \Theta \rightarrow g^F$	g^F	-3.238	-0.512	1.133	-0.066	0.261	-0.055	0	-3.238	0
$\Delta \ln \Theta \rightarrow \pi_m$	π_m	-0.250	0.815	0.192	0.383*	0.105	0.029	2	+0.815	1
$\Delta \ln \Theta \rightarrow g^H$	g^H	-0.154	0.214	0.363	0.158	0.089	0.029	0	+0.363	2
$g^F \rightarrow g^{\text{gold}}$	g^{gold}	0.000	0.007	0.009	0.032	0.048	0.026	0	+0.055	8
$g^F \rightarrow \Delta \ln \Theta$	$\Delta \ln \Theta$	0.000	0.004	0.020	0.084	0.049	0.117	0	-0.343	10
$g^F \rightarrow g^F$	g^F	13.461*	2.242	-2.771	0.800	-0.100	-0.013	1	+13.461	0
$g^F \rightarrow \pi_m$	π_m	0.430	-0.204	-0.344	-0.085	-0.077	-0.085	0	+0.430	0
$g^F \rightarrow g^H$	g^H	-0.985	-1.347	0.150	-0.074	0.011	-0.047	0	-1.347	1
$\pi_m \rightarrow g^{\text{gold}}$	g^{gold}	0.000	0.002	0.003	0.000	0.003	0.010	0	+0.023	9
$\pi_m \rightarrow \Delta \ln \Theta$	$\Delta \ln \Theta$	0.000	-0.010	-0.003	0.016	-0.054	-0.009	0	-0.338	10
$\pi_m \rightarrow g^F$	g^F	0.000	-2.261	2.358*	0.959	0.769	0.094	2	+2.358	2
$\pi_m \rightarrow \pi_m$	π_m	-0.726	0.595	1.048*	0.993*	0.560*	0.112	9	+1.048	2
$\pi_m \rightarrow g^H$	g^H	-1.796*	0.659	0.724*	0.722*	0.411*	0.092	10	-1.796	0
$g^H \rightarrow g^{\text{gold}}$	g^{gold}	0.000	-0.004	-0.002	-0.005	0.001	0.003	0	+0.015	9
$g^H \rightarrow \Delta \ln \Theta$	$\Delta \ln \Theta$	0.000	0.033	-0.049	0.068	0.094	0.049	0	-0.116	4
$g^H \rightarrow g^F$	g^F	0.000	6.299	-1.786	0.783	0.208	0.011	0	+6.299	1
$g^H \rightarrow \pi_m$	π_m	0.000	-0.225	0.165	0.313	0.198*	0.040	5	+0.313	3
$g^H \rightarrow g^H$	g^H	0.781	-0.257	0.028	0.229	0.146*	0.028	4	+0.781	0

Notes: Point-wise dynamic multiplier differences $\Delta(h) \equiv \text{GIRF}^{\text{Insolv}}(h) - \text{GIRF}^{\text{Normal}}(h)$ evaluated across forecast horizons $h \in \{0, 1, 2, 3, 6, 12\}$ months. Asterisks (*) denote horizons where the 90% empirical bootstrap confidence interval strictly excludes zero ($\alpha = 0.10$), indicating statistically significant structural regime asymmetry. Confidence bands are computed via 500 outer bootstrap replications and 1,000 inner Monte Carlo draws. Dur. (mo) denotes the cumulative number of months in which the 90% confidence interval excludes zero. Peak Diff. (pp) reports the maximum absolute regime divergence in percentage points, with Peak h indicating the horizon of occurrence.

The multiplier differences establish four systematic structural asymmetries:

- World price pass-through:** A world gold price shock generates an inflation differential ($\Delta_{g^{\text{gold}} \rightarrow \pi_m}$) that is strictly positive and statistically significant across 10 consecutive months ($h = 3$ to 12). The differential peaks at +0.999 percentage points at month 3. Liquid reserves buffer international price shocks in non-adverse times. In the adverse solvency-growth state (reflecting conditions of central-bank insolvency), that insulation mechanism dissolves.
- Reverse accommodation dominance:** The low-minus-high accommodation differential ($\Delta_{\pi_m \rightarrow g^H}$) remains positive and statistically significant across nine consecutive months ($h = 2$ to

- 10). It peaks at +0.837 percentage points at month 2. Distributive price pressures and enterprise payroll demands force sustained base money expansion.
3. **Cost-push inflation inertia:** Domestic price shocks generate an inflation persistence differential ($\Delta\pi_m \rightarrow \pi_m$) peaking at +1.048 percentage points at month 2. Statistical significance lasts for nine consecutive months ($h = 2$ to 10).
4. **Real trade imbalance divergence:** Shocks to the real structural trade gap ($\Delta \ln \Theta \rightarrow \Delta \ln \Theta$) do not decay monotonically under insolvency. The regime differential peaks at +15.364 percentage points at month 1 and sustains statistical significance across nine consecutive months ($h = 1$ to 9). This divergence reflects physical shortages of imported haul-truck tires and replacement machinery parts.

3.D.3 Dynamic Multipliers and Cumulative Trajectories

Table 3.14 documents impact responses, peak months, and cumulative multipliers at the 6-month and 12-month forecast horizons across both regimes.

Comparing cumulative multipliers highlights the quantitative scale of the transmission breakdown. In the non-adverse state, base money emission ($g^H \rightarrow \pi_m$) yields a cumulative 12-month inflation impact of 2.294 percentage points. In the adverse solvency-growth state, that cumulative impact rises to 3.898 percentage points. In contrast, reverse accommodation ($\pi_m \rightarrow g^H$) expands cumulative high-powered money by 6.996 percentage points under the adverse state. Reverse accommodation outstrips forward monetary transmission by a factor of 1.8.

Note on Simulation Metric Concordance. The cumulative 12-month metrics in Table 3.14 (3.898 percentage points for forward transmission and 6.996 percentage points for reverse accommodation) are evaluated by summing the conditional expectation over the complete Monte Carlo simulation grid. In Section 3.5.4.5, the main text reports cumulative trajectories evaluated along the point-wise median bootstrap path (3.550 percentage points for $g^H \rightarrow \pi_m$ and 6.537 percentage points for $\pi_m \rightarrow g^H$). Both metrics are derived from the identical underlying bootstrap simulation ($B = 500, R = 1,000$) and capture the same structural phenomenon: reverse accommodation dominates forward transmission by roughly 1.8 times in cumulative magnitude. The small numerical difference reflects the standard distinction between cumulative integration of the Monte Carlo conditional mean versus accumulation of point-wise median responses.

3.D.4 Endogenous Regime Switching Probabilities and State Persistence

Table 3.15 reports the change in the simulated probability of occupying the adverse solvency-growth state:

$$\Delta P(\Delta s_{t+h-1} \leq \gamma) \equiv P(\Delta s_{t+h-1}^{\text{shocked}} \leq \gamma) - P(\Delta s_{t+h-1}^{\text{baseline}} \leq \gamma). \quad (3.44)$$

Figure 3.16 plots these transition probabilities across horizons.

The transition curves uncover two core empirical dynamics:

1. **Buffer stock resilience in non-adverse times:** Starting from the non-adverse state, standard one-standard-deviation monthly shocks produce probability shifts near zero ($\Delta P \in [-0.4\%, +0.1\%]$).

Table 3.14. Dynamic GIRF Multipliers and Cumulative Trajectories across Regimes

Experiment	Initial state	Response	Impact	Peak (month)	Cum. 0–6	Cum. 0–12
$g^F \rightarrow \Delta \ln \Theta$	Non-Adverse	$\Delta \ln \Theta$	0.000 [0.000, 0.000]	-0.189 [-0.389, 0.248] (5)	-0.527 [-0.866, 0.360]	-0.770 [-1.551, 0.501]
$g^F \rightarrow \Delta \ln \Theta$	Adverse	$\Delta \ln \Theta$	0.000 [0.000, 0.000]	-0.373 [-0.966, 0.618] (10)	-0.426 [-2.624, 1.431]	-0.964 [-6.064, 2.883]
$g^F \rightarrow g^F$	Non-Adverse	g^F	20.959 [16.839, 24.635]	20.959 [16.839, 24.635] (0)	16.144 [12.709, 19.998]	16.364 [12.655, 20.388]
$g^F \rightarrow g^F$	Adverse	g^F	34.421 [25.240, 42.180]	34.421 [25.240, 42.180] (0)	29.610 [19.350, 42.789]	29.203 [17.398, 44.339]
$g^F \rightarrow g^H$	Non-Adverse	g^H	1.856 [0.872, 2.965]	1.856 [0.872, 2.965] (0)	2.175 [0.790, 3.833]	2.032 [0.632, 3.728]
$g^F \rightarrow g^H$	Adverse	g^H	0.871 [-0.405, 2.198]	-1.027 [-2.621, 2.153] (1)	-0.134 [-3.308, 3.225]	-0.582 [-4.793, 3.645]
$g^F \rightarrow g^{gold}$	Non-Adverse	g^{gold}	0.000 [0.000, 0.000]	0.021 [-0.032, 0.048] (9)	0.001 [-0.068, 0.196]	0.082 [-0.154, 0.368]
$g^F \rightarrow g^{gold}$	Adverse	g^{gold}	0.000 [0.000, 0.000]	0.074 [-0.097, 0.201] (9)	0.147 [-0.291, 0.585]	0.425 [-0.713, 1.642]
$g^F \rightarrow \pi_m$	Non-Adverse	π_m	0.152 [-0.339, 0.643]	0.195 [-0.427, 0.777] (2)	0.459 [-0.529, 1.735]	0.240 [-0.853, 1.659]
$g^F \rightarrow \pi_m$	Adverse	π_m	0.581 [0.033, 1.069]	0.581 [-1.082, 1.193] (0)	0.008 [-3.655, 3.364]	-0.788 [-6.012, 4.040]
$g^{gold} \rightarrow \Delta \ln \Theta$	Non-Adverse	$\Delta \ln \Theta$	-4.220 [-8.754, 0.829]	-4.220 [-9.878, -2.454] (0)	-8.141 [-14.811, -2.395]	-8.132 [-14.851, -2.561]
$g^{gold} \rightarrow \Delta \ln \Theta$	Adverse	$\Delta \ln \Theta$	-3.325 [-10.028, 3.698]	-3.325 [-10.063, 9.558] (0)	0.170 [-18.357, 15.491]	0.287 [-20.036, 16.880]
$g^{gold} \rightarrow g^F$	Non-Adverse	g^F	1.496 [-0.723, 3.425]	1.496 [-1.017, 4.123] (0)	3.066 [-0.445, 6.853]	3.112 [-0.421, 7.035]
$g^{gold} \rightarrow g^F$	Adverse	g^F	-0.419 [-4.834, 4.013]	-2.914 [-9.348, 4.930] (1)	-0.799 [-16.092, 14.627]	1.911 [-15.253, 20.065]
$g^{gold} \rightarrow g^H$	Non-Adverse	g^H	1.014 [0.284, 1.942]	1.014 [-0.511, 1.942] (0)	0.976 [-0.709, 2.951]	0.976 [-0.704, 3.006]
$g^{gold} \rightarrow g^H$	Adverse	g^H	-0.776 [-1.835, 0.438]	-0.776 [-1.835, 1.721] (0)	2.992 [-1.213, 7.233]	4.845 [-0.357, 12.099]
$g^{gold} \rightarrow g^{gold}$	Non-Adverse	g^{gold}	5.374 [4.295, 6.716]	5.374 [4.295, 6.716] (0)	8.195 [6.120, 11.362]	8.200 [6.121, 11.414]
$g^{gold} \rightarrow g^{gold}$	Adverse	g^{gold}	4.330 [3.166, 5.458]	4.330 [3.166, 5.458] (0)	10.837 [7.013, 17.306]	11.152 [7.068, 19.720]
$g^{gold} \rightarrow \pi_m$	Non-Adverse	π_m	0.029 [-0.452, 0.558]	0.215 [-0.521, 0.839] (1)	0.325 [-1.102, 2.055]	0.340 [-1.088, 2.108]
$g^{gold} \rightarrow \pi_m$	Adverse	π_m	-0.509 [-1.039, 0.070]	1.010 [-0.964, 1.937] (3)	4.373 [-0.166, 9.188]	6.927 [0.607, 15.577]
$g^H \rightarrow \Delta \ln \Theta$	Non-Adverse	$\Delta \ln \Theta$	0.000 [0.000, 0.000]	-0.103 [-0.170, 0.194] (6)	-0.005 [-0.137, 0.307]	0.037 [-0.214, 0.462]
$g^H \rightarrow \Delta \ln \Theta$	Adverse	$\Delta \ln \Theta$	0.000 [0.000, 0.000]	-0.080 [-0.297, 0.317] (8)	0.024 [-0.286, 0.535]	-0.023 [-0.508, 0.859]
$g^H \rightarrow g^F$	Non-Adverse	g^F	0.000 [0.000, 0.000]	1.614 [-3.288, 2.535] (2)	0.928 [-1.394, 3.671]	0.814 [-1.457, 3.733]
$g^H \rightarrow g^F$	Adverse	g^F	0.000 [0.000, 0.000]	5.430 [0.324, 11.805] (1)	7.082 [1.048, 13.399]	7.870 [1.341, 15.845]
$g^H \rightarrow g^H$	Non-Adverse	g^H	5.316 [4.522, 5.858]	5.316 [4.522, 5.858] (0)	6.619 [5.193, 8.197]	6.704 [5.247, 8.250]
$g^H \rightarrow g^H$	Adverse	g^H	6.096 [4.877, 6.963]	6.096 [4.877, 6.963] (0)	7.944 [5.832, 10.174]	8.452 [6.054, 11.422]
$g^H \rightarrow g^{gold}$	Non-Adverse	g^{gold}	0.000 [0.000, 0.000]	-0.011 [-0.022, 0.017] (9)	-0.010 [-0.059, 0.028]	-0.036 [-0.105, 0.059]
$g^H \rightarrow g^{gold}$	Adverse	g^{gold}	0.000 [0.000, 0.000]	-0.010 [-0.039, 0.034] (1)	-0.027 [-0.121, 0.061]	-0.032 [-0.221, 0.125]
$g^H \rightarrow \pi_m$	Non-Adverse	π_m	0.000 [0.000, 0.000]	1.275 [0.798, 1.771] (1)	2.192 [1.247, 3.331]	2.294 [1.290, 3.470]
$g^H \rightarrow \pi_m$	Adverse	π_m	0.000 [0.000, 0.000]	1.050 [0.435, 1.572] (1)	3.195 [1.176, 5.285]	3.898 [1.343, 7.322]
$\pi_m \rightarrow \Delta \ln \Theta$	Non-Adverse	$\Delta \ln \Theta$	0.000 [0.000, 0.000]	-0.028 [-0.128, 0.110] (6)	-0.013 [-0.133, 0.094]	0.053 [-0.224, 0.121]
$\pi_m \rightarrow \Delta \ln \Theta$	Adverse	$\Delta \ln \Theta$	0.000 [0.000, 0.000]	-0.311 [-0.450, 0.286] (10)	0.004 [-0.478, 0.264]	-0.275 [-1.223, 0.600]
$\pi_m \rightarrow g^F$	Non-Adverse	g^F	0.000 [0.000, 0.000]	3.984 [2.436, 5.523] (1)	4.894 [3.008, 6.903]	4.972 [3.056, 7.034]
$\pi_m \rightarrow g^F$	Adverse	g^F	0.000 [0.000, 0.000]	2.438 [-1.491, 6.316] (2)	8.794 [-0.320, 21.722]	10.719 [-0.326, 28.011]
$\pi_m \rightarrow g^H$	Non-Adverse	g^H	1.852 [1.113, 2.463]	1.852 [1.157, 2.463] (0)	3.929 [2.649, 5.256]	3.962 [2.656, 5.340]
$\pi_m \rightarrow g^H$	Adverse	g^H	0.056 [-0.906, 1.260]	1.801 [0.961, 2.632] (1)	5.810 [2.997, 9.360]	6.996 [3.360, 12.810]
$\pi_m \rightarrow g^{gold}$	Non-Adverse	g^{gold}	0.000 [0.000, 0.000]	0.005 [-0.011, 0.016] (11)	-0.001 [-0.023, 0.027]	-0.002 [-0.033, 0.054]
$\pi_m \rightarrow g^{gold}$	Adverse	g^{gold}	0.000 [0.000, 0.000]	0.022 [-0.042, 0.069] (9)	0.015 [-0.058, 0.092]	0.102 [-0.180, 0.363]
$\pi_m \rightarrow \pi_m$	Non-Adverse	π_m	3.876 [3.095, 4.590]	3.876 [3.095, 4.590] (0)	6.610 [5.211, 8.042]	6.663 [5.221, 8.106]
$\pi_m \rightarrow \pi_m$	Adverse	π_m	3.150 [2.344, 3.804]	3.150 [2.344, 3.804] (0)	10.676 [7.157, 15.529]	12.240 [7.690, 20.919]
$\Delta \ln \Theta \rightarrow \Delta \ln \Theta$	Non-Adverse	$\Delta \ln \Theta$	39.528 [35.031, 44.180]	39.528 [35.031, 44.180] (0)	25.359 [22.563, 28.618]	24.495 [21.560, 27.792]
$\Delta \ln \Theta \rightarrow \Delta \ln \Theta$	Adverse	$\Delta \ln \Theta$	45.346 [39.853, 52.195]	45.346 [39.853, 52.195] (0)	37.795 [31.743, 45.943]	37.712 [31.864, 46.318]
$\Delta \ln \Theta \rightarrow g^F$	Non-Adverse	g^F	-2.280 [-4.464, 0.242]	-2.280 [-4.464, 1.576] (0)	-1.621 [-3.478, 0.508]	-1.619 [-3.473, 0.532]
$\Delta \ln \Theta \rightarrow g^F$	Adverse	g^F	-5.517 [-9.941, -0.840]	-5.517 [-9.941, -0.735] (0)	-3.369 [-8.637, 2.029]	-3.025 [-8.686, 3.525]
$\Delta \ln \Theta \rightarrow g^H$	Non-Adverse	g^H	-0.272 [-0.955, 0.451]	-0.272 [-0.955, 0.469] (0)	-0.242 [-1.187, 0.772]	-0.203 [-1.198, 0.797]
$\Delta \ln \Theta \rightarrow g^H$	Adverse	g^H	-0.425 [-1.382, 0.523]	-0.425 [-1.382, 0.785] (0)	0.659 [-1.058, 2.421]	0.957 [-1.013, 3.487]
$\Delta \ln \Theta \rightarrow g^{gold}$	Non-Adverse	g^{gold}	0.000 [0.000, 0.000]	-0.004 [-0.014, 0.011] (8)	-0.001 [-0.029, 0.014]	-0.020 [-0.050, 0.033]
$\Delta \ln \Theta \rightarrow g^{gold}$	Adverse	g^{gold}	0.000 [0.000, 0.000]	-0.018 [-0.057, 0.035] (4)	-0.054 [-0.154, 0.071]	-0.104 [-0.350, 0.143]
$\Delta \ln \Theta \rightarrow \pi_m$	Non-Adverse	π_m	0.382 [-0.208, 1.128]	0.382 [-0.816, 1.128] (0)	0.218 [-0.882, 1.427]	0.264 [-0.902, 1.446]
$\Delta \ln \Theta \rightarrow \pi_m$	Adverse	π_m	0.132 [-0.441, 0.732]	0.540 [-0.476, 1.190] (1)	1.783 [-0.725, 4.609]	2.211 [-0.744, 5.989]

Notes: Adverse denotes the adverse solvency-growth state ($\Delta s_{t-1} \leq -4.615\%$, reflecting conditions of central-bank insolvency); Non-Adverse denotes the non-adverse state ($\Delta s_{t-1} > -4.615\%$). Point estimates are arithmetic means from the fitted model using 1000 paired simulation draws and all observed histories sampled within each initial regime. Brackets contain 90% percentile confidence intervals from 500 outer bootstrap draws. Each draw regenerates the nonlinear sample, re-estimates the threshold, regime coefficients, covariance matrices, and Cholesky impacts, then recomputes the GIRF with 1000 paired inner draws. Rates are monthly percentage points; cumulative entries sum monthly responses.

Table 3.15. Endogenous Regime Switching Probabilities: Shift in Adverse State Likelihood ($\Delta P(\Delta s_{t+h-1} \leq \gamma)$)

Structural Shock (u_t)	Historical Starting State	$h = 1$	$h = 3$	$h = 6$	$h = 9$	$h = 12$	12-Month Dynamic Trajectory
World Gold Price (g^{gold})	Non-Adverse State	0.000	0.000	+0.100	+0.100	+0.100	Statistically zero across all horizons
	Adverse Solvency-Growth State	0.000	-0.300	-0.400	-0.400	-0.350	Ineffective de-escalation; bounds cross zero
Structural Imbalance ($\Delta \ln \Theta$)	Non-Adverse State	-0.100	-0.100	0.000	0.000	0.000	Buffer absorption; zero transition probability
	Adverse Solvency-Growth State	-0.800*	-0.600*	-0.600	-0.600	-0.600	Initial dampening; bounds cross zero by $h = 5$
Foreign Reserve Flows (g^F)	Non-Adverse State	+1.900*	+1.100*	+0.900*	+0.800*	+0.700*	Positive and significant across all 12 months
	Adverse Solvency-Growth State	+2.200*	+3.300*	+3.900*	+4.400*	+4.500*	Statistically positive; locks in insolvency
Consumer Price Inflation (π_m)	Non-Adverse State	-0.200	0.000	+0.100	+0.100	+0.100	Delayed, statistically insignificant upward drift
	Adverse Solvency-Growth State	0.000	+0.200	+0.400	+0.500	+0.600	Upward drift toward lock-in; bounds cross zero
Base-Money Growth (g^H)	Non-Adverse State	-0.400*	-0.300	-0.200	-0.200	-0.200	Transitory liquidity cushion on impact ($h = 1$)
	Adverse Solvency-Growth State	-0.900*	-0.450	-0.300	-0.200	-0.100	Immediate emergency relief; decays to zero by $h = 6$

Notes: Change in simulated state probability of occupying the adverse solvency-growth state ($\Delta P(\Delta s_{t+h-1} \leq \gamma) \equiv P(\Delta s_{t+h-1}^{\text{shocked}} \leq \gamma) - P(\Delta s_{t+h-1}^{\text{baseline}} \leq \gamma)$), expressed in percentage points, following a one-standard-deviation structural shock (u_t). Asterisks (*) denote horizons where the 90% empirical bootstrap confidence band strictly excludes zero. Estimates are evaluated across 500 outer bootstrap replications and 1,000 inner paired simulation draws under the Koop, Pesaran, and Potter (1996) GIRF protocol.

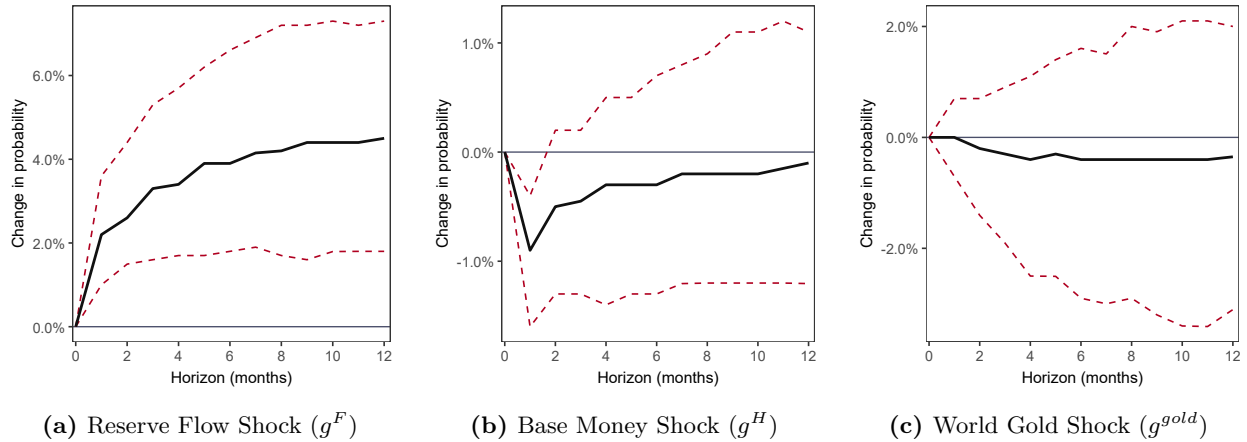


Figure 3.16. Endogenous Regime Switching Probabilities from the Adverse Solvency-Growth State ($\Delta P(\Delta s_{t+h-1} \leq \gamma)$)

Notes: Simulated change in the probability of remaining in the adverse solvency-growth state ($\Delta s_{t-1} \leq -4.615\%$, reflecting conditions of central-bank insolvency) following a one-standard-deviation structural innovation (u_t). Solid lines represent median bootstrap trajectories; dashed crimson lines denote 90% empirical bootstrap confidence intervals ($B = 500, R = 1,000$). The horizontal line marks zero change in transition probability.

A transitory monthly disturbance cannot independently breach the -4.615% threshold. Entering the adverse solvency-growth state required the cumulative drainage of foreign exchange reserves over five consecutive quarters (late 1970 through late 1971). That exhaustion was driven by international credit line cancellations and import surges for capital equipment and food.

2. **State persistence under adverse conditions:** Once the lagged Solvency Growth Gap enters the adverse state ($\Delta s_{t-1} \leq -4.615\%$, reflecting conditions of central-bank insolvency), favorable commodity price shocks fail to reduce the simulated probability of remaining in that state ($\Delta P = -0.35\%$ at month 12, with confidence intervals spanning zero). In this dynamic simulation, the system exhibits substantial state persistence rather than rapid self-correction.

3.D.5 Secondary Structural Pathway Trajectories

Figures 3.17, 3.18, and 3.19 document the secondary transmission pathways in the Threshold VAR.

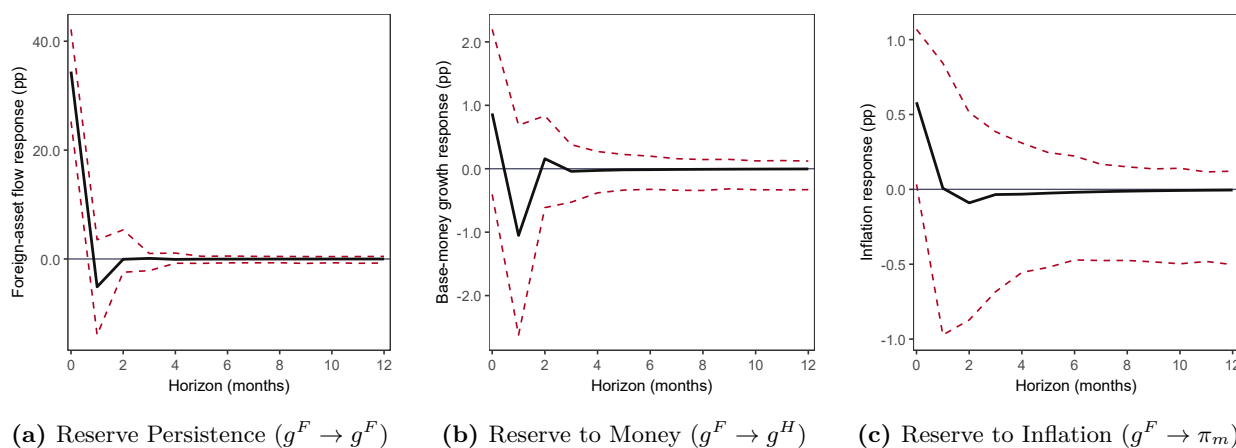


Figure 3.17. Dynamic GIRF Multipliers for Foreign Reserve Shocks in the Adverse Solvency-Growth State
Notes: Generalized Impulse Response Functions over a 12-month horizon in the adverse solvency-growth state ($\Delta s_{t-1} \leq -4.615\%$, reflecting conditions of central-bank insolvency). Solid black lines represent median bootstrap responses to a one-standard-deviation reserve flow innovation; dashed crimson lines depict 90% empirical bootstrap confidence intervals. The horizontal line marks zero.

3.D.6 Diagnostics for Block-Exogeneity and Structural Exclusion Restrictions

Table 3.13 reports diagnostics for the nine block-exogenous and structurally excluded pathways:

1. **World gold price exogeneity diagnostics:** Innovations to domestic international reserves (g^F), consumer price inflation (π_m), Central Bank base money (g^H), and the real trade gap ($\Delta \ln \Theta$) produce no statistically significant response in world gold prices across all 12 forecast horizons. Duration of significance is zero months throughout, with point estimates below 0.055 percentage points. The data do not provide evidence against treating Chile as a price taker in international gold and hard-currency markets.
2. **Structural trade imbalance exogeneity diagnostics:** Domestic nominal shocks to money and inflation produce no statistically significant response in the real structural trade imbalance ($\Delta \ln \Theta$). The trade gap was governed by external terms of trade and historical intermediate

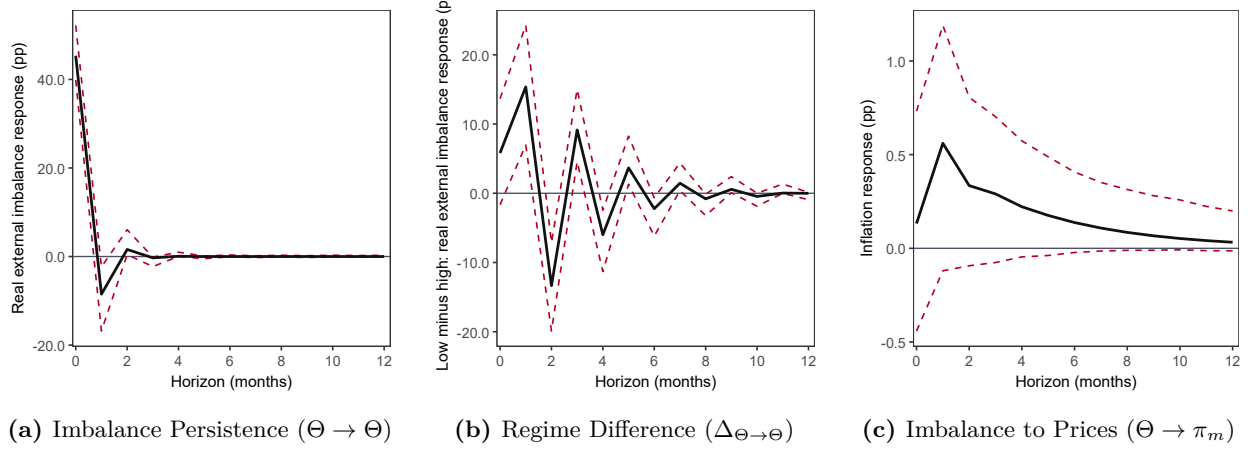


Figure 3.18. Dynamic GIRF Multipliers for Real Structural Imbalance Shocks

Notes: Generalized Impulse Response Functions over a 12-month horizon following a one-standard-deviation real structural trade imbalance shock ($\Delta \ln \Theta$). Solid black lines trace median bootstrap responses; dashed crimson lines denote 90% empirical bootstrap confidence intervals.

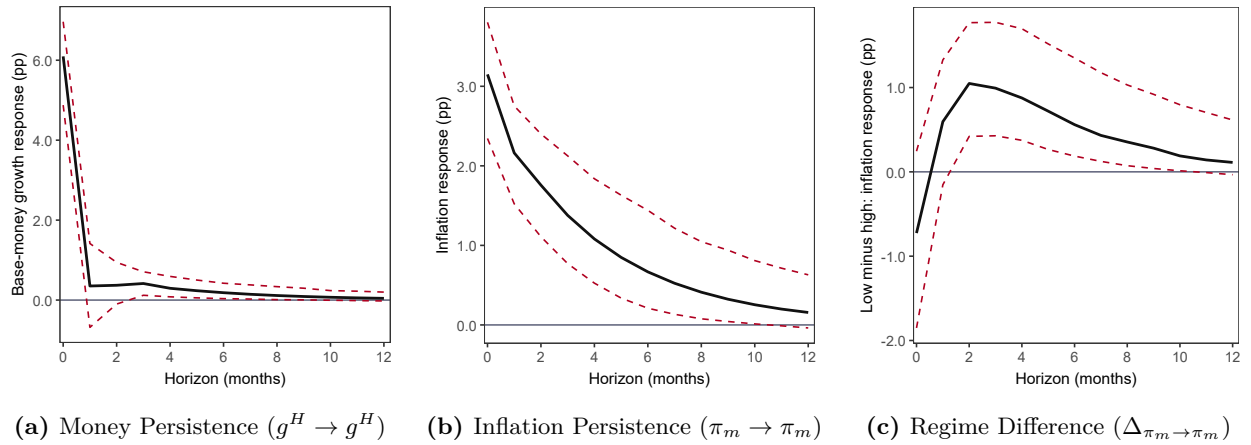


Figure 3.19. Dynamic GIRF Multipliers for Domestic Liquidity and Cost-Push Inertia

Notes: Generalized Impulse Response Functions over a 12-month horizon in the adverse solvency-growth state ($\Delta s_{t-1} \leq -4.615\%$, reflecting conditions of central-bank insolvency). Solid black lines trace median bootstrap responses; dashed crimson lines denote 90% empirical bootstrap confidence intervals.

import requirements for copper mining equipment and industrial fuels.

3. **Direct reserve and credit exclusions:** Imposing the restriction that the real trade gap ($\Delta \ln \Theta$) does not enter directly into contemporaneous reserve flow (g^F) or base money creation (g^H) equations results in exact impact neutrality ($h_0 = 0.000$). This restriction aligns with the directional Granger tests. Structural trade gaps operated sequentially through import bottlenecks and price adjustments ($\Delta \ln \Theta \rightarrow \pi_m$). The Central Bank then accommodated these price increases defensively ($\pi_m \rightarrow g^H$) to keep factories operating and workers paid.

Appendix 3.E Linear VAR Diagnostic Space and Auxiliary Specifications

3.E.1 Diagnostic Admissibility Criteria and Stopping Rule

In accordance with the empirical discipline of this chapter, linear vector autoregressive models are admitted to the main text’s inferential core only when they satisfy three baseline econometric criteria:

1. **Modulus Stability (Criterion A):** All characteristic roots of the companion matrix lie strictly within the unit circle ($\max |\lambda_i| < 1$).
2. **System Residual Whitening (Criterion B):** The multivariate Breusch–Godfrey Lagrange Multiplier test fails to reject the null of no residual autocorrelation at the 5% level ($p > 0.05$).
3. **Equation Residual Whitening (Criterion C):** Individual equation Breusch–Godfrey tests fail to reject the null of no autocorrelation at the 5% level ($p > 0.05$) across all equations.

During the exploratory empirical battery, three modular systems failed to satisfy these criteria: System 4 (Real Sector Dualism), System 5 (External Cost-Push), and the full-sample specification of System 6 (Central Bank Solvency). Rather than searching over ad-hoc lag specifications, non-linear trend adjustments, or arbitrary sub-samples until confirming the preferred theoretical hypotheses, the empirical audit sequence established a formal stopping rule. This appendix documents the diagnostic investigations across the lag search space ($k = 1, \dots, 12$), historical partitions (pre- and post-October 1973), and month-of-year seasonal controls, preserving the empirical record for full transparency.

3.E.2 System 4: Real Sector Dualism (Descriptive Only)

System 4 specifies the five-variable vector:

$$Y_t^{(4)} = \begin{bmatrix} \pi_t & g_{H,t} & \Delta \ln \Theta_t & g_{\text{Manuf},t} & g_{\text{Mining},t} \end{bmatrix}', \quad (3.45)$$

estimated over the full monthly sample (1960:01–1980:12, $N = 250$). Table 3.16 reports the exploratory Granger causality tests evaluated at the SBIC-selected lag ($k^* = 2$).

Table 3.16. Exploratory Granger Tests: Real Sector Dualism ($N = 250$, Full Sample)

Dependent Variable	Predictor	Lag (k^*)	F -Statistic	p -value	$\sum \hat{\beta}_i$
Manufacturing Output ($g_{\text{Manuf},t}$)	Base Money Growth ($g_{H,t}$)	2	3.636**	0.0278	−0.333
Base Money Growth ($g_{H,t}$)	Manufacturing Output ($g_{\text{Manuf},t}$)	2	2.306	0.1018	+0.046
Mining Output ($g_{\text{Mining},t}$)	Base Money Growth ($g_{H,t}$)	2	0.908	0.4046	−0.096
Base Money Growth ($g_{H,t}$)	Mining Output ($g_{\text{Mining},t}$)	2	0.654	0.5211	+0.022
Inflation (π_t)	Manufacturing Output ($g_{\text{Manuf},t}$)	2	7.103***	0.0010	−0.201
Manufacturing Output ($g_{\text{Manuf},t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	2	1.526	0.2195	+0.076

Notes: Evaluated at $k^* = 2$. Status: DESCRIPTIVE ONLY. The system exhibits persistent residual autocorrelation (System Breusch–Godfrey $\chi^2 = 228.74, p < 0.0001$; minimum equation $p < 0.0001$). Residual autocorrelation persists when extending lags up to $k = 12$, under historical calendar partitioning at October 1973, and with seasonal dummy controls. Accordingly, these statistics are not used for formal directional inference.

3.E.3 System 5: External Cost-Push Transmission (Inconclusive)

System 5 specifies the five-variable external transmission vector:

$$Y_t^{(5)} = \begin{bmatrix} g_{P,\text{gold},t} & g_{e,t} & \Delta \ln \Theta_t & g_{H,t} & \pi_t \end{bmatrix}', \quad (3.46)$$

estimated over 1960:01–1980:12 ($N = 250$). Table 3.17 reports exploratory Granger tests at the SBIC-selected lag ($k^* = 1$).

Table 3.17. Exploratory Granger Tests: External Cost-Push Transmission ($N = 250$, Full Sample)

Dependent Variable	Predictor	Lag (k^*)	F -Statistic	p -value	$\sum \hat{\beta}_i$
World Gold Price ($g_{P,\text{gold},t}$)	Domestic Aggregates	1	...	> 0.10	...
Inflation (π_t)	Structural Imbalance ($\Delta \ln \Theta_t$)	1	5.113**	0.0246	−0.019
Devaluation Rate ($g_{e,t}$)	Inflation (π_t)	1	23.305***	< 0.0001	+0.689
Base Money Growth ($g_{H,t}$)	Devaluation Rate ($g_{e,t}$)	1	17.037***	< 0.0001	+0.161
Inflation (π_t)	Devaluation Rate ($g_{e,t}$)	1	1.697	0.1939	+0.046

Notes: Evaluated at $k^* = 1$. Status: INCONCLUSIVE. The system exhibits persistent residual autocorrelation (System Breusch–Godfrey $\chi^2 = 310.16, p < 0.0001$; minimum equation $p < 0.0001$). Extending lag length to $k = 12$, calendar partitioning, and seasonal controls fail to whiten the residuals. Furthermore, key relations display sensitivity to multivariate conditioning. These results are retained for documentation but not used for directional inference.

3.E.4 System 6: Full-Sample Central Bank Solvency Diagnostics

In the full sample (1960:01–1980:12, $N = 250$), System 6 fails residual whitening across all tested lags $k \in \{1, 2, 3, 4\}$ (System Breusch–Godfrey $p < 0.0001$). However, partitioning the sample at October 1973 reveals that residual autocorrelation is concentrated in the post-1973 period under the crawling-peg regime. In the pre-October 1973 subperiod, residual autocorrelation is fully resolved at lag $k = 3$ (System Breusch–Godfrey $p = 0.1262$; minimum equation $p = 0.0557$). As reported in Section 3.5.3.4 and Table 3.8, System 6 is admitted to the main text strictly as a qualified, historically specific result for the pre-1973 era.

Conclusion

The three essays make visible a problem that neither capacity measurement nor a monetary account of crisis can settle alone: the reproduction of production depends on relations that a fixed technical benchmark or an autonomous monetary impulse leaves unexplained. Identifying productive capacity requires an account of how it is formed. Explaining its formation requires attention to accumulation, distribution, technique, and the organization of production. In the periphery, sustaining production also requires access to foreign exchange and international settlement. These connections give the dissertation its unity while preserving the difference between a long-run structural relation and a historically specific crisis.

From the Capacity Benchmark to the Conditions of Reproduction

Chapter 1 establishes the empirical difficulty of separating the productive-capacity denominator from the relations governing its formation. In the full-sample US replication, the bivariate output–capital system does not sustain the required long-run properties. The surviving relation incorporates the rate of exploitation and historical structure. The measurement problem is therefore substantive: changing the treatment of distribution changes the basis on which capacity can be identified. A fitted output path alone cannot establish an autonomous technical ceiling.

Chapter 2 develops the structural account left open by that result. Accumulated capital becomes usable capacity through choices of technique made by decentralized and internally divided firms. Distribution affects those choices, and the composition of the capital stock conditions the resulting transformation. The estimated elasticities below unity mean that capacity formation does not keep pace proportionally with accumulation within the chapter’s specifications. They do not mean that productive capacity must fall in absolute terms or that a crisis follows at a determinate date. The distinction between capacity formation and realized utilization also matters: a changing transformation elasticity does not itself establish how much of the resulting capacity will be used.

Reading the first two essays together thus changes the status of the capacity benchmark. It becomes a relation to be explained and estimated, rather than an engineering constant that can be imported into an analysis of demand or profitability. Functional distribution, labor-process conflict, and capital composition enter this account at different levels. Their connection supplies an interpretation of the macroeconomic relation without making an aggregate coefficient a direct estimate of every organizational process inside the firm.

Peripheral Accumulation, International Money, and Agency

The Chilean extension introduces a material limit to the conversion of an investment incentive into productive capacity. When machinery must be imported, access to foreign exchange conditions whether firms can act on a distributive incentive to mechanize. Chapter 2’s sharply attenuated mechanization response in the constrained state places the balance-of-payments constraint inside the transformation process. The center–periphery comparison therefore concerns the conditions under which accumulation can proceed. Different national specifications and historical samples limit direct numerical comparison, but they do not erase the analytical importance of import dependence.

Chapter 3 shows why the explanation of peripheral crisis cannot stop at that long-run relation. The Unidad Popular confronted the reproduction of enterprise liquidity and international payments as well as productive capacity. The open-economy balance-sheet framework connects domestic monetary liabilities to reserve assets denominated in internationally acceptable means of settlement. Financial subordination and currency hierarchy describe the broader international relations; the solvency measure describes a specific balance-sheet condition within them. Domestic monetary authority does not, by itself, provide command over world money.

The historical analysis distinguishes global monetary instability following the breakdown of Bretton Woods, commodity and import-price movements, targeted bilateral and commercial financial pressure, and domestic distributive and political conflict. Their interaction narrowed the BCCh's policy space without determining one inevitable response. The bank's choice to accommodate enterprise liquidity needs remains an institutional action under constraint. The dynamic estimates provide evidence consistent with that interpretation: in the adverse solvency-growth state, reverse accommodation is more persistent and cumulatively larger than forward monetary transmission over the reported horizon. Forward transmission remains present. Endogenous accommodation therefore qualifies an account centered on autonomous money creation without implying that monetary expansion has no inflationary consequences.

Together, the essays distinguish two roles of the external constraint. It can restrict the acquisition of the means of production, and it can condition the monetary and financial processes required to sustain ongoing production. The long-run mechanization threshold and the monthly solvency-growth threshold address those different questions. Neither should be read as the other's statistical validation. Their connection is an argument about the conditions of peripheral reproduction, not one universal crisis law spanning all three essays.

Limits and Further Inquiry

The dissertation's limits follow from this division of evidence. Chapter 1 uses an aggregate, single-sector framework and a time-invariant transformation parameter. Chapter 2 extends the structural account but does not supply a complete theory of crisis or profitability, and its national specifications are not fully harmonized. Chapter 3 establishes directional precedence and state-dependent conditional responses rather than experimental counterfactual identification. Unobserved common shocks, historical breaks, and the distance between macroeconomic aggregates and institutional decisions remain consequential for interpretation.

Two extensions would deepen the connections developed here. More comparable capital-composition series for the United States and Chile would allow the transformation relation to be estimated on a more common basis. Historical evidence tracing credit allocation between the BCCh, commercial banks, and enterprises would connect the aggregate accommodation patterns more closely to institutional decisions. Comparative research could then examine whether the Chilean solvency dynamics recur elsewhere, without assuming that the estimated threshold transfers unchanged.

Capitalist reproduction cannot be understood by assuming that accumulated capital, productive capacity, and macroeconomic stability expand in step. Connecting capacity measurement, structural explanation, and historical monetary analysis, while retaining the limits of their evidence, is the dissertation's contribution to the political economy of accumulation and crisis in center and periphery.